4008 E / 4008 B / 4008 H / 4008 S

Hemodialysis Machine

Technical Manual

Fresenius Medical Care

Important Information

Organization of the Technical Manual

Page identification

Page identification 1-3 is to be interpreted as: Chapter 1, page 3.

Editorial information

This Technical Manual has been updated to

4/09.03 = 4th edition, September 2003

In case of updates, the pages concerned will be replaced.
Refer to the table below to verify that the Technical Manual is up-to-date.

Page(s)	Current version
0-1	4/09.03
0-2 – 0-4	4/08.03
0-5 – 0-6	4/09.03
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7-1 – 7-2	4/08.03
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Modifications

Manual changes will be released as new editions, supplement sheets or product information. In general, this manual will be subject to modification.

How to use the technical manual

Intended use

This Technical Manual is intended for service technicians. It can be used both for first studies (to acquire a basic knowledge) and for reference purposes (for TSC, maintenance and repair). The study of this manual, however, does not replace the training courses offered by the manufacturer.

Requirements

Knowledge of the current Operating Instructions for the respective system. Experience in mechanics, electrical and medical engineering.

Specifications

For the specifications of the respective system, refer to the current Operating Instructions.

Circuit diagrams and component layouts

The identification on the PCB permits the user/technician to verify if the circuit diagram/component layout matches the PCB actually installed in the system.

Explanation of the Note and Caution symbols used



Note
Informs the operator that in case of a failure to follow the steps as described, a specific function will be executed incorrectly or will not be executed at all, or will not produce the desired effect.



Caution
Advises the operator against certain procedures or actions that could cause damage to the equipment or may have adverse effects on operators and patients.

Instructions to be Observed when Servicing the System

Authorized persons

Assembly, extensions, adjustments, modifications or repairs may only be carried out by the manufacturer or persons authorized by him.

Test equipment and accessories

The activities described in the Technical Manual require the availability of the necessary technical test equipment and accessories.

Precautions

When working on the open system, the following precautions must be respected:
- Protect the components against ingress of fluids.
- Do not touch live parts (e. g. connectors of the power cord or heater).

ESD precautions

When repairing and when replacing spare parts, observe the applicable ESD precautions (e.g. EN 100 015-1).

Addresses

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Local Service

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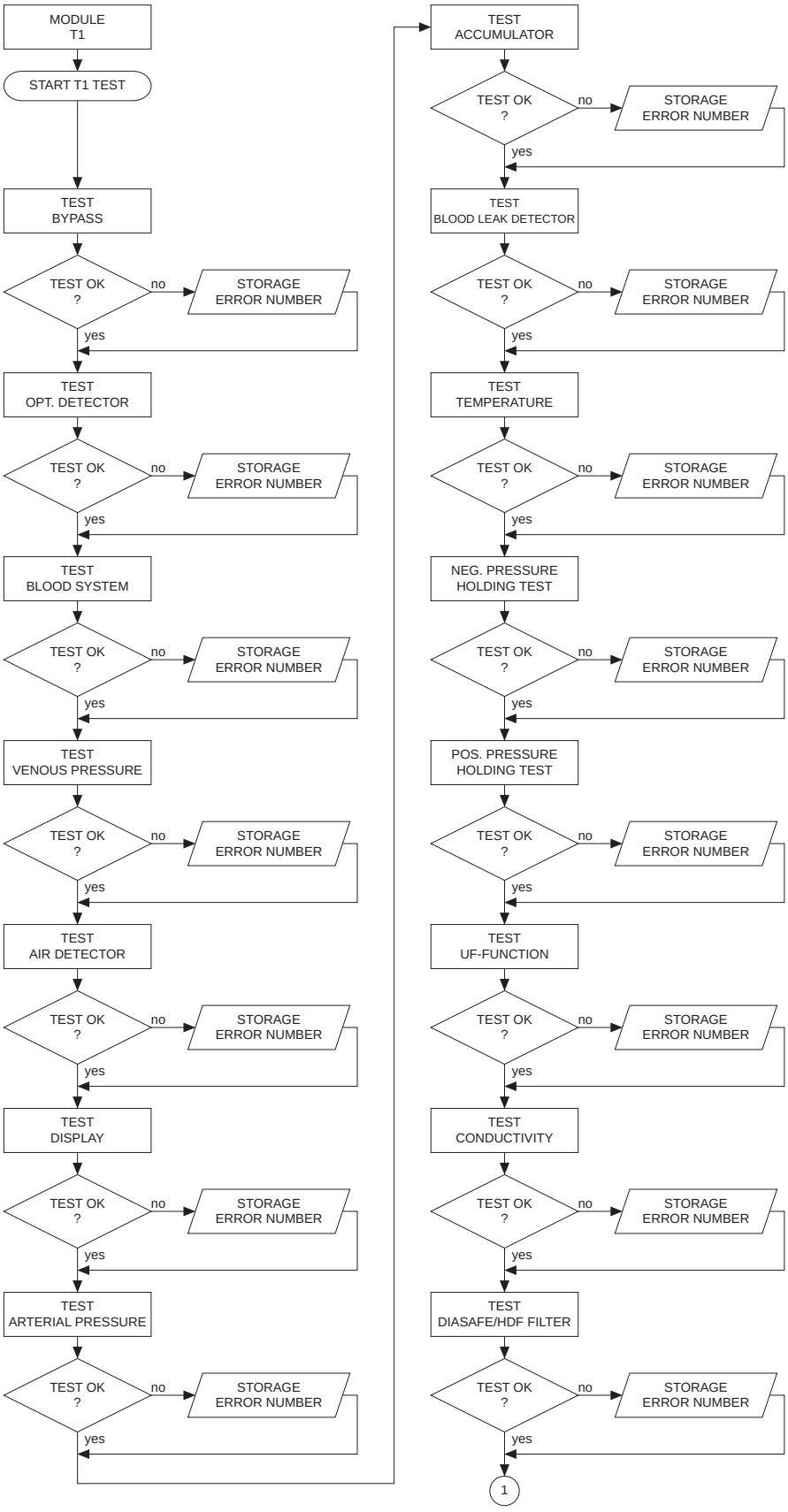
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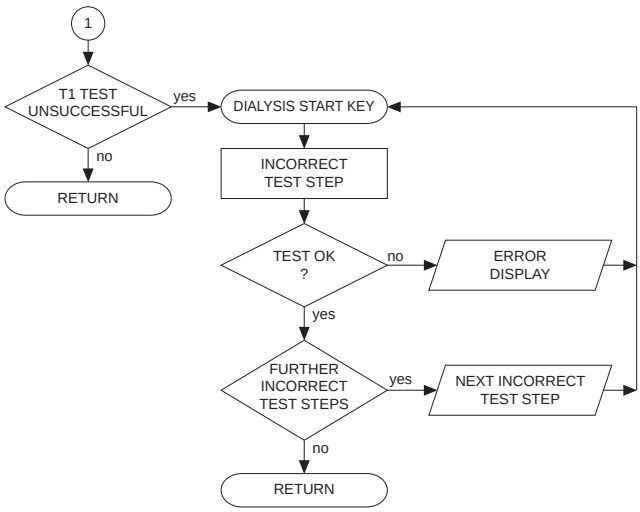
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1.1 Description of the T1 test

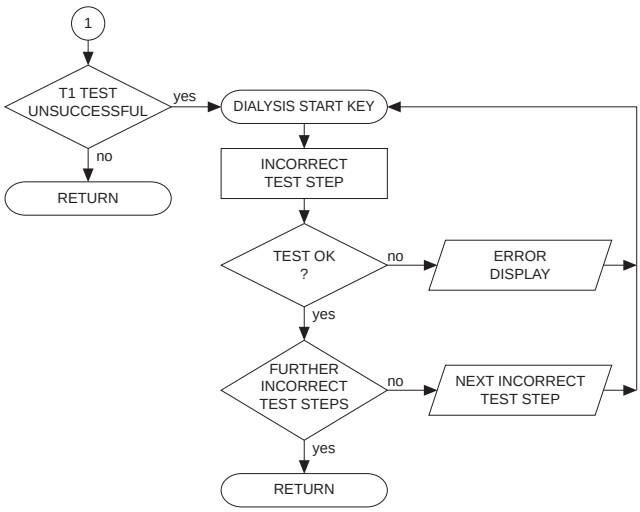
1.1.1 T1 test flow diagram, serial program steps




1
T1 TEST UNSUCCESSFUL
yes
no
RETURN
DIALYSIS START KEY
INCORRECT TEST STEP
TEST OK ?
no
ERROR DISPLAY
yes
FURTHER INCORRECT TEST STEPS
yes
NEXT INCORRECT TEST STEP
no
RETURN

1.1.2 T1 test flow diagram, parallel program steps




1
T1 TEST UNSUCCESSFUL
yes
no
RETURN
DIALYSIS START KEY
INCORRECT TEST STEP
TEST OK ?
no
ERROR DISPLAY
yes
FURTHER INCORRECT TEST STEPS
no
NEXT INCORRECT TEST STEP
yes
RETURN

1.1.3 Description of the T1 test incl. error messages

- **Notes on the descriptions of errors**

The values specified are internal mathematical values, which are used in the program.

Resolutions and tolerances are as follows:

1.	Arterial pressure:	3 mmHg/digit	and ± 1 digit of measured value.
2.	Venous pressure:	3 mmHg/digit	and ± 1 digit of measured value.
3.	Dialysate pressure, coarse:	6.0 mmHg/digit	and ± 1 digit of measured value.
	fine:	0.5 mmHg/digit	and ± 1 digit of measured value.
4.	CD resolution:	0.06 mS/digit	and ± 1 digit of measured value.
5.	Temperature:	0.05 °C/digit	and ± 1 digit of measured value.

- **Prerequisites for starting and running the test**

Error message	*Description*
Power failure	Power failure while the test is in progress
Dialines not conn	The dialysate lines are not in the interlock shunt.
Shunt Cover open	The interlock shunt is open.
Connect Conc.Line **Wrong conc. supply**	The concentrate connector is in the rinse chamber, or concentrate is not connected at all. The error message depends on the central delivery system preselected in the setup menu.
Blood Sensed by OD	The optical detector senses blood in the system.
Flow alarm	Line to or from the dialyzer kinked, malfunctions in the hydraulics.
Water alarm	Water supply interrupted.

- **Overview of the individual test steps**

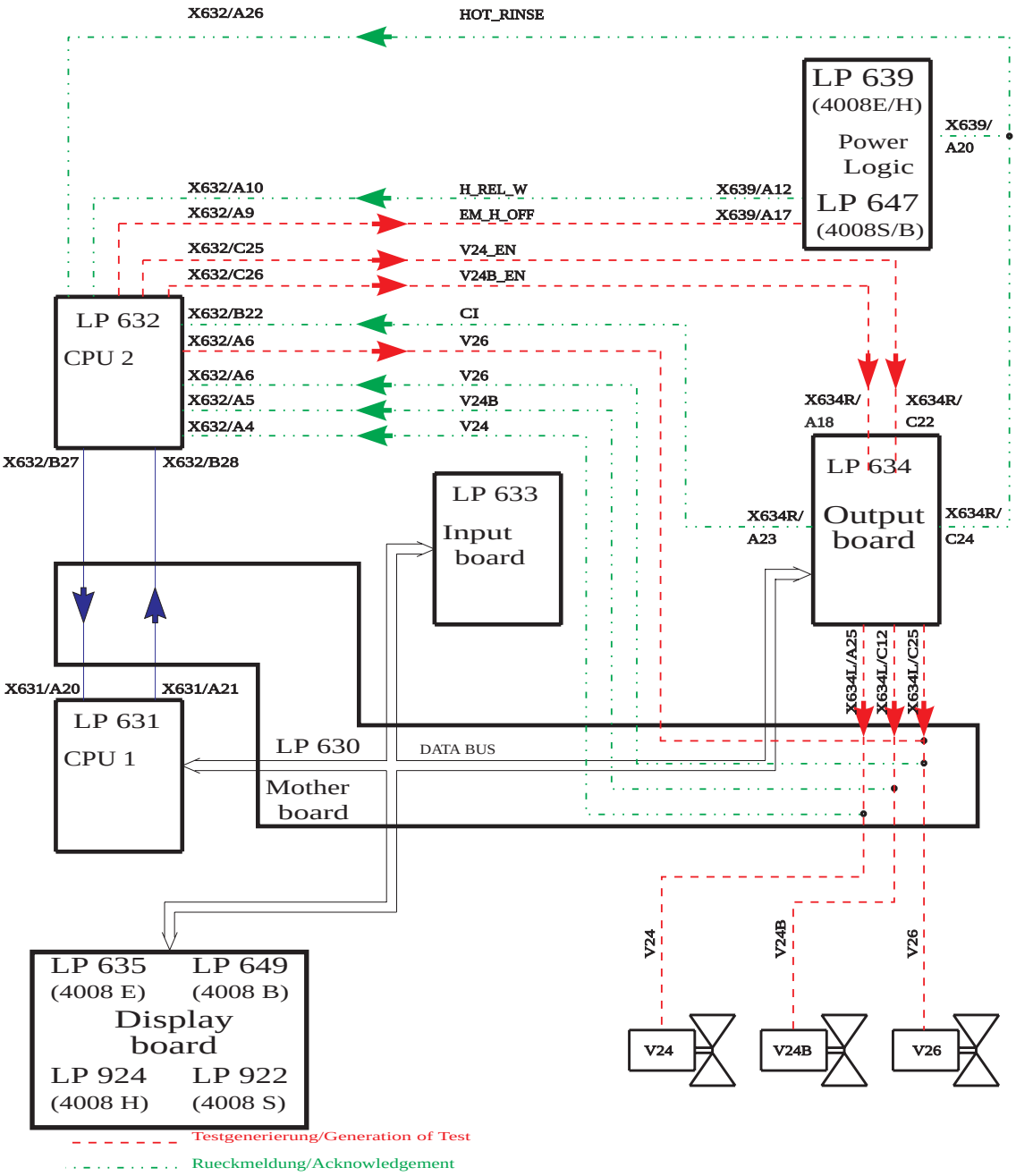
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- **Bypass test**

Test description:

Check of the following functions:
– Heater relay
– Bypass (electric)
– Check of the temperature range changeover

Illustration:



Error description:

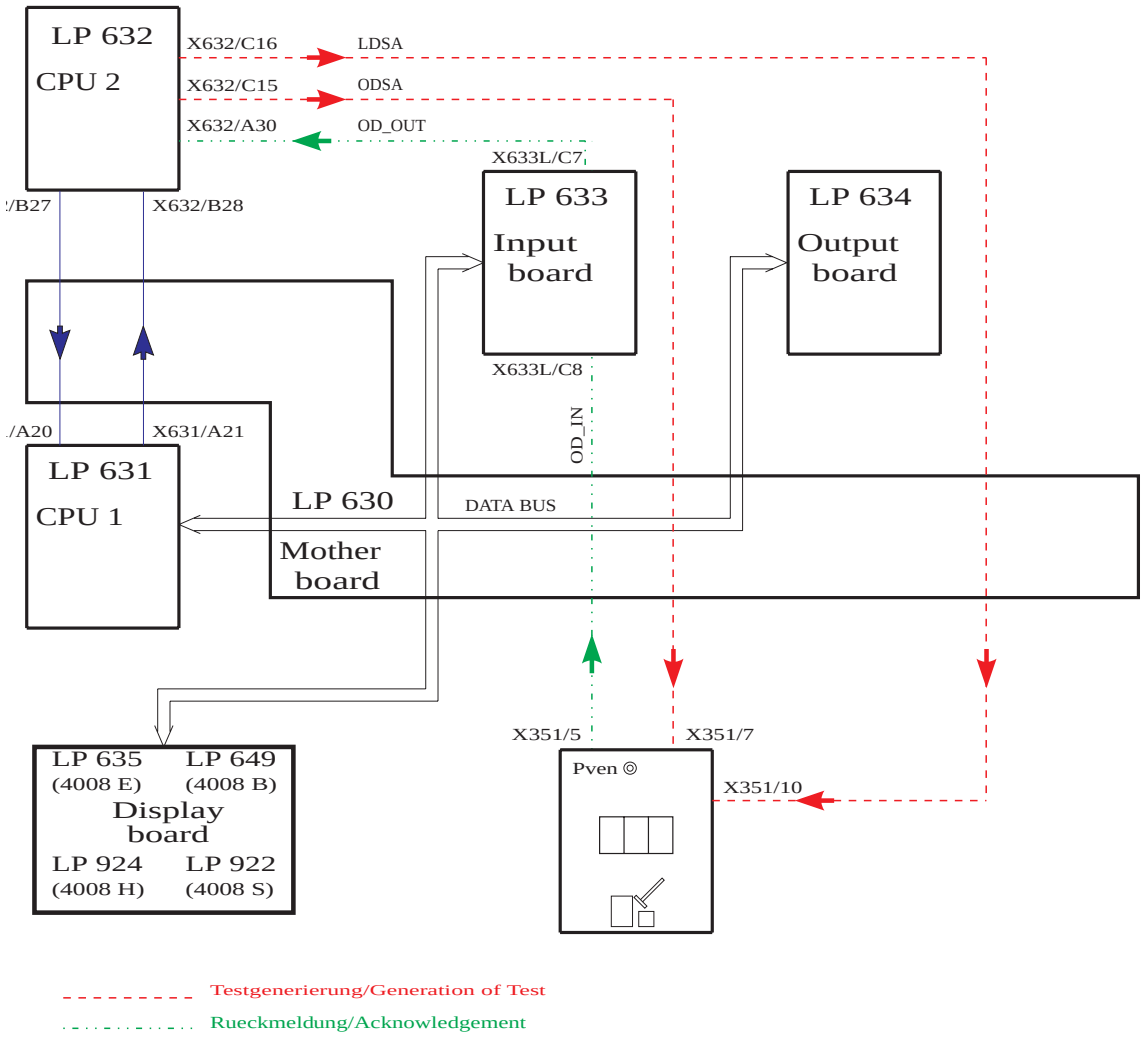
Error message	*Description*
F 01 Bypass	The heater relay is switched off. – Acknowledgement (H_REL_W, X639/A12) → X632/A10, 0 V are missing.
F 02 Bypass	The heater relay cannot be switched off by CPU2. – Acknowledgement (H_REL_W, X639/Y12) → X632/A10, 12 V are missing. – Control line (EM_H_OFF, X632/A9) → X639/A17, 12 V are missing.
F 03 Bypass	The temperature measurement range is set to hot rinse. – Control line (HOTRINSE, X634R/C24) → X639/A20, 0 V are missing. – Acknowledgement (HOTRINSE, X634R/C24) → X632/A26, 0 V are missing.
F 04 Bypass	The extended bypass cannot be correctly switched by CPU2 (V24 = off, V26 = on, V24B = off). – Acknowledgement (V24, X637/C1) → X632/A4, 24 V are missing. – Acknowledgement (V26, X637/C2) → X632/A6, 0 V are missing. – Acknowledgement (V24B, X637/C23) → X632/A5, 24 V are missing.
F 05 Bypass	The extended bypass cannot be correctly switched off by CPU2 (V24 = on, V26 = off, V24B = on). – Acknowledgement (V24, X637/C1) → X632/A4, 0 V are missing. – Acknowledgement (V26, X637/C2) → X632/A6, 24 V are missing. – Acknowledgement (V24B, X637/C23) → X632/A5, 0 V are missing.
F06 Bypass	CPU1 fails to set the temperature control to hot rinse. – Control line (HOTRINSE, X634R/C24) → X639/A20, 12 V are missing. – Acknowledgement (HOTRINSE, X634R/C24) → X632/A26, 12 V are missing.
F 07 Bypass	The extended bypass cannot be correctly switched by CPU1 (V24 = off, V26 = on, V24B = off). – Acknowledgement (V24, X637/C1) → X632/A4, 24 V are missing. – Acknowledgement (V26, X637/C2) → X632/A6, 0 V are missing. – Acknowledgement (V24B, X637/C23) → X632/A5, 24 V are missing.
F08 Bypass	CPU1 fails to reset the temperature control to dialysis. – Control line (HOTRINSE, X634R/C24) → X639/A20, 0 V are missing. – Acknowledgement (HOTRINSE, X634R/C24) → X632/A26, 0 V are missing.
F09 Bypass	The extended bypass cannot be correctly switched off by CPU1 (V24 = on, V26 = off, V24B = on). – Acknowledgement (V24, X637/C1) → X632/A4, 0 V are missing. – Acknowledgement (V26, X637/C2) → X632/A6, 24 V are missing. – Acknowledgement (V24B, X637/C23) → X632/A5, 0 V are missing.
F95 Bypass	System error

- **Optical detector test**

  Test description:

  Attenuation of the optical detector.
  Check of the acknowledgement of the optical detector.

  Illustration:



Error description:

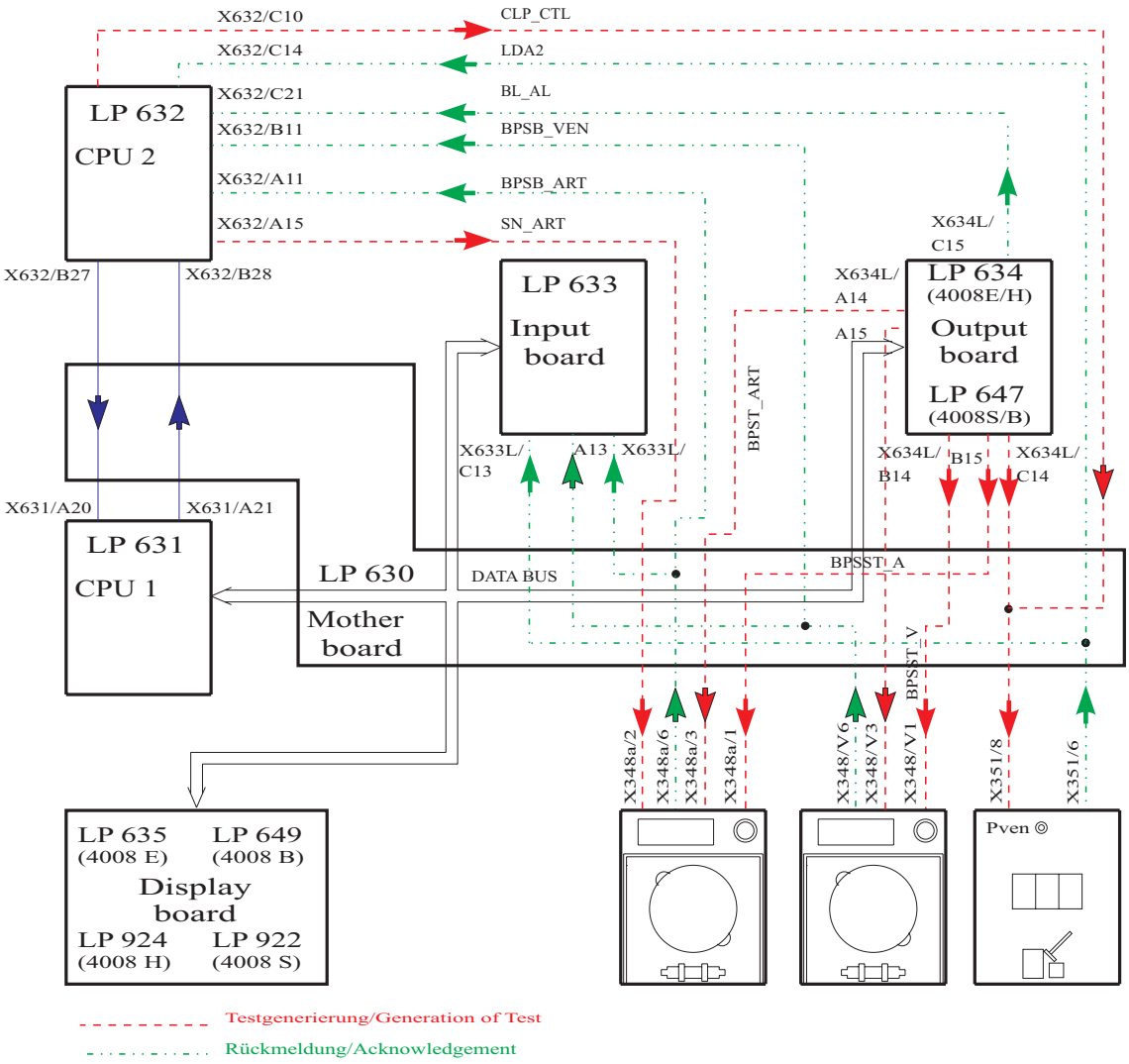
Error message	*Description*
F01 opt. Detector	CPU1 interprets the optical detector in a different way than does CPU2. – Acknowledgement (OD_OUT, X633L/C7) → X632/A30 and the digital input of P.C.B. LP 633 measure different levels.
F02 opt. Detector	CPU2 fails to recognize blood in the system. – Acknowledgement (OD_OUT, X633L/C7) → X632/A30, 0 V are missing. – Detuning (ODSA, X632/C15) → X351/7 not 12V.
F03 opt. Detector	CPU1 fails to recognize blood in the system. – Acknowledgement (OD_OUT, X633L/C7) → digital input on P.C.B. LP 633. – Detuning (ODSA, X632/C15) → X351/7 not 12V.
F04 opt. Detector	CPU2 recognizes that the optical detector senses opaque fluid (required because of the test in the cleaning program). – Acknowledgement X632/A30 not 12V. – AD28 defective.
F96 opt. Detector	System error.

- **Blood system test**

  Test description:

  Check of the following functions:
  – Blood alarm acknowledgement
  – Blood pump switch-off

  Illustration:



Error description:

Error message	*Description*
F09 Bloodsystem	Acknowledgement that CPU2 recognizes that the arterial blood pump is inactive (BP not running). – Acknowledgement (BPSB_ART, X348a/6) → X632/A11, 12 V missing. – Control line (BPSST_ART, X634L/B14) → X348a/1, 12 V missing or (BPST_ART, X634L/A14) → X348a/3, 12 V are missing.
F10 Bloodsystem	Acknowledgement that CPU1 recognizes that the arterial blood pump is inactive (BP not running). – Acknowledgement (BPSB_ART, X348a/6) → X633L/A11, 12 V are missing. – Control line (BPSTT_ART, X634L/B14) → X348a/1, 12 V missing or (BPST_ART, X634L/A14) → X348a/3, 12 V missing. – Level is raised during the T1 test.
F11 Bloodsystem	The arterial blood pump cannot be stopped by CPU1. CPU2 recognizes that the arterial blood pump remains active. – Control line (BPSST_ART, X634L/B14) → X348a/1, 0 V missing, as well as (BPST_ART, X634L/A14) → X348a/3, 0 V missing. – Acknowledgement (BPSB_ART, X348a/6) → X632/A11, 0 V are missing. – The level is raised during the T1 test, or the up/down key on the air detector is blocked and the level is constantly raised.
F12 Bloodsystem	The arterial blood pump cannot be stopped by CPU1. CPU1 recognizes that the arterial blood pump remains active. – Control line (BPSST_A, X634L/B14) → X348a/1, 0 V missing, as well as (BPST_ART, X634L/A14) → X348a/3, 0 V missing. – Acknowledgement (BPSB_ART, X348a/6) → X633L/A11, 0 V are missing.
F13 Bloodsystem	*Applicable for SW 4.91/2.91 and higher if SN, ONLINE-HDF or 4008 HDF pump is connected (= ADKS active)* Acknowledgement that CPU2 detects that the pump is inactive (pump is not running). – Acknowledgement (BPSB_VEN, X348V/6) → X632/ B11, 12V missing – Control line (BPSST_VEN, X634L/B15) → X348V/1, 12V missing or (BPST_VEN, X634L/A15) → X348V/3, 12V missing – Transistor T9 on P.C.B. LP754 defective – IC5 on P.C.B. LP632 defective – In 4008 HDF an HDF treatment was performed, followed by a cleaning program with the substituate pump running, then the T1 test has been re-started. The substituate pump must be switched off because otherwise the test step will fail to be passed (problem was corrected with SW 3.20 in 4008 H/S systems: the substituate pump will be switched off automatically on starting a cleaning program).

F14 Bloodsystem

Applicable for SW 4.91/2.91 and higher if SN, ONLINE-HDF or 4008 HDF pump is connected (= ADKS active)

Acknowledgement that CPU1 detects that the pump is inactive (pump is not running).

- Acknowledgement (BPSB_VEN,X348V/6) → X633L/A13, 12V missing
- Control line (BPSST_VEN, X634L/B15) → X348V/1 not 12V or (BPST_VEN, X634L/A15) → X348V/3 not 12V
- IC16 on P.C.B. LP633 defective
- P.C.B. LP633 recognizes Single-Needle pump although it is not connected.

F15 Bloodsystem

Applicable for SW 4.91/2.91 and higher if SN, ONLINE-HDF or 4008 HDF pump is connected (= ADKS active)

CPU1 fails to stop the corresponding blood pump.

CPU2 detects that the pump remains active.

- Control line (BPSST_VEN, X634L/B15) → X348V/1, 0V missing as well as (BPST_VEN, X634L/A15) → X348V/3 not 0V
- Acknowledgement (BPSB_VEN, X348V/6) → X632/B11, 0V missing
- Transistor T9 on P.C.B. LP754 defective
- IC5 on P.C.B. LP632 defective
- During the test the lines are inserted on the corresponding pump using the Start/Stop key.
- P.C.B. LP633 recognizes Single-Needle pump although it is not connected.

F16 Bloodsystem

Applicable for SW 4.91/2.91 and higher if SN, ONLINE-HDF or 4008 HDF pump is connected (= ADKS active)

CPU1 fails to stop the corresponding blood pump.

CPU1 detects that the pump remains active.

- Control line (BPSST_VEN, X634L/B15) → X348V/1 not 0V as well as (BPST_VEN, X634L/A15) → X348V/3 not 0V
- Acknowledgement (BPSB_VEN, X348V/6) → X633L/A13 not 0V
- IC16 on P.C.B. LP633 defective
- P.C.B. LP633 recognizes Single-Needle pump although it is not connected.

F17 Bloodsystem

Applicable for SW 4.91/2.91 and higher if SN, ONLINE-HDF or 4008 HDF pump is connected (= ADKS active)

Although the recognition of the venous blood pump (ADKS) is not acknowledged, the 24-V supply voltage of the pump can be switched off.

- Acknowledgement line (ADKS, X348V/7) → X633L/A10 not 12V
- Acknowledgement (BPSB_VEN, X348V/6) → X633L/A13 not 12V
- Acknowledgement (BPSB_VEN, X348V/6) → X632/B11 not 12V
- Online-HDF has already been switched on during the T1 test.
- IC16 on P.C.B. LP633 defective.

F18 Bloodsystem

Applicable for SW 5.00/4.10 and higher, check of the BPUS signal (CPU, P.C.B. LP 632)

At the beginning of the test step a maximum of 40s may pass until rotation has stopped.If the blood pump is being activated, the rotation stop alarm must have been cleared.

– Acknowledgement line (BPUS, X348A/8) → X632/A13 not 0V
– Acknowledgement line (BPUS, X348A/8) → X632/A13 not 12V
– Blood pump speed is set to “0”: preset speed during the T1 test.

F19 Bloodsystem

Applicable for SW 5.00/4.10 and higher, check of the BPUS signal (CPU, P.C.B. LP 631 via LP 633)

At the beginning of the test step a maximum of 40s may pass until rotation has stopped. If the blood pump is being activated, the rotation stop alarm must have been cleared.

1. Acknowledgement line (BPUS, X348A/8) → X633L/A12 not 0V
2. Acknowledgement line (BPUS, X348A/8) → X633L/A12 not 12V

F95 Bloodsystem

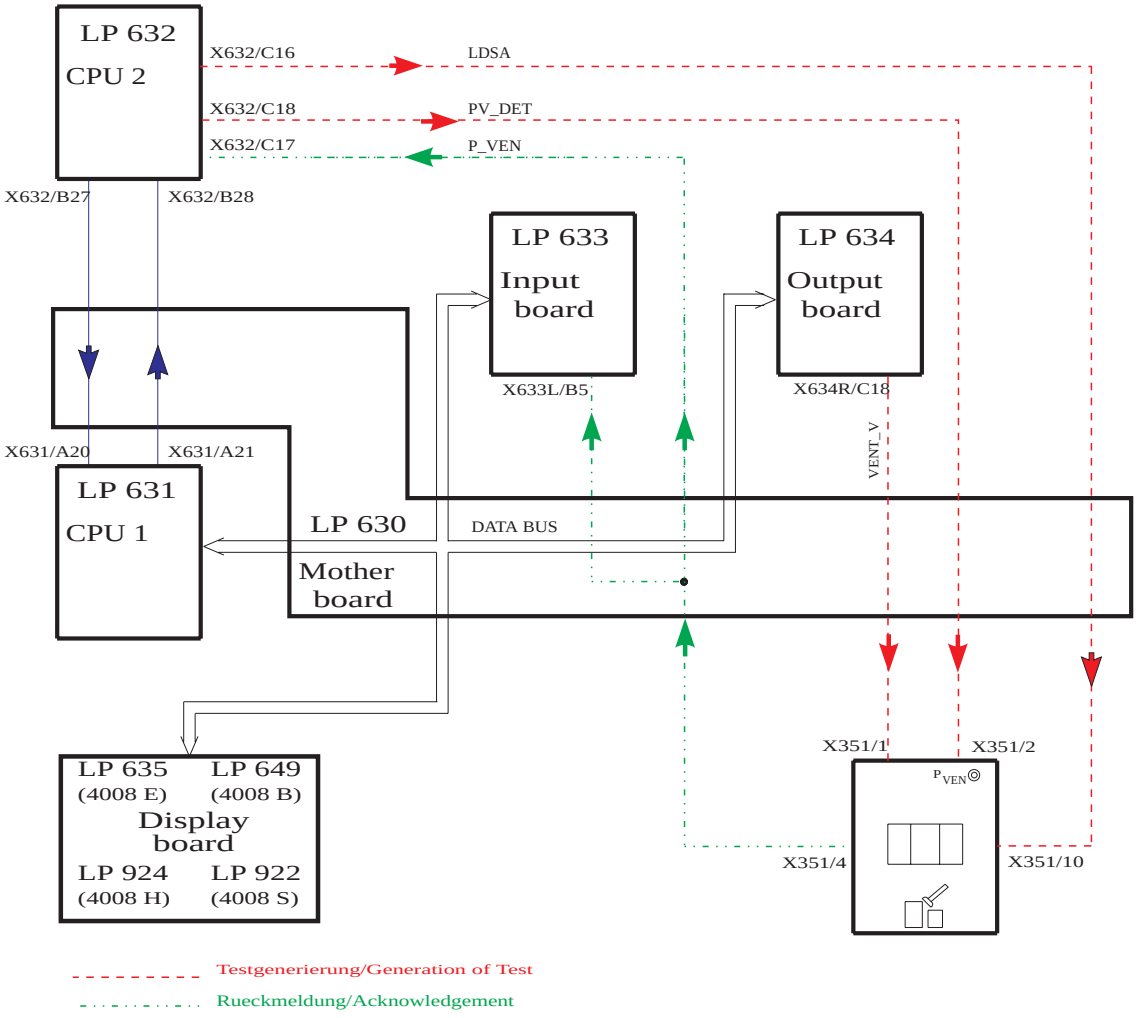
System error.

- **Venous pressure system test**

  Test description:

  Verification of the lower limit by checking the venous zero point.
  The upper limit is tested by detuning the venous pressure unit in positive direction.
  (The venous line clamp is closed during the test.)

  Illustration:



Error description

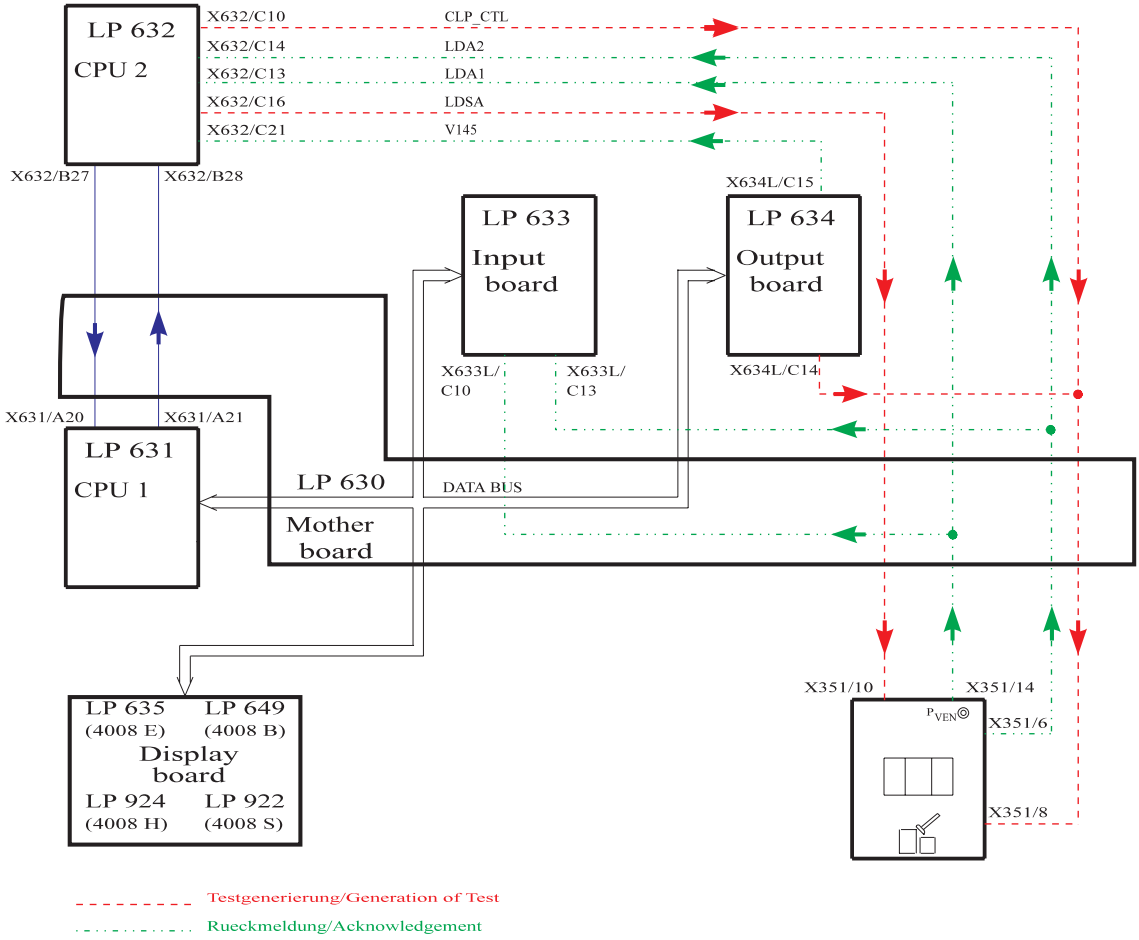
Error message	*Description*
F01 Venous	CPU1 (input board) shows a venous zero point deviation of more than ±12 mmHg (60 s). – Control (VENT_VALVE, X634R/C18) → X351/1 of the vent valve in the LD is defective. – Acknowledgement (P_VEN, X351/4) → X633L/B5 that the voltage value is outside the zero point tolerance. – P-venous has not been calibrated.
F02 Venous	CPU2 shows a venous zero point deviation of more than ±12 mmHg (60 s). – Control (VENT_VALVE, X634R/C18) → X351/1 of the vent valve in the LD is defective. – Acknowledgement (P_VEN, X351/4) → X632/C17, the voltage value is outside the zero point tolerance. – P-venous has not been calibrated.
F03 Venous	With detuning in positive direction, the achieved change in the venous display is less than 100 mmHg (7 s). – The test detuning is defective (PV_DET, X632/C18) → X351/2. – Acknowledgement (P_VEN, X351/4) → X633L/B5, the change in voltage is too low. – P-venous has not been calibrated.
F04 Venous	The deviation in the measured value between CPU1 and CPU2 is higher than ±12 mmHg (if Pven > 100 mmHg). – Acknowledgement (P_VEN, X351/4) → X633L/B5 and X632/C17 measure different voltage values. – P-venous has not been calibrated.
F95 Venous	System error.

- **Air detector test**

  Test description:

  – Test of the air detector by checking the alarm state.
  – Switch-off of the venous line clamp in the air detector module.

  Illustration:



Error description:

Error message	*Description*
F01 Airdetector	CPU1 interprets the air detector signal in a different way than does CPU2. – Acknowledgements (LDA1, X351/14) → X632/C13 and X633L/C10 recognize different signal levels.
F02 Airdetector	The air detector alarm is not recognized by CPU2. – Acknowledgement (LDA1, X351/14) → X632/C13, 0 V are missing. – Transmission weakening (LDSA, X632/C16) → X351/10, 12 V are missing.
F03 Airdetector	Air detector clamps acknowledgement (CPU2) activated (clamp closed). – Acknowledgement (LDA2, X351/6) → X632/C14, 24 V are missing. – Clamp control (CLP_CTL, X634L/C14) → X351/8, 12 V are missing. – Clamp control (CLP_CTL, X632/C10) → X351/8, 12 V are missing.
F04 Airdetector	Air detector clamps acknowledgement (CPU1) activated (clamp closed). – Acknowledgement (LDA2, X351/6) → X633L/C13, 24 V are missing. – Clamp control (CLP_CTL, X634L/C14) → X351/8, 12 V are missing. – Clamp control (CLP_CTL, X632/C10) → X351/8, 12 V are missing.
F05 Airdetector	The blood alarm signal has not been cleared (indicates an alarm). – Acknowledgement (BL_AL, X634L/C15) → X632/C21, 12 V are missing. If the HDF option is used, this signal is not tested (special function).
F06 Airdetector	Closing of the air detector clamp via the CPU2 control line was not possible. – Clamp control (CLP_CTL, X632/C10) → X351/8, 0 V are missing. – Acknowledgement (LDA2, X351/6) → X632/C14, 0 V are missing.
F07 Airdetector	Opening of the air detector clamp via the CPU2 control line was not possible. – Clamp control (CLP_CTL, X632/C10) → X351/8, 12 V are missing. – Acknowledgement (LDA2, X351/6) → X632/C14, 24 V are missing.
F08 Airdetector	Closing of the air detector clamp via the CPU1 control line was not possible, or CPU2 acknowledgement is incorrect. – Clamp control (CLP_CTL, X634L/C14) → X351/8, 0 V are missing. – Acknowledgement (LDA2, X351/6) → X632/C14, 0 V are missing.

F09 Airdetector

Closing of the air detector clamp via the CPU1 control line was not possible, or CPU1 acknowledgement is incorrect.
- Clamp control (CLP_CTL, X634L/C14) → X351/8, 0 V are missing.
- Acknowledgement (LDA2, X351/6) → X633L/C13, 0 V are missing.

F10 Airdetector

The blood alarm message is missing.
- Acknowledgement (BL_AL, X634R/C15) → X632/C21, 0 V are missing.

  If the HDF option is used, this signal is not tested (special function).

F11 Airdetector

Air detector clamps acknowledgement (CPU2) activated (clamp closed).
- Acknowledgement (LDA2, X351/6) → X632/C14, 24 V are missing.
- Clamp control (CLP_CTL, X634L/C14) → X351/8, 12 V are missing.
- Clamp control (CLP_CTL, X632/C10) → X351/8, 12 V are missing.

F12 Airdetector

Air detector clamps acknowledgement (CPU1) activated (clamp closed).
- Acknowledgement (LDA2, X351/6) → X633L/C13, 24 V are missing.
- Clamp control (CLP_CTL, X634L/C14) → X351/8, 12 V are missing.
- Clamp control (CLP_CTL, X632/C10) → X351/8, 12 V are missing.

F13 Airdetector

The blood alarm signal has not been cleared (indicates alarm).
- Acknowledgement (BL_AL, X634L/C15) → X632/C21, 12 V are missing.

  If the HDF option is used, this signal is not tested (special function).

F14 Airdetector

Raise level key on the air detector is constantly active.
- Acknowledgement (LEVEL_UP, X351/3) → X632/C11 not 0V.

F15 Airdetector

Acknowledgement of the supply voltage for the ultrasonic output stage not between 6.5 and 13.5 V after 3 seconds.
- Adapter board AD28 not connected
- Acknowledgement (X351/11 → X633L/25A jumper to X633L/B7) not 12V.
- Relay on AD28 failed to drop.

F16 Airdetector

Acknowledgement of the supply voltage for the ultrasound output stage not >14.5V after 3 seconds.
- Adapter board AD28 not connected.
- Acknowledgement (X351/11 → X633L/25A jumper to X633L/B7) not 16V/24V.
- Relay on AD28 is not controlled.
- No 10-Hz signal at ALARM_REST (X351/12)

F17 Airdetector

Acknowledgement of the supply voltage for the ultrasound output stage not between 6.5 and 13.5 V after 3 seconds.
- Adapter board AD28 not connected
- Acknowledgement (X351/11 → X633L/25A jumper to X633L/B7) not 12V
- Relay on AD28 failed to drop

F95 Airdetector

System error.

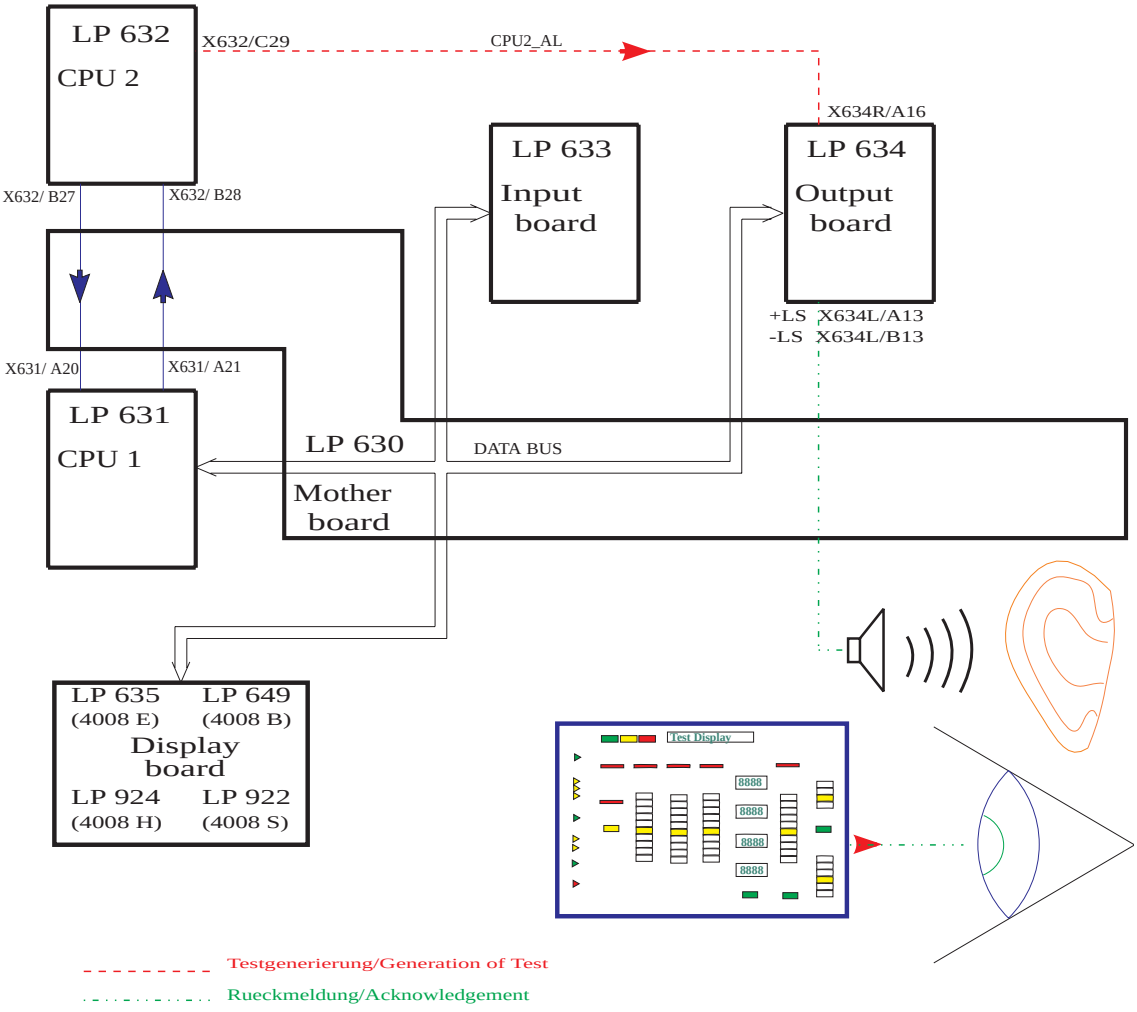
- **Display test**

  Test description:

  Check of all displays and indicators on the monitor front
  – Display test
  – Status LED
  – Alarm LED
  – Seven-segment display, all dark
  – Seven-segment display, all 8888
  – Bar graph
  – CPU1/CPU2 alarm tone

  This display test must be monitored by the user!

  Illustration:



Error description:

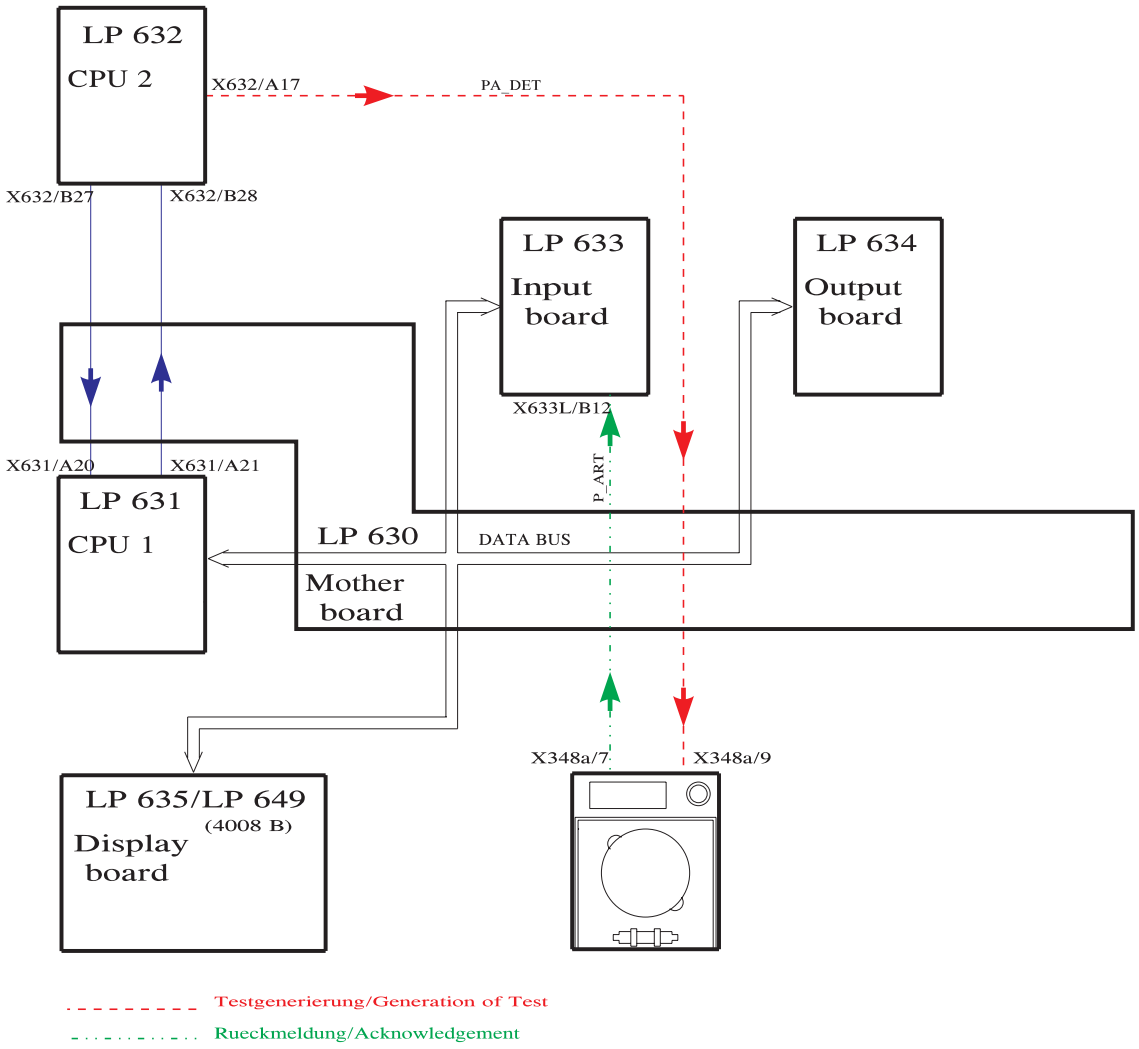
Error message	*Description*
F01 Display	CPU1 failed to start the display test within 5 sec. – The “test started” information transmitted via the serial interface is missing.
F02 Display	CPU1 failed to complete the display test within 120 sec. – The “test completed” information transmitted via the serial interface is missing.
F95 Display	System error.

- **Arterial pressure system test**

  Test description:

  Test of the arterial pressure unit by electronic detuning in positive or negative direction.

  Illustration:



Error description:

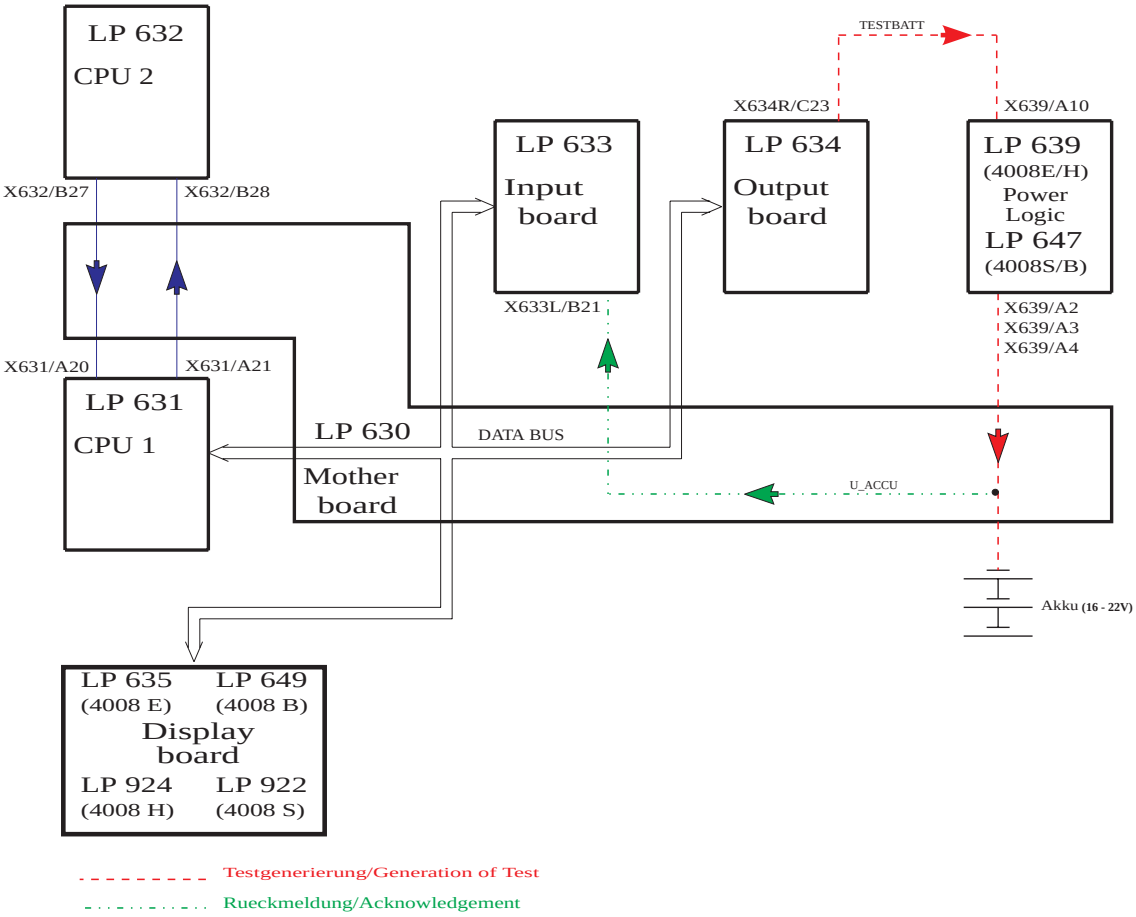
Error message	*Description*
F01 Arterial	With detuning in negative direction, the change achieved on the arterial display is less than 100 mmHg (2 sec). – Acknowledgement (P_ART, X348A/7) → X633L/B12, insufficient voltage change. – Test detuning defective (PA_DET, X632/A17) → X348A/9.
F02 Arterial	With detuning in positive direction, the change achieved on the arterial display is less than 100 mmHg (2 sec). – Acknowledgement (P_ART, X348A/7) → X633L/B12, insufficient voltage change. – Test detuning defective (PA_DET, X632/A17) → X348A/9.
F95 Arterial	System error.

- **Battery test**

  Test description:

  Check of the battery voltage under load.

  Illustration:



Error description:

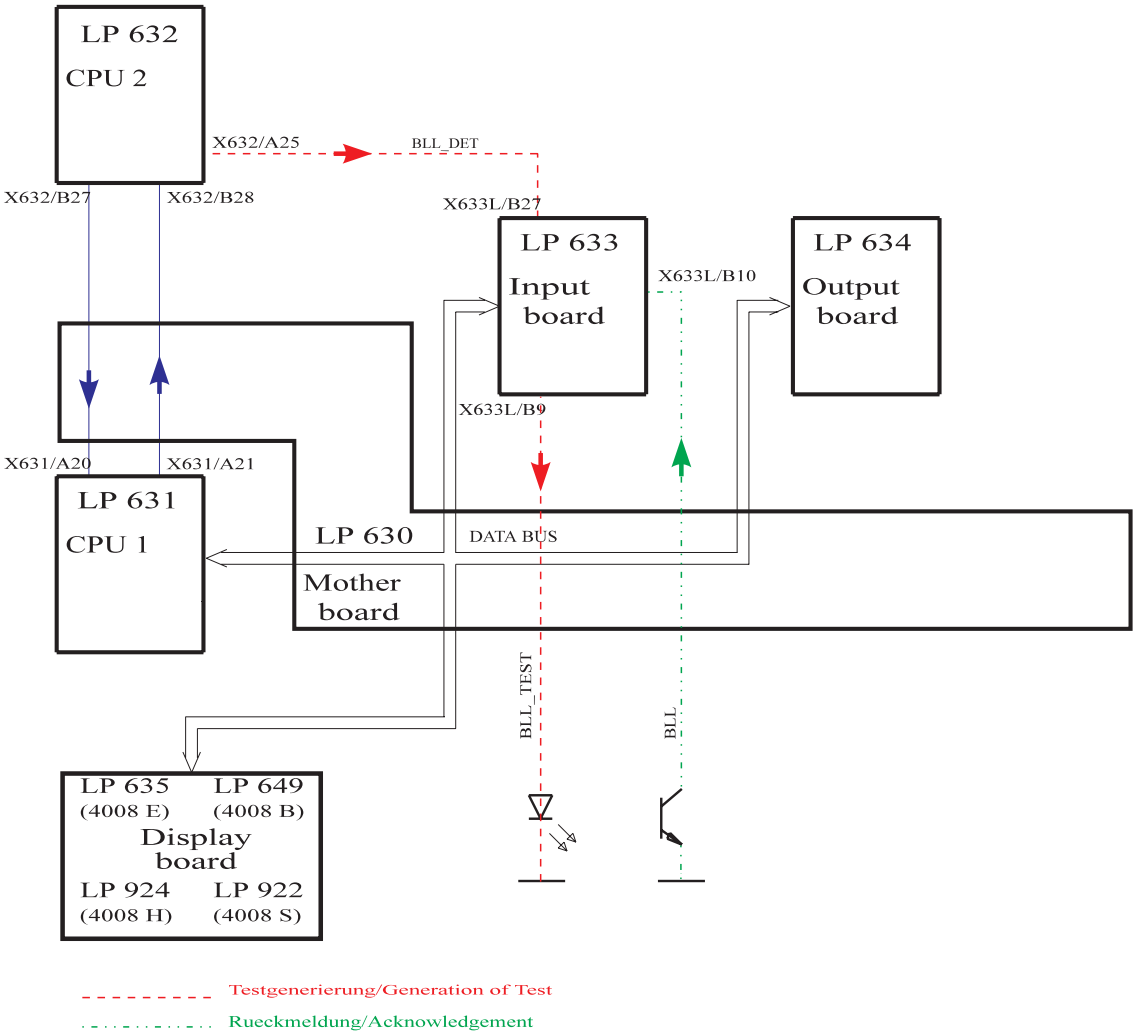
Error message	*Description*
F01 Accumulator	CPU1 failed to complete the battery test within 5 sec. – The “test completed” information transmitted via the serial interface is missing.
F02 Accumulator	The battery charge is insufficient for 15 min emergency operation (maybe no battery connected). – The battery voltage (U_ACCU, ...) → X633L/B21 dropped below 17.6 V. – Acknowledgement (U_ACCU, ...) → X633L/B21 of the battery voltage defective.
F03 Accumulator	The test circuit on P.C.B. LP 639 defective. – The test level is incorrect (TESTBATT, X634R/C23) → X639/A10, the 12-V pulse is missing (100 ms). – Power supply unit LP639 SI5 or in 4008B/S systems fuse in the base defective. – R39 on P.C.B. LP639 (4008E/H) or P.C.B. LP647 (4008B/S) defective, possibly caused by flickering power supply unit.
F95 Accumulator	System error.

- **Blood leak test**

  Test description:

  Test of the blood leak detector by lowering the capacity of the transmitting diode.

  Illustration:



Error description:

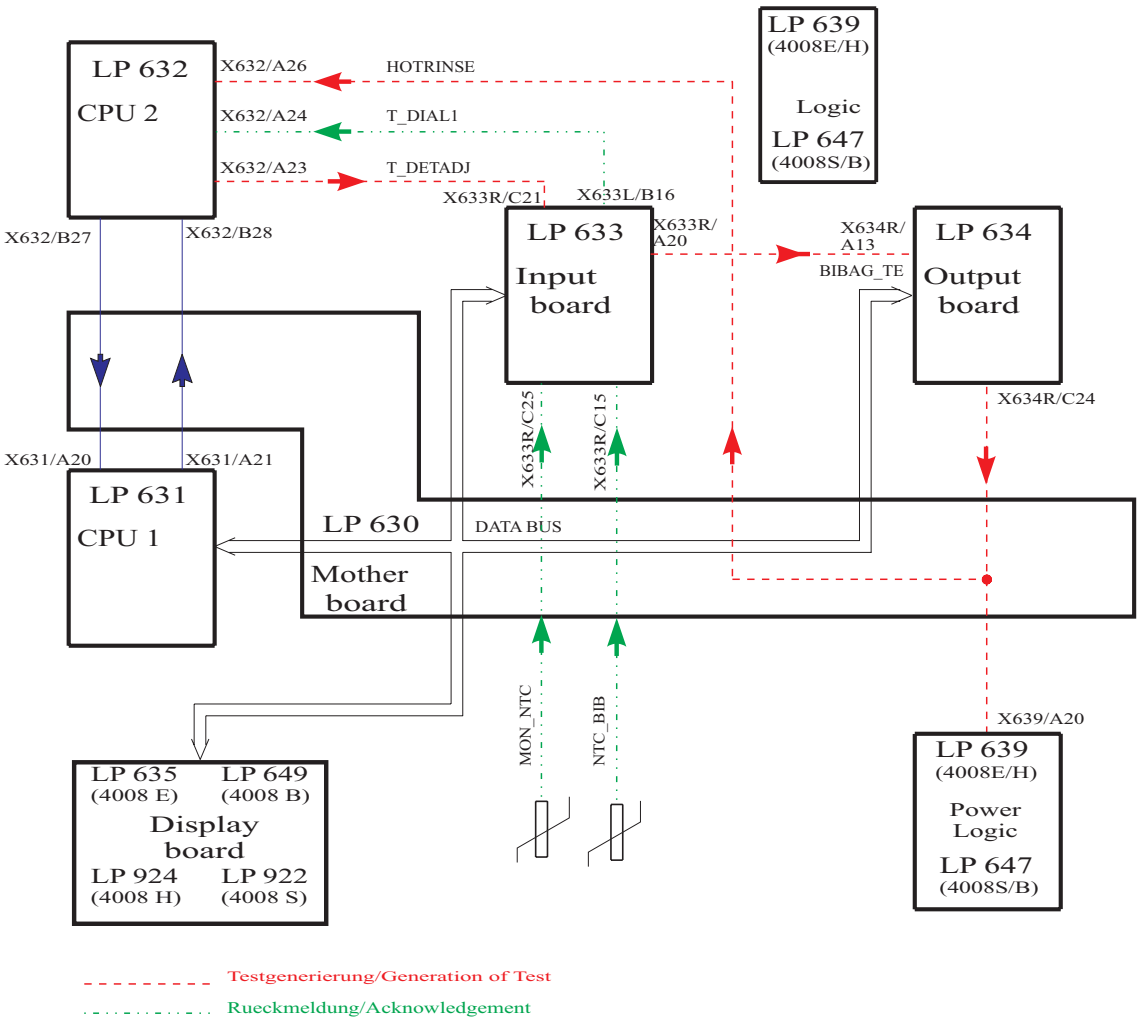
Error message	*Description*
F01 Bloodleak	Blood leak channel and dimness not in alarm-free condition during the T1 test. – Dimness channel contaminated (calcium precipitate, etc.) – Acknowledgement (BLL, X637A/18) → X633L/B10 voltage value within the alarm tolerances (< 3V). – Acknowledgement (BLL_DIM, X637A/21) → X633L/B11 voltage value within the alarm tolerances (<1.5V/ >8V). – DAC_BLL or DAC_DIM not within the tolerances (check calibration)
F02 Bloodleak	The blood leak alarm/dimness alarm is not recognized during test detuning. – Acknowledgement (BLL, X637A/18) → X633L/B10 voltage value not within the alarm tolerances. – Acknowledgement (BLL_DIM, X637A/21 → X633L/B11 voltage value not within the alarm tolerances (<1.5V) – Test detuning (BLL_DET, X632/A25) → X633L/B27 not 5V – Calibration of DAC_BLL or DAC_DIM is too high – Detuning (DAC_DIM, X634R/A11) → X633L/C3 impossible – Dimness calibration is set to potentiometer calibration (BR6 from pos. 1/2 to 2/3).
F03 Bloodleak	After test detuning, the blood leak channel and dimness fail to enter the alarm-free state. – Dimness channel contaminated (calcium precipitate, etc.) – Acknowledgement (BLL, X637A/18) → X633L/B10 voltage value within the alarm tolerances – Test detuning (BLL_DET, X632/A25) → X633L/B27 not 0V. – Acknowledgement (BLL_DIM, X637A/21) → X633L/B11 voltage value within the alarm tolerances (<1.5V / >8V). – DAC_BLL or DAC_DIM not within the tolerances (check calibration)
F95 Bloodleak	System error.

- **Temperature test**

  Test description:

  Test of the upper alarm limit by electronically detuning the temperature display in positive direction.

  Illustration:



Error description:

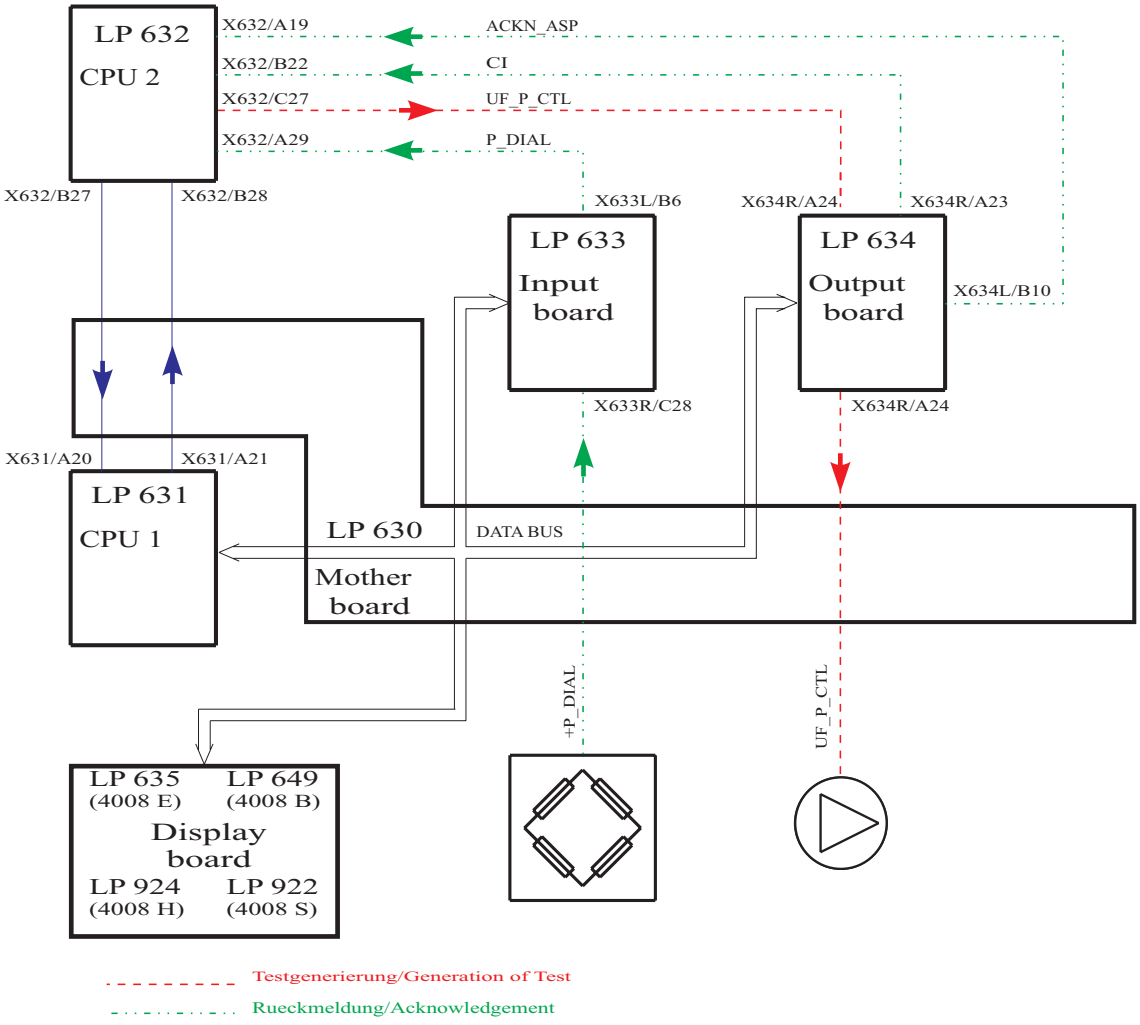
Error message	*Description*
F01 Temperature	The temperature measuring range is not set to hemodialysis. – Control line (HOTRINSE, X634R/C24) → X639/A20, 0 V are missing. – Acknowledgement (HOTRINSE, X634R/C24) → X632/A26, 0 V are missing.
F02 Temperature	The actual temperature is less than 35.0 °C (test running time > 15 minutes). – Calibrate the temperature. – The heater rod failed. – Acknowledgement (T_DIAL1, X633L/B16) → X632/A24, voltage got stuck.
F03 Temperature	The actual temperature is higher than 39.0 °C (test running time > 15 minutes). – Calibrate the temperature. – The regulating sensor (NTC-2) is defective. – Acknowledgement (T_DIAL1, X633L/B16) → X632/A24, voltage got stuck.
F04 Temperature	The temperature failed to stabilize within 15 minutes. – Acknowledgement (T_DIAL1, X633L/B16) → X632/A24 is steadily changing (change > 0.3 °C/15 sec).
F05 Temperature	Detuning in positive direction not higher than 3 °C (10 sec). – Acknowledgement (T_DIAL1, X633L/B16) → X632/A24, change in voltage insufficient. – Detuning (T_DETADJ, X632/A23) → X633R/C21 insufficient.
F06 Temperature	The monitor sensor indicates a constant value. – NTC-3 defective.
F07 Temperature	The test release is missing (max. test running time is 10 minutes). – Run-time problem (software).
F08 Temperature	CPU1 failed to transmit a Bibag status message within 3 sec. – Run-time problem (software).
F09 Temperature	Bibag NTC_BIB detuning not higher than 1 °C. – Acknowledgement (NTC_BIB, X633R/C15) → ADW on P.C.B. LP 633, change in voltage insufficient. – Detuning (BIBAG_TE, X634R/A13) → X633R/A20 insufficient.
F10 Temperature	Bibag temperature display outside of measuring range (15 to 45 °C). – Acknowledgement (NTC_BIB, X633R/C15) → ADW on P.C.B. LP 633.
F95 Temperature	System error.

- **Negative pressure holding test**

  Test description:

  Within a specific time period, the actual value of the dialysate pressure transducer should change within certain limits only.

  Illustration:



Error description:

Error message	*Description*
F01 neg. Pressure	During the start phase a negative pressure of more than 450 mmHg has developed (max. test running time 120 sec), – the hydraulic system is contaminated, – the air separation pump started running.
F02 neg. Pressure	Setting the dialysate pressure to the test pressure (–300 mmHg to –450 mmHg) was not possible (max. test running time 120 sec). Upon repetition of measurement, the range was extended from –260 mmHg to 490 mmHg. – Leakage in the hydraulic system. – The UF pump is defective. – If the HDF filter test was skipped: Clamp the HDF filter.
F03 neg. Pressure	The working point (116 digits) of the differential amplifier cannot be set correctly (max. test running time 120 sec). – Pressure variations are too large. – The D-A converter (IC11) on P.C.B. LP 632 is defective. – The operational amplifier (IC1/IC3) on P.C.B. LP 632 is defective. – The acknowledgement (P_DIAL, X633L/B6) → X632/A29 is defective. – The CI signal is missing (LP 632 → X632/B22).
F04 neg. Pressure	Completion of pressure measurement was not possible (max. test running time 120 sec). – The D-A converter (IC11) on P.C.B. LP 632 is defective. – The operational amplifier (IC1/IC3) on P.C.B. LP 632 is defective. – The acknowledgement (P_DIAL, X633L/B6) → X632/A29 is defetive.
F05 neg. Pressure	The air separation pump started running during the measurement phase. – Acknowledgement (ACKN_ASP, X634L/B10) → X632/A19, 0 V are missing. – ASP has been interrupted electrically.
F06 neg. Pressure	The negative pressure holding test failed to be passed. The dialysate pressure drop exceeds ±40 mmHg (related to ten balancing chamber switching). – Leakage in the hydraulic system.
F07 neg. Pressure	Current increasing pulses were not recognized (min. 2x). – 5-V balancing chamber pulses are missing (CI. X634R/A23) → X632/B22.
F95 neg. Pressure	System error.

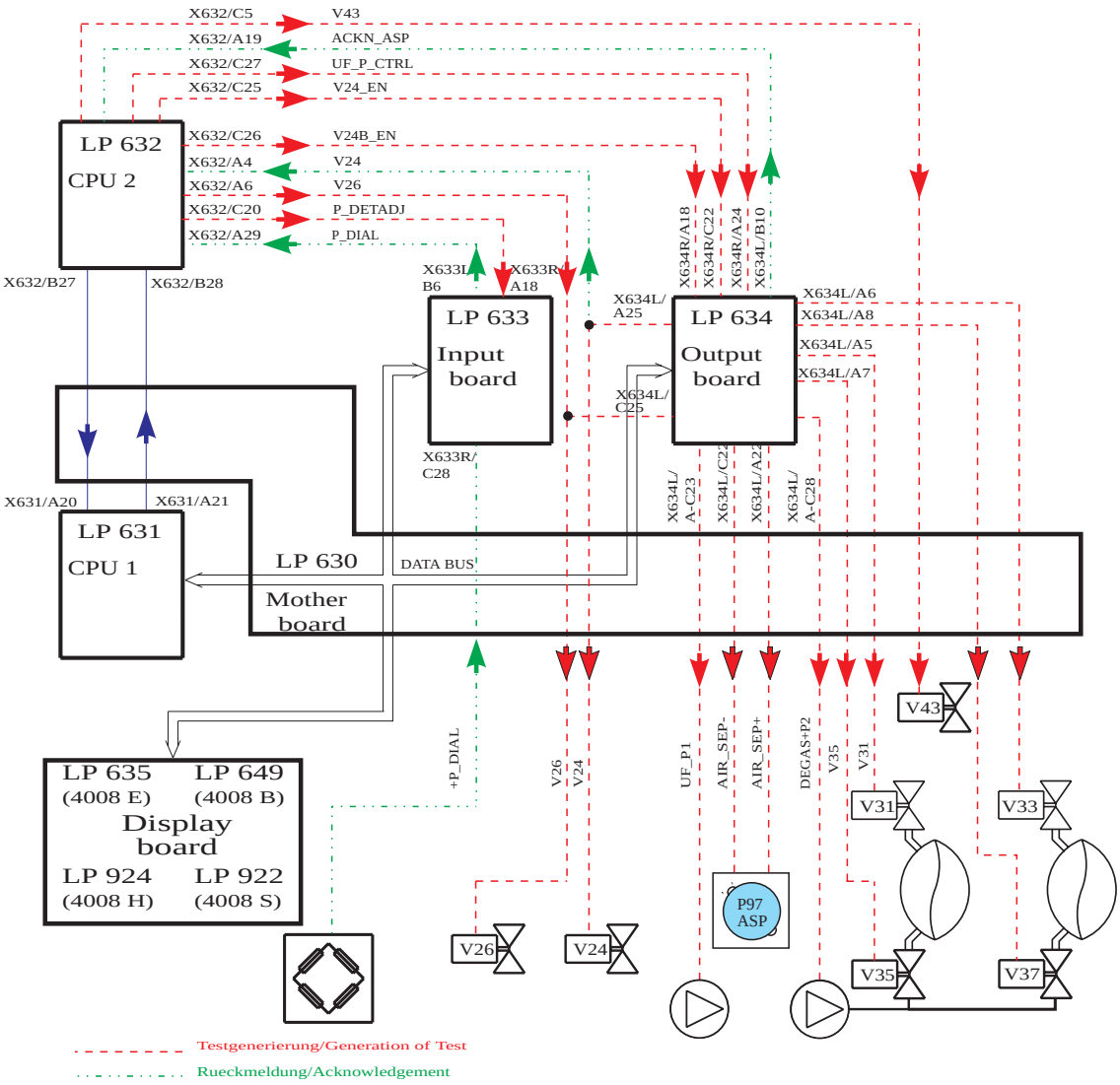
In machines with HDF option, the negative pressure holding test is performed internally only; i.e. V24, V24B are closed and V26 is open.

- **Positive pressure holding test**

  Test description:

  Valves V24, V24B and V26 are checked for proper function (mechanical).
  Test of the TMP unit by detuning it electronically in positive direction.
  With the dialysate flow turned off, positive pressure is applied to the balancing system. The actual value of the dialysate pressure transducer is now monitored for a defined period of time.
  Test of the pump segment of P97.

  Illustration:



Error description:

Error message	*Description*
F01 pos. Pressure	The CPU1 mandatory priming program was not completed (10 sec). – The solenoid valve V43 is not closed.
F24 pos. Pressure	V24 valve error. – Acknowledgement (V24, X637/C1) → X632/A4, 24 V are missing.
F25 pos. Pressure	No pressure increase above 150 mmHg (change in pressure) after valve switching. – Control signals of V24 and V24B mistaken for each other. – Leakage in the external system (shunt interlock, dialysate lines, etc.). – If the HDF filter test was skipped: Clamp the HDF filter.
F26 pos. Pressure	No pressure compensation after opening of V43 (–125 mmHg to 55 mmHg). – V24 got stuck (mechanically open). – V43 not open. – V26 leaking.
F27 pos. Pressure	No pressure compensation after opening of V43 (–125 mmHg to 55 mmHg). – V24 got stuck (mechanically open). – V43 not open. – V189 (retentate valve) leaking.
F02 pos. Pressure	The loading pressure cannot be measured via the solenoid valve V26 in the hydraulic system (P-Dial. < 600 mmHg, 15 sec). – Solenoid valve V26 mechanically not open. – Solenoid valve V43 mechanically not closed. The balancing chamber is switched to passage during this test sequence. V24, V24B and V43 are closed; V26 is open.
F03 pos. Pressure	The hydraulic system cannot be deaerated via the solenoid valve V43; the zero point of –125 to 55 mmHg has not been reached (15 sec). – Solenoid valve V26 mechanically not closed. – Solenoid valve V43 mechanically not open. – Zero point outside the –125 to 55 mmHg range. The balancing chamber is switched to passage during this test sequence. V24, V24B and V26 are closed; V43 is open.
F04 pos. Pressure	The first working point (220 digits) of the differential amplifier cannot be set. – Pressure variations are too large. – The D-A converter (IC11) on P.C.B. LP 632 is defective. – The operational amplifier (IC1/IC3) on P.C.B. LP 632 is defective. – The acknowledgement (P_DIAL, X633L/B6) → X632/A29 is defective.

F05 pos. Pressure

Test detuning results in a change in the measuring range of more than 95 mmHg (60 sec).
- The operational amplifier (IC2) on P.C.B. LP 632 is defective.
- Acknowledgement (P_DIAL, X633L/B6) → X632/A29, change in voltage too large.
- Detuning defective (P_DETADJ, X632/C20) → X633R/C22.
- The balancing chamber valve V36 or V38 (old water valve) is leaking.

F06 pos. Pressure

Test detuning results in a change in the measuring range of less than 85 mmHg (60 sec).
- The D-A converter (IC11) on P.C.B. LP 632 is defective.
- Acknowledgement (P_DIAL, X633L/B6) → X632/A29, change in voltage insufficient.
- Detuning defective (P_DETADJ, X632/C20) → X633R/C22.
- V26 is leaking.

F07 pos. Pressure

After detuning in the test there is a difference (P.diff > ±9 mmHg) between the display and the differential amplifier.
- The voltage divider R23/R9 or the operational amplifier IC2 is defective.
- The operational amplifier IC1/IC3 is defective.
- The balancing chamber valve V36 or V38 (old water valve) is leaking.

F08 pos. Pressure

Test detuning results in a change in the measuring range of more than 400 mmHg (20 sec).
- The operational amplifier (IC2) on P.C.B. LP 632 is defective.
- Acknowledgement (P_DIAL, X633L/B6) → X632/A29, change in voltage too large.
- Detuning defective (P_DETADJ, X632/C20) → X633R/C22.

F09 pos. Pressure

Test detuning results in a change in the measuring range of less than 350 mmHg (20 sec).
- The D-A converter (IC11) on P.C.B. LP 632 is defective.
- Acknowledgement (P_DIAL, X633L/B6) → X632/A29, change in voltage insufficient.
- Detuning defective (DIAL_DET_ADJ, X632/C20) → X633R/C22.

F10 pos. Pressure

The second working point (116 digits) of the operational amplifier cannot be set correctly.
- The D-A converter (IC11) on P.C.B. LP 632 is defective.
- The operational amplifier (IC1/IC3) on P.C.B. LP 632 is defective.

F11 pos. Pressure

Change in the dialysate pressure after closing of the solenoid valve V43 (zero point change of ±20 mmHg within 15 sec).
- The solenoid valve V24B is not closed.
- The balancing chamber valve V36 or V38 (old water valve) is leaking.

The balancing chamber is switched to passage during this test sequence. V43, V24B and V26 are closed; V24 is open.

F12 pos. Pressure

The loading pressure cannot be measured via the solenoid valves V24 and V24B in the hydraulic system (P-Dial. < 600 mmHg, 15 sec).
– Solenoid valve V24 or V24B mechanically not open.

The balancing chamber is switched to passage during this test sequence. V43 and V26 are closed; V24 and V24B are open.

F13 pos. Pressure

The hydraulic system cannot be deaerated via the solenoid valve V43 (P-Dial. not equal to –125 to 55 mmHg, 20 sec).
– The solenoid valve V24 is not closed.
– V43 neither opens electrically nor mechanically.

The balancing chamber is switched to passage during this test sequence. V24 and V26 are closed; V24B and V43 are open.

F14 pos. Pressure

Zero point change after closing of solenoid valve V43 (20 sec).
Standard: P-Dial. not equal to –125 to 55 mmHg.
HDF option: P-Dial. not equal to –125 to 60 mmHg.
– The solenoid valve V24 is not closed.

The balancing chamber is switched to passage during this test sequence. V24, V26 and V43 are closed; V24B is open.

F15 pos. Pressure

The loading pressure is below 780 mmHg ±30 mmHg (10 sec).
– The loading pressure is too low.

F16 pos. Pressure

During the start phase, the pressure dropped below 620 mmHg (measuring tolerance: ±30 mmHg, max. test running time 120 sec).
– Major leakage in the hydraulic system.
– The UF pump spring is defective.
– The loading pressure is too low.
– The air separation pump fails to occlude.
– Relief valve is leaking.

F17 pos. Pressure

During the start phase, it was not possible to reduce the dialysate pressure to a value below 760 mmHg (measuring tolerance: ±30 mmHg, test running time 120 sec).
– The loading pressure is too high.
– The UF pump is defective.

F18 pos. Pressure

The working point (116 digits) of the differential amplifier cannot be set correctly (test running time 120 sec).
– The pressure variations in the system are too large.

F19 pos. Pressure

Completion of the pressure measurement was not possible (max. test running time 120 sec).
– The D-A converter (IC11) on P.C.B. LP 632 is defective.
– The acknowledgement (P_DIAL, X633L/B6) → X632/A29 is defective.

F20 pos. Pressure

The positive pressure holding test failed to be passed. While the flow was off, a pressure drop of more than ±80 mmHg/min was detected in the hydraulic system.
- Leakage in the hydraulic system.
- The UF pump spring is defective.
- ASP fails to occlude.
- Relief valve leaking.
- V84 leaking.

F21 pos. Pressure

The dialysate pressure cannot be set to a value between 460 and 760 mmHg ±30 mmHg (10 sec).
- The heat exchanger is defective.
- Problem in the hydraulic system.

F22 pos. Pressure

The air separation pump is not running during the test phase (2 sec).
- Control line (AIR_SEP+, X634L/A22) → ASP/..., 24 V are missing.
- Control line (AIR_SEP–, X634L/C22) → ASP/..., 0 V are missing.
- Acknowledgement (ACKN_ASP, X634L/B10) → X632/A19, 12 V are missing.

F23 pos. Pressure

Pressure drop in the hydraulic system during the measurement phase (8 sec). Change more than +4 digits or more than –8 digits.
- Leakage in the pump segment of the air separation pump.
- Leakage in the heat exchanger.
- Acknowledgement (P_DIAL, X633L/B6) → X632/A29, change in voltage too large.

F24 – F27

See between F01 and F02

F28 pos. Pressure

ASP functional test (running and delivery test)
- ASP line segment is occluded
- ASP line segment has been incorrectly inserted (check direction of delivery)
- ASP is not running (electrically or mechanically)
- V87 electrically or mechanically closed

F95 pos. Pressure

System error.

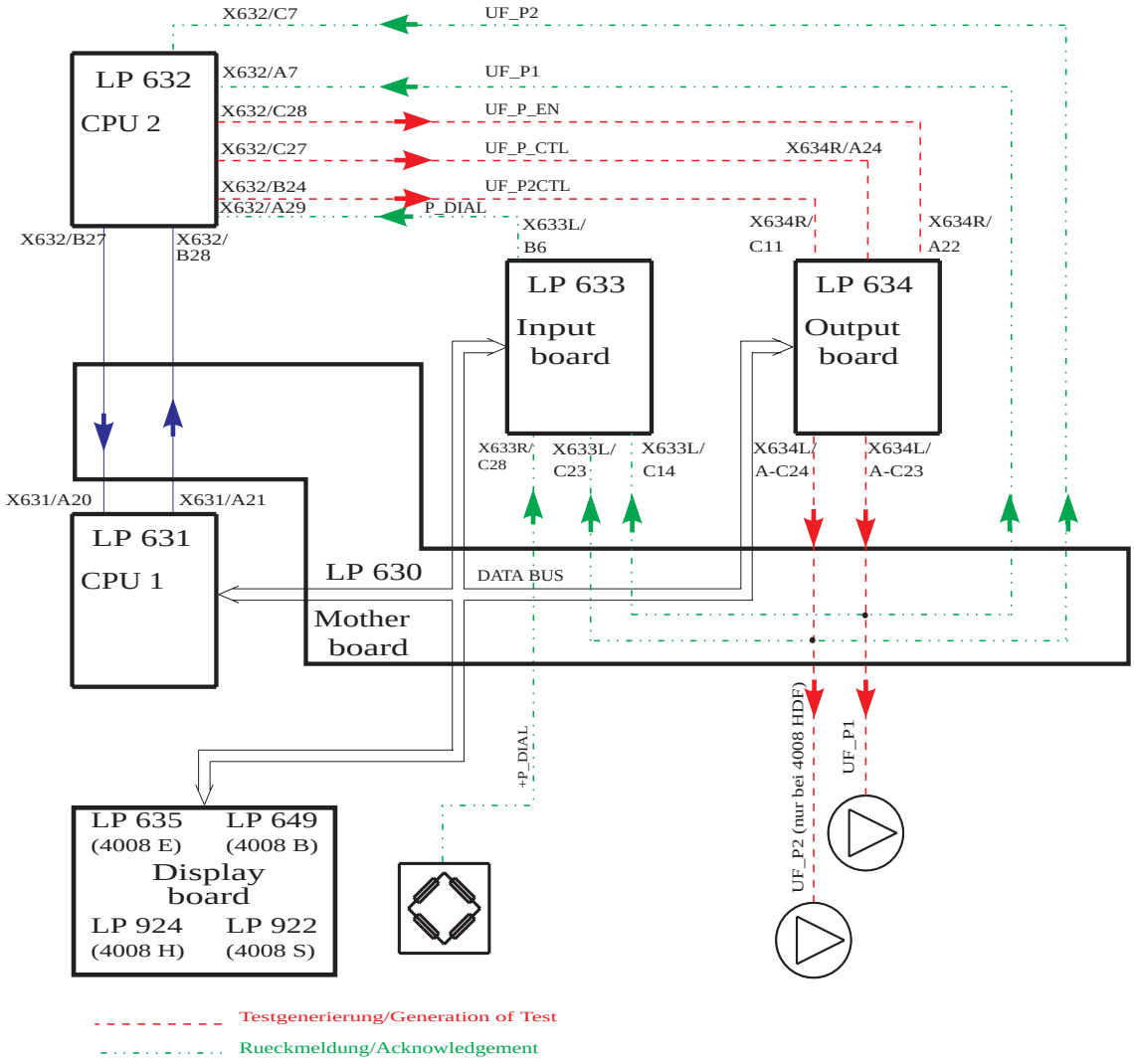
- **UF function test**

Test description:

CPU1 activates the UF pump at a defined rate.
CPU2 checks the UF pump.
CPU2 blocks the control line of the UF pump and checks whether the UF pump stops.
Check of the UF counter.

The following is additionally applicable with built-in 4008 HDF option:
CPU1 activates the UF pump 2 at a defined rate.
CPU2 checks the hydraulic and the electric function of the UF pump 2.
CPU2 blocks the control line of the UF pump 2 and checks whether it stops.
Check of the UF2 counter.

Illustration:



Error description:

Error message	*Description*
F01 UF-Function	The pause between the strokes of the UF pump 1 was shorter than 220 ms. Correct volume delivery is not ensured due to too short a return. – CPU1 issued too high a pump rate.
F02 UF-Function	The pulse time for the UF pump 1 is shorter than 180 ms. Correct volume delivery is not ensured due to too short an emission time. – The monoflop on P.C.B. LP 634 is defective (IC42/R82/C47).
F03 UF-Function	The pulse time for the UF pump 1 is longer than 500 ms. A maximum rate of 5000 ml/h is not possible. – The monoflop on P.C.B. LP 634 is defective (IC42/R82/C47).
F04 UF-Function	No activity of the UF pump 1 during the test (5 sec). – Acknowledgement (UF_P1, X637/B23) → X632/A7, no LOW pulses. – Control line (UF_P1, X634L/ABC23) → X637/B23, no LOW pulses.
F05 UF-Function	The UF pump 1 cannot be stopped by CPU2. – Control line (UF_P_EN, X632/C28) → X634R/A22, 5 V are missing. – The reset input at IC42/pin 3 on P.C.B. LP 634 is defective.
F06 UF-Function	The UF pump acknowledgement of CPU1 is defective. – Acknowledgement (UF_P1, X637/B23) → X622L/C14, no LOW pulses.
F07 UF-Function	The change in pressure after a stroke is less than 20 mmHg. – The UF pump 1 is mechanically defective. – Control line (UF_P1_CTL, X632/C27) → X634R/A24, no LOW pulse.
F09 UF-Function	Dialysate pressure is outside the measuring range (15s). – UF pressure transducer defective – D/A converter (IC11) on P.C.B. LP 632 defective – Operational amplifier (IC1/IC3) on P.C.B. LP 632 defective
F11 UF-Function	The pause between the strokes of the UF pump 2 was shorter than 220 ms. Correct volume delivery is not ensured due to too short a return. – CPU1 issued too high a pump rate.
F12 UF-Function	The pulse time for the UF pump 2 is shorter than 180 ms. Correct volume delivery is not ensured due to too short an emission time. – The monoflop on P.C.B. LP 634 is defective (IC42/R65/C45).

F13 UF-Function

The pulse time for the UF pump 2 is longer than 500 ms. A maximum rate of 5000 ml/h is not possible.

– The monoflop on P.C.B. LP 634 is defective (IC42/R65/C45).

F14 UF-Function

No activity of the UF pump 2 during the test (4 sec).

– Acknowledgement (UF_P2, X637/B26) → X632/C7, no LOW pulses.
– Control line (UF_P2, X634L/ABC24) → X637/B26, no LOW pulses.

F15 UF-Function

The UF pump 2 cannot be stopped by CPU2.

– Control line (UF_P_EN, X632/C28) → X634R/A22, 5 V are missing.
– The reset input at IC42/pin 13 on P.C.B. LP 634 is defective.

F16 UF-Function

The UF pump acknowledgement of CPU1 is defective.

– Acknowledgement (UF_P2, X637/B26) → X633L/C23, no LOW pulses.

F09 UF-Function

Dialysate pressure is outside the measuring range (15s).

– UF pressure transducer defective
– D/A converter (IC11) on P.C.B. LP 632 defective
– Operational amplifier (IC1/IC3) on P.C.B. LP 632 defective

F17 UF-Function

The change in pressure after a stroke of the UF pump 2 is less than 20 mmHg.

– The UF pump 2 is mechanically defective.
– Control line (UF_P2_CTL, X632/B24) → X634R/C11, no HIGH pulse.

F20 UF-Function

The difference in volume between UF pump 1 and UF pump 2 is higher than 25% (range of tolerance 15% to 35%).

– The stroke volume of UF pump 1 or UF pump 2 has been misadjusted.

F95 UF-Function

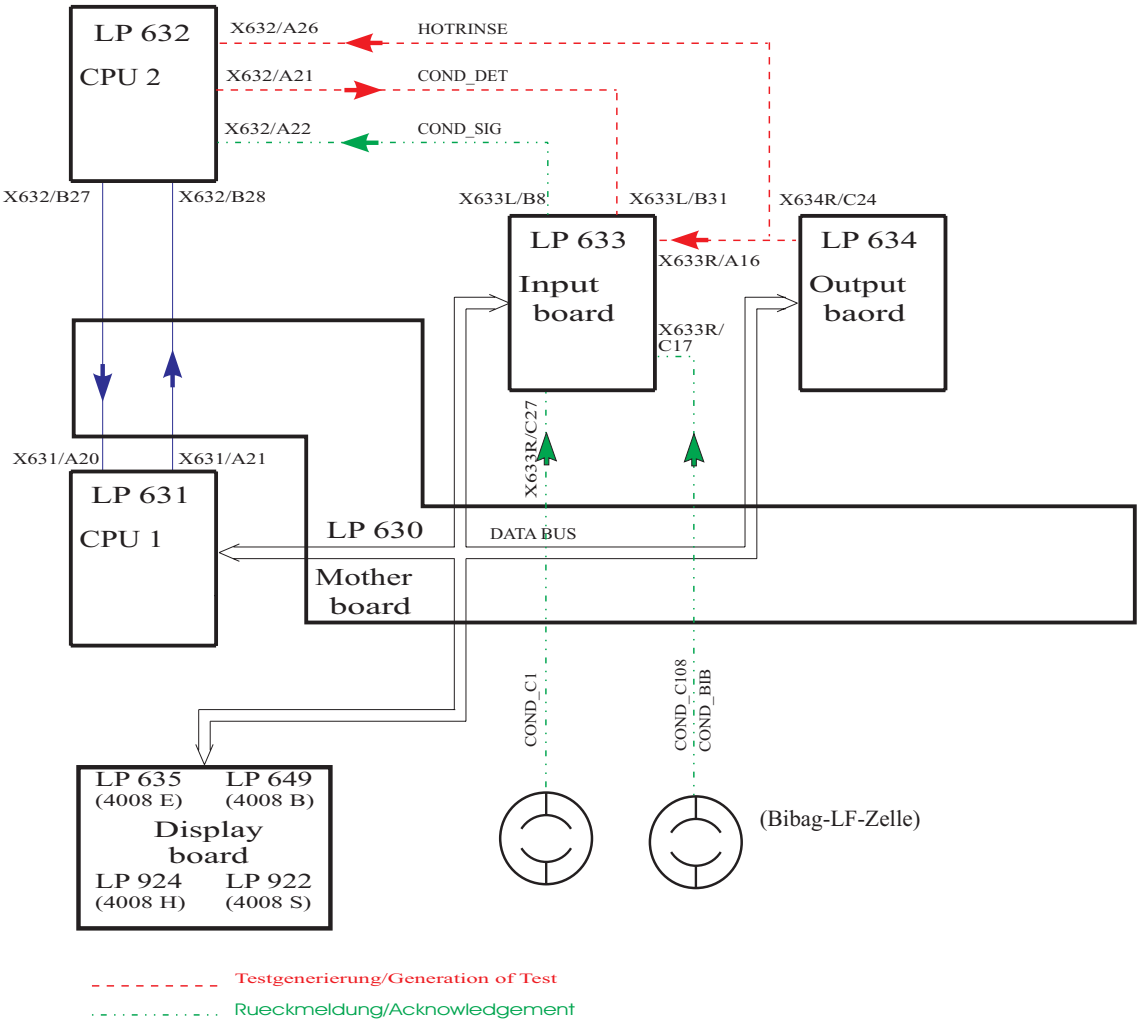
System error.

- **Conductivity test**

  Test description:

  Test of the alarm limits by electronically detuning the conductivity by +5% or by –5%.

  Illustration:



Error description:

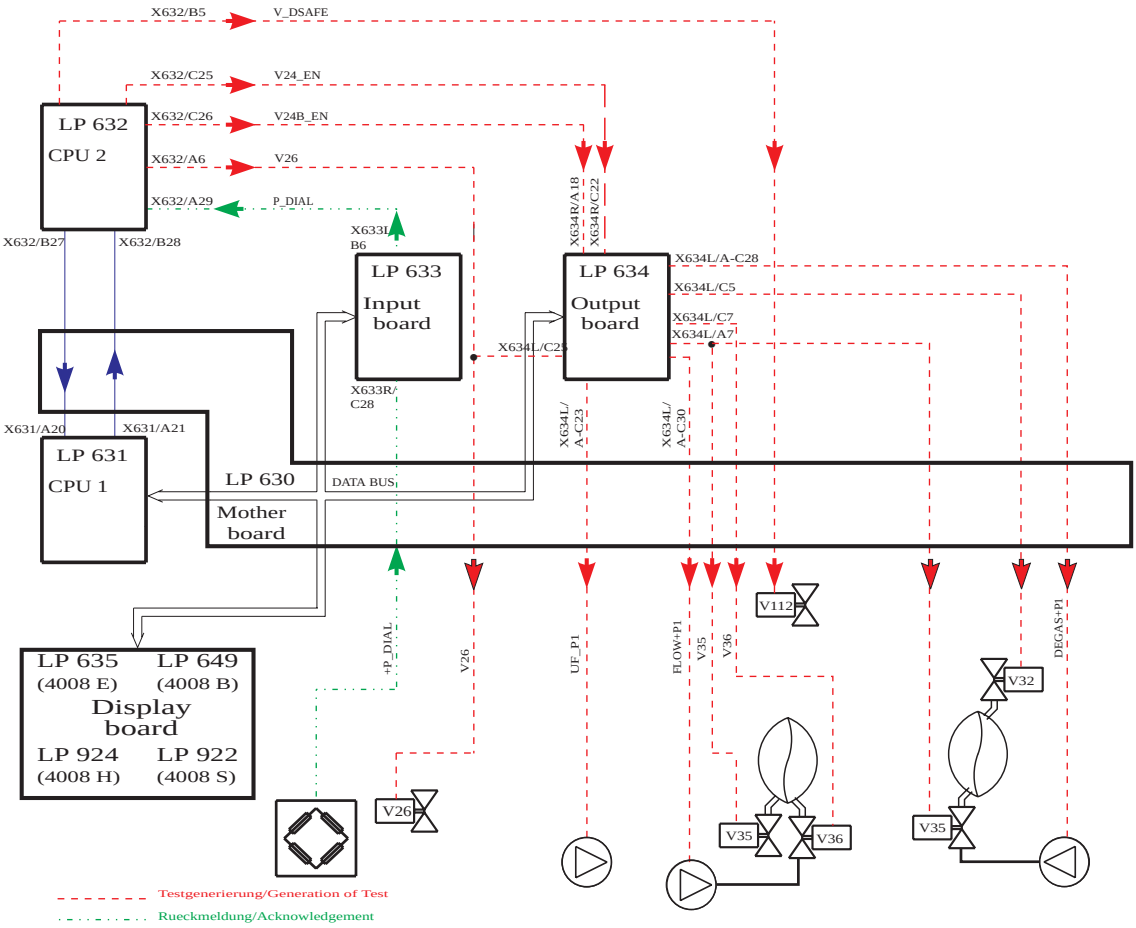
Error message	*Description*
F01 Conductivity	The conductivity failed to be within the scale limits or to stabilize within 10 minutes (±0.1 mS/10 sec). – Concentrate is not connected. – Acknowledgement (COND_SIG, X633L/B8) → X632/A22, voltage outside the measuring range or unstable.
F02 Conductivity	Detuning in positive direction not more than 0.5 mS (10 sec). – Acknowledgement (COND_SIG, X633L/B8) → X632/A22 insufficient. – Detuning (COND_DET, X632/A21) → X633L/B31 insufficient.
F03 Conductivity	Detuning in negative direction not more than 0.5 mS (10 sec). – Acknowledgement (COND_SIG, X633L/B8) → X632/A22 insufficient. – Detuning (COND_DET, X632/A21) → X633L/B31 insufficient.
F04 Conductivity	The conductivity cell indicates a constant value. – The CD cell is defective.
F05 Conductivity	CPU1 failed to transmit a Bibag status message within 3 sec. – Run-time problem (software).
F08 Conductivity	CPU 1 fails to increase the working point (when the conductivity is <40mS/cm uncompensated) for the bibag conductivity by > 5 digits. – Detuning (HOT_RINSE, X634R/C24 → X633R/A16) not 12V – P.C.B. LP633 T2 or IC26 defective
F06 Conductivity	The Bibag CD detuning is not more than 1 mS/cm. – Acknowledgement (COND_SIGNAL3, X633R/A12) → MP TP3 on P.C.B. LP 633, change in voltage insufficient. – Detuning (COND_DET, X632/A21) → X633L/B31 insufficient.
F07 Conductivity	The Bibag CD display is outside of the measuring range (46 to 84 mS/cm). – Acknowledgement (COND_SIGNAL3, X633R/A12) → MP TP3 on P.C.B. LP 633. – Conductivity outside the expected detuning range caused by wrong concentrate on the bicarbonate port or temperature too low.
F95 Conductivity	System error.

- **Diasafe/HDF filter test**

  Test description:

  Test of the filters by testing the volume of the internal capillary and pressure holding test.

  Illustration:



Error description:

Error message	*Description*
F02 Diasafe	The balancing chamber was not stopped by CPU1 (24 sec). – The message via the serial interface from CPU1 to CPU2 is missing. – The current rise pulse is missing (CI, X634R/A23) → X633L/C31, no 5-V pulse.
F04 Diasafe	CPU1 failed to complete one balancing chamber switching within 20 sec (30 ml fluid not removed?). – The message via the serial interface from CPU1 to CPU2 is missing. – The current rise pulse is missing (CI, 634R/A23) → 633L/C31, no 5-V pulse.
F06 Diasafe	During the pressure built-up phase, a negative pressure of less than –450 mHg has developed (24 sec). – Diasafe valve not open, control line (V_DSAFE, X632/B5) → X637/C16, 0 V are missing.
F07 Diasafe	After the maximum fluid volume of 145 ml + 30 ml has been removed, the expected negative pressure of –300 mmHg to –450 mm Hg failed to build up. – Major leakage in the Diasafe filter membrane and/or filter housing. – Major leakage in the O-rings on filter holder/couplings. – V26 electrically or mechanically not closed.
F08 Diasafe	The negative test pressure of more than –300 mmHg has developed before the minimum fluid removal of 145 ml –30 ml has been achieved. – The Diasafe filter is contaminated. – The Diasafe filter was not correctly deaerated upon start of the test. – V112 electrically or mechanically not open.
F09 Diasafe	The zero point for pressure measurement cannot be set. The maximum test time has been exceeded (max. test time 5 min). – Leakage in the Diasafe filter membrane and/or filter housing. – Leakage in the O-rings on filter holder/couplings. – P.C.B. LP 632, IC3/pin 12 not in socket or IC defective (differential amplifier).
F10 Diasafe	The negative pressure to be achieved in the test failed to stabilize within the maximum test time of 5 minutes (change > ±16.7 mmHg/min). – Leakage in the Diasafe filter membrane and/or filter housing. – Leakage in the O-rings on filter holder/couplings. – Leakage in the hydraulic system. – V 26 electrically or mechanically not closed.
F20 Diasafe	It was not possible to prime (deaerate) the dialysate filter within 2 minutes. – Flow problems. – The priming program is permanently active (level sensor, osmosis water, or P.C.B. LP 633, IC36 defective).
F95 Diasafe	System error.

Error message	*Description*
F01 HDF-Filter	The Diasafe option has not been set although ON-LINE HDF has been selected. – CPU 2: DIP switch array 2, switch 1 not set to ON.
F02 HDF-Filter	CPU1 failed to stop the balancing chamber (24 sec). – The message via the serial interface from CPU1 to CPU2 is missing. – The current rise pulse is missing (CI, X634/A23) → X633L/C31, no 5-V pulse.
F04 HDF-Filter	CPU1 failed to comlete one balancing chamber switching within 20 sec (30 ml fluid not removed?). – The message via the serial interface from CPU1 to CPU2 is missing. – Verify the current rise pulse.
F06 HDF-Filter	During the pressure-buildup phase, a negative pressure of less than –370 mmHg has developed (24 sec). – The HDF filter is clamped/clogged. – The Diasafe valve is not open, control line (V_DSAFE, X632/B5) → X637/C16, 0 V are missing.
F07 HDF-Filter	After the maximum fluid volume of 255 ml +60 ml has been removed the expected negative pressure of –220 mmHg up to 370 mmHg failed to build up. – Major leakage in the Diasafe/HDF filter membrane and/or filter housing. – Major leakage in the O-rings on filter holder/couplings. – V26 electrically or mechanically not closed.
F08 HDF-Filter	The negative test pressure of less than –220 mmHg has developed, before the minimum fluid removal of 255 ml –60 ml has been achieved. – The Diasafe/HDF filters are contaminated. – The Diasafe/HDF filters were not correctly deaerated upon start of the test. – V112 electrically or mechanically not open.
F09 HDF-Filter	The zero point for pressure measurement cannot be set. The max. test time has been exceeded (10 min). – Leakage in the Diasafe/HDF filter membrane and/or filter housing. – Leakage in the O-rings on filter holder/couplings.
F10 HDF-Filter	The negative pressure to be achieved in the test failed to stabilize within the maximum test time of 10 minutes (change > ±13.3 mmHg/min). – Leakage in the Diasafe/HDF filter membrane and/or filter housing. – Leakage in the O-rings on filter holder/couplings. – Leakage in the hydraulic system. – V26 electrically or mechanically not closed.

F20 HDF-Filter

It was not possible to prime (deaerate) the Diasafe filter within 2 minutes.
- Flow problems.
- The priming program is permanently active (level sensor, osmosis water, or P.C.B. LP 633, IC36 defective).

F21 HDF-Filter

It was not possible to correctly rinse/prime the HDF filter within 5 minutes (before the test).
- Flow problems.
- No conductivity.
- Conductivity at the upper or lower end of the scale range.
- The HDF pump is not running (e.g. open door).
- The delivery rate of the HDF pump is less than 380 ml/min.
- Line diameter not set to 8 mm.
- NTC6 permanently fails to detect fluid.
- Sieve on V43 clogged.

F22 HDF-Filter

It was not possible to correctly rinse/prime the HDF filter within 5 minutes (after the test).
- Flow problems.
- No conductivity.
- Conductivity at the upper or lower end of the scale range.
- Especially with biBag machines: check filter on V43
- The HDF pump is not running (e.g. open door).
- The delivery rate of the HDF pump is less than 380 ml/min.
- Line diameter not set to 8 mm.
- NTC6 permanently fails to detect fluid.
- Sieve on V43 clogged.

F95 HDF-Filter

System error.

- **Test Online plus / Diasafe plus-Filter**

F01 ONLINE plus
F01 DIASAFE plus
F01 HPU

Present options and DIP switch settings do not match.
CPU1 machine status (MST), HPU status and DIP switch/Array2 changed during the test running time.
– **ONLINE plus:**
  CPU 2: Array 2, DipSw2 not set to OFF
  CPU 2: Array 2, DipSw3 not set to ON
– **DIASAFE plus:**
  CPU 2: Array 2, DipSw2 not set to OFF
  CPU 2: Array 2, DipSw3 not set to OFF
– MST transmitted by CPU1 not matching with the set DIP switch of array 2.
– DIP switch/Array2 changed while the test was in progress.
– HPU logged off.

F34 ONLINE plus
F34 DIASAFE plus

Pressure holding test not passed. Max. number of treatments exceeded?
– Diasafe and HDF filter membranes leaking/worn.

F02 ONLINE plus
F02 DIASAFE plus
F02 HPU

Dialysate outlet pressure (DA1) outside the permissible range (10s).
DA1 test range: –125mmHg ≤ P_dial ≤ 55 mmHg
– Acknowledgement DA 1 (P_DIAL, X633L/B6) → X632/A29
– Acknowledgement line DA 2 (see HPU diagram)

F03 ONLINE plus
F03 DIASAFE plus
F03 HPU

Cross comparison of both pressure transducers (DA1 / DA2) is outside the acceptable tolerance (10s).
P(DA2) == P(DA1) ±20mmHg
– Acknowledgement DA 1 (P_DIAL, X633L/B6) → X632/A29
– Acknowledgement DA 2 (see HPU diagram)

F41 ONLINE plus
F41 DIASAFE plus
F41 HPU

The test valve V183 is leaking. Pressure increase in the system of ΔP(DA2) > 30 mmHg within 4s.
– V183 open, contaminated, or mechanically defective
– HPU, output stage etc. defective

F42 ONLINE plus
F42 DIASAFE plus
F42 HPU

No pressure increase of ΔP(DA2) > 200 mmHg within 4s after opening the test valve V183 in the system.
– V183 fails to open or mechanically defective.
– Air pump defective, is not running
– HPU, V183 and/or air pump output stage etc. defective

F43 ONLINE plus
F43 DIASAFE plus
F43 HPU

The lower pressure test range of ΔP(DA2) > 300 mmHg failed to be achieved within 1s after closing the test valve V183.
– HPU, output stage etc. defective
– ONLINE filter leaking

F44 ONLINE plus
F44 DIASAFE plus
F44 HPU

The upper pressure test range of ΔP(DA2) 750 mmHg was exceeded within 4s after closing of the test valve V183.
– HPU, output stage etc. defective

F04 ONLINE plus
F04 DIASAFE plus

The air pump is running although valve V43 is closed.
– HPU, output stage etc. defective

F05 ONLINE plus

The door on the Online Sys module is open during the rate test.
– Close module door.

F06 ONLINE plus

Port 1 is open during the first pressure build-up phase.
– Close port 1 (substituate port).

F07 ONLINE plus

Port 2 is open during the first pressure build-up phase.
– Close port 2 (rinse port).

F08 ONLINE plus

Failure to reach the test pressure PDIAL2 > 795 mmHg within 12s.
– Calibrate dialysate pressure.
– Replace DA 2 (re-calibration required)
– Air pump (185) or test valve (V183) defective
– Hydraulic system or valve ONL3 (191) leaking
– Air pump control (185) based on V43 status defective (HPU defective)

F09 ONLINE plus

The ONLINE system pump failed to comply with the first test rate of 100 ml/min ± 9 ml/min.
– ONLINE system pump control defective

F10 ONLINE plus

Monitoring unit (Hall sensor) of the pump rotor detects incorrect rotation of the rotor (desired rate 300 ml/min ± 25%).
– ONLINE system pump control defective (outside the tolerance of ±25%)
– Hall sensor /electronics defective

F11 ONLINE plus

The ONLINE system pump failed to comply with the second test rate of 300 ml/min ± 9 ml/min.
– ONLINE system pump control defective

F12 ONLINE plus

After the ONLINE system pump was switched off in the test, the monitoring unit (Hall sensor) detects that the rotor failed to stop correctly.
– Pump stop (output stage) defective
– Hall sensor /electronics defective

F13 ONLINE plus

After the ONLINE system pump was switched off in the test, the actual rate of the module is > 0 ml/min.
– Pump stop (output stage) defective
– Synchro-transmitter electronics defective

F14 ONLINE plus

After activation of the substituate pump the monitoring unit (Hall sensor) of the pump rotor detects incorrect rotation of the rotor (desired rate 300 ml/min).
– Pump control defective (outside the tolerance of ±25%)
– Hall sensor /electronics defective

F15 ONLINE plus

After activation of the substituate pump the system pump failed to comply with the test rate of 300 ml/min ±9ml/min.
– Pump control defective

F16 ONLINE plus

Port 1 open during ONL valve test sequence.
– Close port 1 (substituate port).

F17 ONLINE plus

Port 2 open during ONL valve test sequence.
– Close port 2 (rinse port).

F18 ONLINE plus

Acknowledgement of ONL1 (V193) differs from the desired state of the valve.
– Valve control in the ONLINE Sys module defective
– Valve acknowledgement in the ONLINE Sys module defective

F19 ONLINE plus

Acknowledgement of ONL2 (V192) differs from the desired state of the valve.
– Valve control in the ONLINE Sys module defective
– Valve acknowledgement in the ONLINE Sys module defective

F20 ONLINE plus

Acknowledgement of ONL3 (V191) differs from the desired state of the valve.
– Valve control in the ONLINE Sys module defective
– Valve acknowledgement in the ONLINE Sys module defective

F21 ONLINE plus

Leakage test ONL3 (V191) failed to be passed.
The permitted pressure drop of $\Delta P < -10$ mmHg has been exceeded or the test pressure is $P \leq 710$ mmHg.
– Valve ONL3 (V191) in the ONLINE Sys module leaking
– Leaking system / tubing connections
– Port 1 or 2 in the ONLINE Sys module leaking

F22 ONLINE plus

Leakage test ONL2 (V192) failed to be passed.
The permitted pressure drop of $\Delta P < -10$ mmHg has been exceeded or the test pressure is $P \leq 710$ mmHg.
– Valve ONL2 (V192) in the ONLINE Sys module leaking
– Leaky system /tubing connections
– Port 1 in the ONLINE Sys module leaking

F23 ONLINE plus

Leakage test ONL1 (V193) failed to be passed.
The permitted pressure drop of $\Delta P < -10$ mmHg has been exceeded or the test pressure is $P \leq 710$mmHg.
– Valve ONL1 (V193) in the ONLINE Sys module leaking
– Leaky system /tubing connections

F24 ONLINE plus

After the valves ONL1 to 3 opened, the pressure drop in the system was insufficient (ΔP < –100 mmHg).
- Valve ONL1/ONL2/ONL3 electrically or mechanically not open
- Kinked tubing
- HDF filter strongly contaminated

F25 ONLINE plus

No pressure change of ΔP > 40 mmHg within 15s.
- HDF filter membrane leaking (major leakage)
- No HDF filter installed

F26 ONLINE plus
F26 DIASAFE plus

Insufficient test pressure (P < 750mmHg) in the system.
- HDF filter membrane leaking (major leakage)
- No HDF filter installed
- Hydraulics system leaking

F27 ONLINE plus
F27 DIASAFE plus

After the valve V189 opened, the pressure drop in the system was insufficient (ΔP < –70 mmHg).
- Valve V189 electrically or mechanically not open
- Diasafe filter strongly contaminated
- Filter before/after V43 strongly contaminated

F28 ONLINE plus
F28 DIASAFE plus

Pressure increase in the system fails to exceed P > 760 mmHg.
- Diasafe filter membrane leaking (major leakage)
- No Diasafe filter installed

F29 ONLINE plus
F29 DIASAFE plus

Pressure holding test failed to be passed. Excess pressure drop within a measurement time of 30s (ΔP > –10 mmHg).
- Diasafe and/or HDF filter membrane leaking

F30 ONLINE plus
F30 DIASAFE plus

During the pressure holding test valve(s) ONL1, 2 or 3 and/or V189 was (were) closed (according to electronic acknowledgement).
- Valve control failed

F31 ONLINE plus
F31 DIASAFE plus
F31 HPU

Fill phase has been stopped.
Valve(s) V26 open and/or V24, V24b closed (according to electronic acknowledgement), or failure to perform 25 or 15 balancing chamber switchings within 120s.
- Valve control failed
- Balancing chamber switchings failed (e.g. only “Eigentakt”)

F32 ONLINE plus

Valve(s) ONL1, 2 or 3 closed and/or V24 open or port 1 or 2 open during the rinse phase (according to electronic acknowledgement).
- Valve control failed
- Operator opened ports too early.

F33 ONLINE plus

Rinse phase has been aborted.
Valve V189 open (according to electronic acknowledgement), or failure to perform 34 balancing chamber switchings within 240s.
– Valve control failed
– Failure to detect current rise pulse
– Conductivity not within the scale range
  Possible cause: Concentrate and/or bicarbonate level sensor do not recognize CD, although present.

F34 ONLINE plus
F34 DIASAFE plus

See error message between F01 and F02 ONLINE plus /DIASAFE plus

F41 ONLINE plus
F41 DIASAFE plus

See error message between F01 and F02 ONLINE plus /DIASAFE plus

F42 ONLINE plus
F42 DIASAFE plus

See error message between F03 and F04 ONLINE plus /DIASAFE plus

F43 ONLINE plus
F43 DIASAFE plus

See error message between F03 and F04 ONLINE plus /DIASAFE plus

F44 ONLINE plus
F44 DIASAFE plus

See error message between F03 and F04 ONLINE plus /DIASAFE plus

F95 ONLINE plus
F95 DIASAFE plus
F95 HPU

System error

1.1.4 Description of machine errors during the cleaning programs

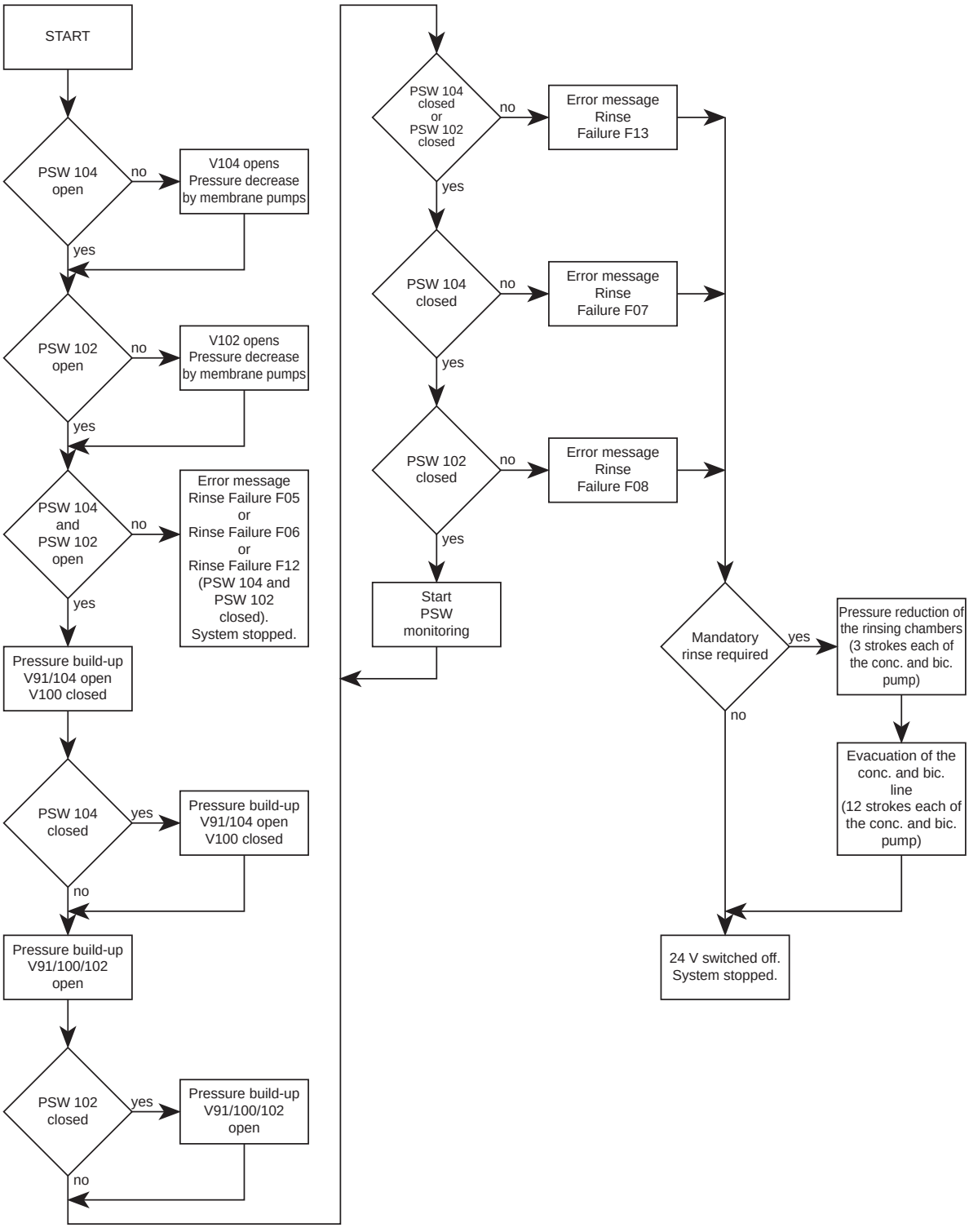
- **V84 monitoring**

Error message	*Description*
Rinse Failure F01	End of the rinse-free program in Dis I to V. Conductivity has been recognized via V84, although the valve is still closed. This error message can be acknowledged by pressing the Rinse key.
Rinse Failure F21	Disinfectant suction phase in Dis I – IV. Maximum permissible UF pump strokes (160) during the suction phase) exceeded. Error message cannot be acknowledged. Turn the system off and on again.
Rinse Failure F02	Disinfectant suction phase in Dis I to IV. Conductivity has not been recognized via V84, and the “Disinfectant empty ?” message has been acknowledged twice. This error message cannot be acknowledged. Switch the machine off and on again. Program Dis V (only on machines with advanced hydraulics) No conductivity detected via concentrate level sensor, and “Disinfectant empty ?” message acknowledged twice. Error message cannot be acknowledged. Turn the system off and on again.
Rinse Failure F03	End of the suction phase in Dis I to IV. Conductivity has been recognized via V84, although the valve is already closed. This error message can be acknowledged by pressing the Disinfection key.
Rinse Failure F04	End of the suction phase in Dis I to IV. The float switch does not recognize any fluid after the disinfectant has been drawn in. Aeration of the disinfectant container! This error message cannot be acknowledged. Turn the system off and on again.

F01, F02 and F03 cause the V84 monitoring flag to be set. I.e. after one of these error messages has occurred, Bergström or ISO-UF dialysis is no longer possible, since it is not possible to switch the flow off. The V84 malfunction can be eliminated by correctly performing Dis I to IV. The problem can also be corrected using the calibration program (by a service technician only), menu item NOVRAM (Reset V84).

- **PSW (pressure switch) monitoring**

  The following requirements must be fulfilled to run the PSW test:
  – DIP switch 8 Dip array 2 on P.C.B. LP631 must be set to ON.
  – Rinse free followed by disinfection or heat disinfection (Dis. I–V)
    or
    Mandatory rinse as individual program



The pressure switches are designed as make contacts.
Specifications:

Delta pressure switch:	Switching point 700 mbar ±20 mbar
Alcatel-SEL-pressure switch:	Switching range 675 – 805 mbar
Envec pressure switch:	Switching range 700 mbar ±20 mbar

Error message	*Description*
Rinse Failure F05	Rinse-free program with following Dis or HDIS or mandatory rinse as individual program in Dis I to V. It was impossible to open the pressure switch for PSW_104 (S124) (bicarbonate). – Pressure on distribution piping > 500 mbar (according to specification, the permissible pressure is max. 500 mbar) pressure peaks on distribution piping: Frequently occurs in distribution pipings with user points if e.g. several patients are disconnected simultaneously and disinfection is started. – Switching point of pressure switch too low: Desired value = 700 mbar ± 20 mbar – Check acknowledgement of pressure switch on P.C.B. LP 633: Bicarbonate: X633L/ A19
Rinse Failure F06	Rinse-free program with following Dis or HDIS or mandatory rinse as individual program in Dis I to V. It was impossible to open the pressure switch for PSW_102 (S123) (concentrate). – Pressure on distribution piping > 500 mbar (according to specification, the permissible pressure is max. 500 mbar) pressure peaks on distribution piping: Frequently occurs in distribution pipings with user points if e.g. several patients are disconnected simultaneously and disinfection is started. – Switching point of pressure switch too low: Desired value = 700 mbar ± 20 mbar – Check acknowledgement of pressure switch on P.C.B. LP 633: Concentrate: X633L/ A20
Rinse Failure F07	Rinse-free program, Dis, HDIS, or mandatory rinse in Dis I to V. Pressure drop during the monitoring phase on PSW_104 (S124) (bicarbonate) or pressure build-up impossible. – Check switching point of pressure switch – Check loading pressure (possibly splinter or contamination in orifice 151, remove tube and purge tube from both ends). – Check negative pressure and test orifice (89). **(For this purpose, remove and purge the tubing from both ends)** – Check check valve (118) and filter (120). – Check CDS valve (104). – Verify tightness of CDS path. – Check acknowledgement of pressure switch on P.C.B. LP 633: Bicarbonate: X633L/ A19 – Cartridge filter upstream of degassing pump clogged or wrong filter (filter for disinfectant container) installed. Filters can be distinguished by different adapters.

Rinse Failure F08

Rinse-free program, Dis, HDIS, or mandatory rinse in Dis I to V.
Pressure drop during the monitoring phase on PSW_102 (S123) (concentrate) or pressure build-up impossible.

- Check switching point of pressure switch.
- Check loading pressure.
  (possibly splinter or contamination in orifice 151, remove tube and purge tube from both ends)
- Check negative pressure and orifice (89).
  (For this purpose, remove and purge the tubing from both ends)
- Check check valve (117) and filter (119).
- Check CDS valve (102).
- Verify tightness of CDS path.
- Check acknowledgement of pressure switch on P.C.B. LP 633:
  Concentrate: X633L/ A20
- Cartridge filter upstream of degassing pump clogged or wrong filter (filter for disinfectant container) installed. Filters can be distinguished by different adapters.

Rinse Failure F09

Five minutes before the end of the mandatory rinse in Dis I to V.
Pressure switch PSW_104 (S124) (bicarbonate) or PSW_102 (S123) (concentrate) did not open after pressure reduction.
See Rinse Failure F12.

Rinse Failure F12

Rinse-free program with following Dis or HDIS or mandatory rinse as individual program in Dis I to V.
The pressure switches for PSW_104 (S124) (bicarbonate) and for PSW_102 (S123) (concentrate) could not be opened.

- Membrane pumps fail to run.
- V 102 or 104 fails to open.
- Pressure on distribution piping > 500 mbar (according to specification, the permissible pressure is max. 500 mbar) pressure peaks on distribution piping: Frequently occurs in distribution pipings with user points if e.g. several patients are disconnected simultaneously and disinfection is started.
- Switching point of pressure switch too low:
  desired value = 700 mbar ± 20 mbar
- Check acknowledgement of pressure switch on P.C.B. LP 633:
  Bicarbonate: X633L/ A19

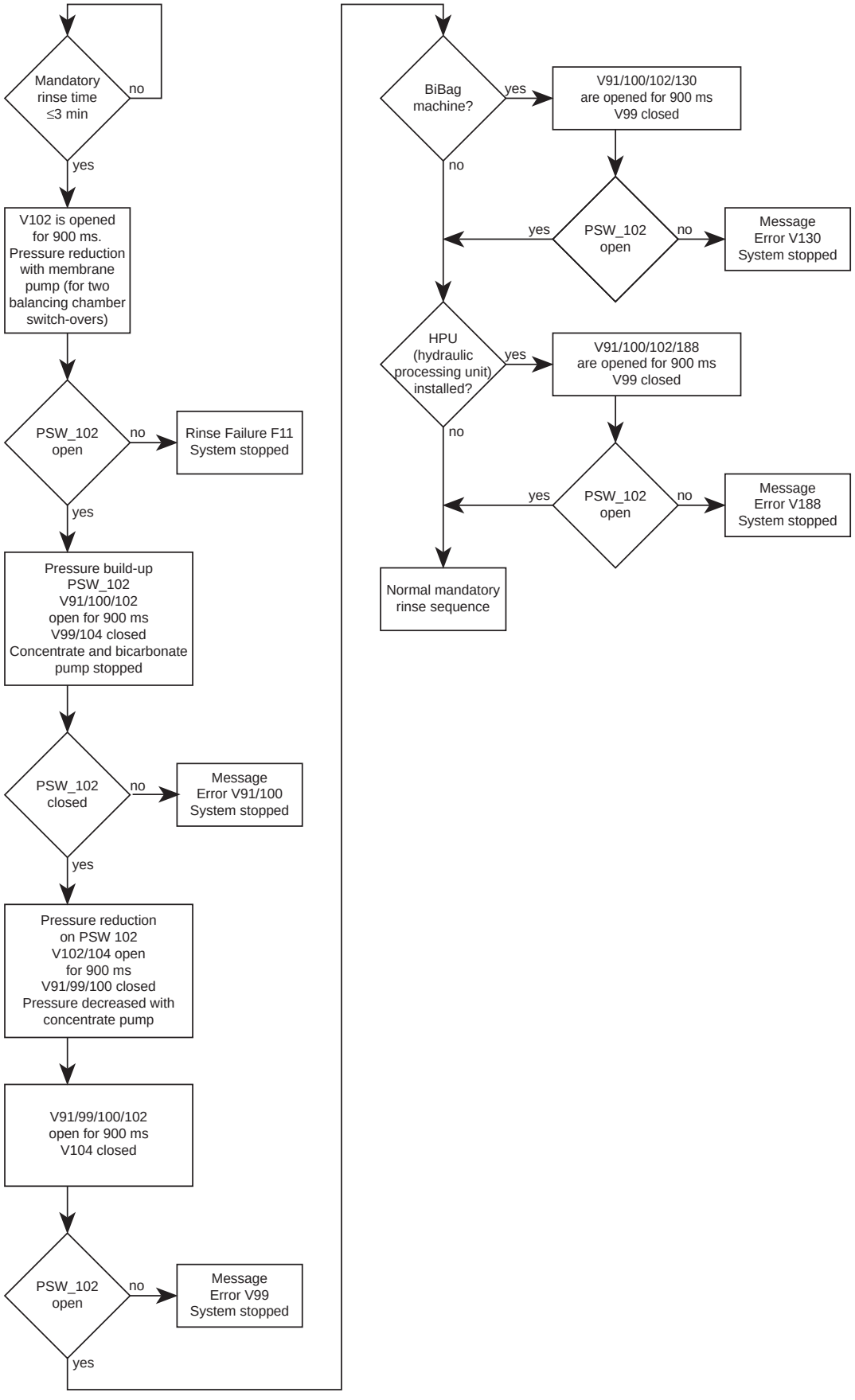
Rinse Failure F13

Rinse-free program with following Dis or HDIS or mandatory rinse as individual program in Dis I to V.
Pressure drop during the monitoring phase on PSW_104 (S124) (bicarbonate) or PSW_102 (S123) (concentrate) or pressure build-up impossible.

- Check switching point of pressure switch.
- Check loading pressure.
  (Possibly splinter or contamination in orifice 151; remove tube and blow through tube from both ends)
- Check negative pressure and orifice (89).
  (For this purpose, remove and purge the tubing from both ends)
- Check check valve (117/118) and filter (119/120).
- Check CDS valve (102/104).
- Verify tightness of CDS path.
- Check acknowledgement of pressure switch on P.C.B. LP 633:
  Bicarbonate: X633L/ A19
  Concentrate: X633L/ A20
- Cartridge filter upstream of degassing pump clogged or wrong filter (filter for disinfectant container) installed. Filters can be distinguished by different adapters.

In case of F07, F08 and F13, the “DO NOT SWITCH OFF !!” message can, in addition, be alternately displayed.
However, this message is displayed only if a mandatory rinse program is requested, since the concentrate and bicarbonate lines still have to be emptied before the machine is switched off.

- **Hydraulics test (check of V91, V99, V100) in machines with central delivery system**



Error message	*Description*
Rinse Failure F11	Three minutes before the end of the mandatory rinse in Dis I to V. The pressure switch PSW_102 (S123) (concentrate) did not open after pressure reduction. – Pressure on distribution piping > 500 mbar (according to specification, the permissible pressure is max. 500 mbar). Pressure peaks on distribution piping: Frequently occurs in distribution pipings with user points if e.g. several patients are disconnected simultaneously and disinfection is started. – Switching point of pressure switch too low: desired value = 700 mbar ± 20 mbar – Membrane pumps fail to run – V102 fails to open electrically or mechanically – Check acknowledgement of pressure switch on P.C.B. LP 633: X633L/ A20
V91/V100 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. V91 or V100 cannot be opened. – V91 or V 100 fail to open electrically: P.C.B. LP 634: V91 = X634L/A12; V100 = X634L/C13 – V 91 or V 100 mechanically not open: check sieve (148) upstream of V100, or valves clogged – V99 constantly open (electrically P.C.B. LP 634: X634L/B12 or mechanically) – V 102 not open – Pressure switch for PSW_102 (S 123) fails to switch
V99 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. V99 cannot be opened. – V 99 fails to open electrically: P.C.B. LP 634: X634L/ B12. – V 99 fails to open mechanically: check sieve (149) before V99, or V99 clogged. – Pressure switch for PSW_102 (S 123) fails to open.
V130 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. V130 cannot be opened (applicable to machines with BIBAG only). – V130 electrically defective: P.C.B. LP 634: X634L/ A4 – V130 mechanically defective or clogged – Pressure switch for PSW_102 (S 123) fails to open. – Check tubing for bicarbonate suction line and bibag block.
V188 Failure	V188 fails to open. – V188 electrically defective. – V188 mechanically defective or clogged – Pressure switch for PSW_102 (S123) fails to open.
F14	Shortly before the end of the mandatory rinse in Dis I to V (CDS: Dis I to IV). The hydraulics test has not been completed correctly, possibly caused by flow problems.

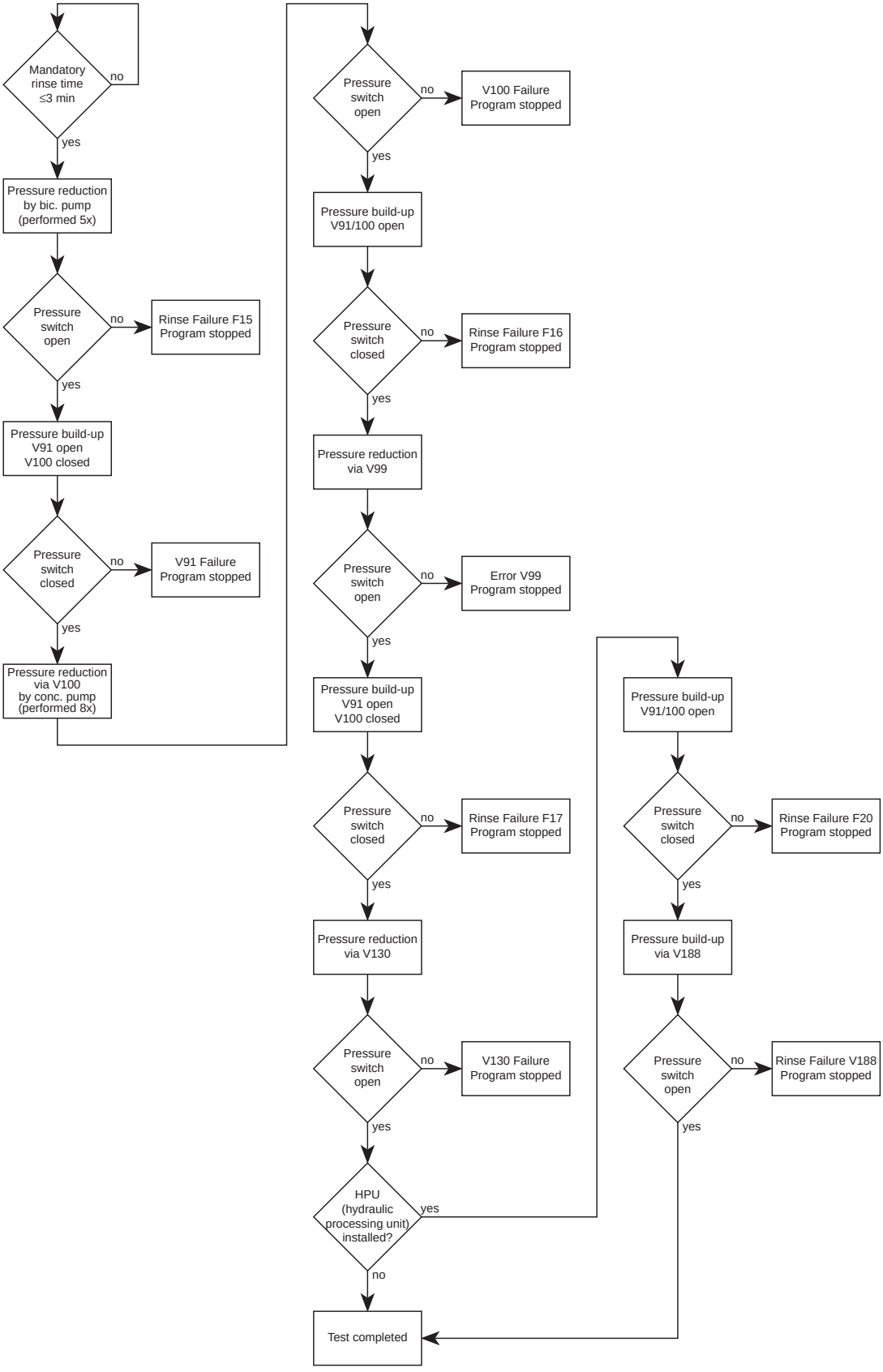
- **Hydraulics test (check of V91 and valve 98) in machines without central delivery system**

Error message	*Description*
F14	Three minutes before the end of the mandatory rinse in Dis I to V. It was not possible to readjust the flow to 750 ml/min ±50 ml/min. V91 defective.
V91 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. After V91 has opened, a flow > 950 ml/min failed to develop. V91 or valve V98 defective.
F14	Shortly before the end of the mandatory rinse in Dis I to V. The hydraulics test has not been completed correctly, possibly caused by flow problems.

- **Hydraulics test (check of V91, V99, V100, V130) in machines with BIBAG and without central delivery system**

  The following requirements must be fulfilled to run the hydraulics test:

  1.The test is run during the last 3 minutes of the mandatory rinse program only.
  2.DIP switch 7 DIP array 2 on PCB LP 631 must be set to ON.



Error message	*Description*
Rinse Failure F15	Three minutes before the end of the mandatory rinse in Dis I to V. DS (BIBAG pressure switch 134) could not be opened at the beginning of the test. – Check pressure switch: Switching point: rated value: 100 mbar + 30 – Suction error of bicarbonate pump – V91 constantly electrically or mechanically open – V99/100 constantly electrically or mechanically closed
V91 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. It is impossible to build up pressure on DS (BIBAG pressure switch 134) via V91. – Pressure switch fails to close mechanically: check switching point. – V91 fails to open electrically: P.C.B. LP 634: X634L/A12. – V91 fails to open mechanically (possibly clogged) – V130 electrically not closed: P.C.B. LP 634: X634L/ A4 – V130 fails to close mechanically (possibly clogged). – Bibag connector leaking (check O rings) – Sealing on the bicarbonate suction tube leaking. – Check acknowledgement of pressure switch on P.C.B. LP 633: X633L/A8. – V99 constantly electrically or mechanically open.
V100 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. It is impossible to build up pressure on DS (BIBAG pressure switch 134) via V100. – V100 fails to open electrically: P.C.B. LP634: X634L/C13. – V100 fails to open mechanically (possibly clogged). – V91 constantly electrically or mechanically open – Concentrate pump fails to pump. – Filter (148) clogged. – Pressure switch fails to open.
Rinse Failure F16	Three minutes before the end of the mandatory rinse in Dis I to V. DS (BIBAG pressure switch 134) cannot be closed. V99 or V130 is leaking. – V91 fails to open electrically or mechanically. – V99 constantly electrically or mechanically open – V130 constantly electrically or mechanically open – Sealing on the concentrate suction tube leaking. – Pressure switch fails to close.
V99 Failure	Three minutes before the end of the mandatory rinse in Dis I to V. DS (BIBAG pressure switch 134) cannot be opened. V99 does not open. – V99 fails to open electrically or mechanically. – V100 fails to open electrically or mechanically. – Pressure switch fails to open. – V91 electrically or mechanically open – Filter (149) upstream of V99 clogged

Rinse Failure F17

Three minutes before the end of the mandatory rinse in Dis I to V.
DS (BIBAG pressure switch 134) cannot be closed.
- V91 fails to open electrically or mechanically.
- V130 electrically or mechanically open
- V100 electrically or mechanically open
- Pressure switch fails to close.

V130 Failure

Three minutes before the end of the mandatory rinse in Dis I to V.
DS (BIBAG pressure switch 134) cannot be opened.
- V130 fails to open electrically or mechanically.
- Pressure switch fails to open.
- Check tubing for bicarbonate suction line and bibag block.
- Bicarbonate line squeezed at strain relief.
- Narrowing in the reducer on the bibag connector

Rinse Failure F 20

Impossible to close the pressure switch (134) via V91/100.
- V91 fails to open electrically or mechanically.
- V130/V188 electrically or mechanically open.
- Pressure switch fails to close.

V188 Failure

The pressure on pressure switch (134) cannot be reduced via V188.
- V188 fails to open electrically or mechanically
- Pressure switch fails to open
- Verschlauchung für Konzentratansaugleitung und Luftabscheide-block prüfen / Check tubing for carbonate suction line and air separator block.
- Concentrate line squeezed at strain relief.

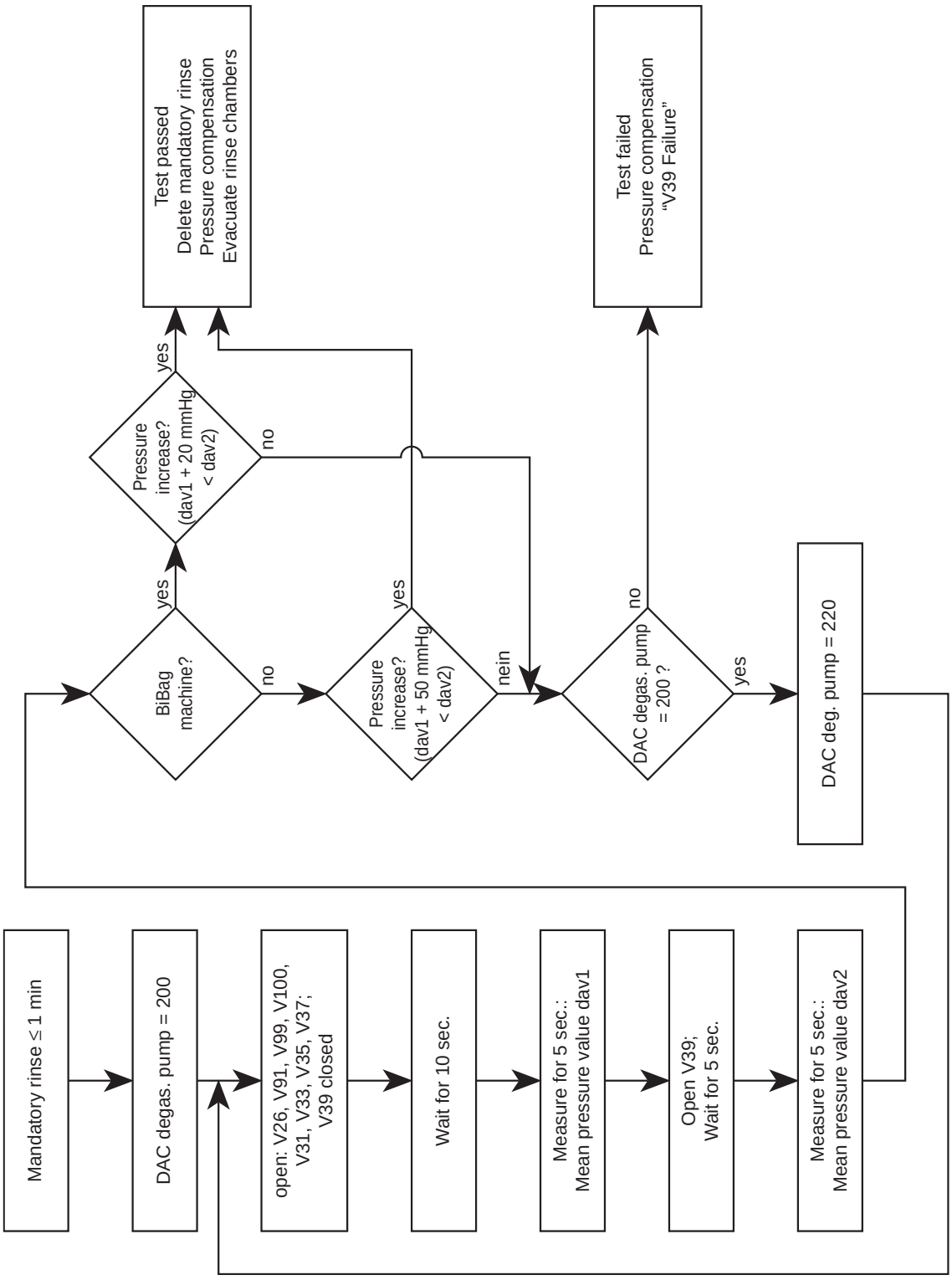
Rinse Failure F14

Shortly before the end of the mandatory rinse in Dis I to V.
The hydraulics test has not been completed correctly, possibly caused by flow problems.

- **V39 test**

  The following requirements must be fulfilled to run the V39 test:

  1. The test is run during the last minute of the mandatory rinse program only.
  2. DIP switch 5 DIP array 2 on PCB LP 632 must be set to OFF.



Error message	*Description*
V39 Failure	On opening V39 a difference in pressure (averaged value V39 open – averaged value V39 closed) is detected on the dialysate pressure transducer (182): Standard system: < 50 mmHg bibag system: < 20 mmHg – V39 fails to open / close electrically or mechanically (possibly hydraulic processing unit defective). – It is impossible to re-adjust the degassing pump (P.C.B. LP 634). – V91, V99, V100 fail to open electrically or mechanically. – Dialysate pressure transducer (182) defective or not calibrated (possibly hydraulic processing unit defective) – Filter 210 (upstream of degassing pump) clogged.

- **Further messages which may be displayed before or during a cleaning program**

Error message	*Description*
Blood Sensed by OD	Start of a cleaning program in RI I to II, HR I to III, Dis I to V. The optical detector in the air detector module recognizes blood.
Shunt Cover open	Start of a cleaning program or during a cleaning program in RI I to II, HR I to III, Dis I to V. The shunt interlock is not closed.
Dialines not conn	Start of a cleaning program in RI I to II, HR I to III, Dis I to V. The dialysate couplings are not connected to the shunt interlock.
No LD alarm	Priming of the blood line system in RI I to II, HR I to III, Dis I to V. The drip chamber in the air detector module does not recognize any alarm.
Conc line not conn	Start of a cleaning program in RI I to II, HR I to III, Dis I to V, or end of the disinfectant suction phase in Dis V. The concentrate plug is not connected to the rinse chamber. Reconnect the concentrate plug to the rinse chamber.
Bic line not conn	Start of a cleaning program in RI I to II, HR I to III, Dis I to V, or end of the disinfectant suction phase in Dis V. The bicarbonate plug is not connected to the rinse chamber. Reconnect the bicarbonate plug to the rinse chamber.
Voltage Failure	During a cleaning program in RI I to II, HR I to III, Dis I to V. The 24-V/12-V supply voltages are drifting. This error can be acknowledged for 8 sec by pressing the respective program key.
CPU-II failed	During a cleaning program in RI I to II, HR I to III, Dis I to V. The watchdog relay has dropped. Communication (RxD or TxD) may be disturbed.
High temperature	During a cleaning program in RI I to II, HR I to III, Dis I to V. Temperature > 41 °C; > 90 °C during HR; > 91 °C during IHR. The machine continues to run. The alarm tone can be acknowledged. Upon error elimination, the message is automatically cleared.
Low temperature	During a cleaning program in RI I to II, HR I to III, Dis I to V. Temperature < 33 °C; < 78.5 °C during HR. The machine continues to run. The alarm tone can be acknowledged. Upon error elimination, the message is automatically cleared.

Water alarm

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
The float switch transmits the “no water available” message for more than 10 seconds.
The balancing chamber has stopped; V41 is permanently open.
Upon error elimination, the message is automatically cleared.

Water alarm

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
For more than 30 seconds, the float switch fails to signal that water is required (not applicable to recirculation programs).
The machine continues to run.
Upon error elimination, the message is automatically cleared.

Flow alarm

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
A current rise pulse is not recognized for more than 12 seconds.
The machine continues to run at “Eigentakt” (10 seconds).
Upon error elimination, the message is automatically cleared.

Upper Flow Alarm

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
The cleaning flow increases to > 1000 ml/min. The program has stopped.
The error can be acknowledged by pressing the respective cleaning program key.

UF-Pump failed

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
The UF pump has stopped or the rate deviates (2800 ml/h < UFR < 6000 ml/h). The program has stopped.
The error can be acknowledged by pressing the respective cleaning program key.

UF-Pump 2 failed

During a cleaning program
in RI I to II, HR I to III, Dis I to VI.
The UF2 pump has stopped (applicable only to machines with 4008 HDF).
The error can be acknowledged by pressing the respective cleaning program key.

Dial. Valve failed

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
V24 or V24B is closed although it should be open.
The program has stopped.
The error message can be acknowledged by pressing the respective program key.

Bypass Valve failed

During a cleaning program
in RI I to II, HR I to III, Dis I to V.
V26 is closed although it should be open.
The program has stopped.
The error message can be acknowledged by pressing the respective program key.

V102 Failure
During a cleaning program
in RI I to II, HR I to III, Dis I to V.
V102 has been opened electrically.
24 V are switched off.
The error cannot be acknowledged.

V104 Failure
During a cleaning program
in RI I to II, HR I to III, Dis I to V.
V104 has been opened electrically.
24 V are switched off.
The error cannot be acknowledged.

HDF-Pump failure
During a cleaning program
in RI I to II, HR I to III, Dis I to V.
The HDF pump has stopped, or the speed deviates (rated value: 400 ml/min, actual value: ≤ 300 ml/min; rated value: 150 ml/min, actual value: ≤ 100 ml/min).
The error message can be acknowledged for one complete cleaning program run by pressing the respective program key. The prompt: “Are you sure ?” is displayed.

Float-Switch Failure
During a disinfectant program in the suction phase in Dis I to V (CDS: Dis I to IV).
The lower switching point of the float switch is not reached within 20 sec.
The program has stopped.

Connect Disinfectant
Disinfectant suction phase in Dis V.
Request to connect the disinfectant.

Press CONFIRM key
Disinfectant suction phase in Dis V.
After the disinfectant has been connected, the Confirm key on the menu panel must be pressed to start the suction procedure.
The program has stopped.

Please Wait
Disinfectant suction phase in Dis V.
Disinfectant is drawn in via the concentrate pump.

Disinfectant empty ?
Disinfectant suction phase in Dis I to V.
Dis V: After the disinfectant has been drawn in, the float switch does not recognize any fluid.
Dis I to IV, Dis VI: The V84 monitoring unit does not recognize any conductivity.

Disinf-Temp. too high
Transition to disinfection in Dis I to V.
Temperature at the end of the rinse-free procedure > 40 °C.
Again and again, the rinse-free procedure is prolonged by 1 minute.
An audible warning is sounded after 4 minutes.
The message is automatically cleared, and it cannot be acknowledged.

Rinse required !	During stored mandatory rinse in Dis I to V. The mandatory rinse has been interrupted (e.g. the machine has been switched off).
Rinse after Disinf.	Selection of a cleaning program, although a mandatory rinse has been requested in HR. A disinfection program has been stopped and subsequently a rinsing or hot rinsing program started.
Power Failure	During a cleaning program in RI I to II, HR I to III, Dis I to V. Line voltage failed.
BIBAG connect. open	Upon start of a cleaning program in RI I to II, HR I to III, Dis I to V. The BIBAG connector is not closed (cap not attached).
Heater error	During the CDS rinsing phase at the end of a hot rinsing program or a hot disinfection program in CDS: HR I to III, Dis II to IV. The heater signal (P.C.B. LP 633: X633R/A26) is not changing for > 40 sec.
Accumulator empty!	Battery voltage <17.2 V ±2.5 % Only in the event of a power failure during the cleaning programs. If the voltage drops below 17 V, the machine will switch off.

1.1.5 Error messages after turning power on

Error message	*Description*
EPROM ERROR	System error. Check the plugs and the EPROM for proper connection. Replace the EPROM, if necessary.
BRAM_#_XXXX_XXXX_XXXX	System error. Switch the machine off and on again. Check the plugs and the BRAM for proper connection. Replace the BRAM, if necessary. Then recalibrate.
RAM ERROR	System error. Switch the machine off and on again. Check the plugs and the RAM for proper connection. Replace the RAM, if necessary.
Keyboard Error	Short-circuit on the keyboard. Switch the machine off and on again. Check the plugs for proper connection. Possible short-circuit on the keys. Replace the front panel, if necessary.
Watchdog Error	This error message can only be displayed shortly after switch-on. Switch the machine off and on again. Check the WD relay and components. Check CPU2/CPU1. Check the plug connectors on the monitor.
XX (not calibrated)	NOVRAM error upon test request. Switch the machine off and on again. Recalibrate the function indicated. Replace the NOVRAM, if necessary.
NTC109 switched off	No valid value has been filed during start in the NOVRAM. The difference in temperature between NTC 109 and NTC 3 is too large. Switch off NTC 109 in the setup menu, or recalibrate the temperature.

1.1.6 Error messages during dialysis

Error message	*Description*
Voltage Failure	The 24-V/12-V supply voltages are drifting. The machine enters the safe state and must be switched off/on. – The 12-V or 24-V operating voltage is outside of the permissible range: 24 V: > 26 V / < 22.5 V 12 V: > 13.5 to 15 V / < 10.5 V – Check the power supply unit. – Power supply unit okay: Check the voltages applied to P.C.B. LP 633: +12 V: X633R/A, C31 +24 V: 24V_EM: X633L/B20
24 V Switched Off	The 24-V supply voltage has fallen below 5 V. The machine enters the safe state and must be switched off/on. – Check the power supply unit. – Power supply unit okay: Check the voltages at P.C.B. LP 633: +24V_EM: X633L/B20 – Remove all plug-in modules. As soon as the machine is running: reconnect each plug-in module individually with the machine switched off; determine the defective module and repair it. – Completely loosen the hydraulic compartment connections. *Caution:* J1 must now be fitted on P.C.B. LP 630 since, without it, the machine would not be able to perform the watchdog test. Be absolutely sure to remove the jumper again for hemodialysis operation. With the machine running, check the short circuit in the hydraulic compartment for 24-V supply and the valves and pumps for short circuit.
CPU-II failed	CPU2 fails to communicate via the serial interface. The machine enters the safe state and must be switched off/on. – The software versions of CPU1 and CPU2 are mismatching. – Hardware defect on CPU2.
Profile time diff.	Deviation in time between CPU1 and CPU2. The error message is emitted 60 seconds after the start of the profile. – The clock module on CPU1 (IC14) is defective; or calibrate the time in case of layout < D.
Cyclical PHT F01	Balancing error. – System leakage. – Applicable to Diasafe machines: On CPU II, the DIP switch array 2, switch 1, is not set to “ON”.
Cyclical PHT F02	Balancing error. – System leakage. – Applicable to Diasafe machines: On CPU II, the DIP switch array 2, switch 1, is not set to “ON”.

Cyclical PHT F03 IC1 or IC3 on P.C.B. LP 632 is defective, or system leakage.

Cyclical PHT F04 It was not possible to complete the test within a specific time interval.

V84 faultiness ! Conductivity is recognized at the V84 electrodes.
This error message is emitted for the first time at the end of the T1 test. The error can be acknowledged for the duration of one hemodialysis procedure by pressing the Dialysis Start key. It is, however, not possible to switch off the flow (Bergström-/ISO-UF operating mode). Should the error occur during Flow OFF, the flow is switched on automatically.

– First of all, it must be verified whether a Rinse Failure F01, F02 or F03 occurred during the previous disinfection procedure (see listing of cleaning program errors). Should this be the case, a disinfection program I to IV (not Dis V) must be completed correctly. The problem can also be corrected using the calibration program, NOVRAM menu item (Reset V84).
– Should this not be possible, the error memory of the machine can be read out.
– Should this neither be possible, the test described below can be performed:
  Remove the disinfectant.
  Switch the machine off and on again.
  Perform or skip the T1 test.
  Should the error message be displayed again at the end of the test, it was generated by a Rinse Failure F01, F02 or F03 and can be cleared only by taking the measures described above.
  Should the message not be displayed again, a second test can be performed:
  Reconnect the disinfectant.
  Set the UF rate and switch on the UF unit.
  Should the error occur at this moment, there is a leakage on V84 (see listing of cleaning program errors).

Shunt Cover open (temporarily)
– P.C.B. LP 633 C24 (100n) temporarily short-circuited.
– Shunt interlock defective (check switches).

Voltage Failure (temporarily) P.C.B. LP 633 C84 (100n) temporarily short-circuited.

UF1 volume - Error
UF2 volume - Error

Failure to pass the test for a UF pump. The fill volume for the secondary air separator is outside the 100 ml ±4 ml tolerance.
Possible cause:
– The UF pump fails to deliver correctly (not calibrated or mechanical defect)
– If the test result is >104 ml, the problem can also be caused by air coming from a poorly deaerated dialyzer.

F327 UF-failure

Pause between two UF1 pump strokes less than 220 ms.
Possible cause:
– CPU-1 defective

F328 UF-failure

Pulse time of one UF1 pump stroke less than 180 ms.
Possible cause:
– Controlling monoflop on LP 634 defective

F329 UF-failure

Pulse time of one UF1 pump stroke exceeds 500 ms
Possible cause:
– Controlling monoflop on LP 634 defective.

F330 UF-failure

Pick-up time of the UF1 pump exceeds 10 sec.
Possible cause:
– Controlling output stage on LP 634 defective.

F331 UF-failure

Theoretical/actual rate of the UF1 pump deviates by more than ±10 %.
Possible cause:
– Communication problem or CPU-1/CPU-2 software problem.

F332 UF-failure

UF1 pump stopped for more than the maximum time period.
Possible cause:
– Controlling output stage on LP 634 defective.
– UF pump interruption
– Communication problem or CPU-1/CPU-2 software problem.

F333 UF-failure

UF1 volume change more than 10 ml although UF is switched off.
Possible cause:
– Communication problem or CPU-1/CPU-2 software problem.

F334 UF-failure

Pause between two UF2 pump strokes less than 220 ms.
Possible cause:
– CPU-1 defective

F335 UF-failure

Pulse time of one UF2 pump stroke less than 180 ms.
Possible cause:
– Controlling monoflop on LP 634 defective.

F336 UF-failure

Pulse time of one UF2 pump stroke exceeds 500 ms.
Possible cause:
– Controlling monoflop on LP 634 defective.

F337 UF-failure

Starting time of the UF2 pump exceeds 10 sec.
Possible cause:
– Controlling output stage on LP 634 defective.

F338 UF-failure

Desired/actual rate of the UF2 pump deviates by more than 10 %.
Possible cause:
– Communication problem or CPU-1/CPU-2 software problem.

F339 UF-failure

UF2 pump stopped for more than the maximum time period.
Possible cause:
– Controlling output stage on LP 634 defective.
– UF pump interruption
– Communication problem or CPU-1/CPU-2 software problem.

F340 UF-failure

UF2 volume change more than 10 ml although UF is switched off.
Possible cause:
– Communication problem or CPU-1/CPU-2 software problem.

F341 UF-failure

Mechanical UF1 pump failure.
Possible cause:
– Broken spring
– Contaminated filter

F342 UF-failure

Mechanical UF2 pump failure.
Possible cause:
– Broken spring
– Contaminated filter

F343 UF-failure

UF1/UF2 pump volume difference
Possible cause:
– Delivery volume altered

- **HPU error**

Error message	*Description*
HPU Error F00	The HPU logs off with index STATUS_ER; no bit is set in the error bit field. – Problem on P.C.B. LP 941 – Problem on CAN distributor board – Problem on P.C.B. LP 763 – Problem on motherboard
HPU Error F01	The cyclic communication has failed for more than 2 seconds. – System error
HPU Error F02	The response to an event violated the time-out. – System error
HPU Error F03	An error occurred in the program sequence. – System error
HPU Error F04	Voltage drop (24V_SW) during HPU operation. – 24V voltage supply on P.C.B. LP 941 failed (watchdog dropped).
HPU Error F05	Watchdog test failed to be passed. – Watchdog circuit on P.C.B. LP 941
HPU Error F06	Reference voltage monitoring detected an error. – Reference voltage circuit on P.C.B. LP941 is defective.
HPU Error F07	The HPU was logged off by the monitor. Will not be displayed since CPU1 has already stopped the communication. – System error
HPU Error F08	General valve malfunction: may occur in HPU SW 2.01 or 3.00. (Software versions before evaluation of the HPU errors). – System error
HPU Error F09	Malfunction of the compressor (185) – MV43 defective or activated – Compressor 185 defective or activated – Error on P.C.B. LP 941
HPU Error F10	Malfunction of valve MV39 – MV39 defective or activated – Error on P.C.B. LP 941
HPU Error F11	Malfunction of test valve (183) – MV43 defective or activated – MV183 defective or activated – Error on P.C.B. LP941

HPU Error F12

Malfunction of evacuation valve (188)
- MV188 defective or activated
- Error on P.C.B. LP 941

HPU Error F13

Malfunction of retentate valve (189)
- MV189 defective or activated
- Error on P.C.B. LP 941

HPU Error F14

Defective component on P.C.B. LP 941
- Error on P.C.B. LP 941

HPU Error F15

Error in the HPU software. Valves are activated incorrectly.
- System error

HPU Error F98

Proceeding to the T1 test is not allowed after restart.
- System error

HPU Error F99

HPU fails without logging off.
- Damaged cable or similar problem
- HPU logged off by CPU1
- CRC error in the transfer HPU → CPU1
- BVM is connected via CAN and software <3.20 is installed in the BVM.
- The VDE test was performed directly after turning the system on. Turn the system on at least 2 minutes before the test.

- **ONLINE module errors**

Error message	*Description*
ONL Error F00	Online module error
ONL Error F01	Watchdog error
ONL Error F02	Watchdog error
ONL Error F03	Watchdog error
ONL Error F04	Error in the program sequence
ONL Error F05	+24V_WD dropped to less than 17V or was switched off
ONL Error F06	Time-out of the communication watchdog exceeded
ONL Error F07	A transmission from the module was not confirmed by the dialysis system
ONL Error F08	General valve error
ONL Error F09	T1 test skipped
ONL Error F10	T1 test for ONLINEplus failed to be passed
ONL Error F11	Reference voltage is outside the tolerance
ONL Error F12	CRC error
ONL Error F13	EEPROM error
ONL Error F14	The monitor disabled the ONLINEplus module
ONL Error F16	Valve error ONL1
ONL Error F17	Valve error ONL1
ONL Error F18	Valve error ONL1

1.2 Functional description of the modules

1.2.1 Blood pump (arterial)

The blood pump ensures a sufficient blood flow in the extracorporeal blood circuit. It is absolutely necessary that sterility is maintained and that the blood is prevented from becoming contaminated.

The blood pump is designed as roller pump provided in an exchangeable plug-in module integrated in the hemodialysis machine. The blood line is installed between a stator, which, with its rolling surface bent in a circle, represents a thrust bearing, and a rotor, which is provided with rollers and pivoted in the stator. The pressure of the rollers causes the development of a narrow or seal. If the rollers are moving in the direction of delivery, the blood is pushed in this direction.

A microprocessor controls the stepper motor with quartz accuracy, depending on the selected delivery rate, the set line diameter, and the monitor signals.

The pressure measuring equipment comprises a piezo-resistive pressure transducer. The pressure-proportional voltage is indicated on the monitor on a quasi-analog LED scale.

Functions of the blood pump:
- RAM and CRC test after turning power on,
- control and monitoring of the function by a dual processor system,
- emergency switchoff in case of an alarm: stop recognition (15 or 30 sec),
- setting of the speed to 180 ml/min during priming,
- measurement of the arterial pressure or the single needle pressure (depending on the model concerned),
- semi-automatic loading and unloading of the line segment.

Error messages:
E.01 Line diameter outside the permissible range
E.02 Undefined hex switch position
E.03 Uncalibrated arterial pressure transducer
E.04 Run-time monitoring error during SN operation
E.05 SN stroke volume outside the permissible range
E.06 SN pressure thresholds outside the range of values of the A-D converter
E.08 Stop alarm
E.09 Error during A-D conversion
E.12 Rotary monitoring error (Hall sensor)
E.13 Monitoring error with regard to current sensing resistors
E.14 Monitoring error with regard to current sensing resistors
E.15 Speed monitoring error

1.2.2 Blood pump (single needle), optional

Essentially, the blood pump (single needle) is identical with the arterial blood pump. The difference lies in the single needle control. During SN operation, the pressure outlet of the compliance vessel is connected to the pressure connector of the SN pump. The pressure transducer is protected by a hydrophobic filter both in the external and the internal tubing system.

The SN stroke volume can be set within a range from 10 ml to 50 ml in increments of 5 ml.

The lower changeover point is fixed to 75 mmHg.

The upper changeover point depends on the stroke volume:

Stroke volume (ml)	10	15	20	25	30	35	40	45	50
Changeover point (mmHg) ± 7 mmHg	110	130	150	172	195	219	244	270	299

1.2.3 Heparin pump

Since the blood flows through an extracorporeal circuit during hemodialysis, coagulation would occur within a short time. The heparin pump allows continuous heparinization of the blood causing the coagulation time to be prolonged. Since the heparin volume required during hemodialysis depends on the respective patient concerned, it must be determined by the attending physician.

A syringe plunger is moved by a drive rod, which is connected to a threaded spindle via a sliding block. A microprocessor-controlled stepper motor causes the spindle to rotate. Depending on the type of activation, the plunger moves up or down. A Hall sensor indicates the upper end position of the plunger. The protective system of the pump comprises a speed monitoring unit (slotted disc with optical sensor) as well as a motor current monitoring unit.

The different syringe types can be selected by means of a coding switch:

0 20-ml syringe
1 30-ml syringe
2 50-ml syringe
3 10-ml syringe
4 – F unused



Caution
Do not change the coding switch position during operation.

Function of the heparin pump:

– RAM and CRC test after turning power on,
– delivery rate adjustable from 0.1 ml to 10 ml in increments of 0.1 ml,
– delivery time preselection (stopwatch) adjustable from 1 min to 9 h 59 min,
– bolus administration.

Error codes:

E01	*Hardware error*, gate array defective
E02	*Hardware error*, reset by spike or test alarm
E03	Checksum error, data loss
E04	First start-up
E05	Incorrect hex switch position
E06	Missing or incorrect data for the variable syringe
E07	Selection of wrong syringe
E11 to E13	*Step error*
E12	Overdelivery during fast return
E14 to E15	*Error in direction of rotation* (software not equal to hardware!)
E16	*Software error*
E19	*Optical sensor error* (stop of syringe holder or optical sensor defective)
E20	*Error in direction of rotation*
E33	*Step error* (impermissible range)
E37	*Slotted disc error*
E40	*Division error* (division by zero)
E41 to E42	*Error in direction of rotation* (fast return)
E43 to E44	*Error in direction of rotation* (slow return)
E45 to E46	*Error in direction of rotation* (fast advance)
E47 to E48	*Error in direction of rotation* (slow advance)
E49	*Step error* (underdelivery during slow advance)
E50	*Step error* (underdelivery during slow return)
E51	*Step error* (overdelivery during fast advance)
E55	*Error in step counting* (optical sensor defective or mechanics too sluggish; no pulses from the slotted disc)
E56	*Error in step counting* (more than 8 pulses during transition of the slotted disc; the slotted disc is oscillating)
E90	*Display error*

1.2.4 Air detector

The penetration of air into the patient's extracorporeal blood circuit may cause an air embolism. In order to catch limited amounts of air and to separate accompanying air bubbles, the venous blood line is expanded (venous drip chamber). A major task of the air detector is to monitor the filling level in the venous drip chamber.

Ultrasonic air detector

The protection system against air infusion uses the method of ultrasonic transmission. Ultrasonic converters are attached on either side of the venous bubble catcher. At periodic intervals of approx. 90 ms, a transmitting resonator generates attenuated ultrasonic vibrations at a natural resonance of approx. 90 kHz, which are absorbed by a receiving resonator. The amplitude of the signal received is dependent upon the medium between the converters. Its value is at its minimum with the bubble catcher empty (air) and at its maximum with bubble-free fluids. The amplitude decreases with increasing air content (foam). The signal path is fail-safe up to and including the receiving resonator, i.e. the failure of any component always leads to a smaller amplitude and, thus, to an alarm. Starting at the receiving resonator, the signal voltage is always sent onto two independent receiver paths. As soon as the signal is too weak, one of these receiver paths causes the blood pump to stop and the other the venous line clamp to close.

The ▲ and ▼ keys are used to both raise and lower the blood level in the venous bubble catcher. As long as the ▲ key is pressed, the venous line clamp closes. The vent valve in the air detector module opens, and the blood level rises. The blood pump runs at reduced speed (180 ml/min). As long as the ▼ key is pressed, the venous line clamp remains open. The vent valve in the air detector module opens, the ventilation pump is running, and the blood level sinks. The blood pump runs at the preselected speed.

Optical detector

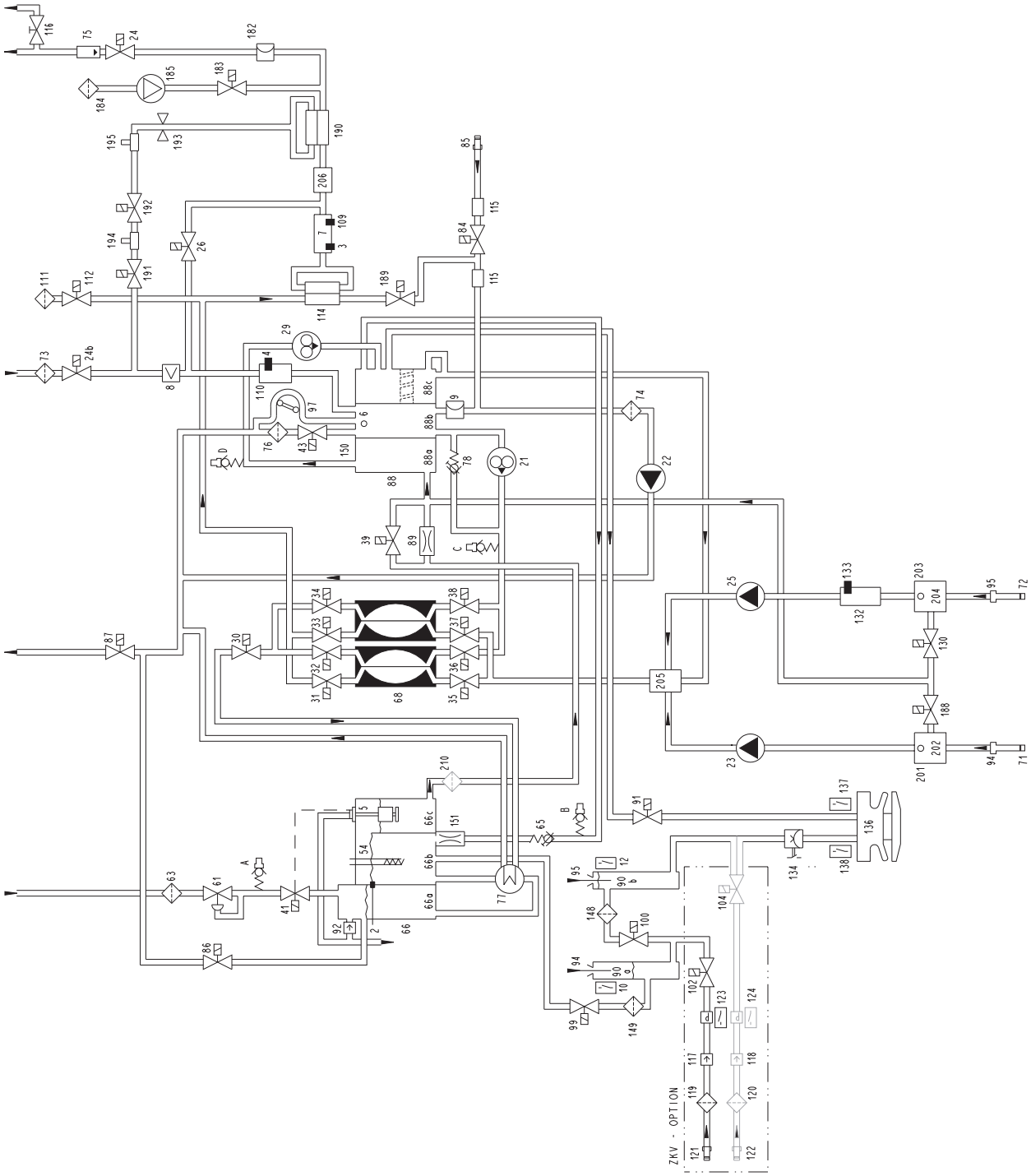
The optical detector serves to detect if there is blood or saline solution or air in the venous return line downstream of the bubble catcher. In the hemodialysis machine, the hemodialysis phase is defined by presence of a dark medium and the preparation phase by presence of a clear medium.

Venous pressure measurement

The venous pressure measuring equipment comprises a piezo-resistive pressure sensor provided on the P.C.B. with following operational amplifier. The pressure-proportional output voltage is supplied onto the logic P.C.B. in the monitor. There, the pressure is indicated on a quasi-analog LED scale, and the transmembrane pressure is computed by determinig the difference between the dialysate pressure and the venous pressure.

1.3 Functional description of the hydraulic unit

Fig.: Flow diagram



Legend

2 Temperature sensor
3 Temperature sensor
4 Temperature sensor (OCM option)
5 Float switch
6 Level sensor
7 Conductivity cell
8 Blood leak detector
9 Pressure transducer
10 Reed contact for concentrate
12 Reed contact for bicarbonate
21 Flow pump
22 UF pump
23 Concentrate pump
24 Dialyzer valve 1
24b Dialyzer valve 2
25 Bicarbonate pump
26 Bypass valve
29 Degassing pump
30 Outlet valve
31 Balancing chamber valve 1
32 Balancing chamber valve 2
33 Balancing chamber valve 3
34 Balancing chamber valve 4
35 Balancing chamber valve 5
36 Balancing chamber valve 6
37 Balancing chamber valve 7
38 Balancing chamber valve 8
39 Negative pressure valve
41 Water inlet valve
43 Fill valve
54 Heater rod
61 Pressure reduction valve
63 Filter/water inlet
65 Loading pressure valve
66 Heater block
66a Water inflow chamber
66b Heater rod chamber
66c Float switch chamber
68 Balancing chamber
71 Filter/concentrate
72 Filter/bicarbonate
73 Filter/dialysate
74 Filter/UF
75 External flow indicator
76 Filter/fill valve
77 Heat exchanger
78 Relief valve
84 Disinfection valve
85 Disinfection connector
86 Recirculation valve
87 Drain valve
88 Multifunction block
88a Degassing chamber
88b Secondary air separator
88c Primary air separator
89 Degassing orifice
90a Acetate rinse chamber
90b Bicarbonate rinse chamber
91 Rinse valve
92 Vent valve
94 Concentrate suction tube
95 Bicarbonate suction tube
97 Air separating pump
98 Rinse valve
99 Rinse valve
100 Rinse valve
102 CDS, concentrate valve
104 CDS, bicarbonate valve
109 Temperature sensor
110 Conductivity cell (OCM option)
111 Hydrophobic filter
112 Vent valve
114 Dialysate filter
115 Disinfection valve sensor
116 Fluid sample valve
117 Check valve (concentrate)
118 Check valve (bicarbonate)
119 Filter (concentrate)
120 Filter (bicarbonate)
121 CDS, concentrate connector
122 CDS, bicarbonate connector
123 Pressure switch for V102
124 Pressure switch for V104
125 Temperature compensation plate
130 Bibag drain valve
132 Bibag conductivity cell
133 Bibag temperature sensor
134 Bibag pressure transducer
136 Bibag connector
137 Bibag microswitch 1
138 Bibag microswitch 2
148 Filter (rinse valve 100)
149 Filter (rinse valve 99)
151 Orifice
182 Pressure transducer 2 (Diasafe plus option)
183 Test valve (Diasafe plus option)
184 Compressor (Diasafe plus option)
188 Evacuation valve
189 Retentate valve
190 Online filter (Online plus option)
191 Online 3 valve (Online plus option)
192 Online 2 valve (Online plus option)
193 Online 1 valve (Online plus option)
194 Rinse port (Online plus option)
195 Substituate port (Online plus option)
201 Concentrate air separator
202 Concentrate level sensor
203 Bicarbonate air separator
204 Bicarbonate level sensor
205 Concentrate / bicarbonate mixing point
206 Buffer volume chamber
210 Filter

Hydraulics measuring points

A Reduced water inlet pressure
B Loading pressure
C Pressure of flow pump
D Pressure of degassing pump

1.3.1 Description of the hydraulic unit

As soon as the inlet valve (41) opens, the water flows through the pressure reducing valve (61) into the chamber (66a) of the heater block and across the heat exchanger (77) into the heater rod chamber (66b).

The concentrate pump (23) admixes concentrate to the inflowing water per balancing chamber phase.

The vent tubing prevents pressure from building up in chambers b and c. In the hot rinse mode, the developing vapor can escape through the vent tubing.

While it is rising, the fluid is warmed up to the preset temperature by the heater (54). The heater is controlled by the temperature sensor (2).

From chamber b, the dialysate flows into the chamber (66c). Incorporated in this chamber is a float switch (5), which controls the solenoid valve (41), thus ensuring the correct fluid level.

The degassing pump (29) draws in the dialysate via the degassing orifice (89). This generates a negative pressure of 0.8 bar.

In the lines and the following chamber (88a), the dialysate is degassed to a level which is sufficient for hemodialysis.

Via the degassing pump (29), dialysate and released air are directed tangentially into the primary air separator (88c), where air bubbles and the airless dialysate are separated. The air accumulates at the top of the chamber (88c). Then, together with the recirculation flow and via the loading pressure valve (65) as well as the chamber (66c), the air escapes into the atmosphere.

Chamber 88c is provided with a separating disc, which serves to prevent bicarbonate, if added, from being recirculated via the heater rod chamber (66b).

At the bottom of chamber 88c, the degassed dialysate is pressed out and into the balancing chamber (68) by means of the loading pressure.

Together with the eight solenoid valves (31 to 38), the balancing chamber (68) constitutes the balancing system. Each of the two sections of the balancing chamber comprises two compartments separated by an elastic membrane each. Hence, there are two chambers with four spaces:
– **F1** and **F2**: fresh fluid
– **A1** and **A2**: waste fluid (used)

As soon as one of the chambers (A1 or A2) is filled with dialysate, the solenoid valves are reversed in groups of four. The valves are reversed by the electronic evaluation of the current rise pulse of the drive motor of the pump (21), which receives this pulse upon membrane abutment. Within the filling phase, F1 or F2 is filled with fresh dialysate by means of the loading pressure. In order to obtain a continuous flow, a second chamber is switched parallel to the first chamber. The second chamber is operated at an inverse sequence.

Each time the chamber is changed over (maximum deflection of the membrane), all valves are closed for approx. 100 ms (dead time).

From the balancing chamber, the dialysate flows through the conductivity cell (7) with integrated temperature sensor (3). The measured conductivity values are indicated on the monitor in ms/cm, related to 25 °C.

The temperature sensor (3) has the following functions:
– temperature compensation of the conductivity display,
– indication of the dialysate temperature.

Should the actual values (temperature or conductivity) of the dialysate exceed or fall below the limit settings, the bypass valve (26) opens, and the dialyzer valve (24) is closed. The machine is now in the bypass mode. The dialysate is discharged into the drain not via the dialyzer, but via the secondary air separator (88b) and the balancing chamber (68).

If the actual conductivity and temperature values of the dialysate are within the set limits, the dialyzer valve (24) opens. The valve (26) is closed. The dialysate flows to the dialyzer.

After the dialyzer, the dialysate which is now loaded with the substances usually eliminated with the urine flows into the secondary air separator (88b) via a filter (73), the valve (24b) and the blood leak detector (8). The secondary air separator (88b) comprises the pressure transducer (9) and the level sensor (6).

With a hematocrit of 0.25, blood losses of 0.5 ml per minute are recognized in the dialysate by the blood leak detector.

Together with the venous back pressure, the signal of the pressure transducer (9) is evaluated and indicated on the monitor as TMP. The fluid level in the secondary air separator (88b) is monitored by the level sensor (6). Due to the secondary air separator (88b), only airless dialysate is always delivered into the balancing chamber (68). Any presence of air bubbles in the balancing chamber (68) would cause balancing errors.

The dialysate is pressed into the balancing chamber (68) by the flow pump (21). As mentioned above, the balancing chamber valves are reversed by the current rise pulses of the drive motor of the flow pump. Using the speed of this pump, the dialysate flow can be adjusted in the dialysis program: 300, 500, and 800 ml/min. In the cleaning programs, the flow of the dialysate is fixed.

The relief valve (78) is used to limit the pressure of the flow pump before the balancing chamber to approx. 2 bar.

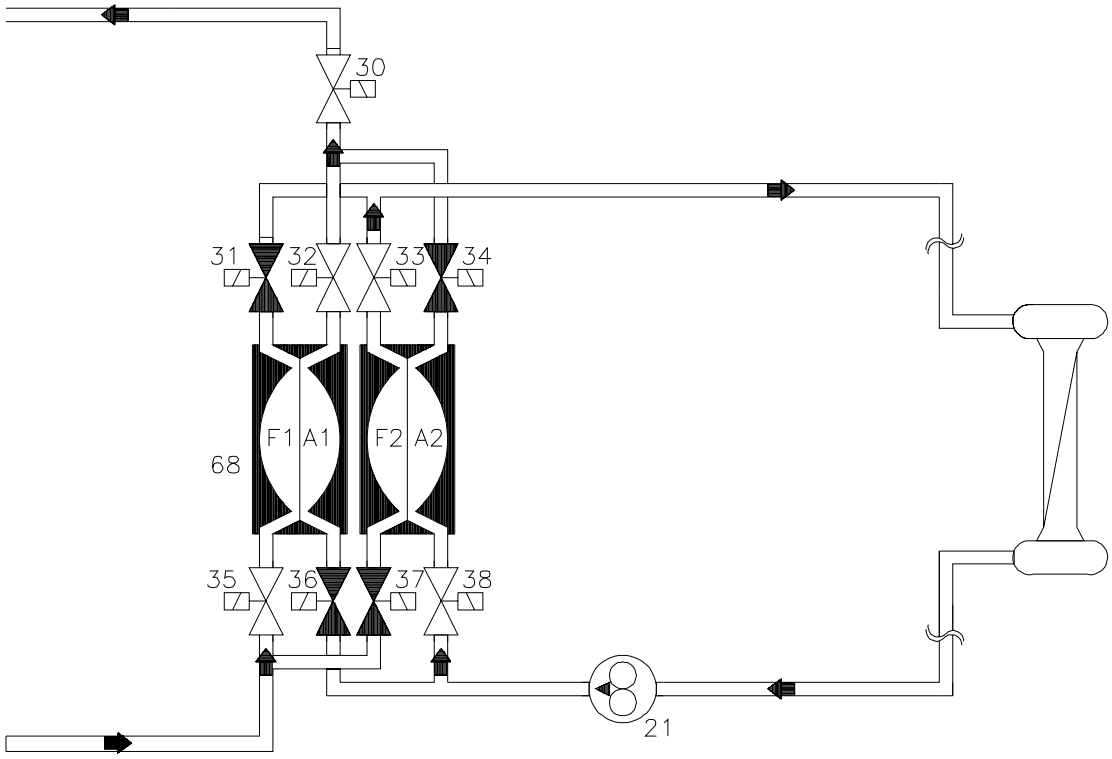
After the balancing chamber, the dialysate flows through the valve (30), the heat exchanger (77) and the valve (87) into the drain.

The valves (86) and (87) serve to recirculate fluid during the hot rinsing and disinfection programs.

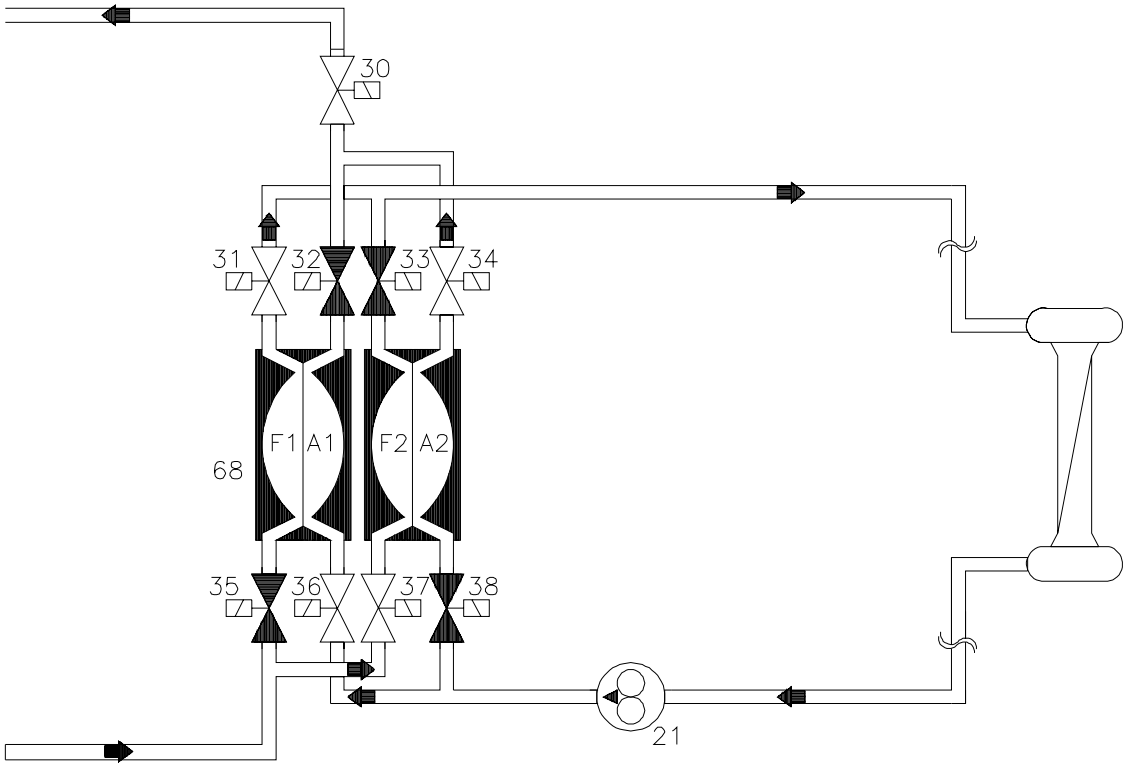
1.3.2 Theory of operation of the balancing chamber

- **(Standard program)**

1st cycle:



2nd cycle:



1st cycle: Closed valves: 31, 34, 36, and 37
F1 is filled with fresh dialysate.
A1 used dialysate is discharged into the drain.
F2 fresh dialysate is forced into the dialyzer.
A2 is filled with used dialysate.

2nd cycle: Closed valves: 32, 33, 35, and 38
F1 fresh solution is forced into the dialyzer.
A1 is filled with used dialysate.
F2 is filled with fresh dialysate.
A2 used dialysate is discharged into the drain.

This system ensures that equal amounts of fluid enter and exit the dialyzer. This leads to an exact balancing of the dialysate and, in conjunction with the ultrafiltration pump (22), a controlled volumetric ultrafiltration.

- **Secondary air purging by the air separation pump 97**

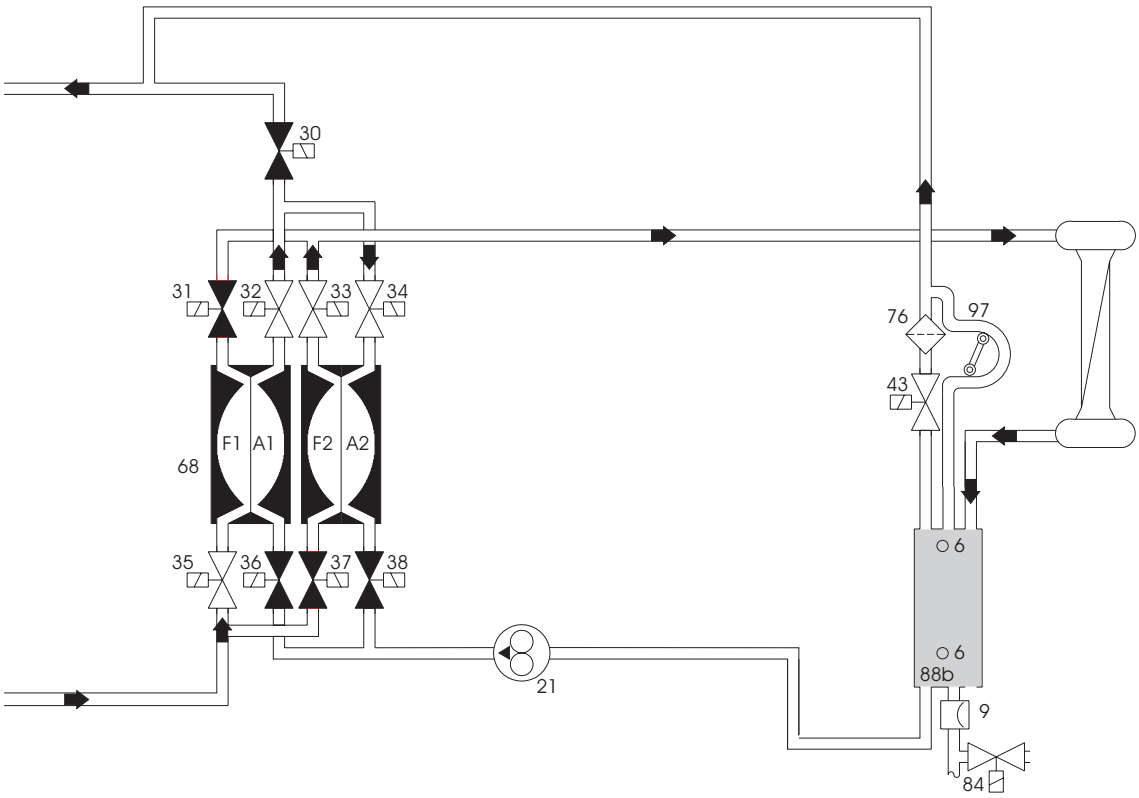
As soon as the fluid level in the secondary air separator (88b) has dropped below the level sensor (6), this sensor activates the air separation pump (97). Should the fluid level not have reached the level sensor (6) within a given time period, the FILL PROGRAM is started.



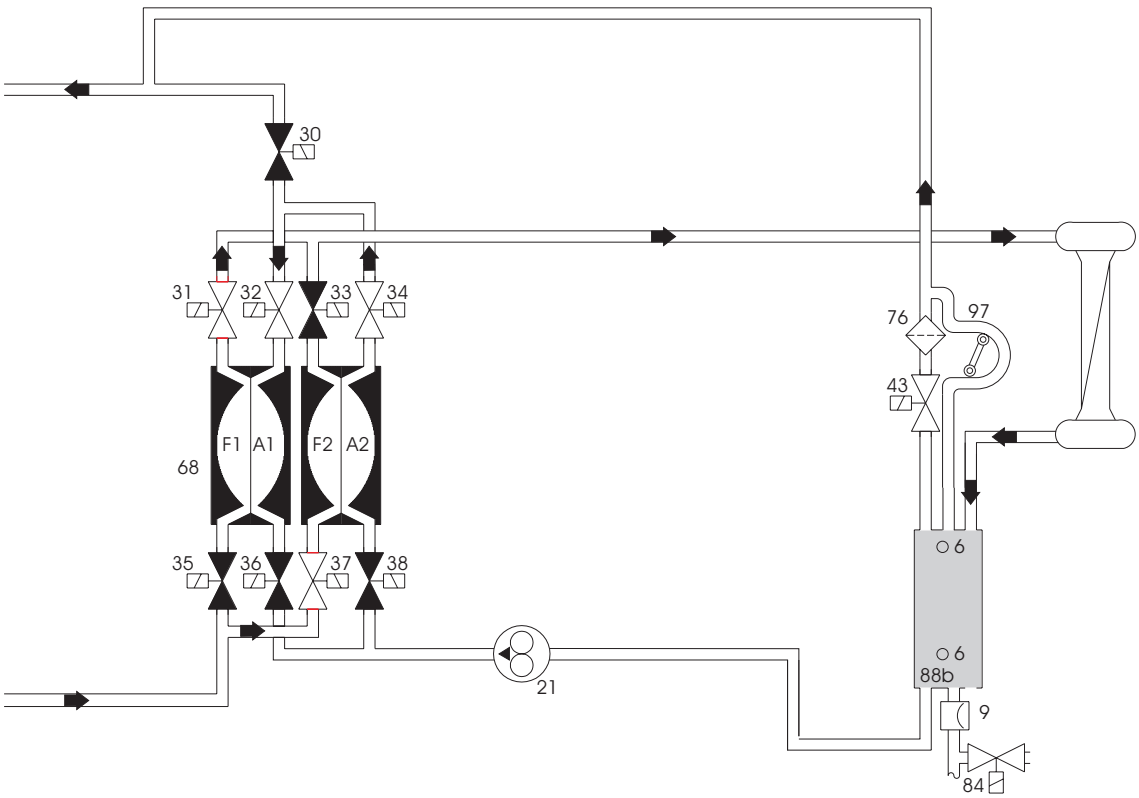
Note
In order to recognize the fluid level, the level sensor (6) requires fluid with a certain minimum conductivity, which is definitely achieved in all dialysis programs. Separation of air is only required in the dialysis programs. In all other programs, the air separation pump (97) and the valve (43) are force-actuated.

- **FILL PROGRAM: air separation by valve 43 at atmospheric pressure**

1st cycle:



2nd cycle:



If not enough air was separated and the fluid level is still below the level sensor (6), the FILL PROGRAM is activated.

The pump (21) fills either chamber A1 or chamber A2. Valves (36), (38), and (30) are closed. The valve (43) opens. The air can escape into the drain.

1st cycle: Chamber F1 is filled. This forces the fluid from chamber A1 into chamber A2. The fluid is then forced into the seconary air separator (88b) by chamber A2 via the dialyzer.

2nd cycle: Chamber F2 is filled. This forces the fluid from chamber A2 into chamber A1. The fluid is then forced into the secondary air separator (88b) by chamber A1 via the dialyzer.

Filling is performed in this way to prevent a change in conductivity. As is the case in the standard program, here as well one stroke of the concentrate pump is still accomplished per balancing chamber cycle (30 ml).

A fill program is always activated at the beginning of hemodialysis (to fill the dialyzer). Should it be activated during the hemodialysis procedure (OD dark), this is shown on the display.



Note
Repeated activation of the fill program during treatment indicates a defect (leakages).

1.3.3 Central delivery system option

The central delivery system is connected to the connectors 121/122. The concentrate flows into the rinse chambers via the inlet filters and the valves 102/104. Through the connected concentrate suction tubes, the concentrate pumps deliver the concentrate to the mixing point.

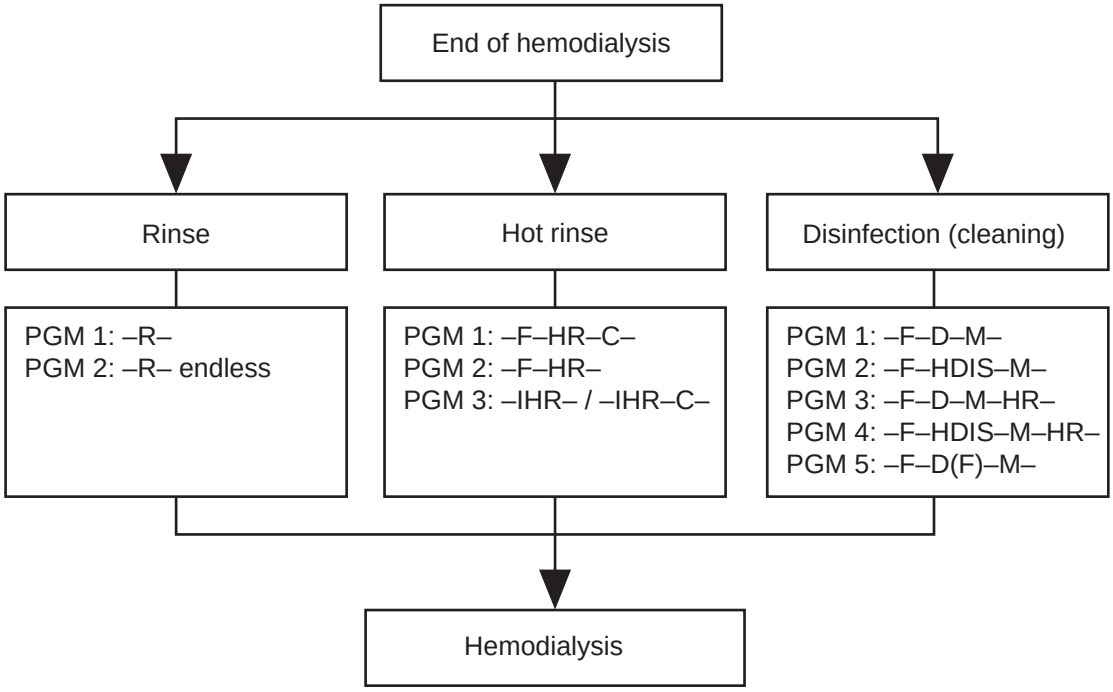
During hemodialysis, the valves 91/99 and 100 are closed. Depending on the central delivery system, V102 and/or V104 are open.

During the cleaning programs, the valves 102 and 104 are closed. During the suction phase of concentrate pump and bicarbonate pump, the valves 91 and 99 open for 500 ms upon each balancing chamber changeover. Valve 100 is open.

In order to check the tightness of the valves 102 and 104, the pressure switch is tested during the rinse-clear phase with following disinfection or hot disinfection or a mandatory rinse. To perform this test, pressure is applied to the two lines between the check valves 117/118 and the valves 102/104. The pressure switches P123 and P124 are used to monitor the pressure. Three minutes before the mandatory rinse program is completed, a functional check of the valves 91/99 and 100 is performed.

1.3.4 Program runs during the cleaning programs

Fig.: Flow chart of cleaning programs – overview



- **Explanation of the abbreviations used**

PGM	Program
R	Rinse
R endless	Endless rinse
F	Rinsing clear
HR	Hot rinsing
C	Cooling rinse
D	Disinfection
D(F)	Disinfection Disinfectant drawn in from the front (concentrate suction tube).
HDIS	Hot disinfection
M	Mandatory rinse
IHR	Integrated hot rinsing

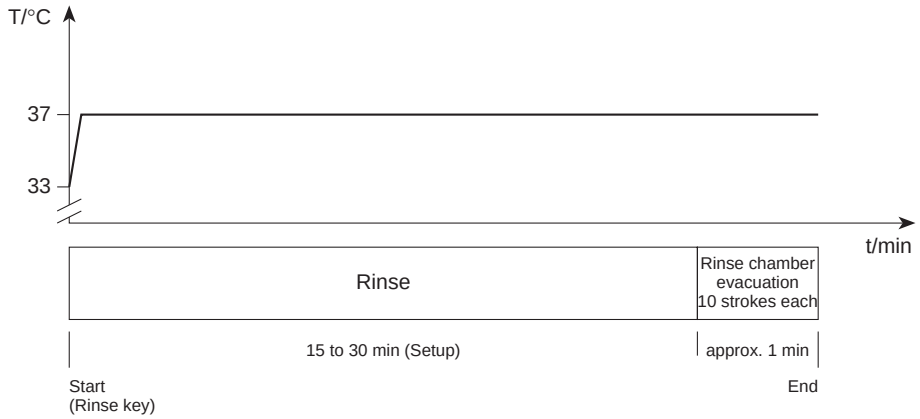
- **Notes on program runs**

  At the end of the set program, the rinse chamber is evacuated for approx. 1 min.

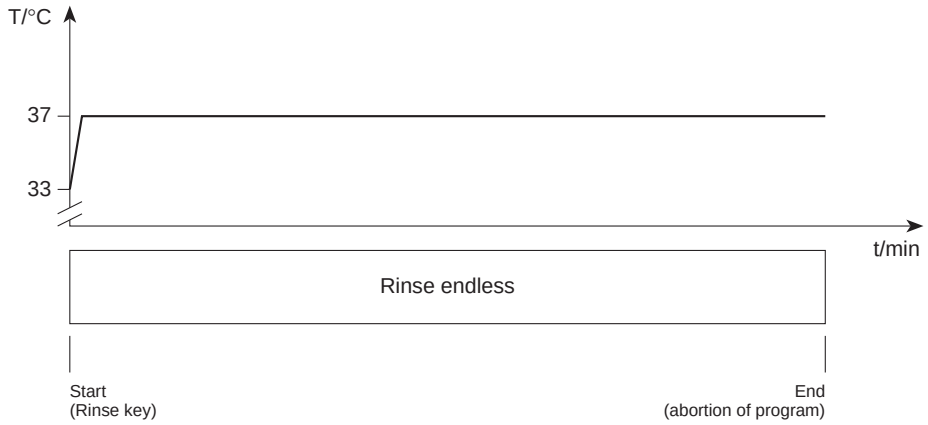
  Any statements on time refer to the factory setting. Shorter or longer program times can be set at any time by means of the SET UP menu (see Technical Manual, Chapter 6).

- **Rinse**

PGM 1: –R–

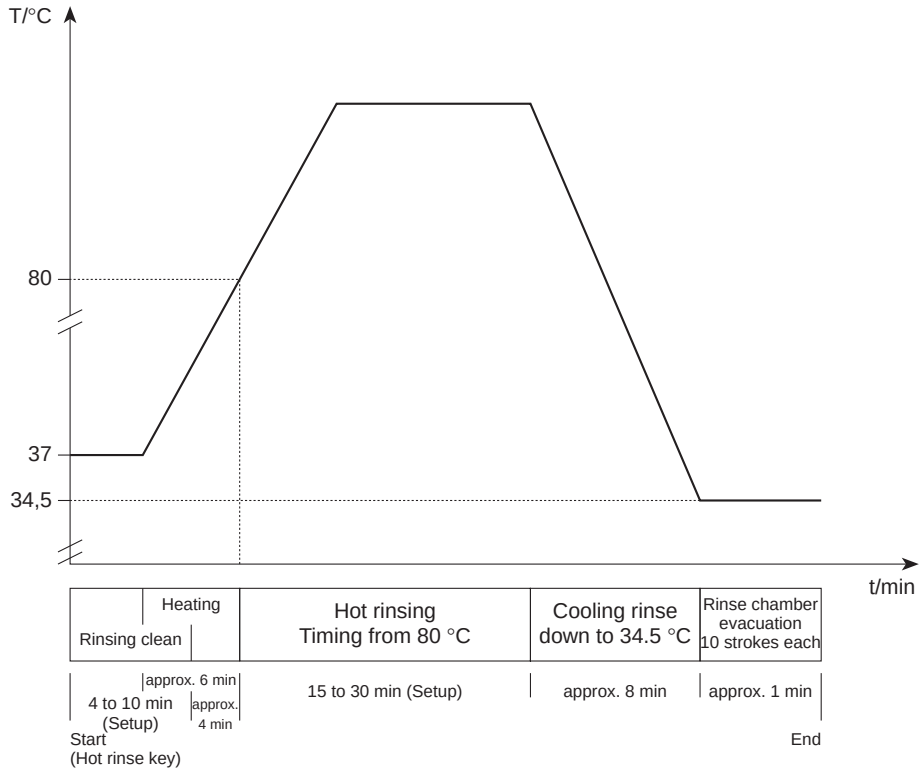


PGM 2: –R– endless

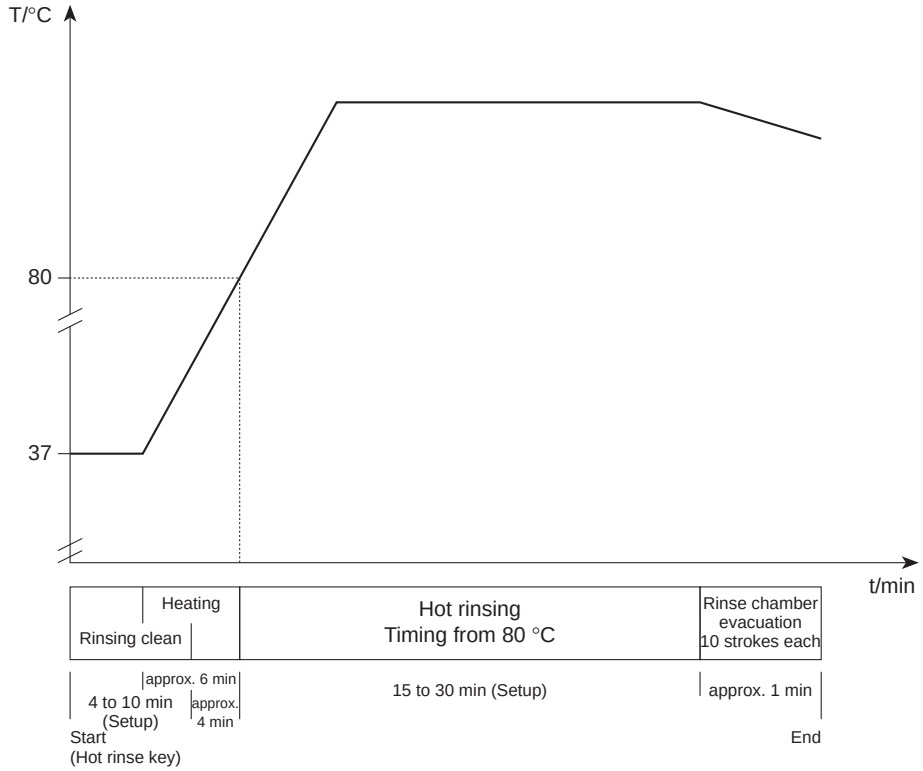


- **Hot rinsing**

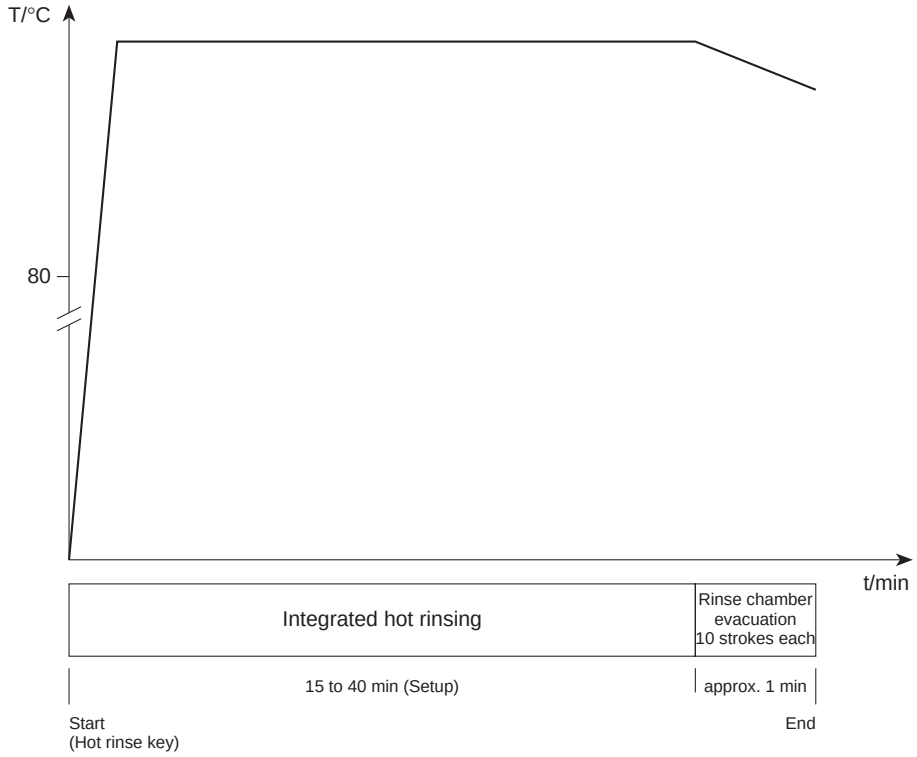
PGM 1: –F–HR–C–



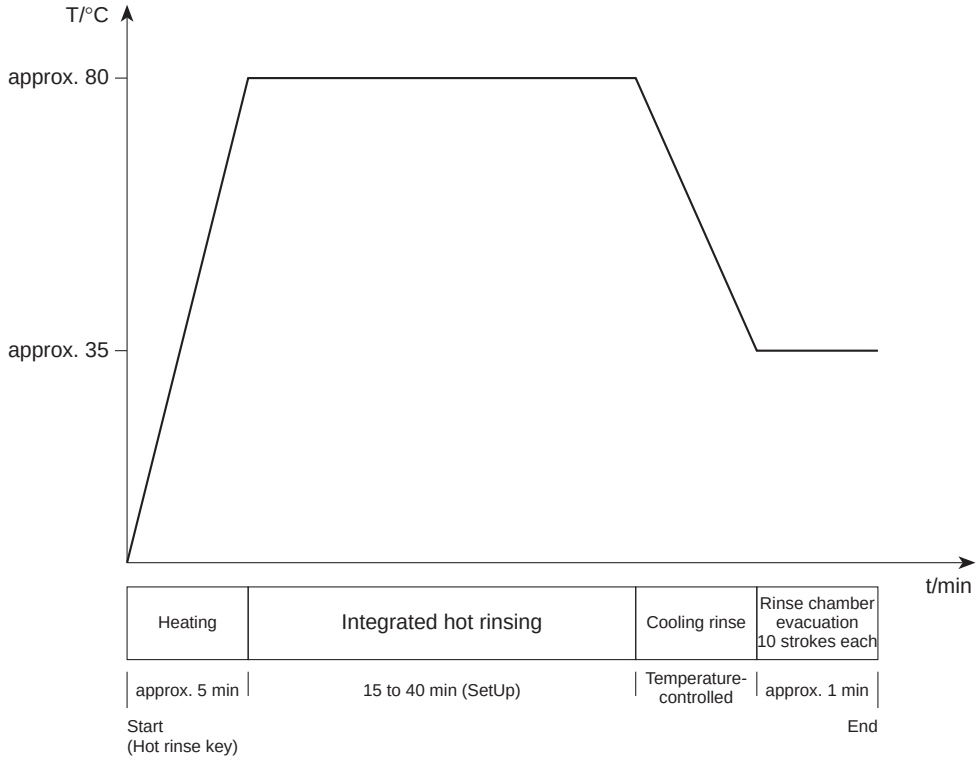
PGM 2: –F–HR–



PGM 3: –IHR–

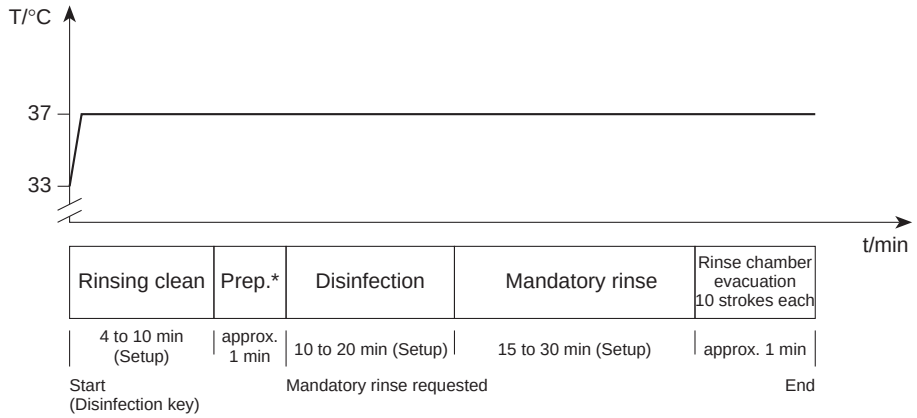


PGM 3: –IHR–C–



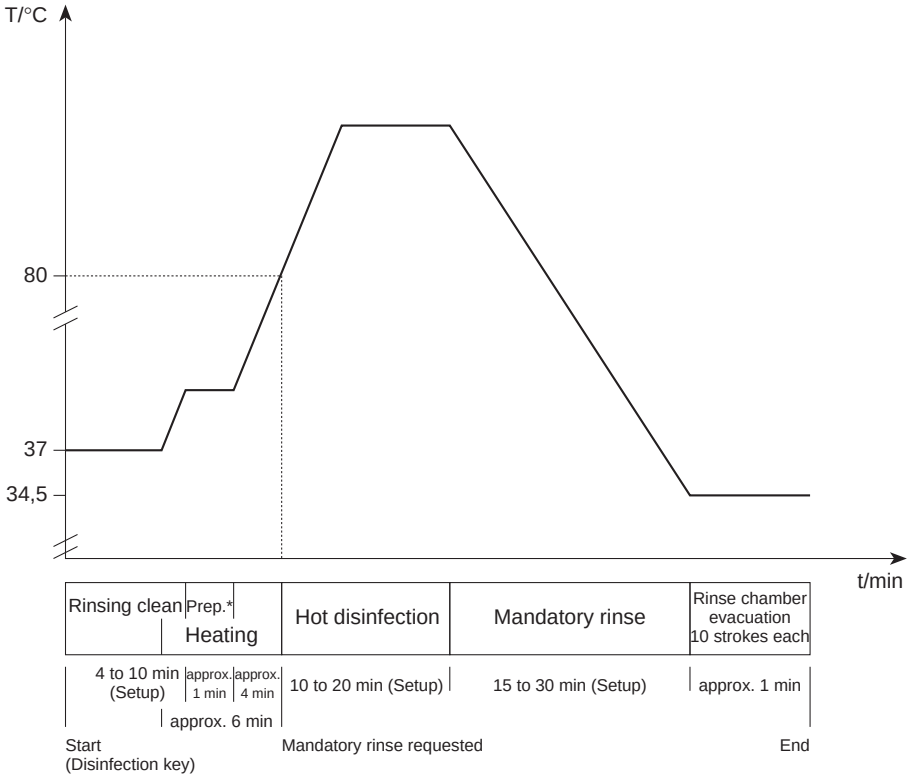
- **Disinfection**

PGM 1: –F–D–M–



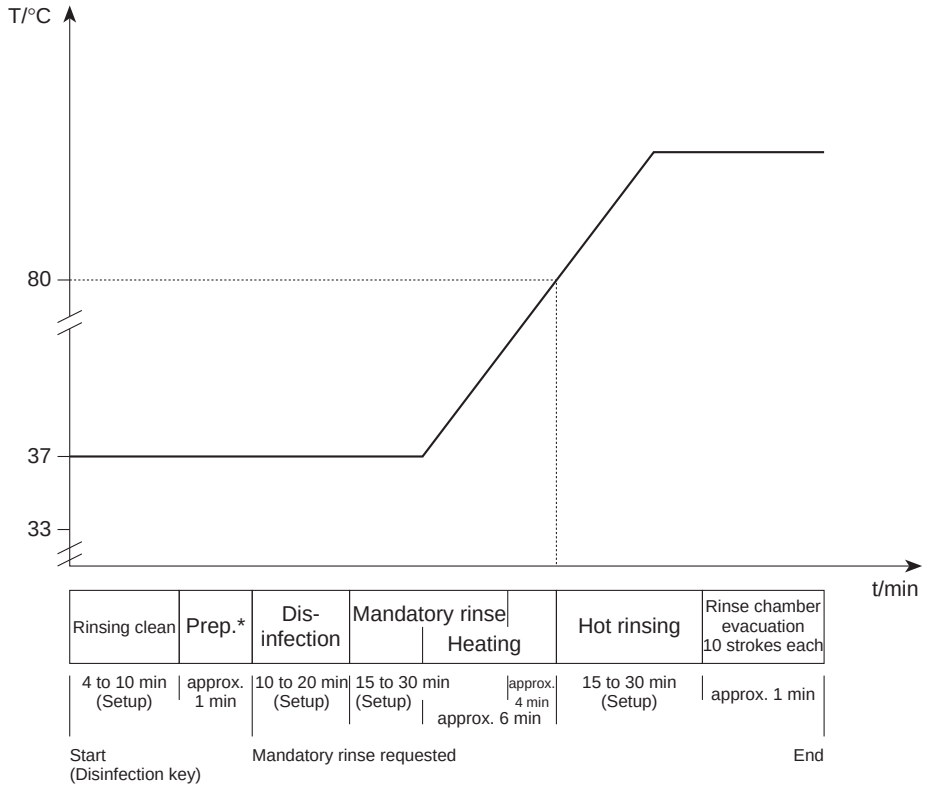
*Prep.: preparation phase:
Heater off
Set the level of the float switch chamber below the lower switching point of the float switch by
1 balancing chamber changeover and
4 UF pump strokes.
Aspiration of disinfectant for 50 UF-pump strokes.

PGM 2: –F–HDIS–M–



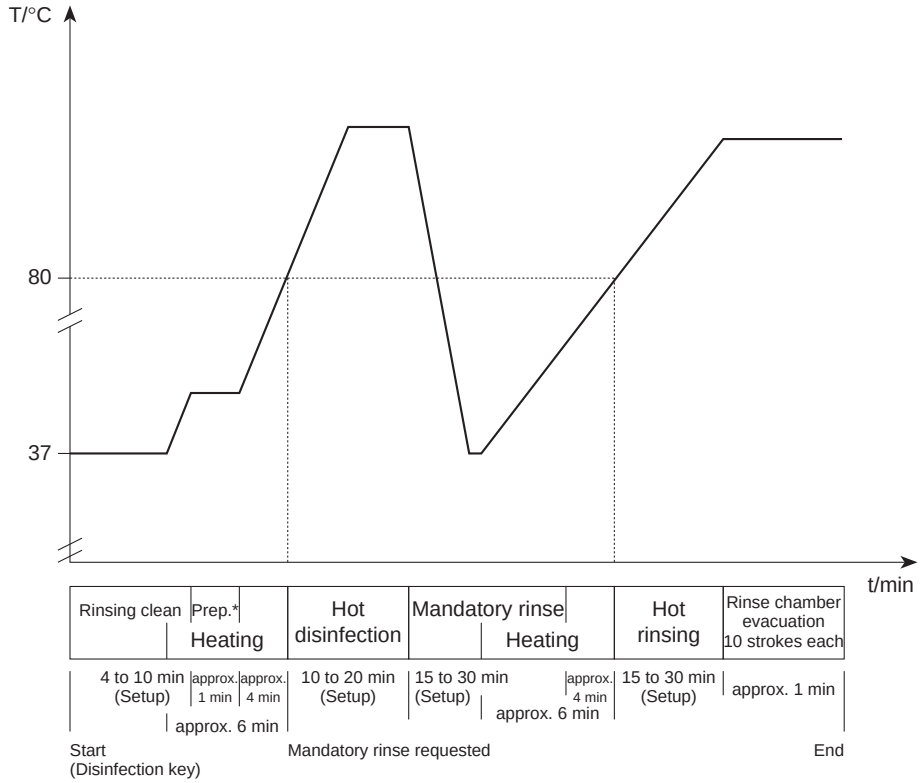
*Prep.: preparation phase:
Heater off
Set the level of the float switch chamber below the lower switching point of the float switch by
1 balancing chamber changeover and
4 UF pump strokes.
Aspiration of disinfectant for 50 UF-pump strokes.

PGM 3: –F–D–M–HR



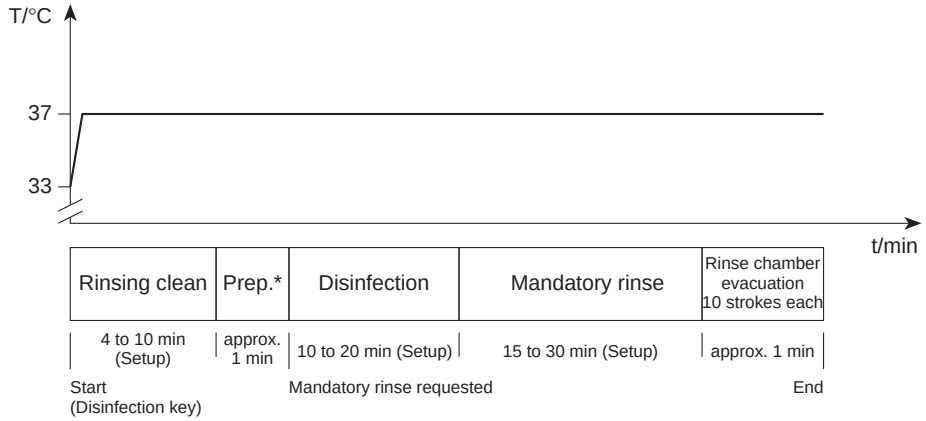
*Prep.: preparation phase:
Heater off
Set the level of the float switch chamber below the lower switching point of the float switch by
1 balancing chamber changeover and
4 UF pump strokes.
Aspiration of disinfectant for 50 UF-pump strokes.

PGM 4: –F–HDIS–M–HR–



*Prep.: preparation phase:
Heater off
Set the level of the float switch chamber below the lower switching point of the float switch by
1 balancing chamber changeover and
4 UF pump strokes.
Aspiration of disinfectant for 50 UF-pump strokes.

PGM 5: –F–D(F)–M–



*Prep.: preparation phase:
Heater off
Set the level of the float switch chamber below the lower switching point of the float switch by
23 UF pump strokes.
Aspiration of disinfectant for
32 concentrate pump strokes à 330 steps.

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2.1 Technical safety checks and maintenance for 4008 hemodialysis systems and options

2.1.1 Important notes

This chapter lists all necessary technical safety checks **(TSC)** and maintenance procedures **(MA)**.

The checks must be performed every 12 months.

Performance of the technical safety checks must be recorded in the Medical Device Register.

Please refer to page 2-5 to page 2-13 for the description of the technical safety checks and maintenance.

Please refer to page 2-15 to page 2-18 for the checklist of the technical safety checks and maintenance.

Instructions to be Observed when Servicing the System

Assembly, extensions, adjustments, modifications or repairs may only be carried out by the manufacturer or persons authorized by him.

The activities described in the Technical Manual require the availability of the necessary technical test equipment and accessories.

When working on the open system, the following precautions must be respected:
– Protect the components against ingress of fluids.
– Do not touch live parts (e. g. connectors of the power cord or heater).

When repairing and when replacing spare parts, observe the applicable ESD precautions (e.g. EN 100 015-1).

2.1.2 Description – Technical safety checks and maintenance

Scope and intervals of technical safety checks INTERVAL: once a year (every 12 months)
The following inspections must be carried out every 12 months at the latest by persons who are qualified to properly perform the specified technical safety checks owing to their educational background and training, their knowledge and experience gained in practice and who are not subject to any directions with regard to this inspection activity.

TSC	MA	No.	Description	Expected value / function
		1	**Visual inspections**	
TSC		1.1	Fuses accessible from the outside	Must comply with the maximum permissible values.
TSC		1.2	Labels and identification	Must be present and legible.
TSC		1.3	Mechanical condition	Must permit further safe use.
TSC		1.4	Damage and contaminations	There must not be any detectable damage or contamination.
TSC		1.5	Power cord	Must not be damaged.
	MA	1.6	Preventive maintenance procedures	
	MA	1.6.1	Replace the sealing in the concentrate/bicarbonate suction tubes and lubricate with silicone paste. Replace the rivet in the suction tubes, if necessary.	
	MA	1.6.2	Check the rubber in the rinse chambers for proper function.	
	MA	1.6.3	Replace the filters of the suction tubes (71/72).	
	MA	1.6.4	Retighten the rinse chamber (90a/90b) screws.	
	MA	1.6.5	Replace the check valve (92). When using CDS, replace the O-rings in the check valves (117/118) or change the valves.	
	MA	1.6.6	Replace the filter sieves; upstream of the UF pump (filter 74), downstream of MV43 (filter 76), between MV99 and rinse chamber (filter 149), between MV100 and rinse chamber (filter 148). When using CDS, replace the filters and O-rings (119/120).	
	MA	1.6.7	Replace the filter sieve in the dialyzer line; replace the complete filter (73), if necessary.	
	MA	1.6.8	Replace the O-rings in the dialyzer couplings.	
	MA	1.6.9	Check the line in the sampling valve (116) dialysate circuit for proper function, replace the complete valve, if necessary.	
	MA	1.6.10	Clean or replace the fan filter in the monitor.	
	MA	1.6.11	Check the air separation pump (97): replace the belt ribbon and the line segment. Observe direction of delivery.	
	MA	1.6.12	MV 84 must be replaced after 2 years. *Only if Puristeril is used.*	
	MA	1.6.13	Replace the connecting piece or equilibration chamber. *Only for systems in which the* ONLINE*plus*™ *option or the* DIASAFE®*plus option is not used.*	
	MA	1.6.14	Replace filter 210 (if present).	
	MA	1.6.15	Replace the filter of the disinfectant suction tube.	
	MA	1.6.16	Replace worn or dirty tubings.	

TSC	MA	No.	Description	Expected value / function
		2	**General checks**	
TSC		2.1	Power failure alarm	Dialysis mode; Continuous sound after removing the power plug. Text displayed: **Emergency operation** The extracorporeal blood circuit incl. all monitoring functions is maintained.
TSC		2.2	Check level sensor	Draw in air via the dialysate couplings. The air separation pump is activated. If more air is detected, the machine will switch to the fill program, depending on the dialysate flow. Text displayed when the OD senses opaque fluid: **Fillprogram**
TSC		2.3	Check valves 91; 99 and 100	Check externally for tightness and proper function. Controlled via the diagnostics program.
			This check will not be applicable if: *– CDS is activated – P.C.B. LP 631 SH2 SW8 set to ON* *– the hydraulics test is activated – P.C.B. LP 631 SH2 SW7 set to ON*	
		3	**Check of the hydraulics**	
			Check all pressures with undamped pressure gauges!	
	MA	3.1	Check the water inlet pressure (reduced) and correct, if necessary.	Connect a pressure gauge before MV41 to measuring point A in the hydraulic unit. With the valve MV41 closed the pressure should range between 0.9 and 1.4 bar.
	MA	3.2	Check the balancing chamber loading pressure and correct, if necessary.	Connect a pressure gauge to the pressure side of the degassing pump (measuring point B in the hydraulic unit). The pressure should be between 1.2 and 1.3 bar.
	MA	3.3	Check the negative degassing pump pressure	Connect a pressure gauge to the suction side of the degassing pump (measuring point D in the hydraulic unit). The negative pressure should be between 0.81 and 0.85 bar.
	MA	3.4	Check the balancing chamber relief pressure at a flow of 800 ml/min (relief valve 78).	Connect a pressure gauge to the pressure side of the flow pump (measuring point C in the hydraulic unit). The maximum pressure should be between 2.0 and 2.1 bar.

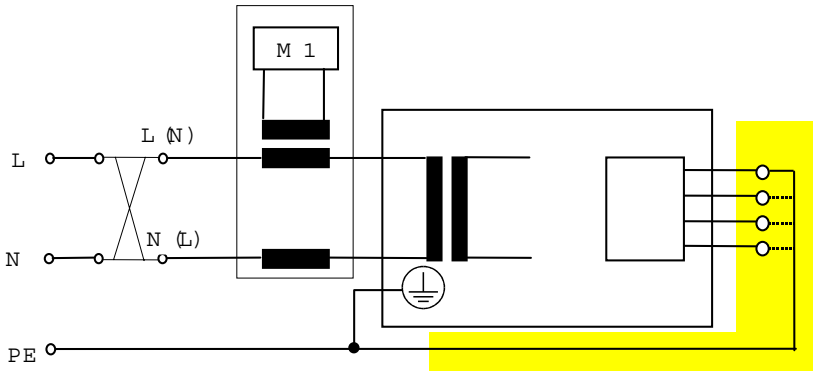
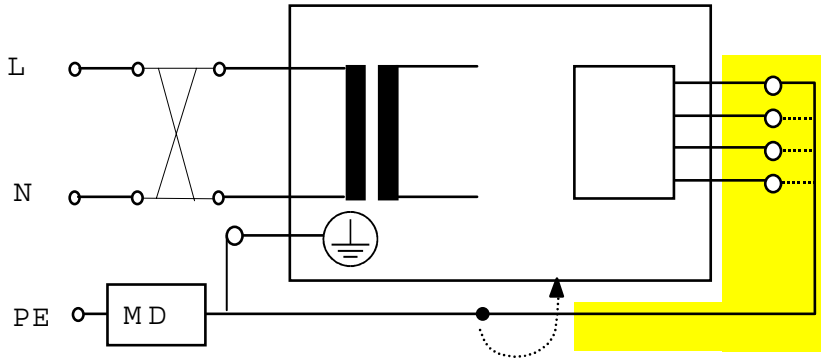
TSC	MA	No.	Description	Expected value / function
		4	**Ultrafiltration system and membrane pumps**	
TSC		4.1	Check the delivery volume of the UF pump.	In the dialysis mode collect 60 ml of dialysate in an appropriate graduated cylinder. 60 strokes = 60 ml (±0.5 ml). Correct the setting of the UF pump, if necessary.
	MA	4.2	Check the balancing chamber volume.	Collect the volume of two consecutive balancing chamber switchings. The average volume must be 30 ml ±1 ml.
	MA	4.3	Test the concentrate pump volume or compare it with an appropriate reference.	Adjust according to calibration instructions, if necessary.
	MA	4.4	Test the bicarbonate pump volume or compare it with an appropriate reference.	Adjust according to calibration instructions, if necessary.
		5	**Dialysis mode**	
	MA	5.1	Temperature	Use a reference meter connected between the dialyzer couplings to verify that the temperature is 37 °C ±0.5 °C. Correct any deviations with the calibration program.
	MA	5.2	Temperature display (not applicable for 4008 B / 4008 S)	The temperature shown on the monitor front panel must be 37 °C ±0.5 °C. Correct any deviations with the calibration program.
	MA	5.3	Verify the dialysate flow 300/500/800 ml/min	Collect fluid on the drain using a measuring cylinder. 800 ml/min (desired value: 765 to 837 ml/min) 500 ml/min (desired value: 471 to 528 ml/min) 300 ml/min (desired value: 279 to 321 ml/min) Adjust according to calibration instructions, if necessary.
	MA	5.4	Dialysate pressure	Perform a TMP test according to the calibration instructions. (part 14 CAL. DIAL. PRESSURE)
TSC		5.5	Verify the conductivity display	When the bi*b*ag® option is used, connect a bi*b*ag®! Measure the conductivity with a reference meter connected bewteen the dialyzer couplings. The conductivity measured must agree with the value on the reference meter. Correct any deviations with the calibration program.

TSC	MA	No.	Description	Expected value / function
		6	**Extracorporeal components**	
	MA	6.1	Arterial pressure transducer	Check the slope of the pressure transducer. After applying a pressure of approx. 200 mmHg to the pressure transducer the value displayed must agree with the reading shown on the external reference meter (tolerance ±mmHg). Correct any deviations with the calibration program.
	MA	6.2	Venous pressure transducer	Check the slope of the pressure transducer. After applying a pressure of approx. 300 mmHg to the pressure transducer the value displayed on the machine must agree with the reading shown on the external reference meter (tolerance ±10 mmHg). Correct any deviations with the calibration program.
TSC		6.3	Arterial and Single Needle blood pump	Check the blood pump rate (calibration program: BP-Rate CHECK).
TSC		6.4	SN switching points	Check the switching points according to the table in the TM.
TSC		6.5	Check the blood pump stop alarm.	Opening the blood pump door will trigger the blood pump stop alarm after 30 s (factory setting).
TSC TSC		6.6 6.7	Air detector	In the event of a blood alarm, the venous line clamp must close. Generate a pressure of about 2 bar in the venous bubble catcher. Ensure that the pressure does not drop by more than 0.1 bar within 3 minutes. (See chapter 3, Adjustment instructions.)

TSC	MA	No.	Description	Expected value / function
		7	**Options**	
		7.1	bi*b*ag®	
	MA	7.1.1	bi*b*ag® connector	Replace the O-rings.
	MA	7.1.2	PSW 134	Check the switching pressure. The maximum switching pressure is 100 mbar + 10 mbar.
		7.2	DIASAFE	
	MA	7.2.1	DIASAFE filter life	Check the filter life. Filter life: 12 weeks
	MA	7.2.2	Hydrophobic filter 111	Replace the filter.
	MA	7.2.3	O-rings in the dialysate couplings of the DIASAFE	Replace the O-rings.
		7.3	DIASAFE® *plus*	
	MA	7.3.1	DIASAFE® *plus* filter life	Check the filter life. Filter life: 12 weeks
	MA	7.3.2	Hydrophobic filter 111	Replace the filter.
		7.4	4008 HDF	
TSC		7.4.1	Check the delivery rate of the 2nd UF pump.	Collect 60 ml of dialysate in the dialysis mode using an appropriate measuring cylinder. 60 strokes = 60 ml (±0.5 ml) If necessary, correct the value.

TSC	MA	No.	Description	Expected value / function
		7.5	ON-LINE-HDF (and DIASAFE)	
	MA	7.5.1	Filter life of DIASAFE and ON-LINE filter	Check the filter life. Filter life of the DIASAFE: 12 weeks Filter life of ON-LINE filter: 8 weeks or 50 treatments
	MA	7.5.2	Hydrophobic filter 111	Replace the filter.
	MA	7.5.3	O-rings in the dialysate couplings of the DIASAFE	Replace the O-rings.
	MA	7.5.4	HDF pump rotor	Check the rotor for smooth running and wear.
	MA	7.5.5	Fastening strap	Check the fastening strap for Luer-lock.
TSC		7.5.6	Substituate pump (part no. 672 521 1) with DC motor ***or*** Substituate pump (part no. 674 982 1) with stepper motor	Speed 150 ml/min To determine the delivery volume: the volume of fluid delivered must agree with the preset value (±10 %). To check the speed: with the above setting the blood pump rotor must turn at 13.5 rpm. (See Technical Manual ON-LINE-HDF, chapter 3). Check the pump rate (calibration program: BP-Rate CHECK).
TSC		7.5.7	Substituate pump stop	Stop the substituate pump by – triggering a blood alarm, – triggering the bypass function, – opening the blood pump door.
TSC		7.5.8	Substituate pump function – Rinse – Hot rinse – Disinfection	 Start the rinse program; the pump will deliver at 400 ml/min. Start the hot rinse program; the pump will deliver at 150 ml/min. Start the disinfection program; the pump will deliver at 400 ml/min.

TSC	MA	No.	Description	Expected value / function
		7.6	ONLINE*plus*™ (and DIASAFE®*plus*)	
	MA	7.6.1	Filter life of DIASAFE®*plus* and ONLINE*plus*™ filter	Check the filter life. Filter life of DIASAFE®*plus* and ONLINE*plus*™ filter: 12 weeks or 100 treatments
	MA	7.6.2	Hydrophobic filters 111 and 184	Replace the filters.
	MA	7.6.3	Substituate port (195) and rinse port (194)	Replace the O-rings.
TSC		7.6.4	Line pinch valve 193 (ONL1)	Replace the line.
TSC		7.6.5	Valve 39 **Note:** This TSC item will not be applicable if DIP switch array 2, P.C.B. LP 632, switch 5 is set to OFF.	Check for proper function.
		7.7	OCM	
			No further technical safety checks and maintenance procedures must be performed.	
		7.8	BPM 4008	
			Perform the technical measurement checks and maintenance procedures every 2 years (see section 2.2).	
		7.9	BTM 4008	
			No further technical safety checks and maintenance procedures must be performed.	
		7.10	BVM 4008	
			No further technical safety checks and maintenance procedures must be performed.	

TSC	MA	No.	Description	Expected value / function
		8	**Checking the electrical safety** In Germany according to DIN VDE 0751 standard – 1st Edition 10/2001. In other countries, observe the local regulations!	
			For measuring points, see 2.1.4 Notes – Checking the electrical safety.	
TSC		8.1	Visual inspections performed	see item 1 Visual inspections
TSC		8.2	Protective earth resistance	Max. 0.3 Ω (with power cord)
TSC		8.3	Measurement of the leakage current (device leakage current) Differential current measurement according to fig. C.6  ***or*** Direct measurement according to fig. C.5  Basic conditions: – Measurement of the protective earth resistance has been completed. – Perform the measurement with the machine being at operating temperature in the Dialysis or Preparation operation mode. – Dialysate: Dialysis temperature > 37°C Dialysate flow 300 ml/min Conductivity 13 mS/min – When performing a direct measurement, the following precautions must be observed: The device must be installed under insulated conditions. All external connections must have been removed from the device.	

TSC	MA	No.	Description	Expected value / function
			Documentation covers the line voltage during the measurement and the maximum device leakage current of both mains polarities scaled to the line voltage of the power supply. Maximum device leakage current: 500 µA Example: Line voltage during measurement: 225 V Device leakage current for mains polarity 1: 180 µA for mains polarity 2: 120 µA Maximum value of both mains polarities: 180 µA Nominal voltage of the power supply: 230 V Scaled to nominal voltage: 184 µA (180 µA : 225 V • 230 V = 184 µA) Device leakage current < 500 µA: OK Additional conditions: If the device leakage current is higher than 90 % of the admissible alarm limit (450 µA), the last measured value or the first measured value must additionally be considered for the rating. If the device leakage current considerably increased since the last measurement or continuously increased since the first measurement (creeping deterioration of the insulation), or if the sum composed of the current value plus the difference since the last measurement is > 500 µA, the measurement has not been passed. Example 1: Leakage current: 470 µA Last measured value: 450 µA 470 + (470 – 450) = 470 + 20 = 490 ➜ OK Example 2: Leakage current: 470 µA Last measured value: 390 µA 470 + (470 – 390) = 470 + 80 = 550 ➜ not passed	
		9	**Functional check**	
TSC		9.1	Perform the functional test	Press the Test key. The machine must successfully pass the T1 test.
	MA	9.2	Hot rinse / disinfection	Run a disinfection program.

2.1.3 Checklist – Technical safety checks and maintenance

Customer/Customer no.:		Service report no.:
Serial no.:	Inventory no.:	Operating hours:
Machine type: 4008 ❑ 4008 B ❑ 4008 H ❑ 4008 S ❑		
With option: SN ❑ bi*b*ag® ❑ 4008 HDF ❑ ON-LINE-HDF ❑ ONLINE*plus*™ ❑ BTM ❑ BPM ❑ BVM ❑ DIASAFE ❑ DIASAFE®*plus* ❑ OCM ❑		

	1	**Visual checks**	
TSC	1.1	Fuses accessible from the outside comply with the indicated values	❑
TSC	1.2	Labels and identifications present and legible	❑
TSC	1.3	Mechanical conditions permits further safe use	❑
TSC	1.4	No damage or contaminations detectable	❑
TSC	1.5	Power cord not damaged	❑
	1.6	Preventive maintenance procedures	
MA	1.6.1	Sealing in the suction tubes changed and lubricated, rivet replaced	❑
MA	1.6.2	Rubber in rinse chambers checked for proper function	❑
MA	1.6.3	Suction tube filters replaced	❑
MA	1.6.4	Rinse chamber screws tight	❑
MA	1.6.5	Check valve replaced	❑
MA	1.6.6	Pre-UF pump filter, filter downstream of MV43, filter between rinse chambers, and on MV99, MV100, CDS and disinfectant port replaced	❑
MA	1.6.7	Dialysate filter replaced or sieve changed	❑
MA	1.6.8	O-rings in dialyzer couplings replaced	❑
MA	1.6.9	Sampling valve functions properly	❑
MA	1.6.10	Fan filter replaced	❑
MA	1.6.11	Ribbon belt and line segment in air separation pump changed	❑
MA	1.6.12	MV84, replaced after 2 years. (Only if Puristeril is used.)	❑
MA	1.6.13	Connecting piece or equilibration chamber replaced. (Only if ONLINE™*plus* or DIASAFE®*plus* option is not used.)	❑
MA	1.6.14	Filter 210 replaced (if present)	❑
MA	1.6.15	Filter of the disinfectant suction tube replaced	❑
MA	1.6.16	No dirty or worn tubings	❑
	2	**General checks**	
TSC	2.1	Power failure alarm – continous sound – display: Emergency operation	❑
TSC	2.2	Air separation by air separation pump activated; display if more air must be separated and OD senses opaque fluid: fill program	❑
TSC	2.3	V91, V99, V100 function properly and do not leak (Check will not be applicable if CDS or hydraulics test is activated)	❑
	3	**Check of the hydraulics**	
MA	3.1	Water inlet pressure (reduced) 0.9 bar to 1.4 bar ... *Measured value:*_____	❑
MA	3.2	Loading pressure 1.25 bar ±0.05 bar ... *Measured value:*_____	❑
MA	3.3	Negative degassing pump pressure 0.81 to 0.85 bar ... *Measured value:*_____	❑
MA	3.4	Balancing chamber relief pressure at 800 ml/min 2.0 bar to 2.1 bar ... *Measured value:*_____	❑

	4	**Ultrafiltration system and membrane pumps**	
TSC	4.1	UF pump, 1 stroke = 1 ml, 60 strokes = 60 ml ±0.5 ml ..... *Measured value:*_____	❑
MA	4.2	Average balancing chamber volume 30 ± 1 ml ..... *Measured value:*_____	❑
MA	4.3	Concentrate pump calibration volume removal / number of strokes ..... *Measured value:*_____	❑
MA	4.4	Bicarbonate pump calibration volume removal / number of strokes ..... *Measured value:*_____	❑
	5	**Dialysis mode**	
MA	5.1	Expected temperature 37 °C ± 0.5 °C ..... *Measured value:*_____	❑
MA	5.2	Temperature display 37 °C ± 0.5 °C ..... *Measured value:*_____	❑
MA	5.3	Dialysate flow check	
		800 ml/min (desired value: 765 to 837 ml/min) ..... *Measured value:*_____	❑
		500 ml/min (desired value: 471 to 528 ml/min) ..... *Measured value:*_____	❑
		300 ml/min (desired value: 279 to 321 ml/min) ..... *Measured value:*_____	❑
MA	5.4	Dialysate pressure	
		– Check zero point with flow off	❑
		– Slope checked	❑
TSC	5.5	Conductivity display checked with reference meter	❑
		If the bi*b*ag® option is used, connect a bi*b*ag®!	
		– CD system ..... *Measured value:*_____	❑
		– CD ref. ..... *Measured value:*_____	❑
	6	**Extracorporeal components**	
MA	6.1	Arterial pressure displayed checked with reference meter	❑
MA	6.2	Venous pressure displayed checked with reference meter	❑
TSC	6.3	Blood pumps: blood pump rate checked (calibration program: BP-Rate CHECK)	❑
TSC	6.4	SN switching pressure checked according to table in TM	❑
TSC	6.5	Blood pump stop alarm checked	❑
TSC	6.6	Venous line clamp closes after blood alarm	❑
TSC	6.7	Pressure of about 2 bar in the venous bubble catcher	❑
		Pressure must not drop by more than 0.1 bar within 3 minutes.	❑
	7	**Options**	
	7.1	bi*b*ag®	
MA	7.1.1	bi*b*ag® connector, O-rings replaced	❑
MA	7.1.2	Switching pressure of PSW134 checked, 100 mbar, + 10 mbar ..... *Measured value:*_____	❑
	7.2	DIASAFE	
MA	7.2.1	DIASAFE filter life checked	❑
MA	7.2.2	Hydrophobic filter 111 replaced	❑
MA	7.2.3	O-rings in the dialysate couplings of the DIASAFE replaced	❑
	7.3	DIASAFE® *plus*	
MA	7.3.1	DIASAFE® *plus* filter life checked	❑
MA	7.3.2	Hydrophobic filter 111 replaced	❑
	7.4	4008 HDF	
TSC	7.4.1	2nd UF pump 1 stroke = 1 ml, 60 strokes = 60 ml ± 0.5 ml ..... *Measured value:*_____	❑
	7.5	ON-LINE-HDF (and DIASAFE)	
MA	7.5.1	Filter life of the DIASAFE and ON-LINE filter checked	❑
MA	7.5.2	Hydrophobic filter 111 replaced	❑
MA	7.5.3	O-rings in the dialysate couplings of the DIASAFE replaced	❑
MA	7.5.4	HDF pump rotor checked (smooth running, wear)	❑
MA	7.5.5	Fastening strap for Luer-lock checked	❑
TSC	7.5.6	Substituate pump	❑
		❑ (part no. 672 521 1) with DC motor: volume delivered by the pump checked ..... *desired/actual:*____/____	
		or	
		❑ (part no. 674 982 1) with stepper motor: pump rate checked (calibration program: HDF-P.-Rate CHECK)	

TSC 7.5.7 Substituate pump stop
– after blood alarm ..... ❑
– after triggering the bypass function ..... ❑
– after opening the blood pump door ..... ❑

TSC 7.5.8 Check substituate pump for proper function
– Rinse program, delivery rate: 400 ml/min ..... ❑
– Hot rinse program, delivery rate: 150 ml/min ..... ❑
– Disinfection program, delivery rate: 400 ml/min ..... ❑

7.6 ONLINE*plus*™ (and DIASAFE®*plus*)

MA 7.6.1 Filter life of DIASAFE®*plus* and ONLINE*plus*™ checked ..... ❑

MA 7.6.2 Hydrophobic filters 111 and 184 replaced ..... ❑

MA 7.6.3 O-rings in substitute port 195 and in rinse port 194 replaced ..... ❑

TSC 7.6.4 Line in the line pinch valve 193 (ONL1) replaced ..... ❑

TSC 7.6.5 Valve V39 checked
(Not applicable if DIP switch array 2, P.C.B. LP 632, switch 5 set to OFF) ..... ❑

7.7 OCM
No further technical safety checks and maintenance procedures must be performed.

7.8 BPM 4008
Perform the technical measurement checks and maintenance procedures every 2 years (see section 2.2).

7.9 BTM 4008
No further technical safety checks and maintenance procedures must be performed.

7.10 BVM 4008
No further technical safety checks and maintenance procedures must be performed.

8 Checking the electrical safety
In Germany according to DIN VDE 0751 standard – 1st Edition 10/2001.
In other countries, observe the local regulations!

For measuring points, see 2.1.4 VDE check.
For 4008 HDF option, check additional measuring point!

TSC 8.1 Visual inspections performed according to item 1 ..... ❑

TSC 8.2 Protective earth resistance max. 0.3 Ω (with power cord) ..... *Measured value:*_____ ❑

TSC 8.3 Measurement of the leakage current ..... ❑
❑ Differential current measurement according to fig. C.6
or
❑ Direct measurement according to fig. C.5

Nominal voltage of power supply ..... _____ *Volt*
Device leakage current mains polarity 1 ..... _____ *µA*
for line voltage ..... _____ *Volt*
scaled to nominal voltage (maximum 500 µA,
see also Additional conditions) ..... *Measured value:*_____

Device leakage current mains polarity 2 ..... _____ *µA*
for line voltage ..... _____ *Volt*
scaled to nominal voltage (maximum 500 µA,
see also Additional conditions) ..... *Measured value:*_____

Test equipment used: _____

	9	**Final checks**
TSC	9.1	T1 test performed ..... ❑
MA	9.2	Hot rinse / disinfection performed ..... ❑

Date:	**Signature:**	**Stamp:**

The system has been released for further use	**❑ Yes ❑ No**

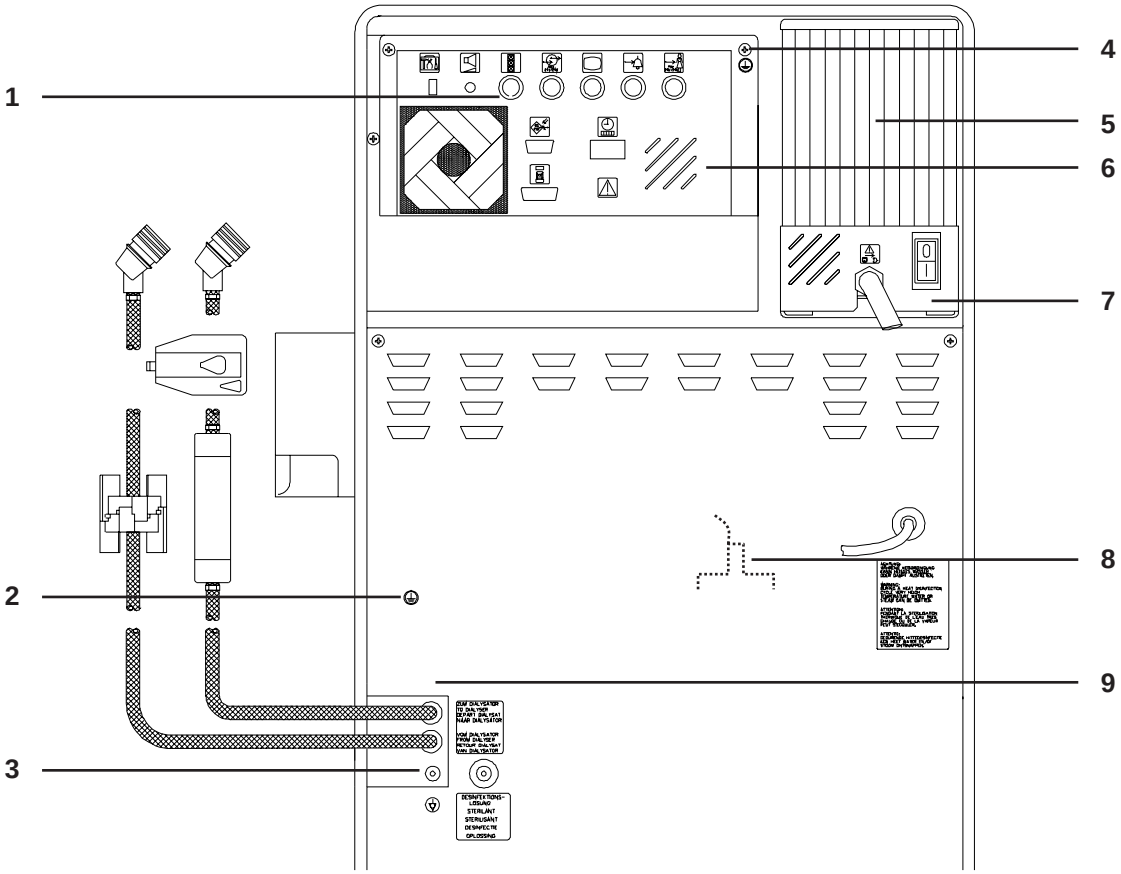
Remarks:

Date:	**Signature:**	**Stamp:**

2.1.4 Notes – Checking the electrical safety

- **Test 4008 E, 4008 H**

1. Protective earth resistance measuring points



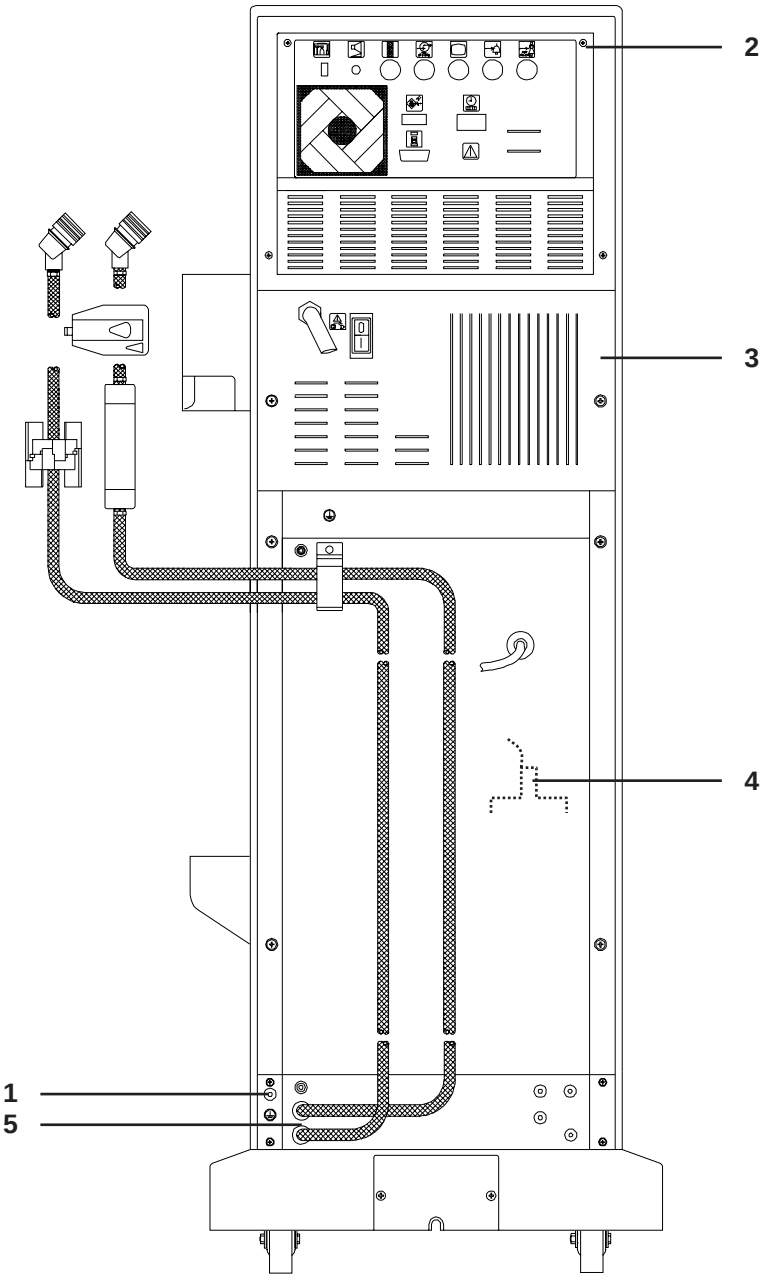
Legend

1 Monitor rear panel (ports housing)
2 Hydraulic unit rear panel (plate on the push-on blade inside)
Caution: The grounding cable must be connected.
3 Ground stud for potential equalization
4 Upper rear panel (screw)
5 Heat sink (power supply unit)
6 Monitor rear panel (plate)
7 Power supply plate
8 Heater rod housing (hydraulic unit open)
9 Dialyzer line ports (hydraulic unit open / earthing screw)

2. Use a meter (e.g. SECUTEST 0701) to check the leakage current.

- **Test 4008 B, 4008 S**

1. Protective earth resistance measuring points



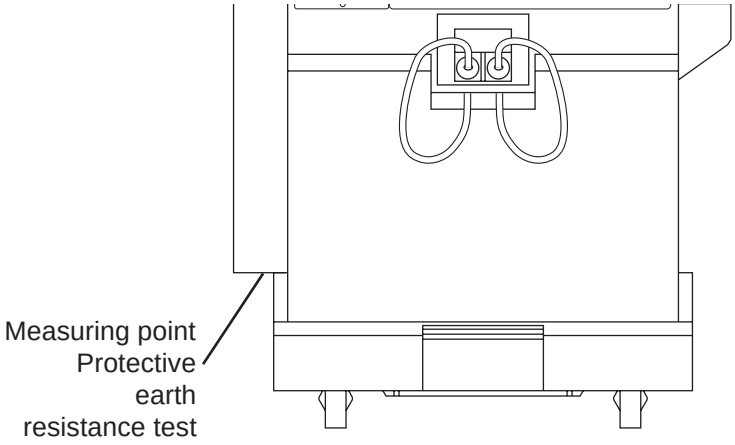
Legend

1	Ground stud for potential equalization	3	Heat sink (power supply unit)
2	Upper rear panel (screw)	4	Heater rod housing (hydraulic unit open)
		5	Dialyzer line ports (adapters)

2. Use a meter (e.g. SECUTEST 0701) to check the leakage current.

- **Test 4008 HDF (option)**

1. Protective earth resistance measuring point



2. Use a meter (e.g. SECUTEST 0701) to check the leakage current.

3. Measurement conditions

The measurements must be taken in the dialysis mode in the “ON phase” of the heater control system.
The scales must be moved out to such an extent that neither of the two end switches are actuated (middle position).

2.2 Technical measurement checks and maintenance for options of 4008 hemodialysis systems

2.2.1 Important notes

This chapter lists all necessary technical measurement checks **(TMC)** and preventive maintenance procedures **(MA)**.

The checks must be performed every 24 months.

After the technical measurement checks have been performed successfully, identify the medical equipment with a mark (label). The year of the next technical measurement checks as well as the authority or person that performed the technical measurement checks must be indicated clearly and traceably.

Please refer to page 2-25 for the description of the technical measurement checks and maintenance.

Please refer to page 2-27 for the checklist of the technical measurement checks and maintenance.

2.2.2 Description – Technical measurement checks and maintenance

- **BPM 4008**

Scope and intervals of technical safety checks **INTERVAL: every 24 months**
The following inspections must be carried out every 24 months at the latest by persons who are qualified to properly perform the specified technical measurement checks owing to their educational background and training, their knowledge and experience gained in practice and who are not subject to any directions with regard to this inspection activity.

TMC	MA	No.	Description	Expected value / function
		1	**Visual checks**	
	MA	1.1	Labels and indications	Must be present and clearly legible. Check of the actual condition.
	MA	1.2	Mechanical condition	Must permit further safe use.
	MA MA MA	1.2.1 1.2.2 1.2.3	Check whether the line connector is correctly attached to the device. Check whether the internal blood pressure module, the printed circuit boards and all cable connections are correctly fixed. Replace damaged lines or cuffs.	
		2	**Functional checks**	
	MA	2.1	Indicating elements	Visual and audible functional checks after turn-on. (See Operating Instructions BPM 4008, Chapter 2.2)
	MA	2.2	Touch panel	Check whether the touch panel is functioning correctly.
TMC		2.3	Leakage test	Perform the leakage test with cuff and pressure line connected. The pressure leakage rate must be less than 6 mmHg/min. (See Technical Manual BPM 4008, Chapter 3.1)
TMC		2.4	Calibration	Calibration: Pressure values: Tolerance: 250 mmHg ±5 mmHg 200 mmHg ±4 mmHg 150 mmHg ±3 mmHg 100 mmHg ±3 mmHg 50 mmHg ±3 mmHg (See Technical Manual BPM 4008, Chapter 3.2)
TMC		2.5	Safety valve	Check the safety valve. The system must be discharged at 320 mmHg ± 10 mmHg. (See Technical Manual BPM 4008, Chapter 3.3)
TMC		2.6	Measuring of blood pressure	Measure the blood pressure in the manual mode. Check the results for plausibility.

2.2.3 Checklist – Technical measurement checks and maintenance

- **BPM 4008**

Customer/Customer no.:	**Service report no.:**
Serial number BPM 4008:	**Installed in system:**
Inventory no.:	**Operating hours:**

	1	**Visual checks**	
MA	1.1	Labels and indications are present and legible	❑
MA	1.2	Mechanical condition permits further safe use	❑
MA	1.2.1	Line connector is correctly fixed to the system	❑
MA	1.2.2	Internal blood pressure module, printed circuit boards, cable connections are correctly fixed	❑
MA	1.2.3	Damaged lines or cuffs have been replaced	❑
	2	**Functional checks**	
MA	2.1	Indicating elements checked	❑
MA	2.2	Touch panel checked	❑
TMC	2.3	Leakage test: pressure leakage rate less than 6 mmHg/min	❑
TMC	2.4	Calibration:	
		Pressure values Tolerance	
		250 mmHg ±5mmHg ..... *Measured value:*_____	❑
		200 mmHg ±5mmHg ..... *Measured value:*_____	❑
		150 mmHg ±3mmHg ..... *Measured value:*_____	❑
		100 mmHg ±3mmHg ..... *Measured value:*_____	❑
		50 mmHg ±3mmHg ..... *Measured value:*_____	❑
TMC	2.5	Safety valve: discharge at 320 mmHg, ±10 mmHg	❑
TMC	2.6	Blood pressure measured	❑

Date:	**Name/Signature:**

2.3 TSC checklist

4008

TSC checklist

for the Technical Safety Checks
to be performed annually (every 12 months)

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Fresenius Medical Care

Name of technician:		**Service report no.:**
Customer/Customer no.:		
Inventory no.:	**Serial no.:**	**Operating hours:**
System type: **With option(s):**		

No.	Description	Measured value	✓
1	**Visual checks**		
1.1	Fuses accessible from the outside comply with the indicated values	–	❑
1.2	Labels and identifications are present and legible	–	❑
1.3	Mechanical condition permits further safe use	–	❑
1.4	No damage or contaminations detectable	–	❑
1.5	Power cord not damaged	–	❑
2	**General checks**		
2.1	Power failure alarm – continuous sound – text displayed: Emergency operation	–	❑
2.2	Air separation by air separation pump activated; text displayed if more air must be separated and OD senses opaque fluid: Fill program	–	❑
2.3	V91, V99, V100 are operational and do not leak. (Check will not be applicable if CDS or hydraulics test activated.)	–	❑
4	**Ultrafiltration system and membrane pumps**		
4.1	UF pump, 1 stroke = 1 ml, 60 strokes = 60 ml ± 0.5 ml	.....	❑
5	**Dialysis mode**		
5.5	Conductivity display checked using a reference meter CD system/CD ref. (If the bi*b*ag® option is used, connect a bi*b*ag®.)	..... /	❑
6	**Extracorporeal components**		
6.3	Blood pumps: check the blood pump rate (calibration program: BP-Rate CHECK)	–	❑
6.4	SN switching pressure checked according to table in TM	–	❑
6.5	Blood pump stop alarm checked	–	❑
6.6	Venous line clamp closes after blood alarm	–	❑
6.7	Pressure of approx. 2 bar in the venous bubble catcher. Pressure must not drop by more than 0.1 bar within 3 minutes.	–	❑
7	**Options**		
7.4	4008 HDF		
7.4.1	Volume delivered by the 2nd UF pump checked: 60 strokes = 60 ml ± 0.5 ml	.....	❑
7.5	Online-HDF		
7.5.6	❑ Substituate pump (part no. 672 521 1) with DC motor: check volume delivered by the pump desired/actual ***or*** ❑ Substituate pump (part no. 674 982 1) with stepper motor: check the pump rate (calibration program:BP-Rate CHECK)	..... / ..... –	❑ ❑
7.5.7	Substituate pump stop: – after blood alarm – after triggering the bypass function – after opening the blood pump door	 – – –	 ❑ ❑ ❑
7.5.8	Check substituate pump for proper function: – Rinse program, delivery rate: 400 ml/min – Hot rinse program, delivery rate: 150 ml/min – Disinfection program, delivery rate: 400 ml/min	 – – –	 ❑ ❑ ❑
7.6	ONLINE*plus*™		
7.6.4	Line in the line pinch valve 193 (ONL1) replaced	–	❑
7.6.5	Valve V39 checked (Not applicable if DIP switch array 2, P.C.B. LP 632, switch 5 set to OFF)	–	❑

TSC checklist

4008

for the Technical Safety Checks
to be performed annually (every 12 months)

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No.	Description	Measured value	✓
8	**Checking the electrical safety** In Germany according to DIN VDE 0751 standard –1st Edition 10/2001. In other countries, observe the local regulations!		
8.1	Visual inspection performed according to item 1.		❑
8.2	Protective earth resistance maximum 0.3 ohms (with power cord)	..... Ω	❑
8.3	Measurement of the leakage current (device leakage current) ❑ Differential current measurement according to fig. C.6 ***or*** ❑ Direct measurement according to fig. C.5 Nominal voltage of power supply: ..... V Device leakage current mains polarity 1 ..... µA for line voltage ..... V scaled to nominal voltage (maximum 500 µA, see also Additional conditions) Device leakage current mains polarity 2 ..... µA for line voltage ..... V scaled to nominal voltage (maximum 500 µA, see also Additional conditions) Test equipment used ……	..... µA ..... µA	❑
9	**Functional test**		
9.1	T1 test performed	–	❑

Date:	**Signature:**	**Stamp:**

The system has been released for further use ❑ **Yes** ❑ **No**

Remarks:

Date:	**Signature:**	**Stamp:**

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3 Adjustment instructions

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Fig.: Measuring equipment



1

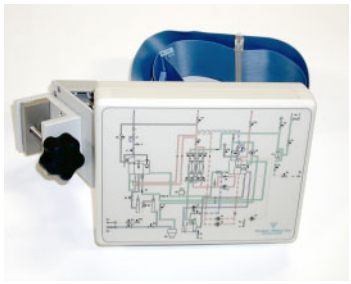


2





3



4



5

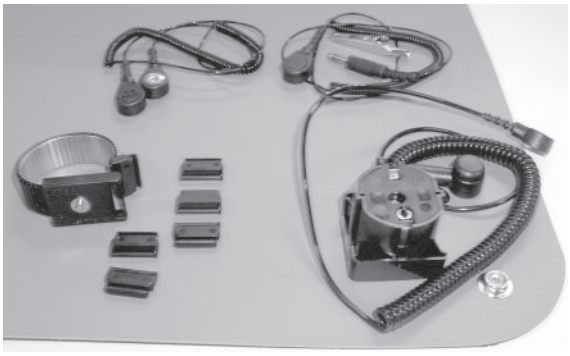


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9

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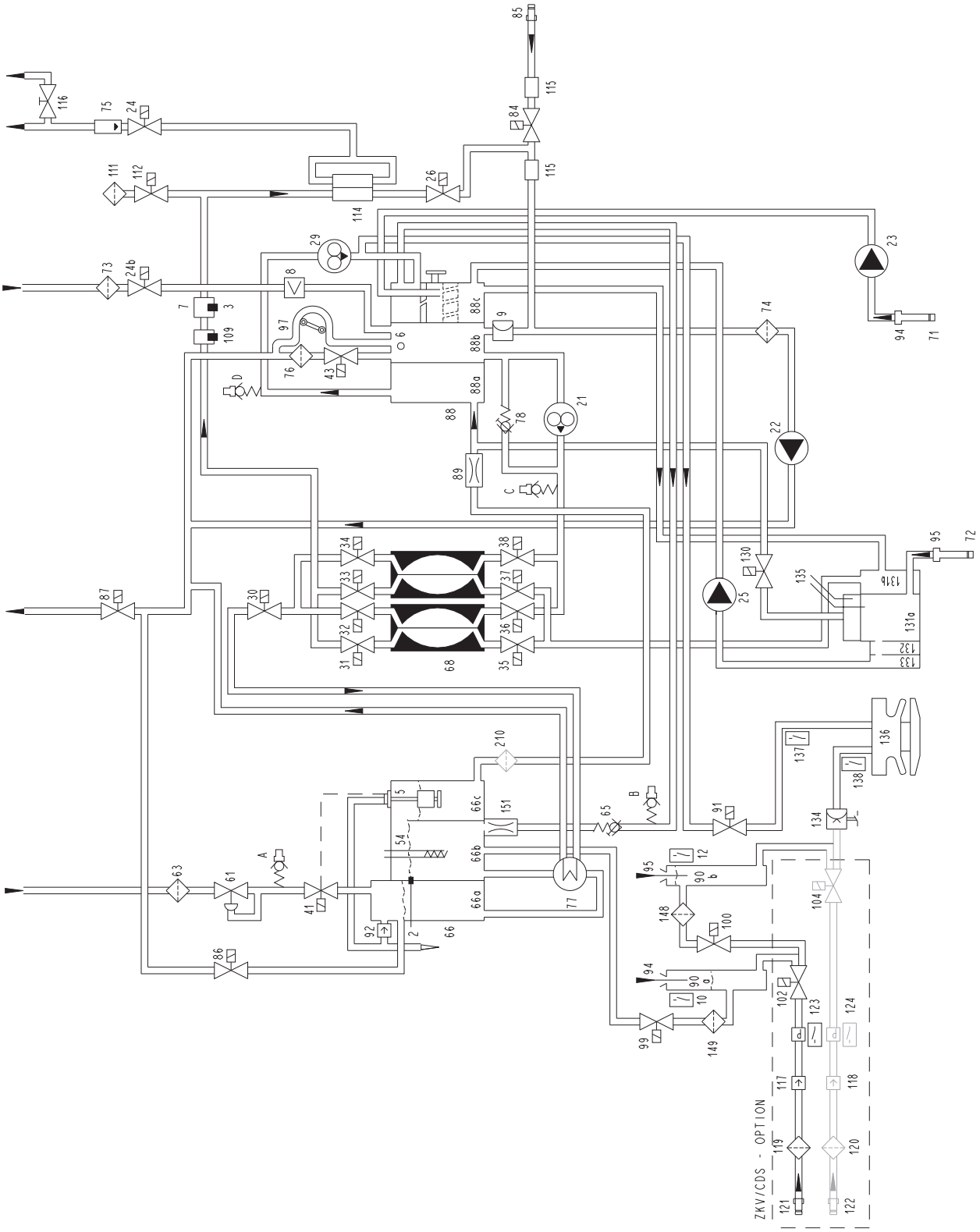


11



Pos.	Measuring equipment	Part number
1	Pressure gauge HMED with carrying case (set)	M30 770 1
2	Universal measuring device UMED (conductivity, pressure, temperature)	M32 403 1
3	Secutest VDE tester (without printer module) Printer module (not ill.) Carrying case (not ill.)	631 064 1 630 652 1 630 648 1
4	Valve control indicator	672 337 1
5	Service software with interface for PC	M30 335 1
6	Electronic pocket scales Test weight with spirit level and certificate	M33 537 1 M33 538 1
7	Measuring cylinder, 100 ml (not ill.)	510 085 1
8	Special tool for installation and removal of modules (not ill.)	671 381 1
9	ESD service kit	630 387 1
10	ESD work station kit	630 388 1
11	IC extraction tool	677 469 1

Fig.: Flow diagram of basic hydraulics incl. DIASAFE®plus (option)



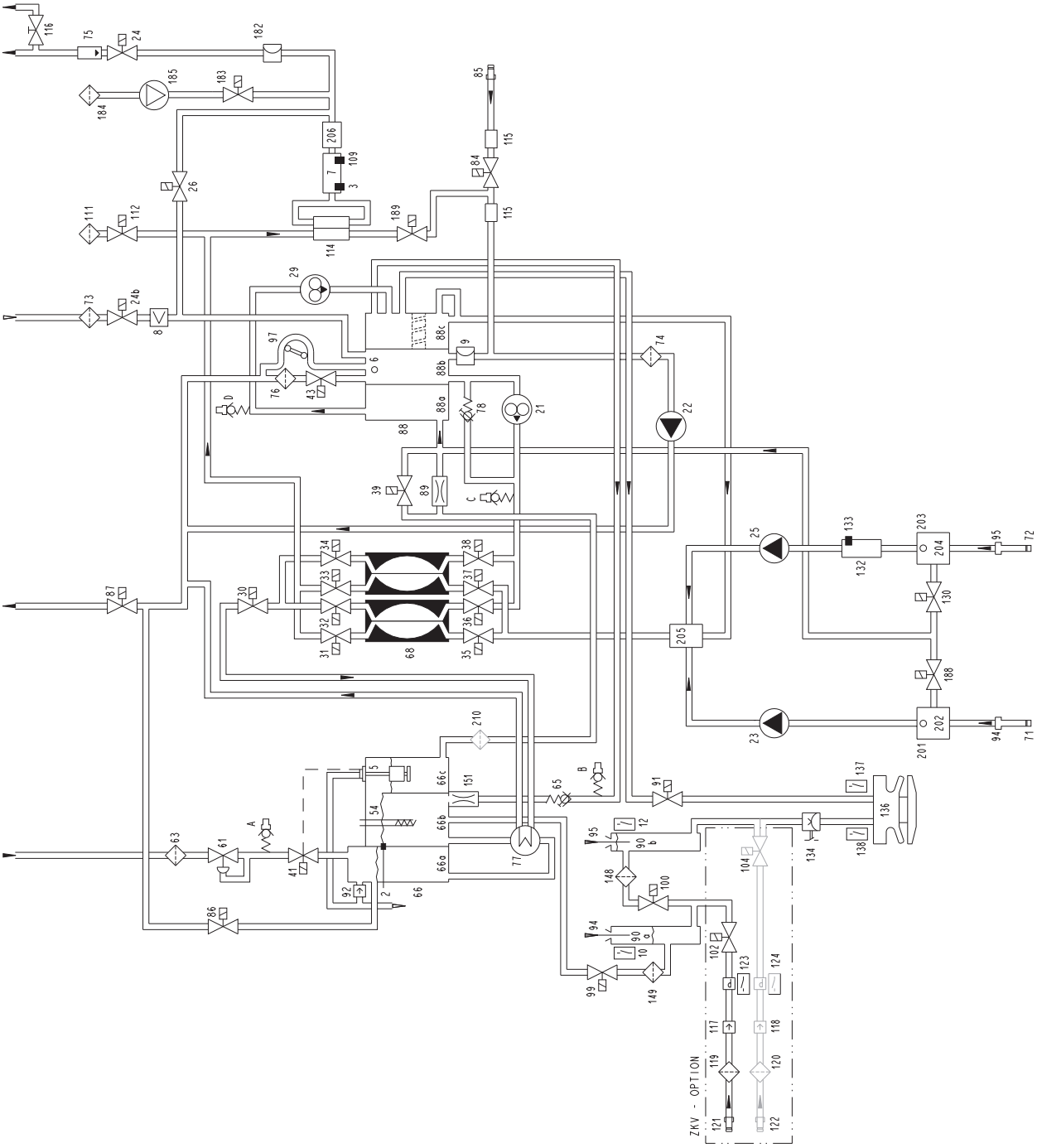
Legend

2 Temperature sensor
3 Temperature sensor
5 Float switch
6 Level sensor
7 Conductivity cell
8 Blood leak detector
9 Pressure transducer
10 Reed contact for concentrate
12 Reed contact for bicarbonate
21 Flow pump
22 UF pump
23 Concentrate pump
24 Dialyzer valve 1
24b Dialyzer valve 2
25 Bicarbonate pump
26 Bypass valve
29 Degassing pump
30 Outlet valve
31 Balancing chamber valve 1
32 Balancing chamber valve 2
33 Balancing chamber valve 3
34 Balancing chamber valve 4
35 Balancing chamber valve 5
36 Balancing chamber valve 6
37 Balancing chamber valve 7
38 Balancing chamber valve 8
41 Water inlet valve
43 Fill valve
54 Heater rod
61 Pressure reducing valve
63 Water inlet filter
65 Loading pressure valve
66 Heater block
66a Water inflow chamber
66b Heater rod chamber
66c Float chamber
68 Balancing chamber
71 Filter/concentrate
72 Filter/bicarbonate
73 Filter/dialysate ext.
74 Filter/UF
75 External flow indicator
76 Filter/fill valve
77 Heat exchanger
78 Relief valve
84 Disinfectant valve
85 Disinfectant connector
86 Recirculation valve
87 Drain valve
88 Multifunction block
88a Degassing chamber
88b Secondary air separator
88c Primary air separator
89 Degassing orifice
90a Rinse chamber concentrate
90b Rinse chamber bicarbonate
91 Rinse valve
92 Vent valve
94 Concentrate suction tube
95 Bicarbonate suction tube
97 Air separation pump
99 Rinse valve
100 Rinse valve
102 Central concentrate delivery valve
104 Central bicarbonate delivery valve
109 Temperature sensor
111 Hydrophobic filter
112 Vent valve
114 Dialysate filter
115 Sensor disinfection valve
116 Sampling valve
117 Check valve (concentrate)
118 Check valve (bicarbonate)
119 Filter (concentrate)
120 Filter (bicarbonate)
121 Central concentrate delivery connector
122 Central bicarbonate delivery connector
123 Pressure switch for V102
124 Pressure switch for V104
130 bi*b*ag® drain valve
131 bi*b*ag® block
131a bi*b*ag® air sep. chamber
131b bi*b*ag® mixing chamber
132 bi*b*ag® conductivity cell
133 bi*b*ag® temperature sensor
134 bi*b*ag® pressure transducer
135 bi*b*ag® level sensor
136 bi*b*ag® connector
137 bi*b*ag® microswitch 1
138 bi*b*ag® microswitch 2
148 Filter/rinse valve 100
149 Filter/rinse valve 99
151 Orifice
210 Filter (degassing orifice)

Hydraulics measuring points

A Reduced water inlet pressure
B Balancing chamber loading pressure
C Flow pump pressure
D Degassing pump pressure

Fig.: Flow diagram 4008 H with advanced hydraulics



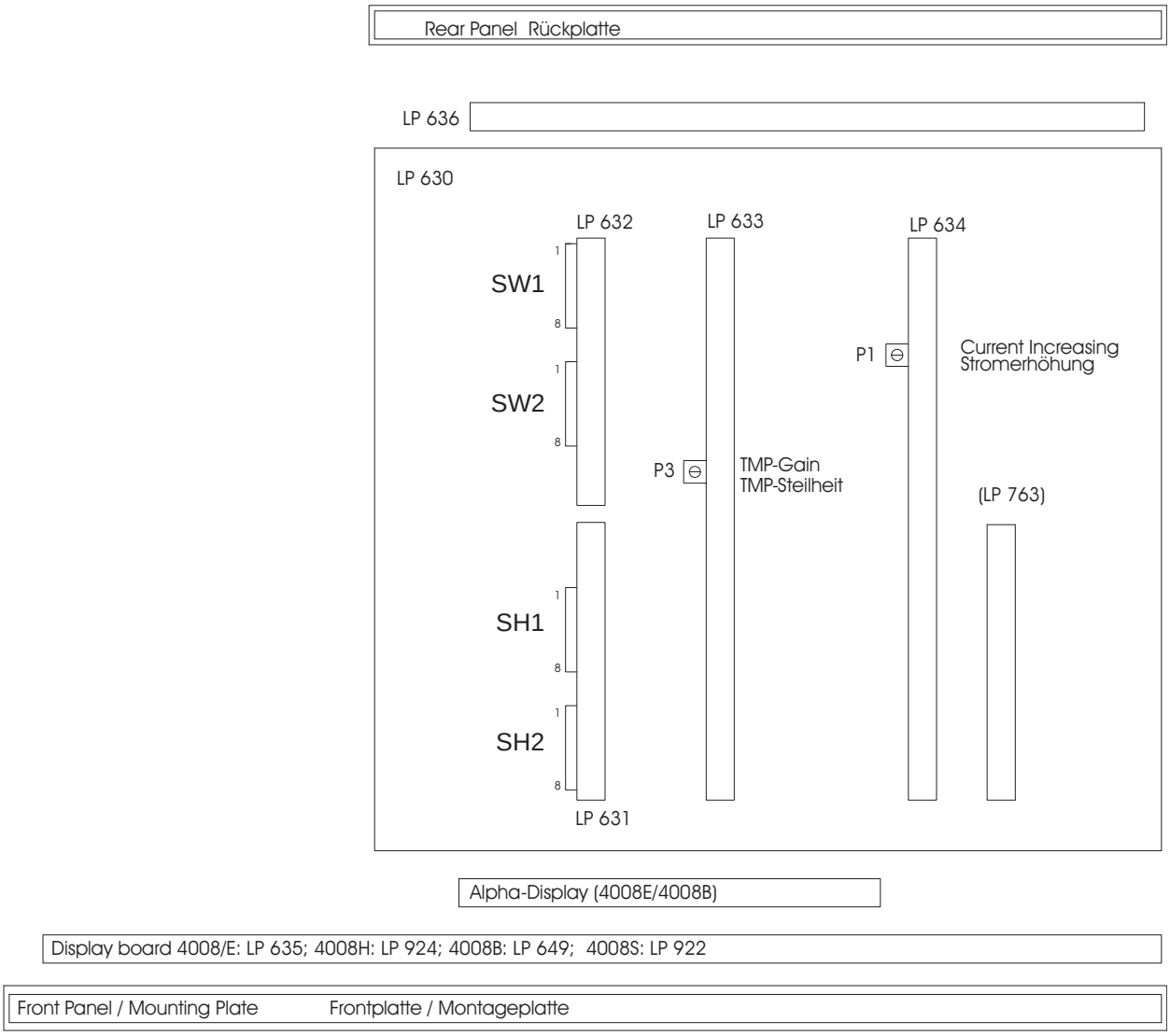
Legend

2 Temperature sensor
3 Temperature sensor
5 Float switch
6 Level sensor
7 Conductivity cell
8 Blood leak detector
9 Pressure transducer
10 Reed contact for concentrate
12 Reed contact for bicarbonate
21 Flow pump
22 UF pump
23 Concentrate pump
24 Dialyzer valve 1
24b Dialyzer valve 2
25 Bicarbonate pump
26 Bypass valve
29 Degassing pump
30 Outlet valve
31 Balancing chamber valve 1
32 Balancing chamber valve 2
33 Balancing chamber valve 3
34 Balancing chamber valve 4
35 Balancing chamber valve 5
36 Balancing chamber valve 6
37 Balancing chamber valve 7
38 Balancing chamber valve 8
39 Negative pressure valve
41 Water inlet valve
43 Fill valve
54 Heater rod
61 Pressure reducing valve
63 Water inlet filter
65 Loading pressure valve
66 Heater block
66a Water inflow chamber
66b Heater rod chamber
66c Float chamber
68 Balancing chamber
71 Filter/concentrate
72 Filter/bicarbonate
73 Filter dialysate external
74 Filter/UF
75 External flow indicator
76 Filter/fill valve
77 Heat exchanger
78 Relief valve
84 Disinfectant valve
85 Disinfectant connector
86 Recirculation valve
87 Drain valve
88 Multifunction block
88a Degassing chamber
88b Secondary air separator
88c Primary air separator
89 Degassing orifice
90a Rinse chamber, concentrate
90b Rinse chamber, bicarbonate
91 Rinse valve
92 Vent valve
94 Concentrate suction tube
95 Bicarbonate suction tube
97 Air separation pump
99 Rinse valve
100 Rinse valve
102 Central concentrate delivery valve
104 Central bicarbonate delivery valve
109 Temperature sensor
111 Hydrophobic filter
112 Vent valve
114 Dialysate filter
115 Sensor disinfection valve
116 Sampling valve
117 Check valve (concentrate)
118 Check valve (bicarbonate)
119 Filter (concentrate)
120 Filter (bicarbonate)
121 Central concentrate delivery connector
122 Central bicarbonate delivery connector
123 Pressure switch for V102
124 Pressure switch for V104
130 bi*b*ag® drain valve
131 bi*b*ag® block
131a bi*b*ag® air sep. chamber
131b bi*b*ag® mixing chamber
132 bi*b*ag® conductivity measuring cell
133 bi*b*ag® temperature sensor
134 bi*b*ag® pressure transducer
135 bi*b*ag® level sensor
136 bi*b*ag® connector
137 bi*b*ag® microswitch 1
138 bi*b*ag® microswitch 2
148 Filter/rinse valve 100
149 Filter/rinse valve 99
151 Orifice
182 Pressure transducer 2
183 Test valve
184 Test valve filter
185 Compressor
188 Evacuation valve
189 Retentate valve
201 Air separator
202 Level sensor
203 Air separator
204 Level sensor
205 Concentrate / bicarbonate mixing point
206 Buffer volume chamber
210 Filter (degassing orifice)

Hydraulics measuring points

A Reduced water inlet pressure
B Balancing chamber loading pressure
C Flow pump pressure
D Degassing pump pressure

Fig.: P.C.B. overview

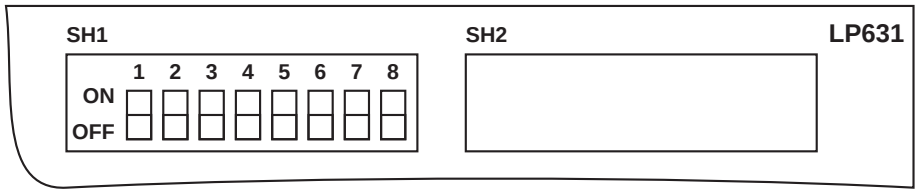


LP 630 Motherboard
LP 631 CPU 1 (operating system)
LP 632 CPU 2 (safety system)
LP 633 Input board
LP 634 Output board
LP 635 Display board
LP 636 External connectors
LP 649 Display board 4008 B
LP 763 Interface board
LP 924 Display board 4008 H
LP 922 Display board 4008 S

3.1 Overview of the DIP switches in the 4008

3.1.1 P.C.B. LP 631 (CPU 1) DIP switch array 1

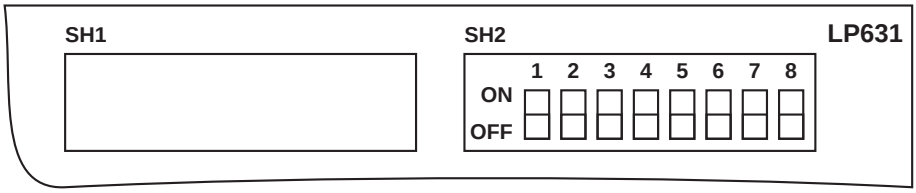
Note
Dip switch 6 is provided for service purposes/troubleshooting only and must be set to the OFF position for dialysis mode.



Switch / Position			Function		
SW 1	**SW 2**		**max. UF rate**		
ON	ON		1000 ml/h		
OFF	ON		2000 ml/h		
ON	*OFF*		*3000 ml/h*		
OFF	OFF		4000 ml/h		
SW 3	**SW 4**	**SW 5**	**Language 1**	**Language 2**	**Language 3**
ON	ON	ON	English	English	English
OFF	ON	ON	German	Finnish	Japanese
ON	OFF	ON	French	Czech	Bulgarian
OFF	OFF	ON	Portuguese	Danish	Greek
ON	ON	OFF	Dutch	Russian	Arabic (only 4008 H/S)
OFF	ON	OFF	Italian	Turkish	Norwegian (only 4008 H/S)
ON	OFF	OFF	Swedish	Polish	Slovenian (only 4008 H/S)
OFF	OFF	OFF	Spanish	Slovakian	Ex-Yugoslavian (only 4008 H/S)
SW 6			**CRC/RAM test**		
ON			skip		
OFF			*perform*		
SW 7			**Heater rod**		
ON			1300 W (at 100 to 120 V)		
OFF			*1600 W (at 220 to 240 V)*		
SW 8			**Test flow**		
ON			500 ml/min		
OFF			*800 ml/min*		

The basic position upon delivery is shown in italics.
For “not used” the switch must be set to OFF.

3.1.2 P.C.B. LP 631 (CPU 1) DIP switch array 2

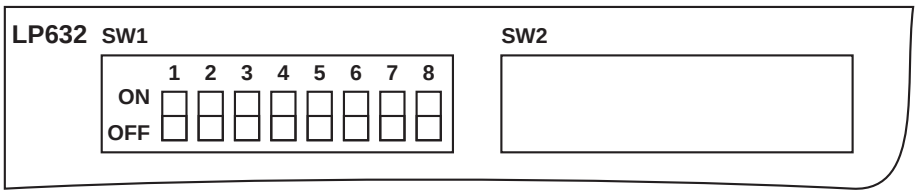


Switch / Position	Function
SW 1 ON *OFF*	**CAL mode** Mode 0 *Mode 1*
SW 2 **SW 3** ON ON OFF ON ON OFF *OFF* *OFF*	**Ext. alarm input** Invalid RO system Patient bell *Ext. alarm*
SW 4 ON *OFF*	**Remote control** System with remote control *System without remote control*
SW 5 ON *OFF*	**COMMCO LP 763 or LP 758 or LP 729** Enabled *Disabled*
SW 6 ON *OFF*	**COMMCO** Special record *Standard record*
SW 7 ON OFF	**Hydraulics test (not CDS)** Active Inactive
SW 8 ON OFF	**Central delivery system** Installed Not installed

The basic position upon delivery is shown in italics.
For “not used” the switch must be set to OFF.

3.1.3 P.C.B. LP 632 (CPU 2) DIP switch array 1

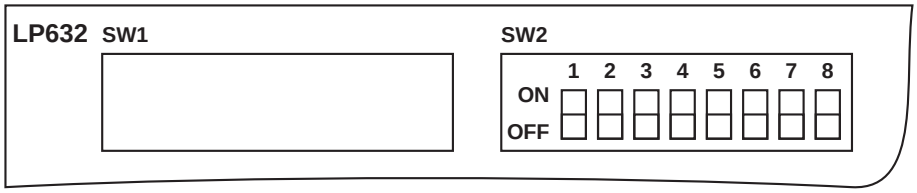
Note
DIP switches 3 and 8 permit to skip test steps which are requested by the machine.
If the switches are set to the “can be skipped” position, it is important to know that the operator can then bypass the automatic test of the safety systems.
The person demanding this switch position shall be solely responsible for such a procedure.



Switch	Position	Function
Switch 1	ON *OFF*	Not used
Switch 2	ON *OFF*	T1 test, serial sequence *T1 test, parallel sequence*
Switch 3	ON *OFF*	Skip T1 test *T1 test mandatory*
Switch 4	ON *OFF*	Test service “ON” (hemodialysis not possible) *Test service “OFF” (automatic T1 test)*
Switch 5	ON *OFF*	Cyclic PHT every 2 minutes and indication of the test result (service) *Cyclic PHT every 12.5 minutes,* *Alarm emission only with cyclical PHT alarm*
Switch 6	*ON* OFF	*Cyclic PHT “ON”* Cyclic PHT “OFF”
Switch 7	ON *OFF*	Air detector with PCB LP 450 without AD28 *Air detector with PCB LP 450-2 or with AD28*
Switch 8	ON *OFF*	HDF test can be skipped *HDF test mandatory*

The basic position upon delivery is shown in italics.
For “not used” the switch must be set to OFF.

3.1.3 P.C.B. LP 632 (CPU 2) DIP switch array 2

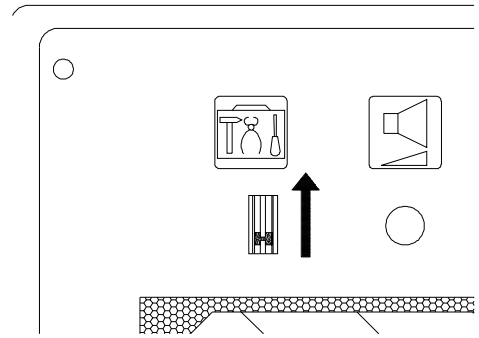


Switch	Position	Function
Switch 1	ON OFF	DIASAFE / DIASAFE®*plus* on DIASAFE / DIASAFE®*plus* off
Switch 2	ON OFF	ON-LINE-HDF on ON-LINE-HDF off
Switch 3	ON OFF	ONLINE*plus*™ on ONLINE*plus*™ off
Switch 4	ON OFF	Advanced hydraulics Basic hydraulics
Switch 5	ON *OFF*	V39 Test off V39 Test on
Switch 6	ON *OFF*	Fast heating HDIS deactivated Fast heating HDIS activated
Switch 7	ON *OFF*	Not used
Switch 8	ON *OFF*	Not used

The basic position upon delivery is shown in italics.
For “not used” the switch must be set to OFF.

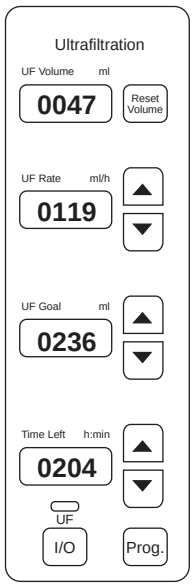
3.2 Calibration mode

3.2.1 Basic conditions



- The hemodialysis machine must be switched off.
- The service switch must be in the **ON** (up) position.
- Turn the hemodialysis machine on.

3.2.2 Messages on the displays on the UF monitor (4008 E/B) or on the screen (4008 H/S)



Watchdog supply voltage (4.5 V to 5.5 V)

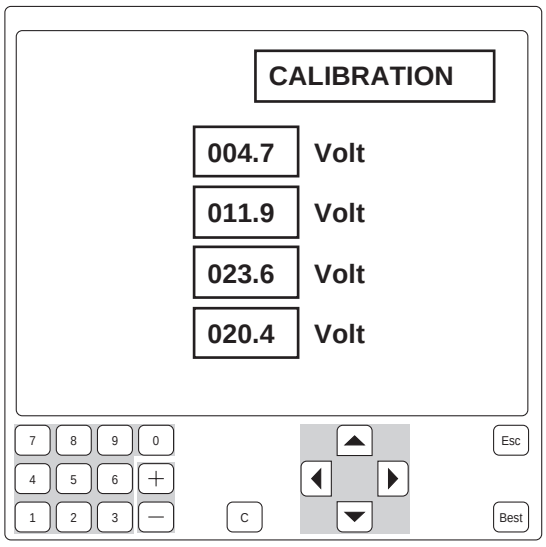
12-V voltage (11.5 V to 12.5 V)

24-V voltage (23 V to 25 V)

Battery charging voltage (>20 V)

The values indicated on the display must be divided by the factor 10.

Values in brackets: tolerances

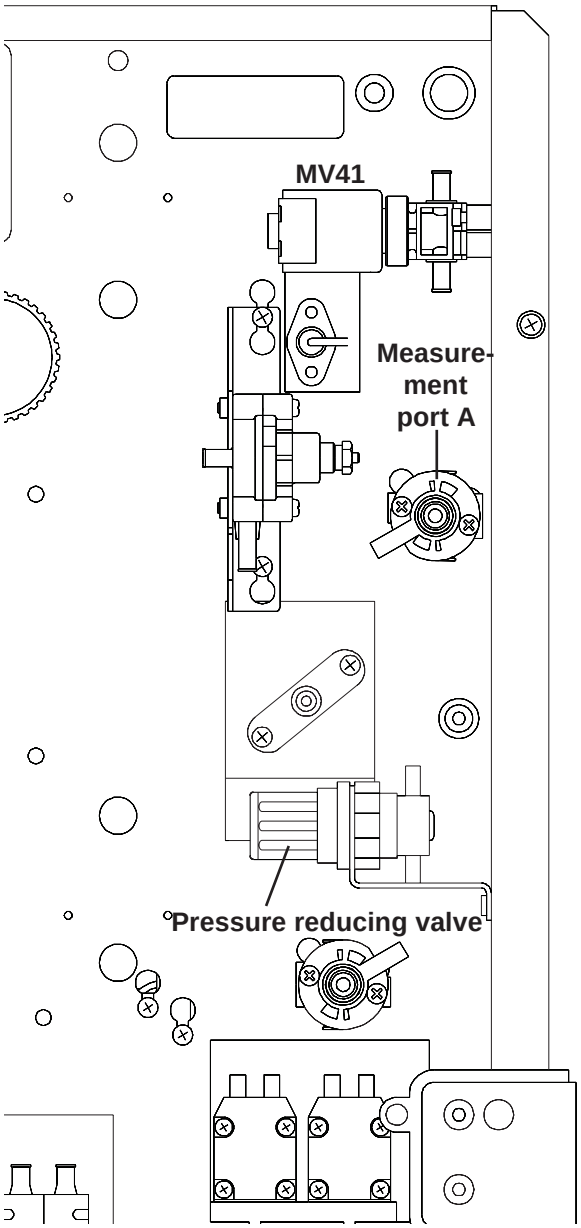


Watchdog supply voltage (4.5 V – 5.5 V)

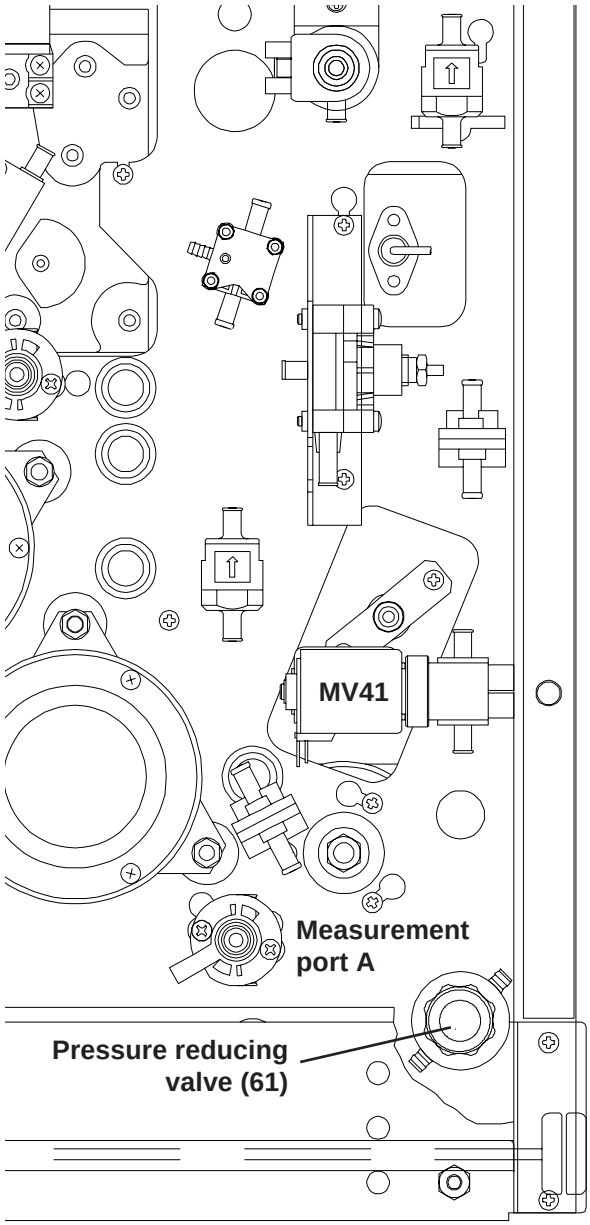
12 V voltage (11.5 V – 12.5 V)

24 V voltage (23 V – 25 V)

Battery charging voltage (> 20 V)



4008 E/H



4008 B/S

3.3 Hydraulics



Note
Measuring equipment for measurement points in hydraulic unit:
UMED, HMED, scales or pressure gauge with a measuring range of –1 to +2.2 bar, min. quality class 1.6.

3.3.1 Reduced water inlet pressure

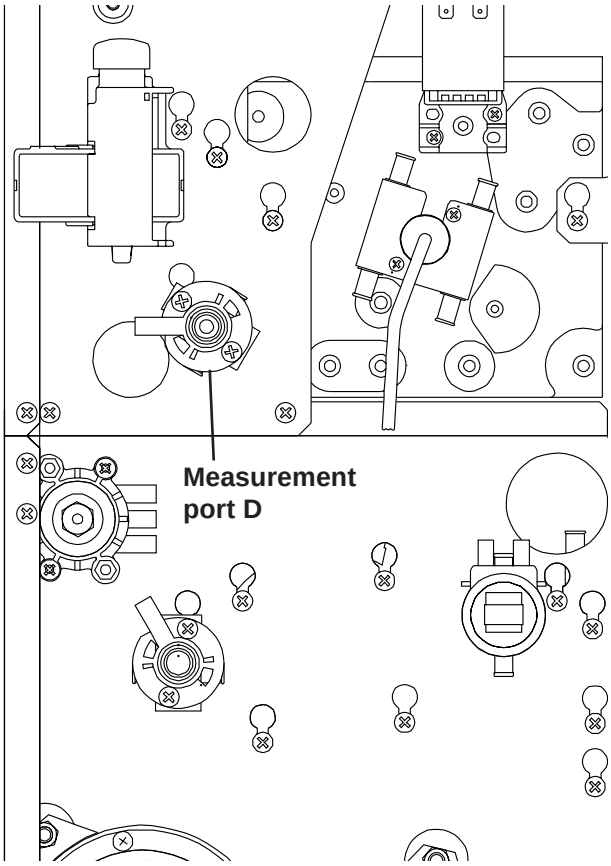
Measuring equipment: UMED, HMED or pressure gauge

Place of measurement: Hydraulics, measurement port A

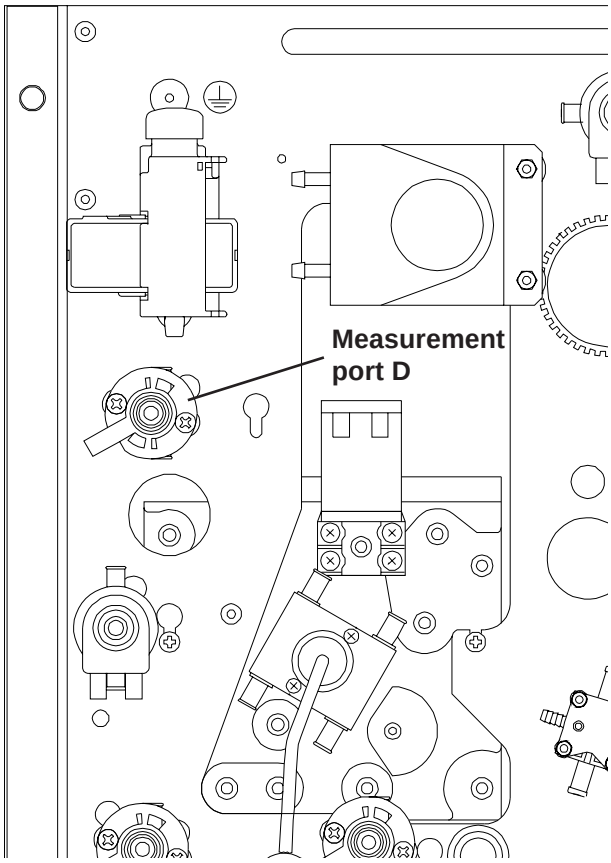
Condition: Flow on

Check/adjustment:

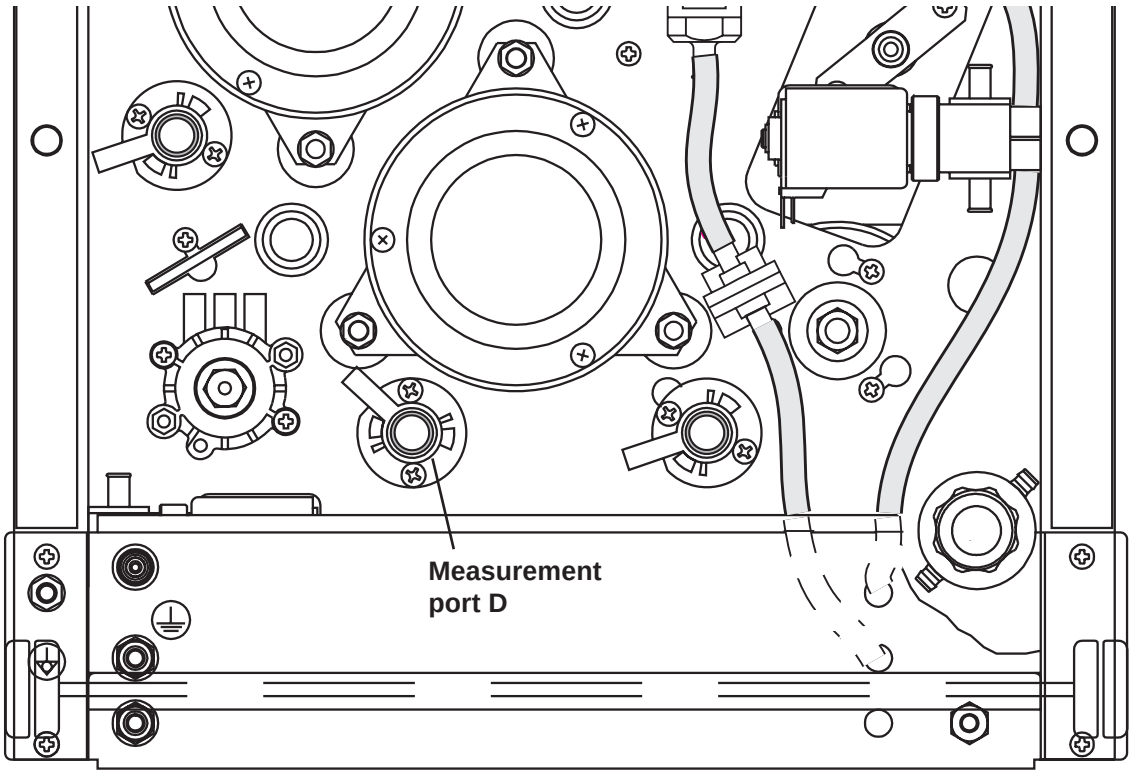
- **Check the reduced water inlet pressure**
  Connect the measuring equipment to measurement port A.
  Measure the water pressure with MV 41 closed.
  Rated value of water inlet pressure: 0.90 to 1.40 bar
  If it deviates from the rated value, the water inlet pressure must be adjusted.
- **Adjust the reduced water inlet pressure**
  Pull back the knurled nut on the pressure reducing valve (16).
  Turn the knurled nut to set the water pressure to the desired value (clockwise: +, counter-clockwise: –).
  Push the knurled nut back in.



4008 E/H



4008 B/S



4008 B/S ONLINE*plus*™

3.3.2 Degassing pump pressure

Measuring equipment: UMED, HMED or pressure gauge

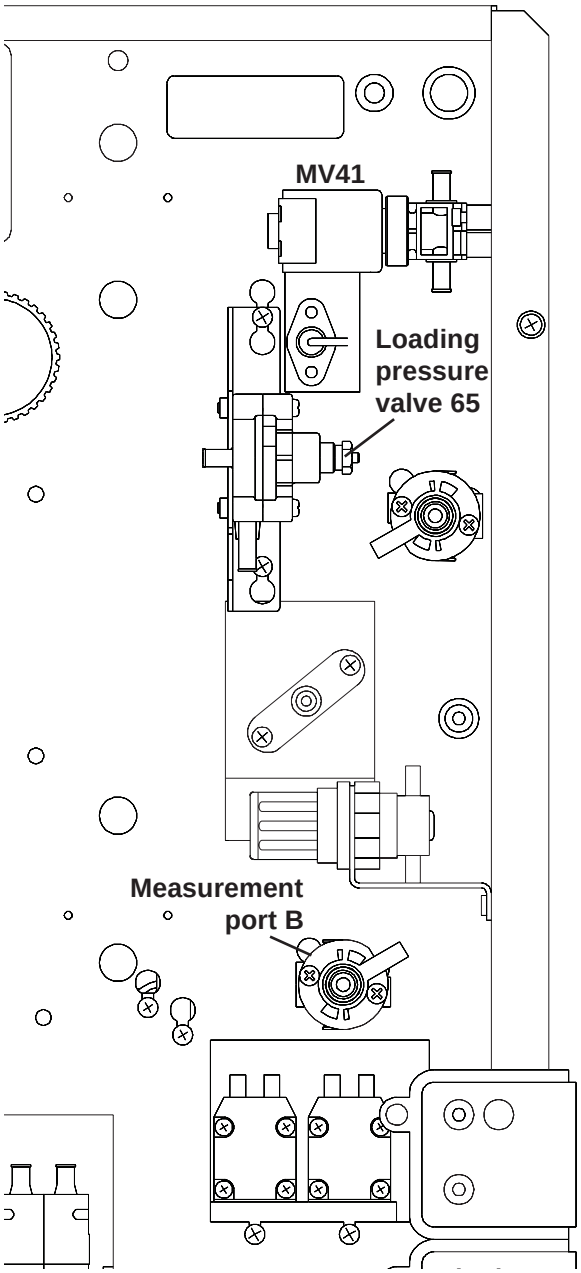
Place of measurement: Hydraulics, measurement port D

Check/adjustment:

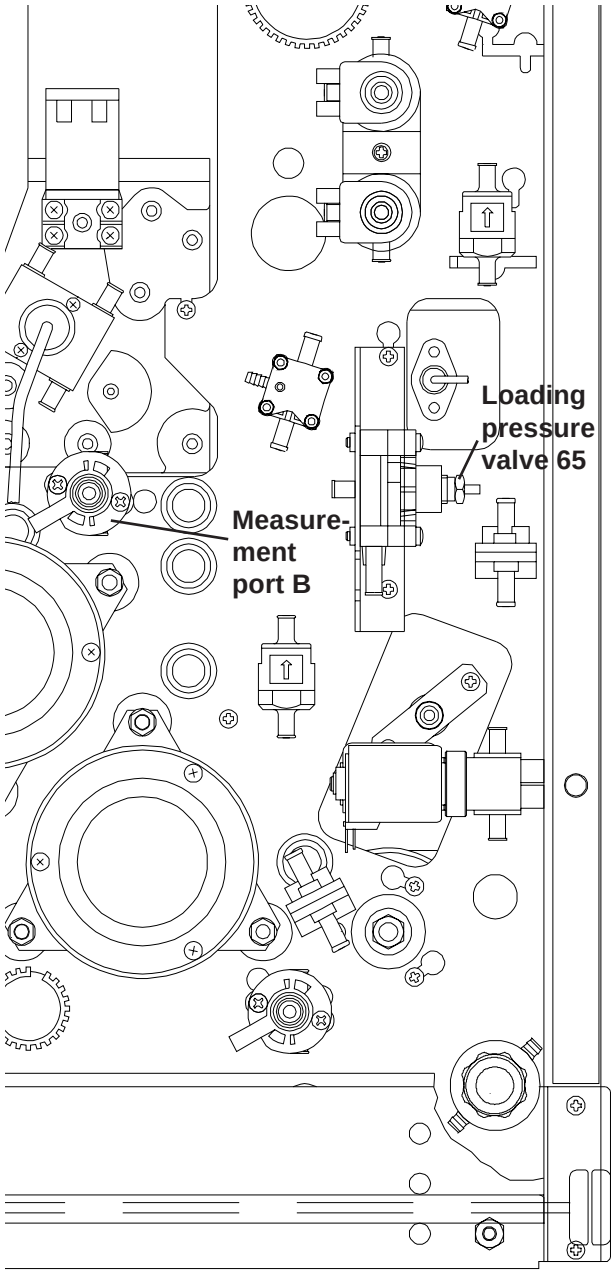
– **Check the pressure of the degassing pump**
  Connect the measuring equipment to the measurement port D.
  Measure the pressure of the degassing pump.
  Rated value of degassing pump pressure: –0.81 to –0.85 bar
  If it deviates from the rated value, the pressure of the degassing pump must be adjusted.
– **Adjust the pressure of the degassing pump**
  Enter the **CALIBRATION** menu, select and start the option **CAL. DEGAS. PRESSURE** (➜ Calibration, Chapter 4, Section 7).



Note
If the pressure of the degassing pump was changed, make sure to check the loading pressure and readjust, if necessary.



4008 E/H



4008 B/S

3.3.3 Balancing chamber loading pressure

Measuring equipment: UMED, HMED or pressure gauge

Place of measurement: Hydraulics, measurement port B

Check/adjustment:

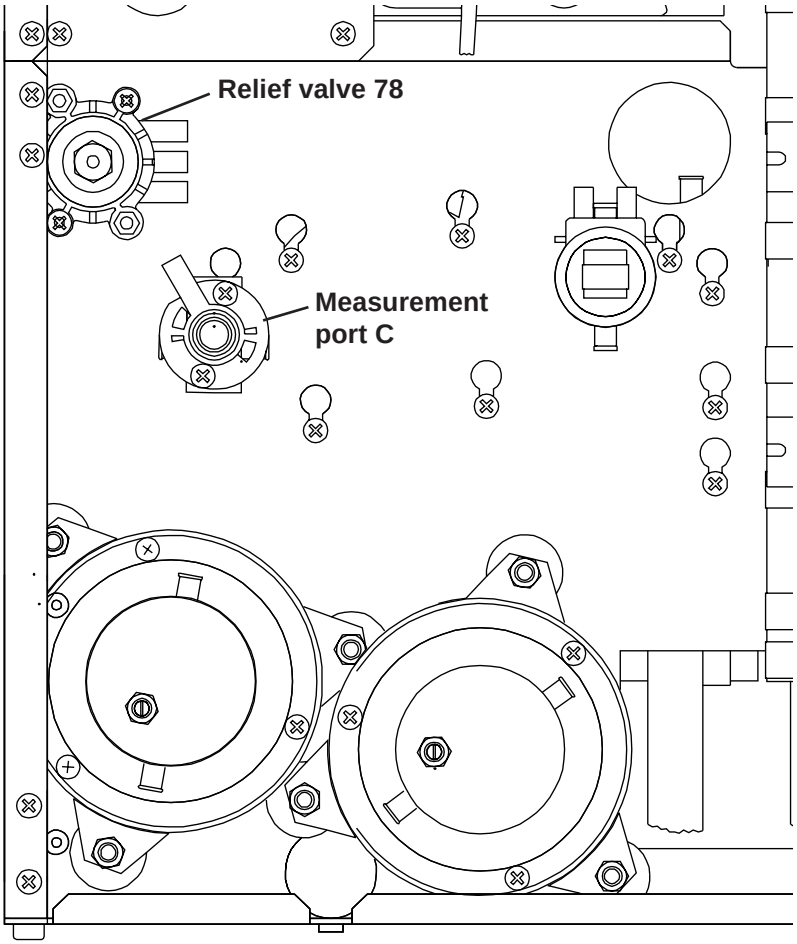
– **Check the loading pressure of the balancing chamber**
  Connect the measuring equipment to the measurement port B.
  Measure the loading pressure of the balancing chamber.
  Rated value of the balancing chamber loading pressure: 1.2 to 1.3 bar
  If it deviates from the rated value, the loading pressure of the balancing chamber must be adjusted.
– **Adjust the loading pressure of the balancing chamber**
  Use the loading pressure valve (65) to adjust the loading pressure to the rated value.
  Turning the adjusting screw clockwise will increase the loading pressure.



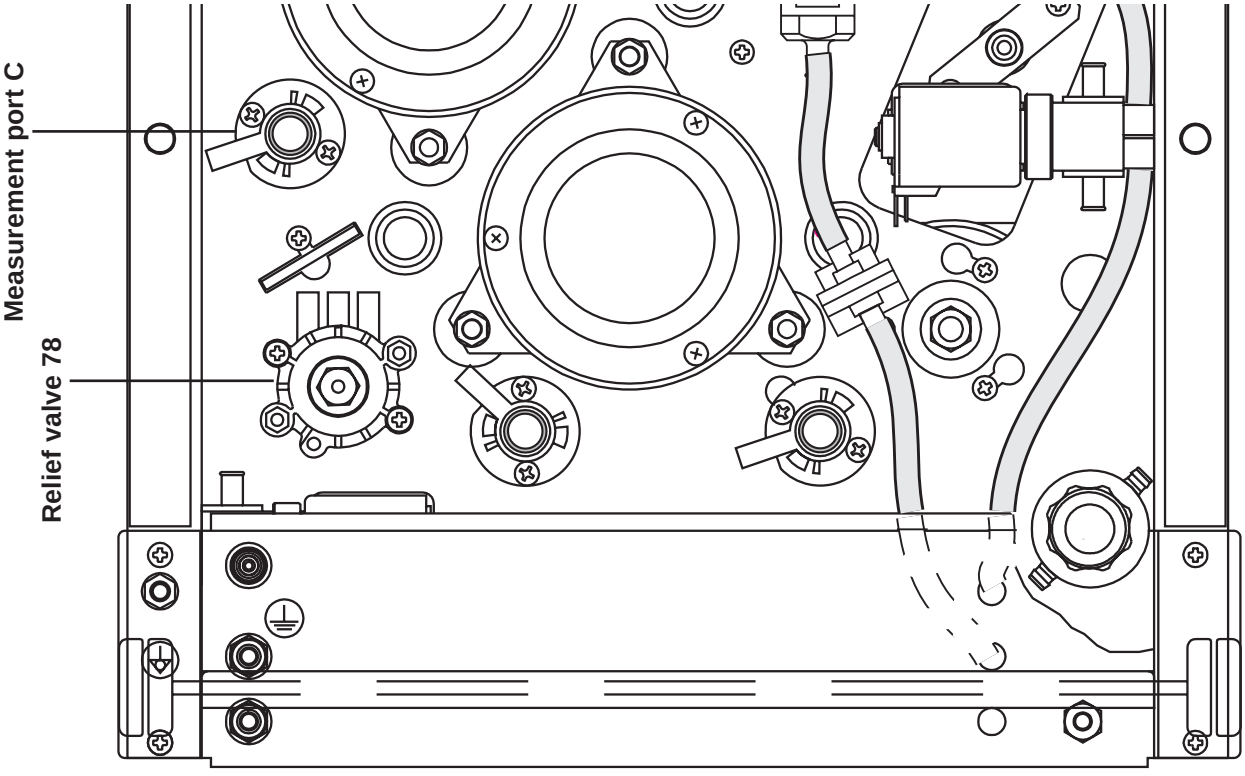
Note
During the balancing chamber fill phase, the loading pressure drops to approx. 1.0 bar.



Note
If the loading pressure was changed, make sure to check the degassing pump pressure and readjust, if necessary.



4008 E/H



4008 B/S ONLINE*plus*™

3.3.4 Flow pump pressure

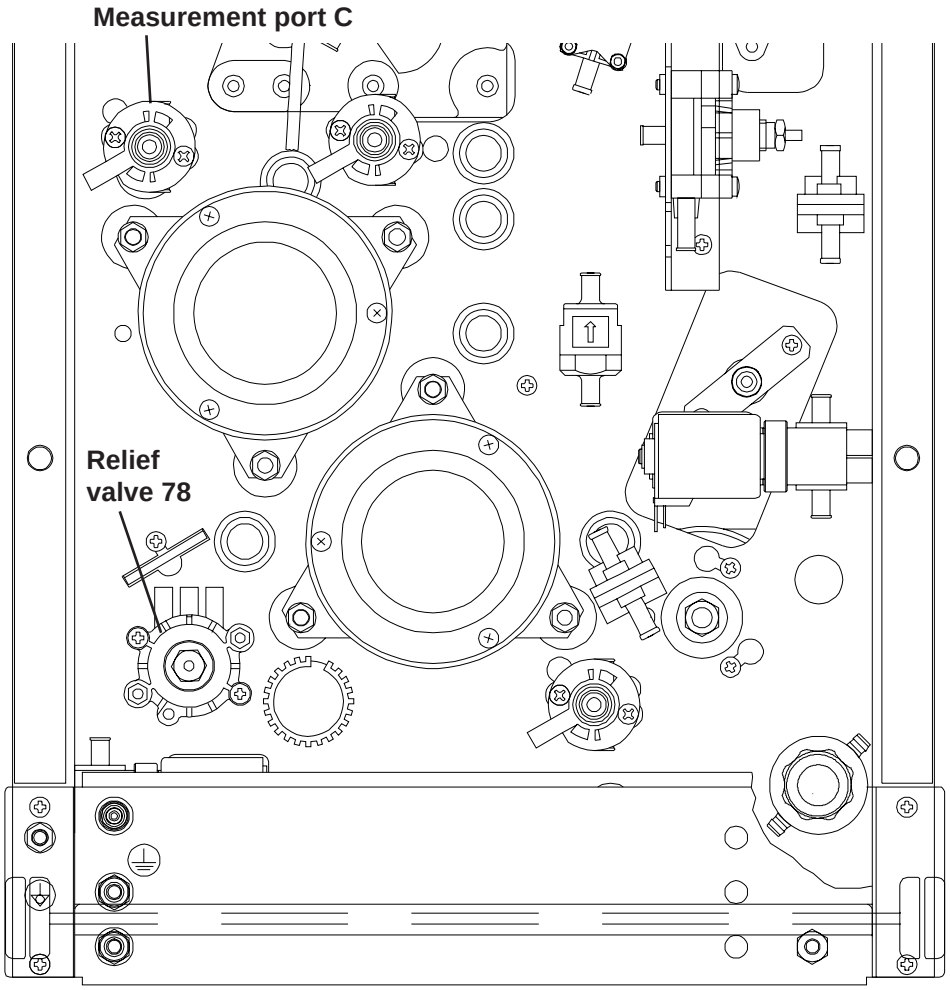
Measuring equipment: UMED, HMED or pressure gauge

Place of measurement: Hydraulics, measurement port C

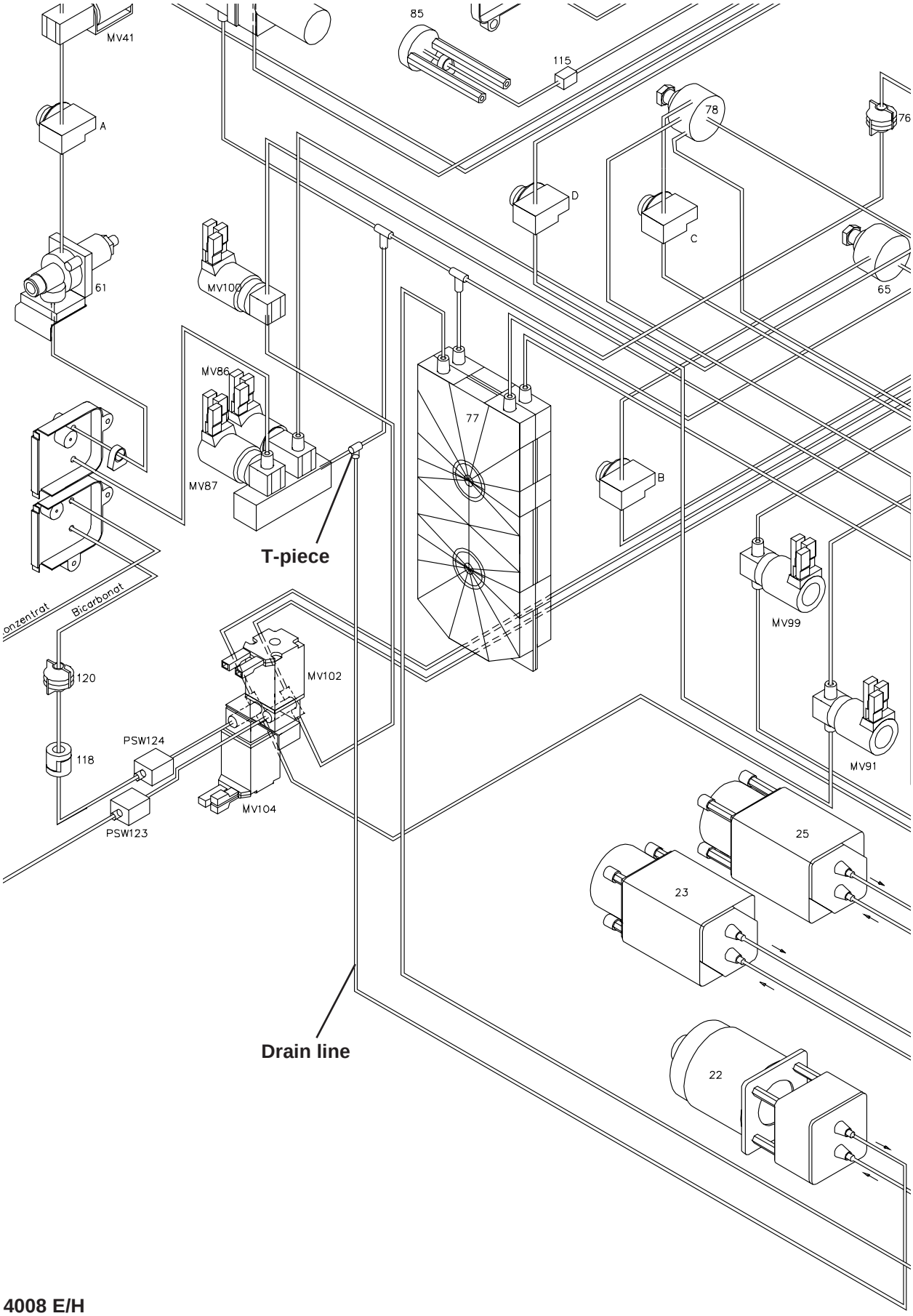
Prerequisite: A dialysate flow of 800 ml/min must have been preselected.

Check/adjustment:

– **Check the pressure of the flow pump**
  Connect the measuring equipment to the measurement port C.
  Turn the water supply off; water alarm; balancing chamber inactive.
  Measure the pressure of the flow pump.
  Rated value of the flow pump pressure: 2.0 to 2.1 bar
  If it deviates from the rated value, the pressure of the flow pump must be adjusted.
– **Adjust the pressure of the flow pump**
  Use the relief valve (78) to adjust the rated value.



4008 B/S


MV41
A
61
MV100
MV86
MV87
T-piece
85
115
78
76
D
C
65
77
B
MV99
MV91
onzentrat
Bicarbonat
120
118
PSW124
PSW123
MV102
MV104
25
23
22
Drain line

4008 E/H

3.3.5 UF pump volume

Note
If scales are used for measuring, it must be ensured that no concentrate is connected.

Measuring equipment: Scales or measuring cylinder, tolerance ±0.5 %

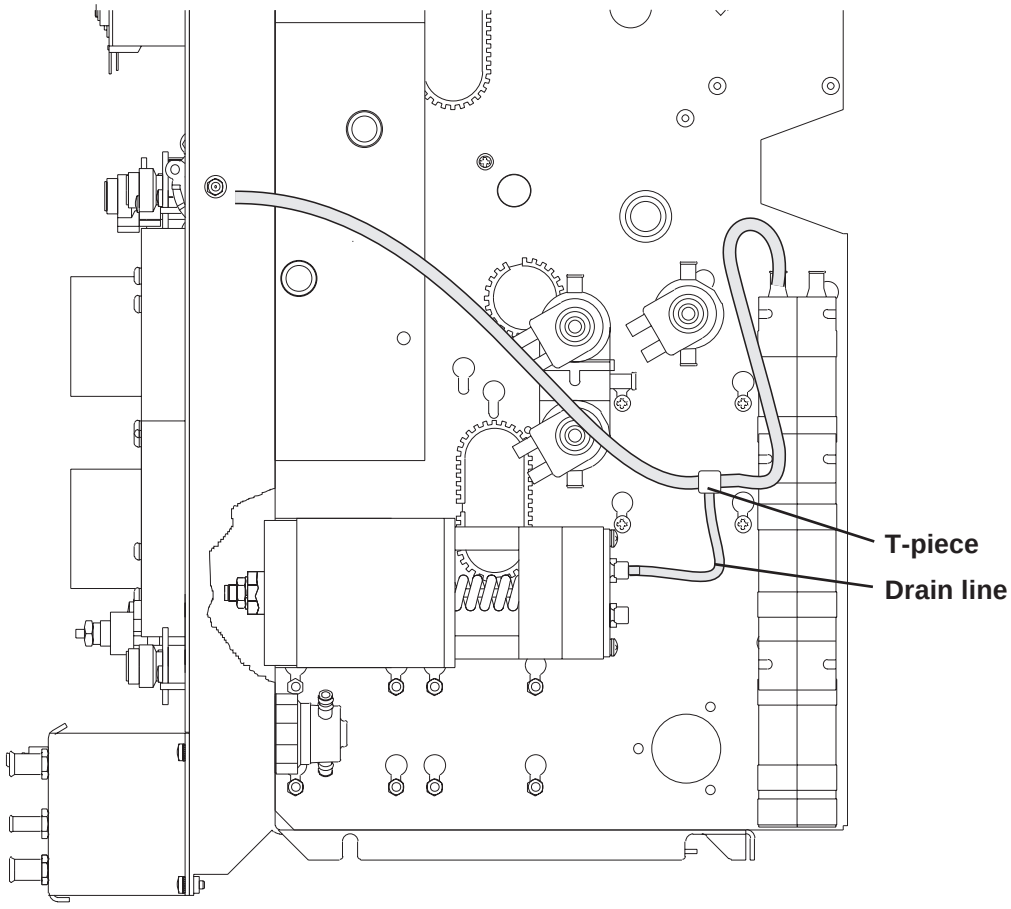
Place of measurement: Hydraulic unit open

Condition: Calibration program selected

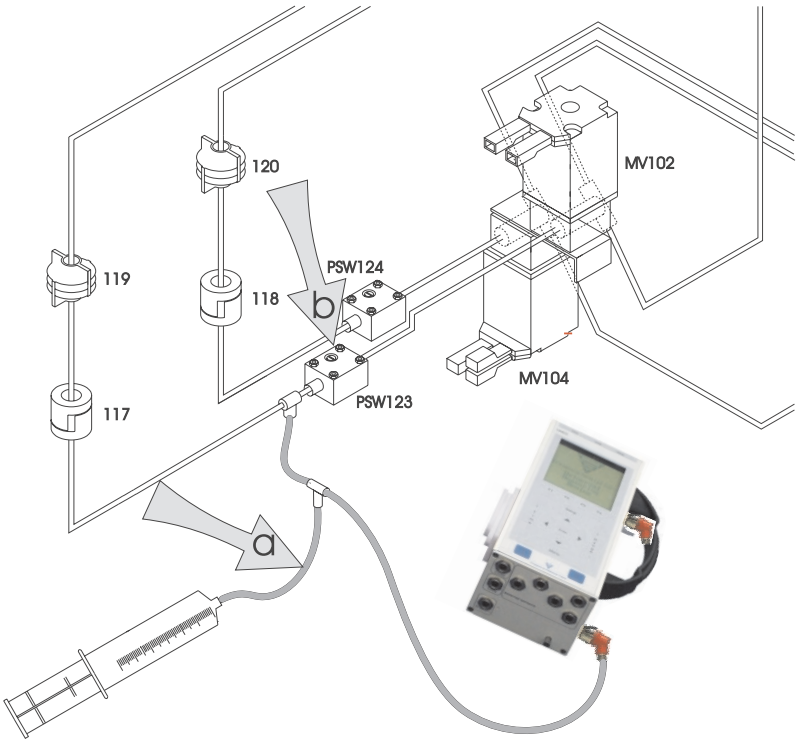
Check/adjustment:

– **Check the UF pump volume**
Remove the drain line of the UF pump from the T-piece (close the T-piece).
Place the drain line in the measuring cylinder.
Access the **CALIBRATION** menu, select and start the **ADJ. UF-PUMP VOLUME** option (➜ Calibration, Chapter 4, Part 6).
Rated value: number of strokes in ml or g ±1%.

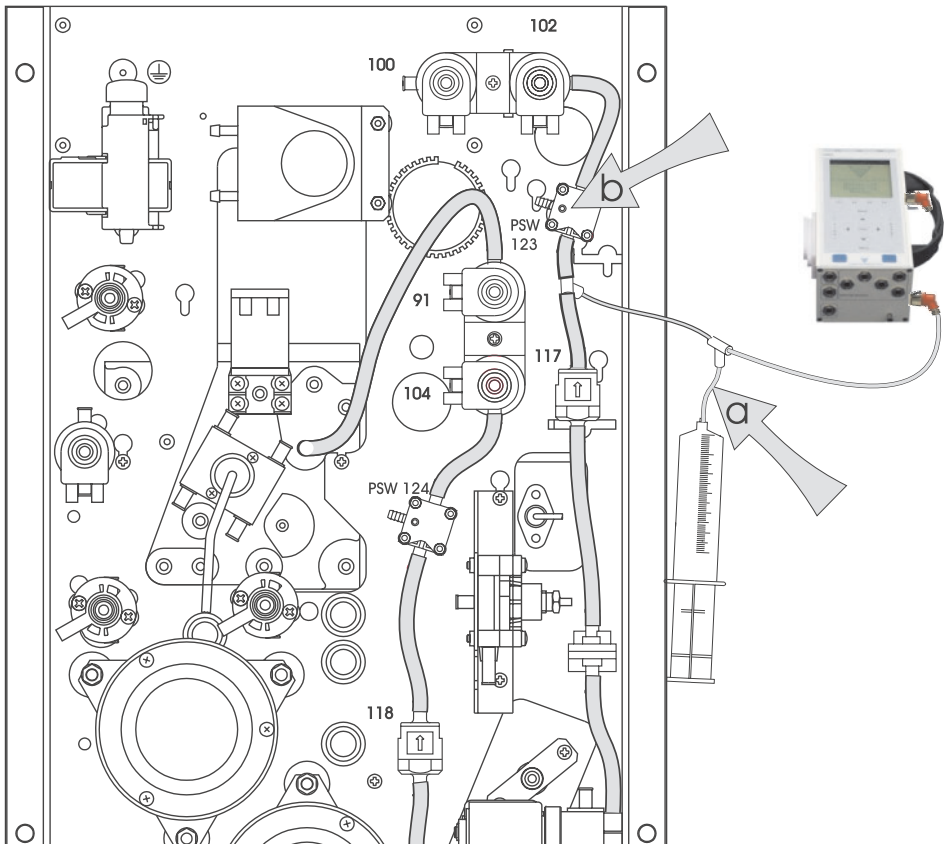
– **Adjust the UF pump**
Remove the protective cover.
Unscrew the lock nut.
Change the delivery volume, using the adjusting screw (turning the adjusting screw clockwise reduces, turning it counter-clockwise increases the stroke volume).
Retighten the lock nut.
Verify the delivery volume.



4008 B/S



4008 E/H



4008 B/S

3.3.6 CDS pressure switch

Measuring equipment: Measurement setup according to diagram,
UMED, HMED or pressure gauge (e.g. 0 to 1 bar, accuracy ±1 %),
syringe

Place of measurement: Hydraulic unit open

Condition: The CDS connectors in position 121 and position 122 must be depressurized.
The pressure compensation port on the pressure switch must be open to air (atmospheric pressure).
The lines of the measuring equipment should be as short as possible.
The service mode must be selected.

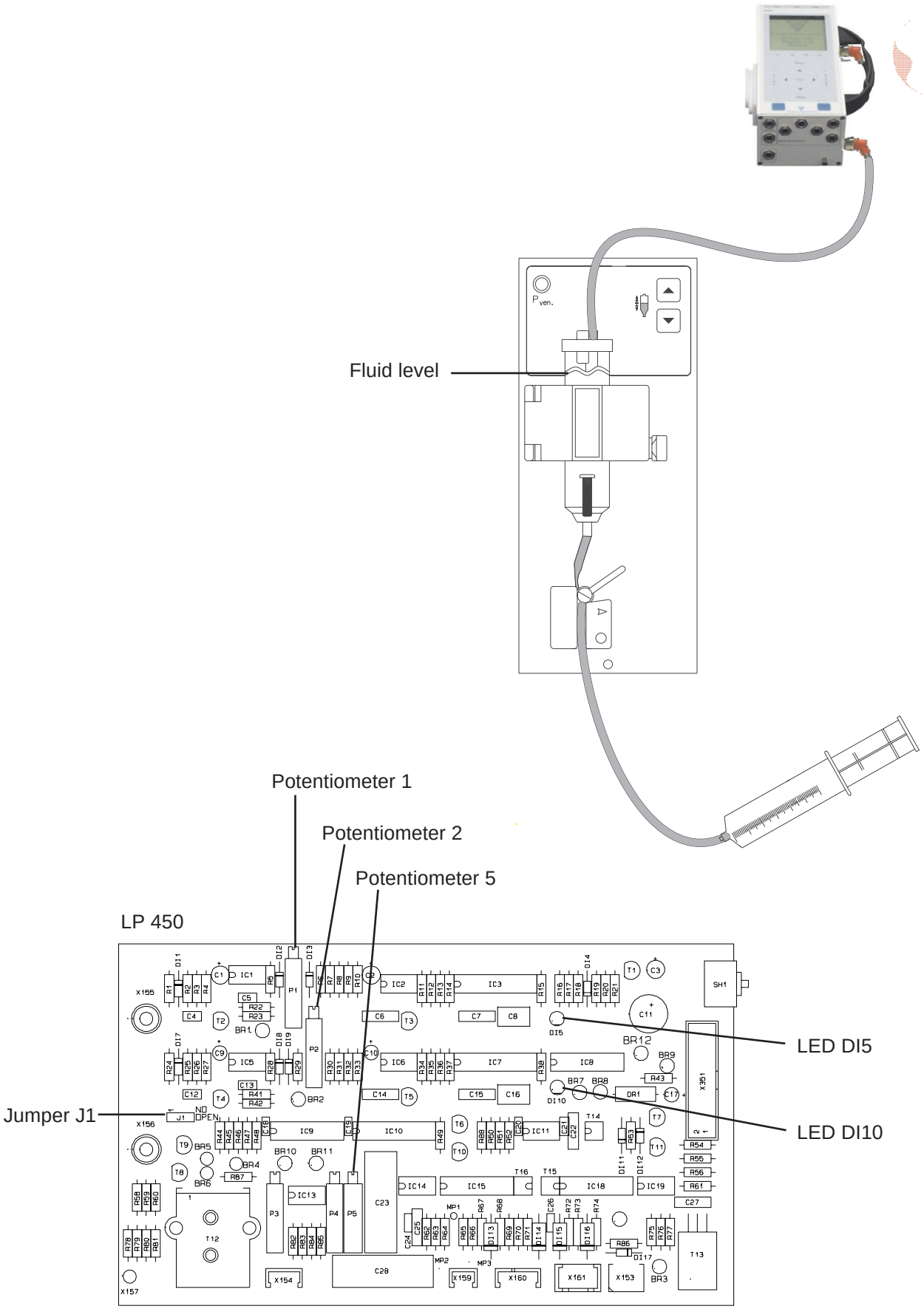
Check/adjustment:

– Connect the measuring equipment as illustrated in the diagram.
– Select the DIAGNOSTICS menu:

READ INPUTS
READ DIGITAL INPUTS
CPU1: RD DIGITAL INP
I: CPU1_PSW_V102 or I: CPU1_PSW_104

– Activate the audible alarm by pressing the (**Alarm**) **Tone Mute** key (depressurized: alarm on).
– During these menu options, the solenoid valves MV102 and MV104 are closed.
– Use the syringe to create a pressure of 0.7 bar.
– Use forceps to clamp the line at point **a**, so that the pressure switch remains loaded with 0.7 bar.
– Verify the switching point by means of the audible alarm
  Rated values:
  Alcatel pressure switch (part no.: 674 322 1) (yellow): 0.68 – 0.80 bar
  Delta pressure switch, dark grey: 0.68 – 0.72 bar
  Envec pressure switch: 0.68 – 0.72 bar
  If the switching point deviates, adjust with the adjusting screw **b** (make sure there is no mechanical load on the pressure switch while adjusting).
– After completed adjustment, depressurize the measuring equipment and repeat the check. If necessary, repeat the adjustment procedure.

This adjustment procedure simultaneously checks the tightness of the check valves 117 and 118 and the solenoid valves 102 and 104.


Fluid level
Potentiometer 1
Potentiometer 2
Potentiometer 5
LP 450
LED DI5
LED DI10
Jumper J1

3.4 Air detector

Caution
For adjusting the air detector, the system must be in “Calibration” mode. The ambient temperature should range between 15 and 35 °C.

Measuring equipment: Measurement setup according to diagram
UMED, HMED or pressure gauge, bubble catcher, syringe filled with degassed water or saline solution

Place of measurement: Air detector

Check/adjustment:

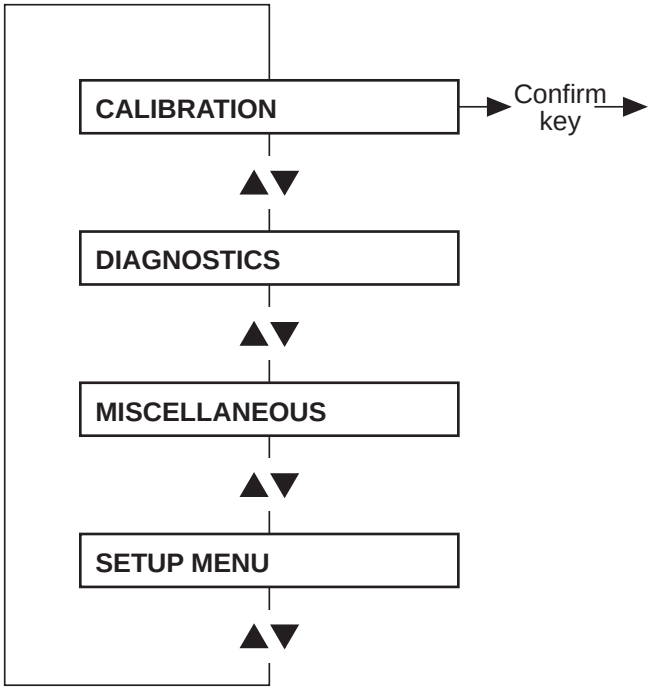
– **Adjust the ultrasonic detector**
Install the measuring set-up before checking/adjusting the air detector. Do not yet place the line in the occlusion clamp.
Jumper J1 / P.C.B. LP 450 set to calibration.
Fill the bubble catcher. The fluid level must be set to approx. 10 mm above the top edge of the sensor holder.
Turn potentiometer 1 and potentiometer 2 on P.C.B. LP 450 clockwise, until the LED DI5 and LED DI10 on P.C.B. LP 450 are dark.
Slowly (attention: time constant) turn potentiometer 1 on P.C.B. LP 450 counterclockwise, until the LED DI5 on P.C.B. LP 450 lights.
Slowly (attention: time constant) turn the potentiometer 2 on P.C.B. LP 450 counterclockwise, until the LED DI10 on P.C.B. LP 450 lights.
After completion of the calibration procedure, set the jumper J1 / P.C.B. LP 450 back to the operation position.

– **Check**
Lower the level in the bubble catcher: an alarm must be emitted.
Raise the level in the bubble catcher: it must be possible to clear the alarm; both LEDs must be off!

– **Check the venous line clamp** (➜ Fig. on p. 3-28)
Place the line in the venous line clamp.
Open the clamp and, using the syringe, generate a pressure of approx. 2 bar.
Close the clamp.
The pressure must not drop by more than 0.1 bar within 3 minutes.

– **Adjust the optical detector**
Use the gray filter, double-laid, part no. 640 560 1.
Diagnostics menu; reading of digital inputs by CPU 1; item I: CPU1_OD_IN.
Install the gray filter, double-laid; close the hinged cover.
Slowly turn potentiometer 5 on P.C.B. LP 450 clockwise, until the UF display indicates 1111.
Slowly turn potentiometer 5 counterclockwise, until the display jumps to 0000. Continue to turn the potentiometer counterclockwise for half a turn.
Avoid incident light from external sources.

4 Calibration program



Adjustments made without display messages:

Adjusting the blood pump stop alarm (blood pump or HDF pump) 4-5

Calibrating the single needle blood pump 4-11

Adjusting the current rise pulse 4-15

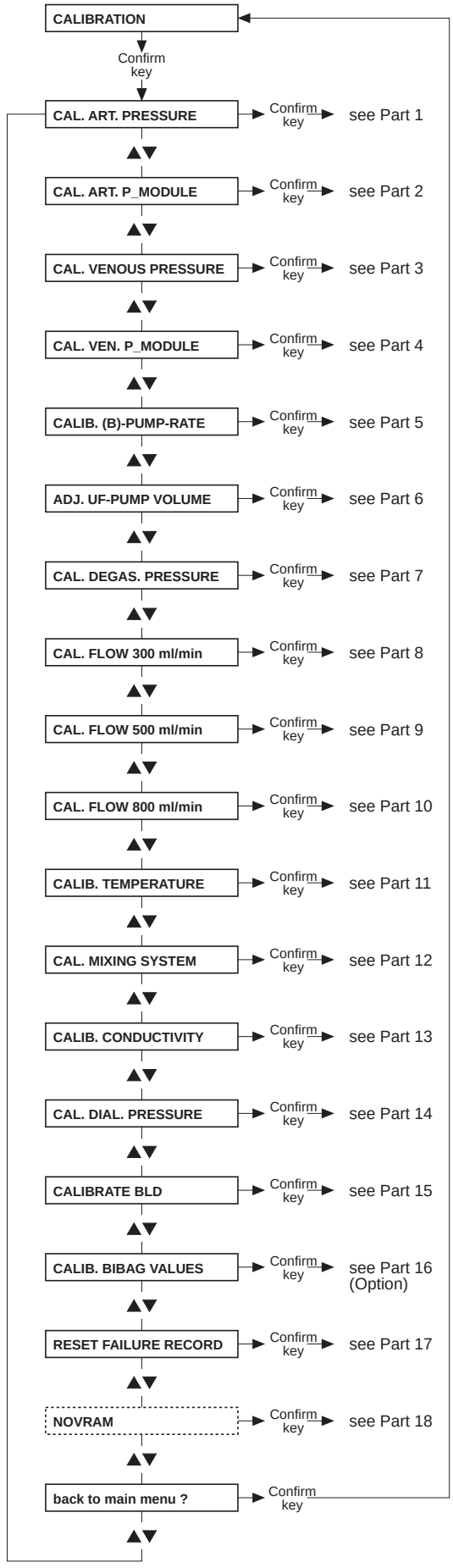
Adjusting the Hall sensor in the heparin pump 4-34

In the Calibration, Diagnostics, Setup and Miscellaneous program the function of the keys differs between 4008 E/B and 4008 H/S machines.

Function	**4008 E/B**	**4008 H/S**
Scrolling through menu options	▲▼	▲▼
Selecting a menu option	Confirm	Conf
Changing values and functions in a menu	▲▼	+/–
Saving changes	Override	Tone Mute
Exiting a menu without saving the data	Select	Esc

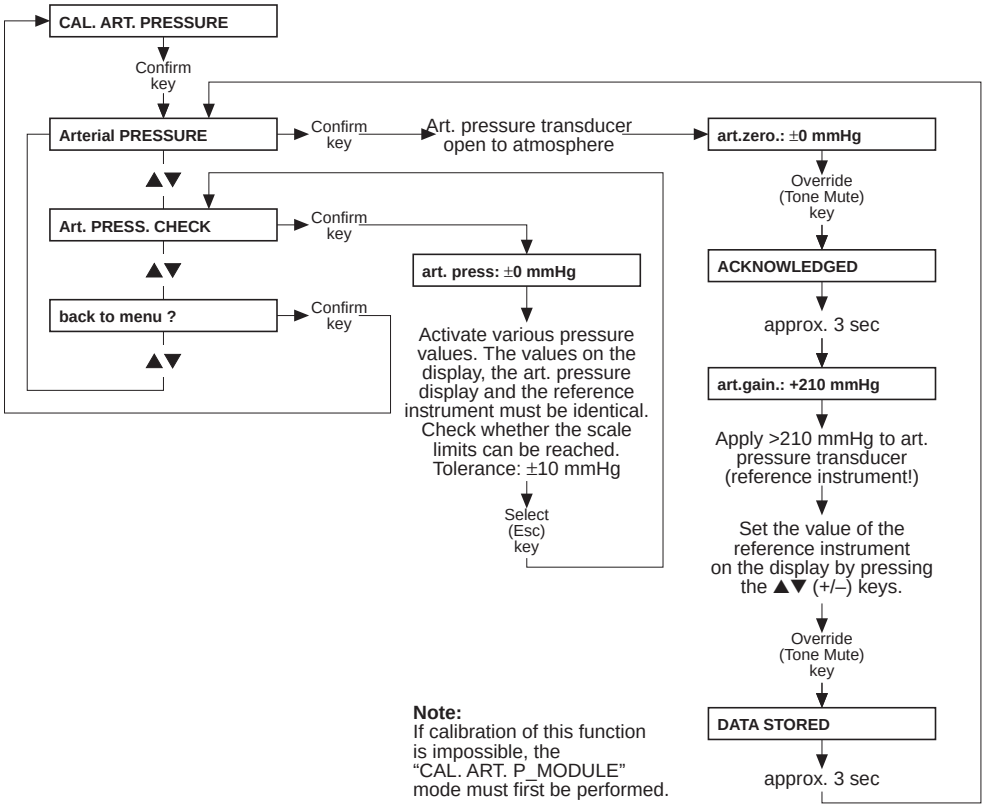
In the description of the steps, the differing keys to be used on 4008 H/S machines are shown in brackets.

- **Main menu**



Note
Before calibrating the hydraulics, remove possibly existing precipitate by running an appropriate disinfection program.

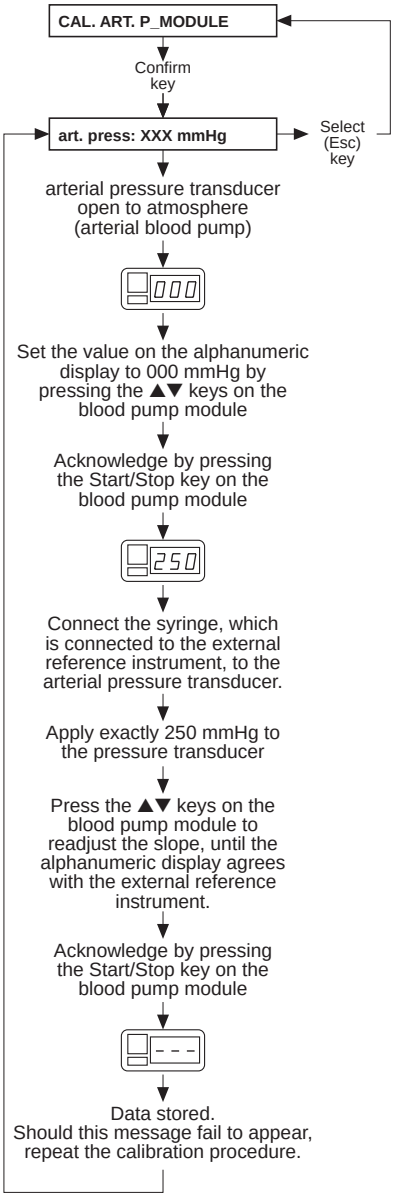
- **Part 1: Calibrate arterial pressure**



Note
Pressure gauge accuracy: ±1 % of the measured value.

- **Part 2: Calibrate the pressure in the arterial blood pump**

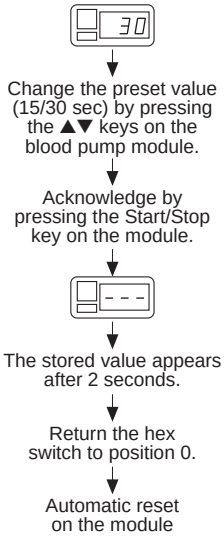
Set the hex switch in the module (P.C.B. LP 624, pos. 1) to position F.
Should the error message E02 appear on the blood pump display, clear the message by pressing the **Start/Stop** key.



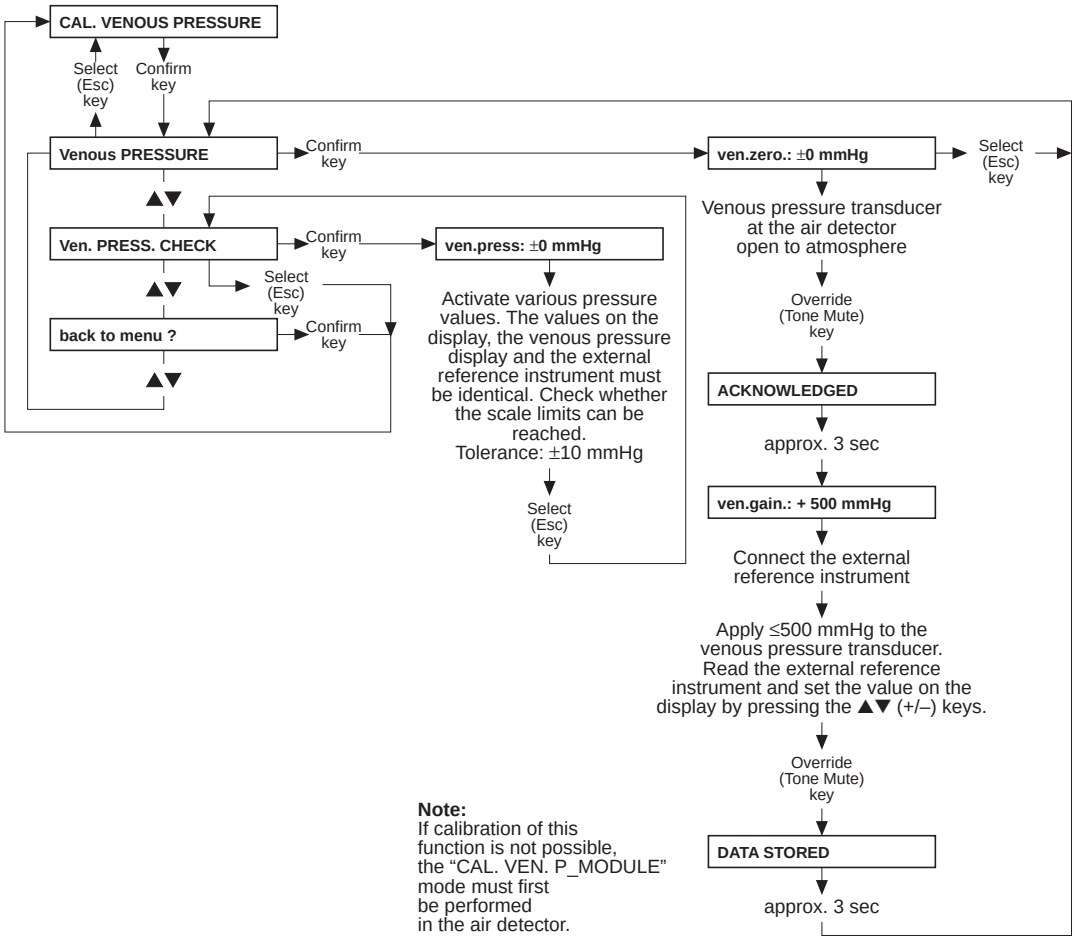
Return the hex switch to position 0.

- **Without display messages: adjusting the blood pump stop alarm (blood pump or HDF pump)**

  Set the hex switch in the module (P.C.B. LP 624, pos. 1) to position B.
  Should the error message E02 appear on the blood pump display, clear the message by pressing the **Start/Stop** key.

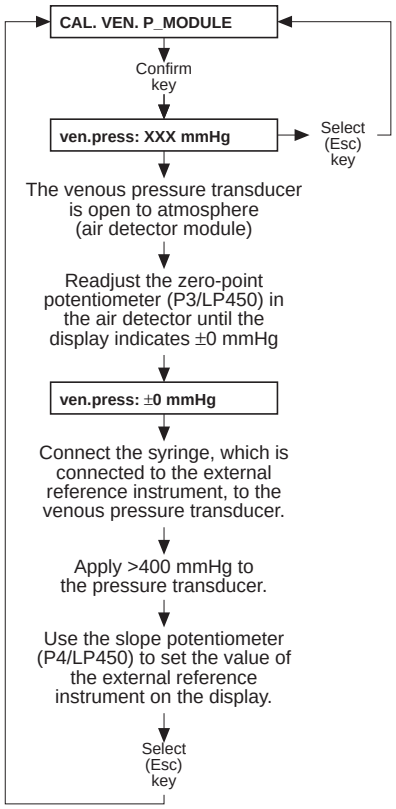


- **Part 3: Calibrate the venous pressure**



Note
Pressure gauge accuracy: ±1 % of the measured value.

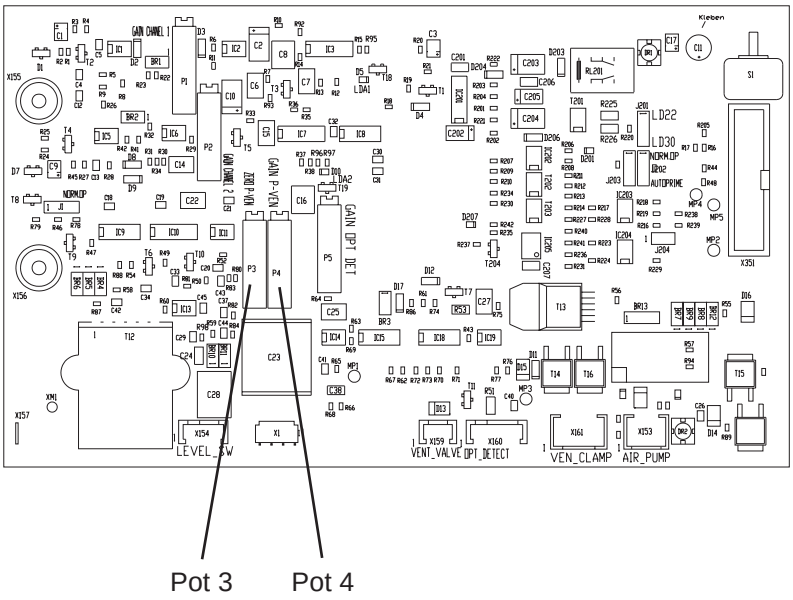
- **Part 4: Calibrate the venous pressure measurement in the air detector**



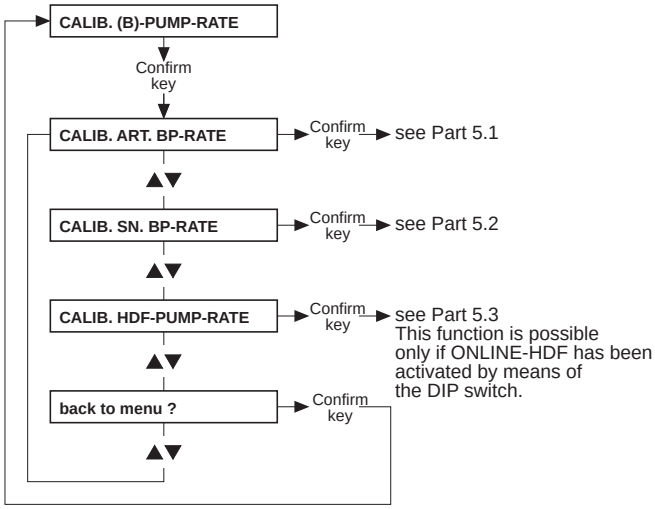
Note:
Check zero point and slope;
if necessary, repeat the procedure.

Note:
When adjusting the
air detector,
execute the
CAL. VENOUS PRESSURE
menu item.

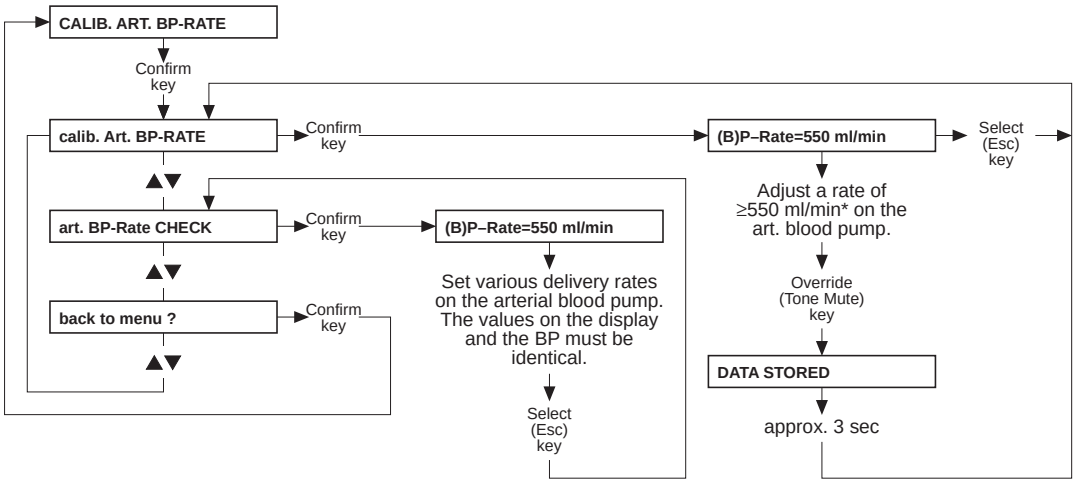
LP 450



- **Part 5: Calibrate the blood pump rates**



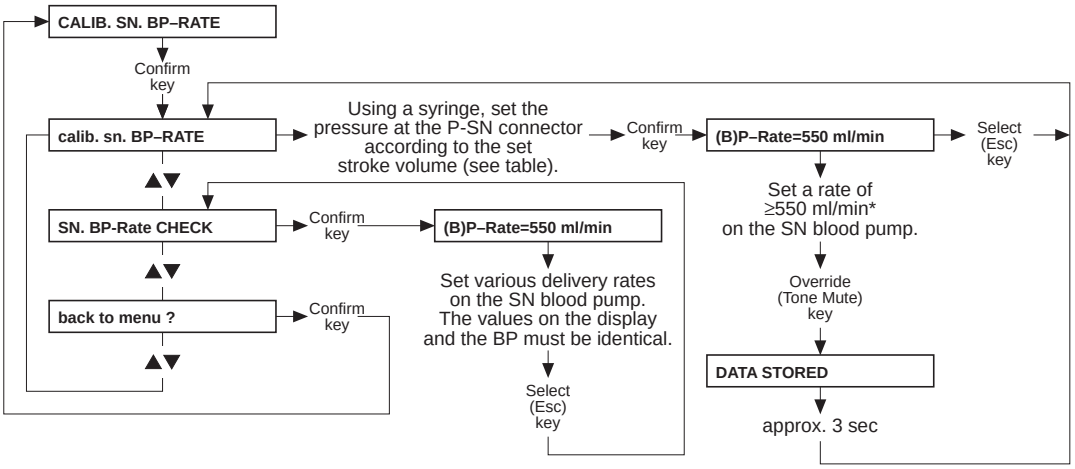
● Part 5.1: Calibrate arterial blood pump



Note: Set the line diameter to 8 mm before starting the calibration procedure and press **Start/Stop** on the blood pump.

* The BP rate of 550 ml/min represents a default value. It can be changed using the ▲▼ (+/–) keys.

● Part 5.2: Calibrate single needle blood pump rate



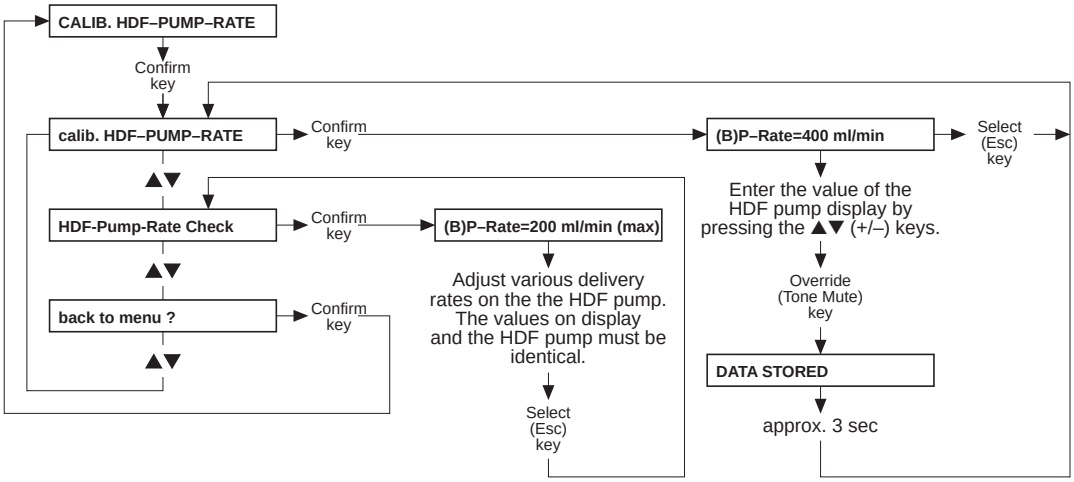
Note: Set the line diameter to 8 mm before starting the calibration procedure and press Start/Stop on the blood pump.

* The BP rate of 550 ml/min represents a default value. It can be changed using the ▲▼ (+/–) keys.

SN pump: lower switching point fixed to 75 mmHg

Stroke volume (ml)	10	15	20	25	30	35	40	45	50
Upper switching point (mmHg) ± 7 mmHg	110	130	150	172	195	219	244	270	299

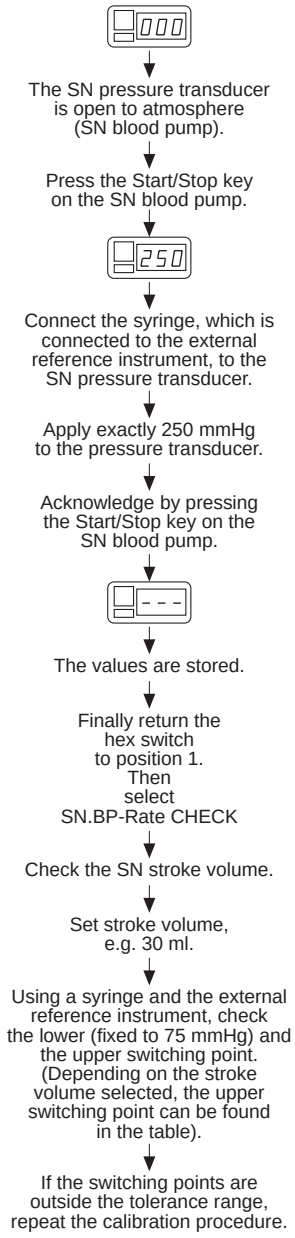
- **Part 5.3: Calibrate ONLINE-HDF pump**



Note: This function is possible only if ONLINE-HDF has been activated by means of the DIP switch.

- **Without display messages: calibrate single needle blood pump (SN pressure)**

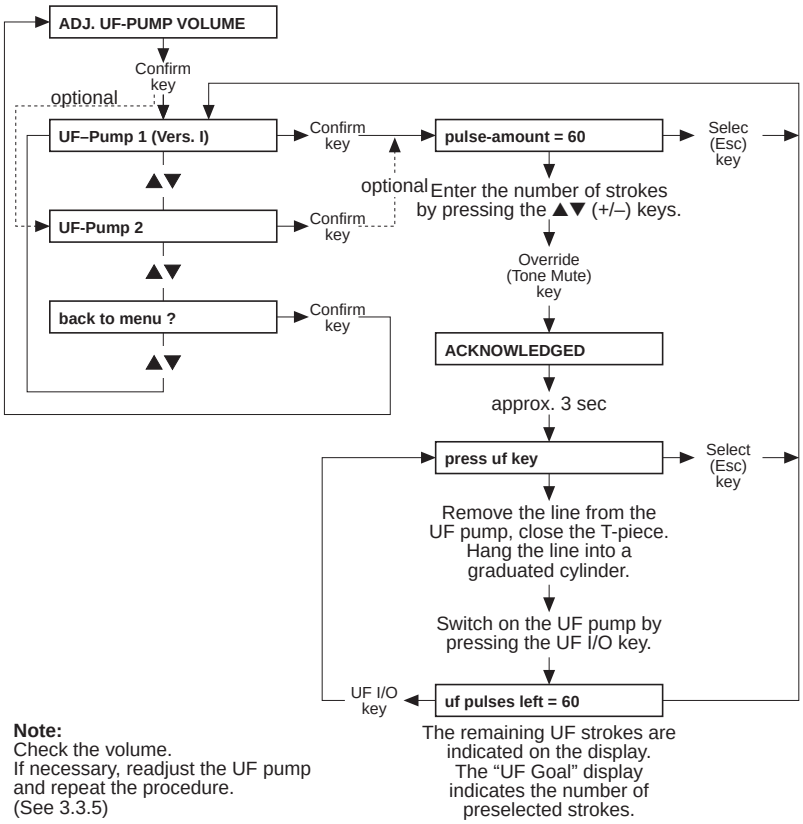
Set the hex switch in the module (P.C.B. LP 624, pos. 1) to position F.
Should the error message E02 appear on the blood pump display, clear the message by pressing the **Start/Stop** key.



SN pump: lower switching point fixed to 75 mmHg

Stroke volume (ml)	10	15	20	25	30	35	40	45	50
Upper switching point (mmHg) ± 7 mmHg	110	130	150	172	195	219	244	270	299

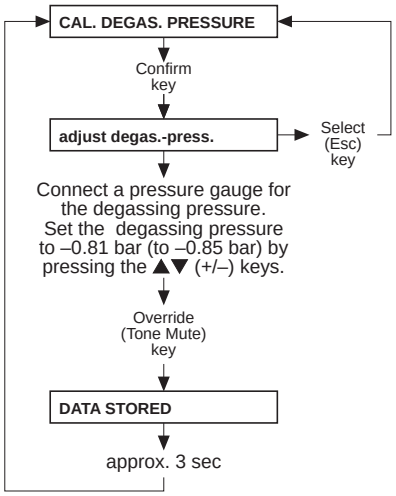
- **Part 6: Adjust UF pump volume**



Note:
Check the volume.
If necessary, readjust the UF pump and repeat the procedure.
(See 3.3.5)

Note
Graduated cylinder accuracy: ±0.5 %.

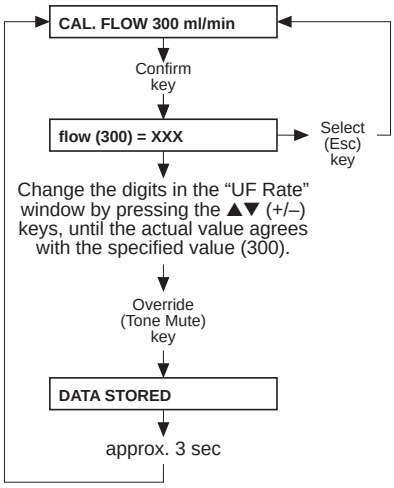
- **Part 7: calibrate degassing pressure**



At this point, the following messages may appear:
– fill program active
– set flow on

See also 3.3.2 Degassing pump pressure

- **Part 8: 300 ml/min flow**

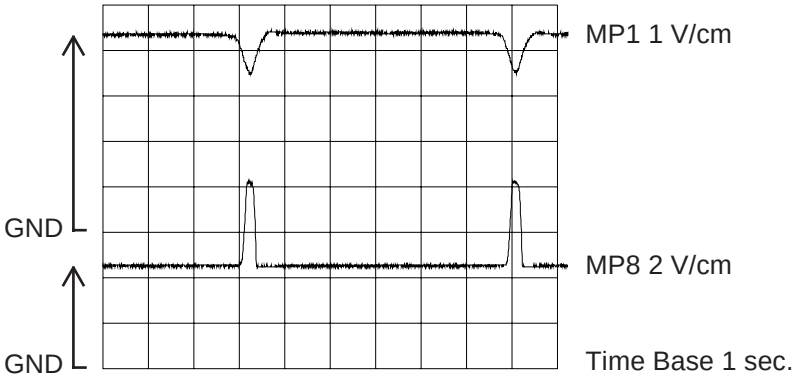


Important:
If it is impossible to adjust the 300 – 500 – 800 flow volumes, or if problems caused by flow alarms occur after the “calibrate flow” message has appeared, this can be caused by the setting of the current rise pulse.

Note:
For the flow selected first, the message “DIASAFE-filling act.” is displayed for 17 balancing chamber switchings.

Adjusting the current rise pulse:

– Select “calibrate flow 300 ml/min”; display: flow (300) = XXX.
– The actual flow XXX must be approx. 300; if necessary, correct it using the ▲ and ▼ keys.
– Connect an oscilloscope to MP8 and MP1, and the ground MP7 to P.C.B. LP 634.
– Use P1 to set the current rise pulse as shown in the diagram below. Make sure that the actual flow (display XXX) remains at approx. 300; if necessary, correct it using the ▲ and ▼ keys.



Alternative adjustment of the current rise (if an oscilloscope is not available):

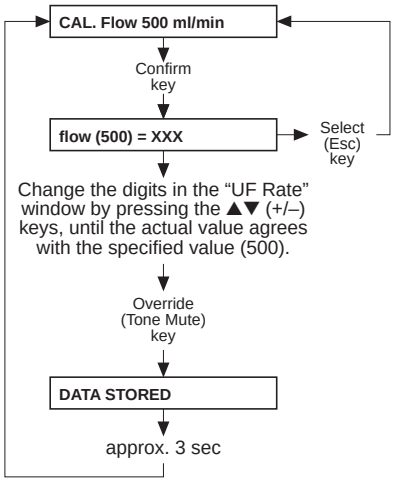
– Select “calibrate flow 300 ml/min”.
– There are two possibilities of reaction by the machine:

1. The machine runs with regular balancing chamber switching.
   Display: flow (300) = XXX.
   – If necessary, correct the flow using the ▲ and ▲ keys, until the actual flow indicates approx. 300.
   – Turn the potentiometer P1 counter-clockwise (wait for at least 10 sec after each rotation!), until the machine switches to “Eigentakt”.
   – Display: flow (300) = 147.
   – Now turn the potentiometer P1 clockwise (wait for at least 10 sec after each half-rotation!), until the actual flow again indicates approx. 300.
   – Turn the potentiometer P1 clockwise for another 2 rotations.

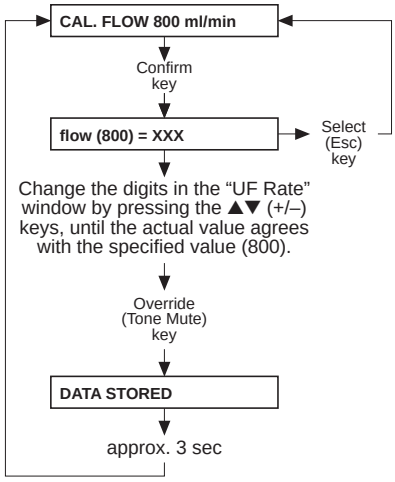
2. The machine is in the “Eigentakt” mode.
   Display: flow (300) = 147.
   – Turn the potentiometer P1 clockwise, until the machine switches from “Eigentakt” to regular balancing chamber switching (wait for approx. 10 sec after each rotation!).
   – Display: flow (300) = XXX.
   – If necessary, correct the flow using the ▲ and ▲ keys, until the actual flow indicates approx. 300.
   – Turn the potentiometer P1 counter-clockwise (wait for at least 10 sec after each rotation), until the machine switches to “Eigentakt”.
   – Display: flow (300) = 147.
   – Now turn the potentiometer P1 clockwise (wait for at least 10 sec after each half-rotation!), until the actual flow again indicates approx. 300.
   – Turn the potentiometer P1 clockwise for another 2 rotations.

After adjusting the current rise pulse, check and, if necessary, readjust the 300/500/800 flow settings.

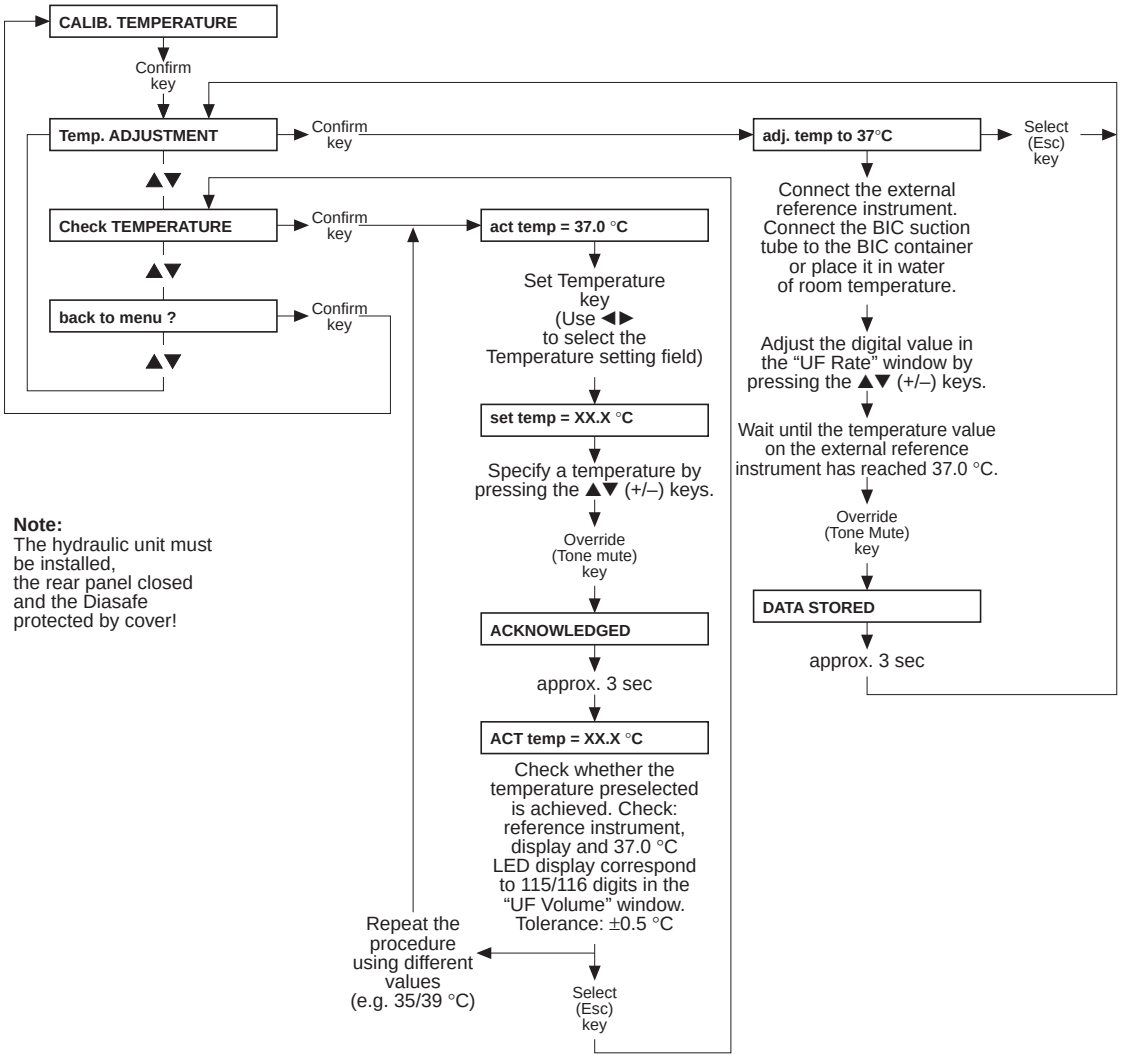
- **Part 9: Calibrate 500 ml/min flow**



- **Part 10: 800 ml/min flow**

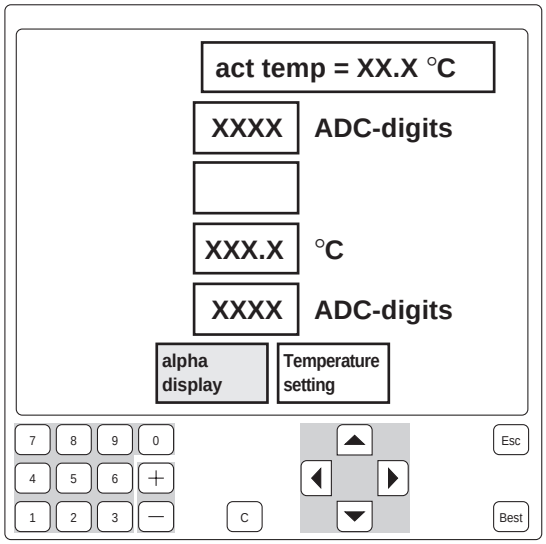


- **Part 11: Calibrate dialysate temperature**

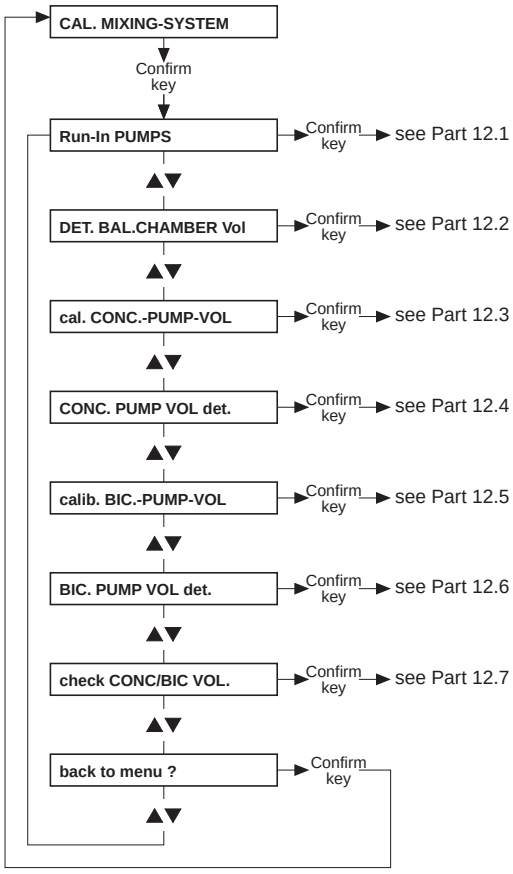


Note
Measuring instrument accuracy: ±0.2 °C.

4008 H/S, display for "Check TEMPERATURE":

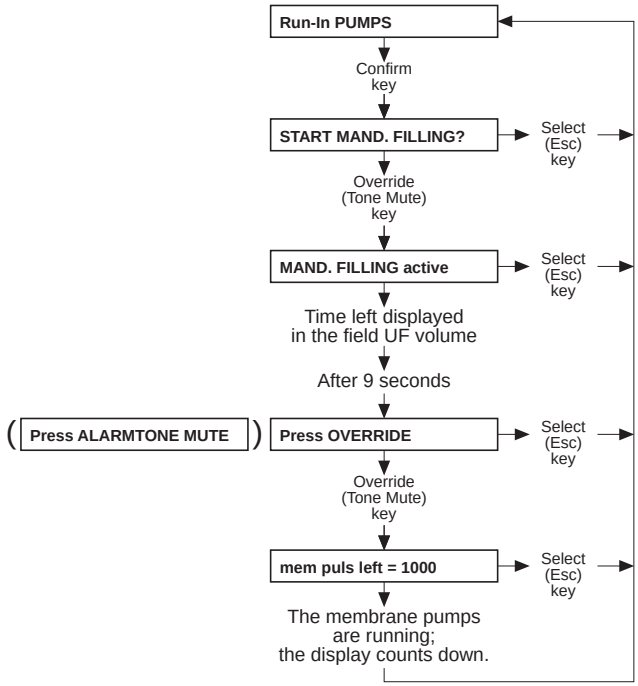


- **Part 12: Calibrate mixing system**

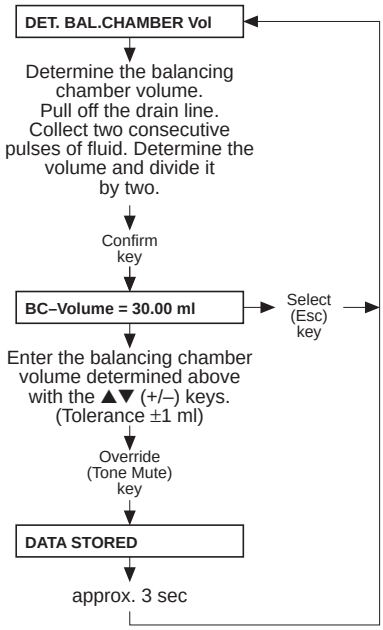


- **Part 12.1: Run-in of membrane pumps**

The concentrate suction tubes are in a container filled with water

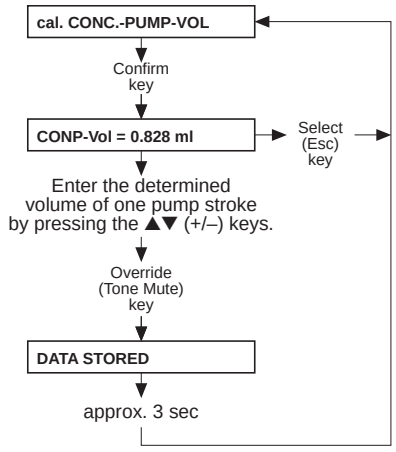


- **Part 12.2: Determine the balancing chamber volume**



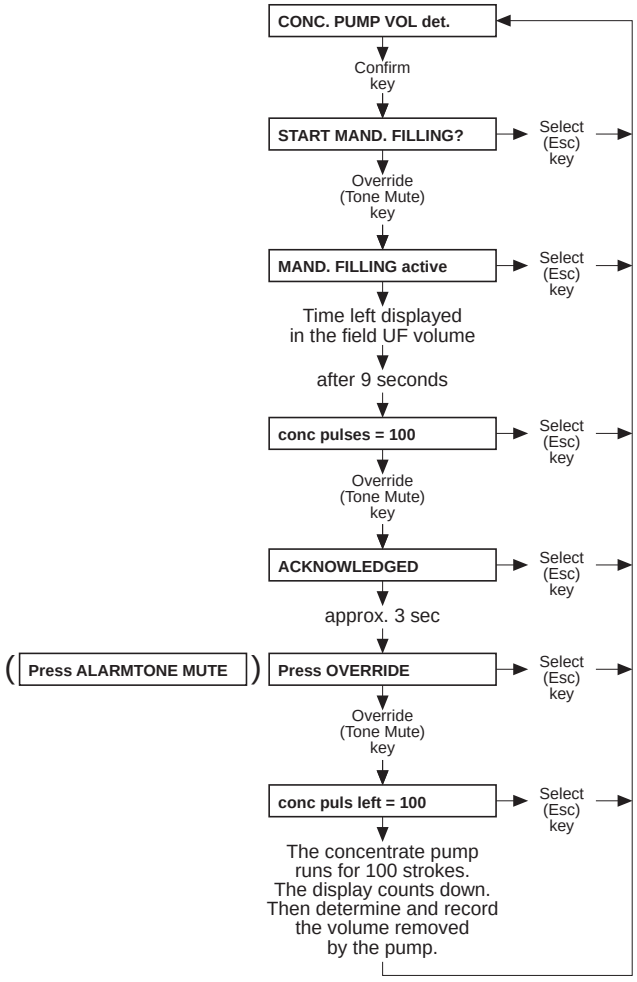
Note
Accuracy of the measuring cylinder: ±0.5 %.

- **Part 12.3: Calibrate the concentrate pump stroke**



Note
Accuracy of the measuring cylinder: ±0.5 %.

- **Part 12.4: Determine the concentrate pump volume**



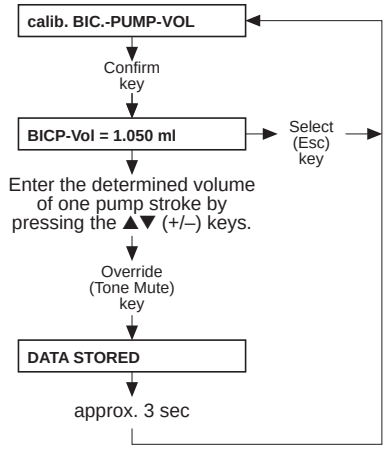
Notes:

100 strokes are factory-set. This setting can be changed by pressing the ▲▼ (+/–) keys (depending on the graduated cylinder used). However, when returning to “CAL. MIXING-SYSTEM”, the display will indicate the factory setting again.

Check the volume and, if necessary, repeat the procedure.

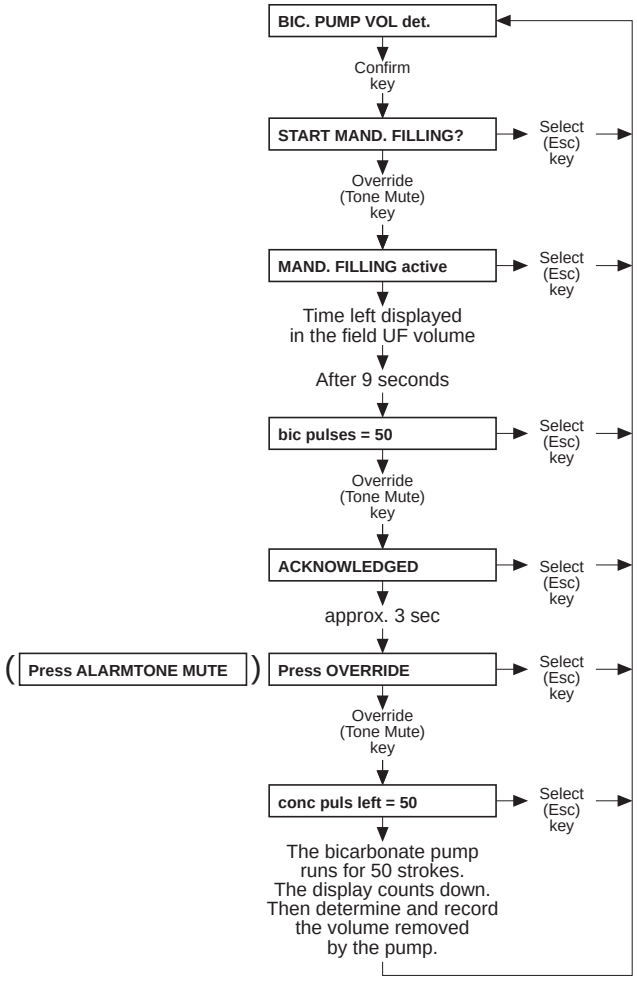
Note
Accuracy of the measuring cylinder: ±0.5 %.

- **Part 12.5: Calibrate bicarbonate pump stroke**



Note
Accuracy of the measuring cylinder: ±0.5 %.

- **Part 12.6: Determine the bicarbonate pump volume**



Notes:

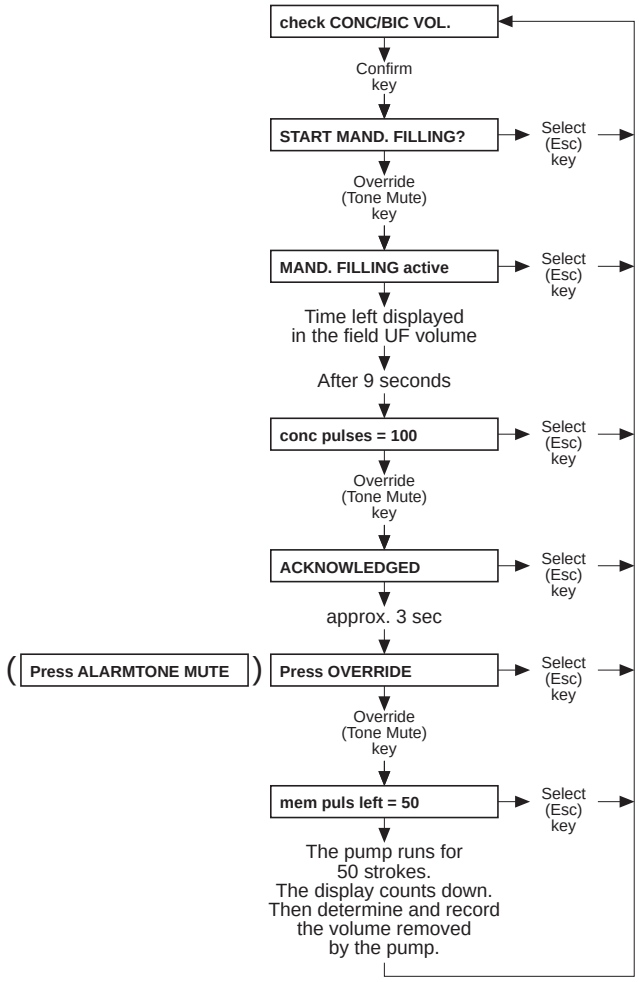
50 strokes are factory-set. This setting can be changed by pressing the ▲▼ (+/–) keys (depending on the graduated cylinder used). However, when returning to “CAL. MIXING-SYSTEM”, the display will indicate the factory setting again.

Check the volume and, if necessary, repeat the procedure.

Note
Accuracy of the measuring cylinder: ±0.5 %.

- **Part 12.7: Check the concentrate and/or bicarbonate volume**

The bicarbonate suction tube is in a container filled with water.



Check mixing ratio

Notes:

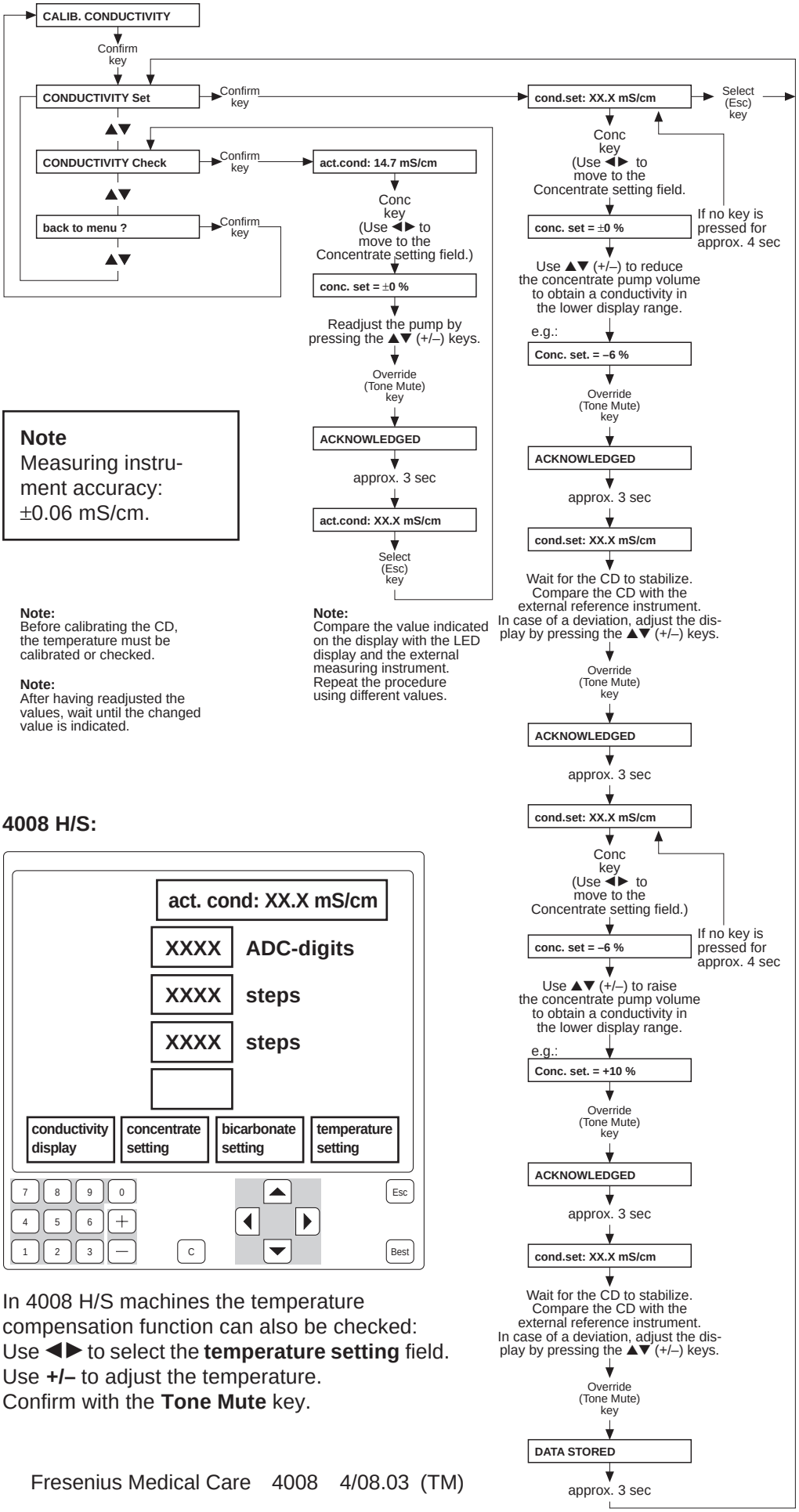
This test step permits verification of the concentrate or bicarbonate pump volumes in accordance with the parameters entered for the mixing system (mixing ratio, BC volume, conc. and bic. pump volume).

The pump whose concentrate suction tube is pulled off is activated.

50 strokes are factory-set. This setting can be changed by pressing the ▲▼ (+/–) keys (depending on the graduated cylinder used). However, when returning to “CAL. MIXING-SYSTEM”, the display will indicate the factory setting again.

Note
Accuracy of the measuring cylinder: ±0.5 %.

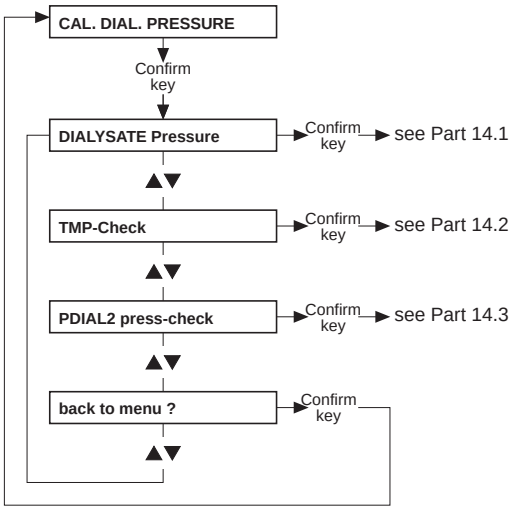
- **Part 13: Calibrate conductivity** (the values stated are examples only)



4008 H/S:

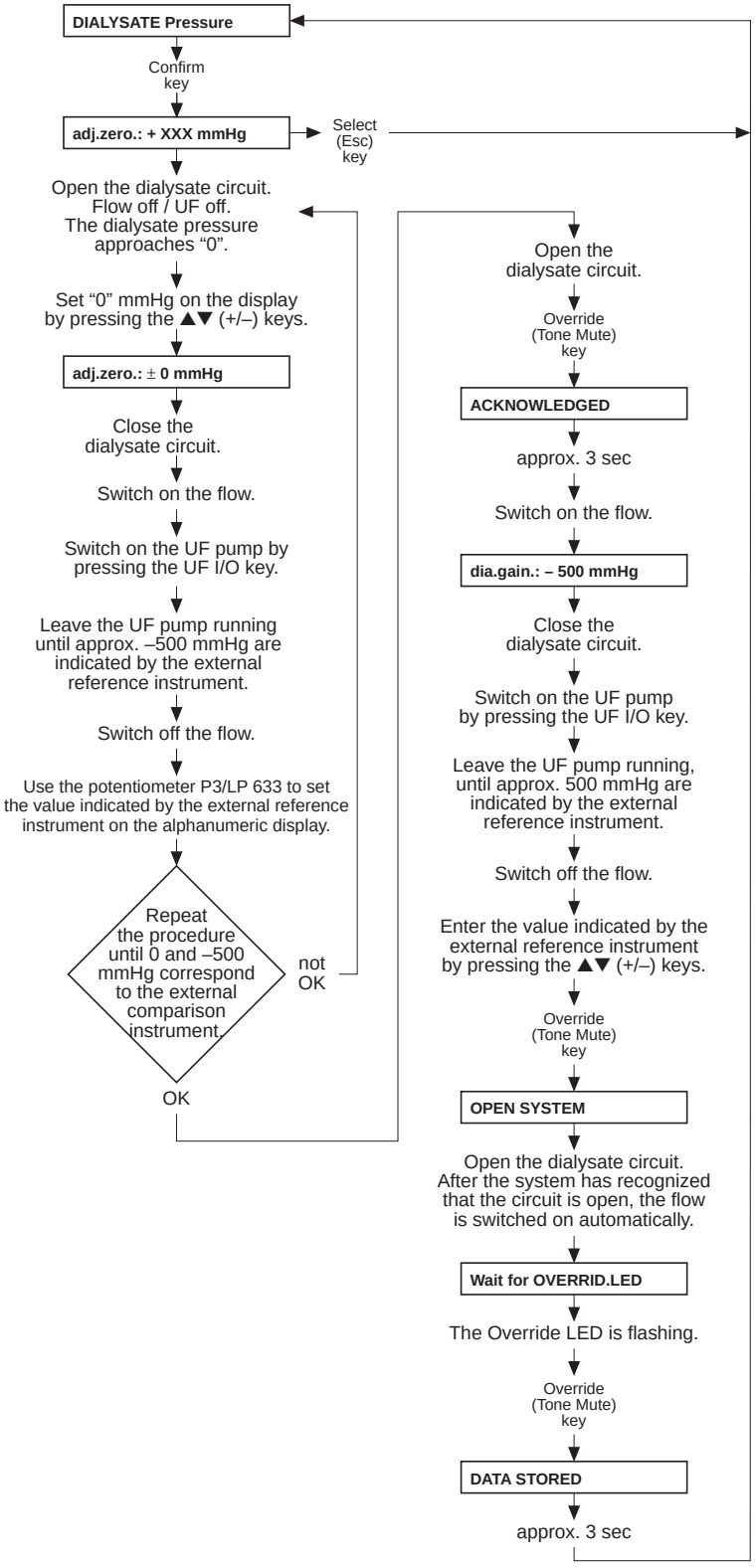
In 4008 H/S machines the temperature compensation function can also be checked:
Use ◀▶ to select the **temperature setting** field.
Use **+/–** to adjust the temperature.
Confirm with the **Tone Mute** key.

- **Part 14: Calibrate the dialysate pressure (stainless steel pressure transducer)**

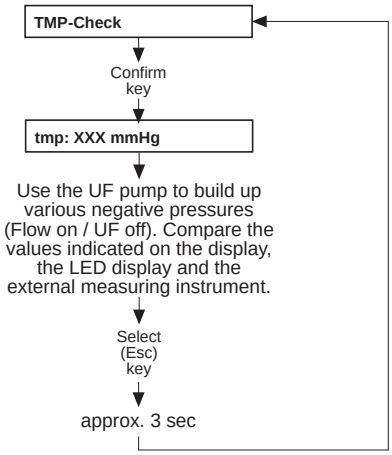


Note
Measuring instrument accuracy: ±1 % of the measured value.

- **Part 14.1: Dialysate pressure**



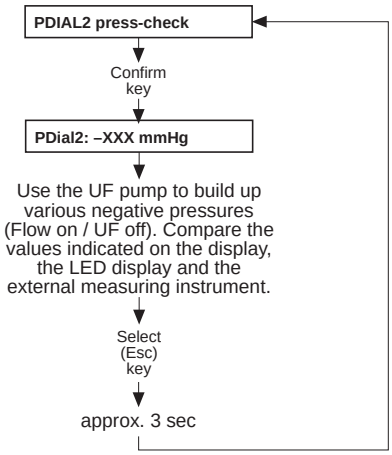
- **Part 14.2: TMP check**



Note:
Observe the venous pressure!

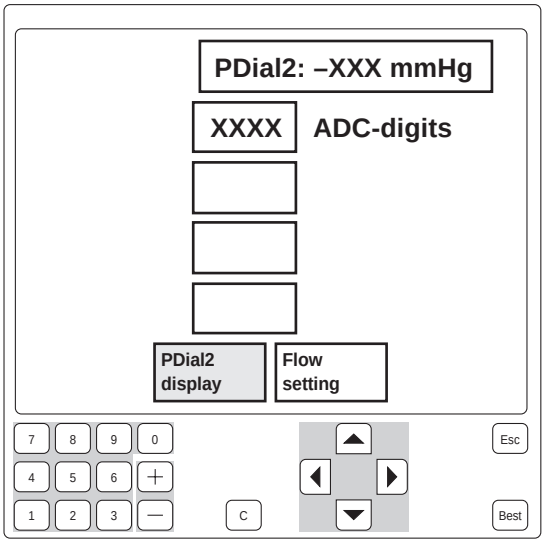
Note:
Switch the flow on and off again once in a while, to maintain the operating temperature of the pressure transducer.

- **Part 14.3: PDIAL2 pressure check**



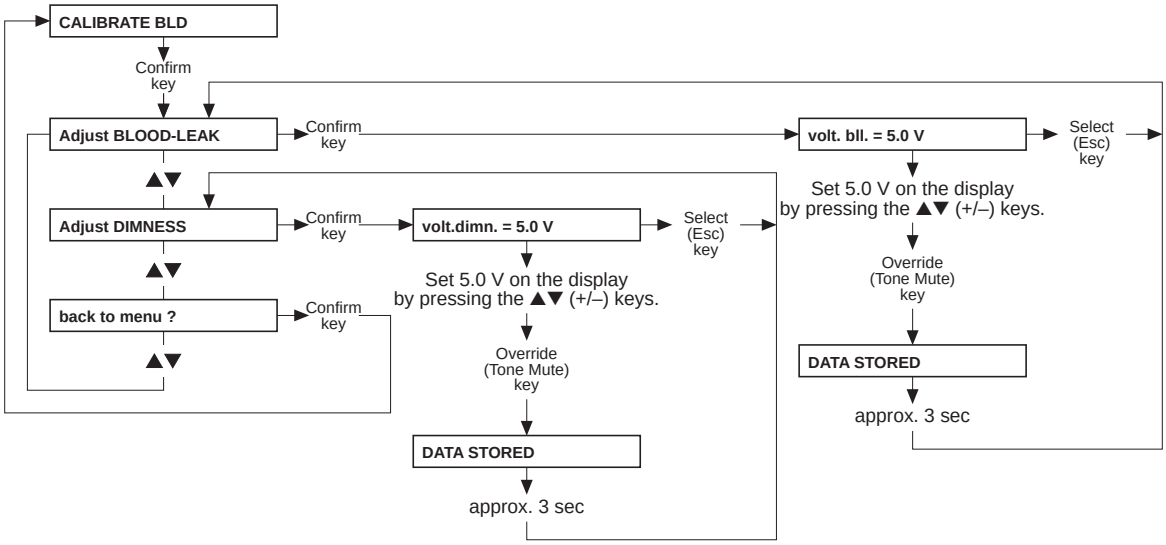
Note:
Observe the venous pressure!

4008 H/S:



In 4008 H/S machines the temperature compensation function can also be checked:
Press ◄► to select the field **Flow setting**.
Adjust the flow by pressing **+/–**.
Confirm by pressing the **Tone Mute** key.

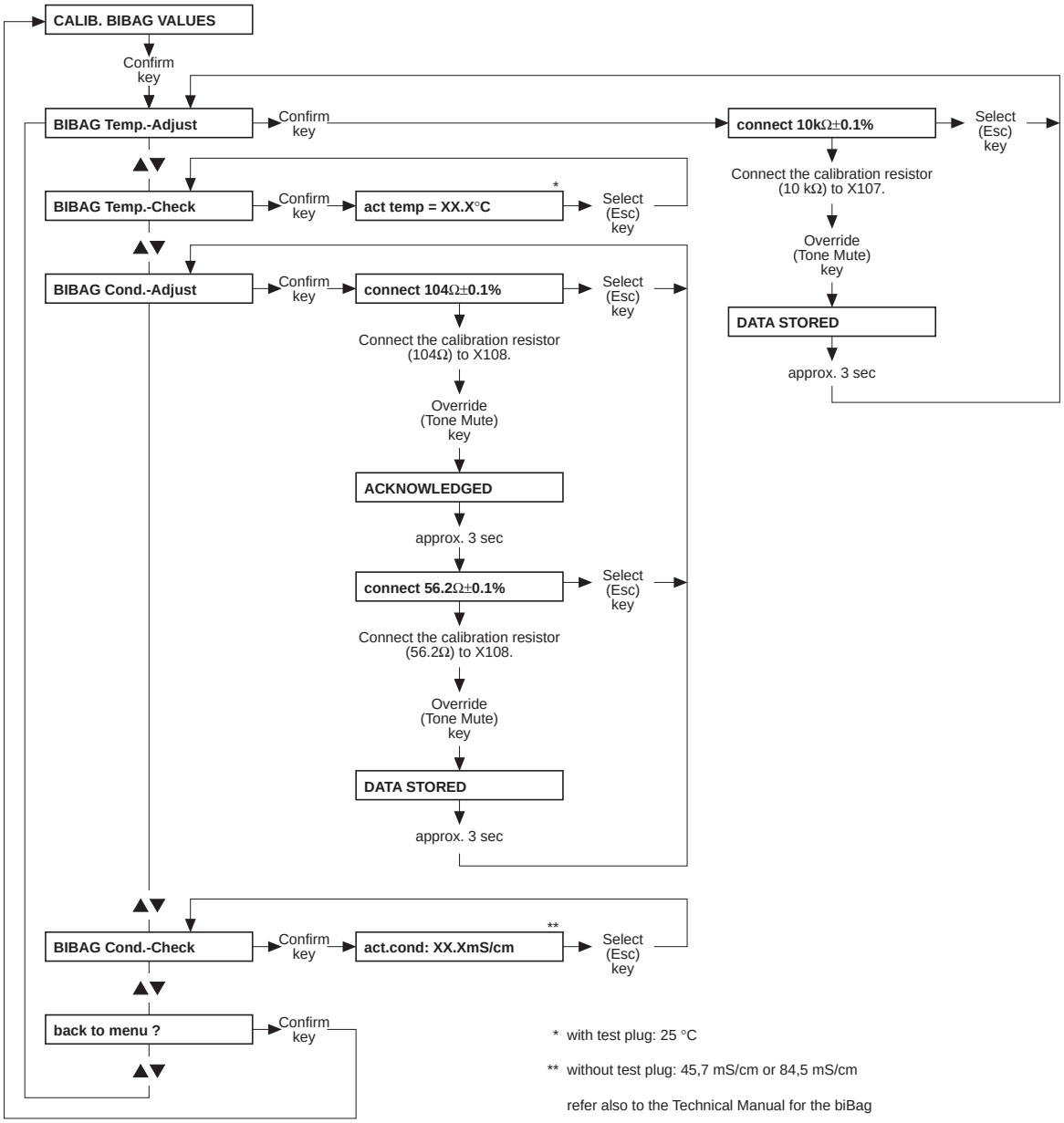
- **Part 15: Blood leak voltage**



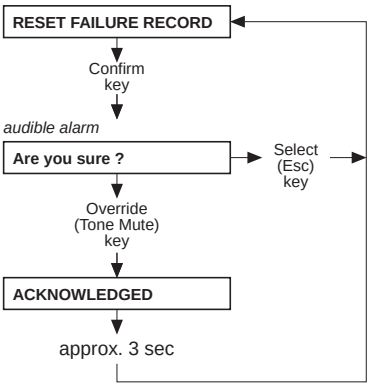
Tolerance for dimness voltage: 4.9 – 5.1 V.

Note: If values deviate check the glass burette for contamination.
Close the housing; temperature 37 °C; avoid incident light from an external source.

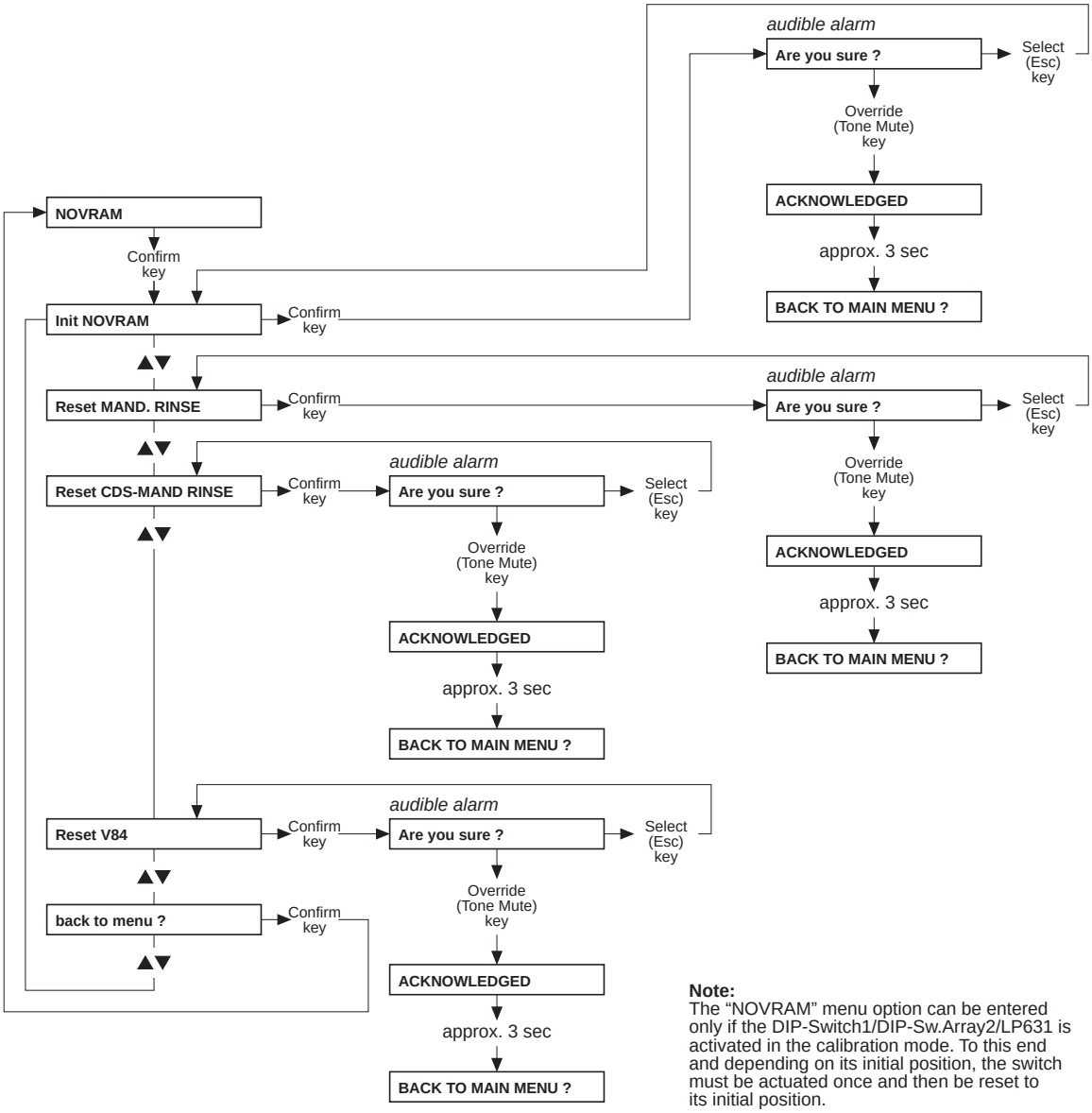
- **Part 16: Calibrate BIBAG values (optional)**



- **Part 17: Reset the failure record**



- **Part 18: Initialize NOVRAM, reset mandatory rinse, reset V84 malfunction**



Note:
The “NOVRAM” menu option can be entered only if the DIP-Switch1/DIP-Sw.Array2/LP631 is activated in the calibration mode. To this end and depending on its initial position, the switch must be actuated once and then be reset to its initial position.

- **Without menu display: adjustment of the Hall sensor inside the heparin pump**

  – Remove plug connector from Hall sensor 2.
  – Move the slide carriage down over Hall sensor 1.
  – Move the slide carriage up to its fully open position.
  – Move the slide carriage down again to the end of its travel.
  – The free motion between the slide carriage and the housing should be approx. 0.5 mm. If necessary, change the position of Hall sensor 1 and repeat the procedure.

  Adjustment of Hall sensor 2
  – Reconnect plug connector for Hall sensor 2.
  – Move the slide carriage down to approx. 2 cm before the end of its travel.
  – Manually turn the threaded spindle approx. 2-3 rotations in delivery direction.
  – Move the slide carriage down.
  – The slide carriage must stop before the mechanical end of its travel. If necessary, change the position of Hall sensor 2 and repeat the procedure several times.

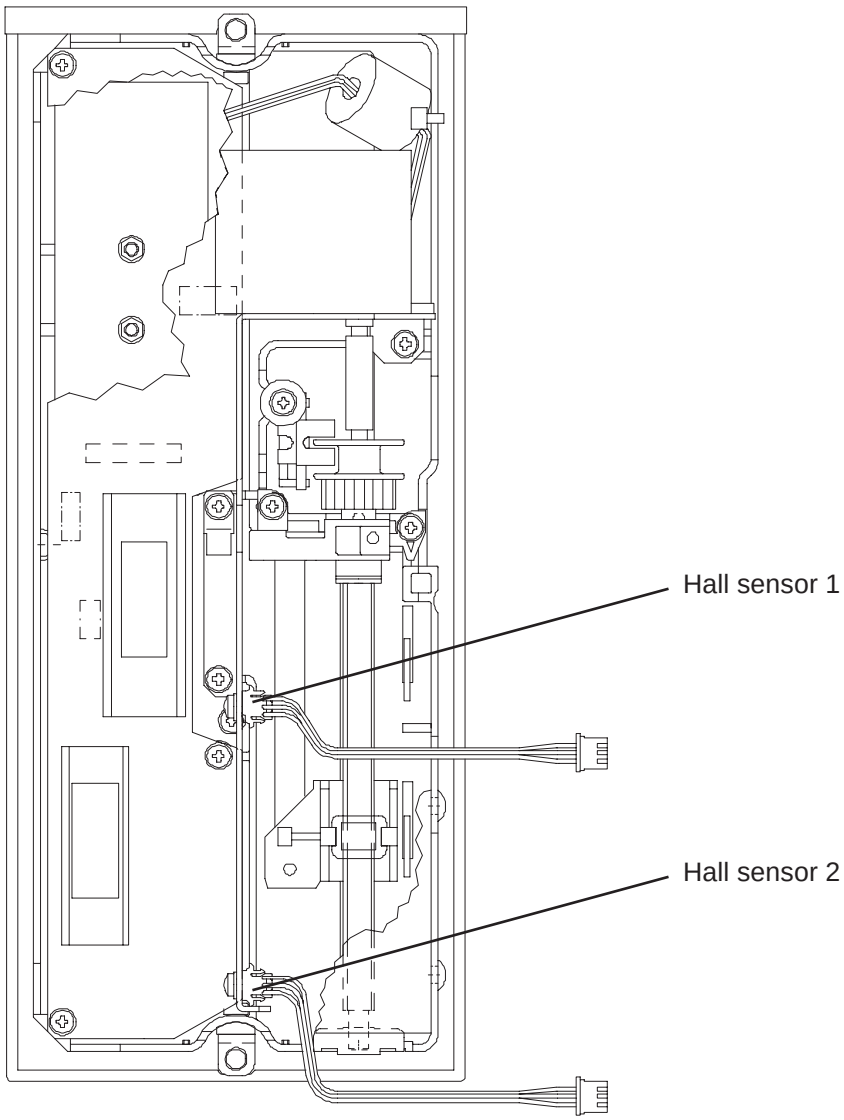


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5 Diagnostics program

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5.12	**ONLINE*plus*™ module**	5-31

5.1 General notes

The diagnostics program serves to activate all inputs and outputs of the hemodialysis machine. Activation is related to CPU1 (P.C.B. LP 631), CPU2 (P.C.B. LP 632), as well as to the output board (P.C.B. LP 634) and the input board (P.C.B. LP 633).

Using this program, the technician is able to program his own settings for testing of error images.

The diagnostics program is divided into the following menus:

- READ INPUTS
  - READ ANALOG INPUTS
    - CPU1: RD ANALOG INP.
    - CPU2: RD ANALOG INP.
  - READ DIGITAL INPUTS
    - CPU1: RD DIGITAL INP
    - CPU2: RD DIGITAL INP
- WRITE OUTPUTS
  - WRITE ANALOG OUTPUTS
    - CPU1: WR ANALOG OUTP
    - CPU2: WR ANALOG OUTP
  - WRITE DIGIT. OUTPUTS
    - CPU1: WR DIGIT. OUTP
    - CPU2: WR DIGIT. OUTP
- INP/OUTP COMBINATION
    - CPU1: COMBINATION
- CAN–COMPONENTS
    - WTR
    - ONLINE–PLUS–MODUL

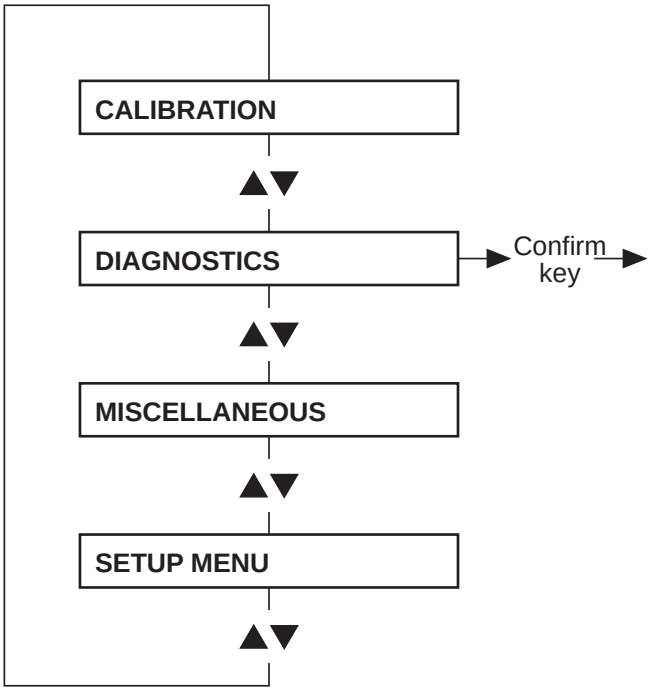
In order to indicate the corresponding levels, *all* UF-seven-segment displays as well as the status indicator (monitor), the external traffic light and the loudspeaker are used in the “READ DIGITAL INPUTS” menu.

The *active* signal state (which may correspond to both present and absent voltage) is indicated by 1111 on the UF displays, activated traffic light (status indicator) and audible signal.
An audible signal can be deactivated by pressing the **(Alarm) Tone Mute** key. With the audible signal deactivated, the **(Alarm) Tone Mute** LED is flashing as a reminder.

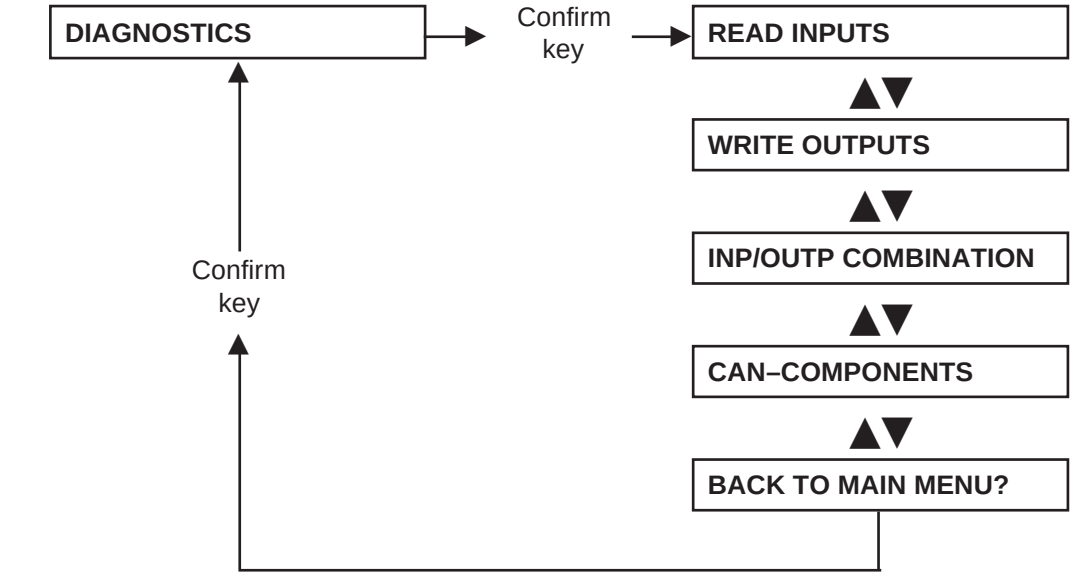
The audible signal indication can be used to evaluate the state of the signal without having to look at the monitor. This may be advantageous in case measurements have to be made behind the machine (e.g. hydraulic unit).

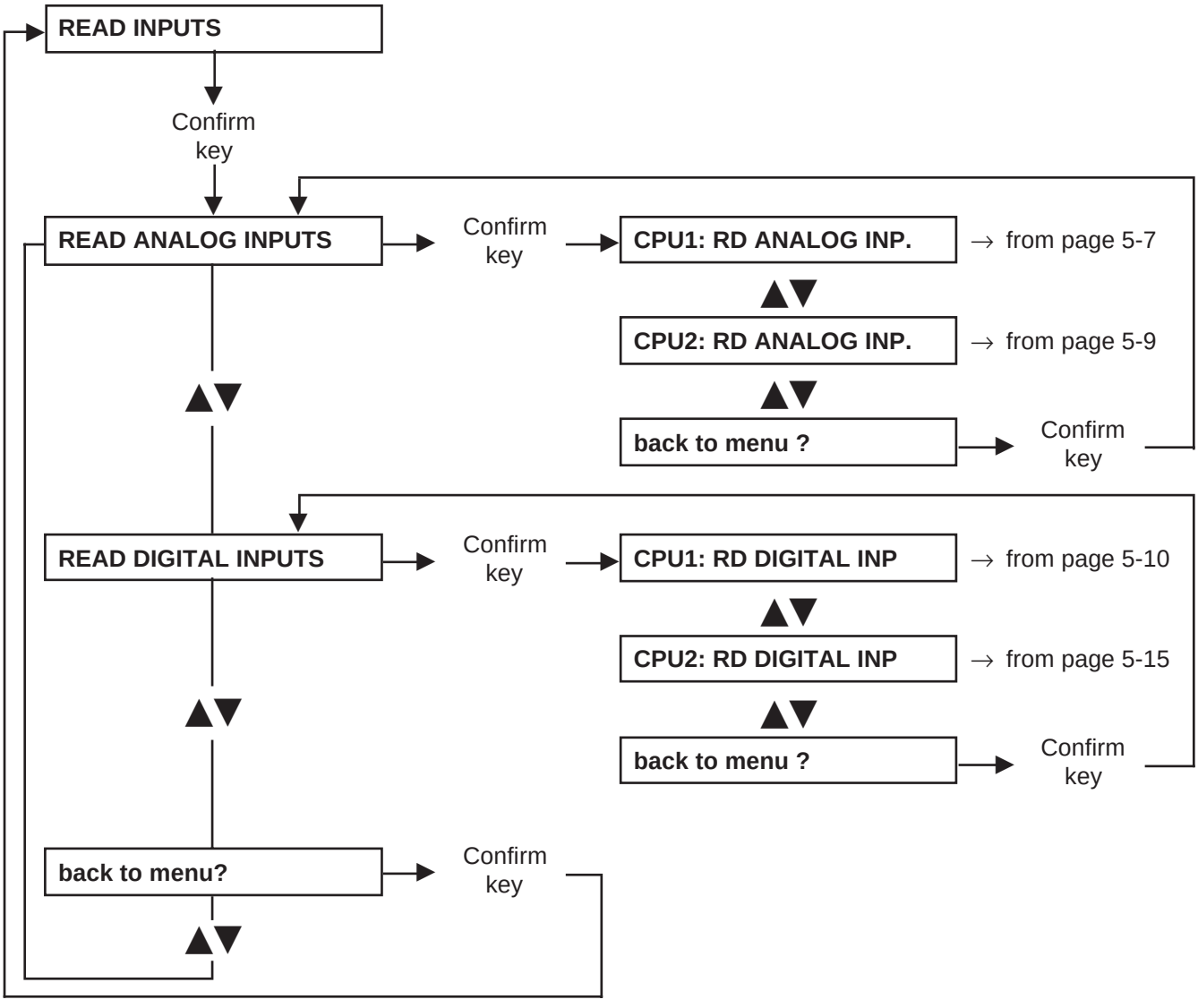
Note
In the diagnostics program, the signals are listed in the order of their electric connection, i.e. in latch groups of 8 signals each, according to the 8-bit data bus and according to the latch numbering on the circuit diagram (e.g. P.C.B. LP 633: CS_LATCH0 – CS_LATCH6).
The are *not* divided into groups of pertinency (e.g. all Bibag signals one after the other). The only exception here is the activation of the solenoid valves. These are listed in the menu in the order of their numbers. This facilitates finding each individual valve since, as a rule, several valves must be simultaneously activated for trouble shooting.
Since the signals are assigned to their respective connections (latch groups), it is possible at any time, by using the circuit diagram, to locate the respective signal in the menu, even if the signal name should have changed. Within one latch group, only *one* known signal suffices to find the renamed signal by counting through the menu.
Deviations of all voltage values indicated are possible due to tolerances and depending on the various systems.

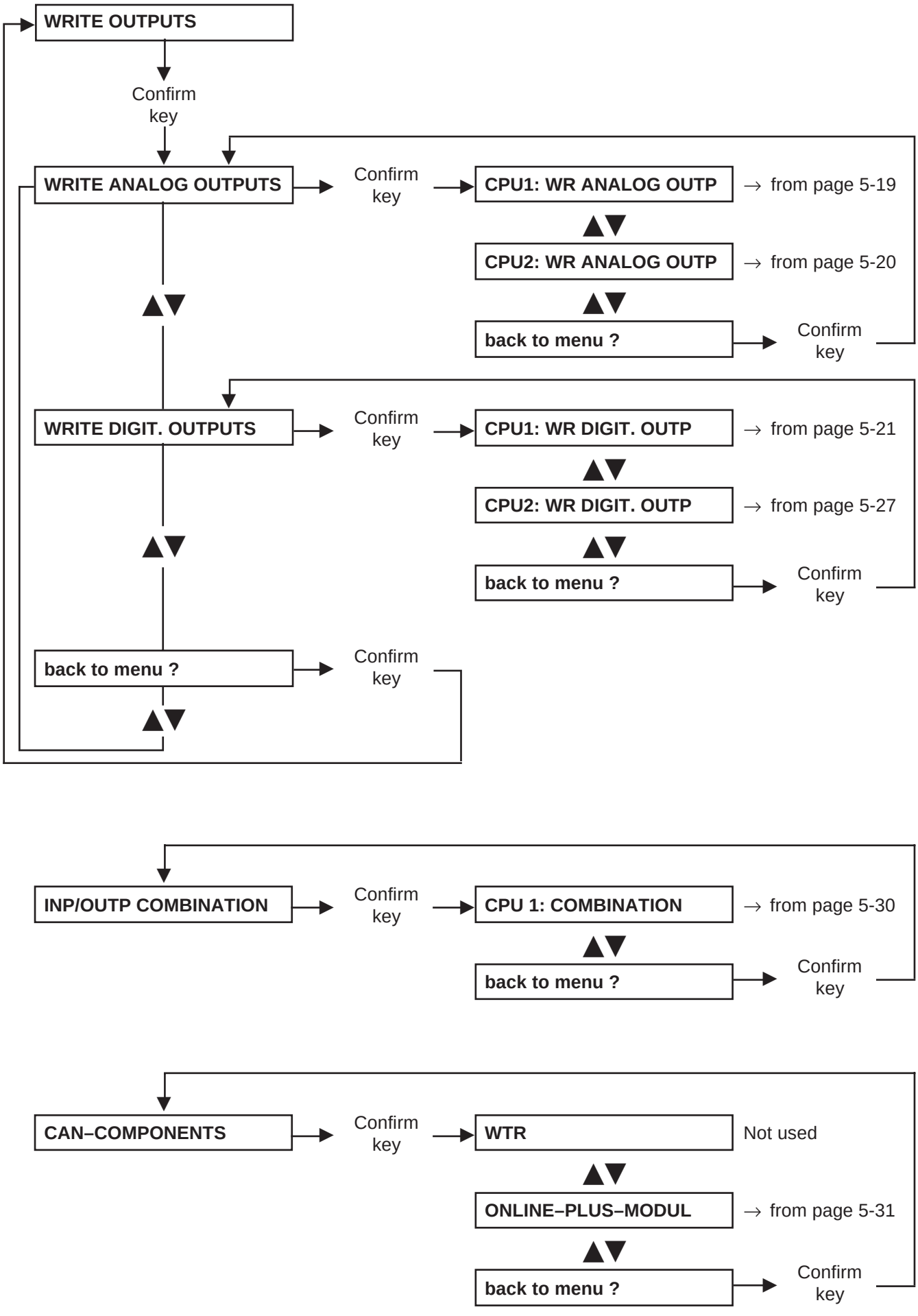
The “CPU1: RD DIGITAL INP” menu item includes the “I:CPU1_KEY_TESTING” item, which serves to perform the key test.
The key actuated is indicated on the alphanumeric display.
The UP, DOWN, CONFIRM, SELECT and I/O keys have not been implemented, since their function can be tested by selecting the corresponding menu.



5.2 Menu structure






WRITE OUTPUTS
Confirm key
WRITE ANALOG OUTPUTS
Confirm key
CPU1: WR ANALOG OUTP
→ from page 5-19
CPU2: WR ANALOG OUTP
→ from page 5-20
back to menu ?
Confirm key
WRITE DIGIT. OUTPUTS
Confirm key
CPU1: WR DIGIT. OUTP
→ from page 5-21
CPU2: WR DIGIT. OUTP
→ from page 5-27
back to menu ?
Confirm key
back to menu ?
Confirm key
INP/OUTP COMBINATION
Confirm key
CPU 1: COMBINATION
→ from page 5-30
back to menu ?
Confirm key
CAN–COMPONENTS
Confirm key
WTR
Not used
ONLINE–PLUS–MODUL
→ from page 5-31
back to menu ?
Confirm key

5.3 Reading the analog inputs of CPU I

Explanation:
UF Volume display: ADC value
Time Left display: Analog voltage (in 0.1 V), converted to the value at the input of P.C.B. LP 633

CPU1: RD ANALOG INP.

↓

Confirm key

↓

I: CPU1_PWR_WD

Voltage for watchdog monitoring
5 V, IC 27/26, ACD 107
(4.5 to 5.5 V)

▲▼

I: CPU1_P_CONC

Not used
0 V, IC 27/27, ACD 0

▲▼

I: CPU1_P_VEN

Venous pressure
0 to 12 V, IC 27/28, ADC 0 to 255

▲▼

I: CPU1_BPR_VEN

Venous blood pump rate
0 to 10 V, IC 27/1, ADC 0 to 215
(8 mm line diameter)

▲▼

I: CPU1_PR_HDF

HDF blood pump rate
0 to 3.3 V, IC 27/1, ADC 0 to 72
(HDF switched on,
8 mm line diameter)

▲▼

I: CPU1_BPR_ART

Arterial blood pump rate
0 to 10 V, IC 27/2, ADC 0 to 215
(8 mm line diameter)

▲▼

I: CPU1_VEN_BPR_SET

Rated value specified for the venous blood pump rate
0 to 8 V, IC 27/3, ADC 0 to 171
(8 mm line diameter)

▲▼

I: CPU1_HDF_PR_SET

Rated value specified for the HDF blood pump rate
0 to 2.7 V, IC 27/3, ADC 0 to 58
(HDF switched on,
8 mm line diameter)

▲▼

I: CPU1_ART_BPR_SET

Rated value specified for the arterial blood pump rate
0 to 8 V, IC 27/4, ADC 0 to 171
(8 mm line diameter)

▲▼

I: CPU1_REF1

Reference voltage AD 0
2.5 V, IC 27/5, ADC 128
(2.3 to 2.5 V)

▲▼

I: CPU1_U_ACCU

Battery voltage
e.g. 22 V, IC 28/26, ADC e.g. 184

▲▼

I: CPU1_P_BIC

Not used
0 V, IC 28/27, ADC 0

▲▼

I: CPU1_24V_EM

24V_EMERGENCY
24 V, IC 28/28, ADC 117
(22.5 to 26 V)

▲▼

I: CPU1_P_ART

Arterial pressure
0 to 10.5 V, IC 28/1, ADC 0 to 225

▲▼

I: CPU1_BLL_DIM

Blood leak dimness voltage
5.0 V, IC 28/2, ADC 108

▲▼

I: CPU1_BLL

Blood leak voltage
5.0 V, IC 28/3, ADC 108

▲▼

I: CPU1_COND_SIGNAL1

CD display
0 to 10.8 V, IC 28/4, ADC 0 to 231

▲▼

I: CPU1_REF2

Reference voltage AD1
2.5 V, IC 28/5, ADC 128
(2.3 V to 2.5 V)

▲▼

I: CPU1_FREE1

Not used
0 V, IC 29/26, ADC 0

▲▼

I: CPU1_TEMP_DIAL2

Temperature NTC 109
0 to 12 V, IC 29/27, ADC 0 to 255

▲▼

I: CPU1_COND_SINGAL2

Not used
0 V, IC 29/28, ADC 0

▲▼

I: CPU1_TEMP_DIAL3

Temperature NTC
(slot X107 / LP 747)
0 – 12 V, IC 29/1, ADC 0 – 255

▲▼

I: CPU1_COND_SIGNAL3

CD cell (slot X108 / LP 747)
0 – 10,8 V, IC 29/2, ADC 0 – 231

▲▼

I: CPU1_FREE2

Not used, open input
IC 29/3

▲▼

I: CPU1_U_BATT_SW

Voltage for alarm tone, if battery relay is off
10.6 V, IC 29/4, ADC 110

▲▼

I: CPU1_REF3

Reference voltage AD2
2.5 V, IC 29/5, ADC 128
(2.3 to 2.5 V)

▲▼

back to menu ?

↓

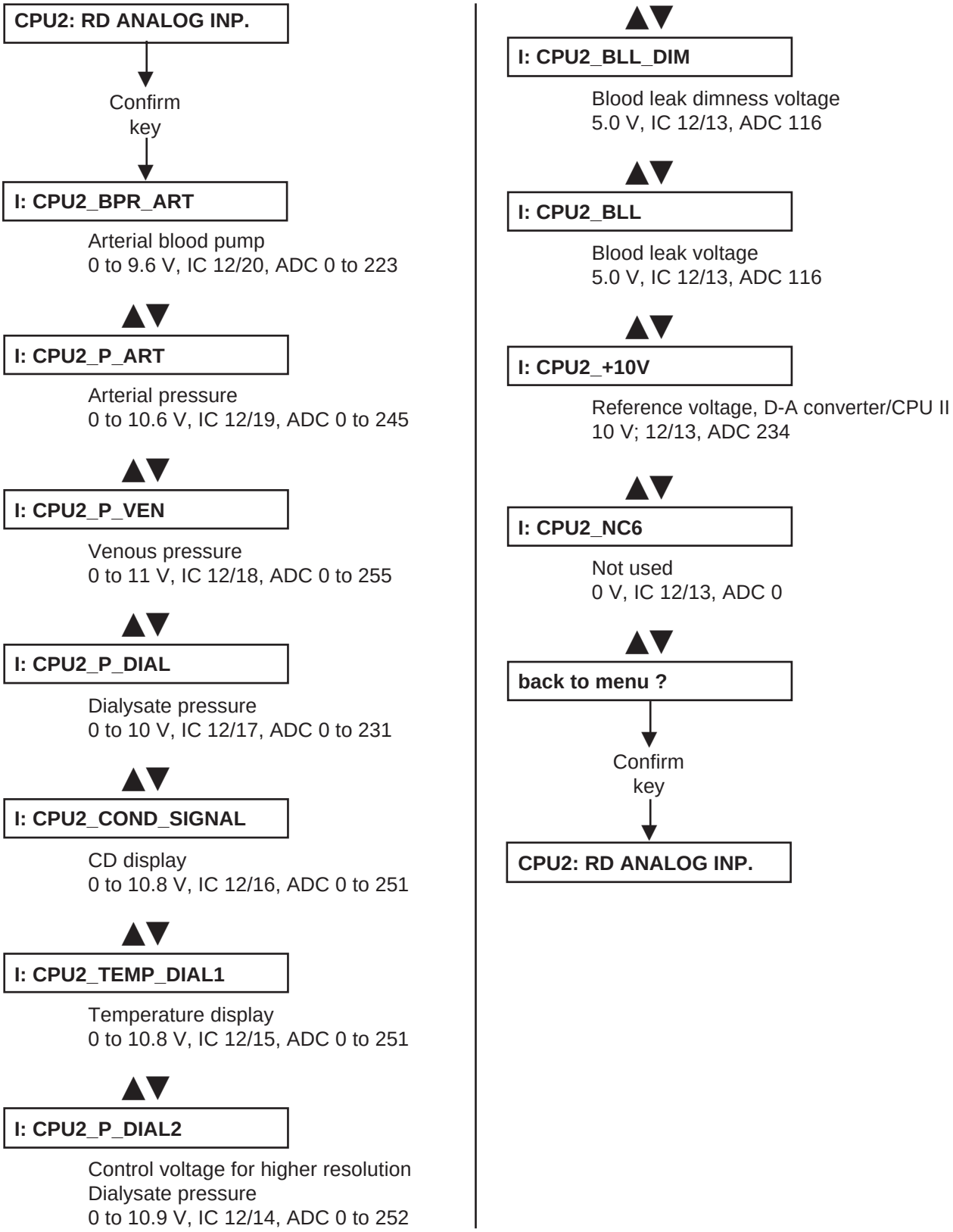
Confirm key

↓

CPU1: RD ANALOG INP.

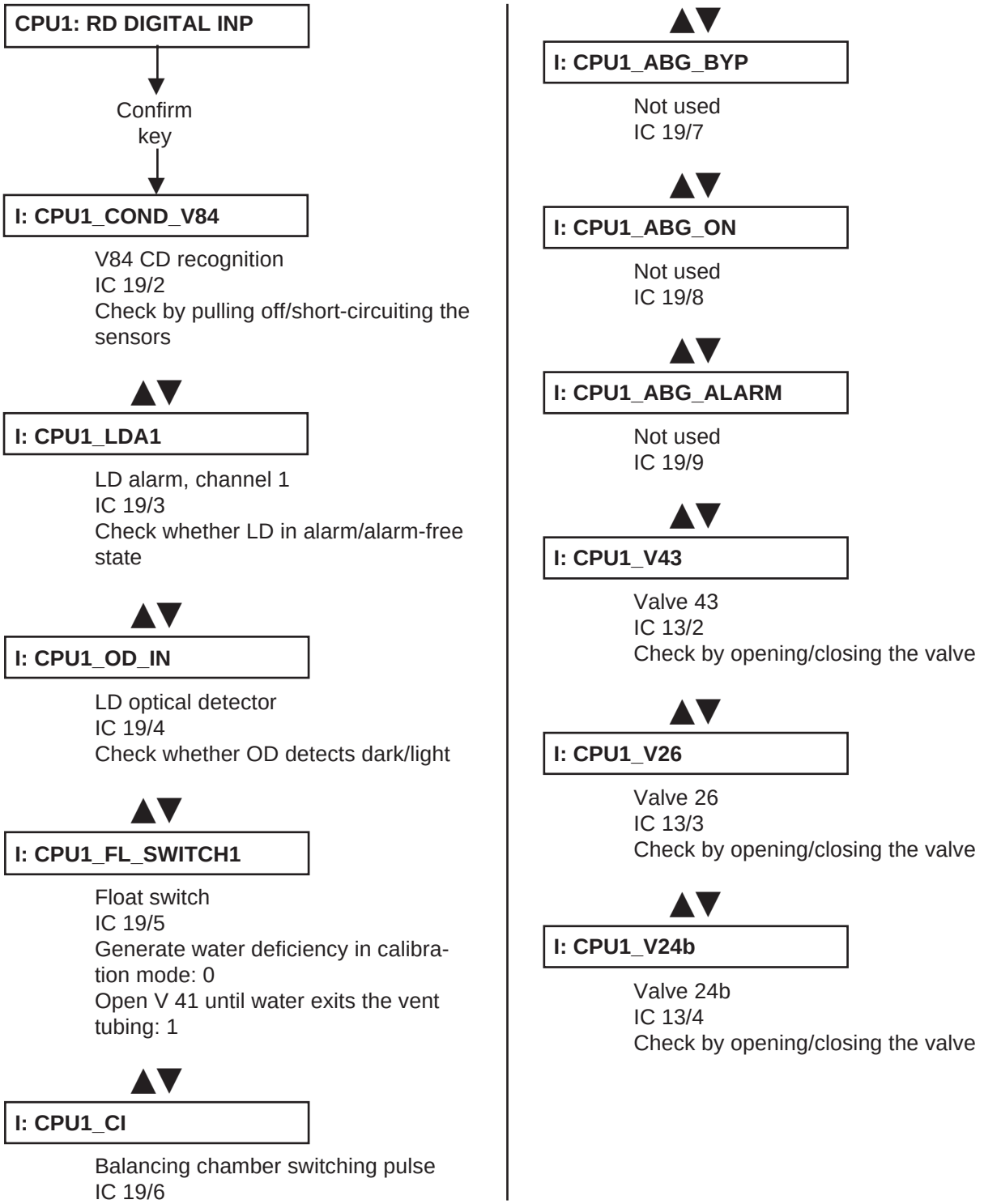
5.4 Reading the analog inputs of CPU II

Explanation:
UF Volume display: ADC value
Time Left display: Analog voltage (in 0.1 V), converted to the value at the input of P.C.B. LP 632



5.5 Reading the digital inputs of CPU I

Explanation:
All UF displays show 0000; red, yellow, green traffic light off: low level at latch on P.C.B. LP 633
All UF displays show 1111; red, yellow, green traffic light on: high level at latch on P.C.B. LP 633
If high level is applied, an audible alarm is simultaneously sounded. This tone can be suppressed by pressing the Alarm Tone Mute key. In this case, the Alarm Tone Mute LED is illuminated.



▲▼

I: CPU1_V24

Valve 24
IC 13/5
Check by opening/closing the valve

▲▼

I: CPU1_UF_P1

Acknowledgement of UF pump 1
IC 13/6
Check possible only in the combination menu

▲▼

I: CPU1_LDA2

LD alarm, channel 2
IC 13/7
Preparation:
LD alarm-free and set CLAMP_CTRL (CPU1: WR DIGIT. OUTP) to 1
Check: initiate an LD alarm

▲▼

I: CPU1_SUB_W_P

Acknowledgement of UF pump 2
IC 13/8

▲▼

I: CPU1_LC11

IC 13/9

▲▼

I: CPU1_REED_BIC

Bicarbonate reed contact
IC 14/2
Actuate rinse chamber/bicarbonate reed contact

▲▼

I: CPU1_BIBAG

Microswitch 137 / connector
IC 14/3
Check by connecting/removing the BIBAG

▲▼

I: CPU1_REED_RINSE

Concentrate reed contact
IC 14/4
Actuate rinse chamber/concentrate reed contact

▲▼

I: CPU1_BIBAG_C

Microswitch 138 / connector
IC 14/5
Check by connecting/removing the cap

▲▼

I: CPU1_PSW_V102

Concentrate pressure switch
IC 14/6
Check by increasing/decreasing the pressure at the pressure switch

▲▼

I: CPU1_PSW_V104

Bicarbonate pressure switch
IC 14/7
Check by increasing/decreasing the pressure at the pressure switch

▲▼

I: CPU1_PWR_OFF

Power off
IC 14/8

▲▼

I: CPU1_HEP_ON

Heparin pump on
IC 14/9
Switch heparin pump on/off

▲▼

I: CPU1_LA32

IC 15/2

▲▼

I: CPU1_SW_ON_OFF

On/off key on machine
IC 15/3
Brief key actuation

▲▼

I: CPU1_PWR_FAIL

Power failure recognition
IC 15/4

▲▼

I: CPU1_SHUNT_OUTP

Microswitch, interlock shunt
IC 15/5
Both lines in the interlock shunt and interlock shunt closed: 0

▲▼

I: CPU1_SHUNT_INP

Microswitch, interlock shunt
IC 15/6
Red line only in the interlock shunt and interlock shunt closed: 0

▲▼

I: CPU1_SHUNT

Microswitch, interlock shunt
IC 15/7
Open/close interlock shunt

▲▼

I: CPU1_SERV_EN

Not used
IC 15/8

▲▼

I: CPU1_LEV_SIGNAL

Level sensor (NTC 6 – replacement)
IC 15/9
Check by removing/short-circuiting the sensor pins

▲▼

I: CPU1_SN

Single needle changeover
IC 16/2
Changeover pressure of SN blood pump reached: 0

▲▼

I: CPU1_ADKS

Single needle blood pump recognition connected
IC 16/3
Connect/remove the blood pump (only with the machine off)

▲▼

I: CPU1_BPSB_ART

Arterial blood pump stop confirmation
IC 16/4
Actuation of Start/Stop key on the arterial blood pump

▲▼

I: CPU1_BPUS_ART

Arterial blood pump revolution stop
IC 16/5
Arterial blood pump alarm field on: 0 (cleared by pressing the Start/Stop key on the blood pump)

▲▼

I: CPU1_BPSB_VEN

Venous blood pump stop confirmation
IC 16/6
Actuation of the Start/Stop key on the venous blood pump

▲▼

I: CPU1_BPUS_VEN

Venous blood pump revolution stop
IC 16/7
Preparation:
Set SNST (CPU1: WR DIGIT. OUTP) to 1 and wait for the alarm field.
Venous blood pump on.
Check: clearing of the alarm field by pressing the Start/Stop key on the venous blood pump

▲▼

E. CPU1_HEP_ALARM

Heparin pump alarm
IC 16/8
Generate a heparin pump alarm
(e.g. by blocking the syringe slide)

▲▼

I: CPU1_BIB_LEVEL

Level sensor 135
IC 16/9
Check by removing/shorting the sensor pins

▲▼

I: CPU1_EXT_ALARM

External alarm
IC 20/2
Release of an external alarm

▲▼

I: CPU1_SERVICE_MODE

Dialysis/calibration changeover
IC 20/3
Reset the service switch

▲▼

I: CPU1_LEVEL_UP

Raise the LD level
IC 20/4
Press the “Raise level” key

▲▼

I: CPU1_LEVEL_DOWN

Lower the LED level
IC 29/5
Preparation:
LD alarm-free, set CLAMP_CTRL
(CPU1: WR DIGIT. OUTP) to 1
Check: press the “Lower level” key

▲▼

I: CPU1_ADS_SN

Not used
IC 20/6

▲▼

I: CPU1_ACKN_CONC

Not used
IC 20/7

▲▼

I: CPU1_ACKN_BIC

Not used
IC 20/8

▲▼

I: CPU1_BIBAG_PSW

BIBAG pressure switch
IC 20/9
Check by increasing/decreasing pressure on the pressure switch

▲▼

I: CPU1_RA21

Not used
IC 21/2

▲▼

I: CPU1_HDF_ON

HDF on
IC 21/3
Actuation of HDF On/Off switch

▲▼

I: CPU1_V102

Acknowledgement, valve 102
IC 21/4
Open/close the valve

▲▼

I: CPU1_V104

Acknowledgement, valve 104
IC 21/5
Open/close the valve

▲▼

I: CPU1_CSS_REED

IC 21/6

▲▼

I: CPU1_HEAT_CLK

IC 21/7

▲▼

I: CPU1_BYP_REQ

IC 21/8

▲▼

I: CPU1_CLP_REQ

IC 21/9

▲▼

I: CPU1_LATCH7_FREE1

Not used
IC 7/2

▲▼

I: CPU1_LATCH7_FREE2

Not used
IC 7/3

▲▼

I: CPU1_LATCH7_FREE3

Not used
IC 7/4

▲▼

I: CPU1_LATCH7_FREE4

Not used
IC 7/5

▲▼

I: CPU1_LATCH7_FREE5

Not used
IC 7/6

▲▼

I: CPU1_LATCH7_FREE6

Not used
IC 7/7

▲▼

I: CPU1_LATCH7_FREE7

Not used
IC 7/8

▲▼

I: CPU1_LATCH7_FREE8

Not used
IC 7/9

▲▼

I: CPU1_DIP1:00011100

DIP switch CPU I/array I
P.C.B. LP 631/IC 12/2 to 9
The position of the DIP switches is shown on the alphanumeric display (1: DIP switch „ON“)

▲▼

I: CPU1_DIP2: 00000000

DIP switch CPU I/array II
P.C.B. LP 631/IC 13/2 to 9
The position of the DIP switches is shown on the alphanumeric display (1: DIP switch „ON“)

▲▼

I: CPU1_KEY_TESTING

Touch panel test
P.C.B. LP 635/IC 73/2 to 6
The key pressed is shown on the alphanumerical display, the LED next to the key is illuminated.

▲▼

I: CPU1_RCU_KEY_TEST

Touch panel test for the remote control RCU 4008.
The key pressed is shown on the alphanumerical display, the LED next to the key is illuminated.

▲▼

back to menu ?

↓

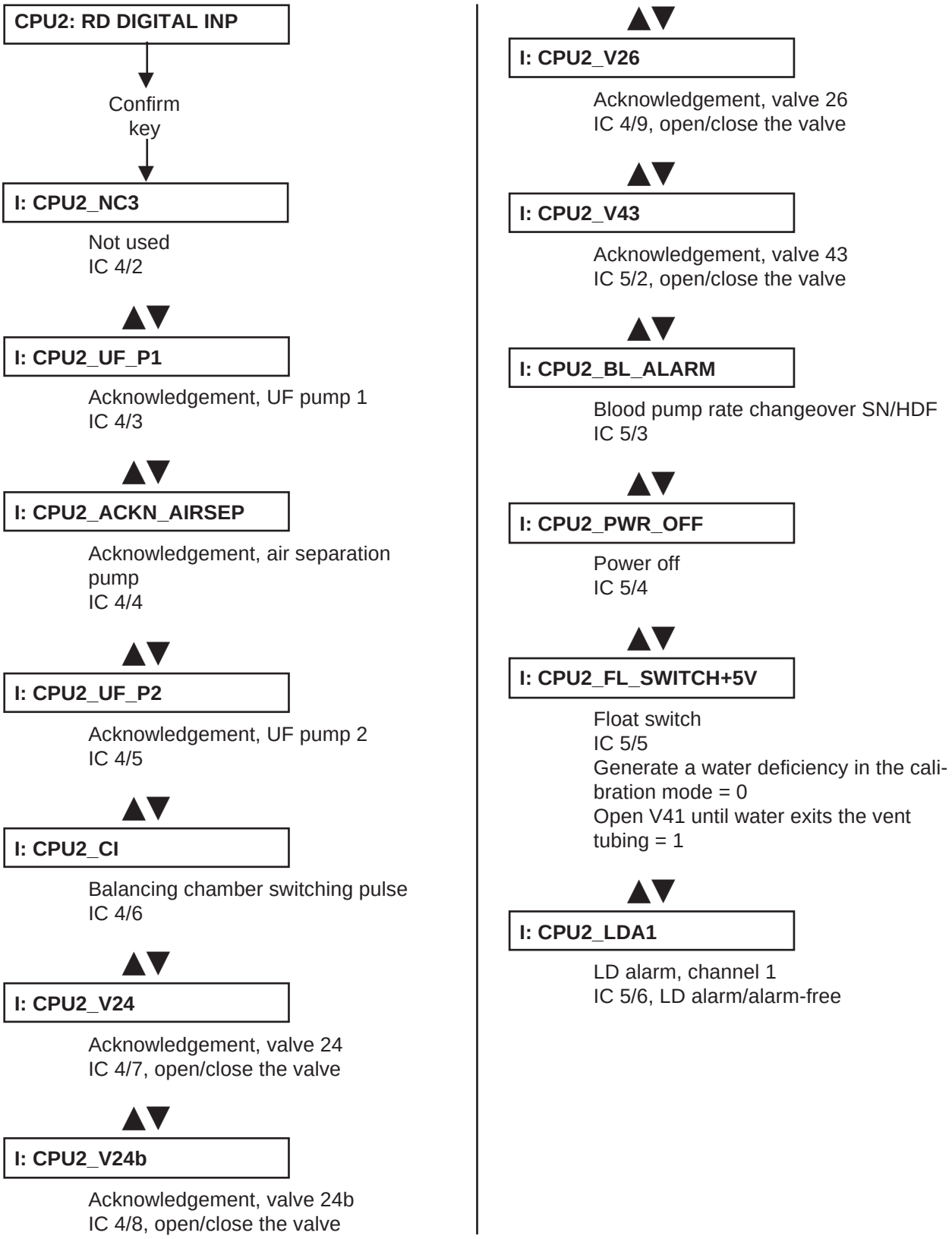
Confirm key

↓

CPU1: RD DIGITAL INP

5.6 Reading the digital inputs of CPU II

Explanation:
All UF displays show 0000; red, yellow, green traffic light off: low level at latch on P.C.B. LP 632
All UF displays show 1111; red, yellow, green traffic light on: high level at latch on P.C.B. LP 632
If high level is applied, an audible alarm is simultaneously sounded. This tone can be suppressed by pressing the Alarm Tone Mute key. In this case, the Alarm Tone Mute LED is illuminated.



▲▼

I: CPU2_LDA2

LD alarm, channel 2
IC 5/7
Preparation:
LD alarm-free and set CLAMP_CTRL (CPU1: WR DIGIT. OUTP) to 1
Check: generate an LD alarm

▲▼

I: CPU2_BPSB_VEN

Venous blood pump stop confirmation
IC 5/8
Actuation of the Start/Stop key on the venous blood pump

▲▼

I: CPU2_BPSB_ART

Arterial blood pump stop confirmation
IC 5/9
Actuation of the Start/Stop key on the arterial blood pump

▲▼

I: CPU2_V42

Not used
IC 6/2

▲▼

I: CPU2_BPST_ART

Acknowledgement, special control of the arterial blood pump
IC 6/3
Check by setting BPST_ART (CPU1: WR DIGIT. OUTP) to 1/0

▲▼

I: CPU2_BPUS_ART

Arterial blood pump revolution stop
IC 6/4
Alarm field of the arterial blood pump on = 0
(cleared by pressing the Start/Stop key on the blood pump)

▲▼

I: CPU2_BPST_VEN

Acknowledgement, special control of the venous blood pump
IC 6/5
Check by setting BPST_VEN (CPU1: WR DIGIT. OUTP) to 1/0

▲▼

I: CPU2_BPUS_VEN

Venous blood pump revolution stop
IC 6/6
Preparation:
Set SNST (CPU1: WR DIGIT. OUTP) to 1 and wait for the alarm field of the venous blood pump to turn on
Check: clear the alarm field by pressing the Start/Stop key on the venous blood pump

▲▼

I: CPU2_ADKS

Single needle blood pump recognition connected
IC 6/7
Connect/remove the blood pump (with the machine off only)

▲▼

I: CPU2_LEVEL_UP

Raise LD level
IC 6/8, press the “Raise level” key

▲▼

I: CPU2_LEVEL_DOWN

Lower LD level
IC 6/9
Preparation:
LD alarm-free, set CLAMP_CTRL (CPU1: WR DIGIT. OUTP) to 1
Check: press the “Lower level” key

▲▼

I: CPU2_RINSE

Concentrate reed contact
IC 7/2
Actuate rinse chamber/concentrate reed contact

▲▼

I: CPU2_V147

IC 7/3

▲▼

I: CPU2_REED_BIC

Bicarbonate reed contact
IC 7/4
Actuate rinse chamber/bicarbonate reed contact

▲▼

I: CPU2_PSW_104

Bicarbonate pressure switch
IC 7/5
Check by increasing/decreasing pressure at the pressure switch

▲▼

I: CPU2_V145

IC 7/6

▲▼

I: CPU2_SHUNT_OUTP

Microswitch, interlock shunt
IC 7/7
Both lines in the interlock shunt and interlock shunt closed: 0

▲▼

I: CPU2_SHUNT_INP

Microswitch, interlock shunt
IC 7/8
Red line only in the interlock shunt and interlock shunt closed: 0

▲▼

I: CPU2_SHUNT

Microswitch, interlock shunt
IC 7/9, open/close the interlock shunt

▲▼

I: CPU2_ABG_ON

Not used
IC 8/2

▲▼

I: CPU2_SERVICE_MODE

Changeover dialysis/calibration
IC 8/3, reset the service switch

▲▼

I: CPU2_HOT_RINSE

Changeover hot rinsing
IC 8/4
Check by setting HOT_RINSE (CPU1: WR DIGIT. OUTP) to 0/1

▲▼

I: CPU2_OD_OUT

LD optical detector
IC 8/5, optical detector light/dark

▲▼

I: CPU2_SNST

Single needle control
IC 8/6

▲▼

I: CPU2_24V_SW

24-V switch
IC 8/7

▲▼

I: CPU2_SN

Single needle changeover
IC 8/8, SN switching pressure reached: 0

▲▼

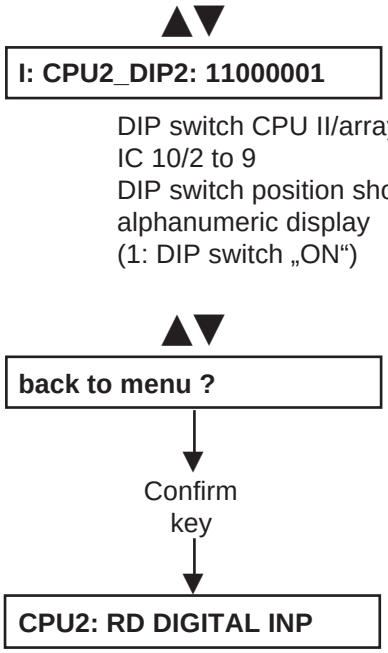
I: CPU2_HEAT_RL_WATCH

Acknowledgement, heater relay
IC 8/9

▲▼

I: CPU2_DIP1: 01100110

DIP switch CPU II/array I
IC 9/2 to 9
DIP switch position shown on the alphanumeric display
(1: DIP switch „ON“)


I: CPU2_DIP2: 11000001
DIP switch CPU II/array II
IC 10/2 to 9
DIP switch position shown on the
alphanumeric display
(1: DIP switch „ON“)
back to menu ?
Confirm
key
CPU2: RD DIGITAL INP

5.7 Writing the analog outputs of CPU I

Explanation:
UF Rate display: DAC value (can be changed with: 4008/E/B: ▲▼ UF Rate, 4008 H/S: +/–)
Time Left display: Analog voltage on P.C.B. LP 634, in 0.1 V

CPU1: WR ANALOG OUTP

↓ Confirm key ↓

O: CPU1_TEMP_SET

Rated value specified for the temperature
IC 53/9
0 to 5 V (X634R/C20, 0.2 to 10 V)

▲▼

O: CPU1_DAC_DIM

Calibration voltage, dimness
IC 53/8
0 to 5 V (C634R/A11, 0 to 5 V)

▲▼

O: CPU1_TEMP_ADJ

Calibration voltage, temperature control
IC 53/7
0 to 5 V (X634R/C21, 0 to 5 V)

▲▼

O: CPU1_DAC_BLL

Calibration voltage, blood leak
IC 53/6
0 to 5 V
(X634R/A12, 0 to 5 V)

▲▼

O: CPU1_BIBAG_TEMP_AJ

Calibration voltage for temperature NTC
(slot X107 / LP 747)
IC 53/5
0 to 5 V (X634R/A13, 0 to 5 V)

▲▼

O: CPU1_DAC_X2

Not used
IC 53/4
0 to 5 V (X634R/C13, 0 to 5 V)

▲▼

O: CPU1_STEUER_EP

Speed setting, degassing pump
IC 53/3
0 to 4.4 V (X634L/ between A, B, C 27 and A, B, C 28, 0 to 21 V)

▲▼

O: CPU1_STEUER_FP

Speed setting, flow pump
IC 53/2
0 to 4.4 V (X634L/ between A, B, C 29 and A, B, C 30, 0 to 21 V)

▲▼

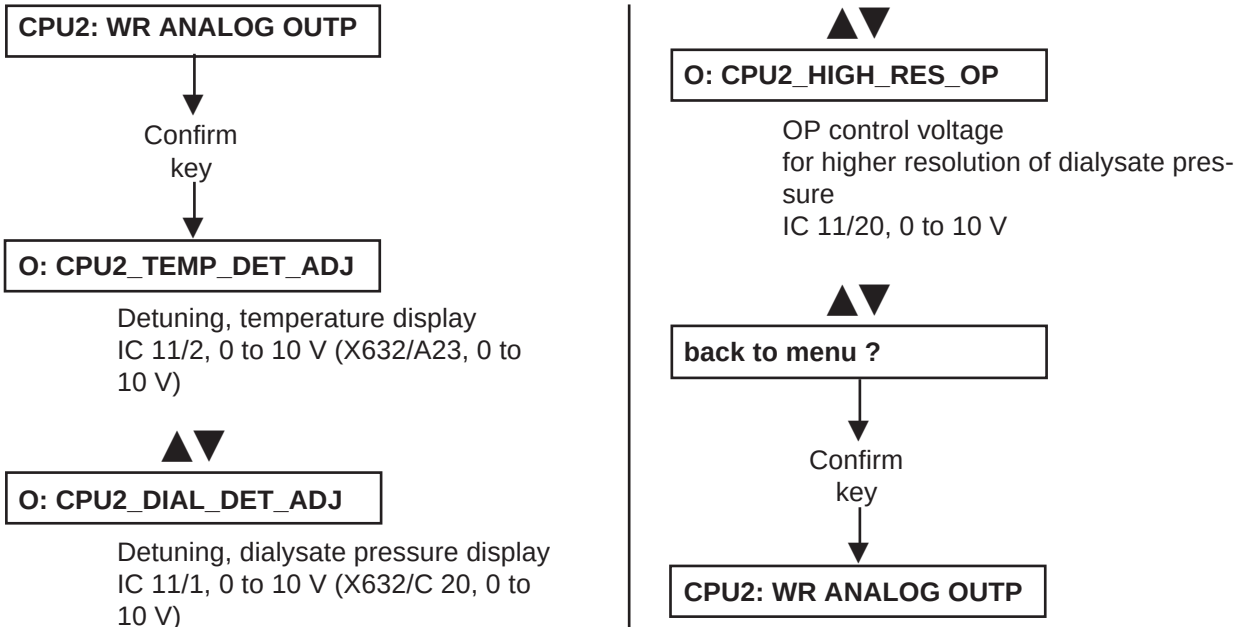
back to menu ?

↓ Confirm key ↓

CPU1: WR ANALOG OUTP

5.8 Writing the analog outputs of CPU II

Explanation:
UF Rate display: DAC value (can be changed with: 4008/E/B: ▲▼ UF Rate, 4008 H/S: +/–)
Time Left display: Analog voltage on P.C.B. LP 632, in 0.1 V



5.9 Writing the digital outputs of CPU I

Explanation:
UF Rate display: 0000 = not active
1111 = active (P.C.B. LP 634 level)
(can be changed with: 4008/E/B: ▲▼ UF Rate, 4008 H/S: +/–)

CPU1: WR DIGIT. OUTP

↓ Confirm key ↓

O: CPU1_V24

Valve 24
IC 10/19

▲▼

O: CPU1_V24B

Valve 24b
IC 10/13

▲▼

O: CPU1_V26

Valve 26
IC 10/18

▲▼

O: CPU1_V130

Valve 130
IC 10/17

▲▼

O: CPU1_V30

Valve 30
IC 7/16

▲▼

O: CPU1_V31

Valve 31
IC 12/19

▲▼

O: CPU1_V32

Valve 32
IC 12/18

▲▼

O: CPU1_V33

Valve 33
IC 12/17

▲▼

O: CPU1_V34

Valve 34
IC 12/16

▲▼

O: CPU1_V35

Valve 35
IC 12/15

▲▼

O: CPU1_V36

Valve 36
IC 12/14

▲▼

O: CPU1_V37

Valve 37
IC 12/13

▲▼

O: CPU1_V38

Valve 38
IC 12/12

▲▼

O: CPU1_V41

Valve 41
IC 7/13
(After a short time interval, the valve closes automatically, to prevent the water from overflowing.)

▲▼

O: CPU1_V43

Valve 43
IC 7/15

▲▼

O: CPU1_V84

Valve 84
IC 7/18
Note: After the valve V84 has been activated, rinsing is mandatory.

▲▼

O: CPU1_V86

Valve 86
IC 7/17

▲▼

O: CPU1_V87

Valve 87
IC 10/15

▲▼

O: CPU1_V91

Valve 91
IC 10/14
Note: When exiting this menu option (return to
„CPU1: WR DIGIT. OUTP“)
the valve will be closed.

▲▼

O: CPU1_V99

Valve 99
IC 7/19
Note: When exiting this menu option (return to
„CPU1: WR DIGIT. OUTP“)
the valve will be closed.

▲▼

O: CPU1_V100

Valve 100
IC 4/17
Note: When exiting this menu option (return to
„CPU1: WR DIGIT. OUTP“)
the valve will be closed.

▲▼

O: CPU1_V102

Valve 102
IC 10/12
(Activation of the valve is possible only if a mandatory rinse is not requested.)

▲▼

O: CPU1_V104

Valve 104
IC 7/14
(Activation of the valve is possible only if a mandatory rinse is not requested.)

▲▼

O: CPU1_V126

Not used

▲▼

O: CPU1_V145

Not used

▲▼

O: CPU1_V147

Not used

▲▼

O: CPU1_V172

Not used

▲▼

O: CPU1_V173

Not used

▲▼

O: CPU1_AIR_SEP_PUMP

Air separation pump
IC 4/18, 19
(1111: clockwise)
Note: When exiting the menu option, the ASP stops

▲▼

O: CPU1_STOP_EP

Stopping of the degassing pump
IC 4/16

▲▼

O: CPU1_STOP_FP

Stopping of the flow pump
IC 4/15

▲▼

O: CPU1_SET_UF1_ON

Activation of UF pump 1
IC 4/14
(0/1 jump = 1 stroke)

▲▼

O: CPU1_SET_UF2_ON

Activation of UF pump 2
IC 4/13
(0/1 jump = 1 stroke)

▲▼

O: CPU1_SET_EN_UF2

Not used
IC 4/12

▲▼

O: CPU1_SET_FLOW_ON

Flow on
Dataword to Gal 23:
0000 0010 (active, V 32 open)
0000 0011 (inactive, V 31, 32 open)

▲▼

O: CPU1_SET_FILL_PRG

Fill program
Dataword to Gal 23:
0000 1010 (V 32, 34 open)

▲▼

O: CPU1_EMPTIING_PRG

Emptying program:
Dataword to Gal 23:
0001 0010 (V 32, 35 open)

▲▼

O: CPU1_FILL_ONE_CHAM

Filling of a balancing chamber compartment
Dataword to Gal 23:
0100 0010 (V 32, 37 open)

▲▼

O: CPU1_EMPTY_ONE_CHA

Emptying of a balancing chamber compartment
Dataword to Gal 23:
1100 0010 (V 32, 37, 38 open)

▲▼

O: CPU1_CO:L:XXXXXXXX

Step number of concentrate pump
The dataword to IC 2 is indicated on the alphanumeric display and can be changed by pressing the UF Rate UP/DOWN keys.
Prerequisite: reed contact of concentrate connector open

▲▼

O: CPU1_BI_L:XXXXXXXX

Step number of bicarbonate pump
Prerequisite: reed contact of bicarbonate connector open

▲▼

O: CPU1_ALARM_SOUND

Alarm tone
IC 5/18, 19 set to 1: active

▲▼

O: CPU1_WARN_SOUND

Warning tone
IC 5/18 set to 1, 19 to 0: active

▲▼

O: CPU1_INFO_SOUND

Infosound
IC 5/18 set to 0, 19 to 1: active

▲▼

O: CPU1_CLK_OVERLAP

Changeover of dead time of balancing chamber
IC 5/17 set to 0: 1 kHz; to 1: 2 Hz

▲▼

O:CPU1_EN_IN_PULSE

Changeover, “Eigentakt”
IC 5/16 set to 0: “Eigentakt”; to 1: changover current rise

▲▼

O: CPU1_BC_PULSE

Balancing chamber switching
IC 5/15

▲▼

O: CPU1_EN_STEP_PULS

Gal changeover
IC 5/14

▲▼

O: CPU1_BC_FUNCTION

Activation of balancing chamber gal
IC 5/13

▲▼

O: CPU1_STEPPER_PULS

“Eigentakt”
IC 5/12

▲▼

O: CPU1_FL_SWITCH_EN

V41 release
IC 7/12
Check:
Set level to 1 and float switch down:
V 41 open
Set level to 0 and float switch down:
V 41 closed

▲▼

O: CPU1_SNST

Single needle control
IC 13/19
Preparation:
LD alarm-free and set CLAMP_CTRL (CPU1: WR DIGIT. OUTP) to 1
Check:
SNST set to 1: as soon as the SN changeover pressure is reached, the venous blood pump is running

▲▼

O: CPU1_CPU_OFF

Automatic switchoff
IC 13/18
In position 1, the machine switches off

▲▼

O: CPU1_EN_PF_AT

Release of audible power failure alarm
IC 13/17
Preparation: set WD_SET (CPU2: WR DIGIT. OUTP)
Check: with EN_PF_AT set to 0/1, the power failure alarm can be switched on/off

▲▼

O: CPU1_PIC_RA3

Not used
IC 13/16

▲▼

O: CPU1_PROG_LOG1

Program logic 1, HDF pump
IC 13/14
(0: speed 200; 1: speed 400)

▲▼

O: CPU1_PROG_LOG2

Program logic 2, HDF pump
IC 13/13
(0: speed 200; 1: speed 150)

▲▼

O: CPU1_VENT_VALVE

LD vent valve
IC 13/12

▲▼

O: CPU1_LOG_42

Not used
IC 11/19

▲▼

O: CPU1_CLR_ALARM

Clearing of the alarm
IC 11/18
Check: generate a heparin pump alarm; by setting CLR_ALARM from 0 to 1, the alarm is cleared.

▲▼

O: CPU1_HOT_RINSE

Changeover, hot rinsing
IC 11/17

▲▼

O: CPU1_TEST_BATT

Battery test
IC 11/16

▲▼

O: CPU1_CPU_AKKU

Battery relay
IC 11/15

▲▼

O: CPU1_HEAT_OFF

Heater blocking
IC 11/14

▲▼

O: CPU1_STAFF_CALL

Staff call
IC 11/13

▲▼

O: CPU1_TL_RED

Traffic light red
IC 11/12
(The pertinent status indicator on the front panel lights simultaneously.)

▲▼

O: CPU1_BPST_ART

Special control of the arterial blood pump
IC 6/19

▲▼

O: CPU1_BPSST_ART

Arterial blood pump system stop
IC 6/18

▲▼

O: CPU1_CLAMP_CTRL

LD clamp control
IC 6/17
Preparation: LD in no alarm state

▲▼

O: CPU1_BPST_VEN

Special control of the venous blood pump
IC 6/16

▲▼

O: CPU1_BPSST_VEN

Venous blood pump system stop
IC 6/15
Preparation:
Set SNST (CPU1: WR DIGIT. OUTP) to 1
Check: apply pressure to the SN blood pump; with BPSST_VEN, the blood pump can be switched on/off.

▲▼

O: CPU1_BL_ALARM

Changeover, blood pump rate
Single needle/HDF
IC 6/14

▲▼

O: CPU1_TL_YELLOW

Traffic light yellow
IC 6/13
(The pertinent status indicator on the front panel lights simultaneously.)

▲▼

O: CPU1_TL_GREEN

Traffic Light green
IC 6/12
The pertinent status indicator on the front panel lights simultaneously.

▲▼

O: PUMP FASTER

4008 H/S: **O: CPU1_BP_FASTER**

Increase blood pump rate +
(RCU 4008)
LP 649/X4.4; LP 635/X4.4;
LP 922/X5.4; LP 924/X6.4

▲▼

O: PUMP SLOWER

4008 H/S: **O: CPU1_BP_SLOWER**

Decrease blood pump rate –
(RCU 4008)
LP 649/X4.5; LP 635/X4.5;
LP 922/X5.5; LP 924/X6.5

▲▼

O: CPU1_OVERLAP_VALUE

Charging of the dead time counter
IC 3/12 to 19
(DAC adjustable from 0 to 255)

▲▼

O: CPU1_V_ADS

Not used

▲▼

O: CPU1_NC_I

Not used

▲▼

O: CPU1_NC_II

Not used

▲▼

O: CPU1_NC_III

Not used

▲▼

O: CPU1_ACKN_FLOW

Not used

▲▼

O: CPU1_ACKN_BL_ALARM

Not used

▲▼

O: CPU1_DISPLAY_TEST

All LED indicators are tested.
The display on the UF monitor counts
from 1 to 0.

▲▼

back to menu ?

↓

Confirm
key

↓

CPU1: WR DIGIT. OUTP

5.10 Writing the digital outputs of CPU II

Explanation:
UF Rate display: 0000 = not active
1111 = active (P.C.B. LP 632 level)
(can be changed with: 4008/E/B: ▲▼ UF Rate, 4008 H/S: +/–)

CPU2: WR DIGIT. OUTP

↓

Confirm key

↓

O: CPU2_WD_RES

Watchdog reset
IC 24/18
Preparation:
Set WD_SET (CPU2: WR DIGIT. OUTP) to 0, then to 1 again.
Check:
Briefly set WD_RES to 0; upon reset to 1, the WD relay is connected.

▲▼

O: CPU2_WD_SET

Watchdog set
IC 24/17
Watchdog relay drops out, 24 V switched off, audible signal is sounded.
To clear the alarm, turn the machine off and on again.
Otherwise calibration impossible.

▲▼

O: CPU2_V24_EN

Release of V 24
IC 24/16
Preparation: V 24
Switch on (CPU1: WR DIGIT. OUTP).
Check: switch the valve on/off with V 24_EN.

▲▼

O: CPU2_V24B_EN

Release of V24b
IC 24/15
Preparation: switch on V 24B (CPU1: WR DIGIT. OUTP).
Check: switch the valve on/off with V 24B_EN.

▲▼

O: CPU2_UF_P_CTRL

Activation of UF pump
IC 24/14, 0/1 level change = 1 stroke
Preparation: set CPU2_UF_P_EN to 1.

▲▼

O: CPU2_UF_P_EN

Release of UF pumps
IC 24/13

▲▼

O: CPU2_CPU2_ALARM

Release of the alarm tone by CPU II
IC 24/12

▲▼

O: CPU2_UF_P2_CTRL

Activation of UF-Pump 2
IC 24/11, 0/1 level change = 1 stroke
Preparation: set CPU2_UF_P_EN to 1.

▲▼

O: CPU2_4066_ENABLE_1

Analog switch for P_ADS_DET
IC 20/13 (X632/A20)

▲▼

O: CPU2_4066_ENABLE_2

Analog switch for +10-V reference voltage
IC 27/5

▲▼

O: CPU2_4066_ENABLE_3

Analog switch for PV_DET
IC 20/6 (X632/C18)

▲▼

O: CPU2_4066_ENABLE_4

Analog switch for PA_DET
IC 20/12 (X632/A17)

▲▼

O: CPU2_4066_ENABLE_5

Analog switch, not used
IC 27/13

▲▼

O: CPU2_4066_ENABLE_6

Analog switch for COND_DET
IC 20/5 (X632/A21)

▲▼

O: CPU2_4066_ENABLE_7

Analog switch for BLL_DIM
IC 27/6

▲▼

O: CPU2_4066_ENABLE_8

Analog switch for BLL
IC 27/12

▲▼

O: CPU2_SN_ART

Single needle control, arterial
X632/A15

▲▼

O: CPU2_LDSA

Attenuation of LD ultrasonic sensor
X632/C16
Preparation:
LD alarm-free and set CLAMP_CTRL (CPU1: WR DIGIT. OUTP) to 1.
Check: by setting LDSA to 1, the clamp at the LD closes.

▲▼

O: CPU2_ODSA

Attenuation of LD optical sensor
X632/C15

▲▼

O: CPU2_CLAMP_CTRL

Clamp control, air detector
X632/C10
Preparation: LD in no alarm state

▲▼

O: CPU2_NC5

Not used
X632/B25

▲▼

O: CPU2_NC7

Not used
X632/B10

▲▼

O: CPU2_BLL_DET

Detuning of blood leak detector
X632/A25

▲▼

O: CPU2_SN_EN

Release of single needle
X632/C19

▲▼

O: CPU2_NC10

Not used
X632/B4

▲▼

O: CPU2_V26

Valve 26
X632/A6

▲▼

O: CPU2_V42

Not used
X632/C4

▲▼

O: CPU2_V43

Valve 43
X632/C5

▲▼

O: CPU2_VENT_DSAFE

Vent valve, Diasafe
X632/B5

▲▼

O: CPU2_EM_HEAT_OFF

Heater relay
X632/A9
For safety reasons, the relay switches off again after having been activated.

▲▼

O: CPU2_NC8

Not used
X632/B9

▲▼

O: CPU2_NC9

X632/C6
IC 29/13

▲▼

O: CPU2_LED1

P.C.B. LP 632, LED 1
IC 21/19

▲▼

O: CPU2_LED2

P.C.B. LP 632, LED 2
IC 21/18

▲▼

O: CPU2_LED3

P.C.B. LP 632, LED 3
IC 21/17

▲▼

O: CPU2_LED4

P.C.B. LP 632, LED 4
IC 21/16

▲▼

O: CPU2_LED5

P.C.B. LP 632, LED 5
IC 21/15

▲▼

O: CPU2_LED6

P.C.B. LP 632, LED 6
IC 21/14

▲▼

O: CPU2_LED7

P.C.B. LP 632, LED 7
IC 21/13

▲▼

O: CPU2_LED8

P.C.B. LP 632, LED 8
IC 21/12

▲▼

back to menu ?

↓

Confirm key

↓

CPU2: WR DIGIT. OUTP

5.11 Writing/Reading the digital outputs of CPU I

Explanation:
UF Volume display: Acknowledgement/input (in case of 1111, the three status LEDs of the traffic light are also illuminated)
UF Rate display: Activation/output (can be changed with: 4008/E/B: ▲▼ UF Rate, 4008 H/S: +/–)

CPU1: COMBINATION

↓ Confirm key

CPU 1_COMBI: V24

Valve 24
Activation, P.C.B. LP 634/IC 10/19
Acknowledgement, P.C.B. LP 633/ IC 13/5

▲▼

CPU 1_COMBI: V24B

Valve 24b
Activation, P.C.B. LP 634/IC 10/13
Acknowledgement, P.C.B. LP 633/ IC 13/4

▲▼

CPU 1_COMBI: V26

Valve 26
Activation, P.C.B. LP 634/IC 10/18
Acknowledgement, P.C.B. LP 633/ IC 13/3

▲▼

CPU 1_COMBI: V43

Valve 43
Activation, P.C.B. LP 634/IC 7/15
Acknowledgement, P.C.B. LP 633/ IC 13/2

▲▼

CPU 1_COMBI: V102

Valve 102
Activation, P.C.B. LP 634/IC 10/12
Acknowledgement, P.C.B. LP 633/ IC 21/4
(Activation of the valve is possible only if a mandatory rinse is not requested)

▲▼

CPU 1_COMBI: V104

Valve 104
Activation, P.C.B. LP 634/IC 7/14
Acknowledgement, P.C.B. LP 633/IC 21/5
(Activation of the valve is possible only if a mandatory rinse is not requested)

▲▼

CPU 1_COMBI: UF1_PUMP

UF pump 1
Activation, P.C.B. LP 634/IC 4/14
Acknowledgement, P.C.B. LP 633/IC 13/6
(when setting from 0 to 1 = 1 stroke; acknowledgement is a brief change to 1)

▲▼

CPU 1_COMBI: AIR_SEP

Switching the ASP on/off

▲▼

back to menu ?

↓ Confirm key

CPU 1: COMBINATION

5.12 ONLINE*plus*™ module

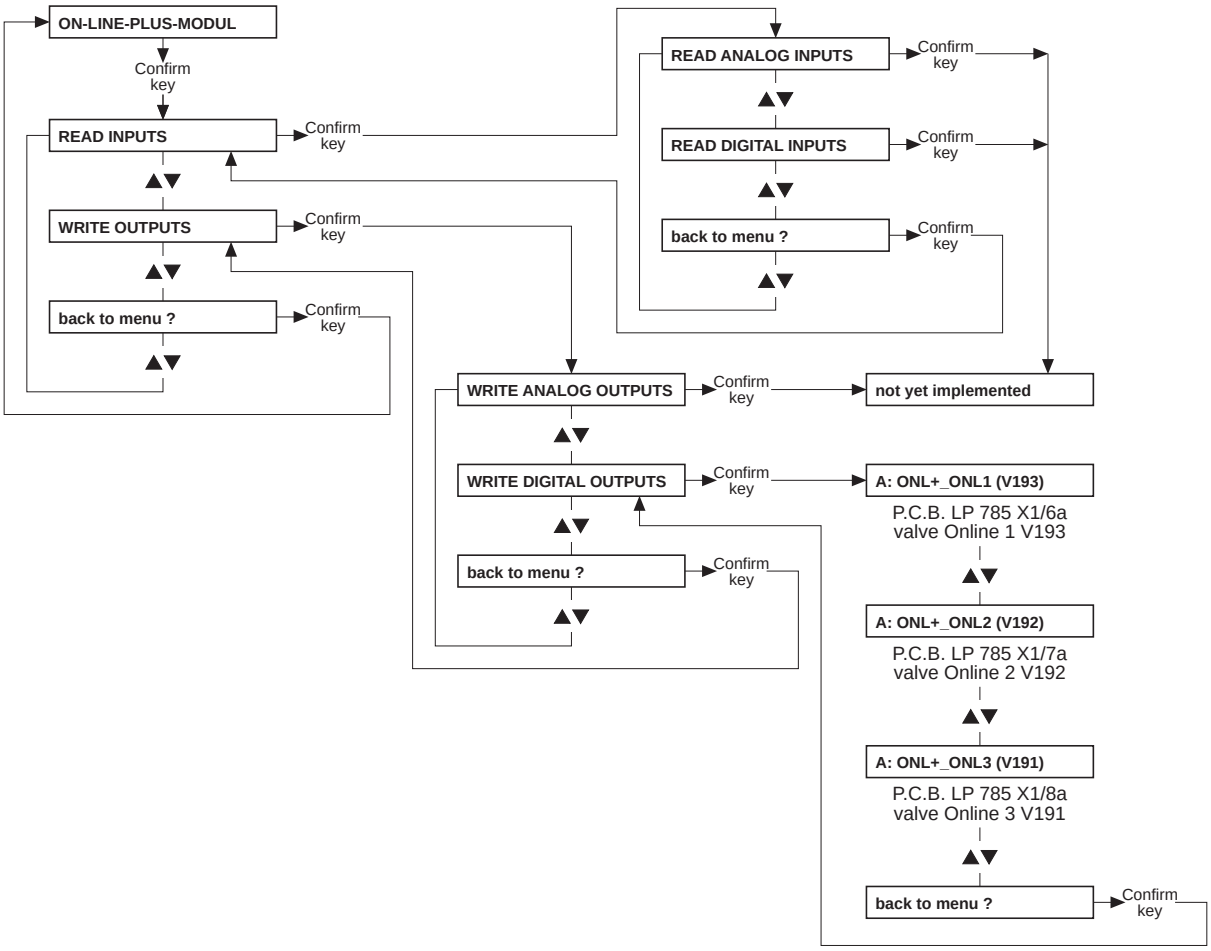


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6 Setup menu

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6.4	**Main menu 4008 H/S Rev. 4.3**	6-35

6.1 Overview Setup menu settings

Menu item	Submenu	Default value	Value range	Resolution
SET ALARM/WARN TIME	**Set ART-AL DELAYTIME**	5 sec	0 – 5 sec	1 sec
	Set VEN-AL DELAYTIME	5 sec	0 – 5 sec	1 sec
	Set FLOW-OFF W-TIME	30 min	30 – 60 min	15 min
	Set UF-WARNING-TIME	10 min	10/30 min	20 min
	Set MUTE-TIME	1 min	1 – 2 min	1 min

Menu item	Submenu	Default value	Value range	Resolution
SETUP CLEANING PGM				
CLEANING Times	**Rinsing TIME**	15 min	5 – 30 min	1 min
	Hotrinse TIME	15 min	15 – 30 min	1 min
	Disinfection TIME	10 min	10 – 20 min	1 min
	Rinsing Free TIME	CPU1: DIP switch array 1, SW8 set to OFF (Test flow 800 ml/min)		
		3 min	3 – 10 min	1 min
		for CDS:		
		5 min	5 – 10 min	1 min
		CPU1: DIP switch array 1, SW8 set to ON (Test flow 500 ml/min)		
		4 min	4 – 10 min	1 min
		for CDS:		
		6 min	6 – 10 min	1 min
	Hot-Disinf TIME	10 min	10 – 20 min	1 min
	Mandatory Rinse TIME	CPU1: DIP switch array 1, SW8 set to OFF (Test flow 800 ml/min)		
		15 min	15 – 30 min	1 min
		for ON-LINE-HDF:		
		20 min	20 – 30 min	1 min
		for ONLINE*plus*™:		
		17 min	17 – 30 min	1 min
		CPU1: DIP switch array 1, SW8 set to ON (Test flow 500 ml/min)		
		15 min	15 – 30 min	1 min
		for ON-LINE-HDF:		
		20 min	20 – 30 min	1 min
		for ONLINE*plus*™:		
		20 min	20 – 30 min	1 min
	CITRO-Mandat-Ri-Time	CPU1: DIP switch array 1, SW8 set to OFF (Test flow 800 ml/min)		
		10 min	10 – 25 min	1 min
		for ON-LINE-HDF:		
		20 min	20 – 25 min	1 min
		for ONLINE*plus*™:		
		17 min	17 – 25 min	1 min
		CPU1: DIP switch array 1, SW8 set to ON (Test flow 500 ml/min)		
		10 min	10 – 25 min	1 min
		for ON-LINE-HDF:		
		20 min	20 – 25 min	1 min
		for ONLINE*plus*™:		
		20 min	20 – 25 min	1 min
	INTEGRATED-HR Time	15 min	15 – 40 min	1 min

Continued on the next page

Menu item	Submenu	Default value	Selectable options
SETUP CLEANING PGM (Continued)			
Pgm COMBINATION only for 4008 E/B	**RINSE Pgm**	**PGM 1: –R–**	**PGM 1: –R–** **PGM 2: –R– endless**
	HOTRINSE Program	**PGM 1: –F–HR–C–**	**PGM 1: –F–HR–C–** **PGM 2: –F–HR–** **PGM 3: –IHR–** **PGM 4: –IHR–C–**
	DISINFECTION Pgm	**PGM 2: –F–HDIS–M–**	**PGM 1: –F–D–M–** **PGM 2: –F–HDIS–M–** **PGM 3: –F–D–M–HR–** **PGM 4: –F–HDIS–M–HR–**
DEFAULT Cleaning Pgm only for 4008 E/B		**PGM 1: –R–**	**PGM 1: –R–** **PGM 2: –R– endless** **PGM 1: –F–HR–C–** **PGM 2: –F–HR–** **PGM 3: –IHR–** **PGM 4: –IHR–C–** **PGM 1: –F–D–M–** **PGM 2: –F–HDIS–M–** **PGM 3: –F–D–M–HR–** **PGM 4: –F–HDIS–M–HR–**

Menu item	Submenu	Default value	Selectable options	
SETUP DILUTION		**1+34**	**1+34** **1+35.83 (NaCl 20)** **1+35.83 (NaCl 26)** **1+35.83 (Belgium)** **1+44 C** **1+44 ACF** **VARIABLE SETTING**	
		Default value	**Value range**	**Resolution**
	VARIABLE SETTING	– – – –	0.800 – 2.500 30.000 – 45.000 25 – 45 25 – 80	0.001 0.001 1 1

Menu item	Submenu	Default value	Selectable options
HDF-DILUTION only for ON-LINE-HDF (option)		**HDF–PRE–dilution**	**HDF–PRE–dilution** **HDF–POST–dilution**

Menu item	Submenu	Default value	Value range	Resolution
SET CONDUCT. LIMIT	**Cd Limit: 12.8 mS/cm**	12.8 mS/cm	12.8 – 14.0 mS/cm	0.1 mS/cm

Menu item	Submenu	Default value	Selectable options
INFO SOUND (C-PGM)		**Info-Sound: ON**	**Info-Sound: ON** **Info-Sound: OFF**

Menu item	Submenu	Default value	Value range	Resolution
SET DIAL PARAMETERS				
SET Flow Parameter	**Flow[ml/min]: 500**	500 ml/min	300 / 500 / 800 ml/min or value set in Dial	
SET Temp. Parameter	**Temp.[°C]: 37.0**	37 °C	35 – 39 °C or value set in Dial	0.5 °C
SET Na/Bic Parameter	**Base Na+ 135mmol**	135 mmol	125 – 150 mmol or value set in Dial	1 mmol
	Prescr. Na+ 135mmol	135 mmol	125 – 150 mmol (±13 mmol around the basic value)	1 mmol
	Bicarbonate ±0mmol	0 mmol	–8 to +8 mmol or value set in Dial	1 mmol
	Limit Na/Base: 13 mmol only for 4008 H/S	13 mmol	0 – 13 mmol however prescribed Na^+, Base Na^+	1 mmol

Menu item	Submenu	Default value	Selectable options
DIALYSIS TIME only for 4008 E/B		**Effect. dialysis time**	**Effect. dialysis time** **UF time**

Menu item	Submenu	Default value	Selectable options
CALC.CUMUL.BLOOD-VOL		**during seq DIAL: YES**	**during seq DIAL: YES** **during seq DIAL: NO**

Menu item	Submenu	Default value	Selectable options
HAEMOGLOBIN UNIT only for BVM (option) and 4008 H/S		**g/dl**	**g/dl** **mmol/l**

Menu item	Submenu	Default value	Value range	Resolution
OCM SETTINGS only for OCM (option) and 4008 H/S	**OCM MEASUREMENT**	**OCM Measurement: OFF**	**OCM Measurement: OFF** **OCM Measurement: ON**	
	OCM ZERO MEASUREMENT	**OCM Zero Measure:ON**	**OCM Zero Measure:ON** **OCM Zero Measure:OFF**	
		Default value	**Value range**	**Resolution**
	OCM MEASURE DEL.TIME	4008 H with advanced hydraulics: 15 sec	1 – 70 sec	1 sec
		4008 H with ONLINE*plus*™: 65 sec	1 – 70 sec	1 sec
		4008 S with advanced hydraulics: 18 sec	1 – 70 sec	1 sec
		4008 S with ONLINE*plus*™: 67 sec	1 – 70 sec	1 sec
	OCM CLR CORR FACTOR	1.000	0.500 – 1.200	0.001
	OCM BASELINE DIFF.	0.015 mS/cm	0.001 – 0.5 mS/cm	0.001 mS/cm
	OCM INTEGRAL VALUE	55	20 – 120	1
	OCM KT/V WARNLEVEL	85 %	0 – 99 %	1 %

Menu item	Submenu	Default value	Selectable options
AUTOM. SN-START		**autom. SN: OFF**	**autom. SN: OFF** **autom. SN: ON**

Menu item	Submenu	Default value	Selectable options
ACTIV. MONIT_NTC109		**MONIT_NTC109: YES**	**MONIT_NTC109: YES** **MONIT_NTC109: NO**

Menu item	Submenu	Default value	Selectable options
ACTIV. STD UF-DATA		**std UF-DATA: NO**	**std UF-DATA: NO** **std UF-DATA: YES**

Menu item	Submenu	Default value	Value range	Resolution
SET STD. PRIME-TIME	**Prime-Time = 2min**	2 min	1 – 5 min	1 min

Menu item	Submenu	Default value	Selectable options
SOUND I/O-SWITCH		**I/O-Warnsound: ON**	**I/O-Warnsound: ON** **I/O-Warnsound: OFF**

Menu item	Submenu	Default value	Selectable options
SET KEY-CLICK only for 4008 H/S		**key-click: ON**	**key-click: ON** **key-click: OFF**

Menu item	Submenu	Default value	Selectable options
BPR/UFR-WARNING		**BPR/UFR-Warning: ON**	**BPR/UFR-Warning: ON** **BPR/UFR-Warning: OFF**

Menu item	Submenu	Default value	Value range	Resolution
SET RINSE-VOLUME	**RINSE-VOL: 1000 ml**	1000 ml	0 – 5000 ml	100 ml

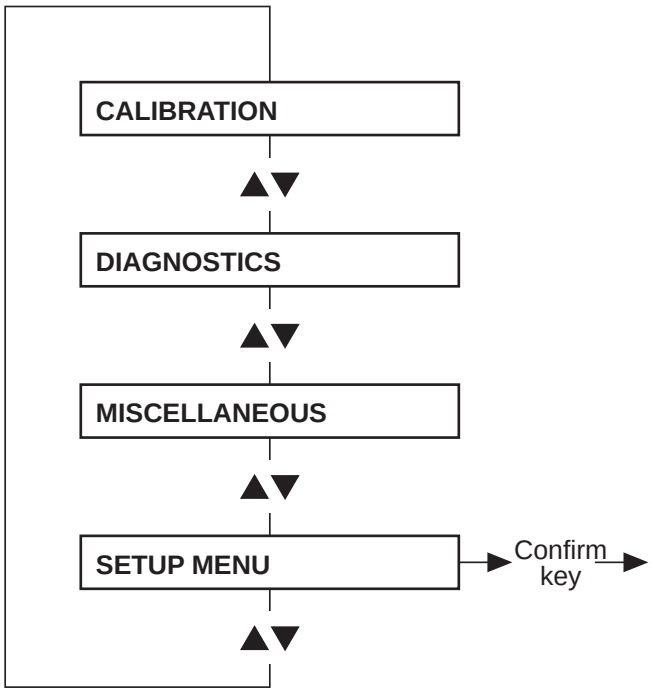
Menu item	Submenu	Default value	Selectable options
T1-TEST AUTOSTART		**T1-T. Autostart: OFF**	**T1-T. Autostart: OFF** **T1-T. Autostart: ON**

Menu item	Submenu	Default value	Value range	Resolution
ONLINE plus SETTINGS only for ONLINE*plus*™ (option)				
ONLINE plus HD	**SET UF-Volume F/R**	0 ml	0 – 1000 ml	100 ml
	SET Rinsing Volume	1000 ml	0 – 5000 ml	100 ml
ONLINE plus HDF	**SET UF-Volume F/R**	500 ml	0 – 1000 ml	100 ml
	SET Rinsing Volume	1000 ml	0 – 5000 ml	100 ml
	SET Substit.-Volume	12 l	0 – 210 l	1 l
ONLINE plus HF	**SET UF-Volume F/R**	1000 ml	0 – 5000 ml	100 ml
	SET Rinsing Volume	1000 ml	0 – 5000 ml	100 ml
	SET Substit.-Volume	20 l	0 – 210 l	1 l
ONLINE plus MISC.	**SET Reinf.-Volume**	240 ml	90 – 480 ml	30 ml

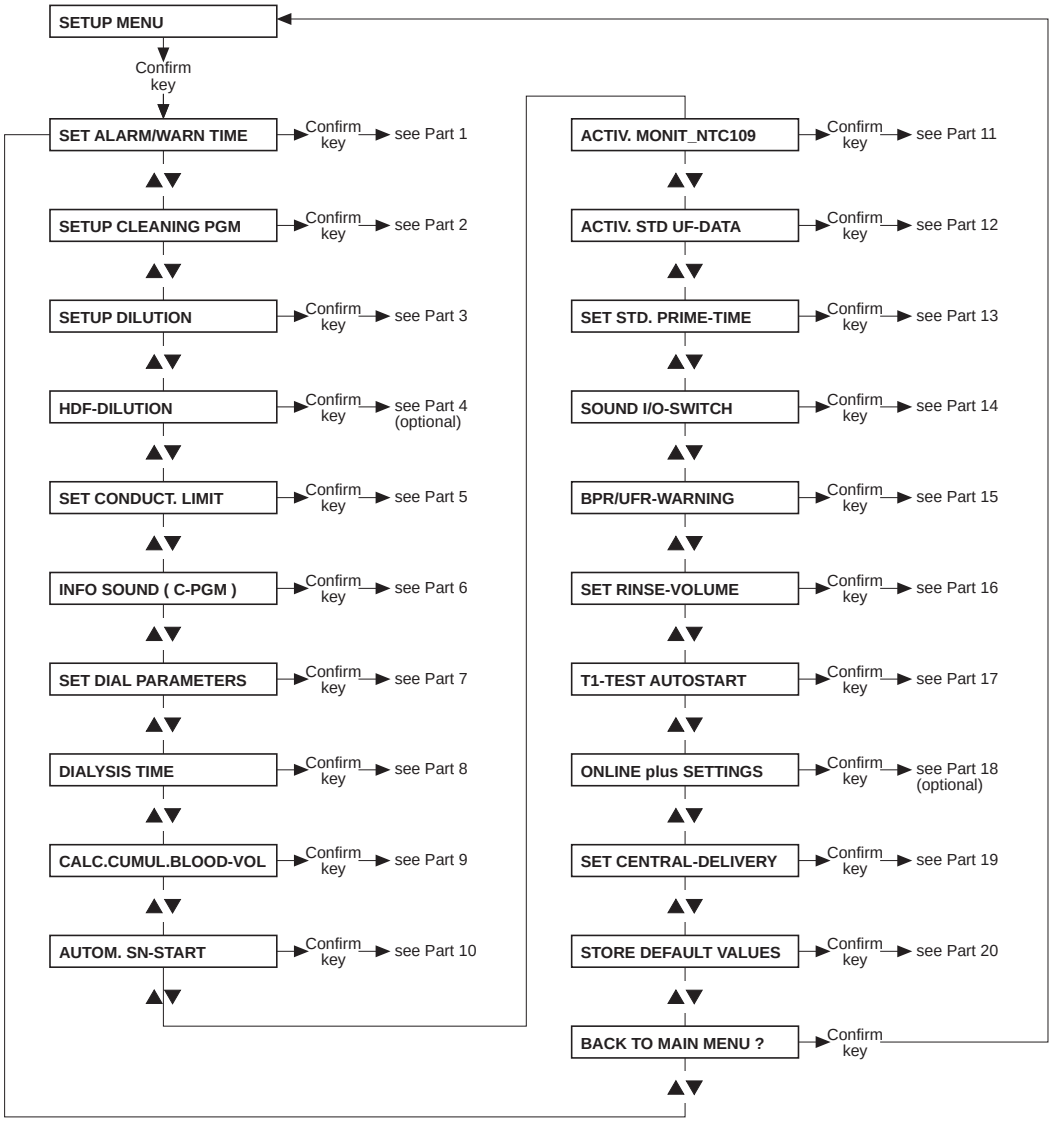
Menu item	Submenu	Default value	Selectable options
SET CENTRAL-DELIVERY		**NO central-delivery**	**NO central-delivery** **central Bic** **central Acid** **central Acid + Bic** **centr acetate-supply**

Menu item	Adjustment
STORE DEFAULT VALUES	**Press OVERRIDE** only for 4008 E/B **Press ALARMTONE MUTE** only for 4008 H/S

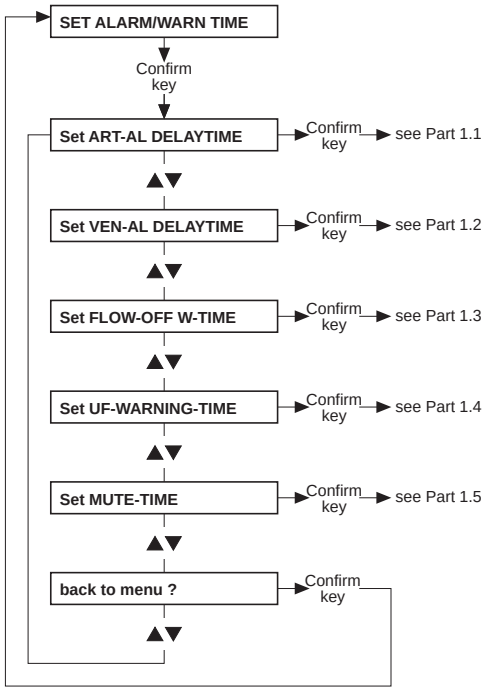
6.2 Overview


CALIBRATION
DIAGNOSTICS
MISCELLANEOUS
SETUP MENU
Confirm key

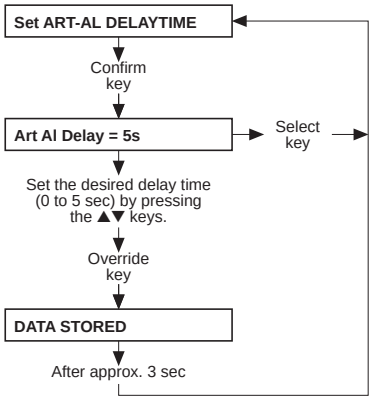
6.3 Main menu 4008 E/B Rev. 5.2



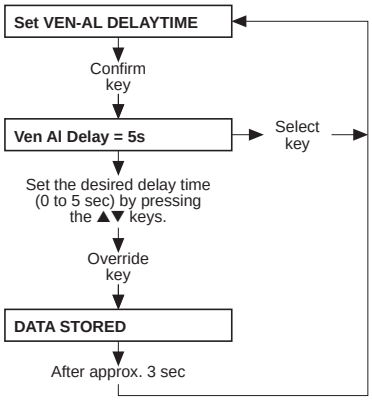
- **Part 1: Set alarm and warning time**



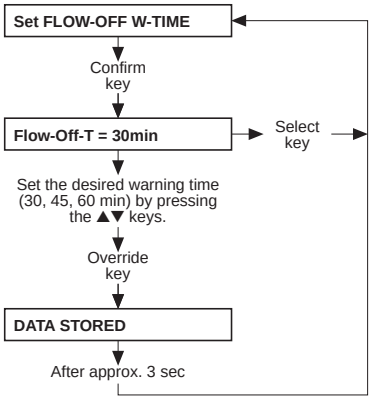
- **Part 1.1: Set the delay time of the arterial alarm**



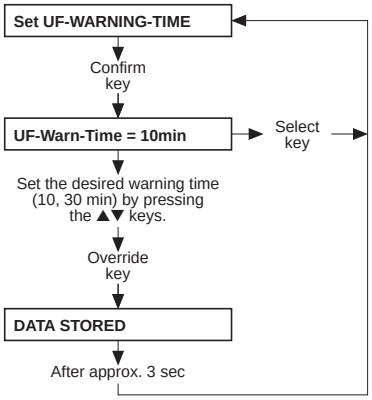
- **Part 1.2: Set the delay time of the venous alarm**



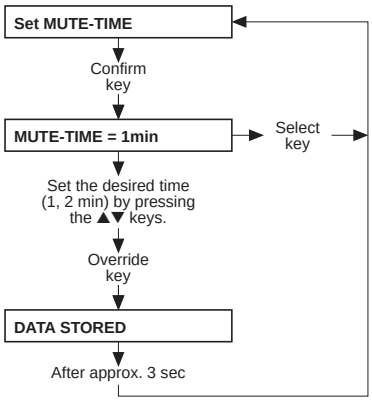
- **Part 1.3: Set the flow-off warning time**



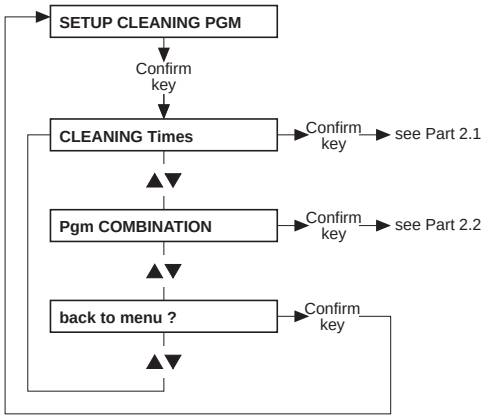
- **Part 1.4: Set the UF warning time**



- **Part 1.5: Set the mute time**

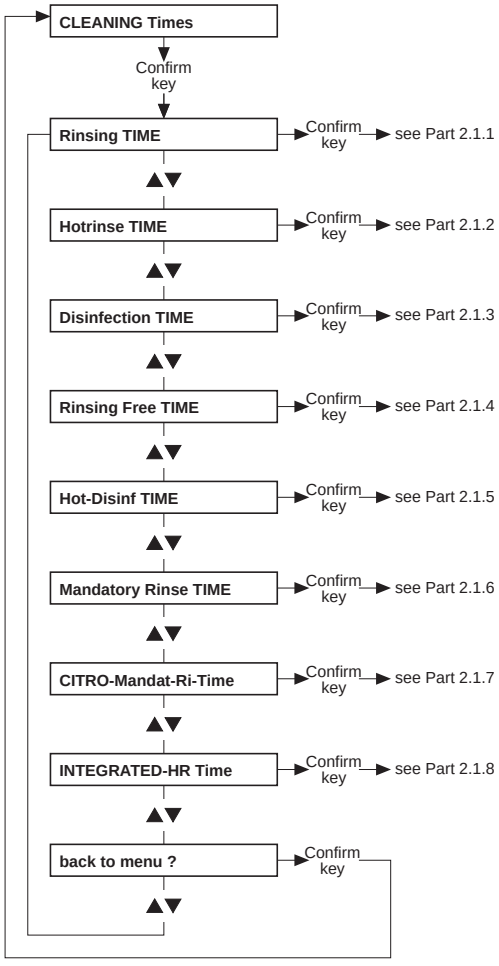


- **Part 2: Set up the cleaning program**

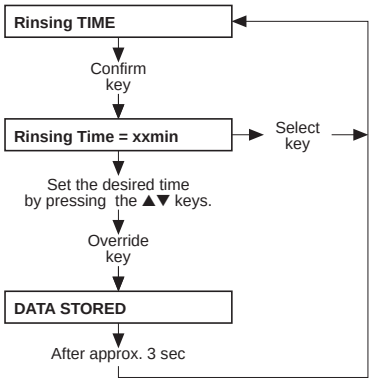


- **Part 2.1: Cleaning times**

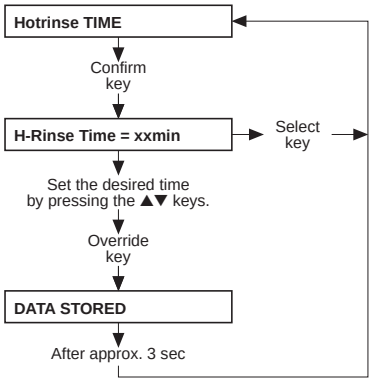
Note
The default values and the adjustable range for the cleaning times are not indicated, as they depend on the particular machine options.



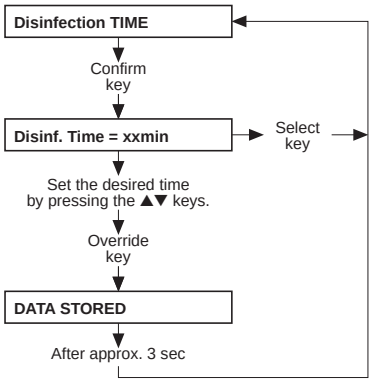
- **Part 2.1.1: Rinsing time**



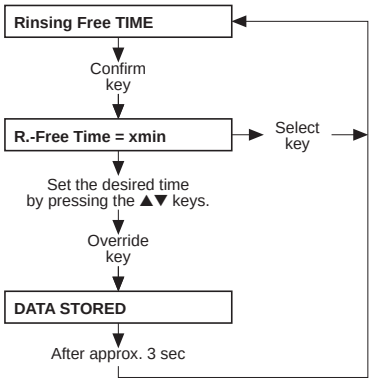
- **Part 2.1.2: Hot rinsing time**



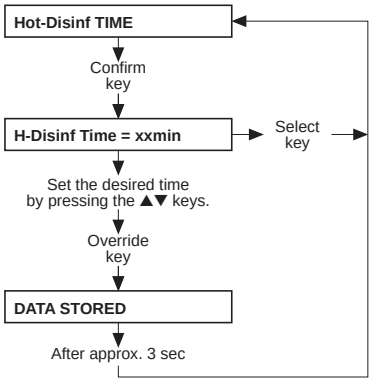
- **Part 2.1.3: Disinfection time**



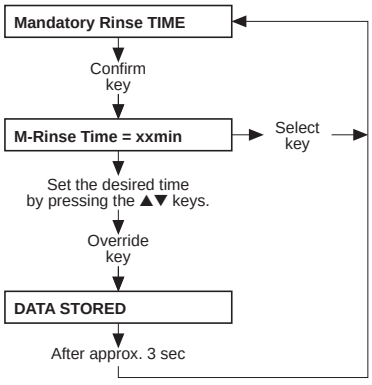
- **Part 2.1.4: Rinsing free time**



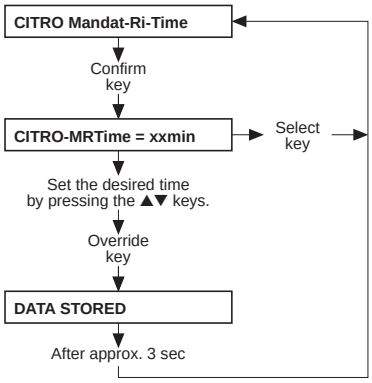
- **Part 2.1.5: Hot disinfection time**



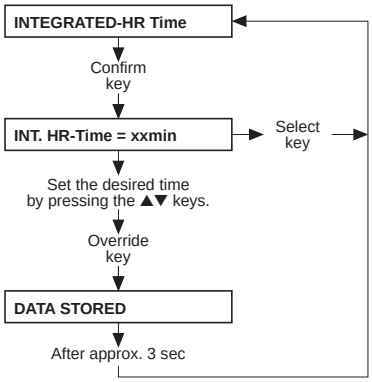
- **Part 2.1.6: Mandatory rinse time**



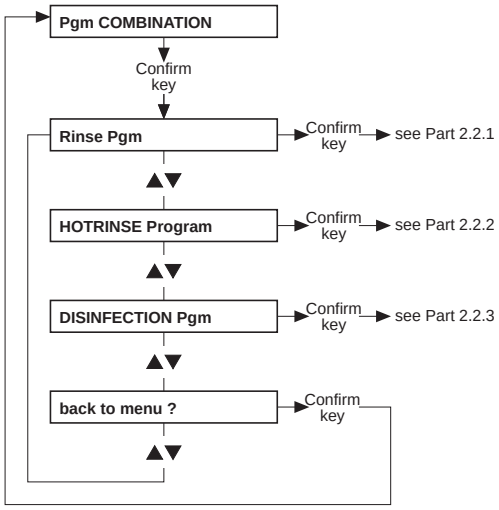
- **Part 2.1.7: Citro mandatory rinse time**



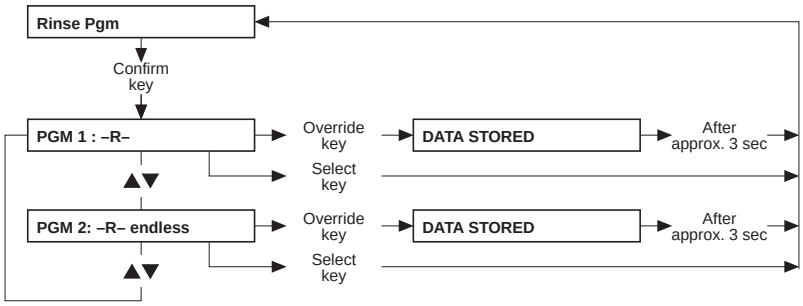
- **Part 2.1.8: Integrated hot rinse time**



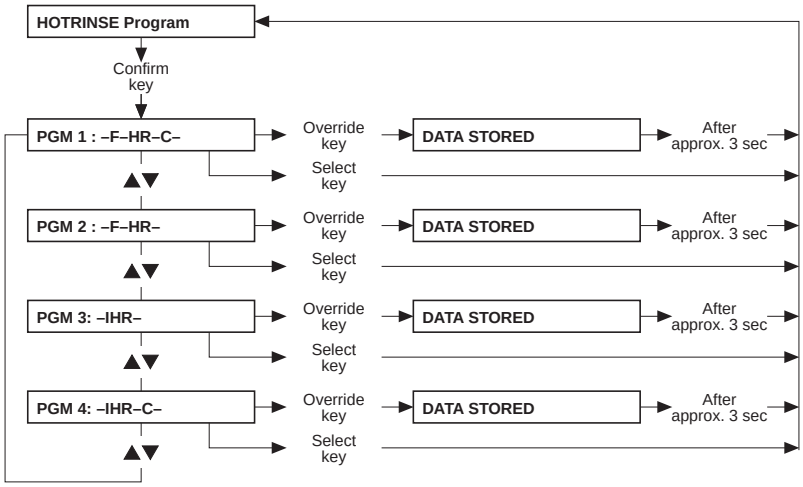
- **Part 2.2: Cleaning program combination**



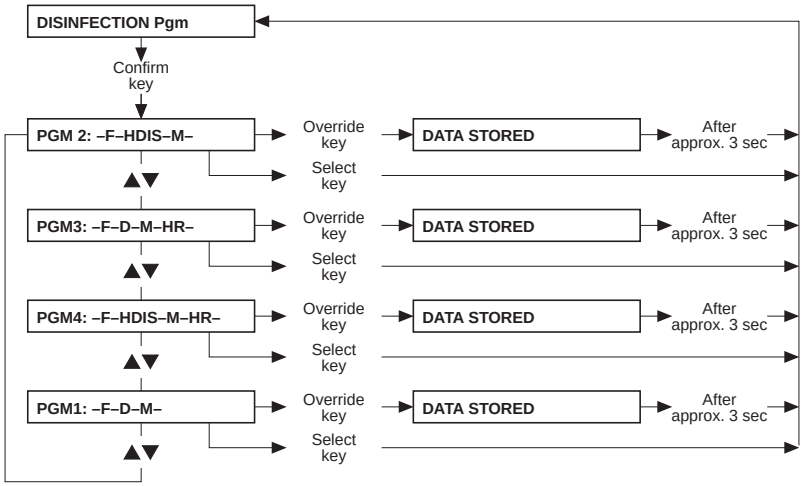
- **Part 2.2.1: Rinse program**



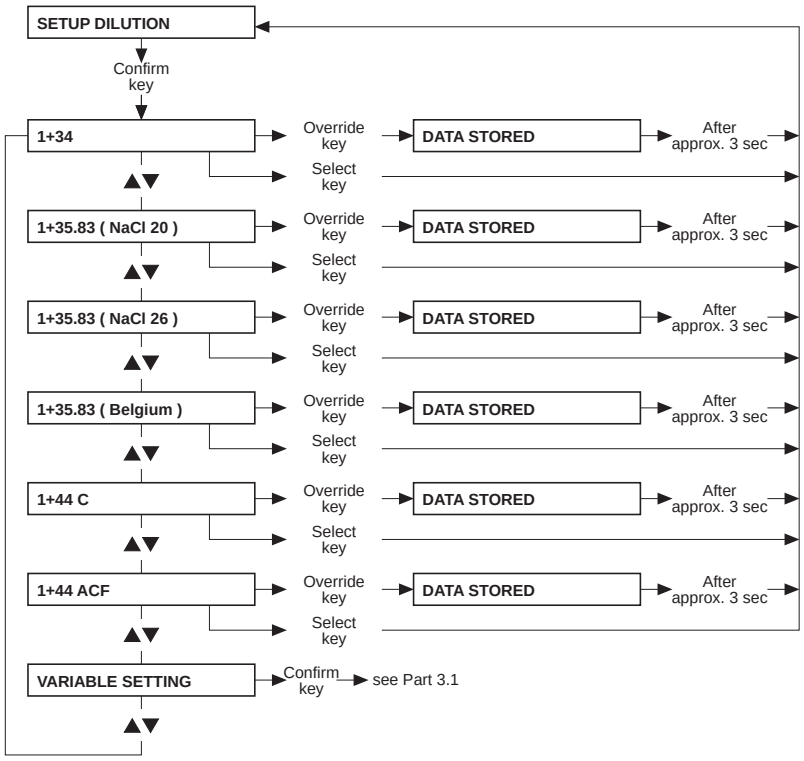
- **Part 2.2.2: Hot rinse program**



- **Part 2.2.3: Disinfection program**

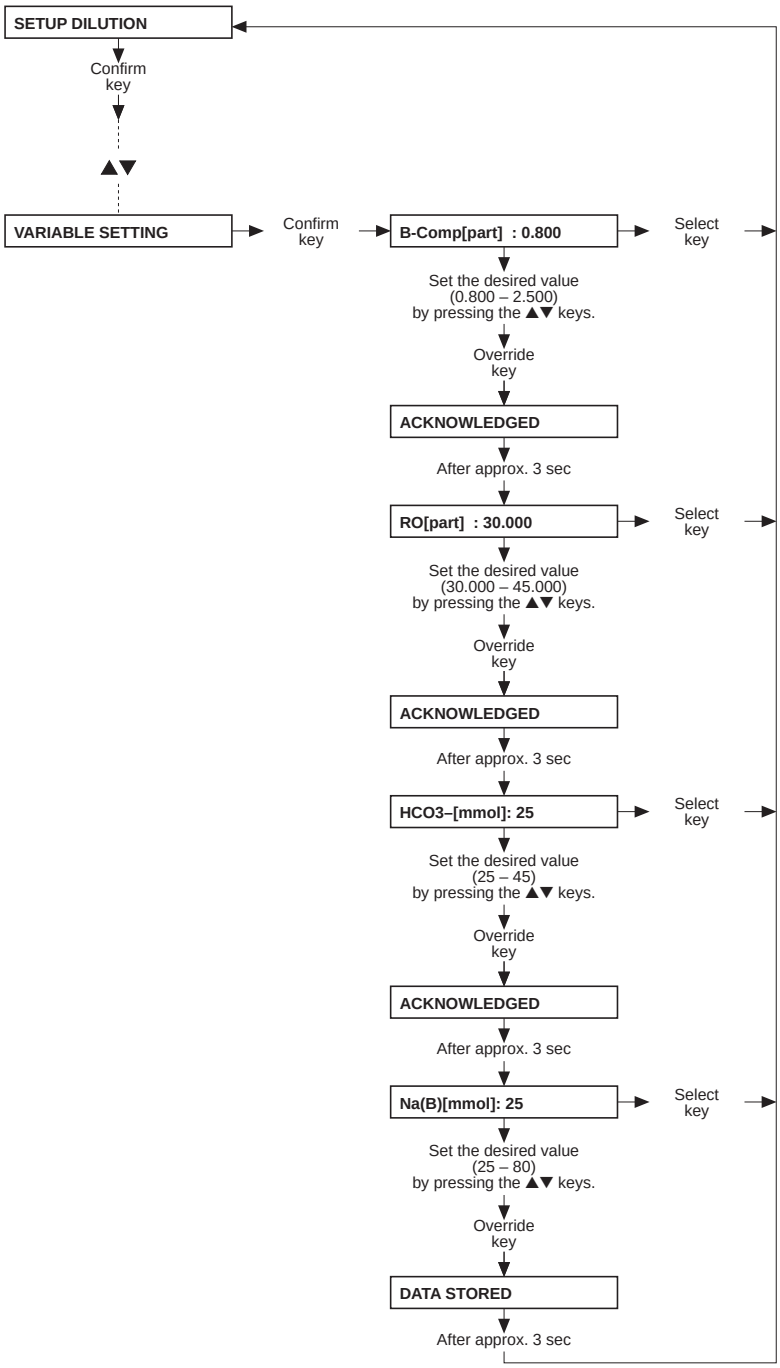


- **Part 3: Dilution**

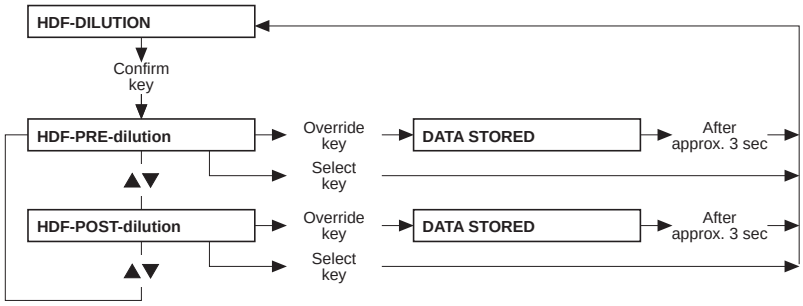


- **Part 3.1: Programmable mixing ratio (dilution)**

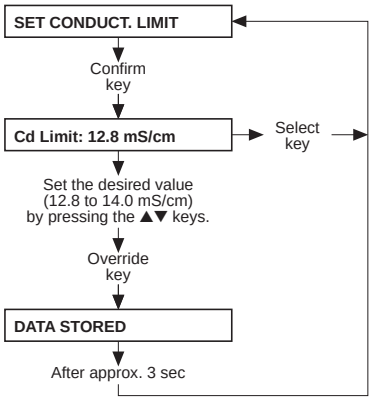
Caution
The operator or technician is informed about his duty of care to enter the component parameters and settings correctly or to check them and to set the CD alarm window to the expected conductivity value.
When using the programmable mixing ratio (dilution), make sure to use the right concentrate.
Using a bi*b*ag® in combination with the programmable mixing ratio (dilution) is not allowed.
Only enter authorized, programmable mixing ratios (dilutions).



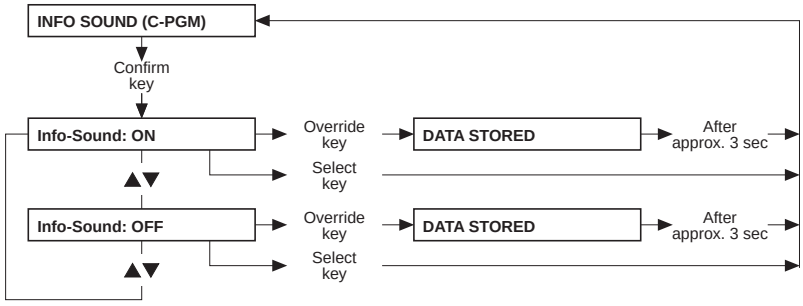
- **Part 4: Set HDF dilution – only for ON-LINE-HDF (option)**



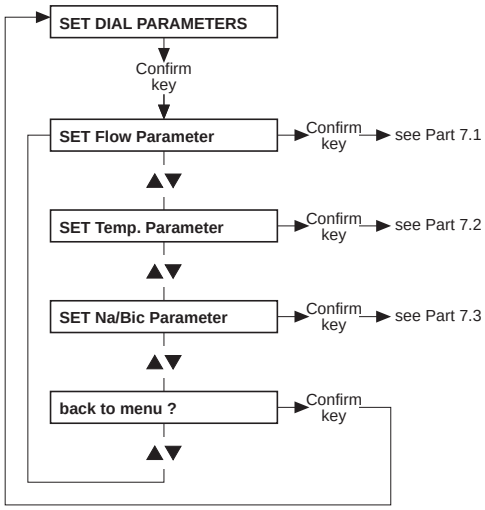
- **Part 5: Set conductivity limit**



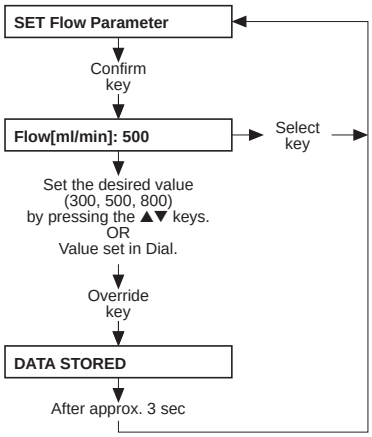
- **Part 6: Infosound cleaning program**



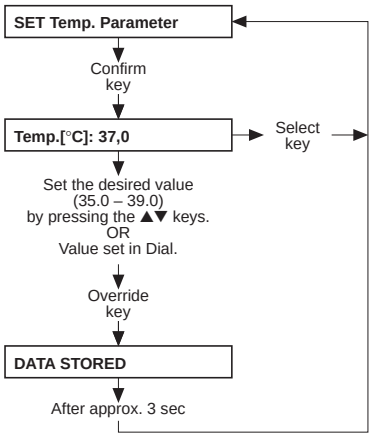
- **Part 7: Set dialysis parameters**



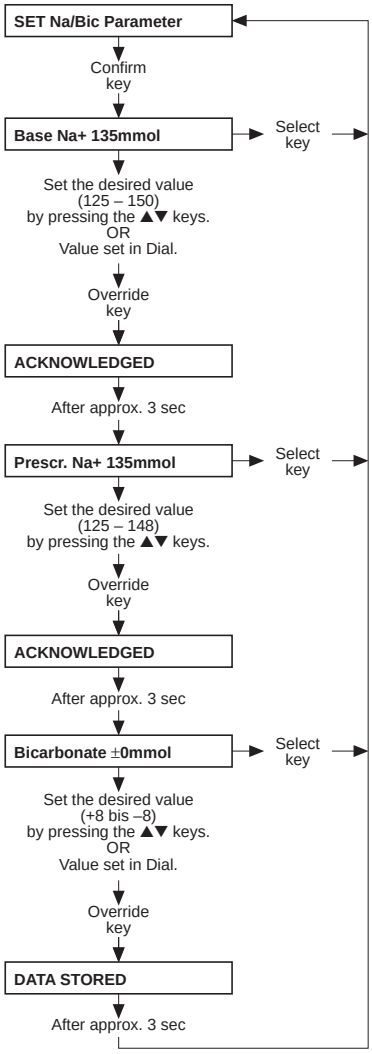
- **Part 7.1: Set dialysate flow**



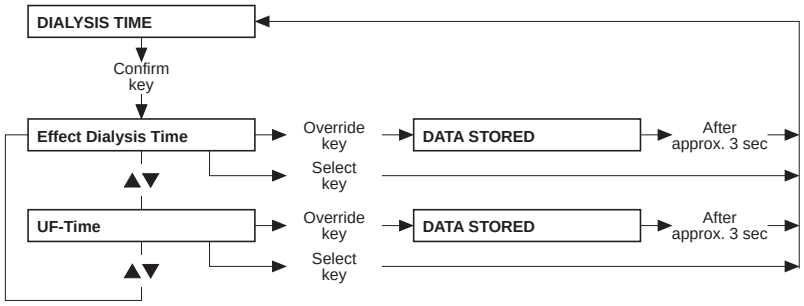
- **Part 7.2: Set dialysate temperature**



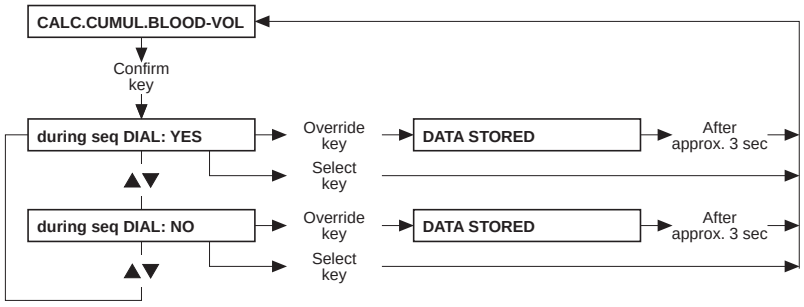
- **Part 7.3: Set Na/Bic**



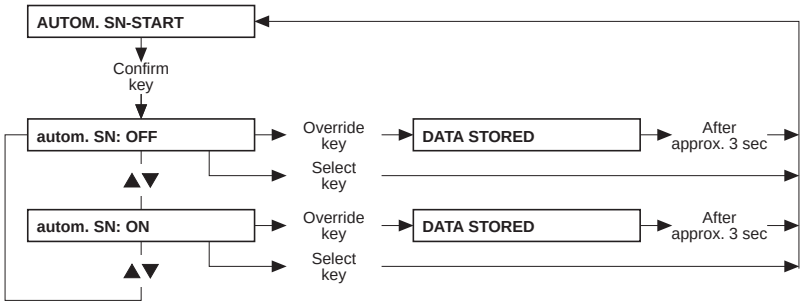
- **Part 8: Dialysis time**



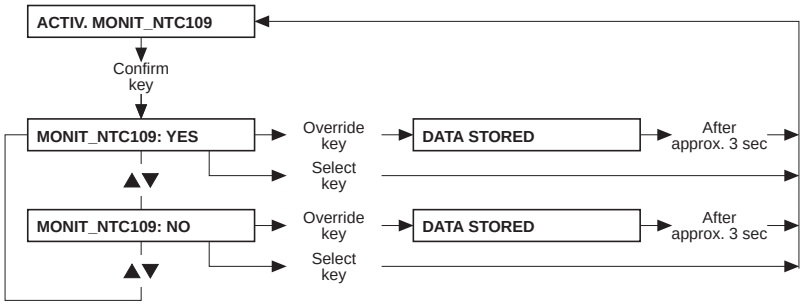
- **Part 9: Cumulated blood volume**



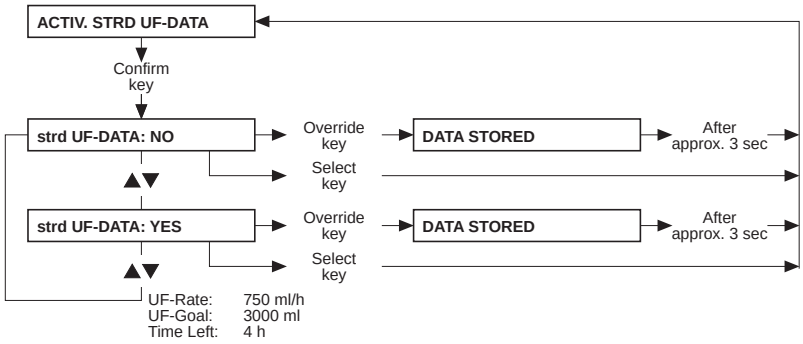
- **Part 10: Automatic single needle start**



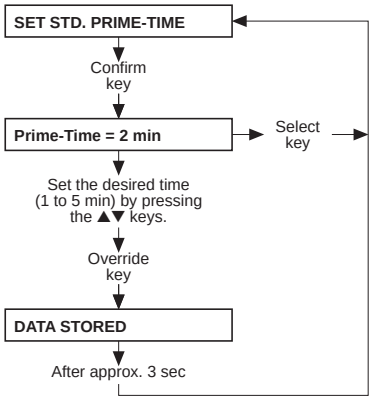
- **Part 11: Activation of Monit_NTC 109**



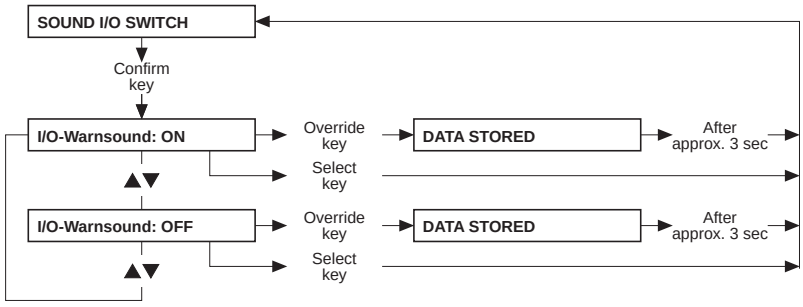
- **Part 12: Activation of standard UF data**



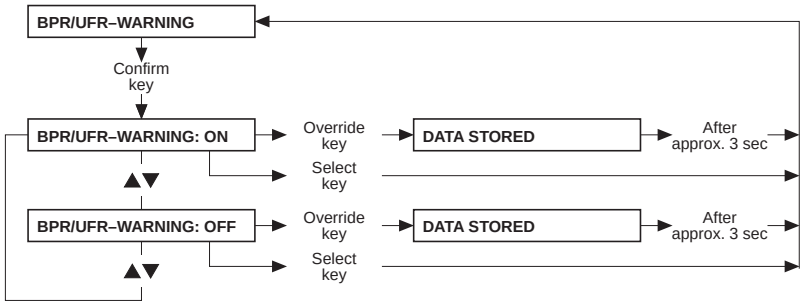
- **Part 13: Set priming time**



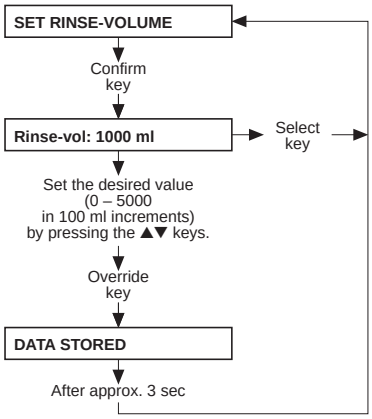
- **Part 14: Warning sound I/O key**



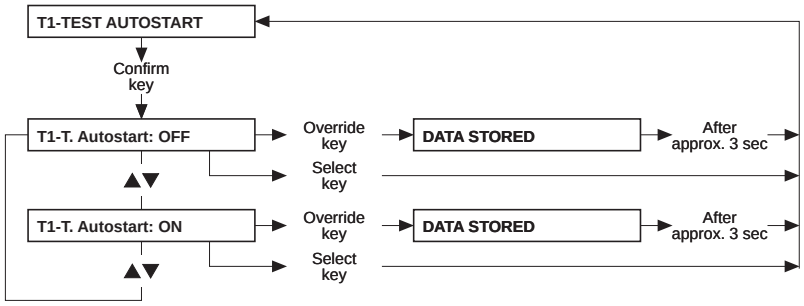
- **Part 15: Set BPR/UFR warning**



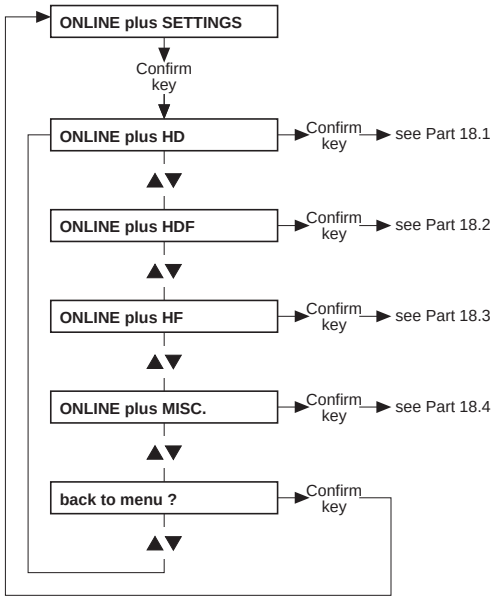
- **Part 16: Set rinse volume**



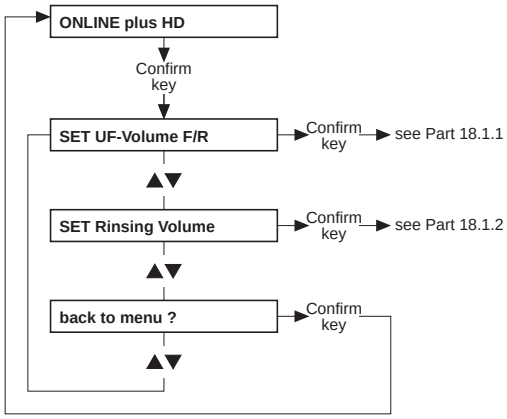
- **Part 17: T1 test autostart**



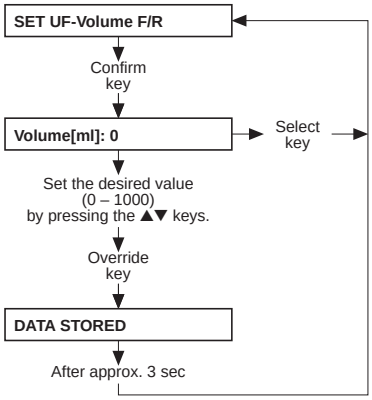
- **Part 18: ONLINE*plus*™ settings – only for ONLINE*plus*™ (option)**



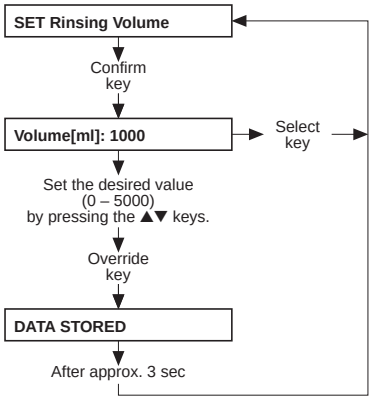
- **Part 18.1: ONLINE*plus*™ HD**



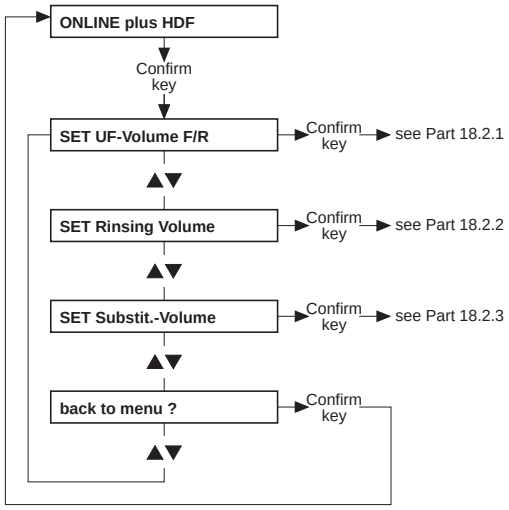
- **Part 18.1.1: Set UF volume (filling/rinsing)**



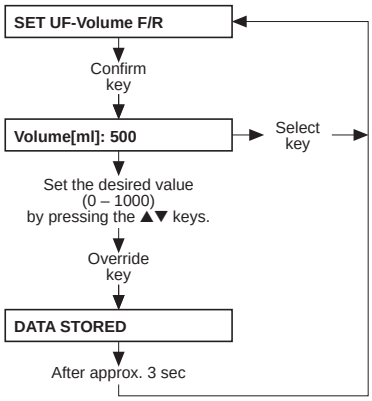
- **Part 18.1.2: Set rinse volume**



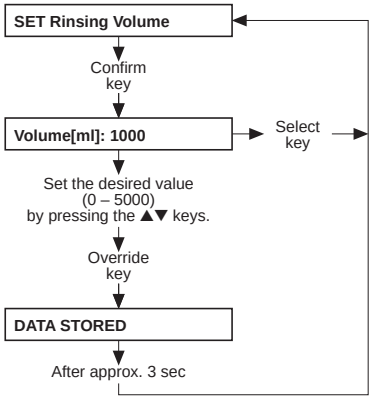
- **Part 18.2: ONLINE*plus*™ HDF**



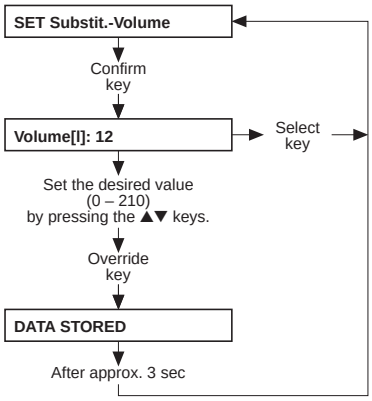
- **Part 18.2.1: Set UF volume (filling/rinsing)**



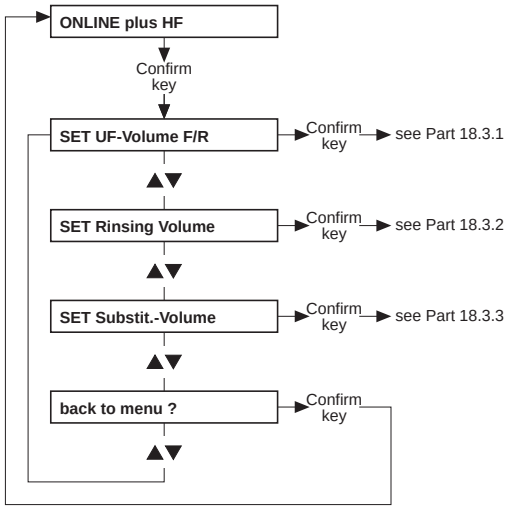
- **Part 18.2.2: Set rinse volume**



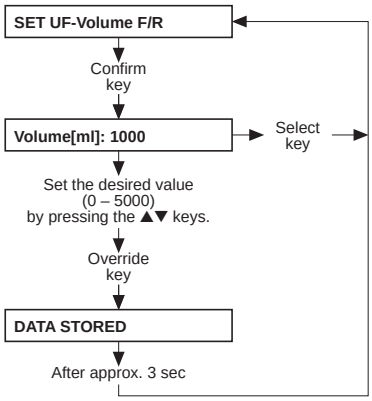
- **Part 18.2.3: Set substituate volume**



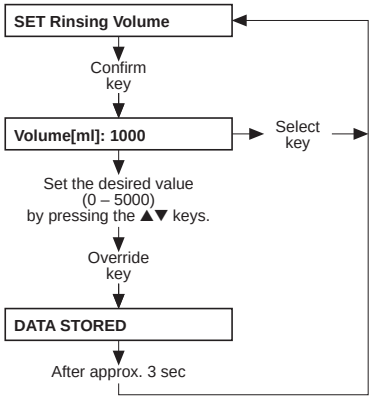
- **Part 18.3: ONLINE*plus*™ HF**



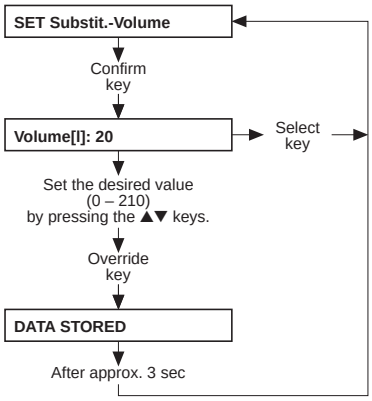
- **Part 18.3.1: Set UF volume (filling/rinsing)**



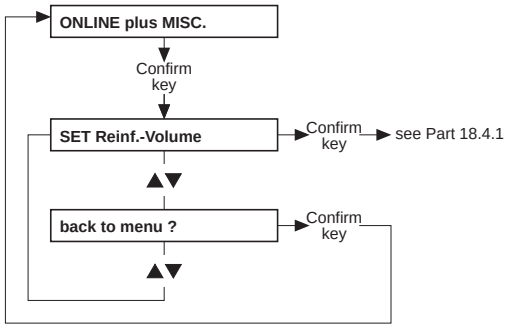
- **Part 18.3.2: Set rinse volume**



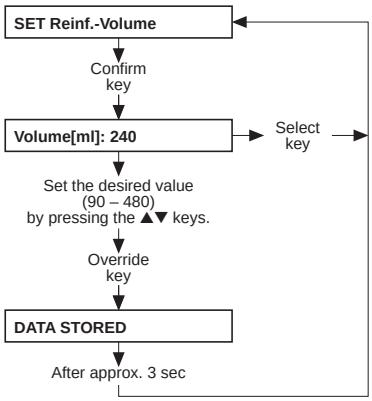
- **Part 18.3.3: Set substituate volume**



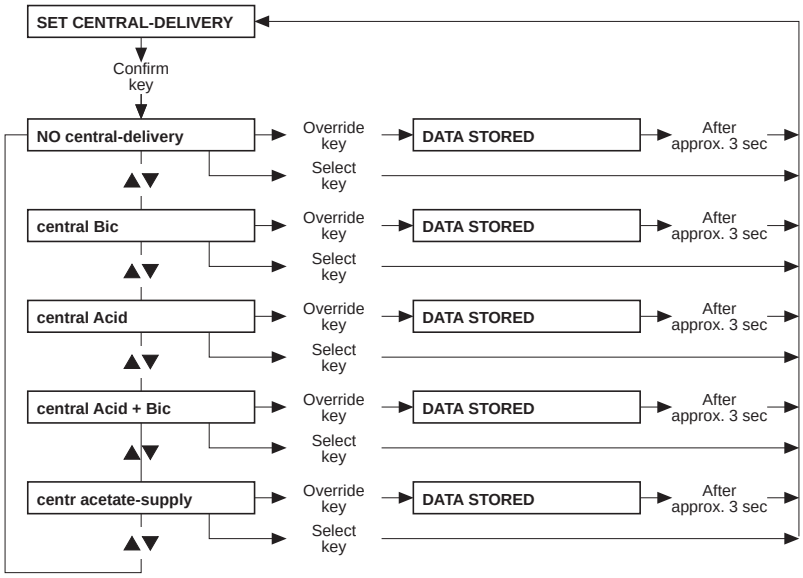
- **Part 18.4: ONLINE*plus*™ miscellaneous**



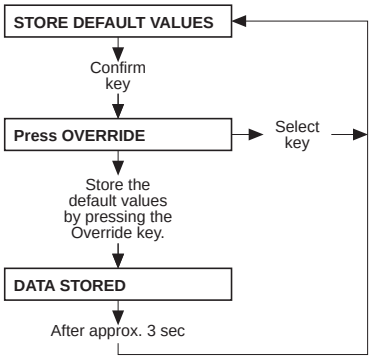
- **Part 18.4.1: Set reinfusion volume**



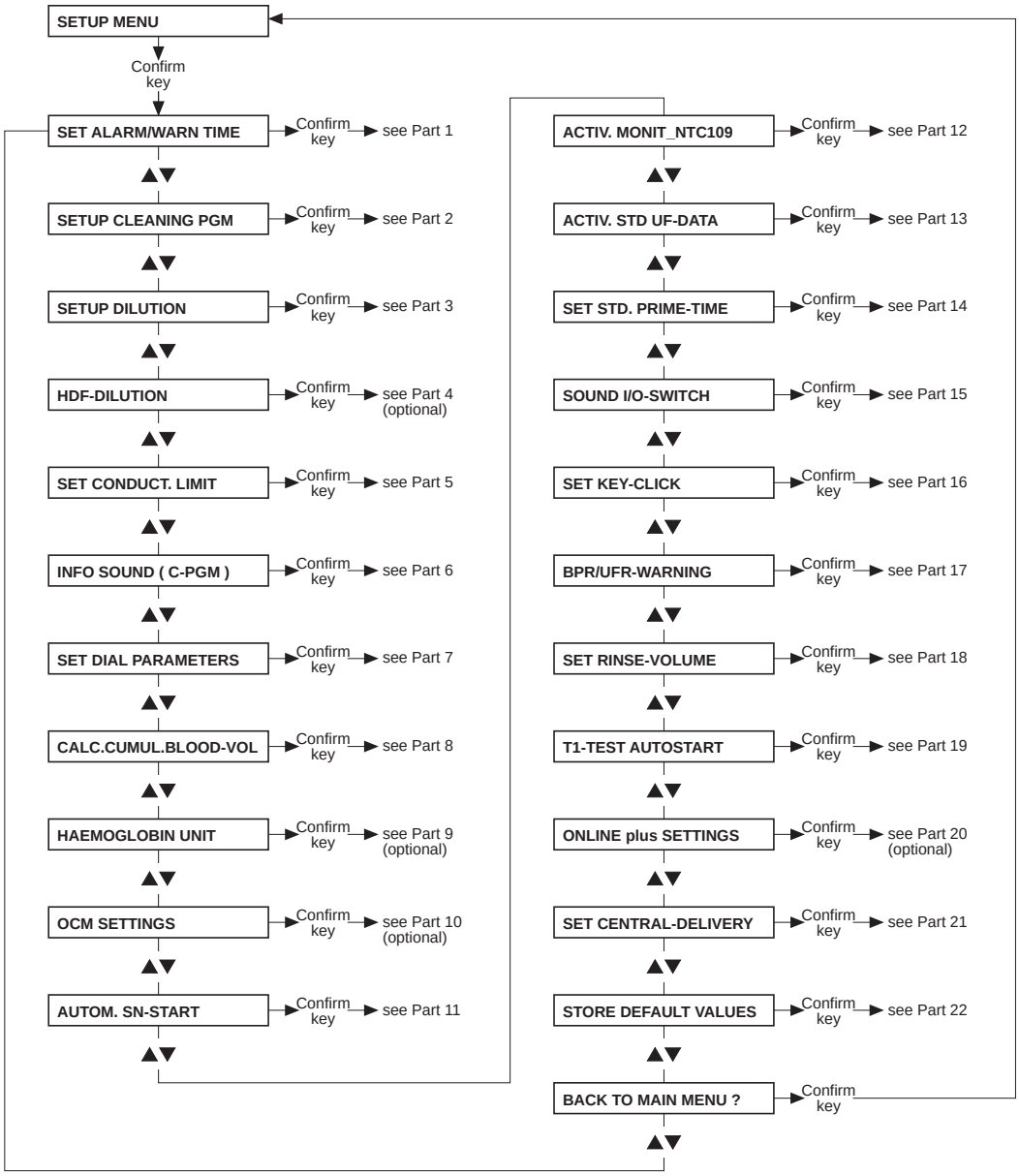
- **Part 19: Set parameters for central delivery system**



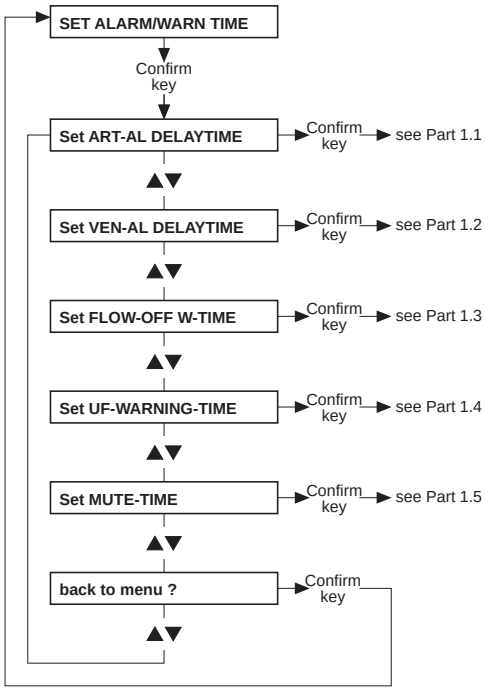
- **Part 20: Store default values**



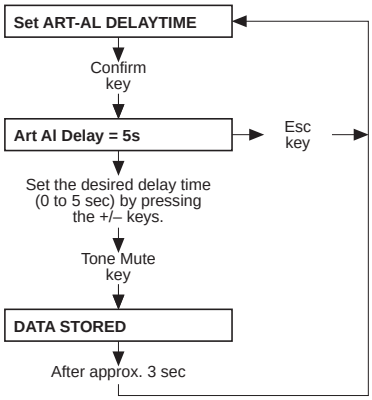
6.4 Main menu 4008 H/S Rev. 4.3



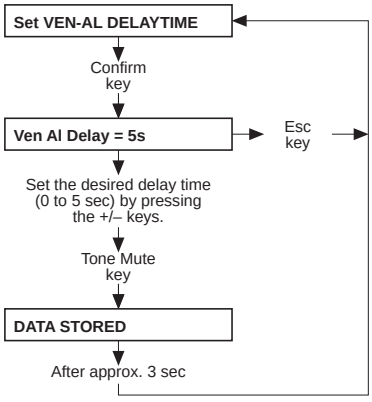
- **Part 1: Set alarm and warning time**



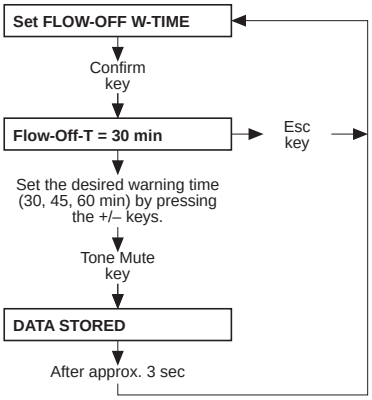
- **Part 1.1: Set the delay time of the arterial alarm**



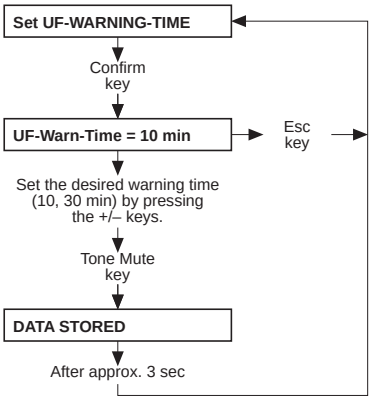
- **Part 1.2: Set the delay time of the venous alarm**



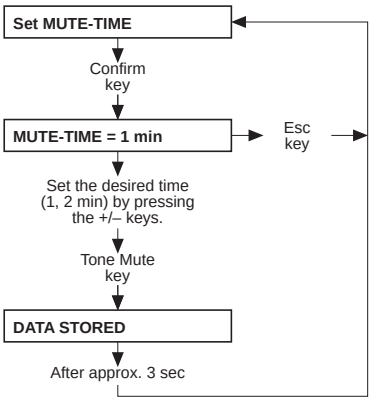
- **Part 1.3: Set flow-off warning time**



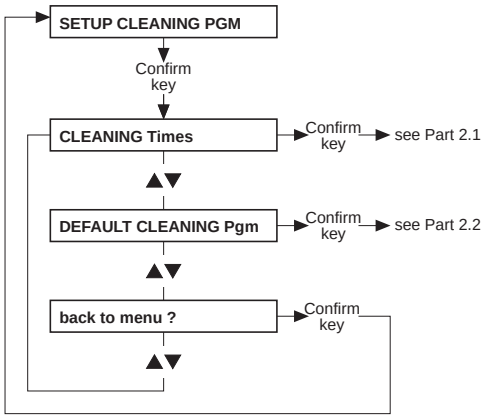
- **Part 1.4: Set UF warning time**



- **Part 1.5: Set mute time**

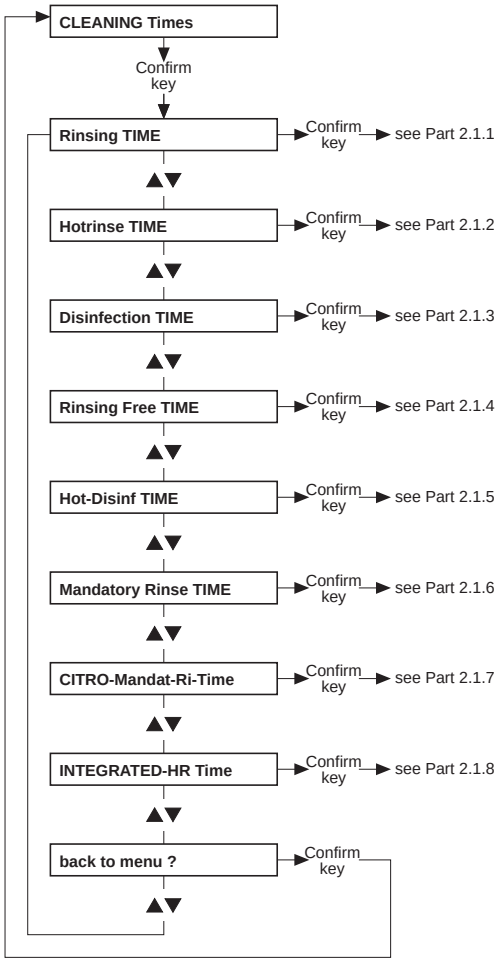


- **Part 2: Set up the cleaning program**

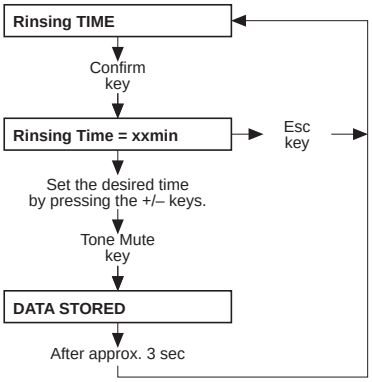


- **Part 2.1: Cleaning times**

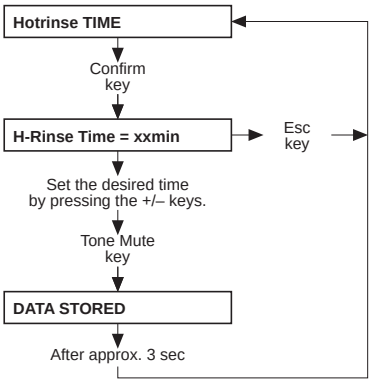
Note
The default values and the adjustable range for the cleaning times are not indicated, as they depend on the particular machine options.



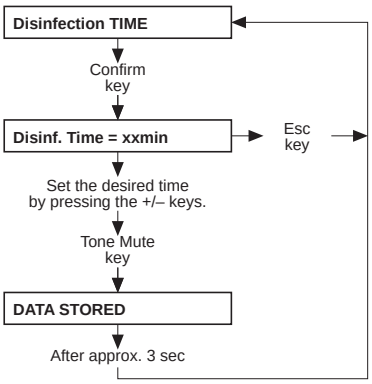
- **Part 2.1.1: Rinsing time**



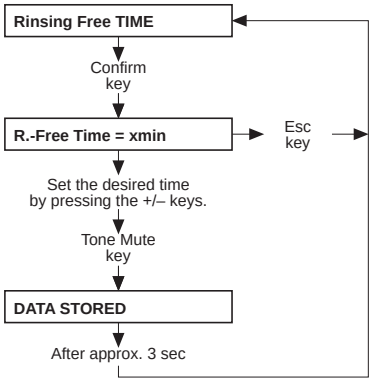
- **Part 2.1.2: Hot rinsing time**



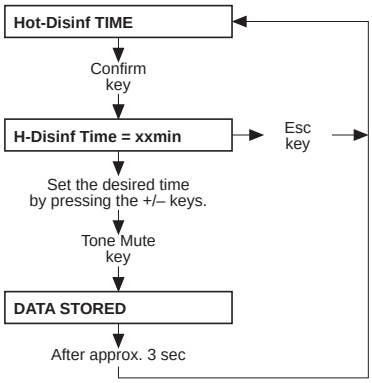
- **Part 2.1.3: Disinfection time**



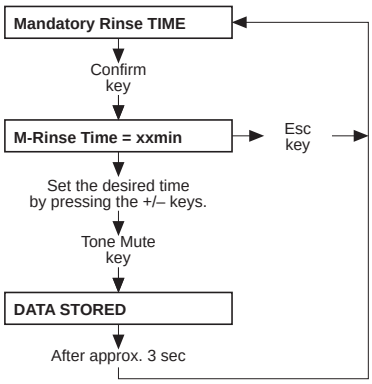
- **Part 2.1.4: Rinsing free time**



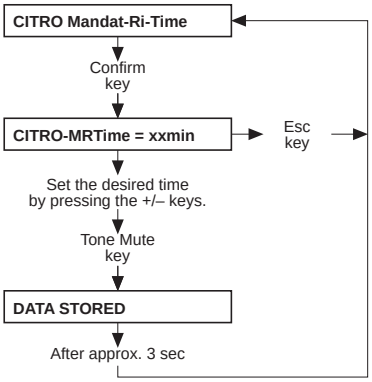
- **Part 2.1.5: Hot disinfection time**



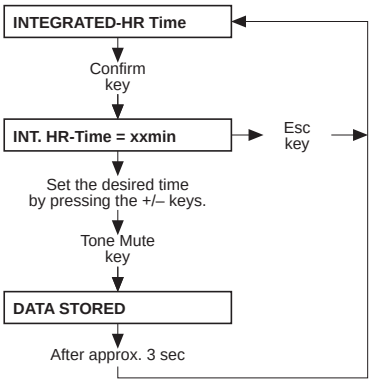
- **Part 2.1.6: Mandatory rinse time**



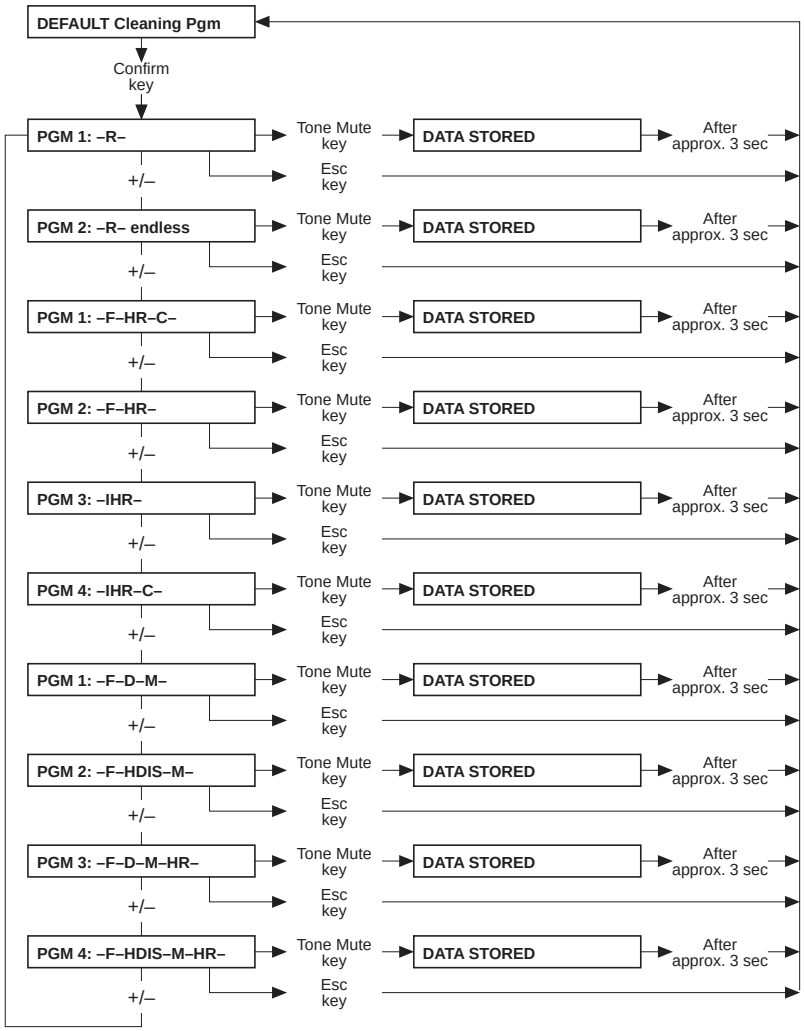
- **Part 2.1.7: Citro mandatory rinse time**



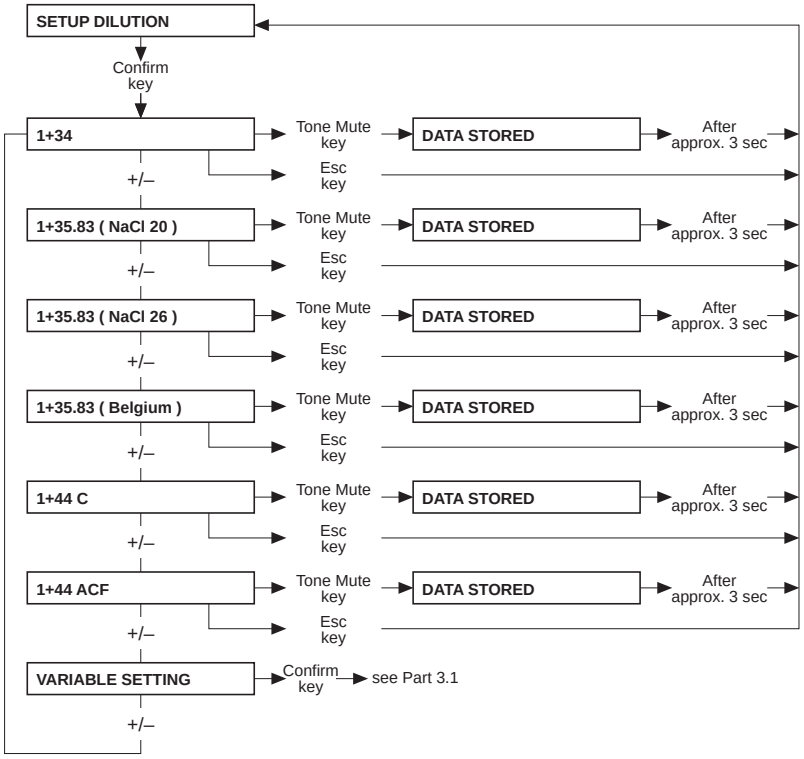
- **Part 2.1.8: Integrated hot rinsing time**



- **Part 2.2: Set Default cleaning program**

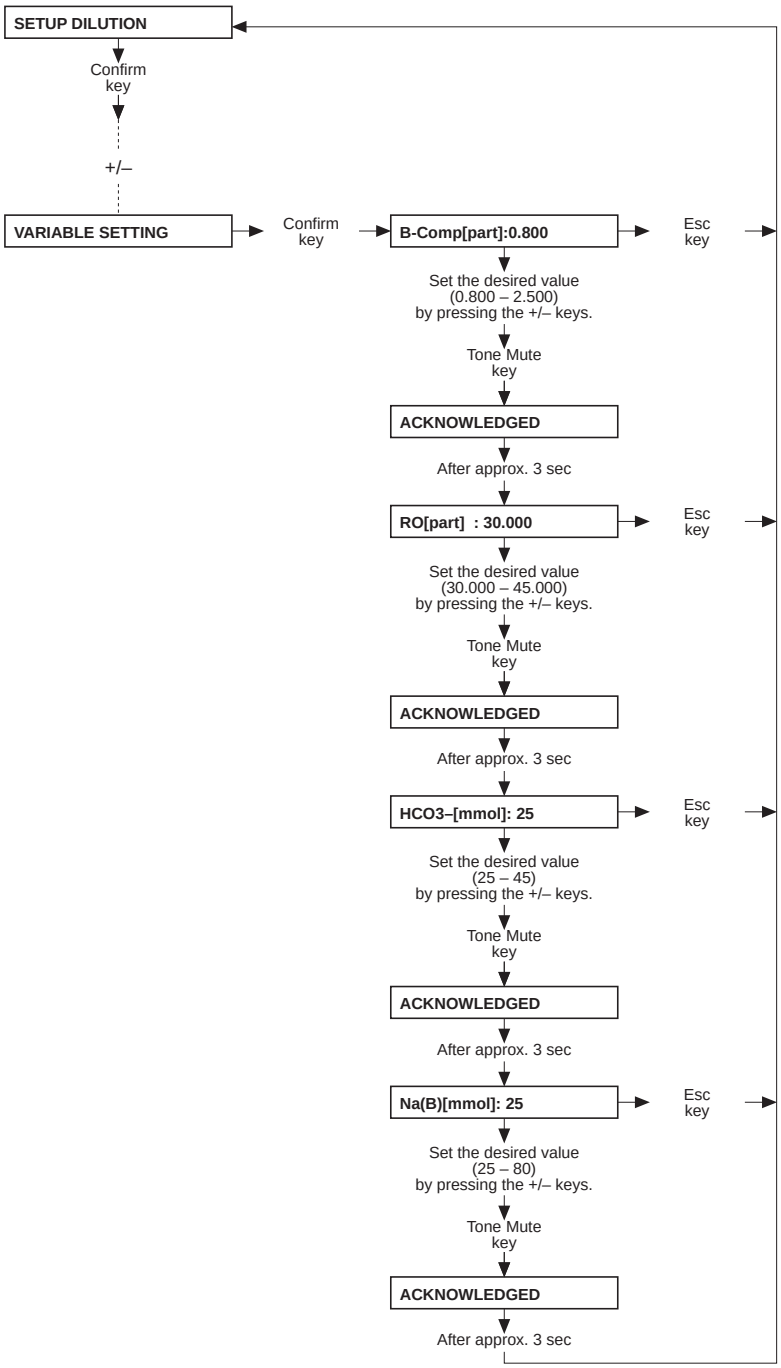


- **Part 3: Dilution**

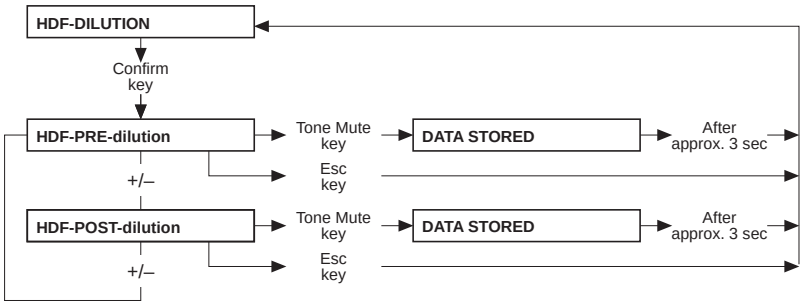


- **Part 3.1: Programmable mixing ratio (dilution)**

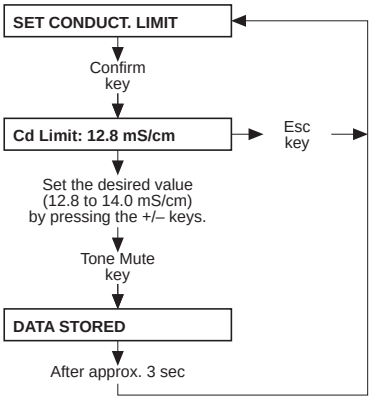
Caution
The operator or technician is informed about his duty of care to enter the component parameters and settings correctly or to check them and to set the CD alarm window to the expected conductivity value.
When using the programmable mixing ratio (dilution), make sure to use the right concentrate.
Using a bi*b*ag® in combination with the programmable mixing ratio (dilution) is not allowed.
Only enter authorized, programmable mixing ratios (dilutions).



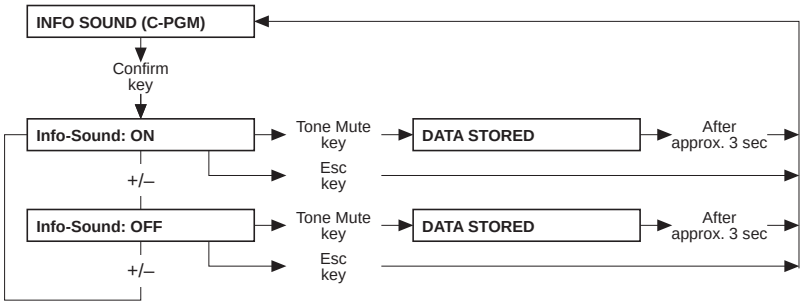
- **Part 4: Set HDF dilution – only for ON-LINE-HDF (option)**



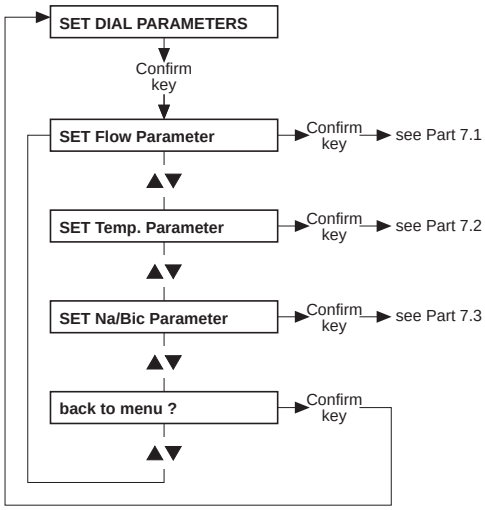
- **Part 5: Set conductivity limit**



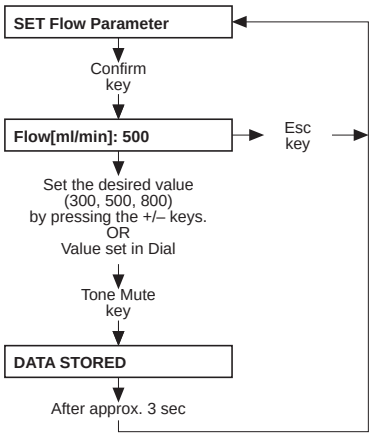
- **Part 6: Infosound cleaning program**



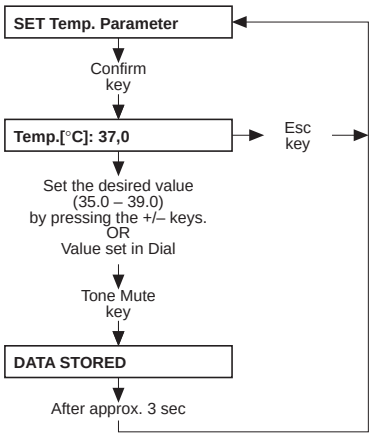
- **Part 7: Set dialysis parameters**



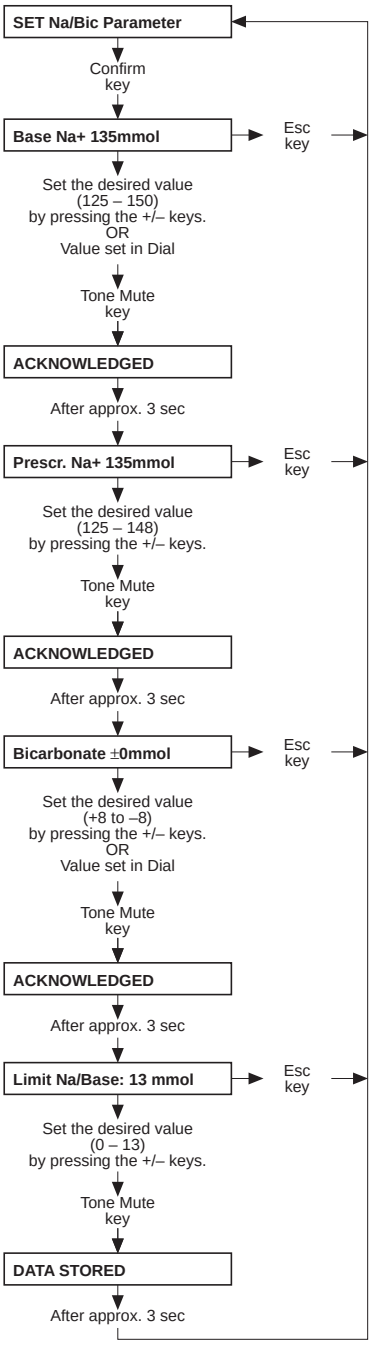
- **Part 7.1: Set dialysate flow**



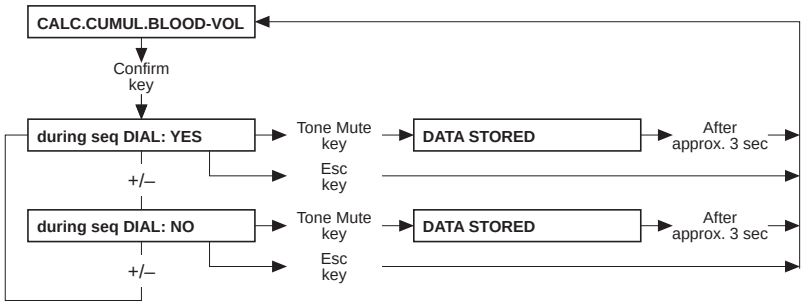
- **Part 7.2: Set dialysate temperature**



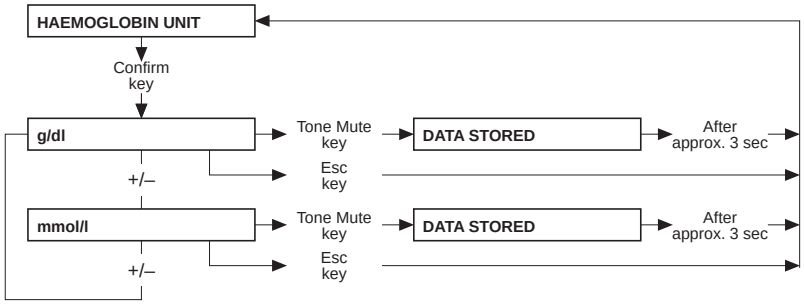
- **Part 7.3: Set Na/Bic**



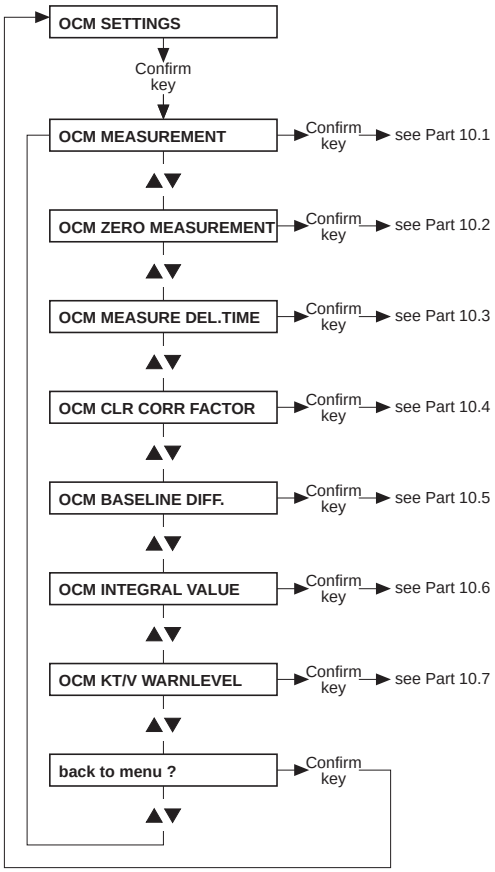
- **Part 8: Cumulated blood volume**



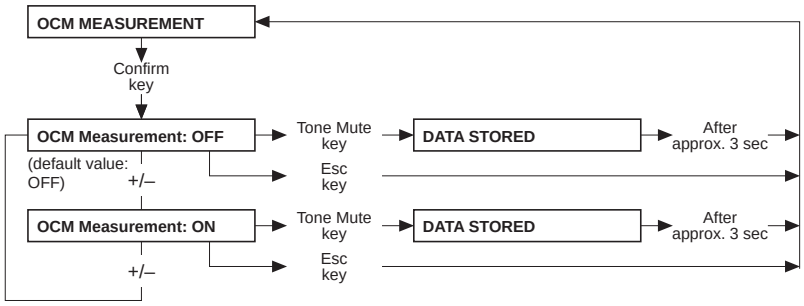
- **Part 9: Haemoglobin unit – only for BVM (option)**



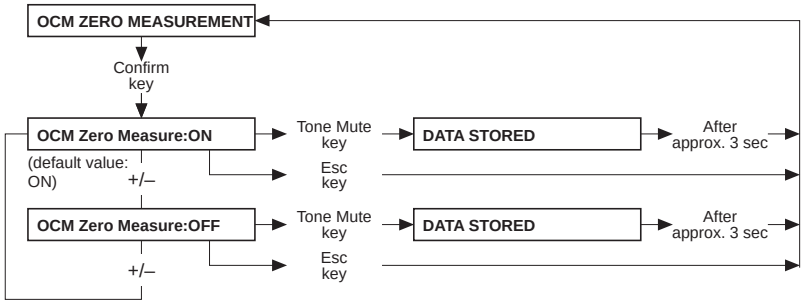
- **Part 10: OCM settings – only for OCM (option)**



- **Part 10.1: Activate OCM measurement**

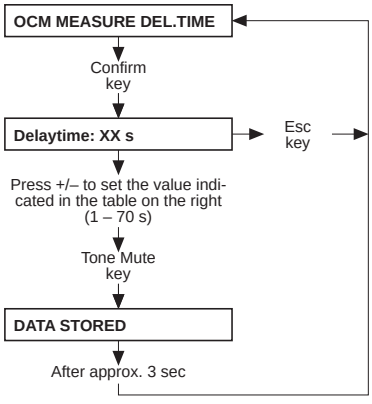


- **Part 10.2: Activate OCM zero measurement**



Note
If the OCM zero measurement is set to “OFF”, the OCM option is deactivated. If the OCM option is reactivated (“ON”), an OCM pulse calibration must be performed.

- **Part 10.3: Set the OCM measurement delay time**

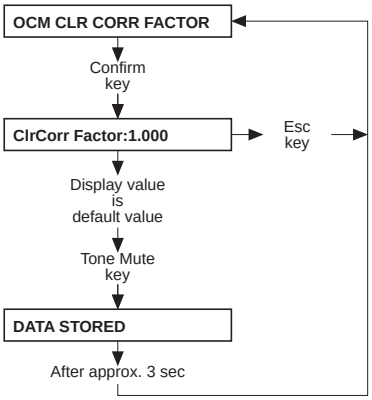


System	Time
4008 H with advanced hydraulics	15 Sec
4008 H with ONLINE*plus*™	65 Sec
4008 S with advanced hydraulics	18 Sec
4008 S with ONLINE*plus*™	67 Sec

- **Part 10.4: Set OCM correction factor**

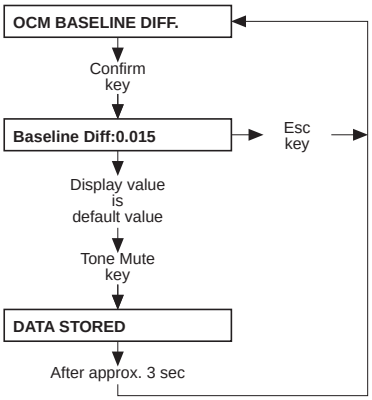
Note
Do not change the default value.

Note
The 4008 H/S offers the theoretical possibility of defining a correction value for all clearance measurements. The intention is to be able to adapt the system to to the latest state of technology resulting from the on-going scientific discussion. According to the current standard of knowledge, this correction value has to be set to “1” (factory setting) for Fresenius polysulfone membranes. The indicated accuracy specifications are valid only for Fresenius polysulfone membrane combined with this correction value setting of “1”.



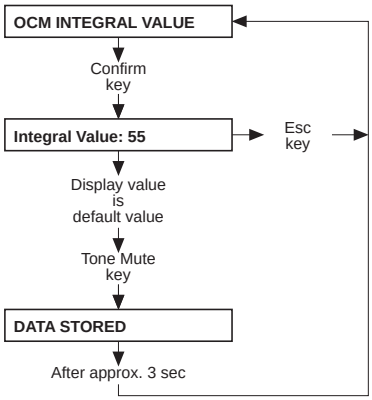
- **Part 10.5: Set OCM baseline difference**

Note
Do not change the default value.

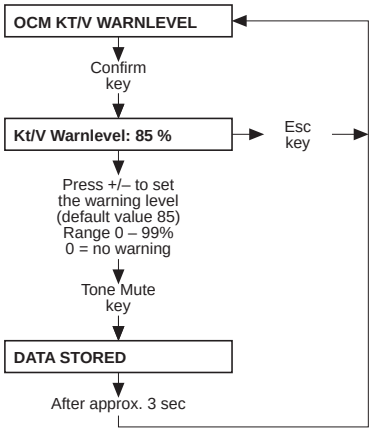


- **Part 10.6: Set OCM integral value**

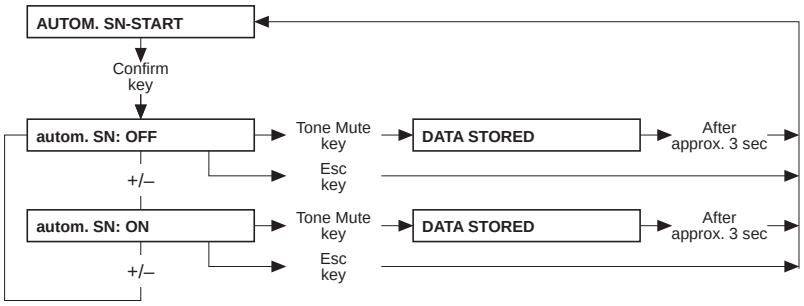
Note
Do not change the default value.



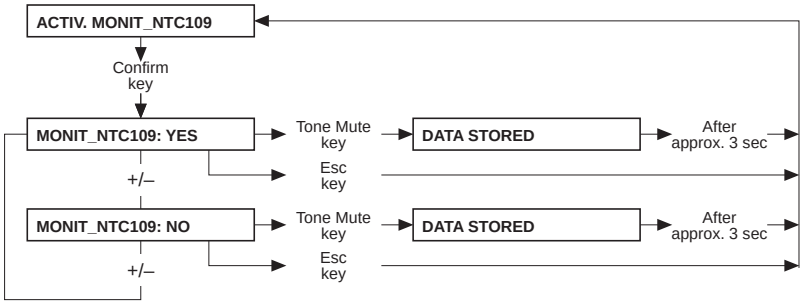
- **Part 10.7: Set OCM Kt/V warning level**



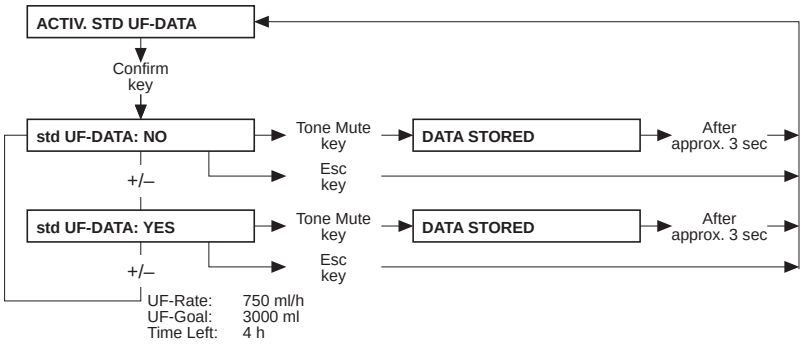
- **Part 11: Automatic single needle start**



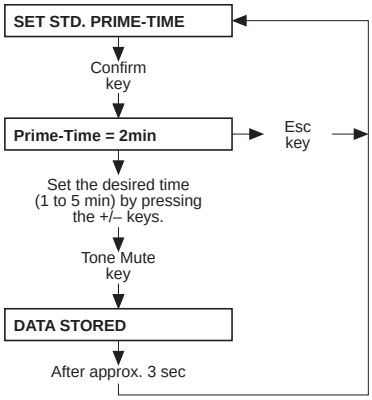
- **Part 12: Activation of Monit_NTC 109**



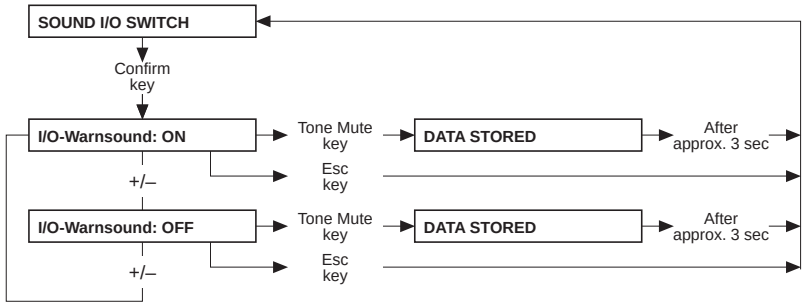
- **Part 13: Activation of standard UF data**



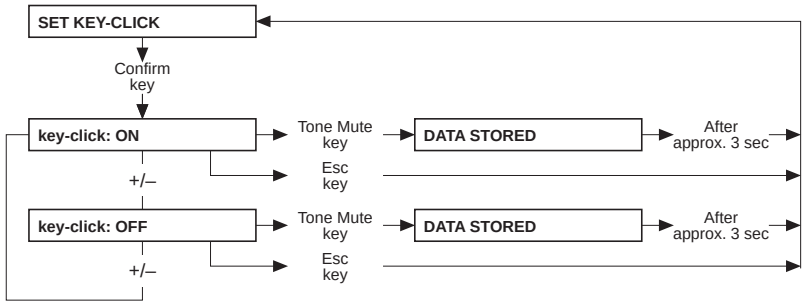
- **Part 14: Set priming time**



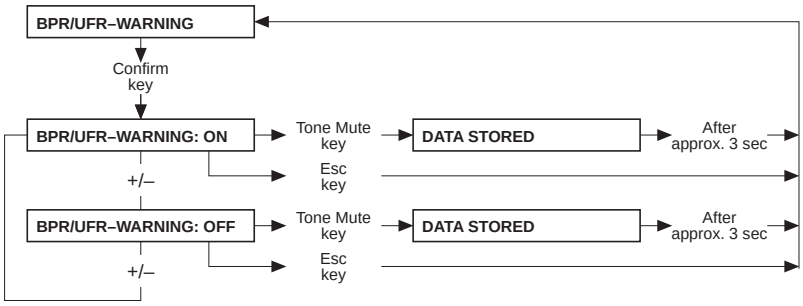
- **Part 15: Warning sound I/O key**



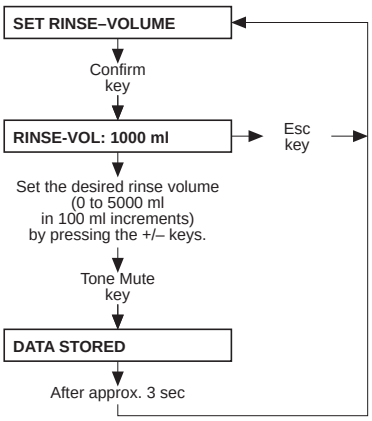
- **Part 16: Set key-click**



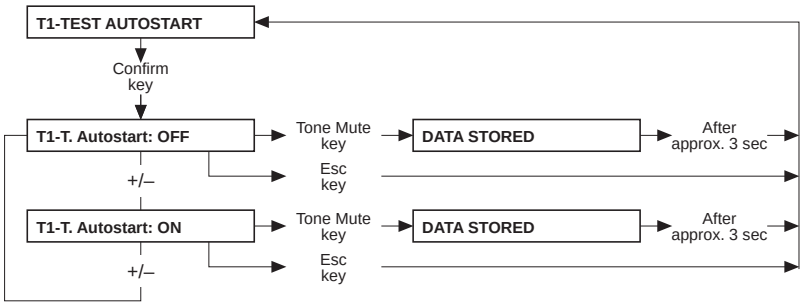
- **Part 17: Set BPR/UFR warning**



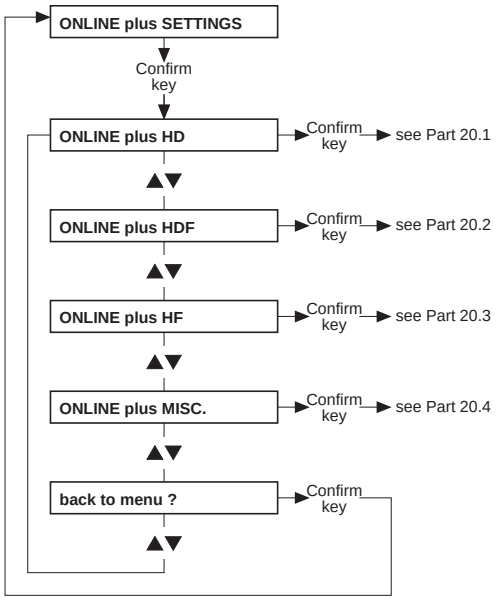
- **Part 18: Set rinse volume**



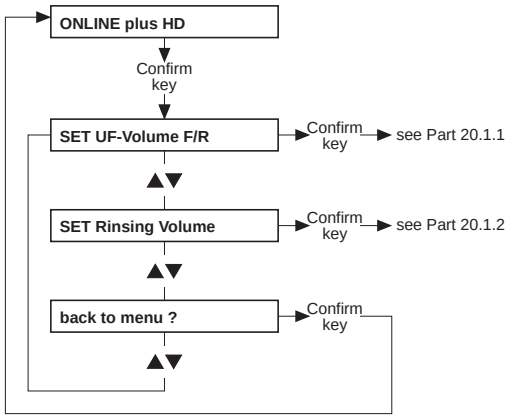
- **Part 19: T1-Test Autostart**



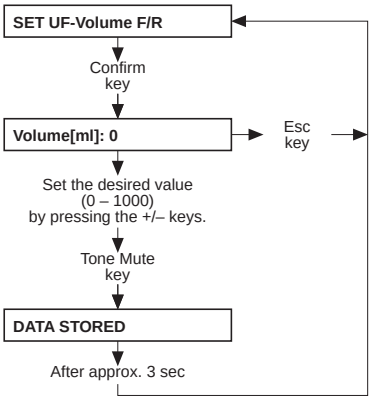
- **Part 20: ONLINE*plus*™ settings – only for ONLINE*plus*™ (option)**



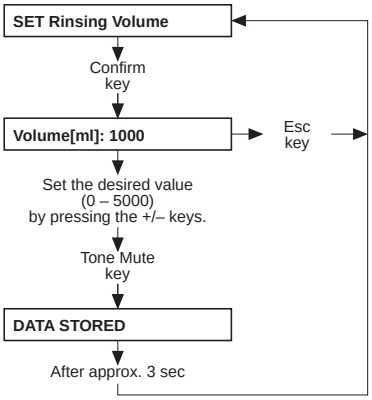
- **Part 20.1: ONLINE*plus*™ HD**



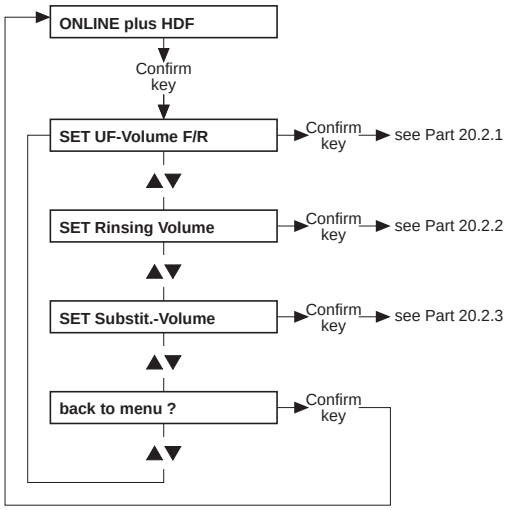
- **Part 20.1.1: Set UF volume (filling/rinsing)**



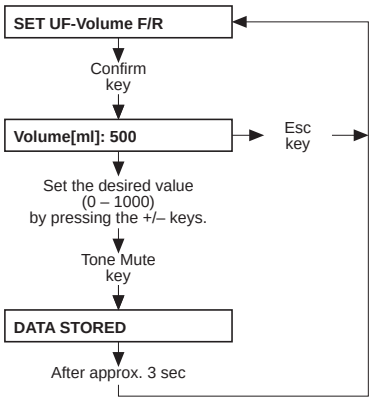
- **Part 20.1.2: Set rinse volume**



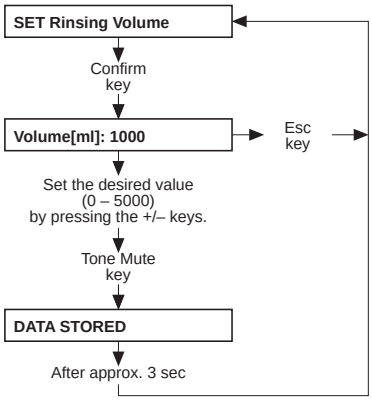
- **Part 20.2: ONLINE*plus*™ HDF**



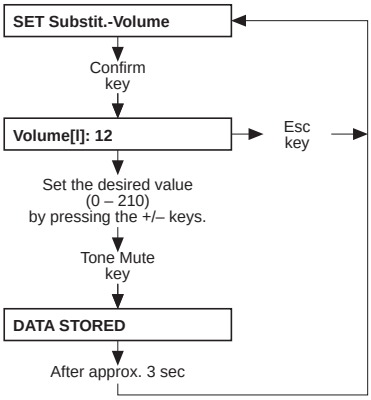
- **Part 20.2.1: Set UF volume (filling/rinsing)**



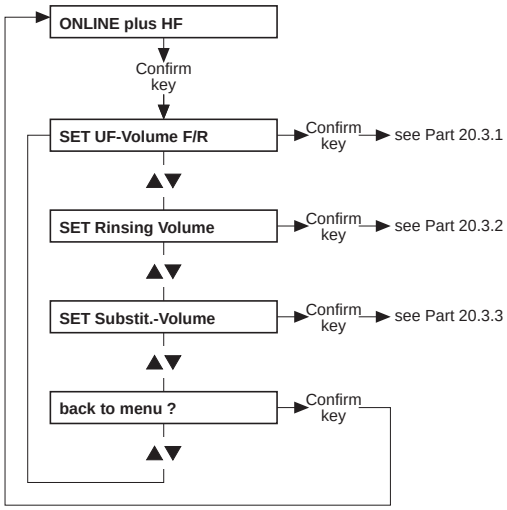
- **Part 20.2.2: Set rinse volume**



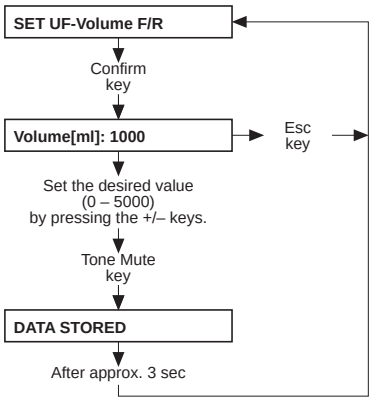
- **Part 20.2.3: Set substituate volume**



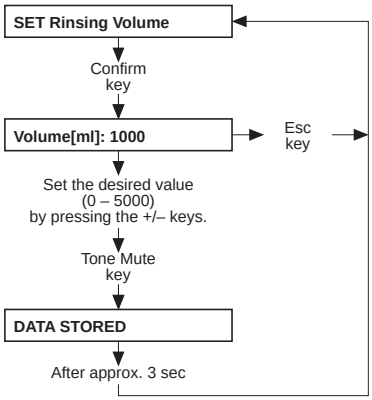
- **Part 20.3: ONLINE*plus*™ HF**



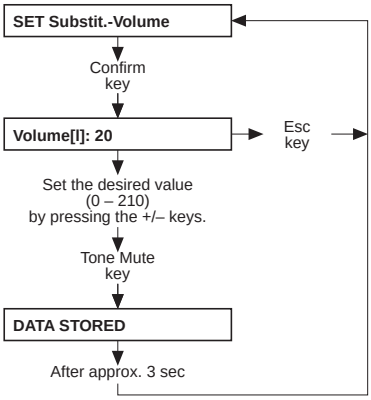
- **Part 20.3.1: Set UF volume (filling/rinsing)**



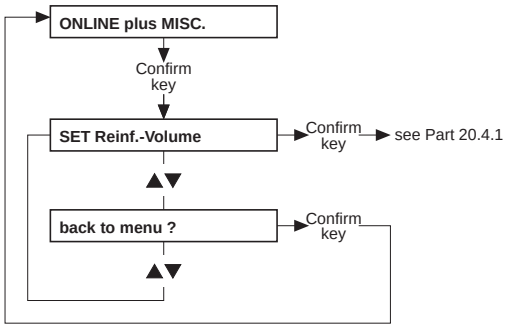
- **Part 20.3.2: Set rinse volume**



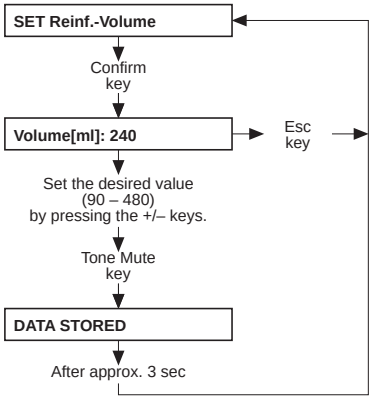
- **Part 20.3.3: Set substituate volume**



- **Part 20.4: ONLINE*plus*™ miscellaneous**

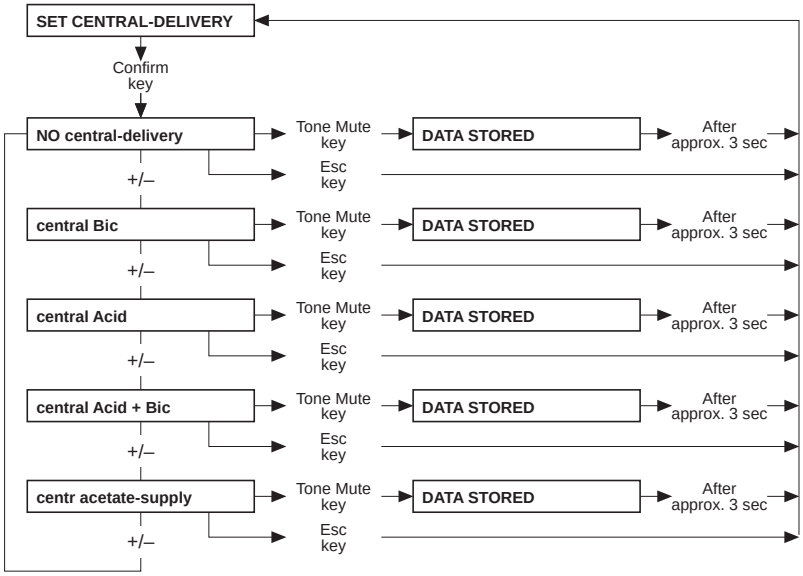


- **Part 20.4.1: Set reinfusion volume**

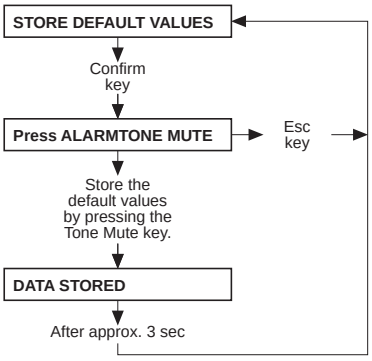


- **Part 21: Set parameters for central delivery system**

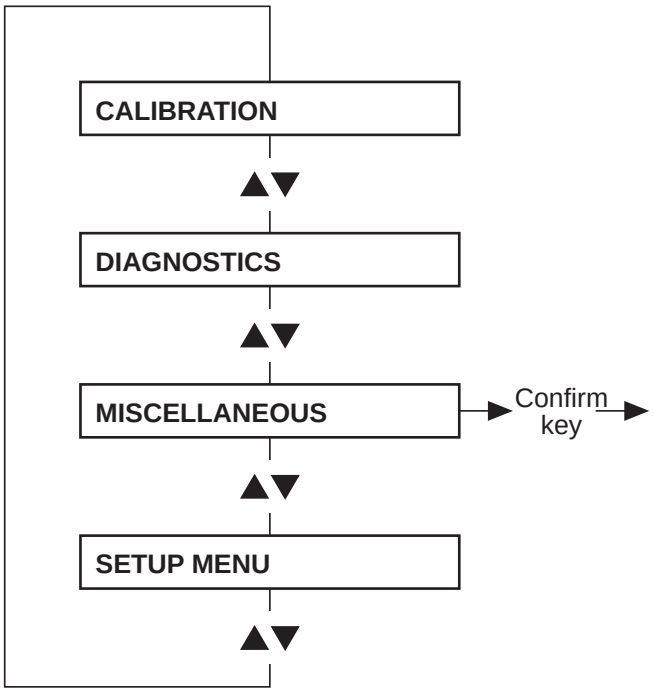
Note
Central delivery is not available for the 3mix option.



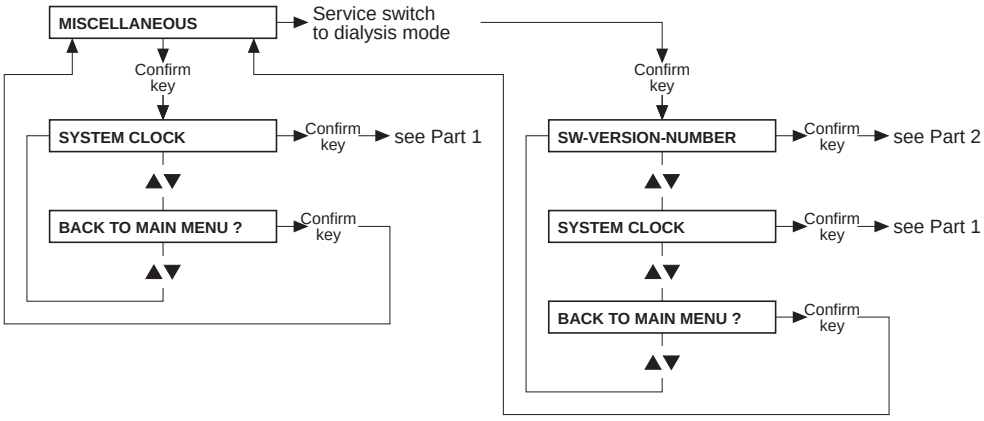
- **Part 22: Store default values**



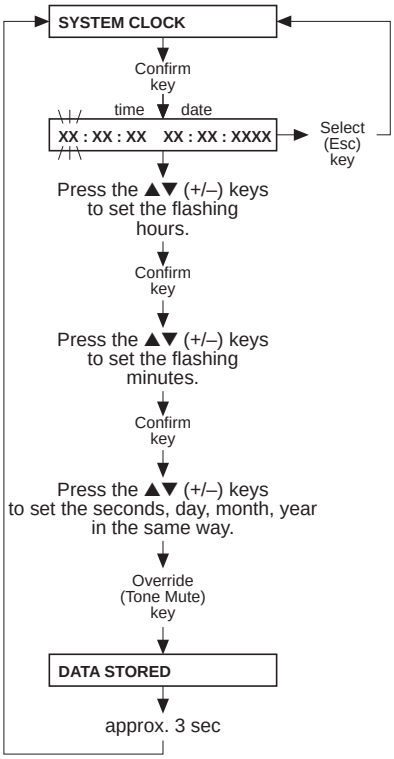
7 Miscellaneous



- **Main menu**



- **Part 1**


SYSTEM CLOCK
Confirm key
time
date
XX : XX : XX XX : XX : XXXX
Select (Esc) key
Press the ▲▼ (+/–) keys to set the flashing hours.
Confirm key
Press the ▲▼ (+/–) keys to set the flashing minutes.
Confirm key
Press the ▲▼ (+/–) keys to set the seconds, day, month, year in the same way.
Override (Tone Mute) key
DATA STORED
approx. 3 sec

- **Part 2**

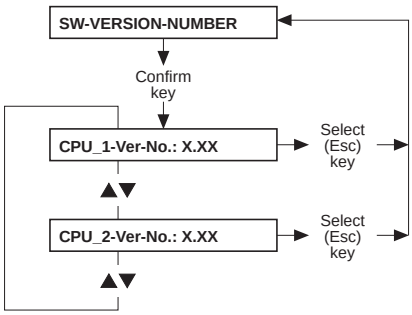

SW-VERSION-NUMBER
Confirm key
CPU_1-Ver-No.: X.XX
Select (Esc) key
▲▼
CPU_2-Ver-No.: X.XX
Select (Esc) key
▲▼

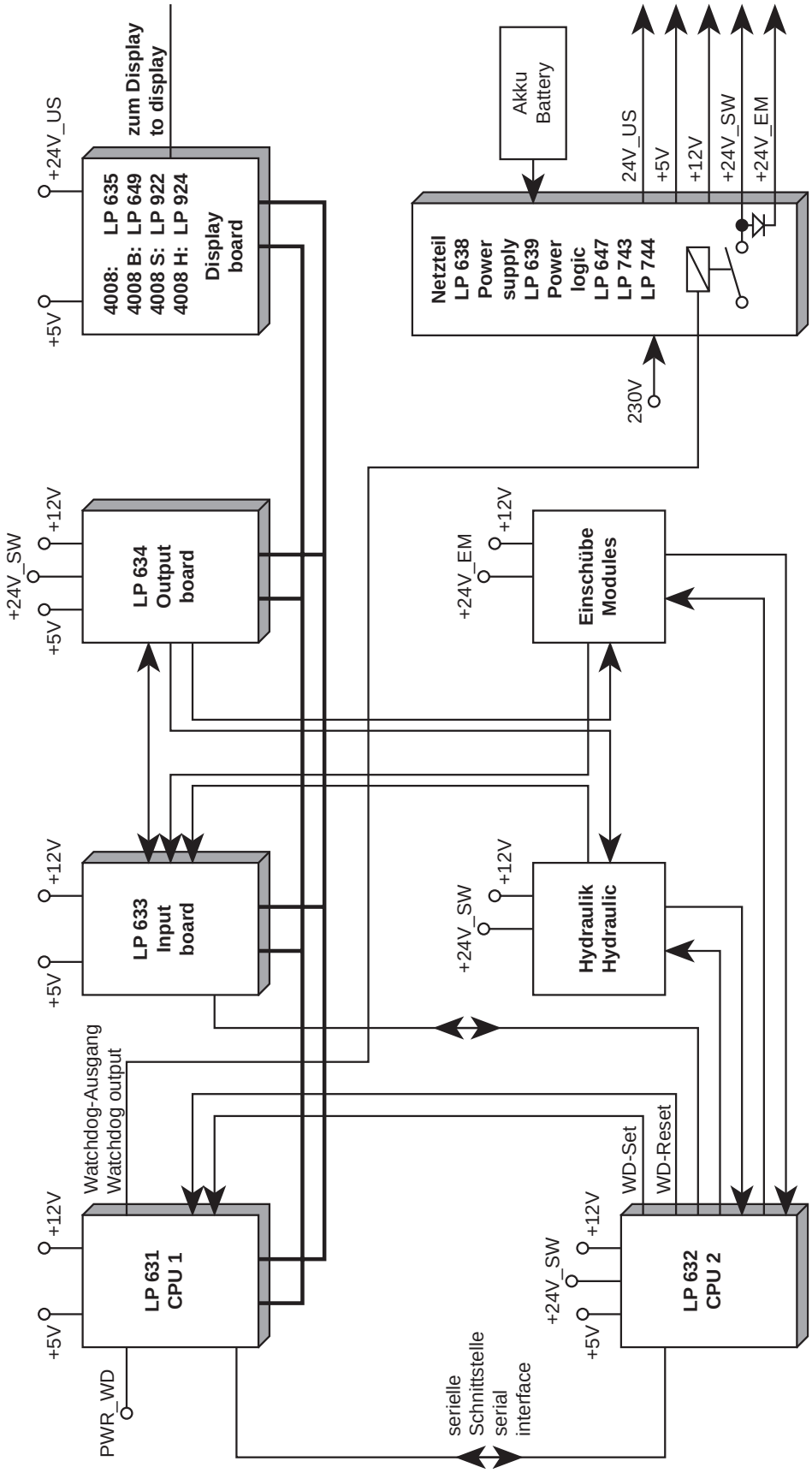
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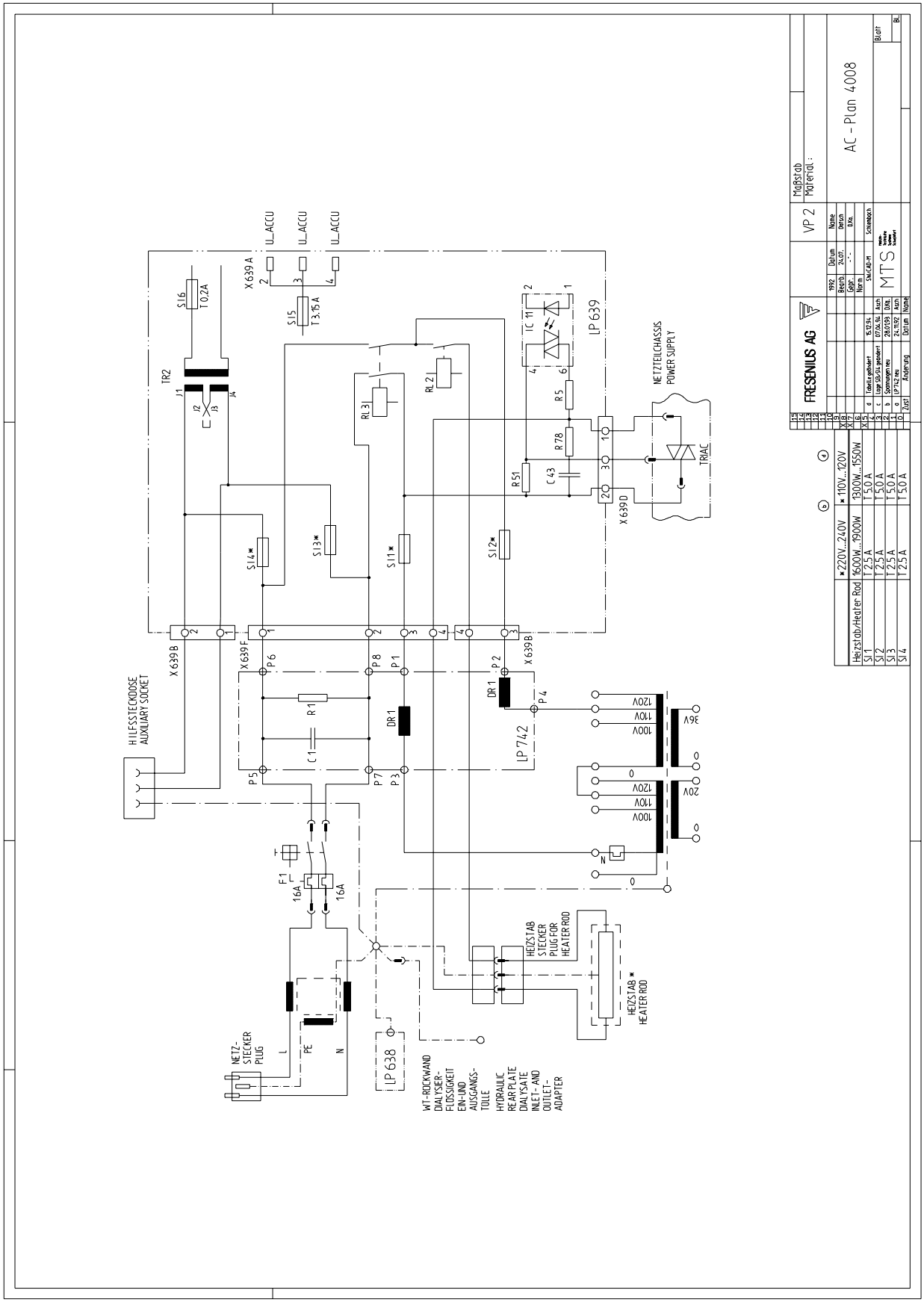
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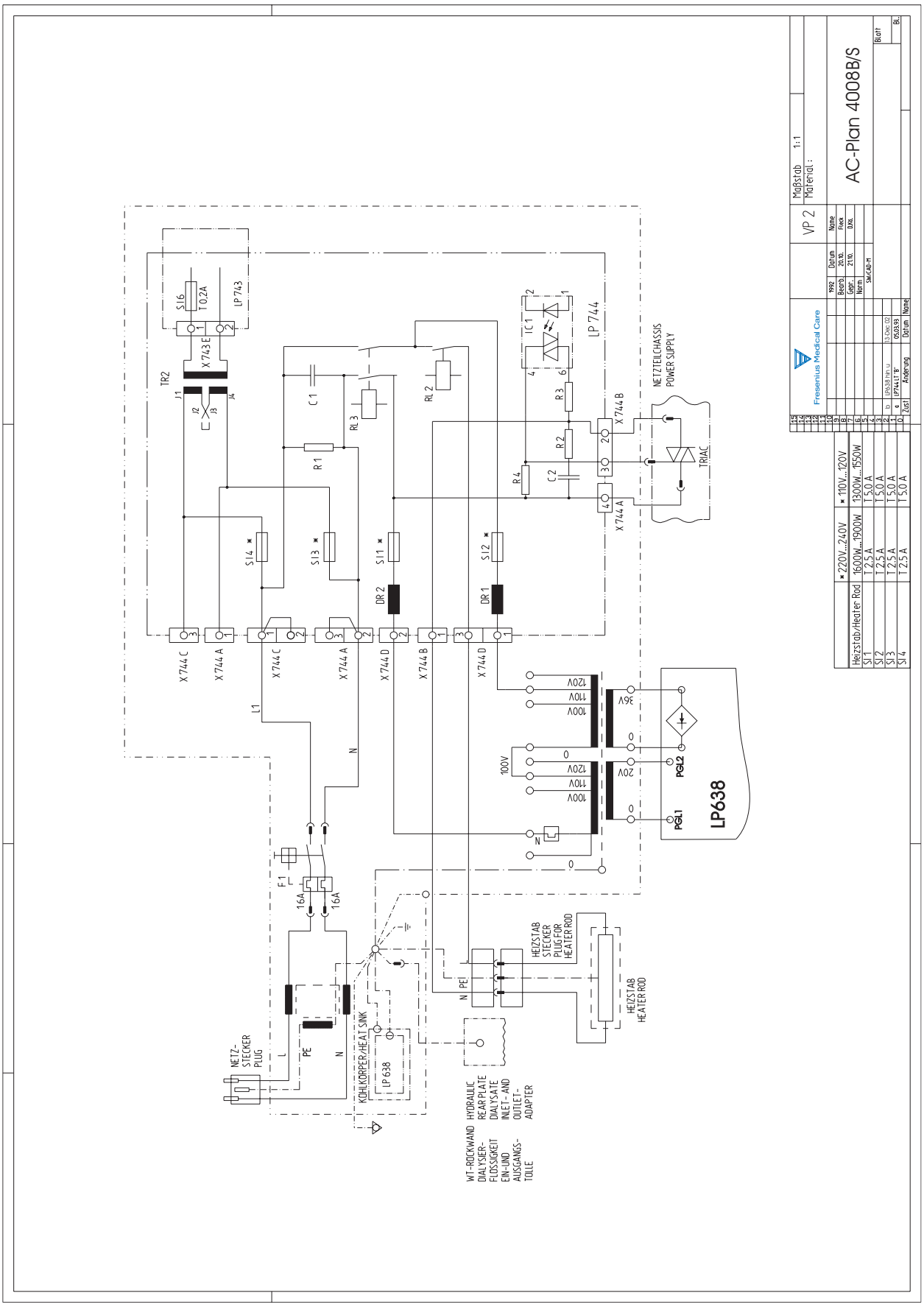
8.1 Block diagram 4008



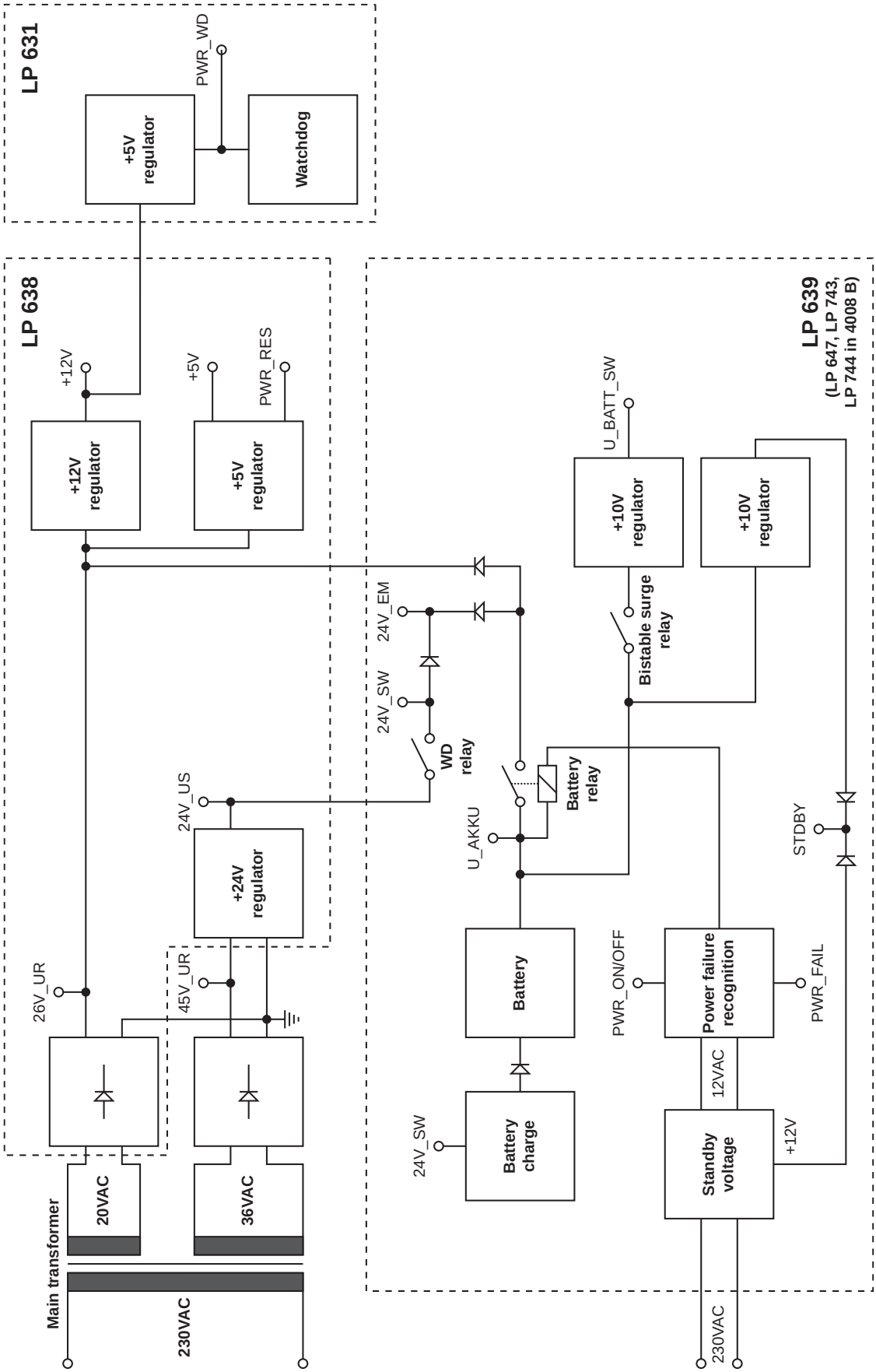
8.2 AC diagram 4008 E/H



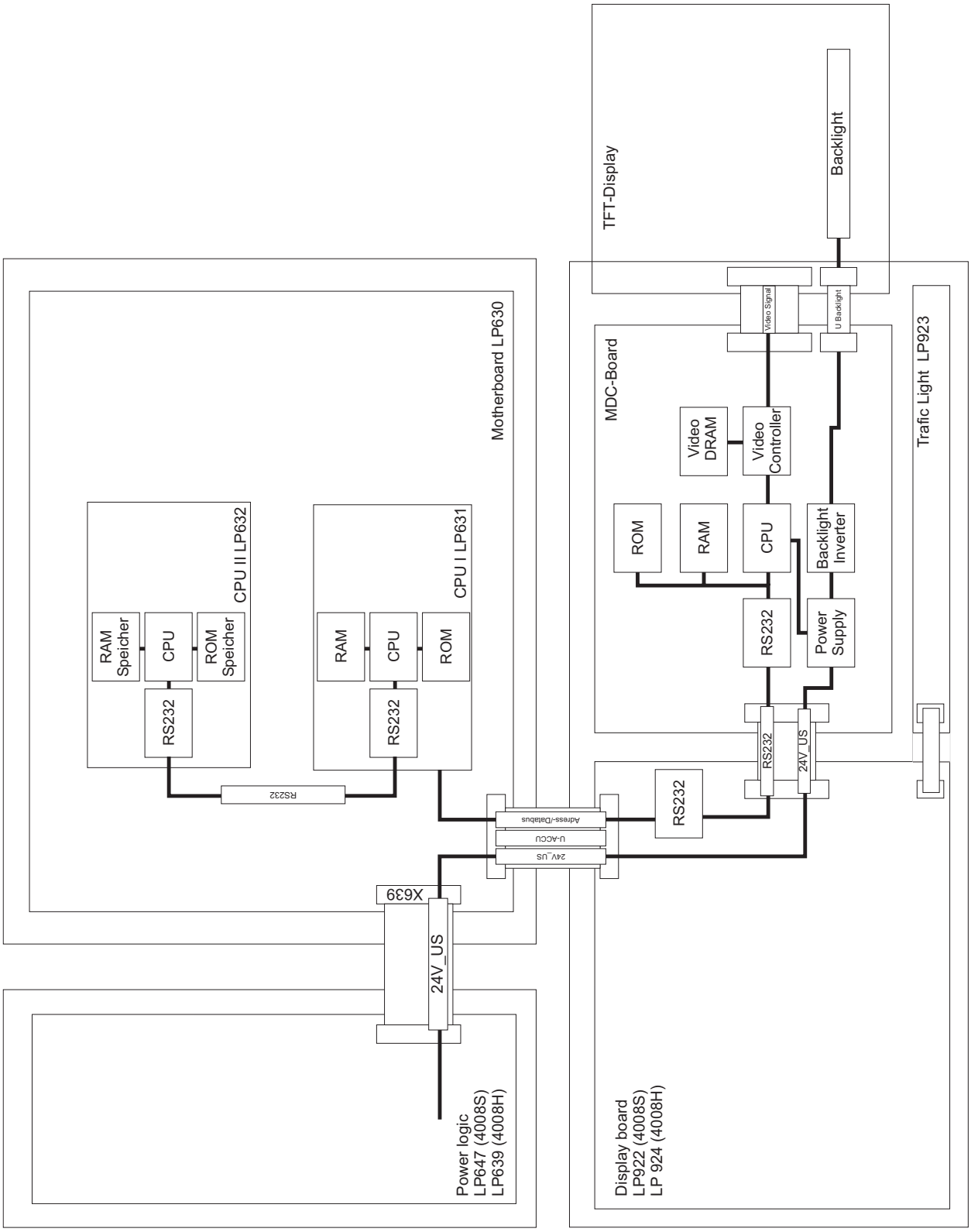
8.3 AC diagram 4008 B/S



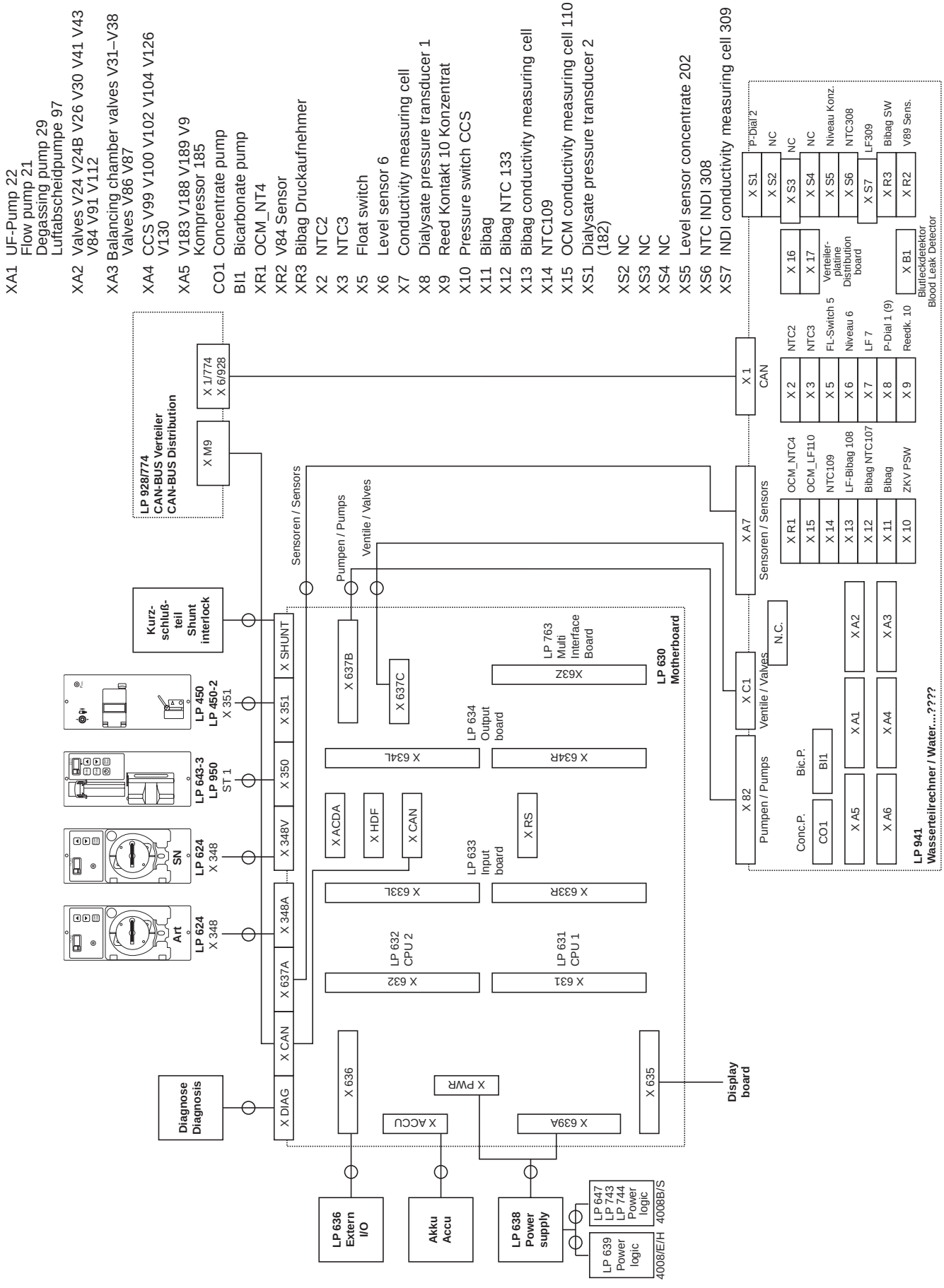
8.4 Block diagram of voltage supply



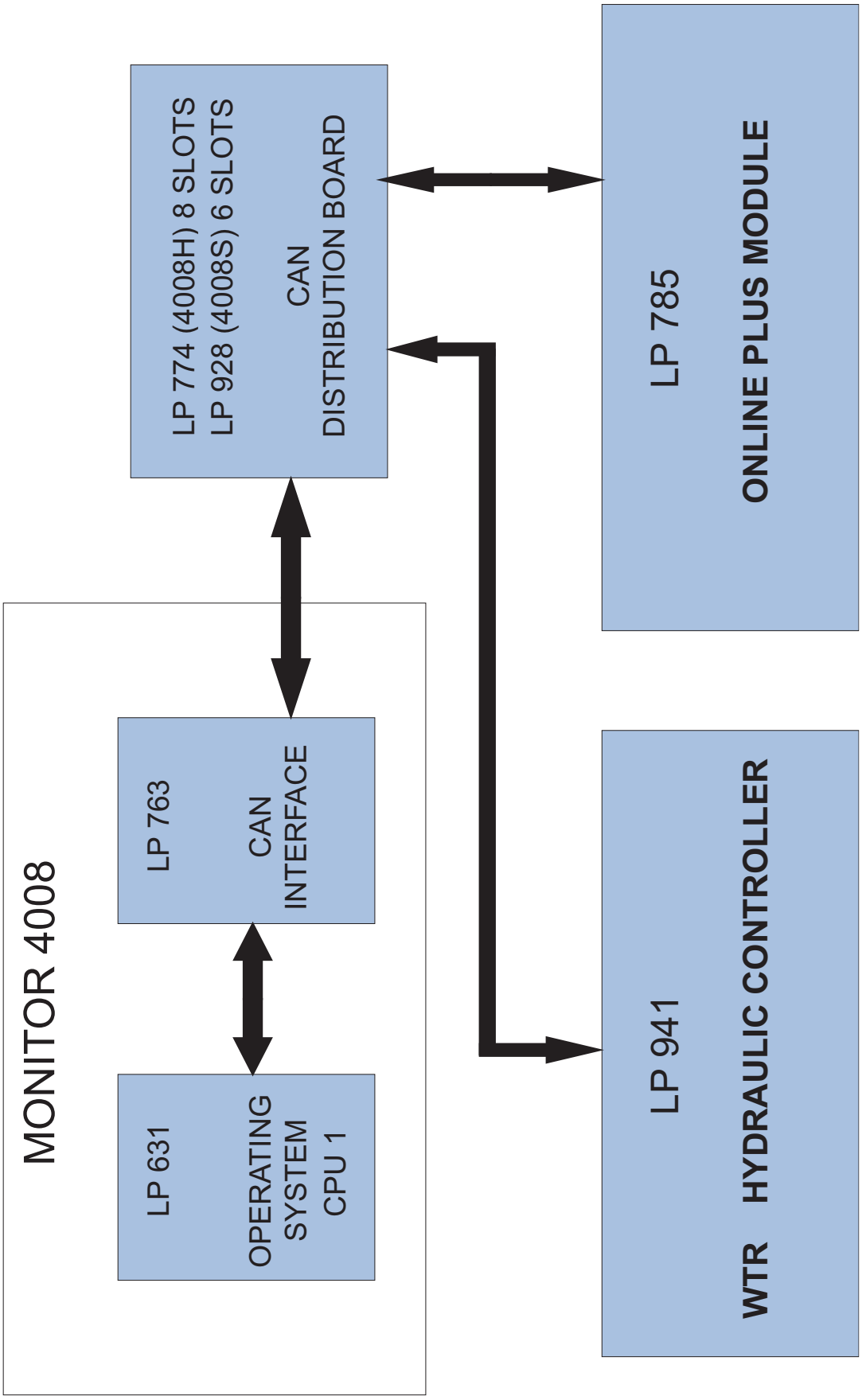
8.5 Block diagram of Screen 4008 H/S



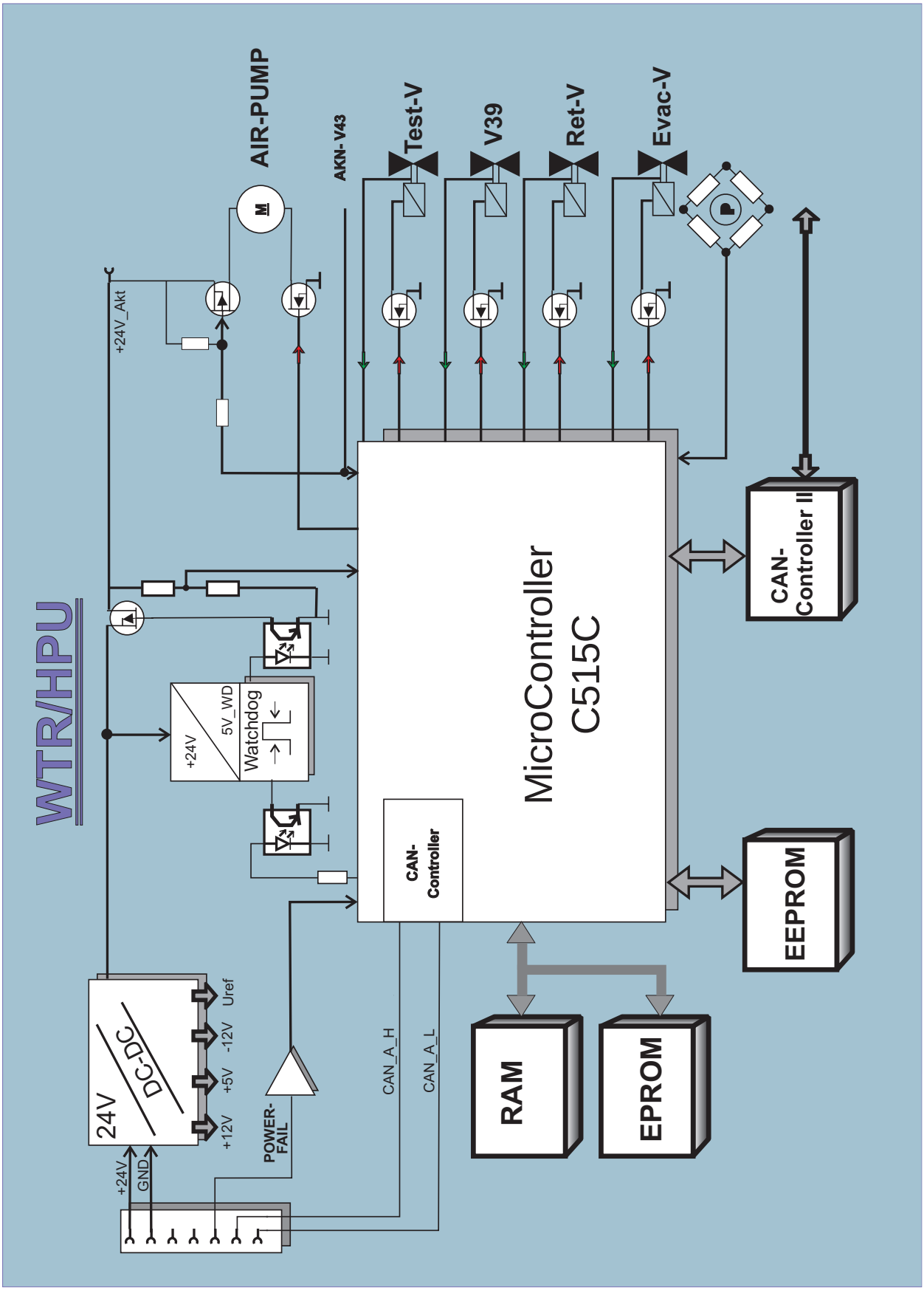
8.6 Connection layout diagram



8.6.1 Connection diagram CAN communication



8.6.2 Hydraulics processor block diagram



8.7 P.C.B. LP 450-2 Level detector control (LD)

8.7.1 Description

- **Audible section (transmitter)**

The self-capacitance of the transmitting converter is charged up to +12 V via R78 and discharged periodically via T9. The discharging current incites the oscillator to a dampened ultrasonic oscillation. With bridge BR1 being inserted in the calibration position (1-2), the converter is charged up to +6 V only, owing to the voltage division via R78 and R46. This reduces the transmitting amplitude to half its size. With transistor T8 being activated by the LDSA signal, the transmitting amplitude is reduced to a third of its size (test) due to the voltage division via R78 and R79.

- **Audible section (receiver)**

Up to the outputs of the monostable alarm circuits IC 3 and IC 7 respectively, the two receiver channels are identical. This is the reason why only the upper channel will be described. The voltage emitted by the receiving converter is amplified by T2 in the amplifier stage and delivered to OP IC 1, which is connected as precision rectifier. The amplified positive half waves, which are emitted to pin 3 of comparator IC 2 via the RC element R6 and C6, appear at the cathode of DI3. As soon as the peaks of the signal envelope pending there exceed the reference voltage at pin 2, output 1 emits H pulses at 90-ms intervals, which, inverted via T3, are applied to the trigger input pin 5 of the first monostable circuit IC 3. The two monostable circuits of IC 3 can be retriggered. Due to their loading, the first one has a time constant of 60 ms and the second of 470 ms. The first 60-ms time constant has been selected such that, after each trigger pulse on the input side, the output can return to its initial position, until the next trigger pulse arrives. If this is the case, the second monostable circuit is retriggered at 90-ms intervals, while its output LDA1 remains at H level (alarm-free). Should comparator IC 2 not emit trigger pulses due to too small an input signal, the second monostable circuit returns to its initial position after 470 ms, while inducing L level at pin 14 of plug X351. The monitor interprets this L level as level detector alarm, which stops the blood systems.
The working method of the second receiver channel is the same with the exception of the longer time constant of 700 ms of the second monostable circuit IC 7. Here, however, the output pin 7 sets the storage flip-flop IC 8 by means of H level in case of an alarm. The output pin 2 of IC 8 then inhibits FET T14 by means of L level so that the venous line clamp located in the drain branch becomes dead and closes. Upon T14 being inhibited, DI11 is inhibited as well, so that a 24-V level is applied to pin 6 of plug X351 via R55. During the test, this voltage is sampled in order to determine whether the venous line clamp has been turned off by the second receiver channel. If the alarm condition has been eliminated meanwhile, the flip-flop IC 8 can be reset by means of the positive edge of the dialysis start pulse from pin 12 of plug X351. In this case, the H level at pin 2 of IC 8 turns FET T14 on again, and the venous line clamp is opened.

- **Clamp control**

  The clamp magnet can be controlled via transistor T13. This method is used for turn-off in case of all blood alarms and for single needle control. If an H level is applied to the CLAMP-CTRL signal at pin 8 of plug X351, the output voltage at pin 7 of OP IC 19, which is connected as integrator, drops linearly. Should this voltage fall below the reference voltage at pin 3 of IC 19, the output pin 1 turns to H level and controls T13 via T7. This delayed opening of the clamp magnet is coordinated with the starting delay of the blood pump and requisite for a trouble-free single needle operation. If the CLAMP-CTRL signal turns to L level, the integrator is rapidly recharged again via DI12 and R86, and the venous line clamp is actually closing without delay.

- **Pulse generation**

  Timer IC 11 is connected as astable multivibrator. It generates a symmetrical square-wave frequency of approx. 2.8 kHz, which is divided by the binary divider IC 10. 90 Hz are available at pin 3 for energizing the infrared emitter of the optical detector. The unbalanced square-wave voltage for energizing the ultrasonic transmitter (90 ms L level, 0.2 ms H level) is generated via the flip-flop IC 9.

- **Visual section**

  The infrared emitter of the optical detector is energized by pin 3 of IC 10 at a frequency of 90 Hz via transistors T10 and T11. The phototransistor on the opposite side of the optical sensor absorbs the modulated light and controls IC 14, which is connected as current-voltage converter, at pin 6. A square-wave voltage, whose amplitude is proportional to the transparency of the medium in the optical sensor, is applied to pin 7 of IC 14. This voltage reaches pin 3 of comparator IC 14 via voltage divider R65 and R66. If the positive half waves exceed the reference level at pin 2 of IC 14, a square-wave voltage is delivered by output pin 1; otherwise, the output is provided with L potential. Via pin 5 of plug X351, the square-wave voltage is delivered to the monitor, where it is used to generate a logic level.

- **+12-V monitoring**

  In order to avoid malfunctions of the level detector and other circuits, the +12-V voltage is monitored by transistor T1. Via C3, any possible AC voltage superposition of the +12-V voltage is coupled onto the base emitter path of T1, which is biased via R20 and R21. Should the AC portion exceed approx. 0.5 Vss, T1 is controlled by means of the AC voltage. In addition, positive +12-V pulses, which set the alarm flip-flop IC 8 via DI4, are applied to the collector. The venous line clamp is now closing.

- **Venous pressure measurement**

  The pressure-proportional diagonal voltage of pressure sensor T12 is amplified to a voltage of 0 to +10 V by IC 13, which is connected as instrument amplifier. This voltage reaches the monitor via pin 4 of plug X351.

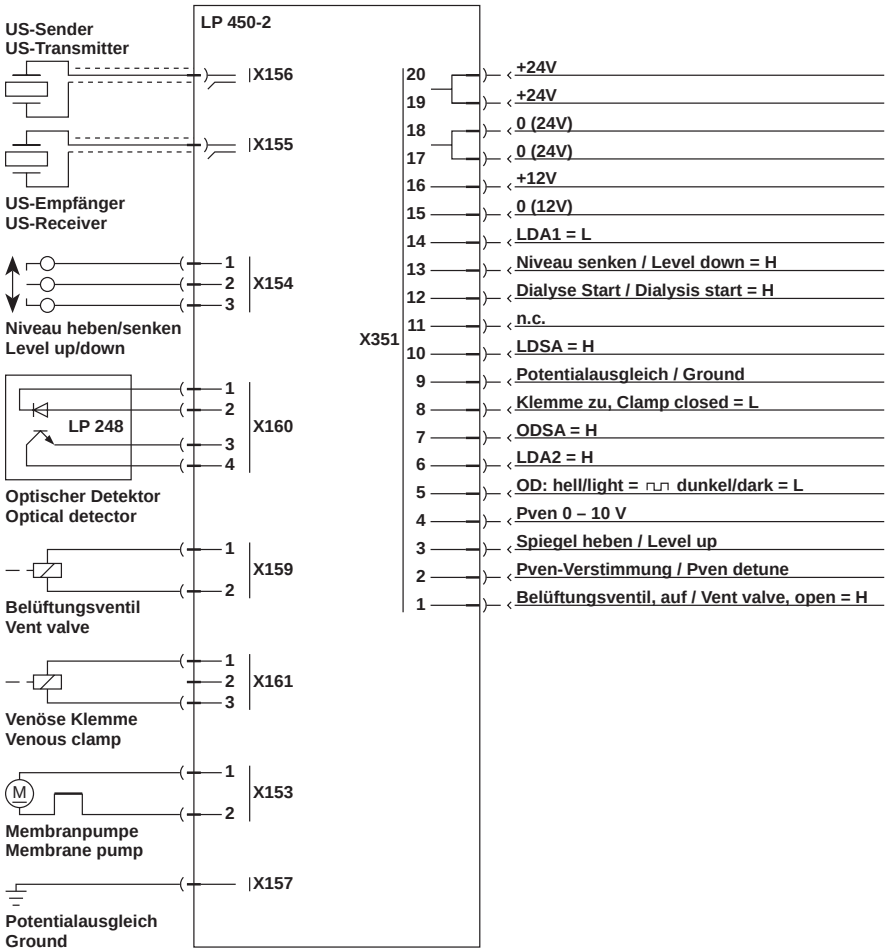
- **Raising the level in the venous bubble catcher**

  The H level at pin 1 of X351 is inverted by IC 18, pins 3 and 6; or the L level at pin 3 of X154 energizes FET T16 via IC 15, pins 5, 6, 4. The vent valve is opened by FET T16. Simultaneously, the LEV-UP-OUT signal is inverted via IC 15, pins 8, 9, 10. The venous line clamp is blocked via DI17, and level raising is blocked at IC 15, pin 12.

- **Lowering the level in the venous bubble catcher**

  An L level at X154, pin 1, energizes FET T15 via IC 18, pins 2, 9. FET T15 activates the ventilation pump and, in addition, the vent valve via DI16. Level lowering can be blocked via an H level applied to pin 1 or pin 8 of IC 18. This is the case, if an L level is applied to at least one of pins 1, 2, 12, 13 of IC 15 (in case of a level detector alarm from channel 1 or 2; if the venous line clamp is being closed by external control; or in case of LEV-UP).

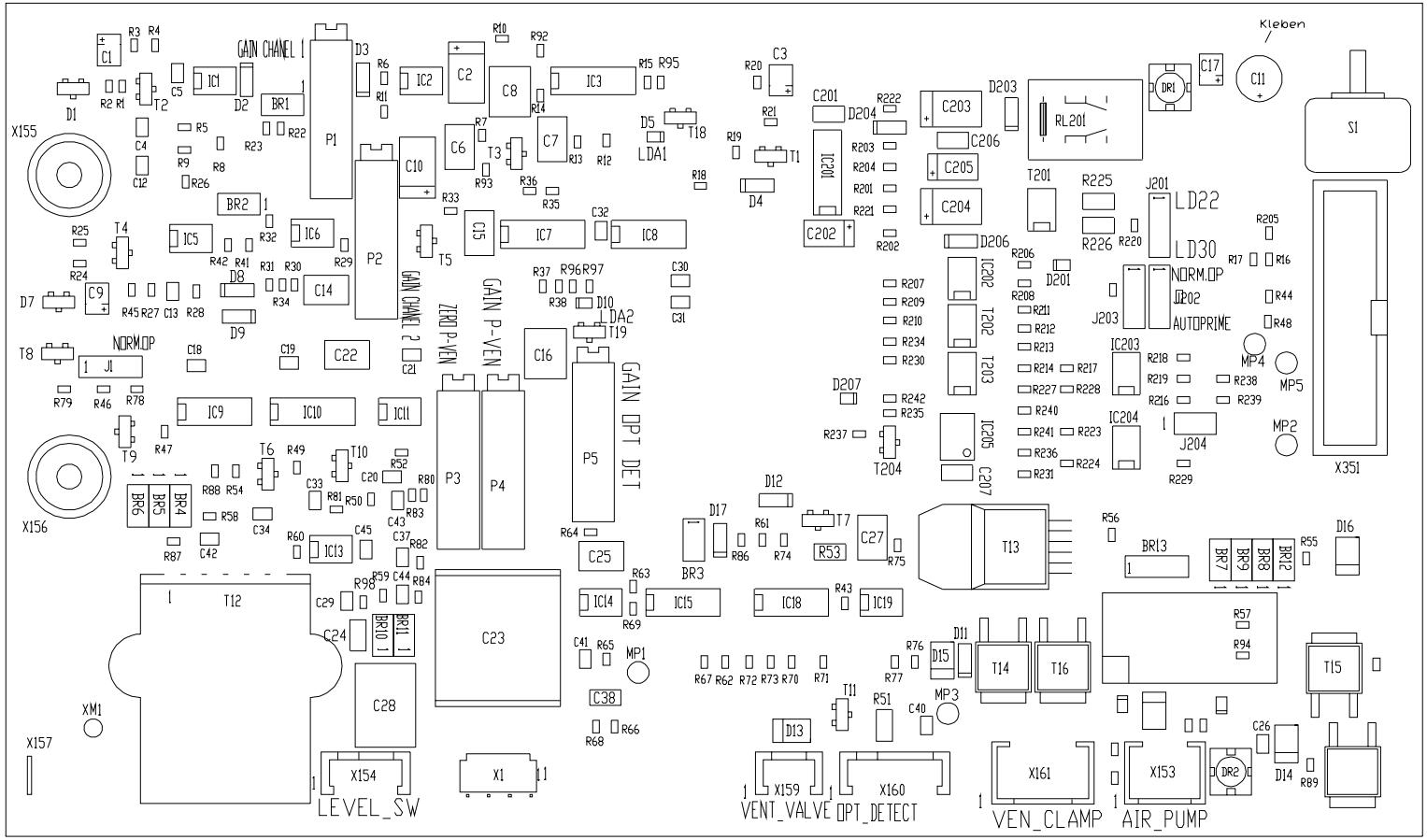
Fig.: Signal plan P.C.B. LP 450-2



8.7.2 Circuit diagram and component layout diagram
P.C.B. LP 450-2 Level detector control (LD)

P.C.B. LP 450-2

Component layout diagram



BRÜCKEN: X=GESCHLOSSEN /0=OFFEN
LINKS : X=FITTED /0=NOT FITTED

BR3: nur geschlossen für Funktion Heben/Senken
BR3: only fitted for function level up/down

Ersatzlieferung:
Generell uncodiert, Brücken je nach Einsatz (Luftdetektortyp)

Spare Parts:
Generally not coded, set link depending on use (level det. type)

BR 1	BR 2	TK
0	0	30mm
X	X	22mm

JUMPER 1 MODE	
CALIBRATION	NORMAL OPERATION
1-2	2-3

SYSTEM	BR 4	5	6	7	8	9	10	11	12	13	
A 1008 D	X	0	0	0	X	X	0	X	0	0	1/2
A 2008 D	0	0	X	0	X	X	X	0	0	0	1/2
A 2008 D,E/USA	0	0	X	0	X	X	X	0	0	0	3/2
A 2008 BSS	0	0	X	0	X	X	0	0	X	0	1/2
2008 E	0	0	X	X	0	0	0	X	0	0	1/2
4008	0	0	X	X	0	0	X	0	X	0	1/2
4008 ADS	0	0	X	X	0	0	X	0	X	0	1/2
ART	0	0	X	X	0	0	X	0	X	0	1/2

POS.	LP450	LP450-1	LP450-2	
R59	1M33	2M43	1M33	
J202/J203	1-2 NORMAL OPERATION	1-2 NORMAL OPERATION	2-3 AUTOPRIME OPERATION	
J201 nur bei Autoprime/ only for Autoprime	nicht bestückt not fitted	nicht bestückt not fitted	1-2 LD 22	2-3 LD 30
J204 nur bei Autoprime/ only for Autoprime	nicht bestückt not fitted	nicht bestückt not fitted	bestückt fitted	

				DATUM	NAME
			BE	08.06.00	Schießer
			GE	14.11.00	Gerhard
b	C11 ALU ELKO	20.05.03			
a	J204 hinzu	07.02.01	GAn		
	ANDERUNG	DATUM	NAME		

Fresenius Medical Care Deutschland GmbH

Luftdetektorsteuerung
Level-Detektor-Control

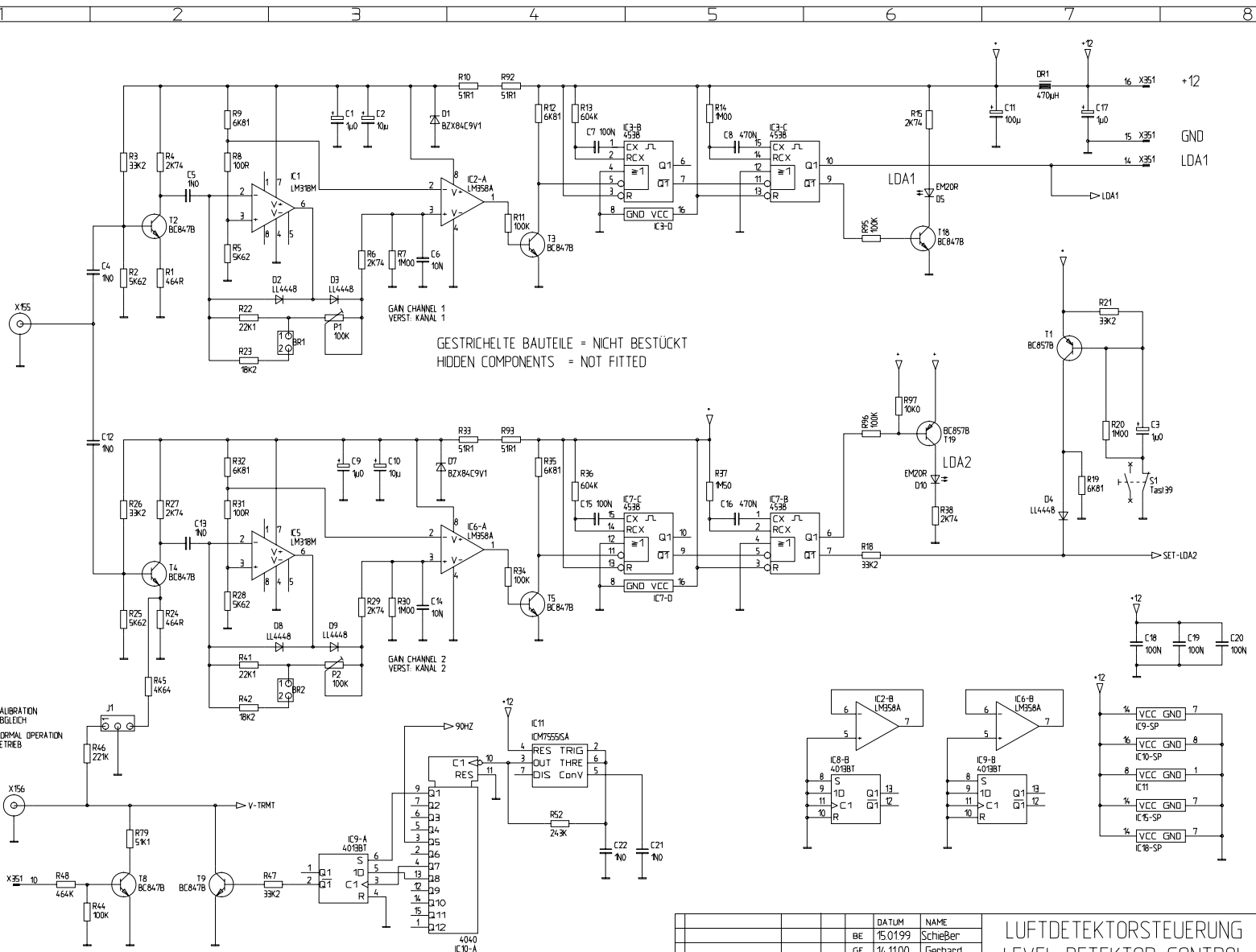
LP450-2 M31449

BLATT BP

ERSATZ ERSATZ FORM.A3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN



					DATUM	NAME	LUFTDETEKTORSTEUERUNG LEVEL-DETEKTOR-CONTROL	BLATT 1 /3
				BE	15.01.99	SchieBer		
				GE	14.11.00	Gerhard		
					Layout	450 L		
c	R202/C202 geändert	30.08.01						
b	J204 hinzu	07.02.01	GAn				LP450-2 M31449	SP
a	R64/R69 geändert	19.12.00	GAn	Fresenius Medical Care Deutschland GmbH				
	ANDERUNG	DATUM	NAME				ERSATZ ERSATZ	FORM.A3

P.C.B. LP 450-2

Circuit diagram 1/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN

GESTRICHELTE BAUTEILE = NICHT BESTÜCKT
HIDDEN COMPONENTS = NOT FITTED

CLAMP-CTRL
KLEMMENSTEUERUNG
+24V
+24V
GND24V
GND24V
EARTHPOTENTIAL
POTENTIALAUSGLEICH
ALARM-RESET
ALARM LOSCHEN
LEV-DN-SW
GND
LEV-UP-SW
GND
LEVEL-DOWN-SW
TASTER SPIEGEL SENKEN
LEVEL-UP-SW
TASTER SPIEGEL HEBEN
VENT-VALVE
ENTLÜFTUNGSVENTIL
TES-P-VEN

LDA2
VEN.LINE-CLAMP (+)
VEN.ABSPERRKLEMME (+)
VEN.LINE-CLAMP (-)
VEN.ABSPERRKLEMME (-)
VENTILATIONPUMP(+)
BELÜFTUNGSPUMPE (+)
VENTILATIONPUMP(-)
BELÜFTUNGSPUMPE (-)
VENTVALVE (+)
ENTLUFTUNGSVENTIL (+)
VENTVALVE (-)
ENTLÜFTUNGSVENTIL (-)
SIGNAL LEVEL DOWN
SIGNAL SPIEGEL SENKEN
SIGNAL LEVEL UP
SIGNAL SPIEGEL HEBEN
P-VEN
OD

P-VEN DETUNING
P-VEN VERSTIMMUNG
ZERO P-VEN
NULLPUNKT P-VEN
GAIN P-VEN
VERST.P-VEN
GAIN OPT DET
VERST.OPT.DET
FOR P-VEN TIME CONSTANT
FUEA P-VEN ZEITKONSTANTE

LP248

					DATUM	NAME
				BE	15.01.99	Schießer
				GE	14.11.00	Gerhard
					Layout	450 L
c	R202/C202 geandert	30.08.01				
b	J204 hinzu	07.02.01	GAn			
a	R64/R69 geandert	19.12.00	GAn			
	ANDERUNG	DATUM	NAME			

Fresenius Medical Care Deutschland GmbH

LUFTDETEKTORSTEUERUNG
LEVEL-DETEKTOR-CONTROL

LP450-2 M31449

BLATT 2 /3

SP

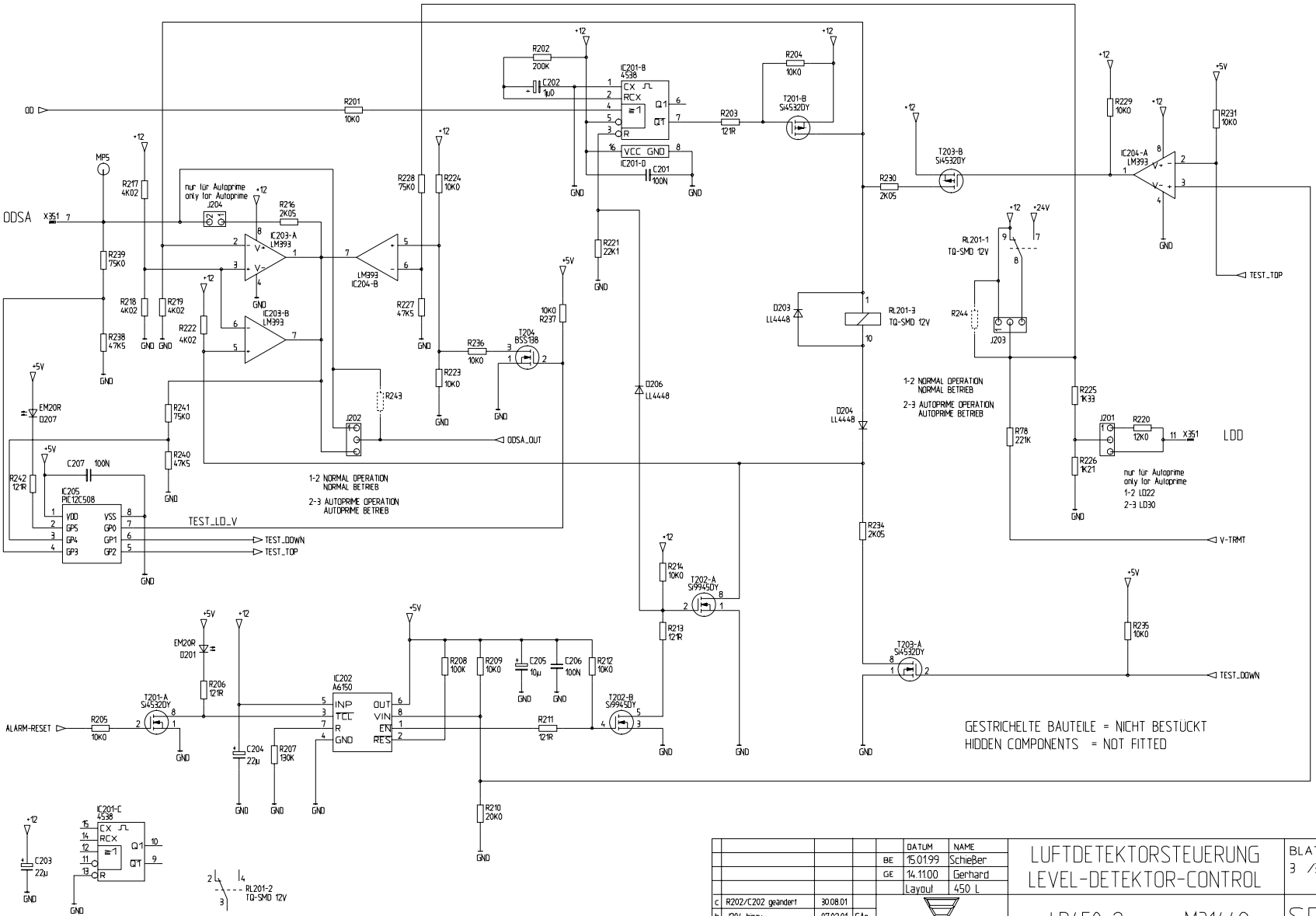
ERSATZ ERSATZ FORM A3

P.C.B. LP 450-2

Circuit diagram 2/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN



				DATUM	NAME	LUFTDETEKTORSTEUERUNG LEVEL-DETEKTOR-CONTROL
			BE	15.01.99	SchieBer	
			GE	14.11.00	Gerhard	
				Layout	450 L	
c	R202/C202 geändert	30.08.01				LP450-2 M31449
b	J204 hinzu	07.02.01	GAn			
a	R64/R69 geändert	19.12.00	GAn			
	ANDERUNG	DATUM	NAME	Fresenius Medical Care Deutschland GmbH		

BLATT 3 /3		
SP		
ERSATZ	ERSATZ	FORM.A3

P.C.B. LP 450-2

Circuit diagram 3/3

8.8 P.C.B. LP 493 Blood leak detector

8.8.1 Description

- **Operational principle**

A 2-color LED alternately emits green and red light through a measuring cuvette. A photoelectric cell converts the light intensity received into a voltage logarithmic to the light intensity, so that the square-wave amplitude generated from the alternating green/red illumination is proportional to the logarithm of the green/red quotient and that non-specific types of dimness, which equally attenuate both colors, do not affect the signal.

- **Circuit description**

Blood leak channel
Together with crystal Q1 and the capacitive voltage divider C8/C9, IC 6 generates a frequency of 32768 Hz in a three-point circuit. Via R23/R22, the binary divider IC 6 supplies transistor T1 with pulses of 32 Hz from Q10 for energizing the green LED and, inverted via the joint emitter resistor R34 and transistor T2, for energizing the red LED.
The output voltage of the photoelectric cell is amplified via C2 and OP IC 1a and then inverted via OP IC 3a. The channel switch IC 5 scans the 2nd half (from IC 6-13 to IC 5-10) of the transient output amplitude of OP IC 1a and OP IC 3a respectively in phases synchronous to the green/red energization (from IC 6-15 to IC 5-9). The voltage is filtered via R12/C3 and amplified via OP IC 4b.

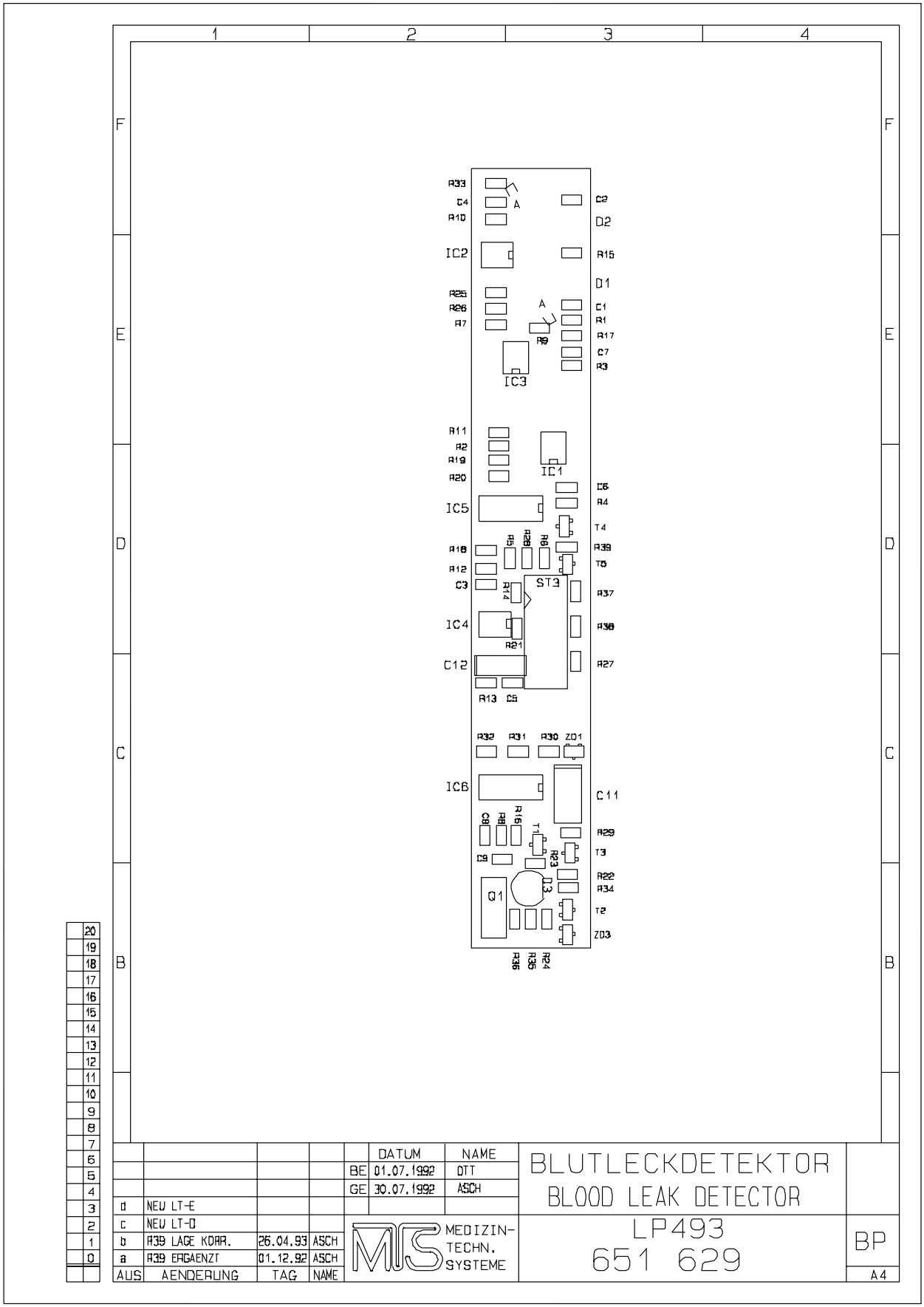
Temperature and aging regulation
A second photoelectric cell, which evaluates a reference ray of light not passing through the cuvette, readjusts the current of the red LED (via C10, OP IC 1b, OP IC 3b, IC 5, OP IC 4a, and T3) so that the different temperature coefficients or agings of the green and the red LED are compensated. The Zener diode raises the potential so that T3 can act as current-decreasing element. Since the difference is measured, the temperature coefficient of the photoelectric cells has no effects whatsoever.

Calcium warning
During bicarbonate dialysis, calcium is deposited in the measuring cuvette. Due to color-specific attenuation, this calcium delivers a signal opposite to that of a blood leak. This is the reason why the difference between cuvette channel and reference channel is evaluated via OP IC 2a/2b for calcium warning.

Test
By means of H level, transistor T4 becomes conductive via R5, thus detuning, by means of voltage division at R21/R28, the control loop for the green/red ratio, so that a blood leak signal is simulated.

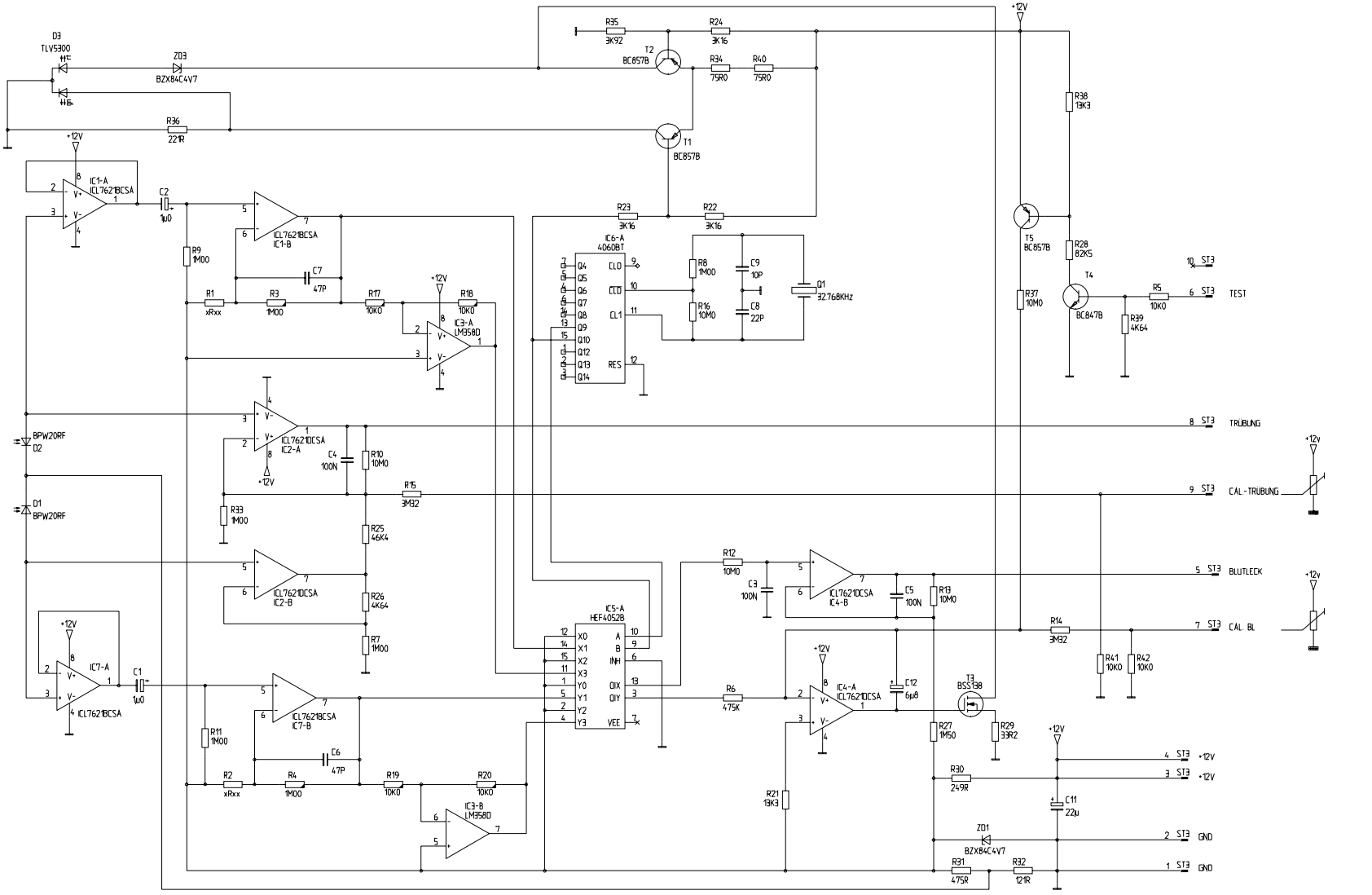
P.C.B. LP 493

Component layout diagram



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SCHUTZVERMERK NACH DIN 34 BEACHTEN



					DATUM	NAME	Blutleckdetektor Blood Leak Detector	BLATT 1 /1
				BE	18.01.89	Ott		
				GE	30.07.89	Asch		
					Layout:	LP493F		
b	IC7/R40-42 hinzu						LP493 651629	SP
a	R5/R39	011292	Asch	Fresenius Medical Care Deutschland GmbH				
	ANDERUNG	DATUM	NAME		ERSATZ	ERSATZ		FORM.A3

P.C.B. LP 493

Circuit diagram

8.9 P.C.B. LP 624 Control board (BP)

8.9.1 Description

- **General information**

This printed circuit board comprises both the control and the power part.

Plug connections on P.C.B. LP 624:
– X186, connection to the position sensor
– X188, connection to the stepper motor
– X189, connection to P.C.B. LP 748
– X190, connection to the lid switch
– X192, connection to the pressure transducer
– X348, connection to the dialysis monitor

Hex switches on P.C.B. LP 624:
0 Arterial blood pump
1 Single needle blood pump
2 – A Without function
B Adjustment of BP stop alarm (15 or 30 sec)
C Without function
D Inquiry of time meter (display x 100 = number of hours)
E Test operation (for the manufacturer only)
F Calibration of pressure transducer

- **Voltage generation**

Both the +24-V and the +12-V voltage supply are made available to the blood pump by the monitor. From the +24-V voltage, the +5-V supply voltage is generated on the module by the switching regulator IC 20, in order to minimize the dissipated power.

- **Stepper motor activation**

For noise reduction purposes, the stepper motor is operated in the microstep mode. Resolution amounts to 60 microsteps per step. The RISC processor alternately transmits an 8-bit dataword to pins 3 and 5 of the two D-A converters of IC 7. At the output of the converters, two sine-wave voltages are available, which are phase-shifted by 90 degrees. Together with the current direction signals, these voltages are supplied to the stepper motor controller IC 2. Together with the two SM drivers IC 1/IC 22 and the current sensor resistors R58/R59, IC 2, as feedback, forms a closed control loop, which impresses two sine-wave currents (phase-shifted by 90 degrees) in the two windings of the SM.

- **Microprocessor**

  The crystal Q2 between the processor connections 39 and 40 determines the pulse frequency of the microprocessor. The crystal starts to oscillate by means of the capacitors C5 and C6.
  The keyboard signals and the BSST, BPST signals are read in via port P4.
  IC 9 is provided as intermediate storage of the addresses A0 to A7.
  The ALE signal at pin 50 of the microprocessor represents the control line of the data-address latch.
  The signal from the revolution and rotational direction sensor (position sensor) in the pump bed is read in via port T1.
  When the hemodialysis machine is being switched off or if the voltage fails, the operating data of the pump are stored in the NOVRAM IC 21 via ports 5.1 to 5.3. The comparator IC 23 recognizes any low voltage (Power-Down).
  The WR line is used to control the data acceptance for the display into the external data latch IC 14.

- **Intermediate PLL circuit**

  Together with the counter IC 19, the PLL module IC 4 causes the frequency coming from the processor to be multiplied by a factor of 4, to activate the stepper motor, because the processor would be too slow for generating this frequency.

- **RISC processor**

  The RISC processor IC 5 is supplied with a pulse signal by the processor. Upon each pulse cycle, it alternately reads an 8-bit dataword from a look-up table for each phase of the stepper motor. This dataword contains both the current direction and the rated current value.
  In addition, the RISC processor contains a watchdog for the CPU.
  Should the pulse at pin 8/IC 5 be missing, the RISC processor releases a reset at the CPU via pin 7.

- **Display control**

  The dataword for energization of the display is stored in IC 14. Multiplexed operation of the individual digits is made possible by the decoder IC 18.

- **Speed and flow**

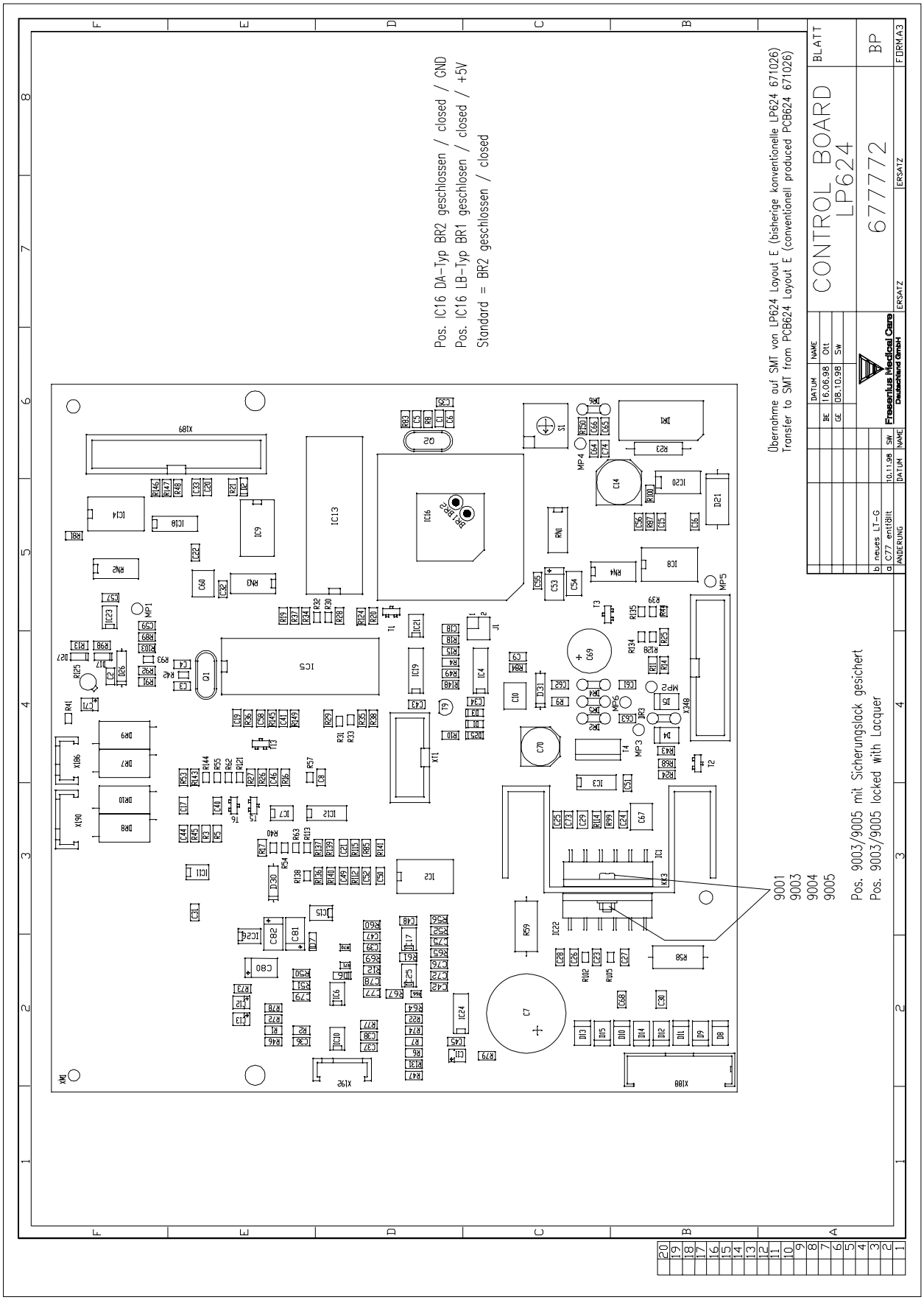
  The processor emits the speed via port P1.1 and the flow via port P1.2 to the dialysis monitor.

- **Pressure measurement**

  The differential measuring amplifier for the pressure transducer is formed by IC 6 (1/2/3) and IC 6 (5/6/7). Via the A-D converter input AN7, the measuring signal is read in by the processor and calibrated to zero point and slope via the software. Subsequently, the measuring signal is pulse-width-modulated via port P1.3 and emitted to the monitor via a following D-A converter IC 11 (5/6/7).

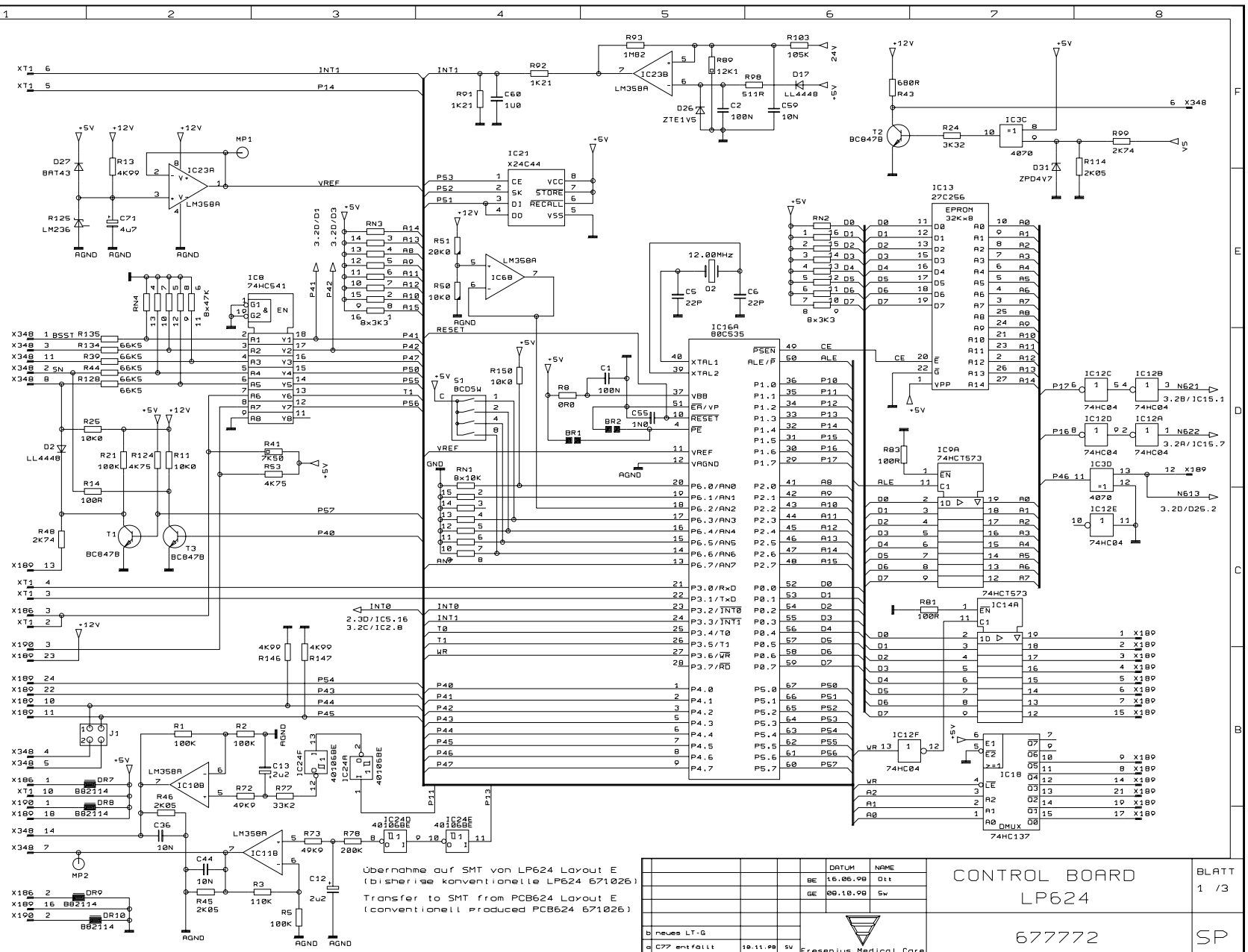
P.C.B. LP 624

Component layout diagram



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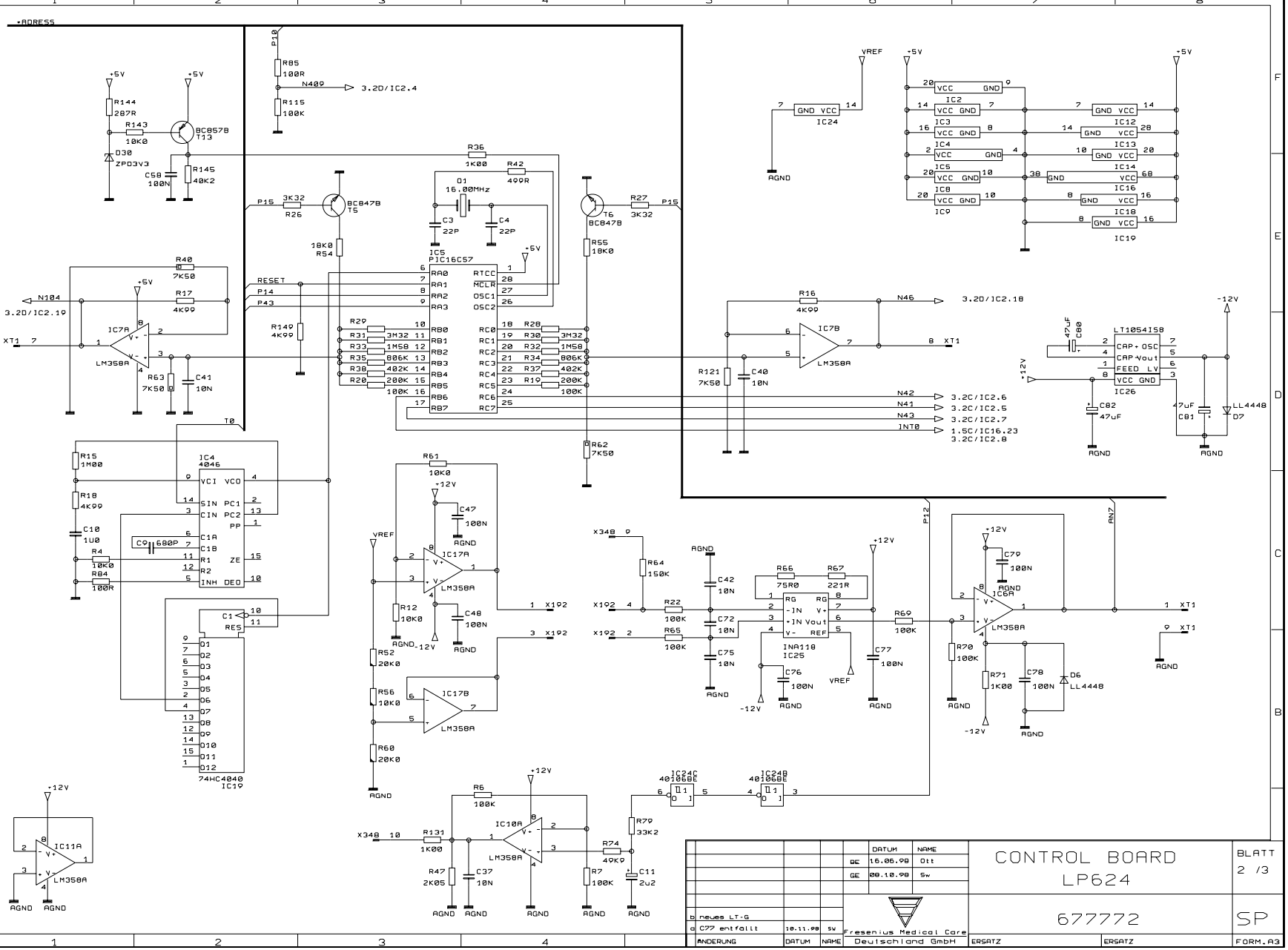


P.C.B. LP 624

Circuit diagram 1/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



	DATUM	NAME
BE	16.06.98	Ott
GE	08.10.98	Sw

b	neues LT-G		
a	C72 entfällt	10.11.98	Sw
ÄNDERUNG		DATUM	NAME

Fresenius Medical Care Deutschland GmbH

CONTROL BOARD LP624

677772

BLATT 2 /3

SP

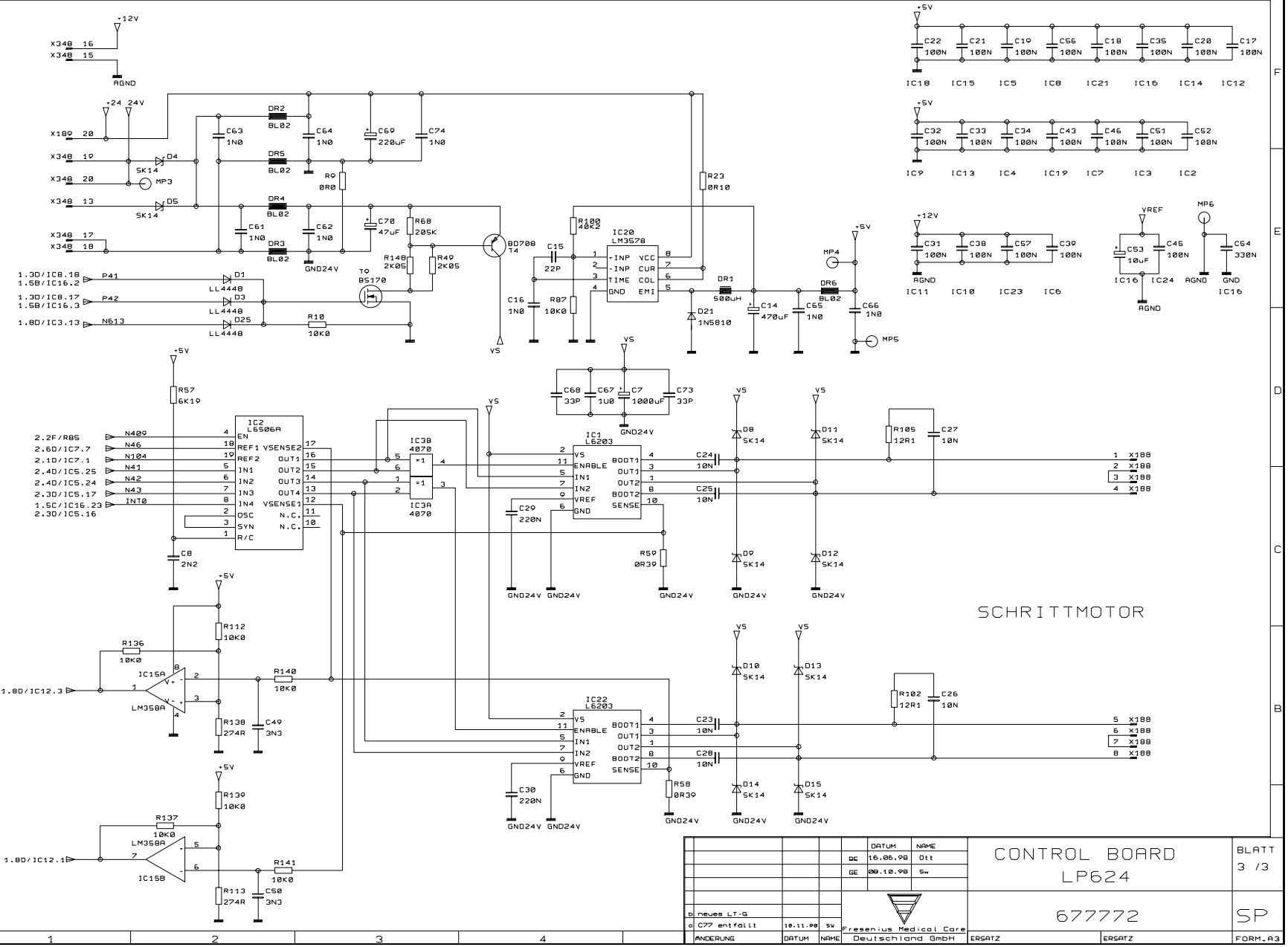
ERSATZ | ERSATZ | FORM.A3

P.C.B. LP 624

Circuit diagram 2/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



	DATUM	NAME		
	DE 16.06.98	Ott	CONTROL BOARD LP624	BLATT 3 /3
	GE 00.10.98	Sw		
b neues LT-G			677772	SP
c C77 entfällt	10.11.98	Sw	Fresenius Medical Care Deutschland GmbH	
ÄNDERUNG	DATUM	NAME	ERSATZ	ERSATZ FORM.A3

P.C.B. LP 624

Circuit diagram 3/3

8.10 P.C.B. LP 630 Motherboard

8.10.1 Description

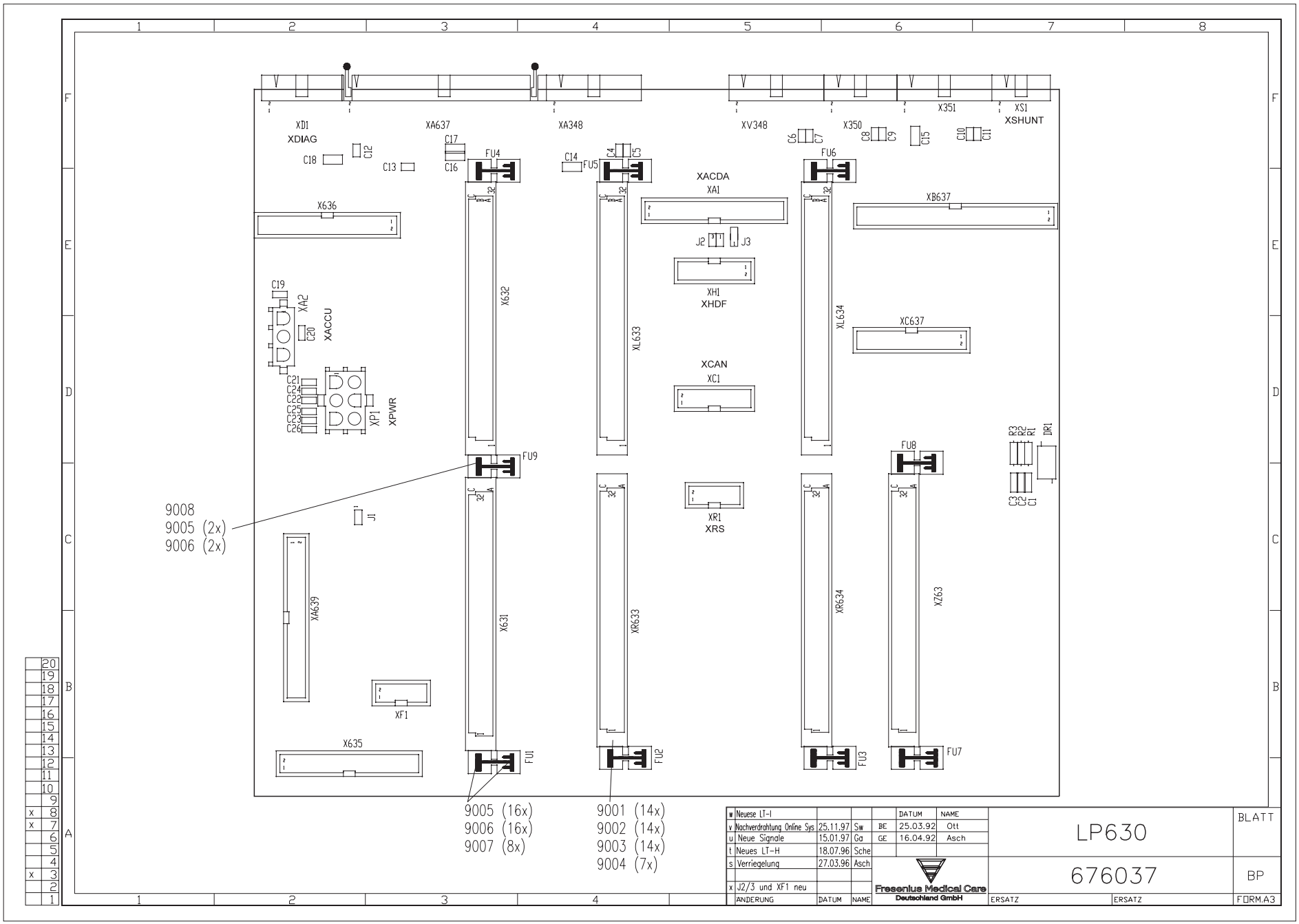
P.C.B. LP 630 is the main P.C.B of the monitor. It comprises the board locations for the control P.C.B.s and the display P.C.B. All control signals of the 4008 E (up to 24 V) are distributed from this P.C.B.

It is provided with the following connection sockets, which are accessible from the monitor rear:
- Auxiliary socket (e.g. HDF)
- RS232 (optical interface)
- ACDA (optional: connection of an LDL adsorption equipment)
- Diagnostics plug (connection of auxiliary service equipment)
- Sensors
- Blood pump (arterial)
- Blood pump (SN)
- Heparin pump
- Level detector
- Pumps
- Valves
- HDF (optional: connection of the ON-LINE HDF system)

8.10.2 Circuit diagram and component layout diagram P.C.B. LP 630 Motherboard

P.C.B. LP 630

Component layout diagram



w	Neuese LT-I				DATUM	NAME
v	Nachverdrahtung Online Sys	25.11.97	Sw	BE	25.03.92	Ott
u	Neue Signale	15.01.97	Ga	GE	16.04.92	Asch
t	Neues LT-H	18.07.96	Sche			
s	Verriegelung	27.03.96	Asch			
x	J2/3 und XF1 neu					
	ANDERUNG	DATUM	NAME			

Fresenius Medical Care Deutschland GmbH

LP630

676037

BLATT

BP

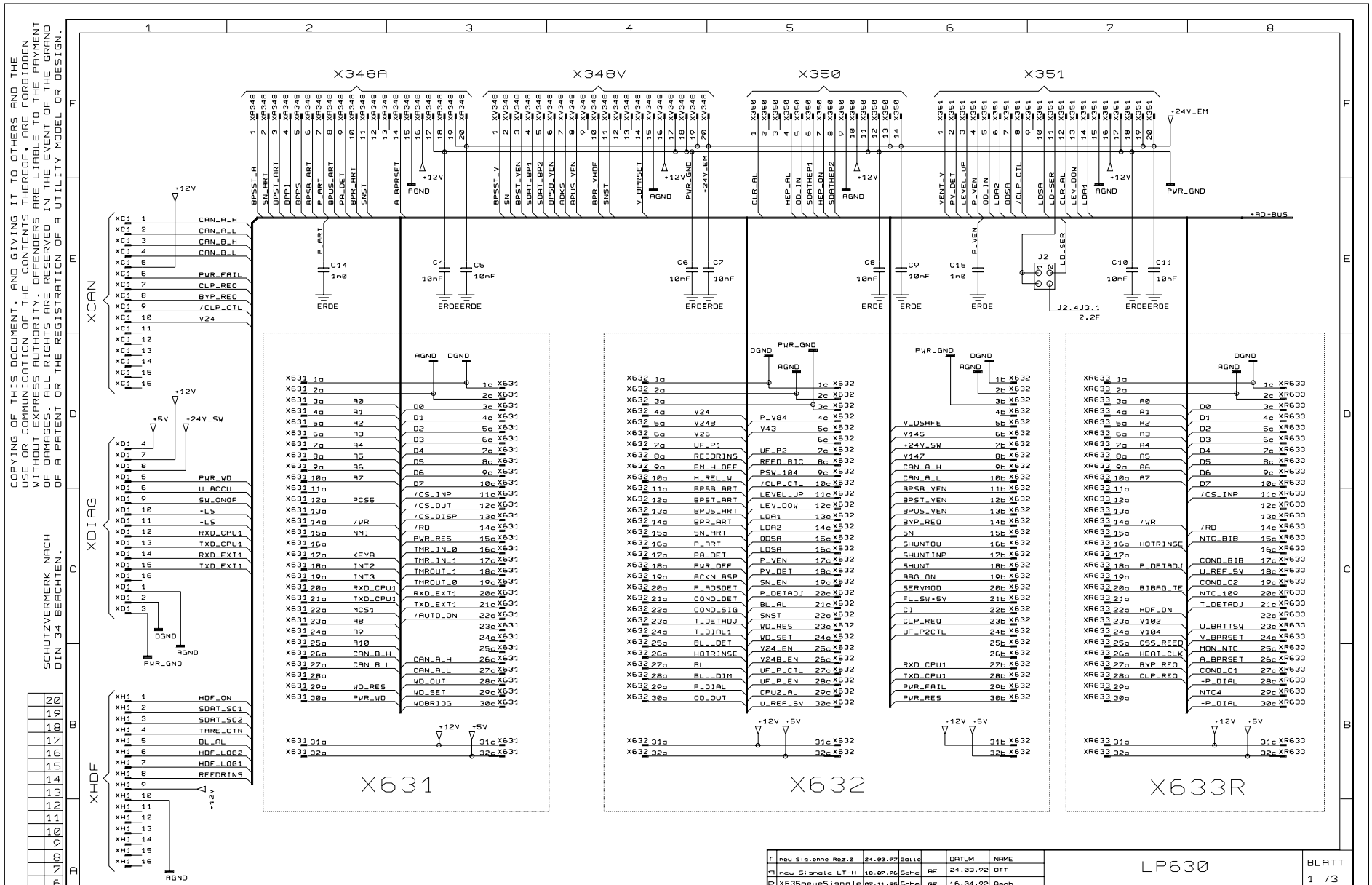
ERSATZ

ERSATZ

FORM.A3

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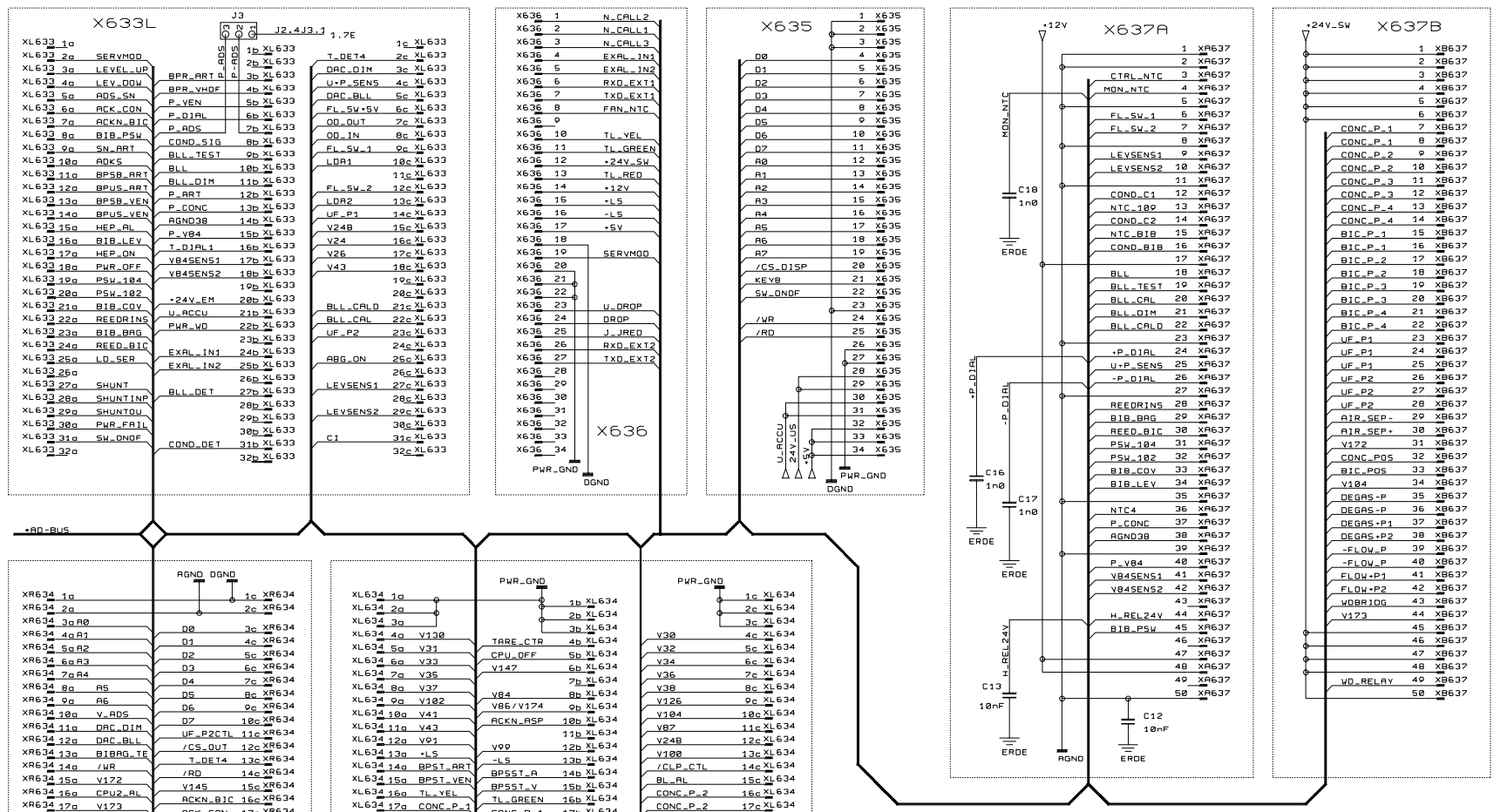
r	neu Sig.ohne Rez.2	24.03.92	Galle		DATUM	NAME	LP630 MOTHER-BOARD
q	neu Signale LT-H	18.07.96	Sche	BE	24.03.92	OTT	BLATT 1 /3
p	X635neueSignale	07.11.96	Sche	GE	16.04.92	Asch	
o	XF1 neu						676037 SP
t	neues LT-I						
s	Signal Anpassung	26.11.97	Sw	Fresenius Medical Care Deutschland GmbH			
	ANDERUNG	DATUM	NAME		ERSATZ	ERSATZ	FORM.A3

P.C.B. LP 630

Circuit diagram 1/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



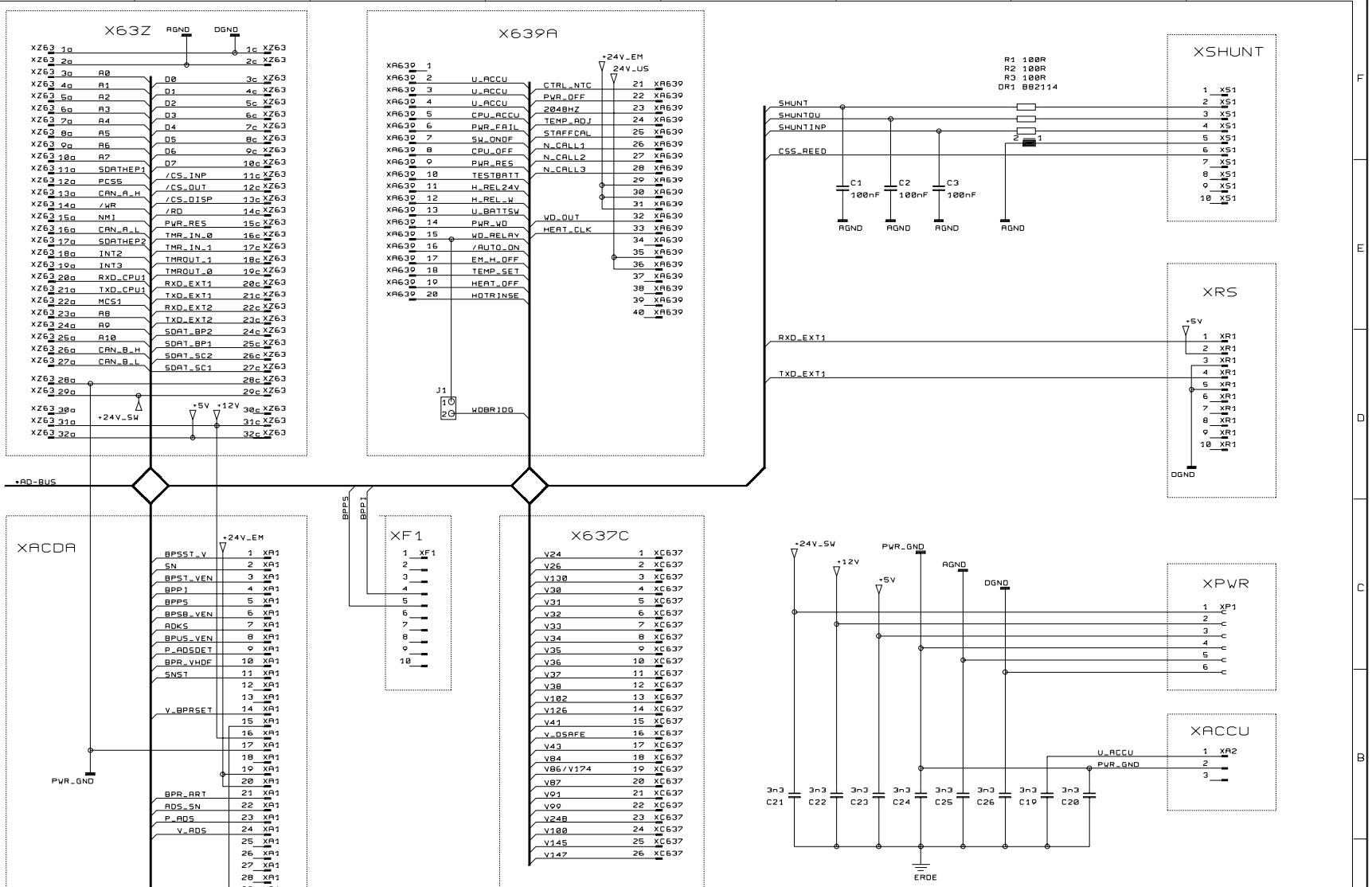
r	neu sig.anne Rez.2		DATUM	NAME	LP630 MOTHER-BOARD	BLATT 2 /3
q	neu Signale LT-H	BE	24.03.92	OTT		
P	X635neueSignale	GE	16.04.92	Asch		
u	XF1 neu					
t	neues LT-I				676037	SP
s	Signal Anpassung					
ANDERUNG		Fresenius Medical Care Deutschland GmbH			ERSATZ	ERSATZ FORM.A3

P.C.B. LP 630

Circuit diagram 2/3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



r	neu sig.ohne Rez.2		DATUM	NAME	LP630 MOTHER-BOARD	BLATT 3 /3
q	neu Signale LT-H	BE	24.03.92	OTT		
p	X635neueSignale	GE	16.04.92	Asch		
u	XF1 neu				676037	SP
t	neues LT-I					
s	Signal Anpassung					
	ANDERUNG		Fresenius Medical Care Deutschland GmbH		ERSATZ ERSATZ	FORM.A3

P.C.B. LP 630

Circuit diagram 3/3

8.11 P.C.B. LP 631 CPU 1

8.11.1 Description

- **General information**

P.C.B. LP 631 is a part of the safety-critical computer control.
This assembly group represents a complete single-board computer which, in conjunction with input and output boards and a second computer system, performs the entire control of the hemodialysis machine.
The use of large-scale integration modules manufactured in CMOS technology results in a high-capacity and reliable control unit with relatively small-size dimensions.

The P.C.B. comprises the following components:
– 8/16-bit microcontroller 80C188
– BUS logic
– EPROM
– Static RAM
– Nonvolatile RAM
– Serial interface module
– Real-time clock with battery buffering
– Watchdog
– DIP switch arrays
– Status indicator LEDs
– Multifunction PAL

- **8/16-bit microcontroller 80C188**

The 80C188 module, which has been developed from the INTEL 8088 microprocessor, represents a large-scale integration 16-bit microcontroller in CMOS technology.

The following components are already integrated in the chip:
– Clock generator
– Two independent DMA channels
– Programmable interrupt controller
– Three programmable 16-bit counters/timers
– DRAM refresh controller
– Programmable storage and peripheral equipment chip-select logic
– Programmable wait-state generator
– Local bus controller

Internally, the 80C188 operates with 16 bits, and externally with a multiplexed data-address bus with a data bus width of 8 bits. In conjunction with the integrated system components, the external 8-bit data bus width permits a simple, yet high-capacity circuit design.
With the exception of the DMA and the DRAM refresh controller, all of the components integrated in the IC are used during processing of the program.

- **BUS logic**

  In conjunction with the signals -DEN, DT/-R and ALE, three TTL latches (74HC573) and a data bus driver (74HC245) assume the task of editing the multiplexed data-address bus.
  The address bus is fully decoded and provided with a physical address range of 1 MByte.
  The edited data and address bus is conducted to the internal peripheral modules of CPU1 and to the 64-pin contact strip.
  The chip-select signals for external input/output boards are edited by the programmable storage and peripheral equipment chip-select logic incorporated in the microcontroller.

- **EPROM**

  The program to be performed by the machine is programmed in the EPROM. The 32-pin EPROM base can be equipped with program memory EPROMs of a size of up to 512 kBytes. The CS line of the EPROM is connected to the -UCS (Upper Memory Chip Select) signal of the microcontroller.
  This output of the microcontroller becomes active as soon as the microcontroller reads a program code or table data out of the EPROM.
  The scope of range, within which this line is intended to be active, can be programmed.

- **Static RAM**

  During operation of the machine, the 128-kByte read-write memories installed on the P.C.B. are used as data memories for temporary data, tables and branch vectors.
  The CS line of the RAM is connected to the _LCS (Lower Memory Chip Select) signal of the microcontroller.
  This output of the mcirocontroller becomes active, if the microcontroller requires or intends to store RAM data. The direction of data transfer is determined in conjunction with the -RD and -WR signals.
  The scope of the range, within which this line is intended to be active, can be programmed.

- **Nonvolatile RAM**

  The 48Z02 nonvolatile read-write memory (2 kBytes, no more than 8 kBytes) is provided for storing important data, which shall remain preserved after the machine has been switched off.
  A battery incorporated in the nonvolatile RAM as well as a voltage-monitoring circuit guarantee that the RAM cells are supplied with voltage and, consequently, that the data is preserved after the machine has been switched off.

  The following data is stored in the nonvolatile RAM:
  – Calibration data
  – Operating parameters
  – Data for power failure protection

  The CS line of the NOVRAM is connected to the -MCS0 (Midrange Memory Chip Select 0) signal of the microcontroller.
  This output of the microcontroller becomes active, if the microcontroller requires or intends to store NOVRAM data. The direction of the data transfer is determined in conjunction with the -RD and -WR signals.

  The scope of the range, within which this line is intended to be active, can be programmed.

- **Serial interface module**

  An 85C30 serial interface module SCC is mounted on the P.C.B. for the purpose of serial communication with external peripheral equipment. This module is designed so that it can control two serial transmission channels in the full-duplex mode.

  The transmission parameters for channel A and channel B are as follows:
  – Transmission rate of 9600 baud
  – 1 start bit
  – 8 data bits
  – 1 stop bit
  – No parity bit

  Channel A is reserved for the communication with CPU 2 (P.C.B. LP 632). It operates with TTL level.
  Channel B is reserved for connections to external peripheral equipment. The insertion of a V.24 level converter (MAX232) into the transmitter and receiver line permits unrestricted operation on the V.24 level.
  For channel B, the XON/XOFF method is used as handshake record.
  Since the processing speed of the microcontroller during the transmission sequences should not be reduced too strongly by status requests (Polling Mode), the major part of the communication is performed in the interrupt operating mode.
  To this end, the interrupt input of the SCC is monitored by the interrupt controller provided in the 80C188 (INT0). Upon the arrival of an interrupt request, i.e. a data byte is ready for being retrieved or the transmitter buffer is empty, the SCC unit which has released the interrupt is determined, and the respective function will be called up.
  The data transfer between the transmitter buffers in the RAM and the respective registers in the SCC is now performed by means of the interrupt treatment function. As soon as the data transfer has been completed, the interrupt treatment function will be terminated and the machine returns to the program which has been interrupted.

  In addition, the communication via channel A is used as technical safety device, i.e. the software watchdog. The software watchdog operates as a complement to the hardware watchdog. It is provided for monitoring the cooperation between the two processor boards CPU 1 and CPU 2.
  Here, the communication between the two computers via a serial interface and according to a fixed communication protocol is taken as a basis.
  Communication consists of the transmission of data records (10 bytes length) and the evaluation of a response from the receiver.

  Detailed structure of a data record with the venous pressure taken as an example:

Identification	Blank	Alarm status	Sign	Value	Checksum
(3 bytes)	(1 byte)	(1 byte)	(1 byte)	(3 bytes)	(1 byte)
VEN	–	H	+	400	X

Rules for communication:
Changes in value and/or status within the monitoring range of the respective computer result in the transmission of a data record to the other computer.

Each data record transmission must be acknowledged by the receiver.

Data records are acknowledged positively only if the comparison of the identification, the check of the value range and the comparison of the transmitted and computed checksum turned out to be positive.

Any negative acknowledgement by the receiver causes the transmission of the data record to be repeated.

Should a specific number of successive negative acknowledgements be exceeded, the system enters the “safe state”, i.e. either CPU 1 stops the periodical triggering of the hardware watchdog, or CPU 2 activates the hardware watchdog by emitting a WD_SET signal.

Should changes in value not cause any data record to be transmitted within a specific period of time, a data record is transmitted compulsorily with the reaction of the receiver being evaluated.

In order to prevent the regular program run from being inhibited by one computer permanently transmitting to the other, at least 50 ms must elapse between two transmissions.

- **Real-time clock with battery buffering**

  The integrated module IC M 7170 or RTC 6591 represents a real-time clock (RTC), which is required for the machine to switch on automatically and which serves as a reference time base for time-critical functions during processing of a program.
  Since the modules are buffered by lithium batteries, the system time continues to run correctly even after the machine has been switched off, thus preserving the programmed switch-on time.
  As soon as the programmed switch-on moment has been reached, the -INT output of the IC M 7170 emits a signal, which generates the -AUTO_ON signal via a switching transistor. This signal is conducted to the POWER-LOGIC P.C.B. (LP 639), causing the machine to be switched on.
  A variable capacitor permits the fine adjustment of the clock module (required with IC M 7170 only).

- **Watchdog**

  The watchdog monitors the correct function of the CPU 1 board.
  Should the watchdog detect a malfunction, the 24-V supply voltage for system slide-in modules and hydraulics is switched off via a relay.
  Before the 24-V supply voltage is switched on, an initial test (functional check) is performed. During this check, the correctness of the time window, the communication with CPU 2 and the switch-off possibility by CPU 2 are tested.
  Should one or more of these prerequisties not be fulfilled, switch-on of the 24-V supply voltage is inhibited, with the system emitting the message “watchdog error” and remaining in the “safe state”.
  In order to further increase operating safety, the hardware watchdog is equipped with its own supply voltage; in addition, all signals transmitted to the hardware watchdog are optically decoupled. Thus, a resistance against external voltages of >35 V is achieved.

  Structure:
  The hardware watchdog comprises two one-shot multivibrators of type CD4538, which are responsible for the generation of the time window, as well as a D-flip-flop of type 74HC74 which, in conjunction with a driver transistor, establishes the output stage. The multivibrator combination is triggered by pulses (every 500 ms), which are generated by the software on CPU 1.
  Control of the output stage is realized by means of two signals emitted by CPU 2 (WD_SET, WD_RESET).

Regular operation:
After a program runtime of 500 ms has elapsed, a function is called in, which releases a short-time pulse at a digital output (PAL 16V8). This pulse causes the first monoflop (MF1) to be triggered and the inverted output pin 7 to turn to L level. This, in turn, causes monoflop 2 (MF2) to be triggered with its output pin 10 turning to H level. As a result, the reset input of the D-flip-flop becomes inactive.
The D-flip-flop is set by a pulse emitted by CPU 2 (WD_RESET). The 24-V relay is switched on by the H level at output pin 9 (WD_OUT) via a following amplifier stage.
The watchdog is activated.
After the switch-on time of MF1 (approx. 400 ms) has elapsed, the inverted output (pin 7) turns to H level, thus deactivating the trigger input of MF2 (pin 11).
If the next trigger pulse for MF1 arrives within the time span prescribed, the output of MF1 returns to L level, thus retriggering MF2. Since the output pin 10 remains at H level, the D-flip-flop remains set.

Triggering too rapid:
Should MF1 be triggered too frequently, i.e. repeatedly during a specified time span, the switch-on time of MF1 will not elapse (permanent retriggering). The inverted output of MF1 continuously remains at L level. As a result, MF2 no longer receives any trigger edges and de-energizes after approx. 670 ms. Consequently, the 24-V relay will not be energized. It will de-energize (“safe state”).

Triggering too slow or missing:
Should too long a time span elapse until the trigger pulse for MF1 arrives, MF2 also de-energizes after approx. 670 ms, thus turning its output to L level.
Consequently, the 24-V relay will not be energized. It will de-energize (“safe state”).

Initial watchdog test:
After the hemodialysis machine has been switched on and the input/output modules have been initalized by the computer, the hardware watchdog is checked for correct function.
The watchdog test is divided in several phases.

1st phase:
The watchdog is triggered by CPU 1 in the correct time-slot pattern, and a trigger request is transmitted to CPU 2.
After having recognized the trigger request (prerequisite: correct communication between CPU 1 and CPU 2), CPU 2 tests the permissible voltage level (<5 V) and, with all prerequisites being fulfilled, emits a release pulse for the output-stage flip-flop (WD_RESET). Switch-on of the 24-V voltage is monitored via an analog input. If the voltage has achieved the correct value within a given period of time, the 1st phase is completed successfully.

2nd phase:
The watchdog is triggered too slowly; switch-off of the 24-V voltage is monitored. As soon as the voltage has fallen below a specific value (<5 V), the watchdog is again triggered by CPU 1 in the correct time-reference and a trigger request is transmitted to CPU 2.
After having recognized the trigger request, CPU 2 checks the permissible voltage level and, with all prerequisites being fulfilled, emits a release pulse for the output-stage flip-flop (WD_RESET).
With the function being correct, the 24-V voltage is again switched on. The increase of the 24-V voltage is monitored.
The 2nd phase is now completed successfully.

3rd phase:
The watchdog is triggered too rapidly; switch-off of the 24-V voltage is monitored. As soon as the voltage has fallen below a specific value (<5 V), the watchdog is again triggered by CPU 1 in the correct time-reference and a trigger request is transmitted to CPU 2.
After having recognized the trigger request, CPU 2 checks the permissible voltage level and, with all prerequisites being fulfilled, emits a release pulse for the output-stage flip-flop (WD_RESET).
With the function being correct, the 24-voltage is again switched on. The increase of the 24-V voltage is monitored.
The 3rd phase is now completed successfully.

4th phase:
The watchdog is triggered by CPU 1 in the correct time-reference. The 24-V voltage must be applied.
CPU 2 emits an inhibit pulse (WD_SET) to the reset input of MF2. The output of MF2 acts on the output stage and resets the output-stage flip-flop, thus inhibiting the energization of the relay. The 24-V voltage is switched off, this being monitored by CPU 2.
As soon as the voltage has fallen below a specific value (<5 V), the cut-off route of CPU 2 is tested, and CPU 2 emits a release pulse for the output-stage flip-flop (WD_RESET).
With the function being correct, the 24-V voltage is again switched on. Achieving of the 24-V voltage level is being monitored.
The 4th phase is now completed successfully.

Only after the above sequence has been completed correctly will the 24-V voltage for the further function run be switched on.

- **DIP switch arrays**

  Two eight-fold switch arrays permit various operating parameters to be predefined.

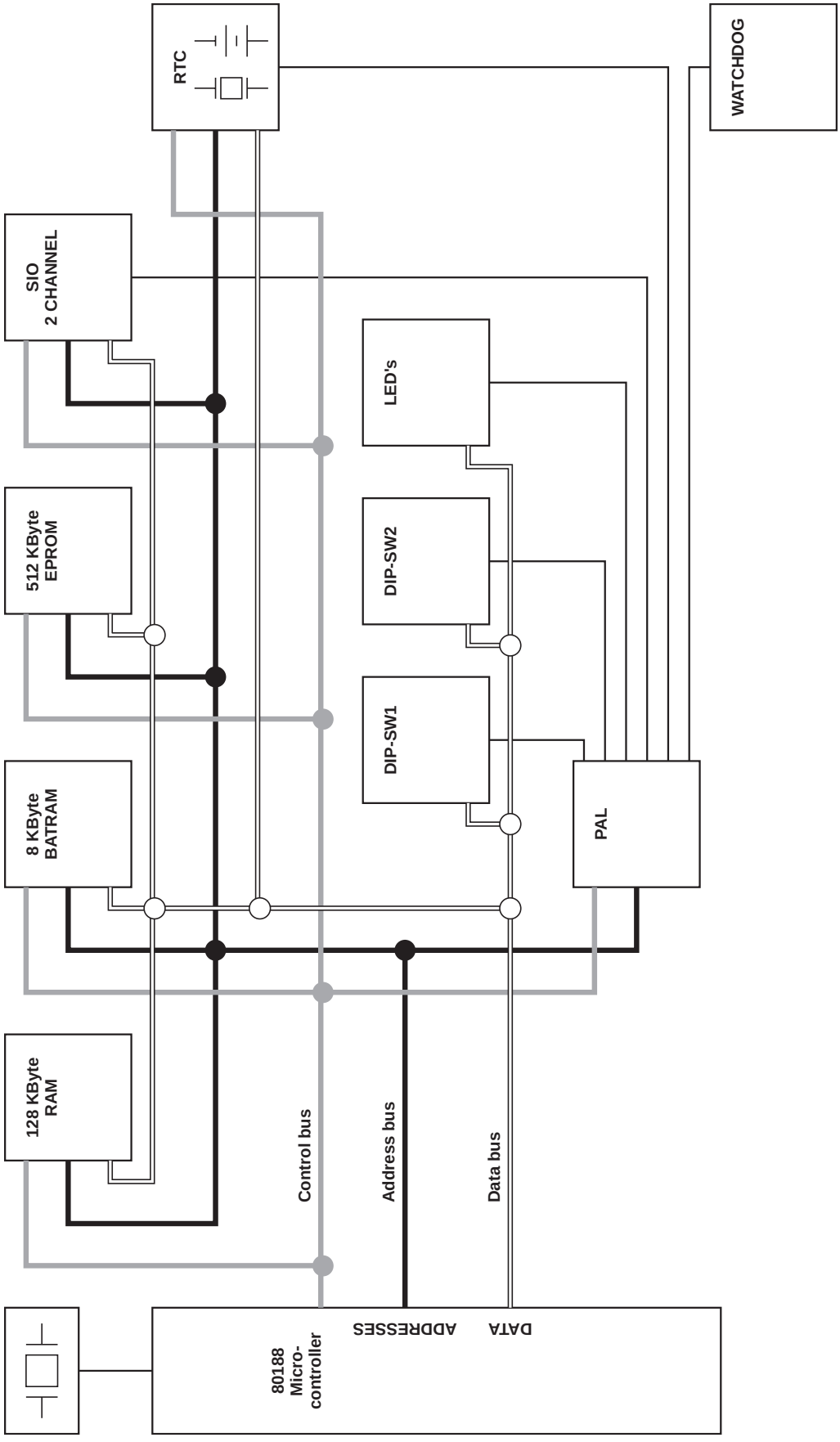
- **Status indicator LEDs**

  During the program run, an eight-fold LED line can be used for the indication of status values. After the machine has been switched on, various system initializations are performed by the program. Altogether, this procedure is performed four times, if the system is correctly initialized. Subsequently, the CRC check and the RAM test are performed. In addition to the text on the alphanumeric display, various LEDs are turned on. After these tests have also been completed successfully, the light-emitting diodes are used for indicating interrupt events and their processing time during the further course of operation.

- **Multifunction PAL**

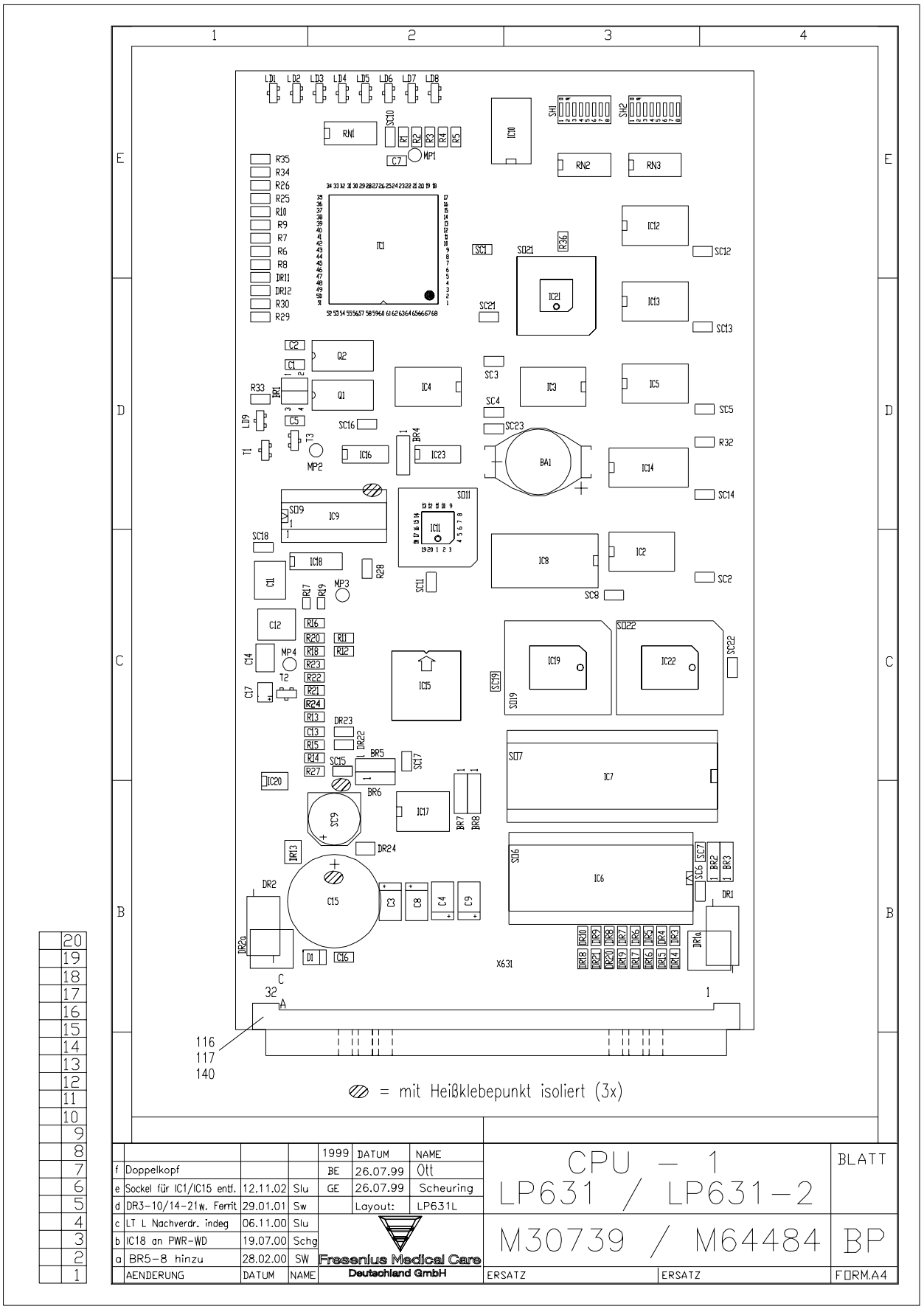
  A programmable logic module PAL16V8 or GAL16V8 is provided for decoding the chip-select signals and for various logical operations for various components of the CPU 1 board.
  Boolean operation tables and logical equations filed in the PAL determine the behavior of the outputs in relation with the input signals.
  By the use of a PAL, it was possible to reduce the number of the integrated logic ICs to a large extent.

Fig.: Block diagram P.C.B. LP 631 CPU 1



P.C.B. LP 631

Component layout diagram



				1999	DATUM	NAME
f	Doppelkopf			BE	26.07.99	Ott
e	Sockel für IC1/IC15 entf.	12.11.02	Slu	GE	26.07.99	Scheuring
d	DR3-10/14-21w. Ferrit	29.01.01	Sw		Layout:	LP631L
c	LT L Nachverdr. indeg	06.11.00	Slu			
b	IC18 an PWR-WD	19.07.00	Schg			
a	BR5-8 hinzu	28.02.00	SW			
	AENDERUNG	DATUM	NAME			

Fresenius Medical Care Deutschland GmbH

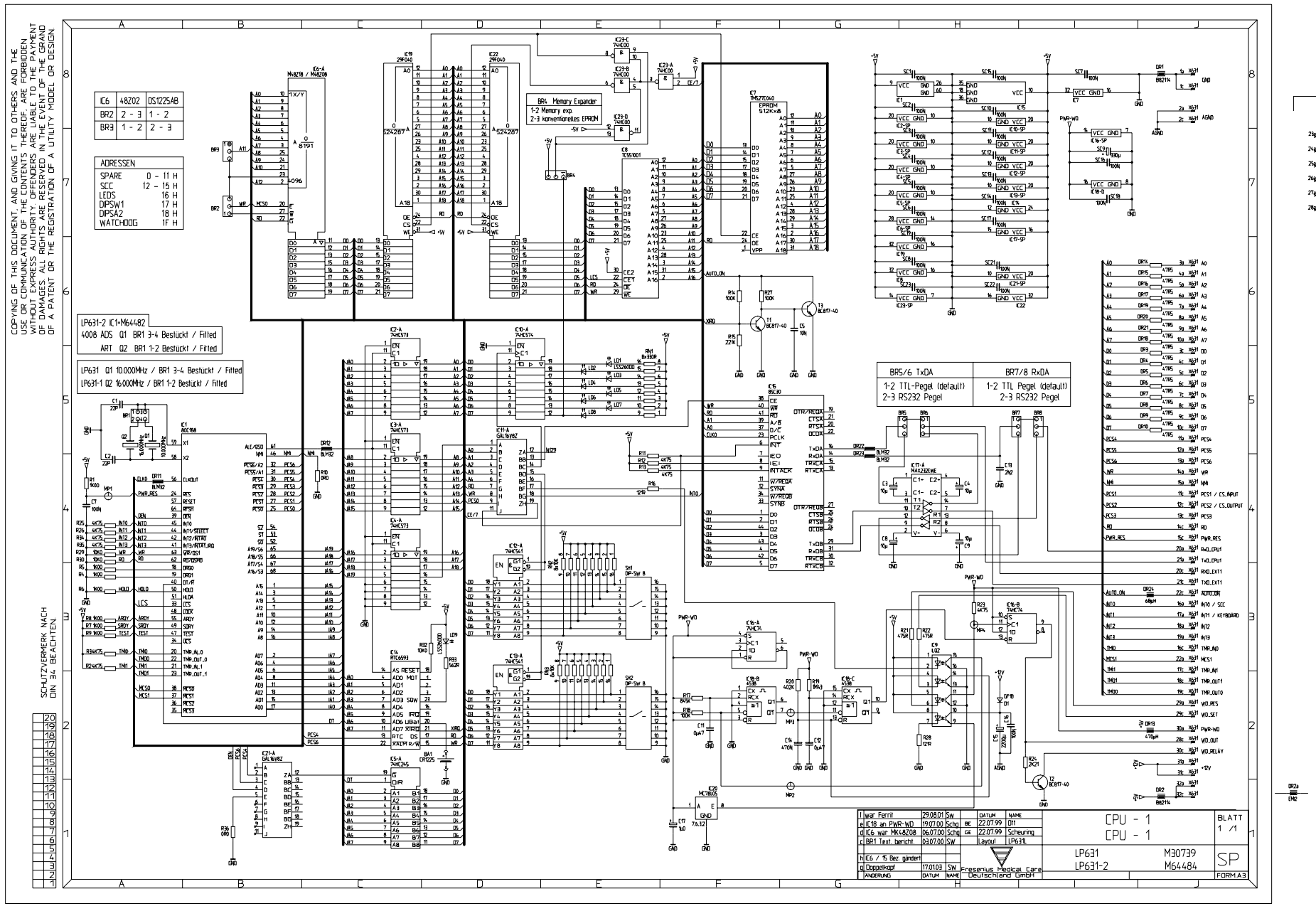
CPU – 1
LP631 / LP631-2

BLATT

M30739 / M64484

BP

ERSATZ | ERSATZ | FORM.A4


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SCHUTZVERMERK NACH DIN 34 BEACHTEN
IC6 | 48Z02 | DS1225AB
BR2 | 2 - 3 | 1 - 2
BR3 | 1 - 2 | 2 - 3
ADRESSEN
SPARE 0 - 11 H
SCC 12 - 15 H
LEDS 16 H
DIPSW1 17 H
DIPSA2 18 H
WATCHDOG 1F H
BR4 Memory Expander
1-2 Memory exp.
2-3 konventionelles EPROM
LP631-2 IC1=M64482
4008 ADS Q1 BR1 3-4 Bestückt / Fitted
ART Q2 BR1 1-2 Bestückt / Fitted
LP631 Q1 10.000MHz / BR1 3-4 Bestückt / Fitted
LP631-2 Q2 16.000MHz / BR1 1-2 Bestückt / Fitted
BR5/6 TxDA
1-2 TTL-Pegel (default)
2-3 RS232 Pegel
BR7/8 RxDA
1-2 TTL Pegel (default)
2-3 RS232 Pegel
CPU - 1
CPU - 1
LP631
LP631-2
M30739
M64484
SP
BLATT 1 /1
FORM-A3
Fresenius Medical Care Deutschland GmbH

P.C.B. LP 631

Circuit diagram

8.12 P.C.B. LP 632 CPU 2

8.12.1 Description

- **Technical data**

  – 80C535 processor
  – 11-MHz clock pulse frequency
  – 128 Kbyte EPROM
  – 8 Kbyte NOVRAM + 2 Kbyte SRAM
  – Serial interface
    (Baud rate 9600, 8 data bits, 1 stop bit, no parity bit)
  – 9 analog inputs (0 to 10 V)
  – 40 digital inputs (24 V, 12 V, TTL level)
  – 6 analog outputs (0 to 10 V)
  – 24 digital outputs (TTL level, Darlington output stages)

- **General information**

  P.C.B. LP 632 (CPU 2) has the following functions in the hemodialysis machine:
  – T1 test generation
  – Monitoring functions:
    a) Transmembrane pressure
    b) Conductivity
    c) Temperature
    d) Hardware and software watchdog
    e) UF pump monitoring
  – Calibration voltages for:
    a) Dialysate pressure (TMP display)
    b) Temperature display

  P.C.B. LP 632 (CPU 2) constitutes a closed self-contained processor system (stand-alone system). Basically, the electronics of the CPU 2 comprises three essential functional blocks:
  – Analog and digital inputs
  – Microcontroller system
  – Analog and digital outputs

- **Analog inputs (circuit diagram 1/6)**

  All of the analog inputs are designed for an operating range of 0 to 10 V. Each of these input signals is adapted to the integrated analog-to-digital converter (AN0 to AN7) of the 80C535 via a voltage divider with 12.1-kOhm/10-kOhm resistors and an impedance transformer (IC1 and IC2, LM324N – quadruple OP).

  The 5-V reference voltage (UREF_5V) required for the A-D converters is derived from the 12-V voltage supply via voltage divider R1/R2 (43.2 K/30.1 K). The two following operational amplifiers (IC 3, LM324N) are used as impedance transformers. They deliver the current required for the integrated A-D converter of the 80C535 microcontroller and supply further A-D converters on the input board P.C.B. LP 633 via the contact strip x632/C30.

- **Digital inputs (circuit diagram 2/6)**

  All of the digital inputs are each connected to a voltage divider. These voltage dividers are used to adapt different signal voltages (24 V, 12 V, 5 V) to the system of P.C.B. LP 632.

  Values are as follows:
  – 178.0 kOhms/47.0 kOhms for the 24-V signals,
  – 66.5 kOhms/47.0 kOhms for the 12-V signals.
  – The values for the 5-V signals (TTL level) are directly applied to the inputs of the drivers. Here, the resistors provided have been replaced by bridges.

  Via IC 4 and IC 5 (74HC541), the input signals are directly made available to the microcontroller at port 4 and port 1. IC 6, IC 7, IC 8, IC 9, and IC 10 (74HC541, TRI-STATE BUFFER) are used as bus drivers in the system. The processor has access to these drivers and inputs via specific addresses (chip-select signals).

  The two DIP SWITCHES SW1 and SW2 are provided for the basic adjustment of the hemodialysis machine or for the adjustment of its optional equipment.

- **Microcontroller system (circuit diagram 4/6)**

  Microcontroller 80C535 (IC 12) is used as CPU. The microcontroller is provided with a multiplexed 8-bit data and address bus as well as freely available ports. The clock pulse frequency of the processor is deducted from a 11.0592-MHz crystal oscillator. A serial interface (TxD/pin 22 and RxD/pin 21) is directly made available by CPU 80C535. This interface is used for the communication between CPU 2 and CPU 1 (P.C.B. LP 631).

  The peripheral module PSD313 (Programmable System Devices) contains all elements necessary for the microcontroller system. The interface to the microcontroller consists of the multiplexed 8-bit bus and address lines A8 to A15 mentioned above, as well as of the following control lines /RD-, /WR-, ALE-, /PSEN- and /RST. A further element of the PSD313 is represented by the 128-Kbyte EPROM (read only memory), where the program is filed, as well as a 2-Kbyte SRAM, which is used by the microcontroller. In addition, the module contains a programmable logic (PLD) for the generation of the necessary chip-select signals, which are emitted at ports B and C, and an address latch, which is available via port A. This address latch serves to separate data signals (D0 to D7) and address signals (A0 to A7) and/or to store the address bytes.

  Tasks of the control lines:

/RD	ReaD	Reading of the addressed RAM point
/WR	WRite	Writing into the addressed RAM point
ALE	AddressLatchEnable	Storing of the address bytes A0 to A7 into the latch
/PSEN	ProgramStorENable	Reading of operations/commands from the EPROM
/RST	ReSeT	Microcontroller defined at zero address or new system start

  The storage module MK48Z08 (IC 13) constitutes an 8-Kbyte NOVRAM (nonvolatile read access memory). This NOVRAM is required for protecting the calibration data. A part of the NOVRAM, which is not needed for the calibration data, is used as normal RAM during the program run.

- **Analog and digital outputs (circuit diagram 5/6)**

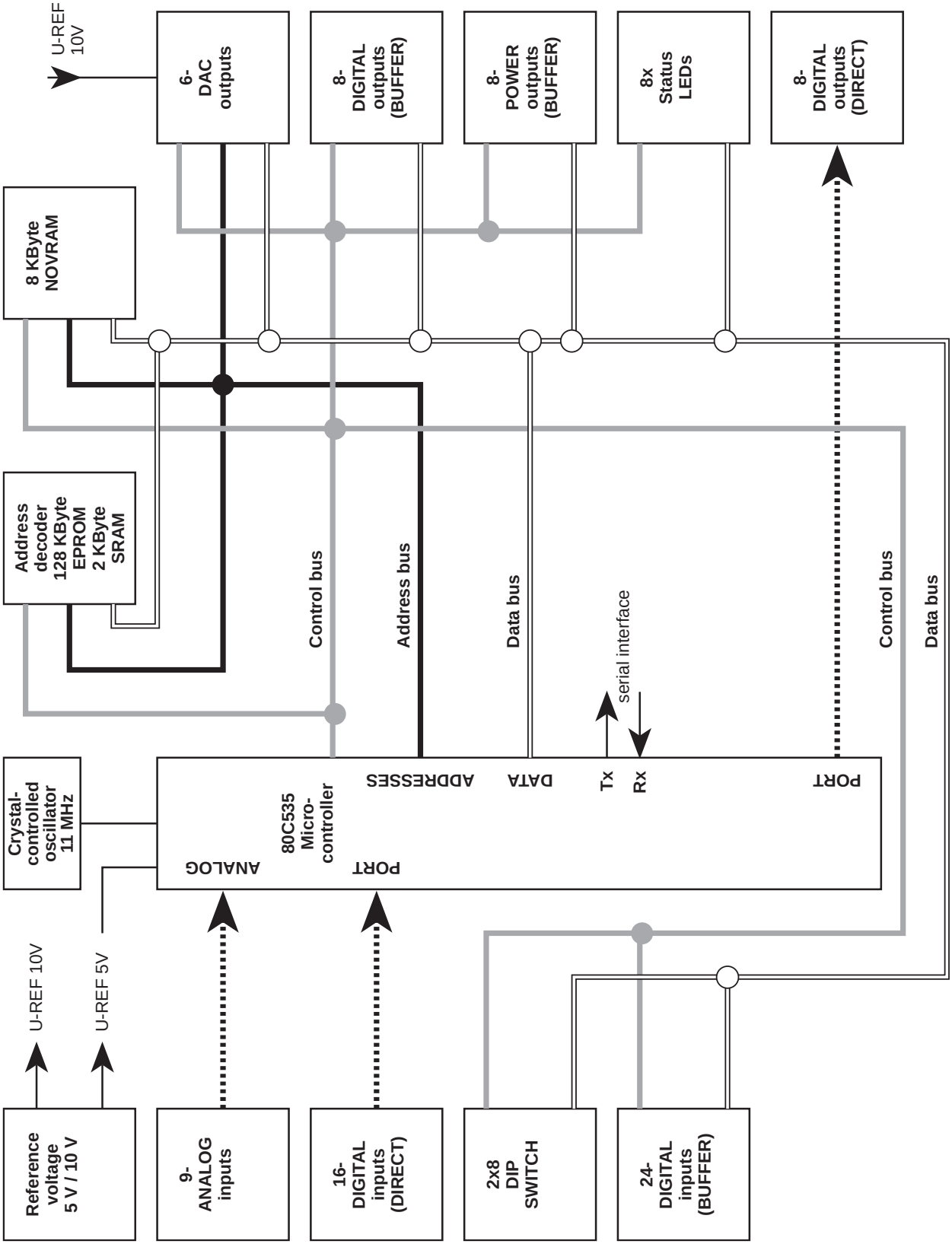
  The analog outputs are required for detuning the sensors during the test generation. The digital-to-analog converter IC11 (AD7226) is provided with four analog outputs. During the test, channels C and D are switched onto the sensor lines via analog switches (IC 20, CD40066). The analog switches are activated via level converters (IC 32, 4505). During hemodialysis, channel C is additionally used to adjust the working point of the high-resolution dialysate pressure amplifier (IC 3, LM324N). Channels A and B of the D-A converter simultaneously serve to generate the calibration voltages for the temperature display and the dialysate pressure (TMP display).

  The 10-V reference voltage is deducted from the temperature-stabilized zener diode IC 28 (ZN458B). IC 3/OP-2 operates as non-inverting amplifier and generates the +10-V reference voltage for the D-A converter IC11.

  Since it might become necessary to activate various peripheral equipment (control signals, actors, etc.), P.C.B. LP 632 is provided with TTL level outputs IC 24 / 74HC541/74HC574 and IC 30 / 74HC574 as well as with level converters (IC 33, 4504), which generate the 12-V signals. The IC 29 (UCN2803) with Darlington output stage (max. 600 mA) is used to control actuators (e.g. solenoid valves V43).

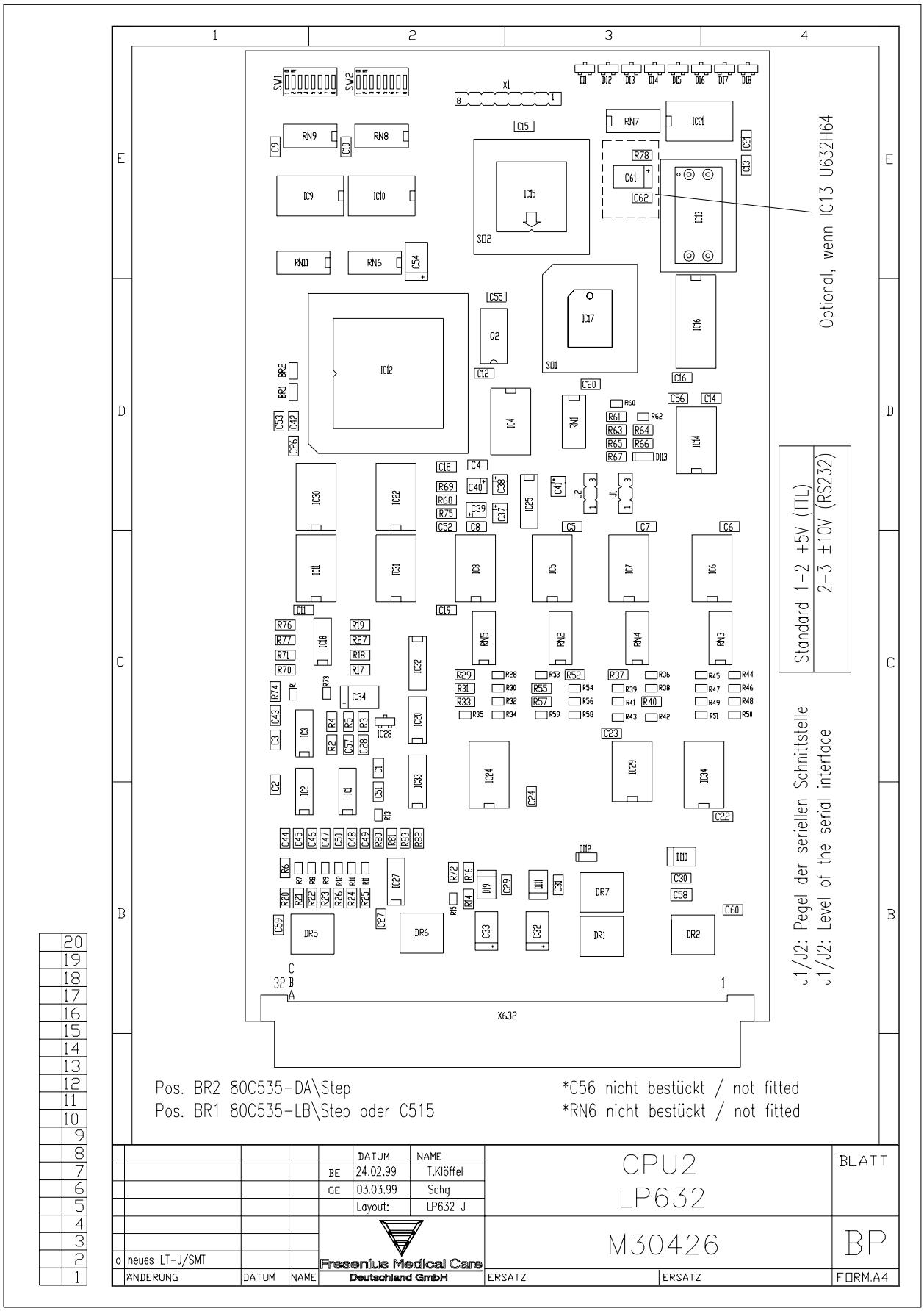
  In addition, 8 status LEDs are available, which are switched by the microcontroller via IC 21 (74HC574).

Fig.: Block diagram P.C.B. LP 632 CPU 2



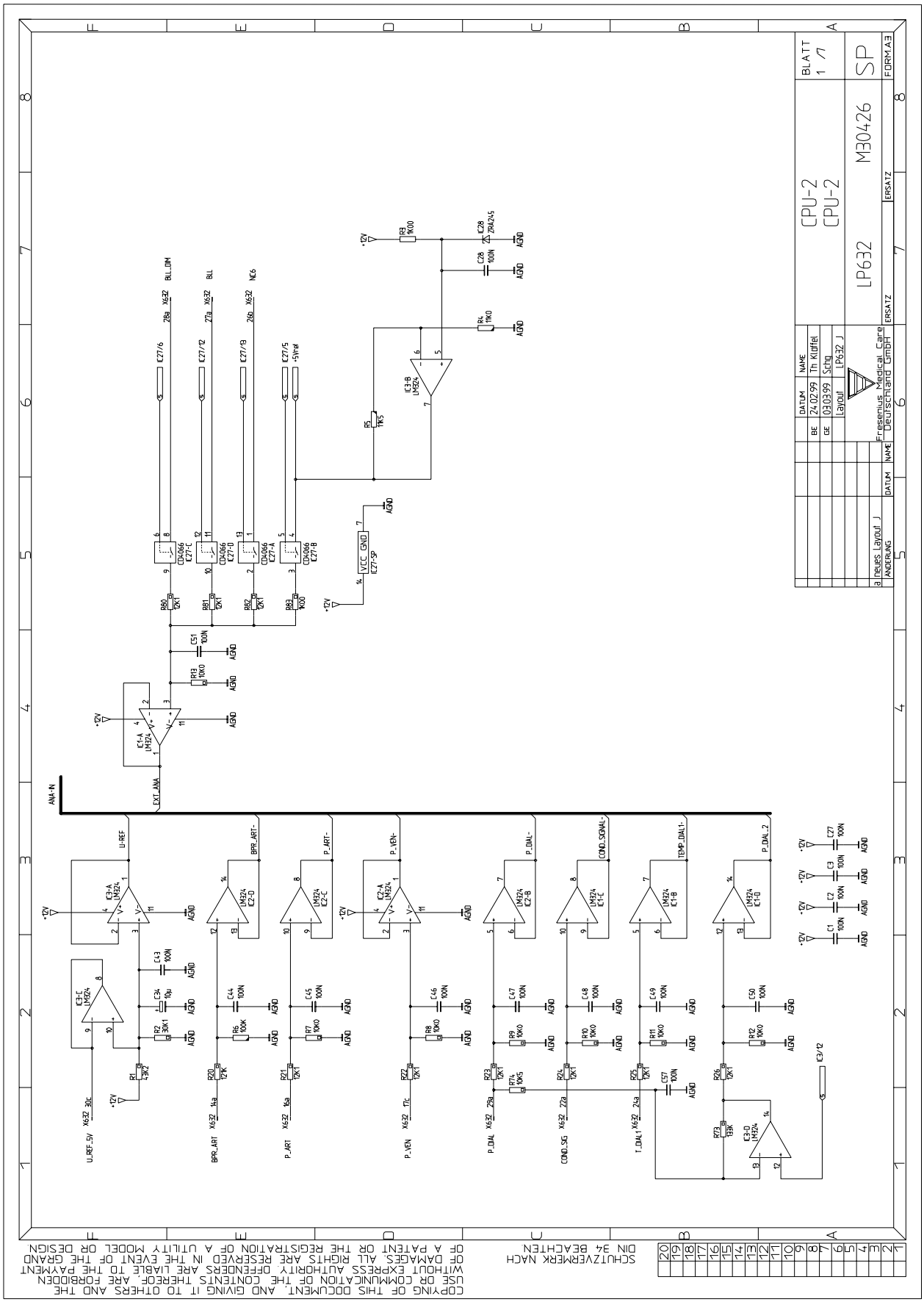
P.C.B. LP 632

Component layout diagram



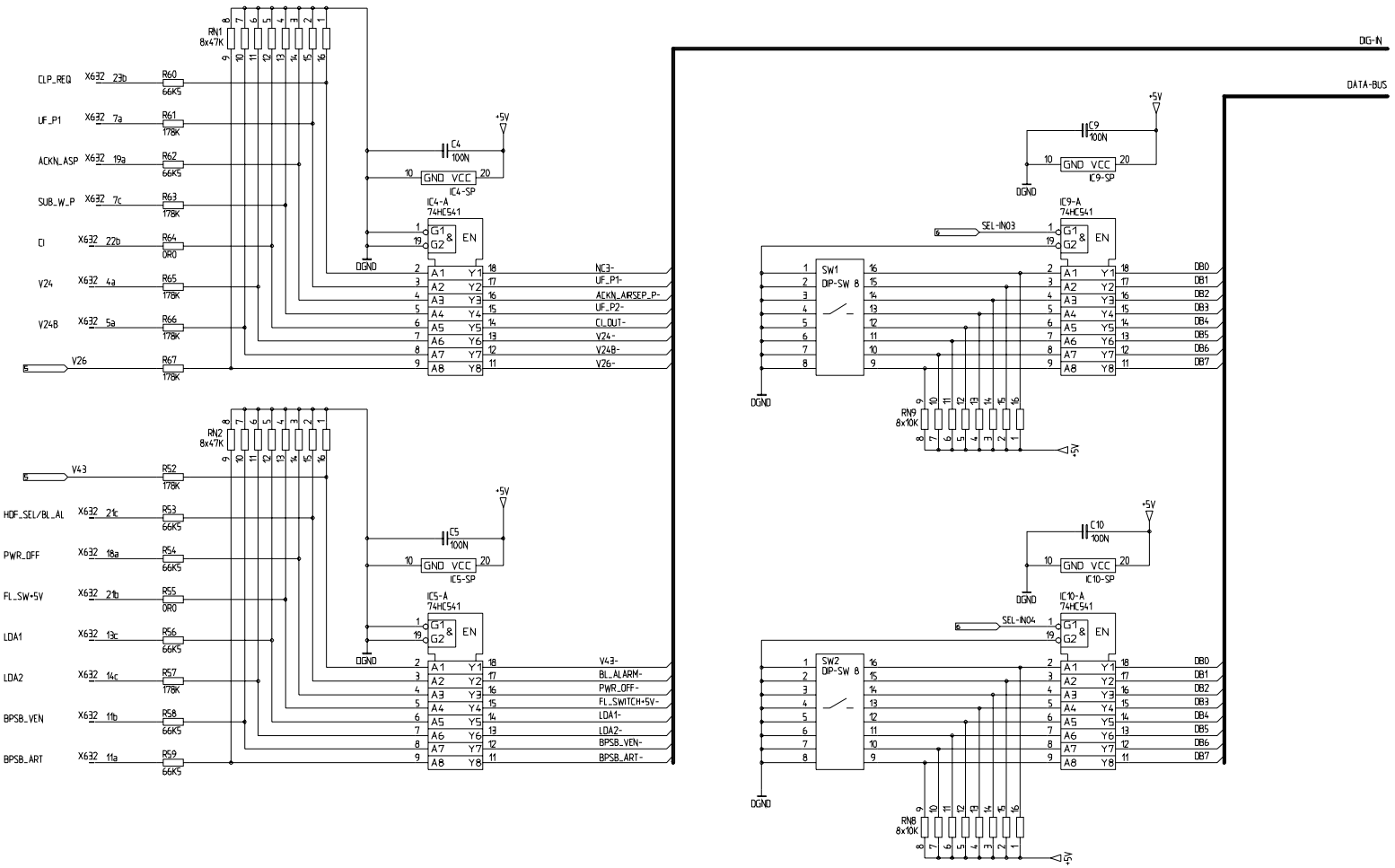
P.C.B. LP 632

Circuit diagram 1/7



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SCHUTZVERMERK NACH
DIN 34 BEACHTEN



	DATUM	NAME
BE	24.02.99	Th Klaffel
GE	03.03.99	Schg
Layout		LP632 J

a neues Layout J

ANDERUNG | DATUM | NAME

Fresenius Medical Care Deutschland GmbH

CPU-2
CPU-2

LP632 M30426 SP

BLATT 2 / 7

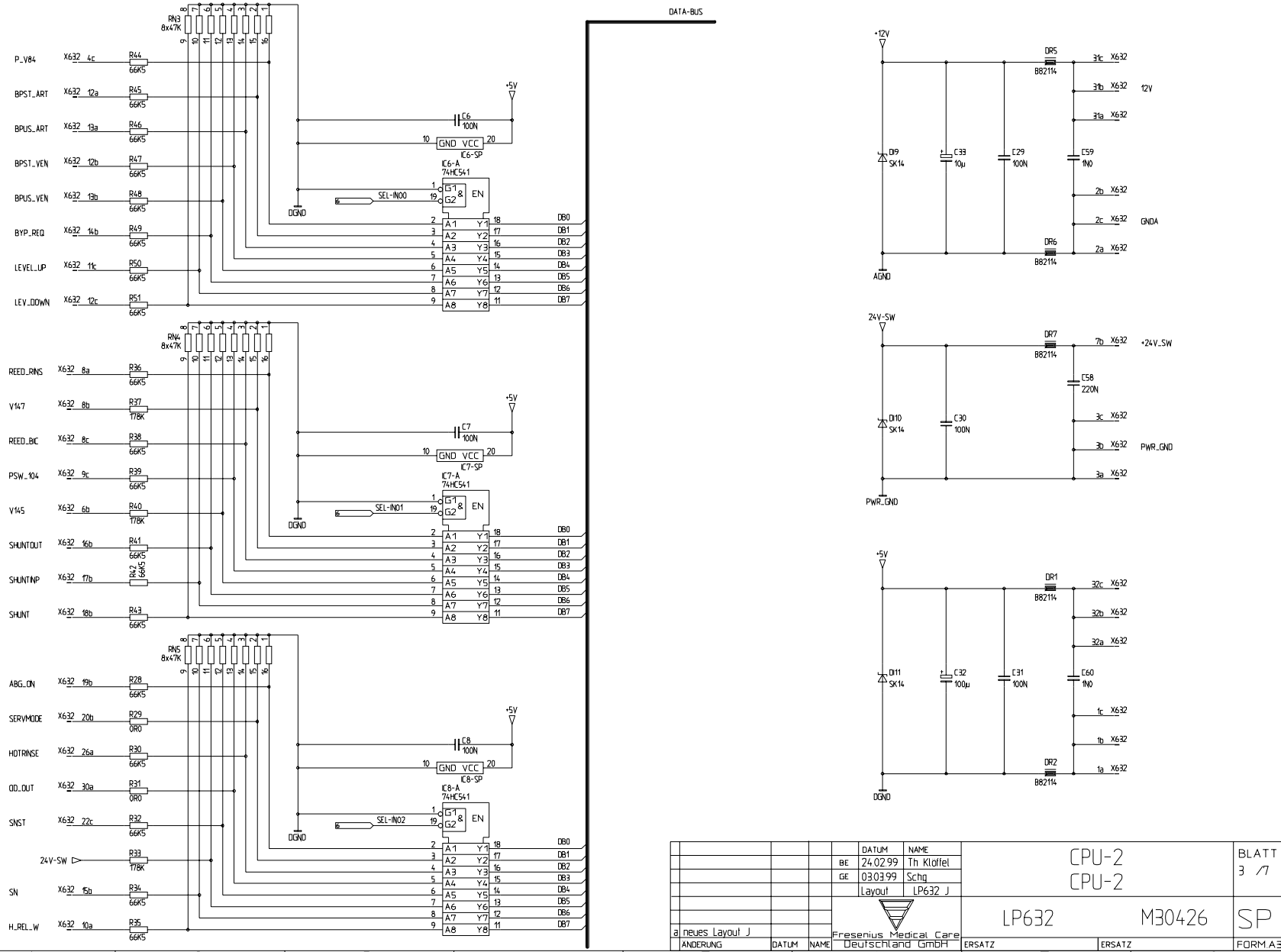
ERSATZ | ERSATZ | FORM.A3

P.C.B. LP 632

Circuit diagram 2/7

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN



	DATUM	NAME
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GE	03.03.99	Schg
Layout		LP632_J

a neues Layout J

Fresenius Medical Care Deutschland GmbH

CPU-2
CPU-2

LP632 M30426 SP

BLATT 3 /7

ANDERUNG DATUM NAME ERSATZ ERSATZ FORM A3

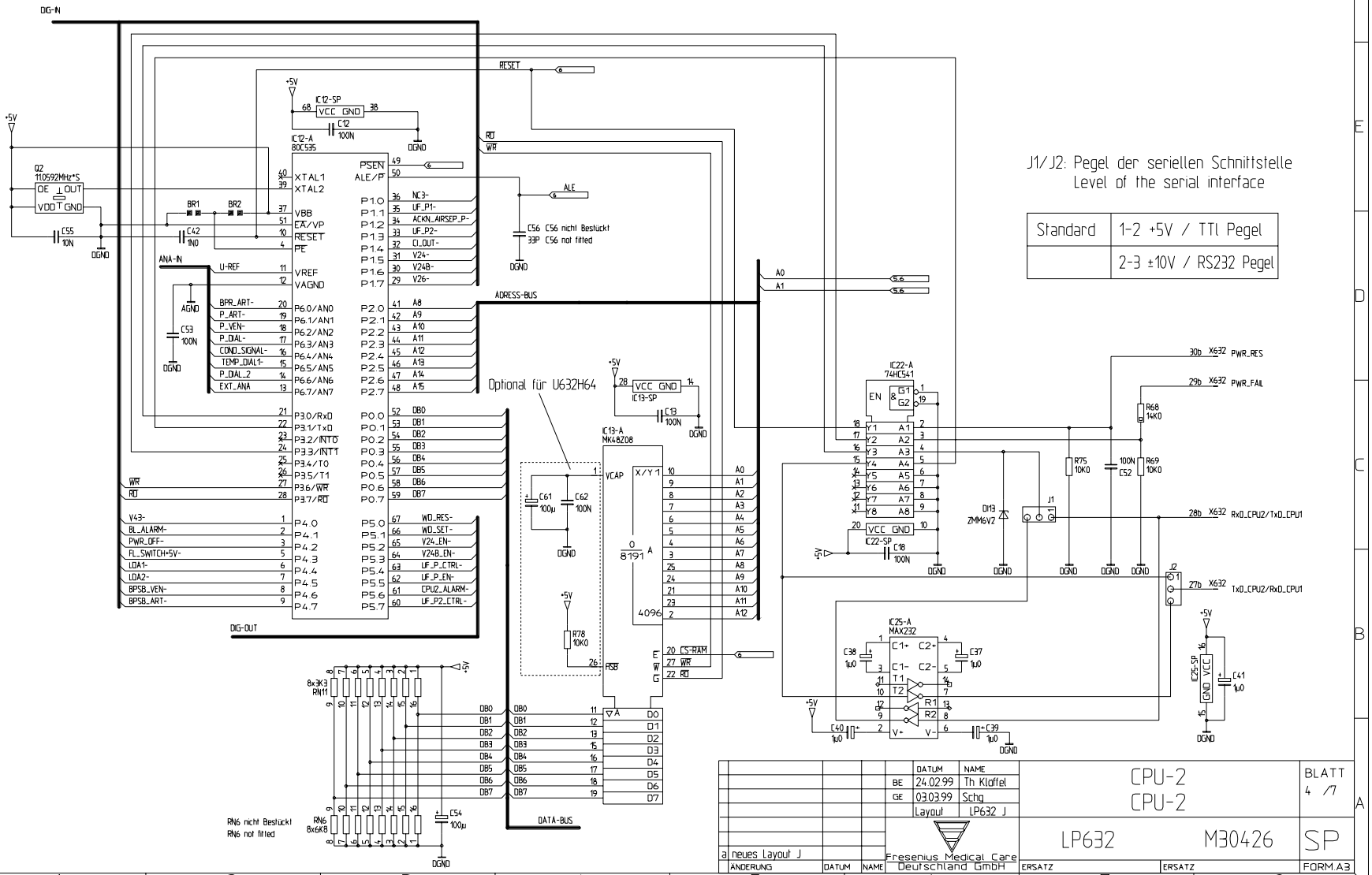
P.C.B. LP 632

Circuit diagram 3/7

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SCHUTZVERMERK NACH DIN 34 BEACHTEN

Pos. BR1 80C535-DA/Step
Pos. BR2 80C535-LB/Step oder C515

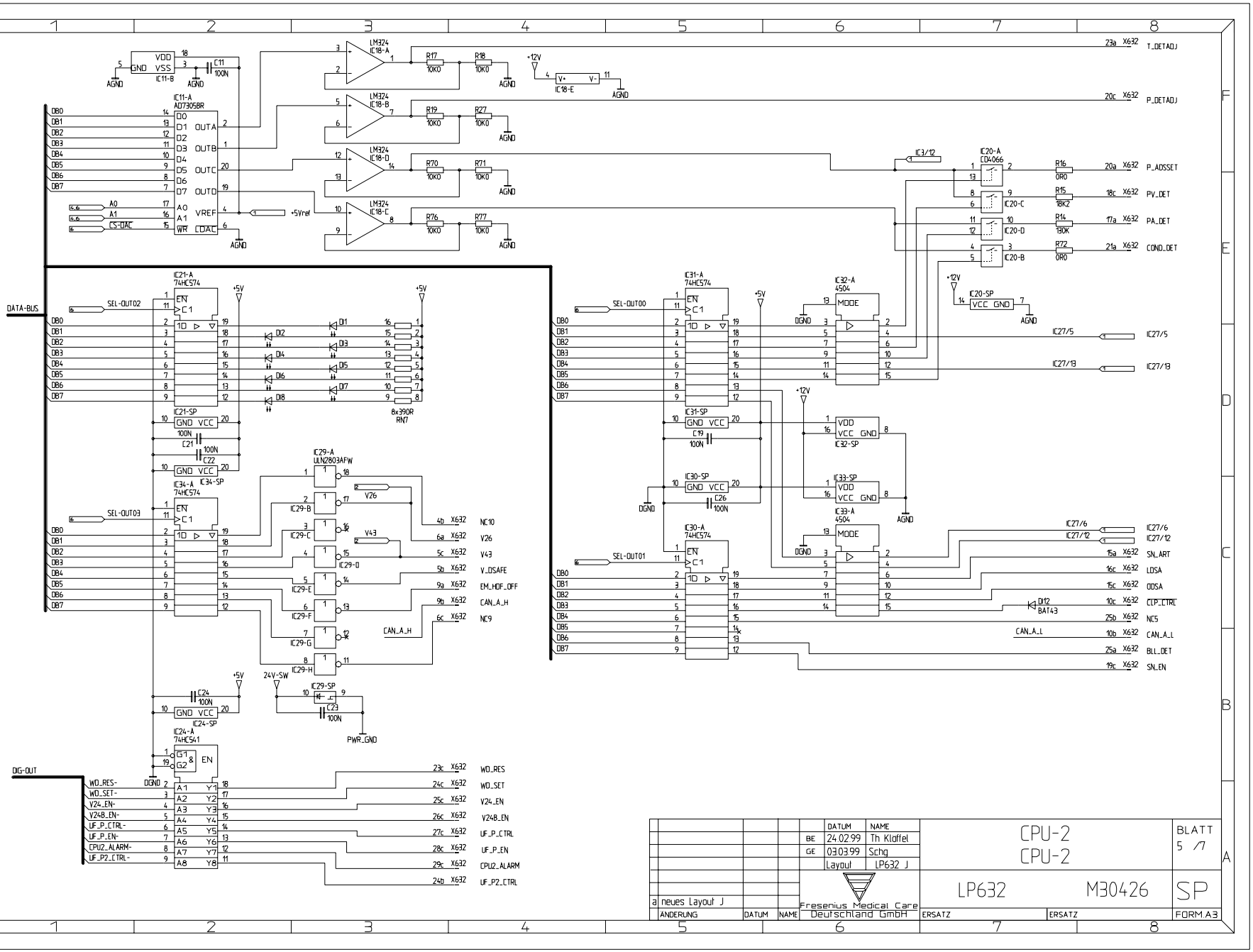


P.C.B. LP 632

Circuit diagram 4/7

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			GE	03.03.99	Schg
				Layout	LP632 J
a	neues Layout J				
	ANDERUNG	DATUM	NAME		

Fresenius Medical Care Deutschland GmbH

CPU-2
CPU-2

LP632 M30426 SP

ERSATZ ERSATZ FORM.A3

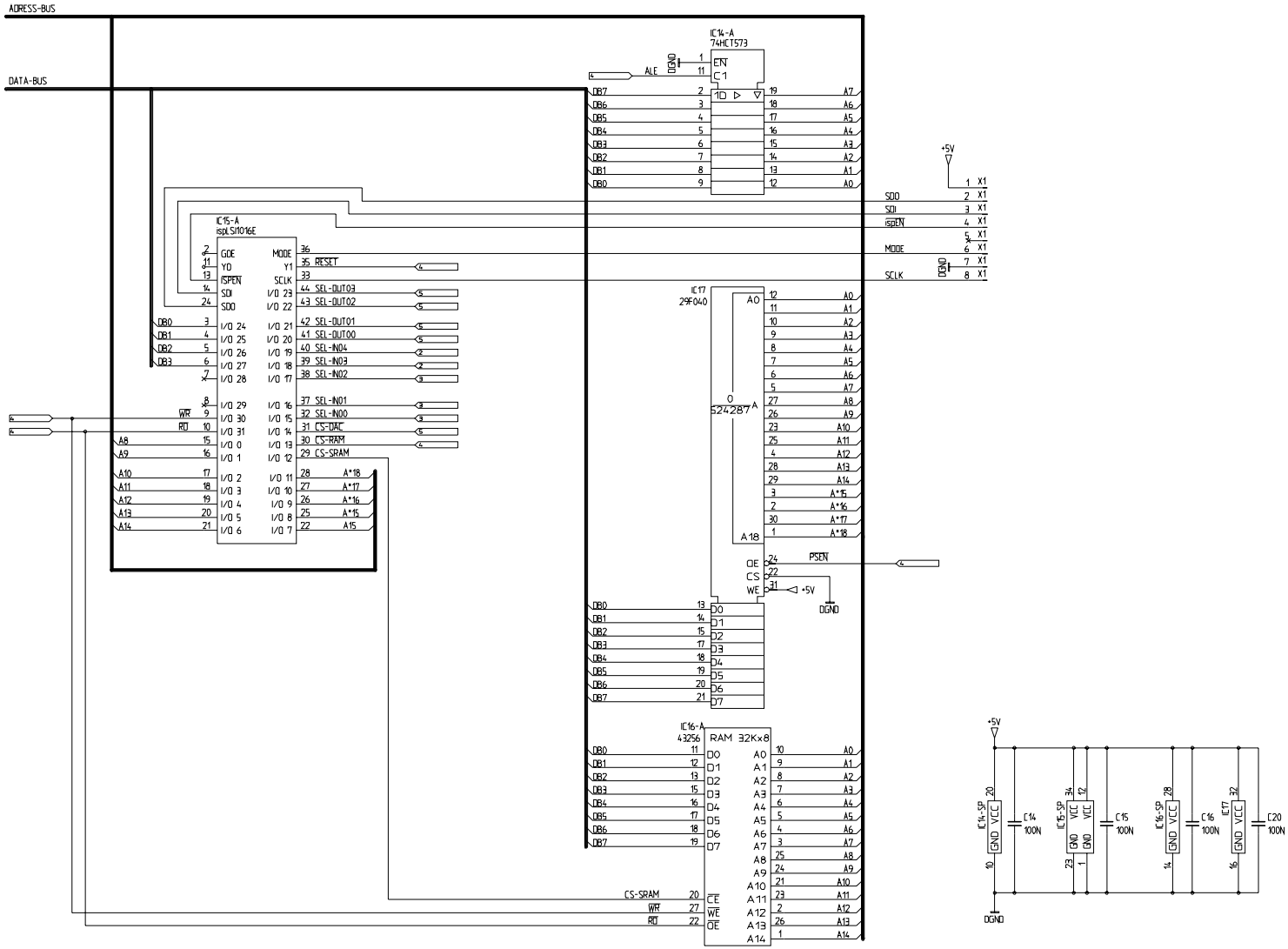
BLATT 5 /1

P.C.B. LP 632

Circuit diagram 5/7

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN



				DATUM	NAME
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			GE	03.03.99	Schg
				Layout	LP632 J
a	neues Layout J				
	ÄNDERUNG	DATUM	NAME		

Fresenius Medical Care Deutschland GmbH

CPU-2
CPU-2

LP632 M30426 SP

ERSATZ ERSATZ FORM A3

BLATT 6 / 7

P.C.B. LP 632

Circuit diagram 6/7

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

Pinbelegung X632

Pin	A	B	C
1	DGND	DGND	DGND
2	AGND	AGND	AGND
3	PWR_GND	PWR_GND	PWR_GND
4	V24	NC10	P_V84
5	V24B	V_DSAFE	V43
6	V26	V145	NC9
7	UF_P1	+24V_SW	SUB_W_P
8	REED_RINS	V147	REED_BIC
9	EM_H_OFF	CAN_A_H	PSW_104
10	H_REL_W	CAN_A_L	$\overline{\text{CLP_CTRL}}$
11	BPSB_ART	BPSB_VEN	LEVEL_UP
12	BPST_ART	BPST_VEN	LEV_DOWN
13	BPUS_ART	BPUS_VEN	LDA1
14	BPR_ART	BYP_REQ	LDA2
15	SN_ART	SN	ODSA
16	P_ART	SHUNTOUT	LDSA
17	PA_DET	SHUNTINP	P_VEN
18	PWR_OFF	SHUNT	PV_DET
19	ACKN_ASP	ABG_ON	SN_EN
20	P_ADSDET	SERVMODE	P_DETADJ
21	COND_DET	FL_SW+5V	HDF_SEL/BL_AL
22	COND_SIG	CI	SNST
23	T_DETADJ	CLP_REQ	WD_RES
24	T_DIAL1	UF_P2_CTRL	WD_SET
25	BLL_DET	NC5	V24_EN
26	HOTRINSE	NC6	V24B_EN
27	BLL	TxD_CPU2/RxD_CPU1	UF_P_CTRL
28	BLL_DIM	RxD_CPU2/TxD_CPU1	UF_P_EN
29	P_DIAL	PWR_FAIL	CPU2_AL
30	OD_OUT	PWR_RES	U_REF-5V
31	+12V	+12V	+12V
32	+5V	+5V	+5V

	DATUM	NAME
BE	24.02.99	Th Klöffel
GE	03.03.99	Schg
	Layout	LP632 J

a neues Layout J

ANDERUNG | DATUM | NAME

Fresenius Medical Care Deutschland GmbH

CPU-2
CPU-2

LP632 M30426

BLATT 7 /7

SP

FORM.A4

8.13 P.C.B. LP 633 Input board

8.13.1 Description

- **Address coding (circuit diagram 1/7)**

  P.C.B. LP 633 (input board) is directly connected to P.C.B. LP 631 (CPU 1) via the address bus, the data bus, and the control bus.
  IC2 buffers the control bus, which consists of the /WR/RD and CS_INPUT signals.
  The address bus and the data bus are buffered by IC 1 and IC 3. They are cut off highly resistive, as soon as H level is applied to the CS_INPUT signal at IC 2, pin 9. An L signal at IC 2, pin 9, releases ICs 1, 3 and 4. IC 4 decodes the addresses (4-to-16-line decoder). The decision, which outputs turn to L level, is taken by the address data at inputs A to D. The L signal releases the connected module. The ADCx_ALE and ADCx_OE signals are generated by IC 9 and IC 10 by means of the CS_ADCx, WR and RD signals.

- **ADC oscillator (circuit diagram 1/7)**

  IC 8 constitutes the oscillator for the analog-to-digital converters.

- **Serial receiver (circuit diagram 1/7)**

  At the moment, IC 6 is not needed. It will be used for data transmission (serial/parallel conversion) with the new level detector module.

- **External alarm (circuit diagram 2/7)**

  A voltage (AC voltage or DC voltage of 20 V to 40 V) can be applied to pins X633L/24b and X633L/25b. Rectified by GL1, this voltage generates the EXT_ALARM signal at OC1, pin 5.

- **Driver (circuit diagram 2/7)**

  The digital input signals are applied to driver modules IC 13 to IC 16 as well as IC 19 to IC 21. The driver modules are activated by the RD and /CS_LATCHx signals being applied.
  IC 17 and IC 18 are provided as voltage converters (12 V $\rightarrow$ 5 V).
  IC 45 constitutes a retriggerable single-shot multivibrator, which is triggered by the optical detector in the level detector module via the OD_IN signal at X633L/8c (frequency approx. 90 Hz).

- **Analog-to-digital converters (ADCs) (circuit diagram 3/7)**

  Via voltage dividers, the analog input signals are applied to the inputs of the A-D converters. The reference voltage for the ADCs, which is generated on CPU 2, is driven by IC 22 (5, 6, 7). Conversion of the analog input signals is started by the H-active ALE signal and can be read in after the conversion by applying the H-active OE signal.

- **Analog signal processing (circuit diagram 4+5/7)**

Dialysate pressure transducer (sheet 4)
The bridge voltage of the pressure transducer applied to the contact strip X633/R, pin 28c and pin 30c, is amplified by IC 33 (5, 6, 7 and 1, 2, 3) (OP1 and OP2). The zero-point offset can be adjusted by means of the DIAL_DET_ADJ signal (from output board DAC) via R183. The output voltage is proportional to the pressure applied. The supply voltage (between 4 V and 5 V, depending on the solder strap selected) for the pressure transducer is generated by IC 32 (1, 2, 3) and transistor T1.

Temperature (sheet 4)
The circuit provided for measuring the temperature of the dialysate by means of the monitor NTC (3) constitutes a modified inversion adding stage with IC 32 (5, 6, 7). The nonlinearity of the monitor NTC (3) is compensated by resistor R55. The conversion from one range of measured values to the other is performed by transistors T8 and T6. For the hot rinsing procedure, the value of the amplification as well as of the offset of the circuit is modified. For the test procedure, the output signal is modified by means of resistor R56. This signal is also used for calibrating the temperature offset.

Conductivity (sheet 5)
Together with R112, C69, C71, and R113 as well as with R110, R111, and the channel resistor of transistor T4, IC 34 (5, 6, 7) constitutes an amplitude-controlled 1-kHz Wien bridge generator. The amplitude of the sinusoidal oscillation is controlled at transistor T4 via comparator IC 34 (1, 2, 3) as well as diode D1, R107 and R108. The cell current at pin X633R/27c is rectified via diodes D3, D4, D5, and D6. The differential voltage is measured at R114 and amplified by IC 39 (1, 2, 3). R114 is also used to measure the currents of the oscillator and the control unit, compensated by R120. The ripple differential voltage and any possible superimposed common-mode AC voltage is filtered by capacitor C72. The initial range from 0 to 15 mS/cm (actual conductivity) is suppressed by resistors R121 and R126. The input voltage is amplified by IC 39 (5, 6, 7) (OP2) to an output voltage ranging from 0 to 10 V. Further filtering is accomplished by capacitor C73. Diode D2 prevents negative selections of the circuits connected to pin X633L/8b. In order to test both the lower and the upper CD alarm limit, a voltage (0 to 10 V) is applied to pin X633L/31b, so that the amplitude of the measuring signal will be modified.

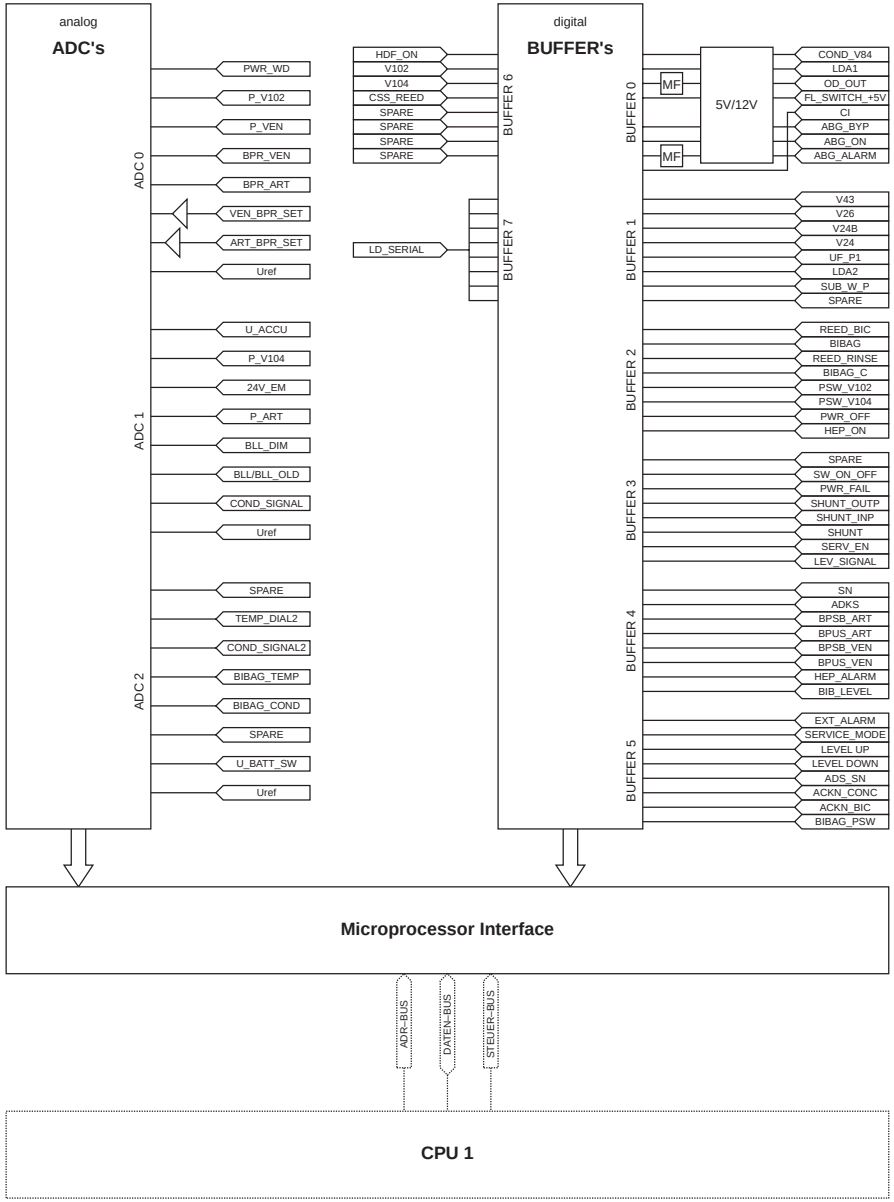
Level sensor (sheet 6)
IC 36 constitutes the integrated fluid level indicator LM 1830. This IC comprises a circuit, which is used to differentiate between two conditions:
– presence of fluid
– absence of fluid

The circuit is provided with two sensing electrodes, to which an AC voltage generated in IC 36 is applied. A comparator circuit then compares the resistance of the fluid with an internal resistance or a reference resistance. With the resistances being different from each other, a frequency is generated in relation to the difference in the resistances. This causes the output of IC 36, pin 12, to turn to L level. Resistor R25 constitutes the reference resistance of the fluid. The 6-kHz oscillator frequency is determined by means of C43. In order to prevent the electrodes from becoming galvanized, the two capacitors C62 and C60 are used as coupling capacitors.



Attention
When operated with de-ionized water (osmosis water), the hemodialysis machine can constantly remain in the prime program during preparation for hemodialysis, until a specific basic conductivity value has been achieved.

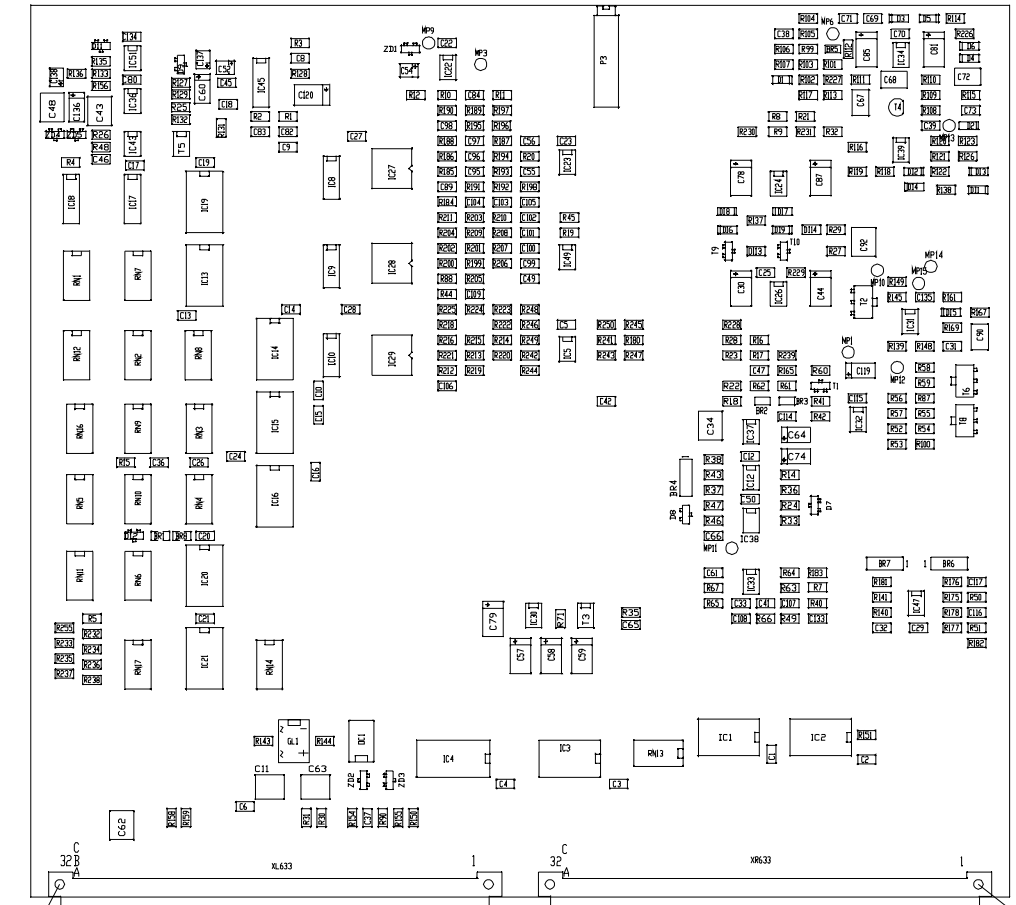
Fig.: Block diagram P.C.B. LP 633



8.13.2 Circuit diagram and component layout diagram
P.C.B. LP 633 Input board

P.C.B. LP 633

Component layout diagram



Lötbrücken / solder links

	Druckaufnehmer / pressure senor	
	Keramik ceramic	● Data Instruments data instruments
BR2	offen / open	● geschlossen / closed
BR3	geschlossen / closed	● offen / open
BR5	(Für LF-Verstimmungsbereich muß offen sein (for conductivity shift range) has to be open	

Lötbrücken / solder links
auf Lötseite / on solderside

	● 4008	Auto-Prime-*Plus*
BR1	● offen / open	geschlossen / closed
BR8	● offen / open	geschlossen / closed

Jumper		Function
BR4	1-2	Puristeril
	2-3	Citrosteril / Sporotal
BR6	2-3	BLL_CAL_DIM
BR7	1-2	BLL_CAL

● = Standard

					DATUM	NAME	
				BE	26.01.97	Ott	
				GE	26.02.98	Schwegler	
					Layout:	LP633 H	
b	IC36 war LM1830						
a	Nachverdr. C79 integriert	12.05.98	Sw	Fresenius Medical Care Deutschland GmbH			
	ANDERUNG	DATUM	NAME				

INPUT-BOARD
LP633

677211

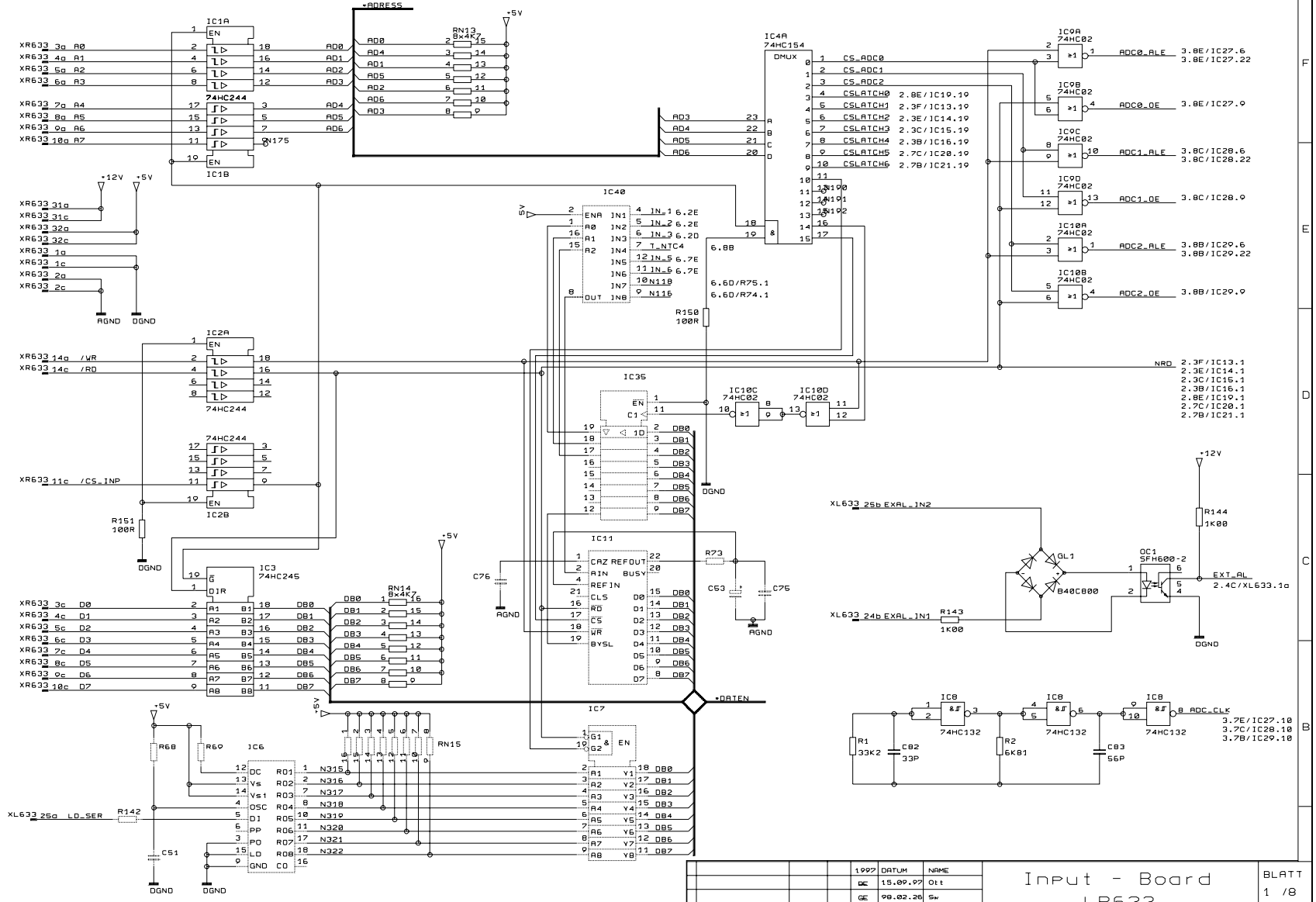
BLATT
BP

ERSATZ ERSATZ

FORM.A3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



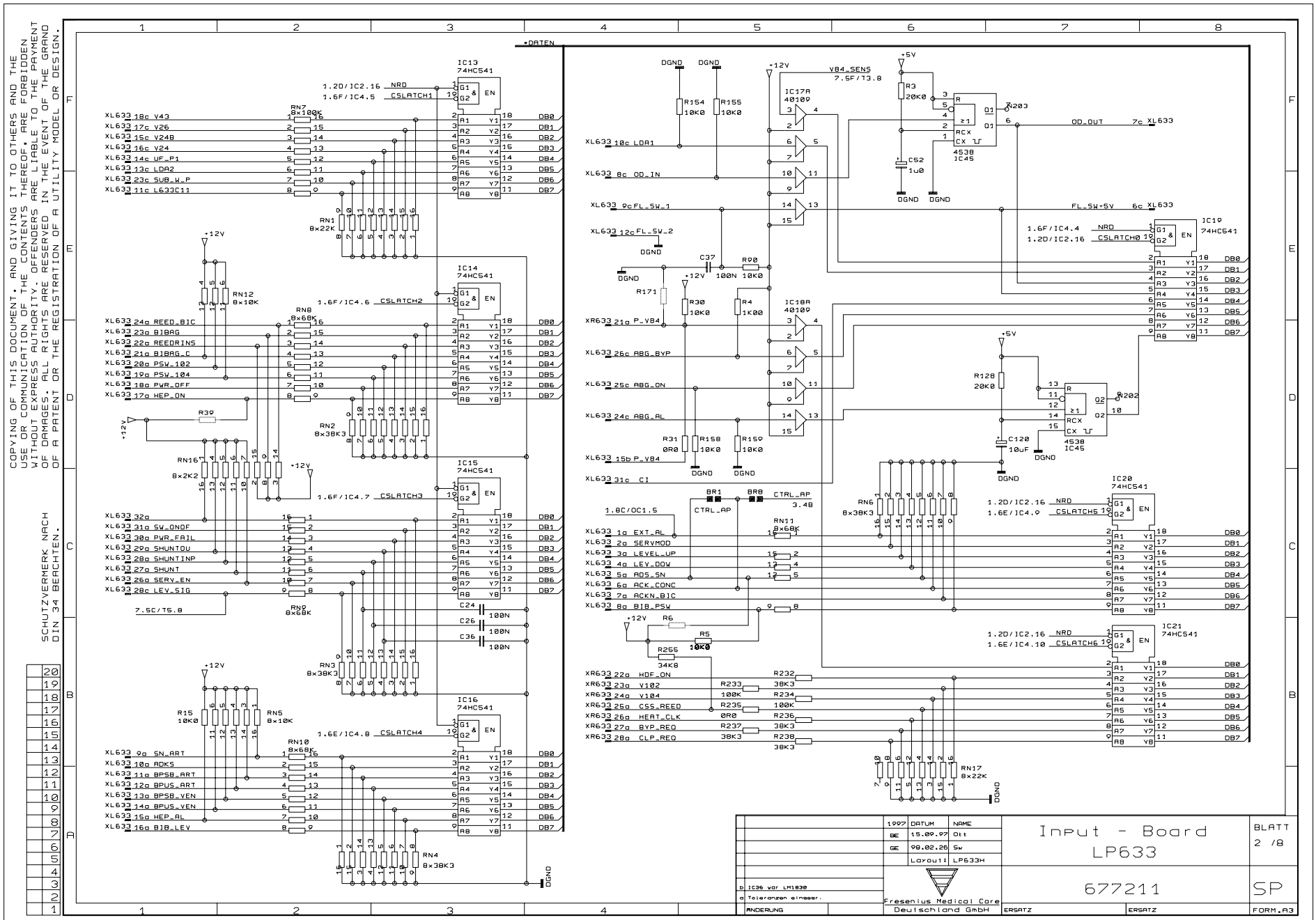
		1997	DATUM	NAME	Input - Board LP633
		DE	15.09.97	Ott	BLATT 1 /8
		GE	98.02.26	Sw	
			Layout:	LP633H	
D	IC36 war LM1030				677211
C	Toleranzen eingegr.	99.04.99	Sw	Fresenius Medical Care	SP
	ÄNDERUNG	DATUM	NAME	Deutschland GmbH	ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 633

Circuit diagram 1/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



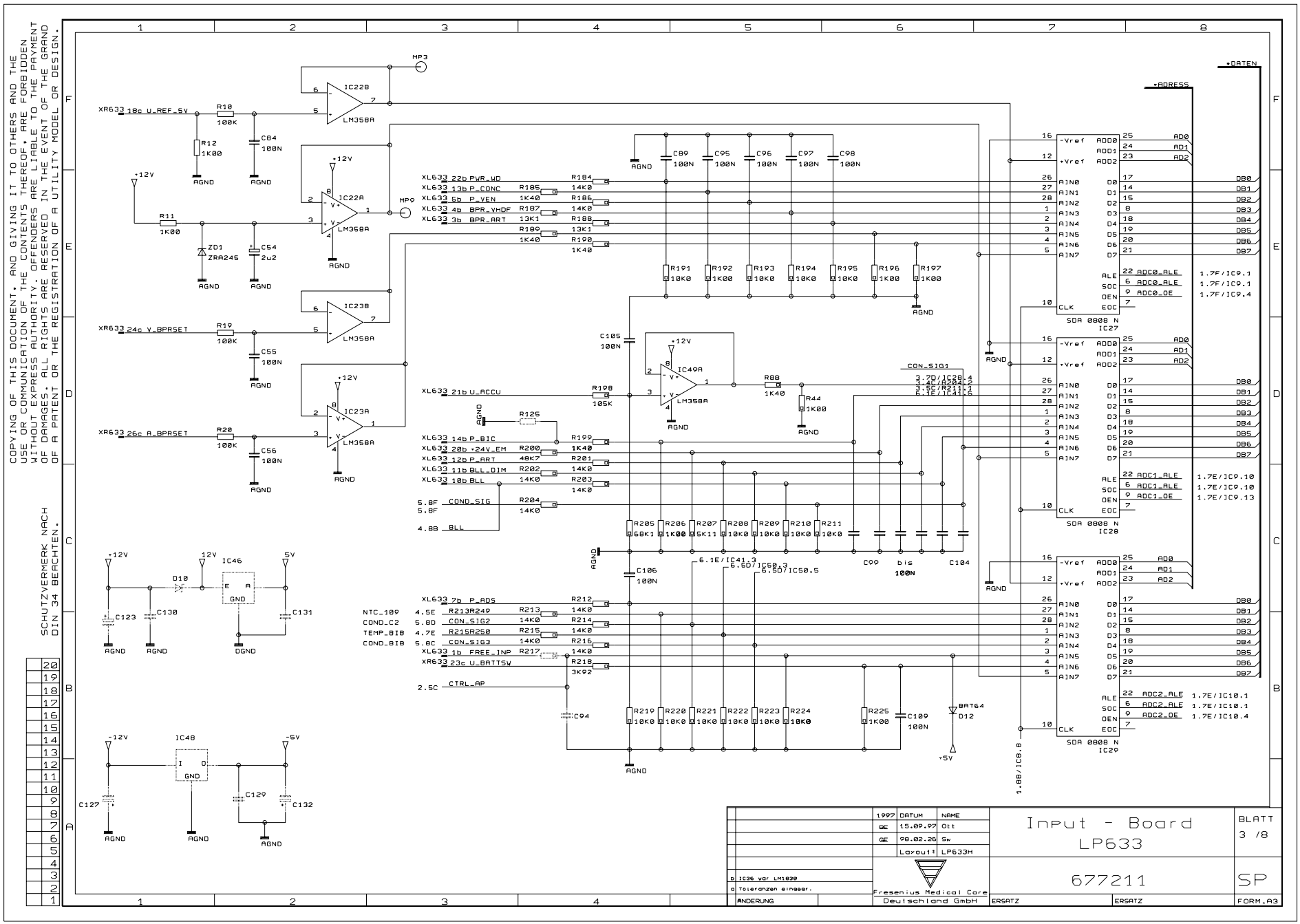
	1992	DATUM	NAME	Input - Board LP633
	BE	15.09.97	Ott	BLATT 2 /8
	GE	98.02.26	Sw	
		Layout:	LP633H	677211 SP
b IC36 war LM1830				
a Toleranzen eingear.				
ANDERUNG	Fresenius Medical Care Deutschland GmbH		ERSATZ	ERSATZ FORM.A3

P.C.B. LP 633

Circuit diagram 2/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



1997	DATUM	NAME	Input - Board LP633
RE	15.09.97	Ott	BLATT 3 /8
GE	98.02.26	Sw	
	Layout:	LP633H	677211 SP

b IC36 war LM1830
a Toleranzen eingegr.

Fresenius Medical Care Deutschland GmbH

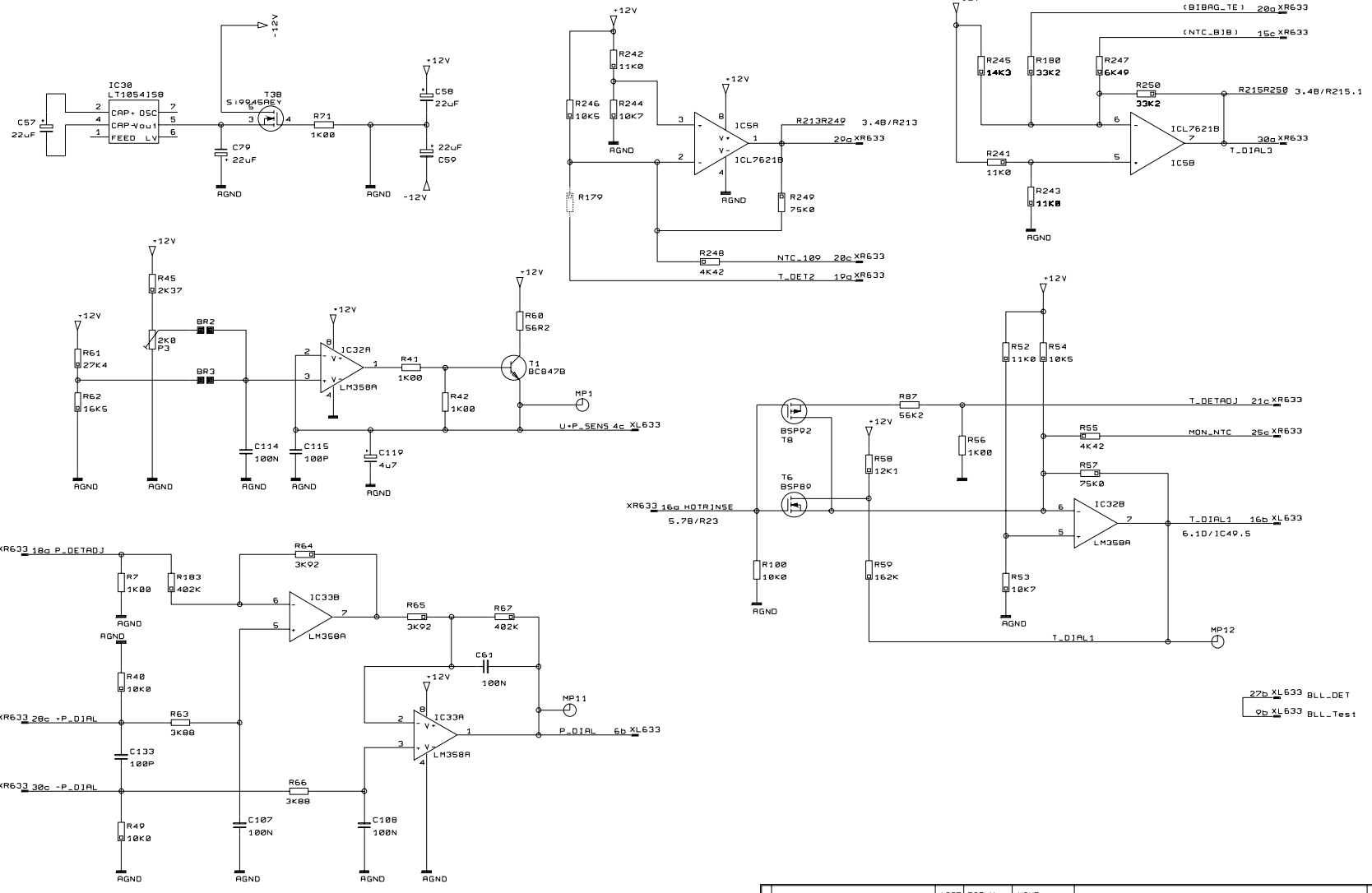
ANDERUNG | ERSATZ | ERSATZ | FORM.A3

P.C.B. LP 633

Circuit diagram 3/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



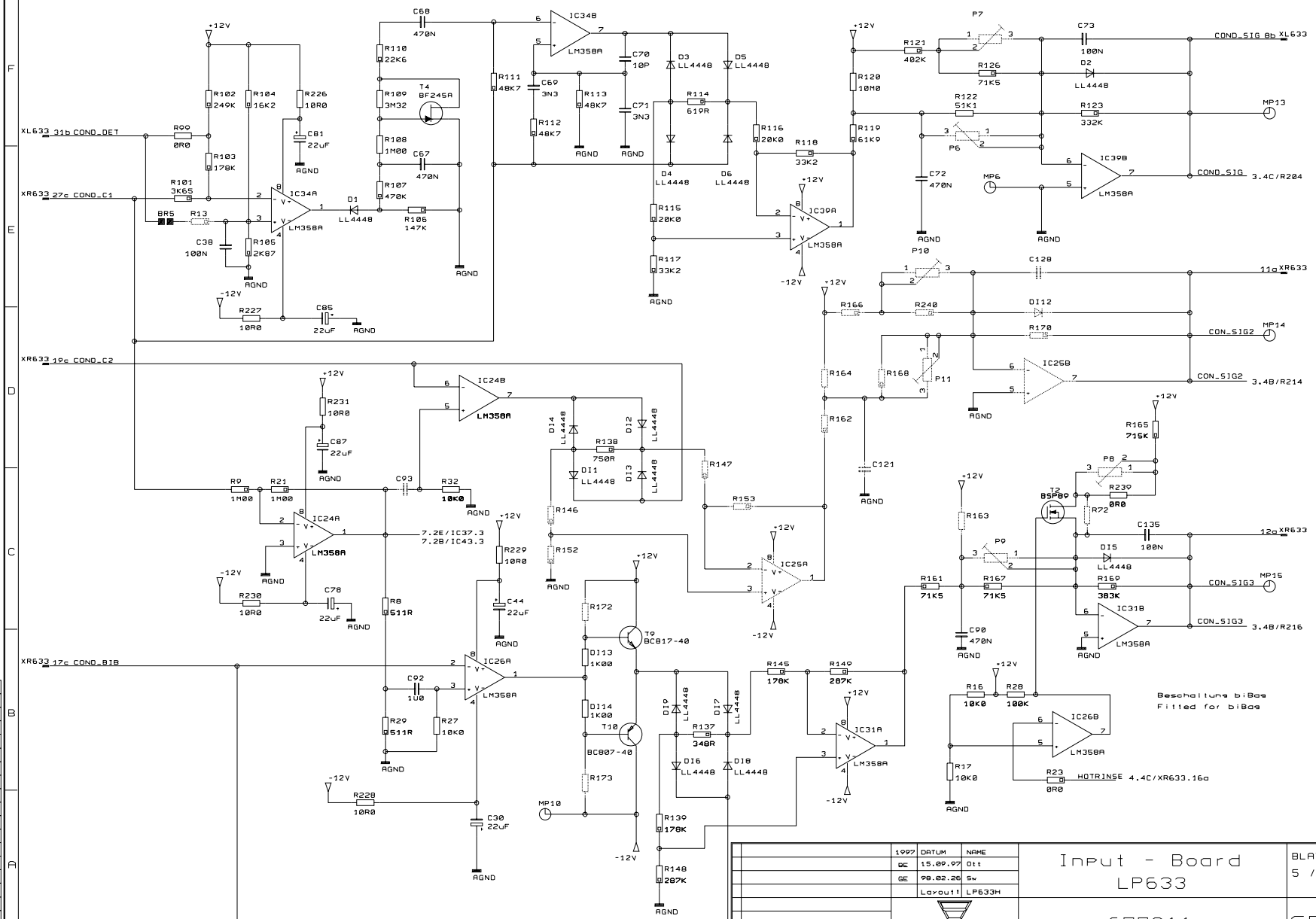
	1997 DATUM	NAME	Input - Board LP633
	BE 15.09.97	Ott	BLATT 4 /8
	GE 98.02.26	SW	
	Layout:	LP633H	677211 SP
D IC36 war LM1830		Fresenius Medical Care Deutschland GmbH	
C Toleranzen eingegr.			
ANDERUNG		ERSATZ	ERSATZ FORM.A3

P.C.B. LP 633

Circuit diagram 4/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



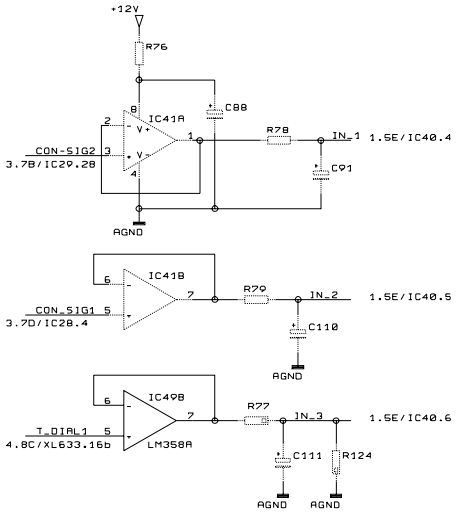
		1997	DATUM	NAME	Input - Board LP633
		DE	15.09.97	Ott	BLATT 5 /8
		GE	98.02.26	Sw	
			Layout!	LP633H	677211
b IC36 war LM1830		Fresenius Medical Care Deutschland GmbH			SP
a Toleranzen eingegr.					
ANDERUNG		ERSATZ		ERSATZ	FORM.A3

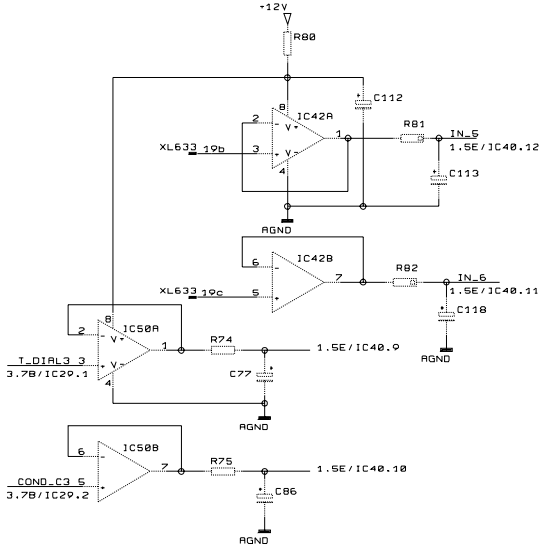
P.C.B. LP 633

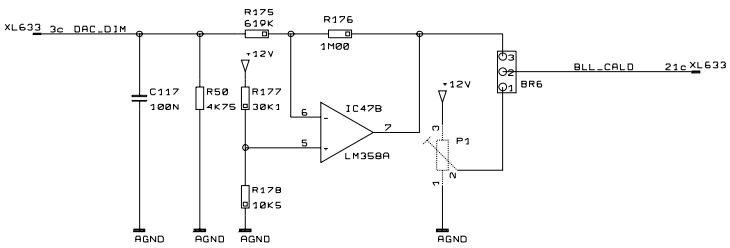
Circuit diagram 5/8

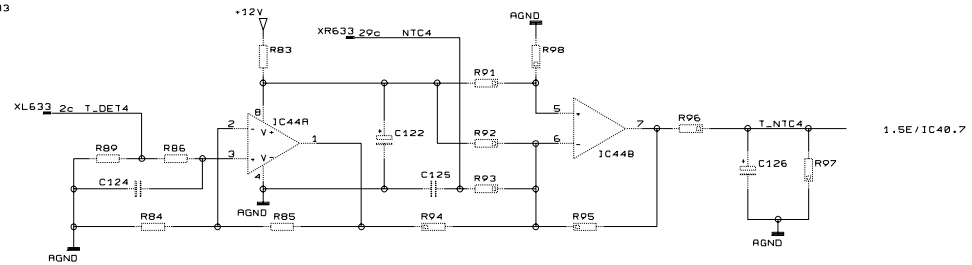
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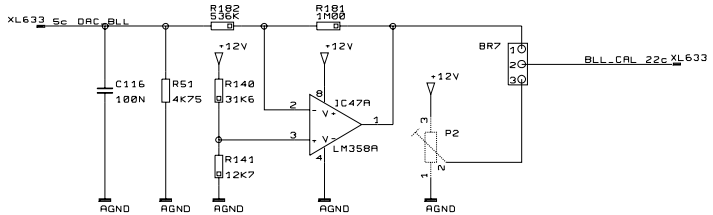
SCHUTZVERMERK NACH DIN 34 BEACHTEN.











		1997	DATUM	NAME	Input - Board LP633
		BE	15.09.97	Ott	BLATT 6 /8
		GE	98.02.26	Sw	
			Layout:	LP633H	
					677211 SP
					ERSATZ / ERSATZ / FORM.A3
D	IC36 war LM1830				
C	Toleranzen eingegr.				
B	ANDERUNG				

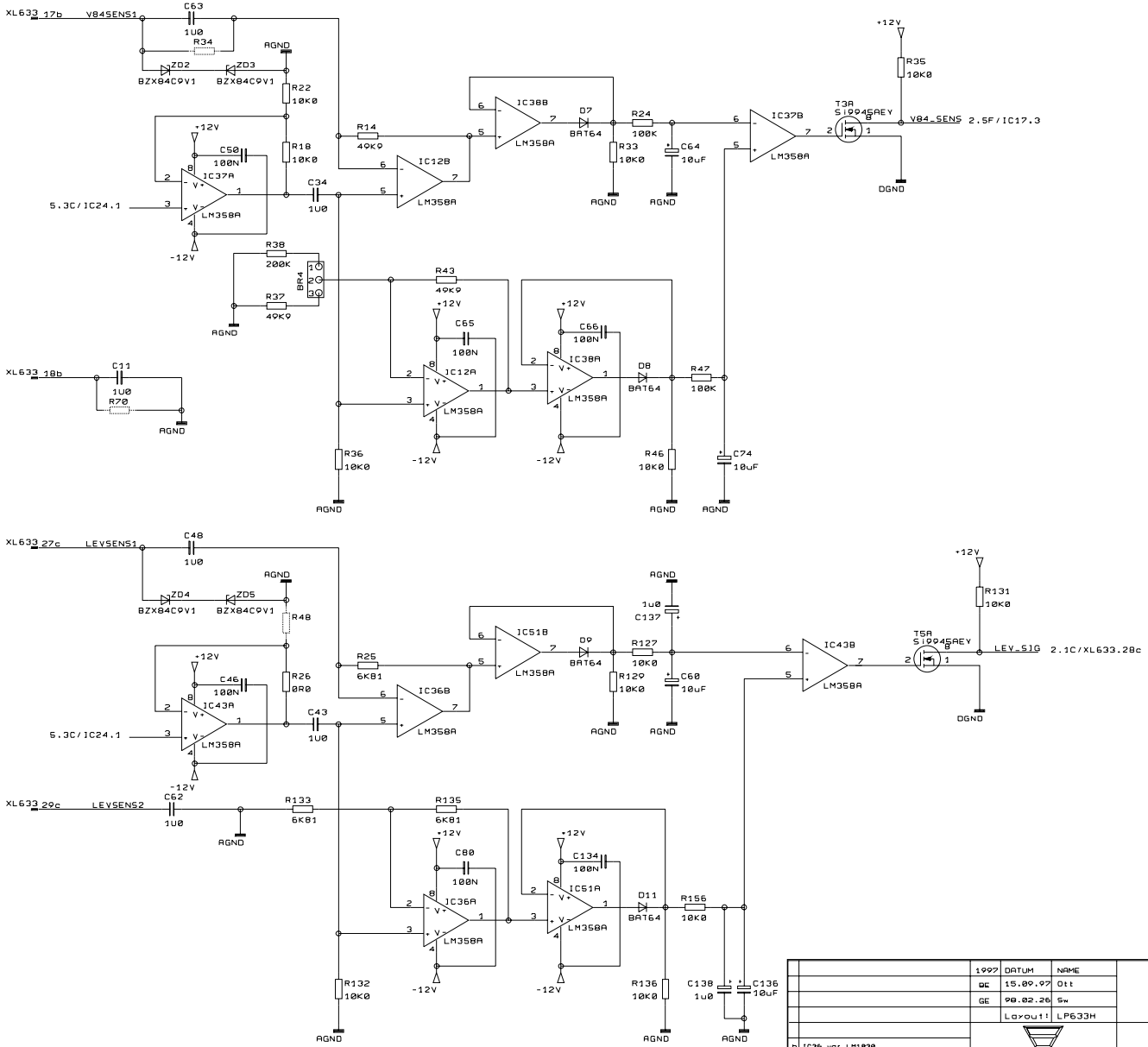
Fresenius Medical Care Deutschland GmbH

P.C.B. LP 633

Circuit diagram 6/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



		1997	DATUM	NAME	Input - Board LP633
		DE	15.09.97	Ott	BLATT 7 /8
		GE	98.02.26	Sw	
			Layout!	LP633H	
b	IC36 war LM1830				677211 SP
a	Toleranzen eingegr.				
	ANDERUNG	Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 633

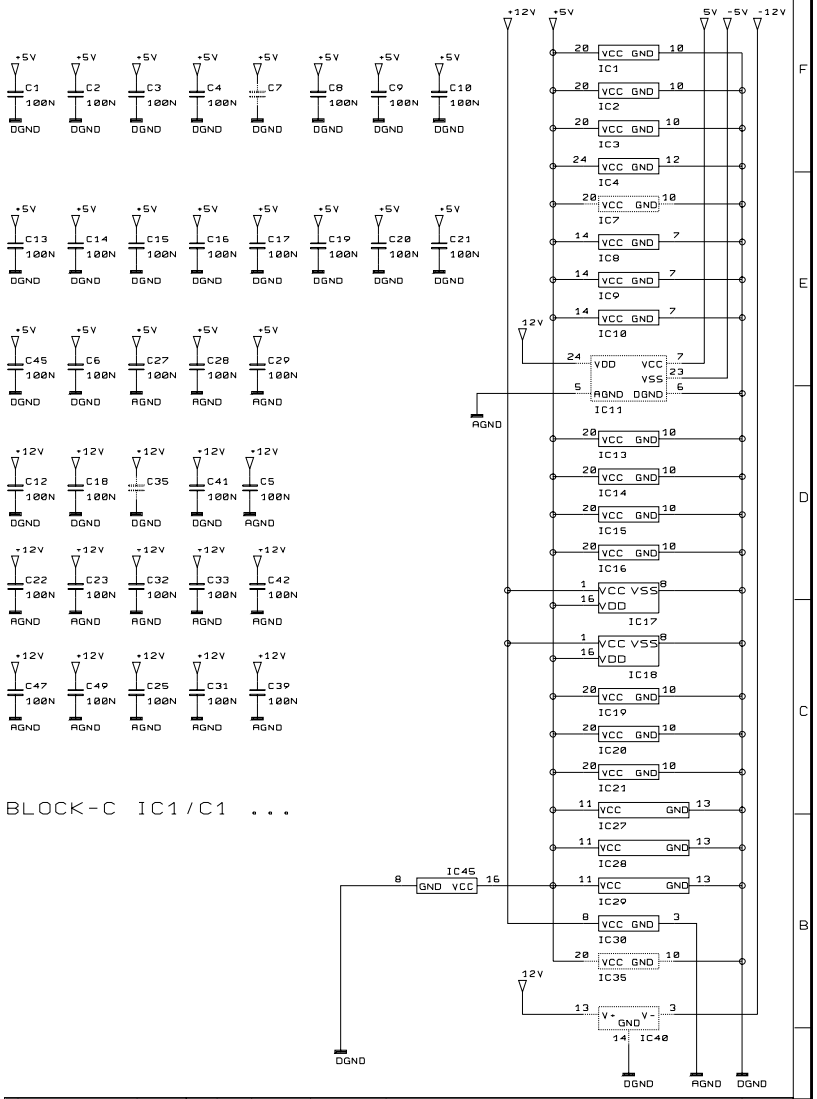
Circuit diagram 7/8

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

PIN-SIGNAL: XL633			
PIN	A	B	C
01	..UNDEF..	..UNDEF..	..UNDEF..
02	SERVMOD	..UNDEF..	T_DET4
03	LEVEL_UP	BPR_ART	DAC_DIM
04	LEV_DOW	BPR_VHDF	U+P_SENS
05	ADS_SN	P_VEN	DAC_BLL
06	ACK_CON	P_DIAL	FL_SW+5V
07	ACKN_BIC	P_ADS	OD_OUT
08	BIB_PSW	COND_SIG	OD_IN
09	SN_ART	BLL_TEST	FL_SW_1
10	BYP_REQ	BLL	LDA1
11	BPSB_ART	BLL_DIM	..UNDEF..
12	BPUS_ART	P_ART	FL_SW_2
13	BPSB_VEN	P_CONC	LDA2
14	BPUS_VEN	AGND38	UF_P1
15	HEP_AL	P_V84	V24B
16	BIB_LEV	T_DIAL1	V24
17	HEP_ON	V84SENS1	V26
18	PWR_OFF	V84SENS2	V43
19	PSW_104	..UNDEF..	..UNDEF..
20	PSW_102	+24V_EM	..UNDEF..
21	BIB_COV	U_ACCU	BLL_CALD
22	REEDRINS	PWR_WD	BLL_CAL
23	BIB_BAG	..UNDEF..	UF_P2
24	REED_BIC	EXAL_IN1	..UNDEF..
25	LD_SER	EXAL_IN2	ABG_ON
26	..UNDEF..	..UNDEF..	..UNDEF..
27	SHUNT	BLL_DET	LEVSENS1
28	SHUNTINP	..UNDEF..	..UNDEF..
29	SHUNTOU	..UNDEF..	LEVSENS2
30	PWR_FAIL	..UNDEF..	..UNDEF..
31	SW_ONOFF	COND_DET	CI
32	..UNDEF..	..UNDEF..	..UNDEF..

PIN-SIGNAL: XR633		
PIN	A	C
01	DGND	DGND
02	AGND	AGND
03	A0	D0
04	A1	D1
05	A2	D2
06	A3	D3
07	A4	D4
08	A5	D5
09	A6	D6
10	A7	D7
11	..UNDEF..	/CS_INP
12	..UNDEF..	..UNDEF..
13	..UNDEF..	..UNDEF..
14	/WR	/RD
15	..UNDEF..	NTC_BIB
16	HOTRINSE	..UNDEF..
17	..UNDEF..	COND_BIB
18	P_DETADJ	U_REF_5V
19	..UNDEF..	COND_C2
20	BIBAG_TE	NTC_109
21	..UNDEF..	T_DETADJ
22	HDF_ON	..UNDEF..
23	V102	U_BATTSW
24	V104	V_BPRSET
25	CSS_REED	MON_NTC
26	HEAT_CLK	A_BPRSET
27	BYP_REQ	COND_C1
28	CLP_REQ	+P_DIAL
29	..UNDEF..	NTC4
30	..UNDEF..	-P_DIAL
31	+12V	+12V
32	+5V	+5V



				1997	DATUM	NAME	Input - Board LP633
				BE	15.09.97	Ott	BLATT 8 /8
				GE	98.02.26	Sw	
					Layout:	LP633H	
b	IC36 var LM1830			Fresenius Medical Care Deutschland GmbH			677211 SP
a	Toleranzen eingegr.						
	ANDERUNG	DATUM	NAME		ERSATZ	ERSATZ	FORM.A3

P.C.B. LP 633

Circuit diagram 8/8

8.14 P.C.B. LP 634 Output board

8.14.1 Description

- **Address decoding (circuit diagram 1/9)**

  The binary code applied to the address inputs is decoded via IC 14. It energizes the respective latches via the enable line 1-11 and 14 and the D-A converter (IC 53) via pin 13.

  Eight output voltages within a range from 0 to 5 V are delivered by the D-A converter. Out1 and Out 5 are amplified to a factor of 2 via OP27 and OP36 respectively. The 5-V reference voltage for the D-A converter is delivered by IC 27, pin 1.

- **Degassing pump control (circuit diagram 2/9)**

  Via the ST-EP control line at IC 8, which is used to compensate mass distortions, the rated voltage for the degassing pump is directed to the control input of the PWM module IC 50, pin 2. Via voltage dividers R100 and R97, the output voltage for the degassing pump is directed to the input of amplifier IC 50, pin 1, where it is compared with the rated voltage. The resultant signal controls the duration for starting the power transistor T4.
  The output current is measured at the variable 0.1-ohm multiplier R110, resulting in a maximum current of 2 A.

- **Flow pump control and end position recognition (circuit diagram 3/9)**

  Speed control:
  The dialysate flow is adjusted by means of a speed-controlled gear pump.
  Via the ST-FP control line at IC 8, which is used to compensate mass distortions, the rated voltage for the flow pump is directed to the control input of the PWM module IC 38, pin 2.
  Via voltage dividers R68 and R67, the output voltage for the flow pump is directed to the input of amplifier IC 38, pin 1, where it is compared with the rated voltage. The resultant signal controls the duration for starting the power transistor T1.
  The output current is measured at the variable 0.1-ohm multiplier R70, resulting in a maximum current of 2 A.

  End position recognition:
  As soon as the balancing chamber is filled on the drain side, the dialysate flow is stopped. This leads to an increase in pressure which, in turn, causes a current increase in the motor of the gear pump and is measured at R70.
  The motor current is amplified via OP32, pins 6 and 7, and is filtered by collector frequencies by means of C31, C44, and C32.
  The motor current is differentiated via C51, R42, and OP32. This differentiated current pulse is shaped into a square-wave signal by Schmitt trigger OP32.

  The signal, which is depending upon the current increase, is available at the output of the differentiating network C38 and R33.

- **Balancing chamber control (circuit diagram 4/9)**

  Upon each positive edge, the D-FF IC 17 is energized via the Clk-Bc signal line. At the Q./Q outputs, D-FF IC 17 informs the GAL module IC 23, which of the balancing chamber valves is to be activated.
  The data word of IC 12 decides upon the PRG type of valve switching.

	Data word IC 12
Flow “off”	0000 0011
Hemodialysis operation	0000 0010
Prime program	0000 1010
Emptying program	0001 0010

  Switching to single-valve control is possible via the FBKU signal line (the respective balancing chamber valve is controlled by the data word of IC 12).

- **ASP control (air separating pump) (circuit diagram 5/9)**

  UF pump control:
  At pin 4, the monoflop IC 42 is triggered by H level from IC 4, pin 14, via IC 39, pins 1 and 3. Via the power transistor T8, the output of IC 42, pin 6, controls the UF pump 1, with the starting time being determined by the RC element R82 and C47.
  At pin 12, the monoflop IC 42 is triggered by H level from IC 4, pin 13, via IC 39, pins 4 and 6. Via the power transistor T3, the output of IC 42, pin 10, controls the UF pump 2, with the starting time being determined by the RC element R65 and C45.

  The UF pumps can also be activated by CPU 2 via the UF-P-CTRL and UF_P2_CTRL signal lines.

- **Digital control outputs (circuit diagram 6/9)**

  Via the open collector drivers IC 34, IC 21, and IC 20, or via level converters IC 41, the respective signal lines are controlled by the data residing at latch IC 13, IC 11, and IC 6.

- **Stepper motor control (circuit diagram 7/9)**

  Initial control:
  After a reset at IC 24, pin 28, the level at IC 24, pin 8, is checked. H level corresponds to bicarbonate operation; with L level, only the concentrate pump is activated.
  The first trigger pulse at IC 24, pin 7, is used to initialize the pumps.

  Operation:
  The concentrate or bicarbonate pump respectively is activated by a trigger pulse at IC 24, pin 7. The step-number memory ICs (latch IC 2 or IC 1) are energized and read in via the control line IC 24, pin 6.

  L level applied to pin 6 = concentrate pump step number
  H level applied to pin 6 = bicarbonate pump step number

  The direction of rotation is reversed at pin 5 of IC 37 or at pin 5 of IC 26 respectively.
  The optical sensor signals (bic pos/conc pos) are shaped via the Schmitt trigger circuit IC 16 and read in at IC 24, pin 19, and IC 24, pin 18.
  Between the individual strokes, the motor current must be reduced (L level at IC 37, pin 10, and IC 26, pin 10).

- **Valve control (circuit diagram 8/9)**

  The data residing at latch IC 7 and IC 10 are directed to the driver modules of type 2068 and activate the individual factors.

  Dialysate valve control:
  Only by H level via the V24-en/V24b-en release line of CPU 2 can valves 24 and 24b be activated via IC 51 and IC 40.

  Water inlet control:
  Via IC 51, the H level of IC 7, pin 12, releases the float switch signal and controls valve 41 via IC 18, pin 3 and pin 2. Valve 41 can also be directly controlled via IC 7, pin 13, and IC 31, pin 12.

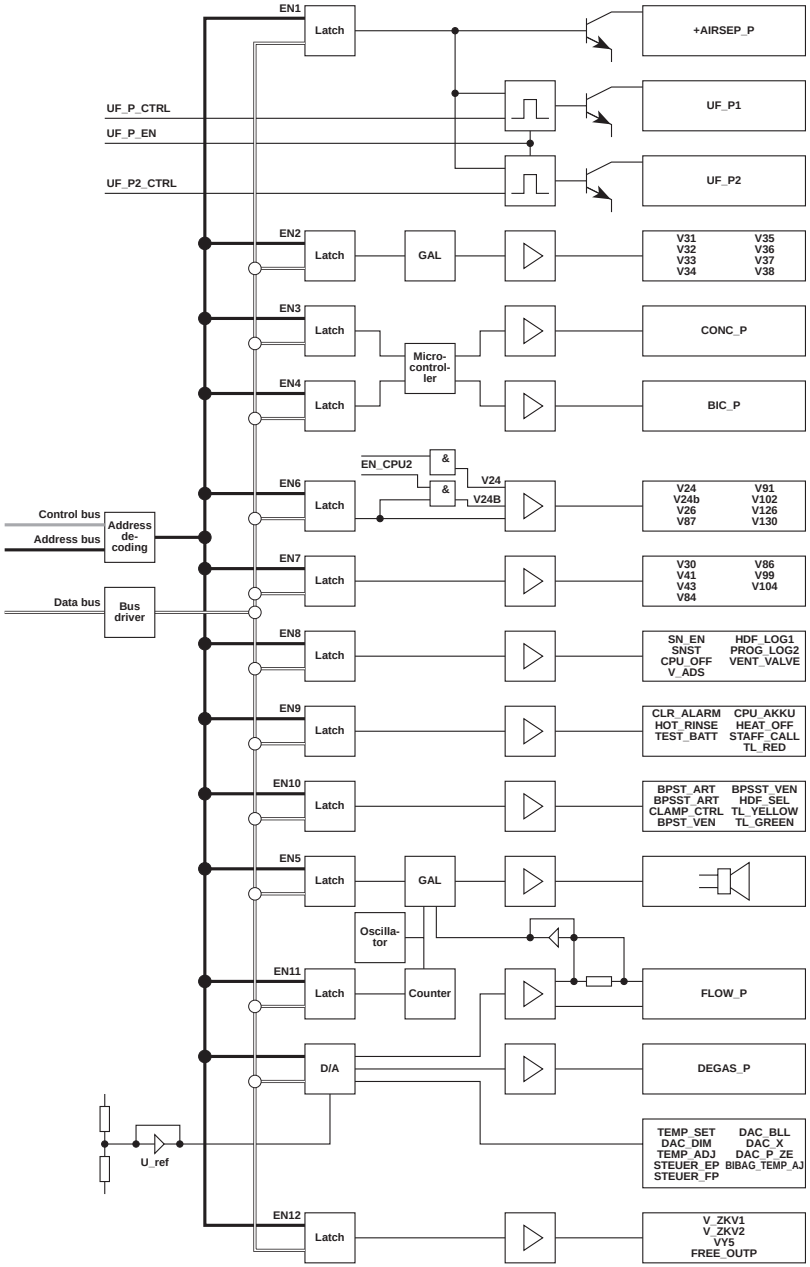
- **Alarm tone control and signal processing for both balancing chamber and stepper motor trigger (circuit diagram 9/9)**

  Oscillator circuit:
  The oscillator circuit consists of a binary counter IC 35, R53, R56, C35, C39, and Q2.
  The output frequencies are active even with power failure, since U-Bat is fed to the 12-V supply network via D18.

  Power failure alarm:
  In case of power failure, L level is applied to IC 31, pin 7. Since the output of IC 31, pin 5, becomes highly resistive, approx. 9 V as well as the power failure frequency of 2048 Hz are applied to IC 31, pin 9, and are then directed to the output stage IC 45 (type TDA 7052) via IC 31, pin 20 and pin 11.
  Via the EN-PF-AT signal line, the power failure alarm tone is suppressed by means of H level at IC 31, pin 7.

  Balancing chamber dead time:
  By means of the positive edge, the D-FF IC 17 is set (H level at pin 5) and the count-down unit IC 15 released at pin 9. The counter is decreased by 1 increment upon each positive clock pulse edge. As soon as the counter reaches 0, pin 14 turns to L level, resetting the D-FF IC 17. By means of L level at pin 9 of IC 15, the data of IC 16 is again called into the counter.

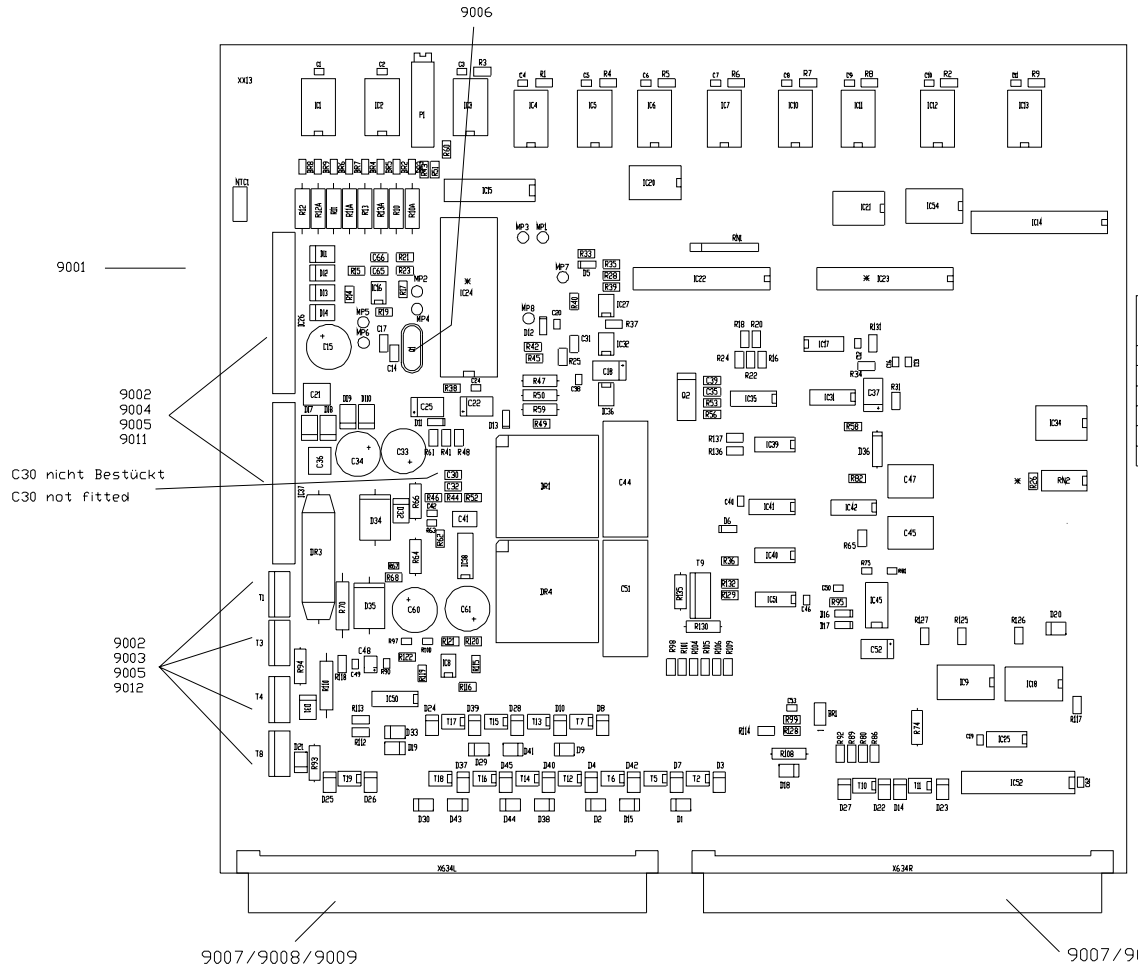
Fig.: Block diagram P.C.B. LP 634 Output board



8.14.2 Circuit diagram and component layout diagram
P.C.B. LP 634 Output Board

P.C.B. LP 634

Component layout diagram



X = Brücke geschlossen
X = bridge closed

MOTOR	CONC				BIC			
	IR2	OR68	IR2	OR68	IR2	OR68	IR2	OR68
BR	2	3	4	5	6	7	8	9
C5833-9212 or C6431-9212		X		X		X		X
Standard C7080-9212K	X		X		X		X	
LP634-2 (400B Mono) C5833-9212 or C6431-9212		X		X		X		X
LP634-2 (400B Mono) Standard C7080-9212K		X		X	X		X	
LP634-1	nicht benutzt / not used							

* LP634-1 nicht bestückt / not fitted

BR1	Geschlossen / Closed Ohne Bic. / Without Bic.
	Offen / Open Mit Bic. / With Bic.

			1999	DATUM	NAME	OUTPUT-BOARD
			BE	16.09.99	Ott	
			GE	16.09.99	Scheuring	
				Layout	LP634J	
a	IC53 ers. CTL_AP hinzu	01.02.00				
	ÄNDERUNG	DATUM	NAME	Fresenius Medical Care Deutschland GmbH		

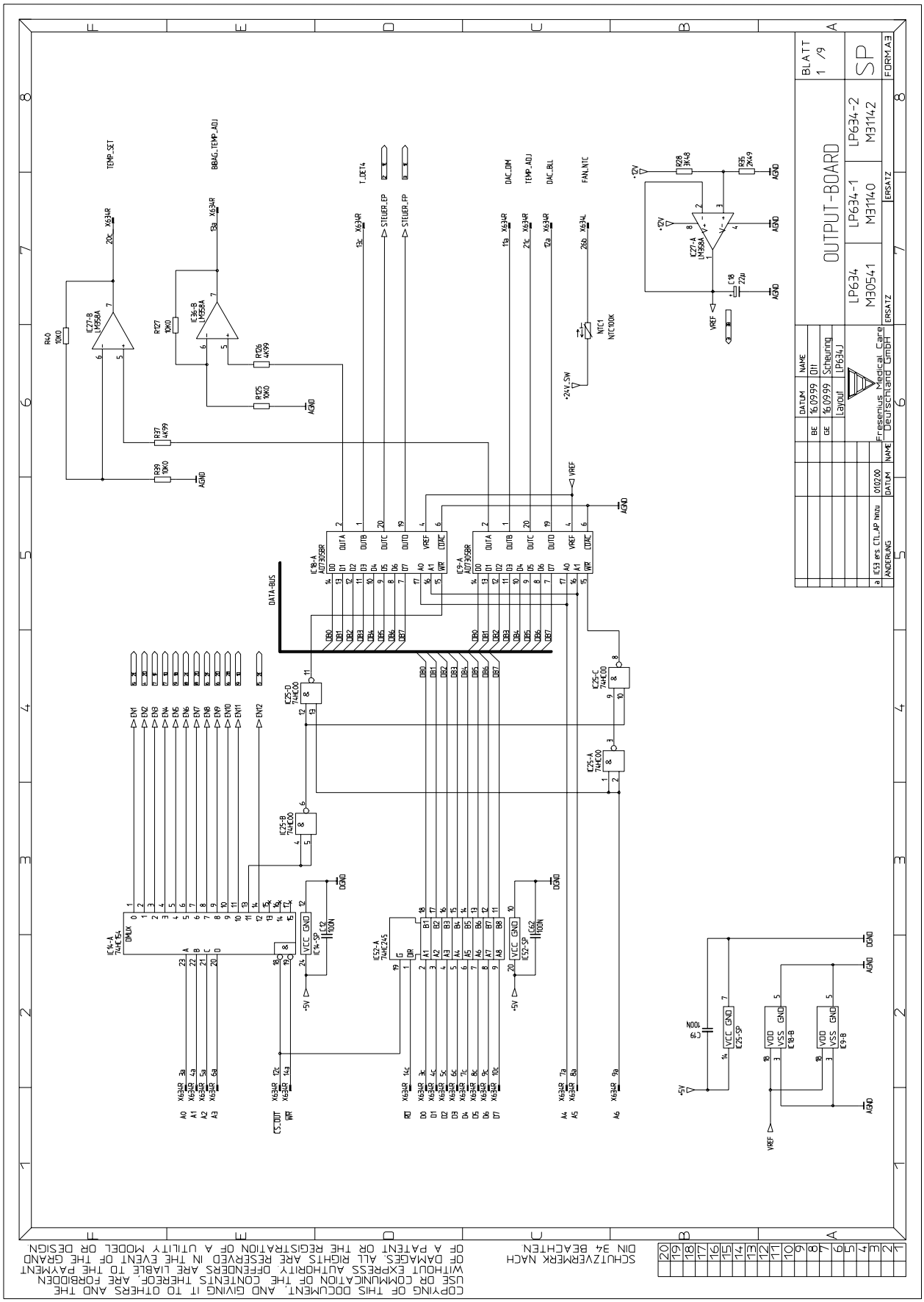
LP634 M30541	LP634-1 M31140	LP634-2 M31142
ERSATZ	ERSATZ	

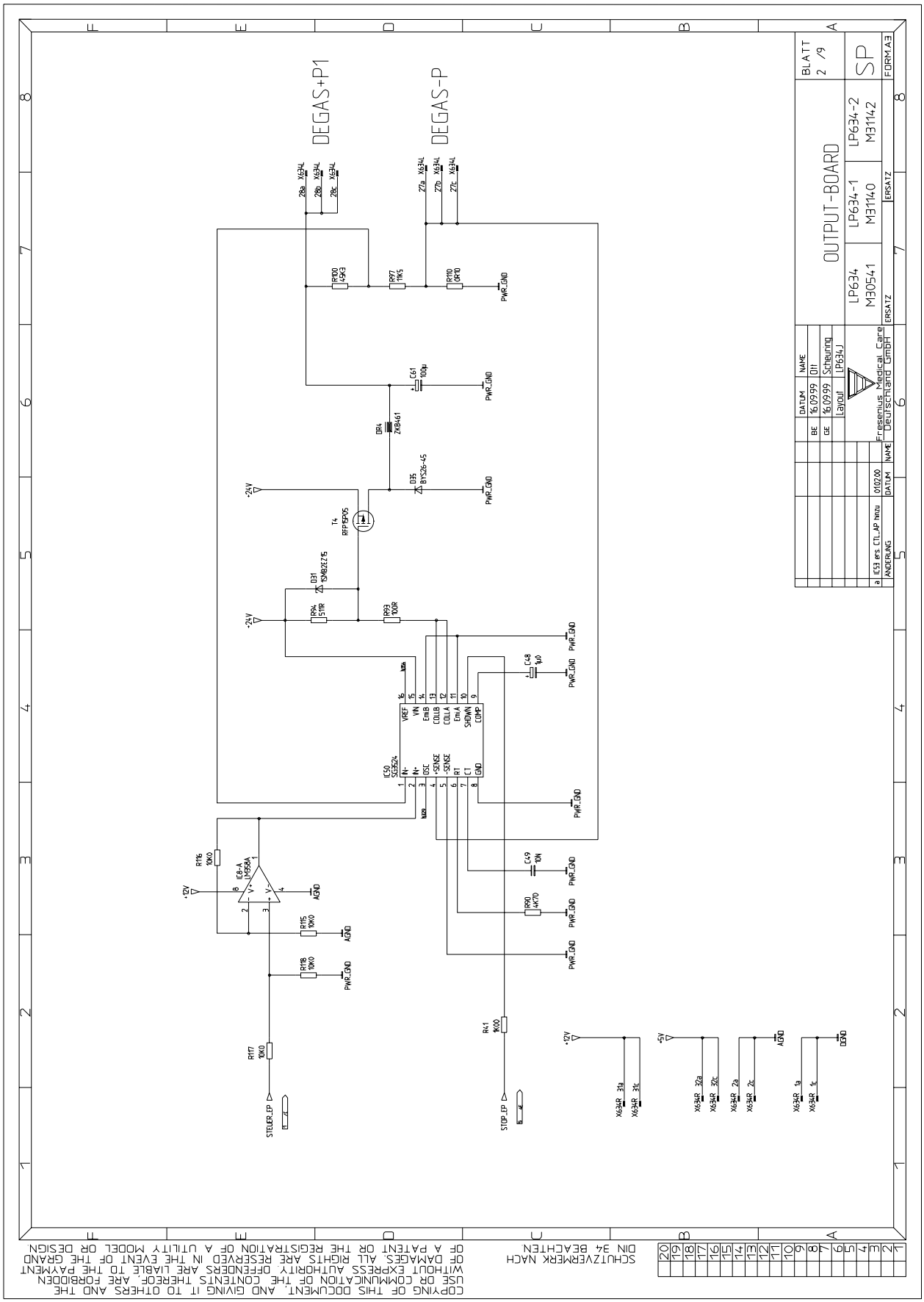
BLATT BP

FORM.A3

P.C.B. LP 634

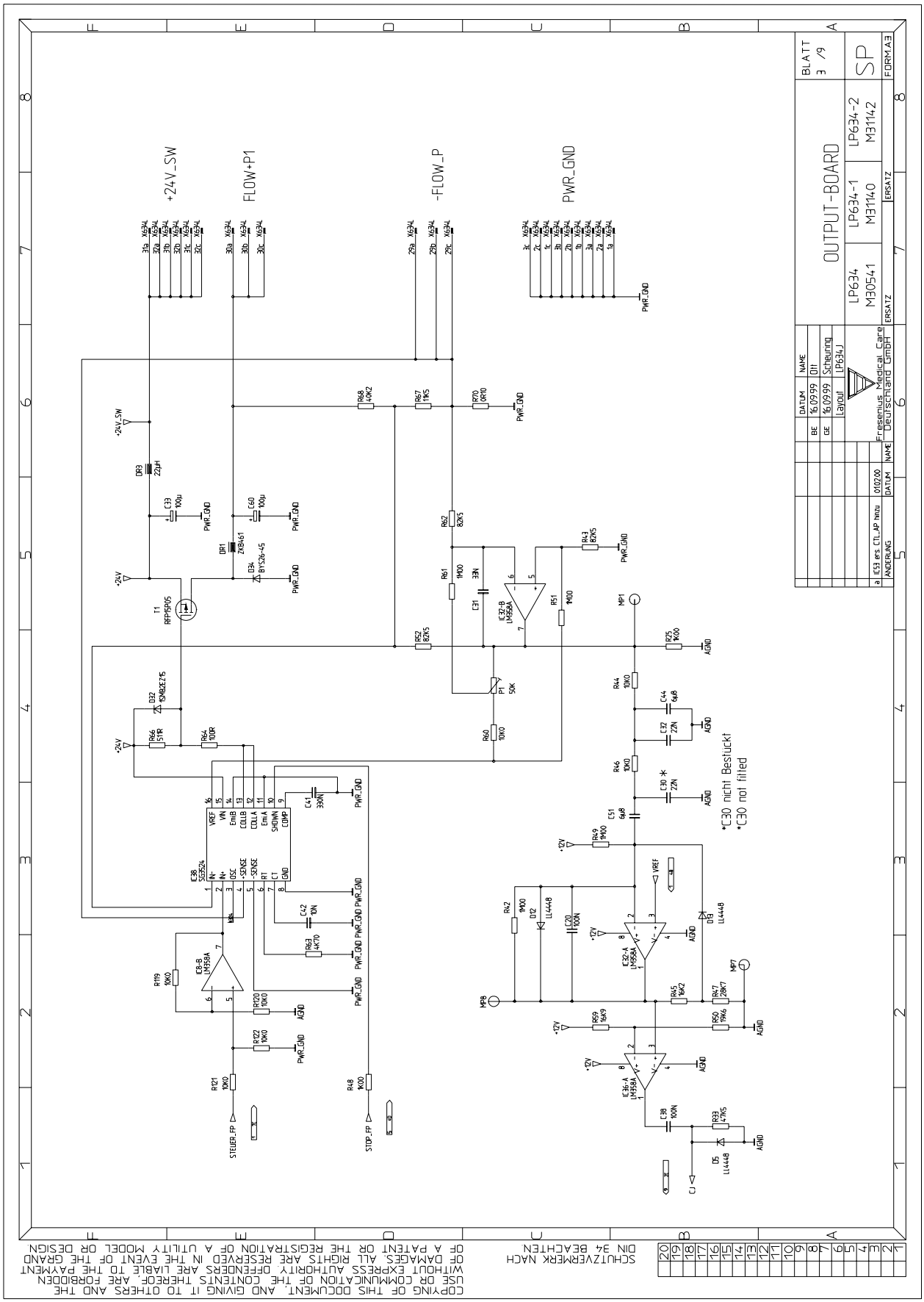
Circuit diagram 1/9

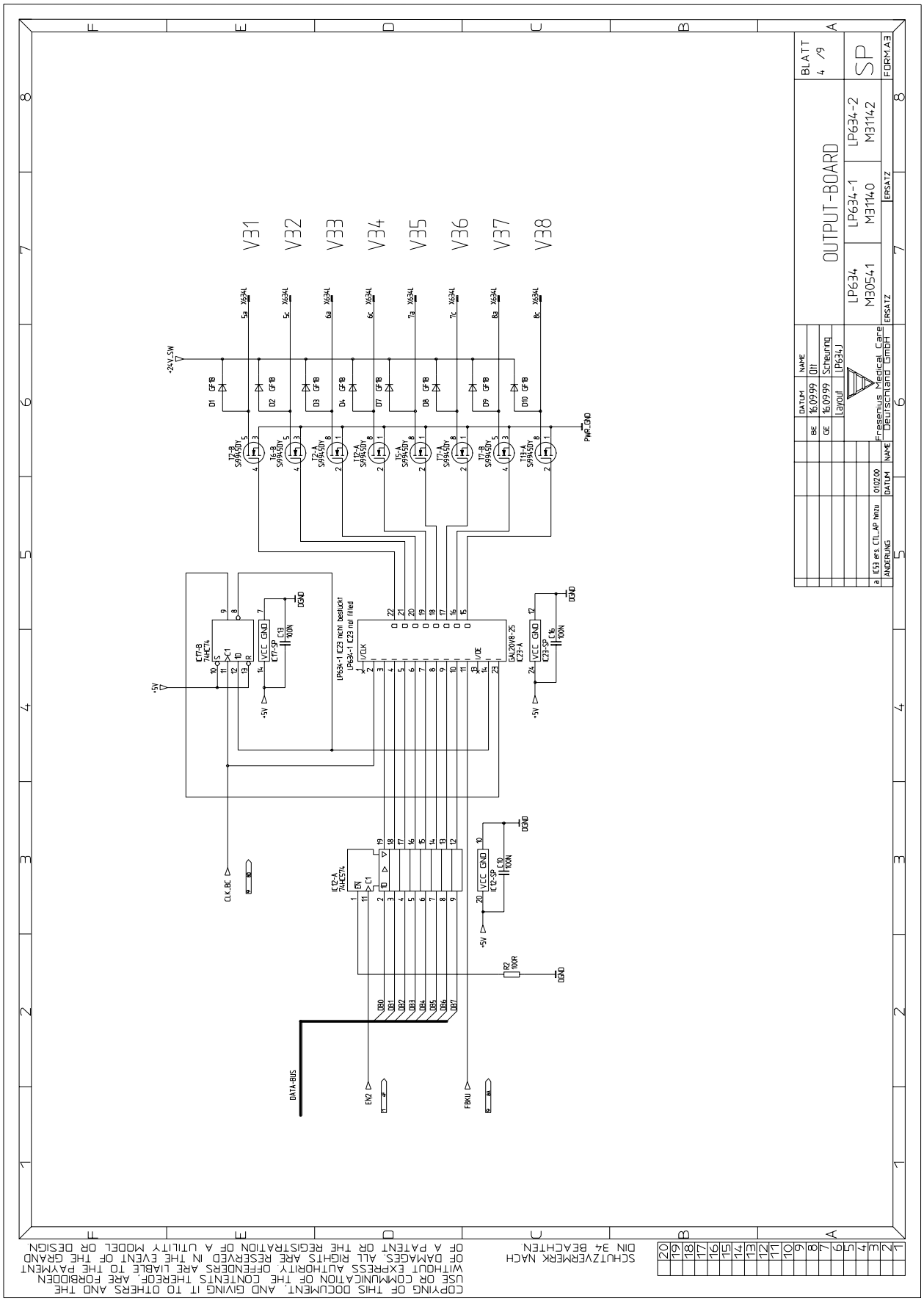



OUTPUT-BOARD
LP634
M30541
LP634-1
M31140
LP634-2
M31142
SP
BLATT 2 /9
FORMAT A3
Fresenius Medical Care Deutschland GmbH
DEGAS+P1
DEGAS-P
PWR_GND
AGND
DGND
+24V
+12V
+5V
STEUER_FP
STOP_FP
IC50 SG3524
IC8-A LM358A
D31 SMBZ12.75
D35 BYS26-45
T4 RFP15P05
DB4 Z40A41
R116 10K0
R115 10K0
R118 10K0
R117 10K0
R141 1K00
R94 511R
R93 100R
R90 4K70
C48 1µ0
C49 10N
C51 100µ
SCHUTZVERMERK NACH DIN 34 BEACHTEN
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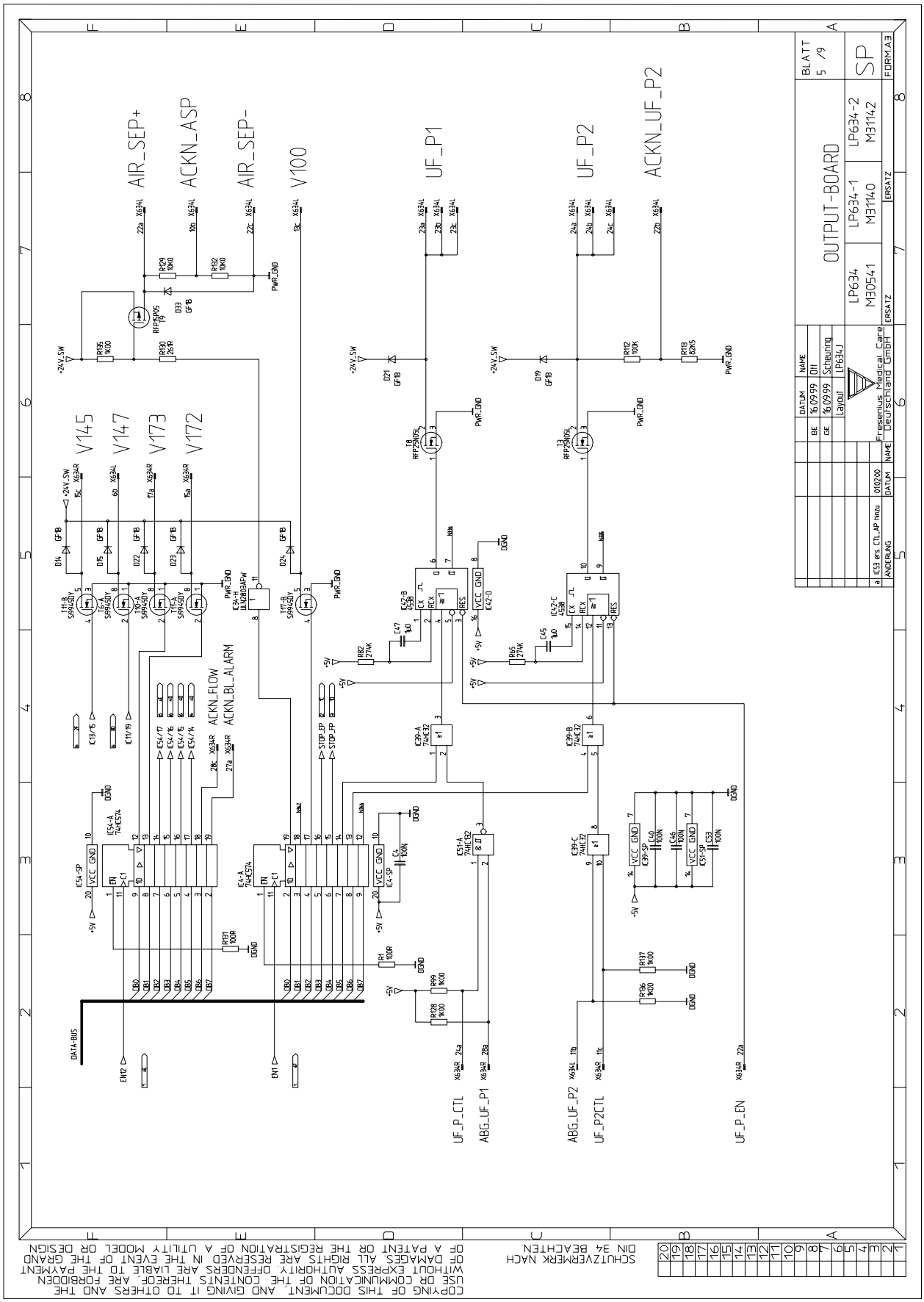
P.C.B. LP 634

Circuit diagram 3/9



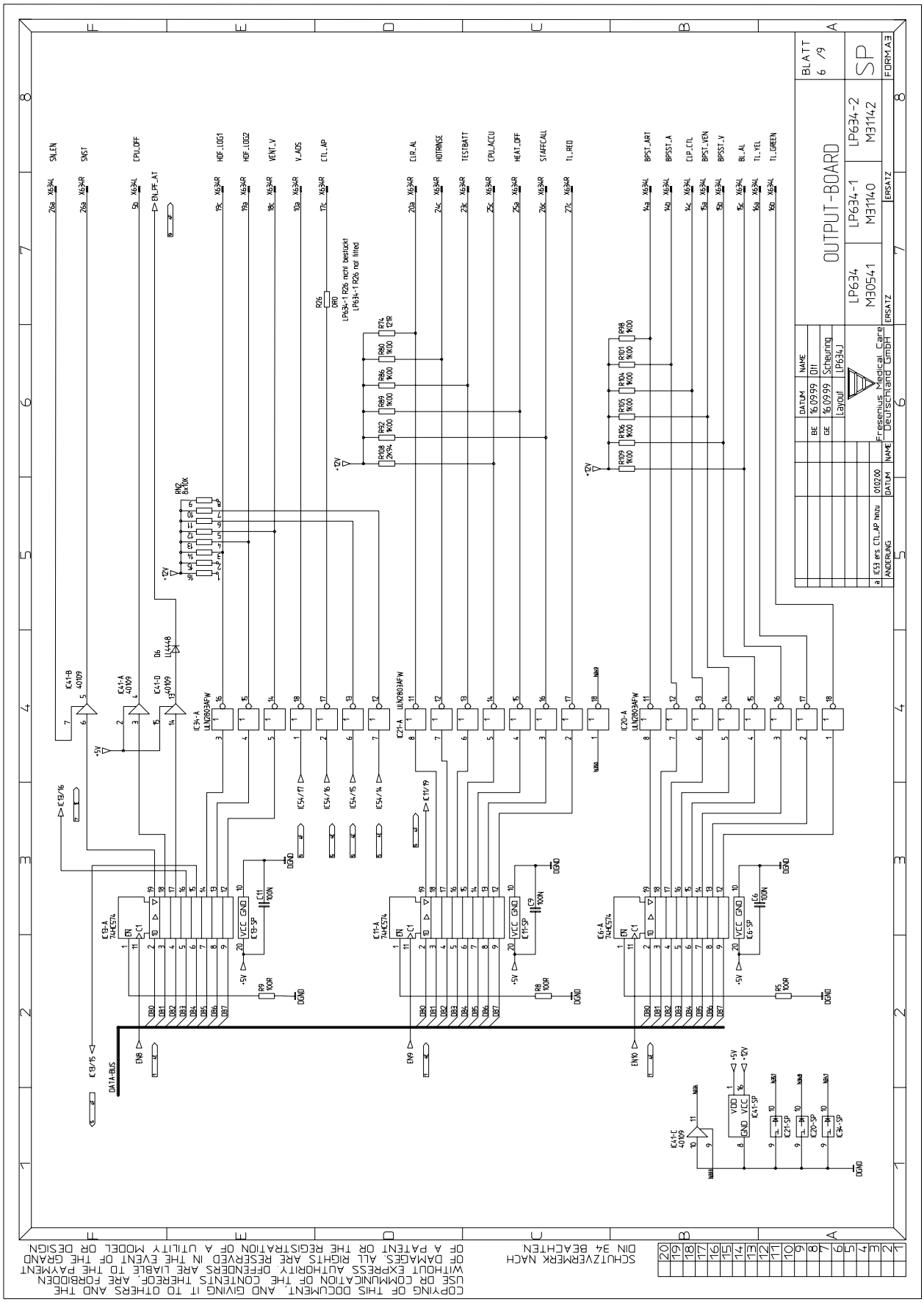


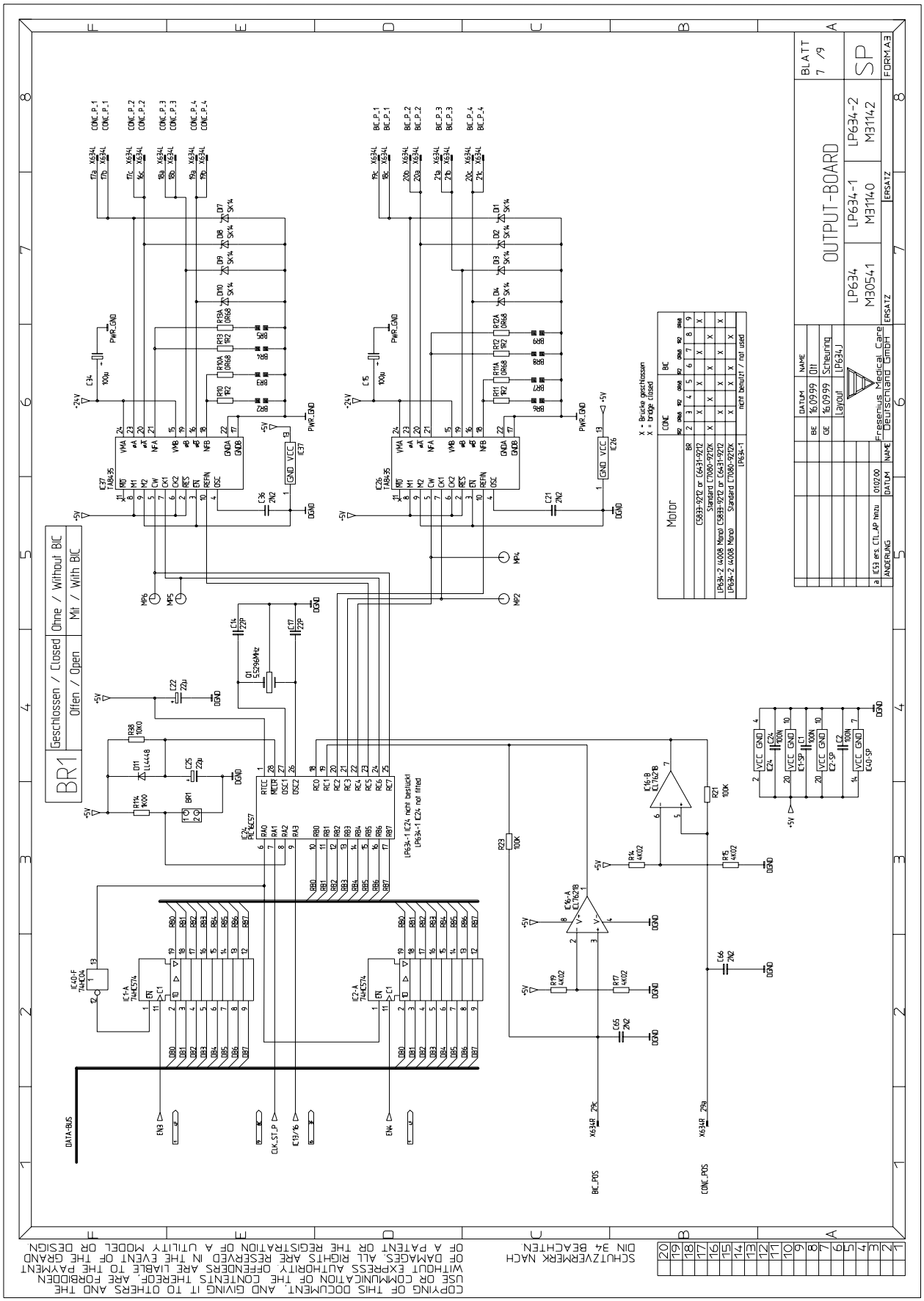
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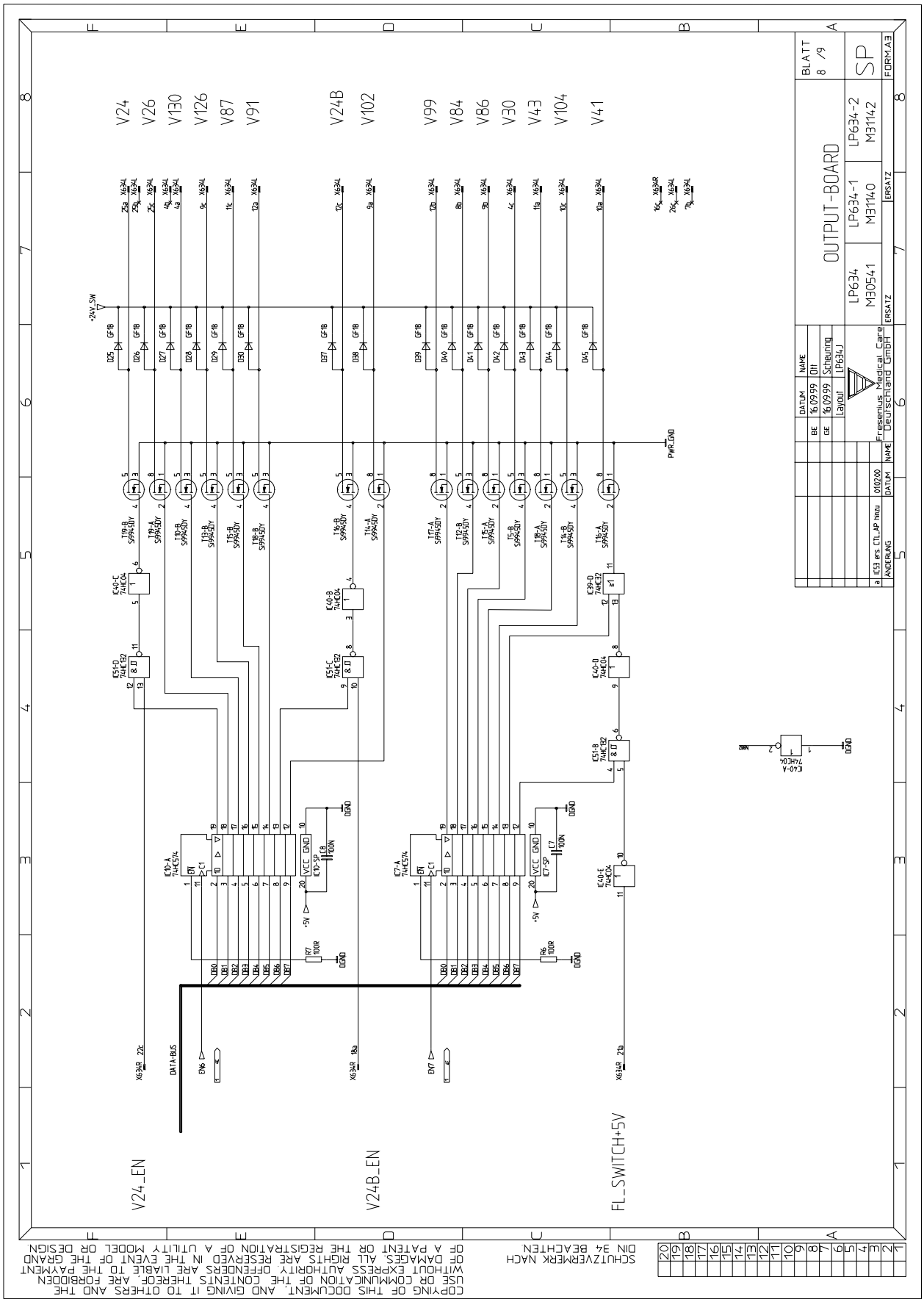

OUTPUT-BOARD
LP634
M30541
LP634-1
M31140
LP634-2
M31142
SP
BLATT 5 /9
FORMAT A3
Fresenius Medical Care Deutschland GmbH
AIR_SEP+
ACKN_ASP
AIR_SEP-
V100
UF_P1
UF_P2
ACKN_UF_P2
V145
V147
V173
V172
ACKN_FLOW
ACKN_BL_ALARM
DATA-BUS
UF_P_CTL
ABG_UF_P1
ABG_UF_P2
UF_P2CTL
UF_P_EN
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SCHUTZVERMERK NACH DIN 34 BEACHTEN

P.C.B. LP 634

Circuit diagram 6/9

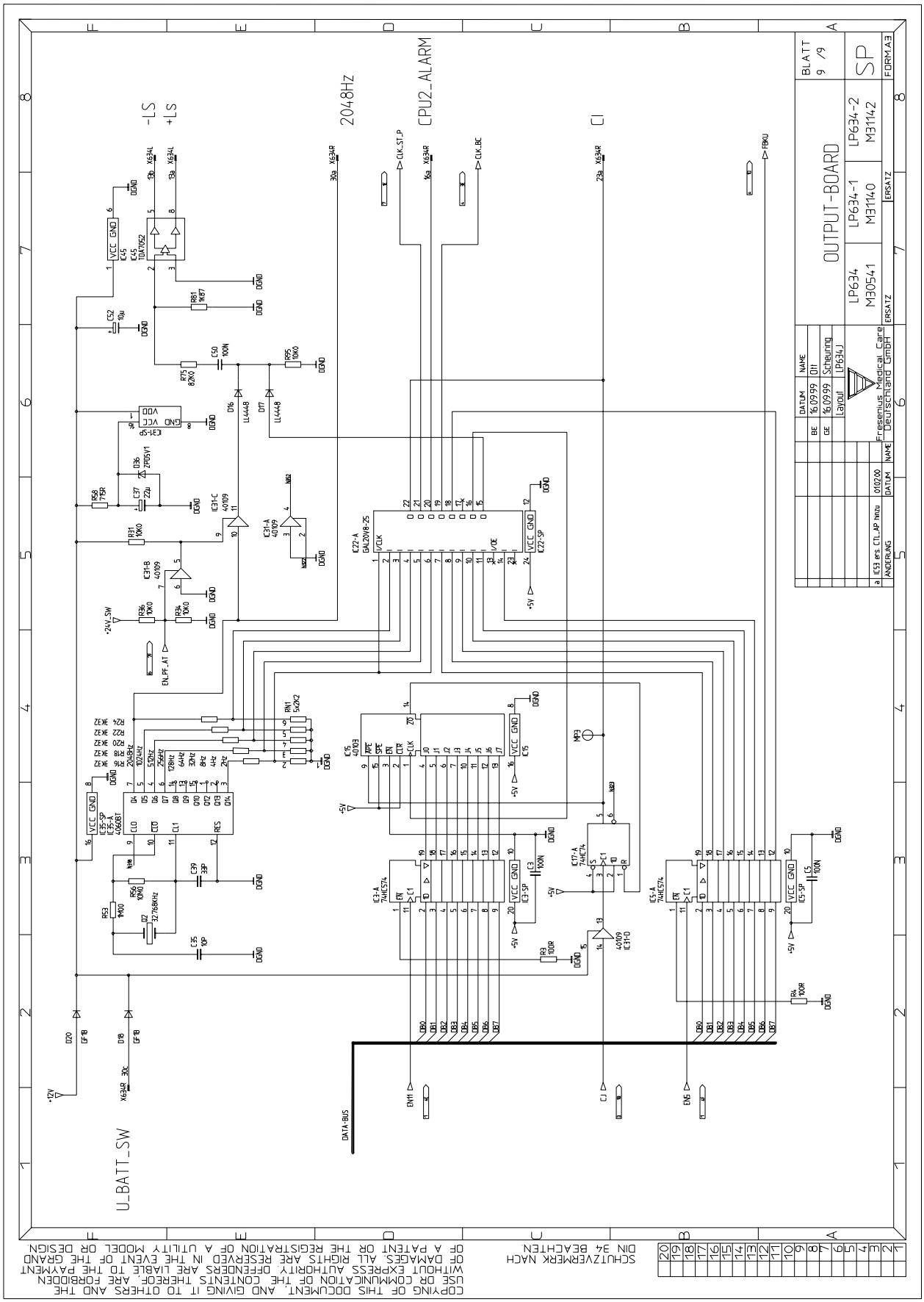



OUTPUT-BOARD
LP634
M30541
LP634-1
M31140
LP634-2
M31142
BLATT 7 /9
FORMAT A3
Fresenius Medical Care Deutschland GmbH
Motor
Geschlossen / Closed
Offen / Open
Ohne / Without BIC
Mit / With BIC
BR1
X = Brücke geschlossen
X = bridge closed
nicht benutzt / not used
SCHUTZVERMERK NACH DIN 34 BEACHTEN
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OUTPUT-BOARD
LP634
M30541
LP634-1
M31140
LP634-2
M31142
BLATT 8 /9
FORMAT A3
Fresenius Medical Care Deutschland GmbH
V24
V26
V130
V126
V87
V91
V24B
V102
V99
V84
V86
V30
V43
V104
V41
V24_EN
V24B_EN
FL_SWITCH+5V
DATA-BUS
+24V_SW
PWR_GND
DGND
+5V
SCHUTZVERMERK NACH DIN 34 BEACHTEN

P.C.B. LP 634

Circuit diagram 9/9



8.15 P.C.B. LP 635 Display board

8.15.1 Description

4008 E hemodialysis machine – monitor

- **Address decoder (circuit diagram 1/7)**

  IC 1 and IC 2 are provided as 4-in-16-bit address decoders. The 22 chip selects are generated by means of addresses A0 to A3 and A4 for IC 1 as well as A4 inverted for IC 2. The 22 chip selects are generated only if, via IC 9, either X635/24 (write) or X635/25 (read) is at L level. IC 71 is used as data bus driver and is reversed in its direction by WR and RD.

- **Keyboard matrix (circuit diagram 5/7)**

  Pins N1 and GND of plug X2 are directly connected to the circuits in the power supply unit. These circuits are used to switch the machine on and off. Via IC 54, the remaining keys on the touch sensitive keyboard are connected to a matrix at the keyboard encoder IC 53 and an additional data line DB4. After the CPU has received the interrupt, the keyboard code is read out by means of CS20 via buffer IC 73.

- **Brightness control (circuit diagram 1+2/7)**

  The clock pulses required for the bar graph displays are delivered by IC 74 and the following gate logic. Superimposed by a higher frequency and a variable mark-to-space ratio, these clock pulses are used to modulate the brightness. To this end, the shift register IC 6 is provided. IC 6 is loaded in parallel with the desired mark-to-space ratio via a data word at DB0 to DB7 (CS21) and is cyclically rotating due to clock generator IC 7/11, 12, 13, and IC 7/8, 9, 10.

- **Status indicator LEDs (circuit diagram 6/7)**

  All status indicators as well as the condition indicator are switched by means of IC 21 and IC 25 (CS16, CS17).
  Both modules are
  – connected to the brightness control by R1,
  – not connected to the brightness control by R2 (maximum brightness).
  The I/O LED (power on/off) is directly connected to the 5-V supply network.

- **Alarm indicator LEDs (circuit diagram 7/7)**

  All alarm indicators are energized by IC 23 and CS18. The key click is controlled by output 19 of IC 23 via IC 46/5.14.

- **Text display (circuit diagram 1/7)**

  The text display is an intelligent, 20-character display module. It is provided with its own processor. Texts are loaded by means of ASCII characters via the data bus CS22. By means of CS20 and via IC 73 and the data bus, the CPU can read whether the text display is ready to receive new data.

- **UF displays (circuit diagram 2/7)**

  IC 3 and IC 4 are 8-digit seven-segment display drivers. IC 3 operates the UF Volume and UF Rate displays. IC 4 operates the UF Goal and UF Time Left displays. These display drivers are loaded in series with 16-bit packets upon each rising clock edge. This information is buffered into the digit registers by means of the IC 5/16 and IC 5/18 load signals. Included in this information is the brightness control.

- **Bar graph displays for arterial and venous pressure (circuit diagram 3/7)**

  IC 19, IC 16, and IC 22 are provided as data memories for the arterial bar graph display, whose 30 LEDs are organized within a 5x6 matrix. The 5 lines are directly energized via the emitter sequence IC 30 by data buses DB0 to DB4 and the 6 columns via the 8-out-of-3 decoder IC 28 by data buses DB5 to DB7. The actual value (CS1) is stored by IC 19; the upper limit (CS2) is stored by IC 16; and the lower limit (CS3) is stored by IC 22. The respectively applicable LED is energized via the clock inputs CL1, CL2, and CL3. The clock pulses of CL1 have a longer duration than those of CL2 and CL3. Thus, the actual value is represented more brightly than the limits.
  IC 18 (CS4), IC 15 (CS5), and IC 20 (CS6) are provided as data memories for the venous bar graph display. The working method of this display is identical with that of the arterial bar graph display.

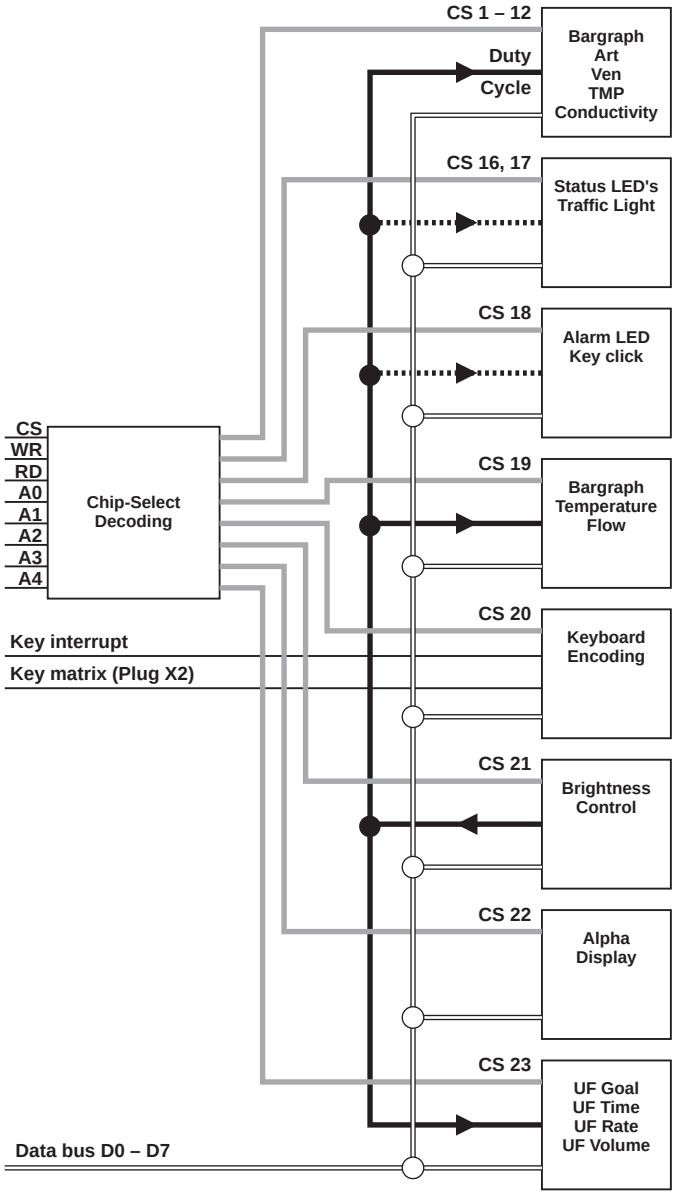
- **Bar graph displays for TMP and conductivity (circuit diagram 4/7)**

  IC 59 (CS7), IC 55 (CS8), and IC 57 (CS9) are provided as data memories for the TMP bar graph display.
  IC 58 (CS10), IC 60 (CS11), and IC 56 (CS12) are provided as data memories for the CD bar graph display.
  The working method of both displays is identical with that of the arterial bar graph displays.

- **Bar graph displays for temperature and flow (circuit diagram 7/7)**

  The bar graph displays for temperature and flow are energized by of IC 24 via CS19. The data lines DB4 to DB7 contain the value for the temperature display. Only 16 of the 20 LEDs of the temperature display are actually visible. The data lines D0 to D3 contain the value for the flow display. IC 72/19 is responsible for the brightness control of the temperature display, and IC 72/18 serves to turn the display dark.

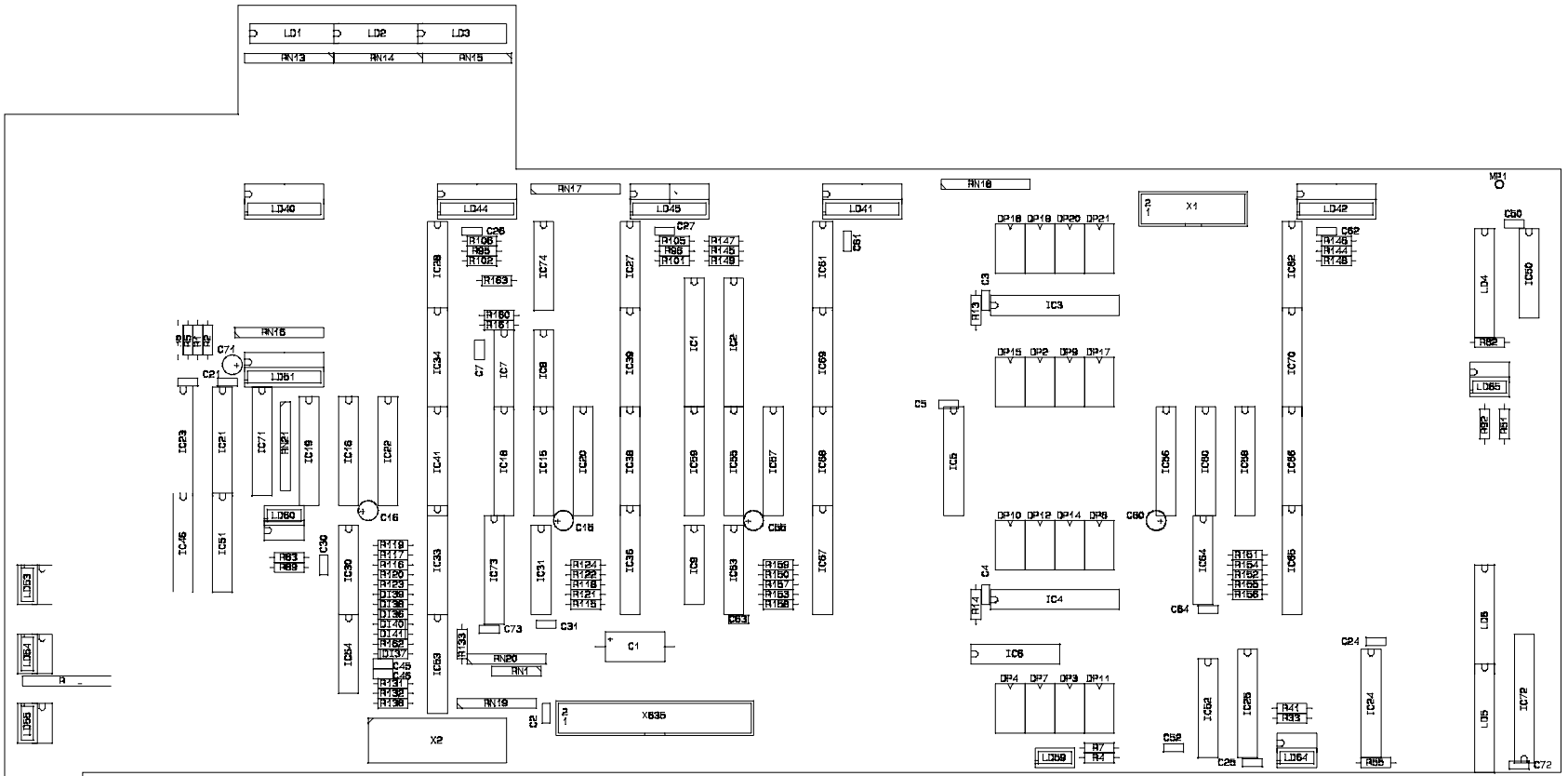
Fig.: Block diagram P.C.B. LP 635 Display board



8.15.2 Circuit diagram and component layout diagram
P.C.B. LP 635 Display board

P.C.B. LP 635

Component layout diagram



IC-Sockel Pos.: Für

2001	DP2/3/4/6/7/9-12/14/15/17-21
2002	LD40-42/44/45/51
2003	LD1-3
2004	LD53-57/59/60/63-66/68/70
2005	IC33/34/36/38/39/41/65-70 und LD4-6

R1 und R5 nicht bestückt. R1 and R5 not fitted.

g	LT-D	31.03.93	Asch	1991	DATUM	NAME	Display Board
f	LT-C	05.02.93	Asch	BE	07.11.91	Dtt	
				GE	08.11.91	Asch	
k	C75 entfällt						
j	RN22 neu / LT-E	12.09.96	Sch				LP635 670436
i	RN1 4x hinzu	03.11.94	Asch				
h	C75 hinzu	21.03.94	Asch				
	ANDERUNG	DATUM	NAME	Fresenius Medical Care Deutschland GmbH			ERSATZ ERSATZ

BLATT BP

FORM.A3

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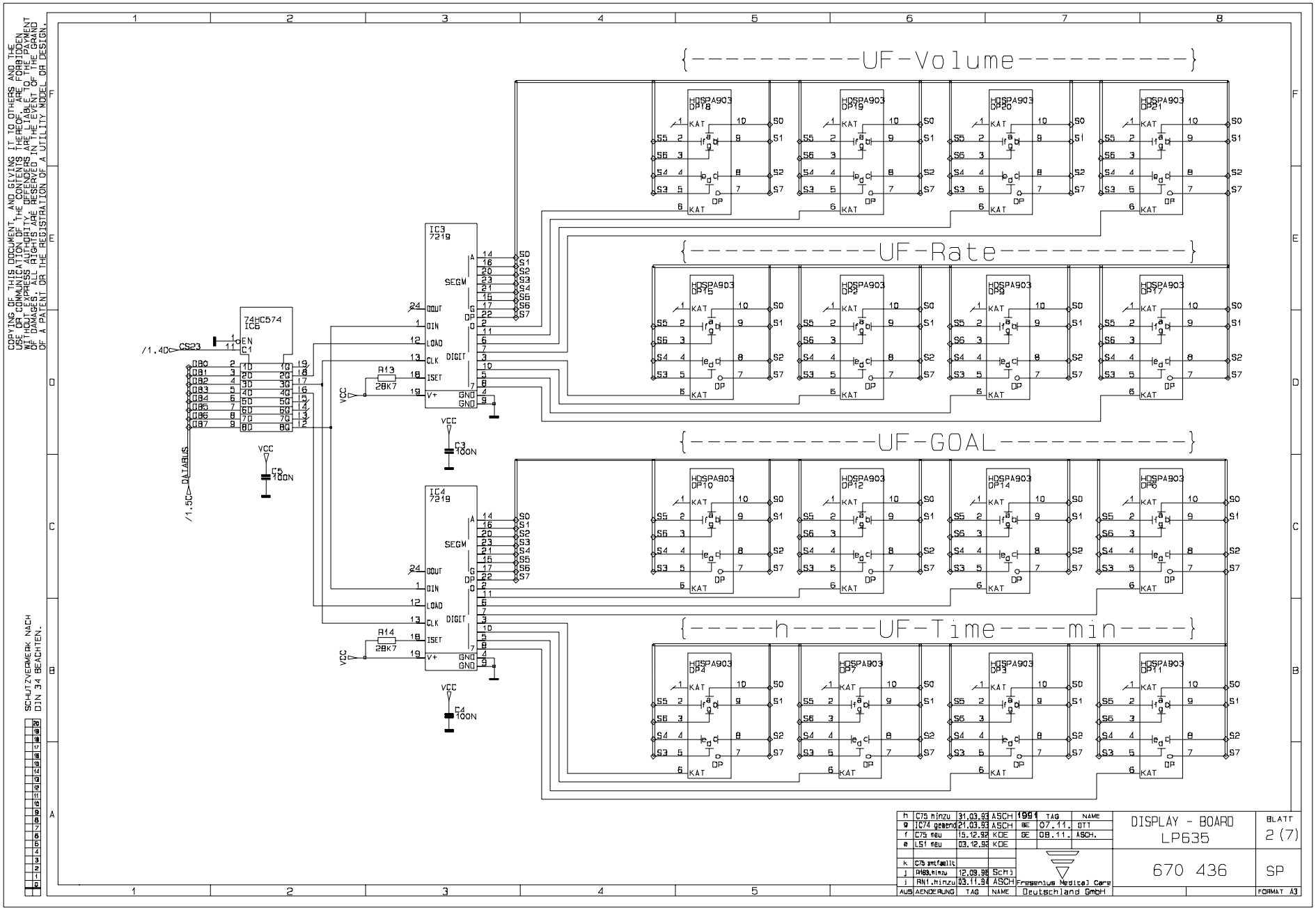
SCHUTZVERMERK NACH
DIN 34 BEACHTEN.

ALPHA-DISPLAY

h	C75 hinzu	31.03.93	ASCH	1991	TAG	NAME	DISPLAY - BOARD LP635	BLATT 1 (7)
g	IC74 geaend	21.03.93	ASCH	BE	07.11.	DT1		
f	C75 neu	15.12.92	KOE	GE	08.11.	ASCH.		
e	L51 neu	03.12.92	KOE					
k	C75 entfaellt						670 436	SP
j	R169,hinzu	12.09.96	Sch1					
i	RN1,hinzu	03.11.94	ASCH	Fresenius Medical Care Deutschland GmbH				
AUS	AENDERUNG	TAG	NAME					FORMAT A3

P.C.B. LP 635

Circuit diagram 1/7



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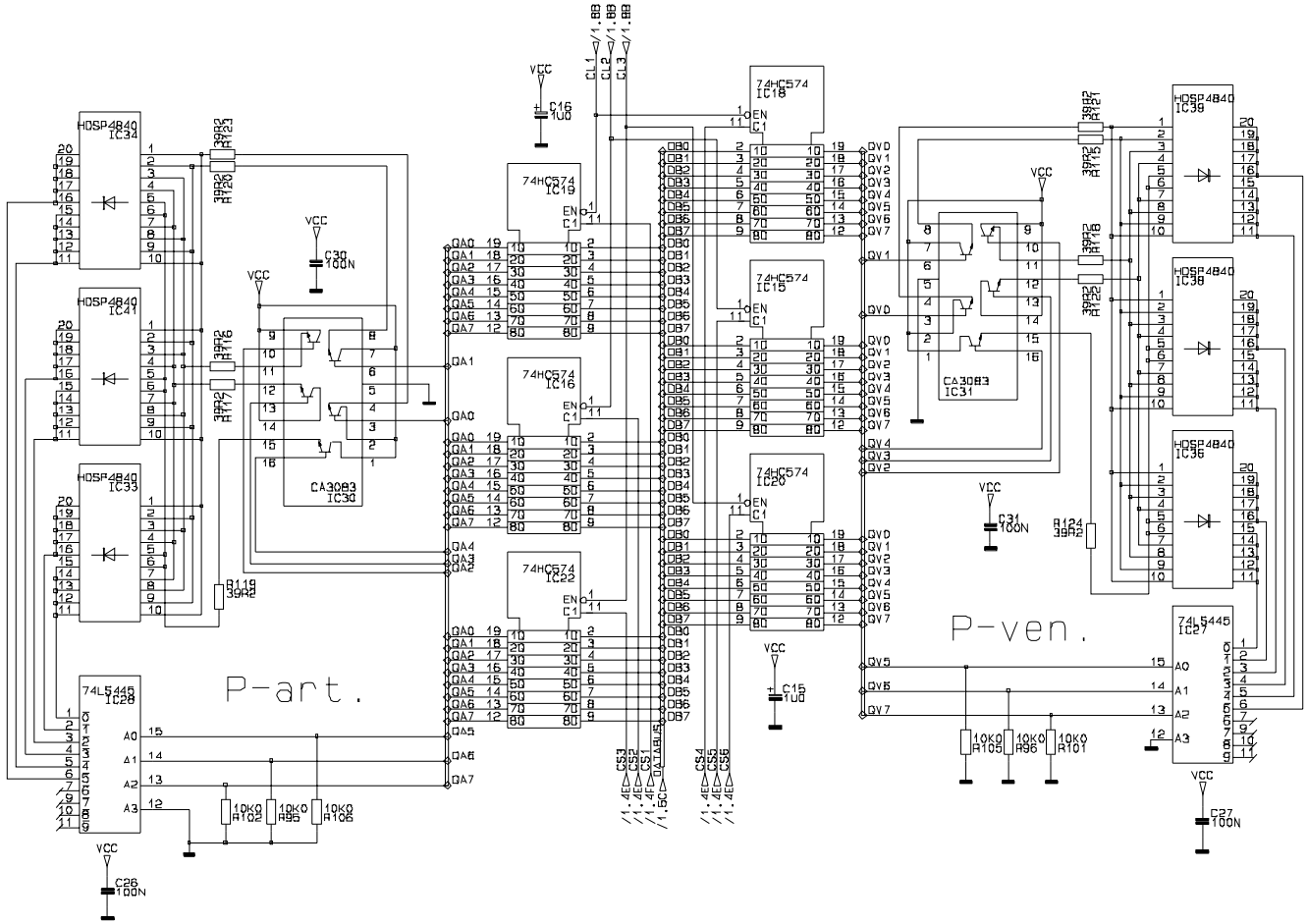
h	C75 hinzu	31.03.93	ASCH	1991	TAG	NAME	
g	IC74 geaend	21.03.93	ASCH	BE	07.11.	DTT	
f	C75 neu	15.12.92	KOE	GE	08.11.	ASCH.	
e	L51 neu	03.12.92	KOE				
k	C75 entfaellt						
j	R69,hinzu	12.09.96	Schi				
i	RN1,hinzu	03.11.94	ASCH				
AUS	AENDERUNG	TAG	NAME				

P.C.B. LP 635

Circuit diagram 2/7

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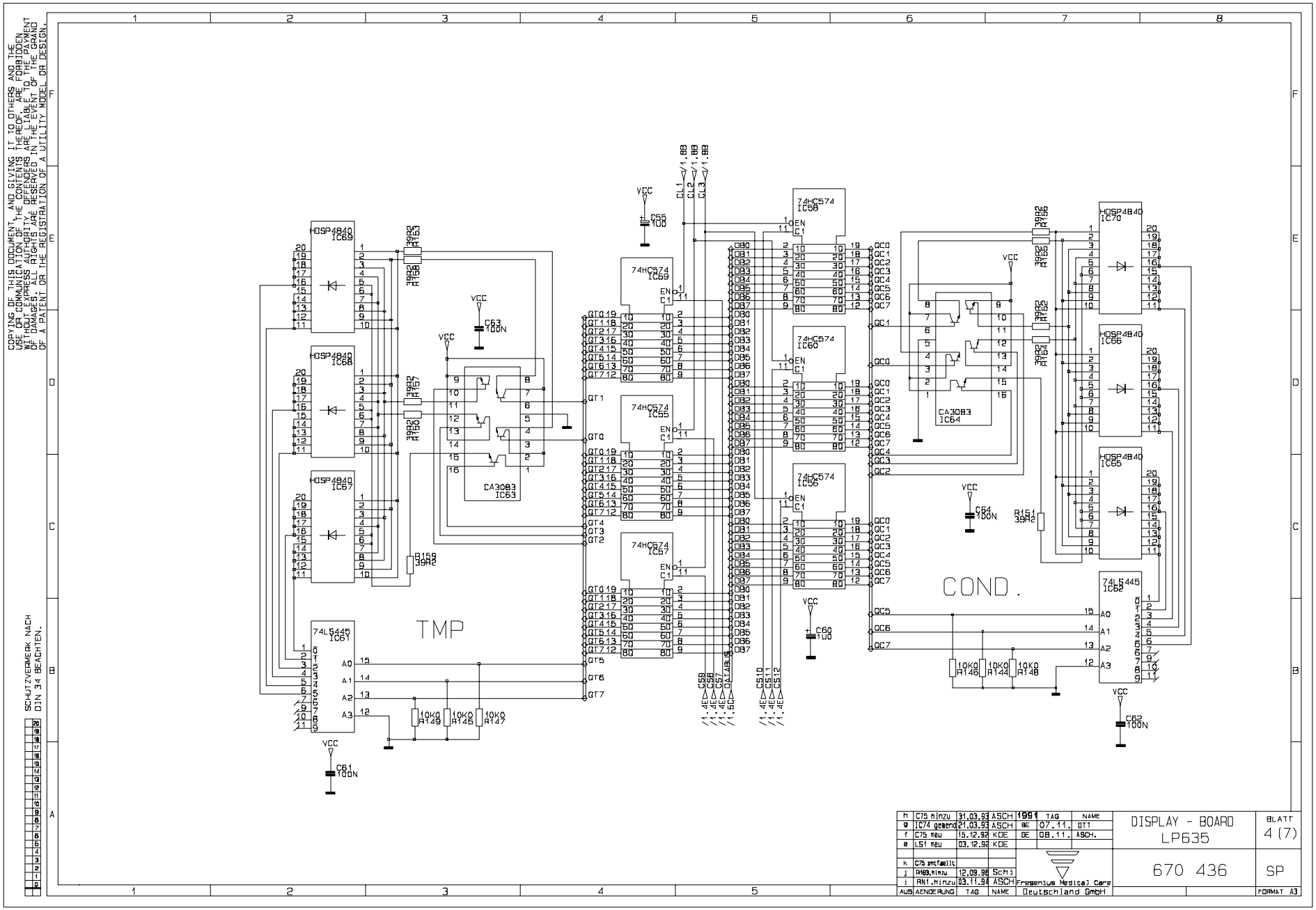
SCHUTZVERMERK NACH
DIN 34 BEACHTEN.



h	C75 hinzu	31.03.93	ASCH	1991	TAG	NAME	DISPLAY - BOARD LP635	BLATT 3 (7)
g	IC74 geaend	21.03.93	ASCH	BE	07.11.	DTI		
f	C75 neu	15.12.92	KOE	GE	08.11.	ASCH.		
e	L51 neu	03.12.92	KOE					
k	C75 entfaellt						670 436	SP
j	R63,hinzu	12.09.95	Sch1	Fresenius Medical Care Deutschland GmbH				
i	RN1,hinzu	03.11.94	ASCH					
AUS	AENDERUNG	TAG	NAME					FORMAT A3

P.C.B. LP 635

Circuit diagram 3/7

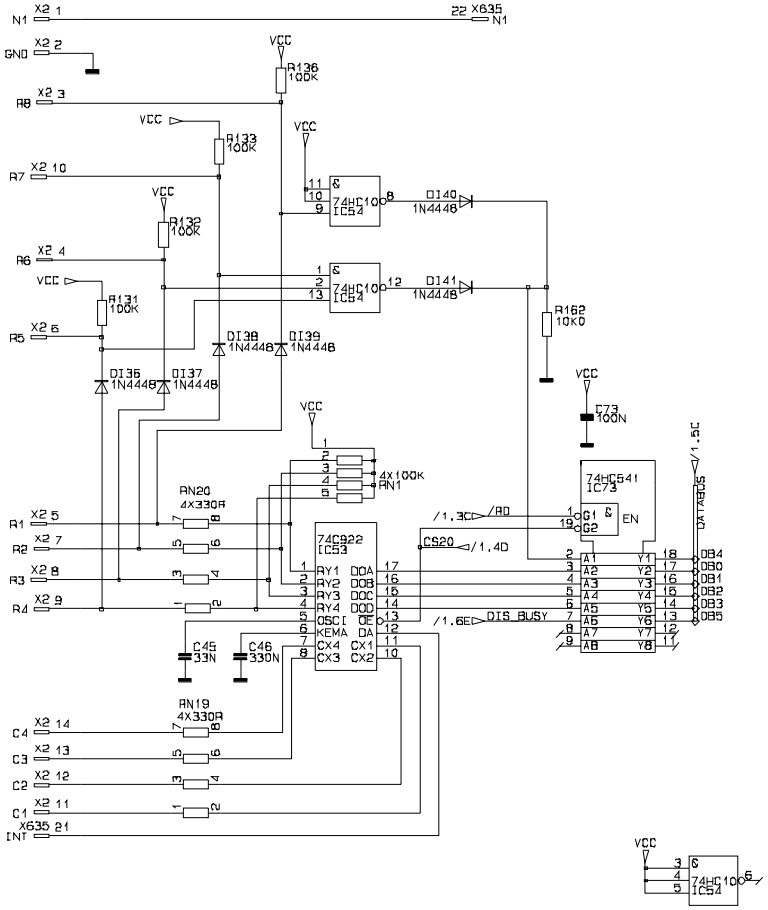

TMP
COND.
HDSP4840 IC69
HDSP4840 IC68
HDSP4840 IC67
74L5445 IC61
CA3083 IC63
74HC574 IC69
74HC574 IC65
74HC574 IC67
74HC574 IC68
74HC574 IC60
74HC574 IC56
CA3083 IC64
HDSP4840 IC70
HDSP4840 IC66
HDSP4840 IC65
74L5445 IC62
VCC
C55 100
C63 100N
C61 100N
C64 100N
C62 100N
C60 100
R153 3992
R152 3992
R151 39R2
R159 39R2
R157 3992
R158 3992
R156 3992
R155 3992
R154 3992
10KΩ R149
10KΩ R145
10KΩ R147
10KΩ R146
10KΩ R144
10KΩ R148
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SCHUTZVERMERK NACH DIN 34 BEACHTEN.
DISPLAY - BOARD LP635
670 436
BLATT 4 (7)
SP
FORMAT A3
Fresenius Medical Care Deutschland GmbH

P.C.B. LP 635

Circuit diagram 4/7

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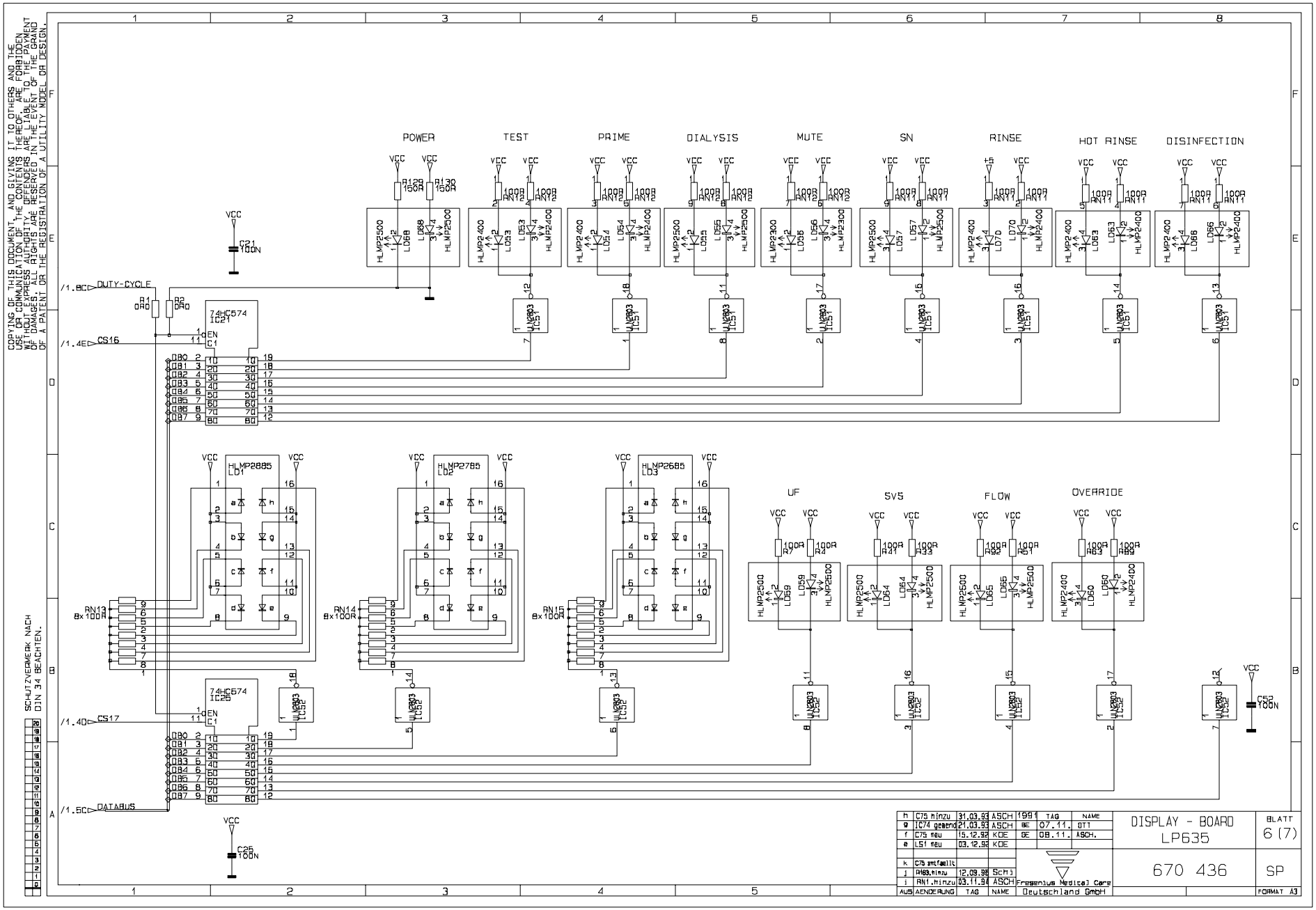
SCHUTZVERMERK NACH DIN 34 BEACHTEN.



h	C75 hinzu	31.03.93	ASCH	1991	TAG	NAME	DISPLAY - BOARD LP635	BLATT 5 (7)
g	IC74 geaend	21.03.93	ASCH	BE	07.11.	DTI		
f	C75 neu	15.12.92	KOE	DE	08.11.	ASCH.		
e	L51 neu	03.12.92	KOE					
k	C75 entfaellt						670 436	SP
j	R63,hinzu	12.09.95	Schi					
i	RN1,hinzu	03.11.94	ASCH	Fresenius Medical Care Deutschland GmbH				
AUS	AENDERUNG	TAG	NAME					FORMAT A3

P.C.B. LP 635

Circuit diagram 5/7

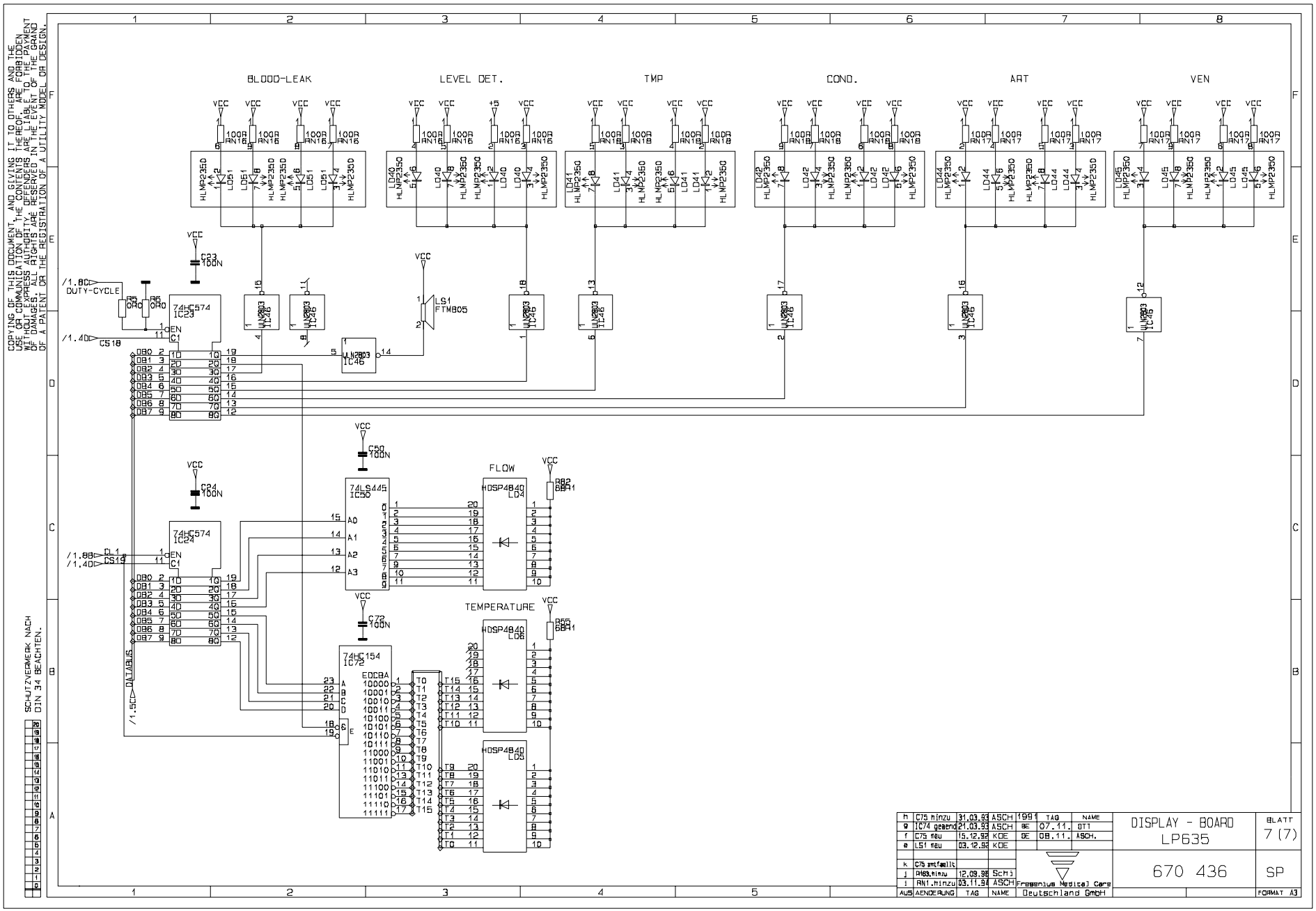


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P.C.B. LP 635

Circuit diagram 6/7



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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

h	C75 hinzu	31.03.93	ASCH	1991	TAG	NAME	DISPLAY - BOARD LP635
g	IC74 gesend	21.03.93	ASCH	BE	07.11.	DTI	
f	C75 neu	15.12.92	KOE	DE	08.11.	ASCH.	
e	LS1 neu	03.12.92	KOE				
k	C75 entfällt						670 436
j	R63,hinzu	12.09.95	Schi	Fresenius Medical Care Deutschland GmbH			
i	RN1.hinzu	03.11.94	ASCH				
AUS	AENDERUNG	TAG	NAME				

BLATT 7 (7)

SP

FORMAT A3

P.C.B. LP 635

Circuit diagram 7/7

8.16 P.C.B. LP 636 External connectors

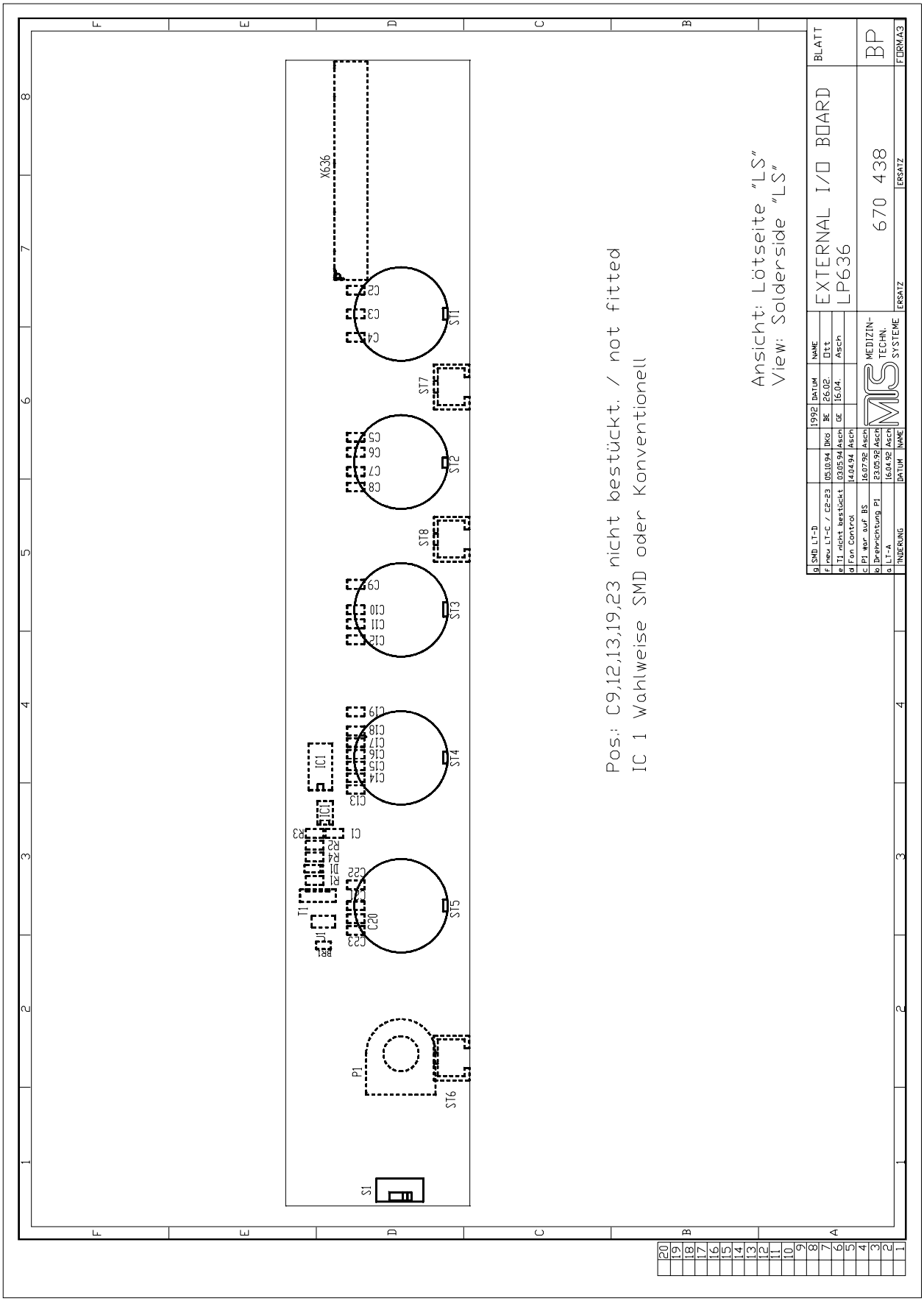
8.16.1 Description

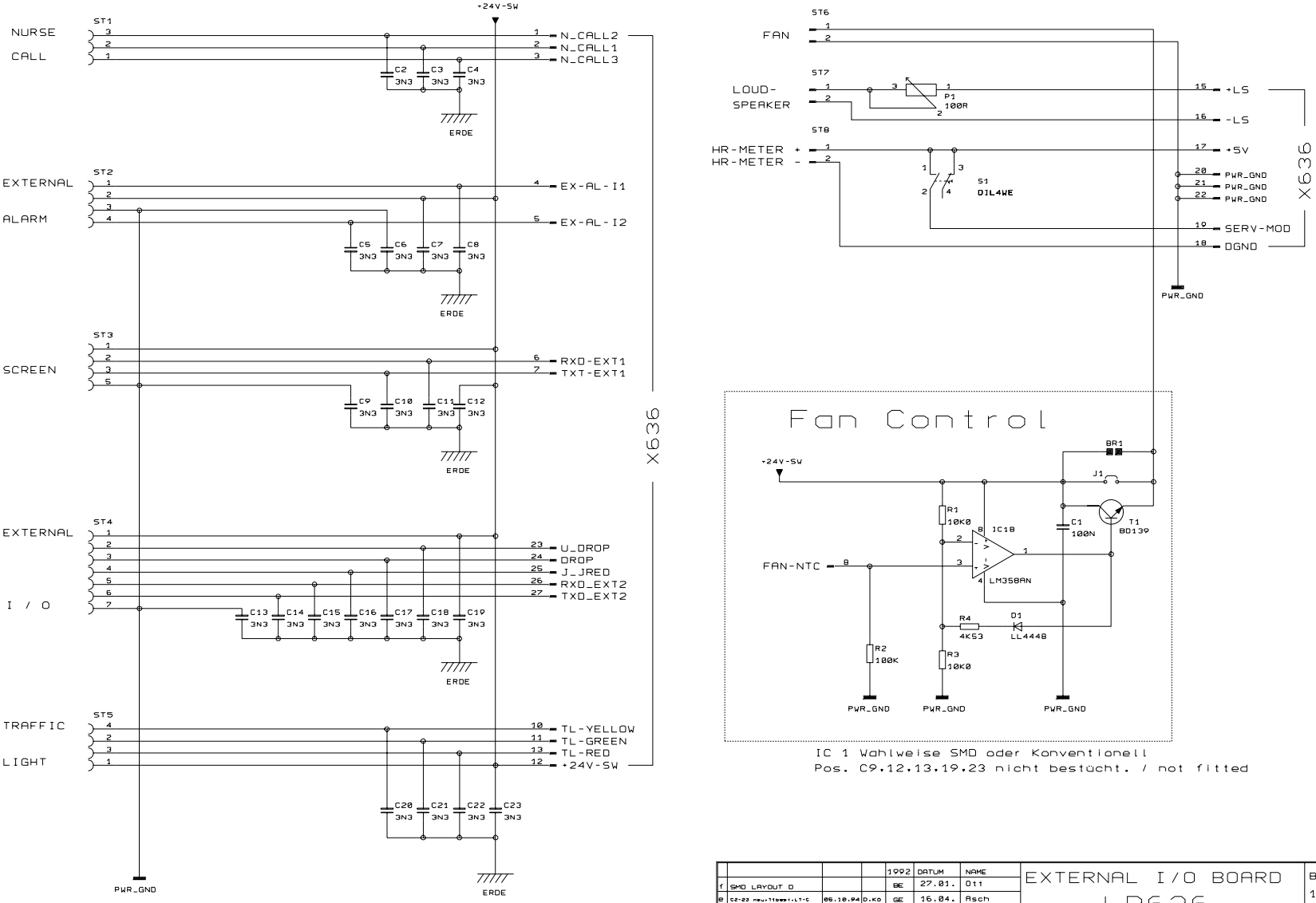
P.C.B. LP 636 is provided with the following connection sockets:
- Loudspeaker
- Alarm volume control
- Traffic light
- Input/output connection socket (for additional equipment available as option, such as a drop counter for LDL adsorption)
- Screen (optional)
- ALARM IN connection socket
- ALARM OUT connection socket

In addition, P.C.B. LP 636 is provided with a service switch.

P.C.B. LP 636

Component layout diagram





IC 1 Wahlweise SMD oder Konventionell
Pos. C9.12.13.19.23 nicht bestücht. / not fitted

				1992	DATUM	NAME	EXTERNAL I/O BOARD LP636
f	SMD LAYOUT D			BE	27.01.	Ott	BLATT 1 /1
e	C2-23 neu "Bezeichn.LT-C	05.10.94	D.KO	GE	16.04.	Asch	
d	T1 nicht bestückt	03.05.94	Asch				
c	Fan Control	14.04.94	Asch				670 438 SP
b	Drehr.P1	22.03.92	Asch	MEDIZIN-TECHN. SYSTEME			
a	LT-A	16.04.92	Asch				
	ÄNDERUNG	DATUM	NAME		ERSATZ	ERSATZ	FORM.A3

P.C.B. LP 636

Circuit diagram

8.17 P.C.B. LP 638 Power supply

8.17.1 Description

- **General information**

  Three regulated voltages are available at the outputs of P.C.B. LP 638:
  – +5 V/2.5 A
  – +12 V/1.5 A
  – +24 V/15 A

  One switching regulator is provided to generate each of these voltages.
  Due to its higher capacity, the +24-V regulator is made out of discrete components (switching transistor, current limitation).
  One integrated type-L-296 switching regulator is used to generate the +5-V voltage and one to generate the +12-V voltage.

- **+5-V regulator (circuit diagram 2/2)**

  Via points PGL1 and PGL2, the AC voltage of the 20-V winding of the main transformer is directed onto the P.C.B. After having been rectified via GL1 and smoothed by means of C26, a non-regulated voltage of approx. 26 V is available as input voltage for the regulator. High-frequency disturbances are filtered via the input filter comprising DR100 and C101. Capacitors C100 and C102 are mainly used for additional smoothing of the input voltage, which is applied to IC 100, pin 3, of the switching regulator. The storage reactor DR101, the freewheeling diode D100 as well as output capacitors C106 and C107 are applied to the output of the regulator IC 100, pin 2. Capacitor C108 is used to suppress high-frequency spikes, which might be caused by the switching procedure. The maximum current is limited to approx. 2.5 A by resistor R101. The switching frequency of the regulator is determined by C110 and R103. It amounts to approx. 50 kHz. To be regulated, the output voltage is fed back via IC 100, pin 10. Via input IC 100, pin 1, any overvoltage is recognized at output IC 10, pin 2 (e.g. caused by a defect of the L-296 switching transistor). Via IC 100, pin 15, the thyristor TR100 is energized and the output, thus, short-circuited. Due to the resultant high current, fuse SI2 is released.
  In addition, IC 100 is used to generate the power reset (e.g. upon start of the machine). The power reset is available at IC 100, pin 14. Resistor R102 is provided as pull-up resistor to +5 V.

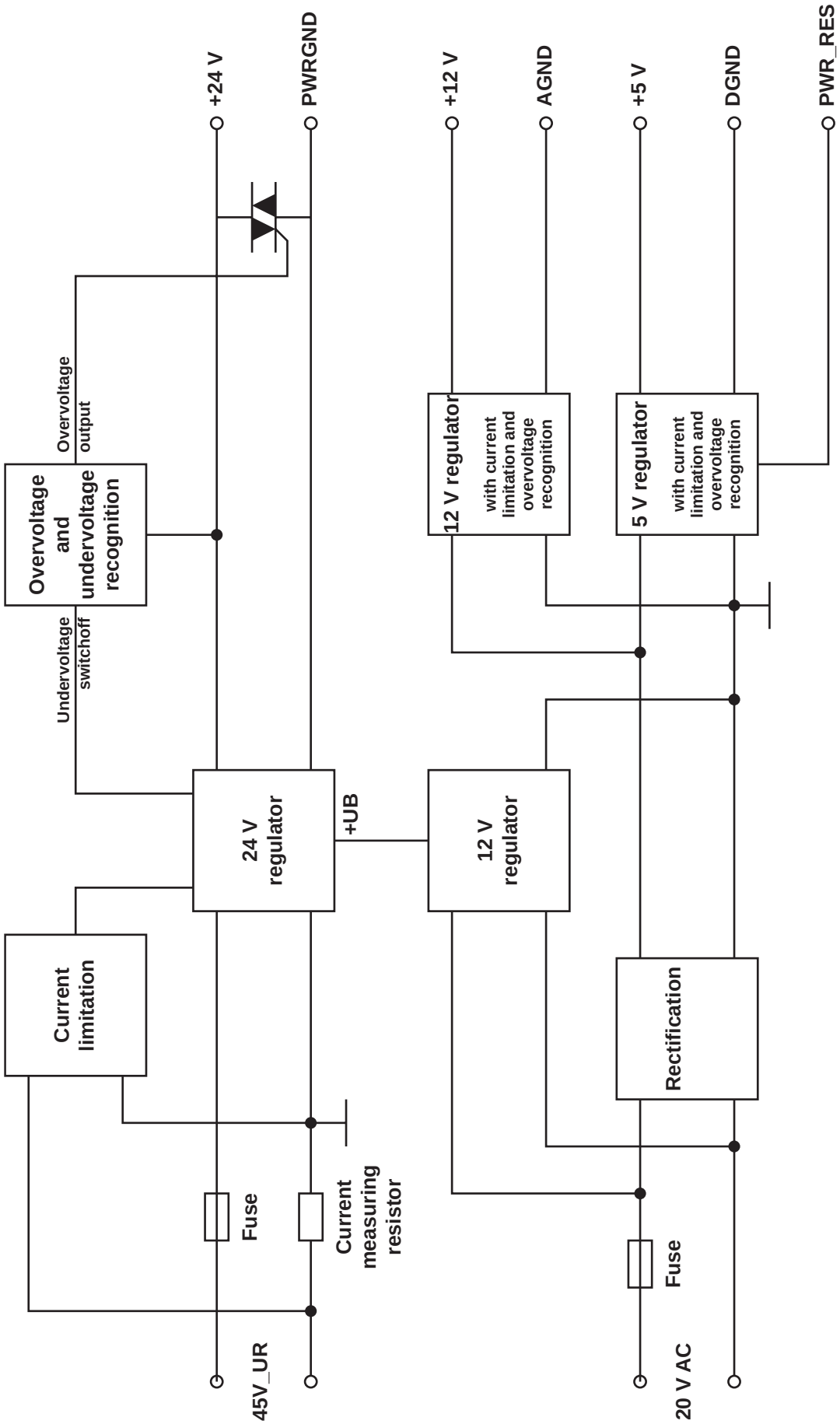
- **+12-V regulator (circuit diagram 2/2)**

  The circuit is designed in the same manner as that of the +5-V regulator. However, pin 14 of IC 200 is not utilized. In addition, only a part of the output voltage is fed back via voltage dividers R203 and R204, in order to generate the +12-V voltage.

- **+24-V regulator (circuit diagram 1/2)**

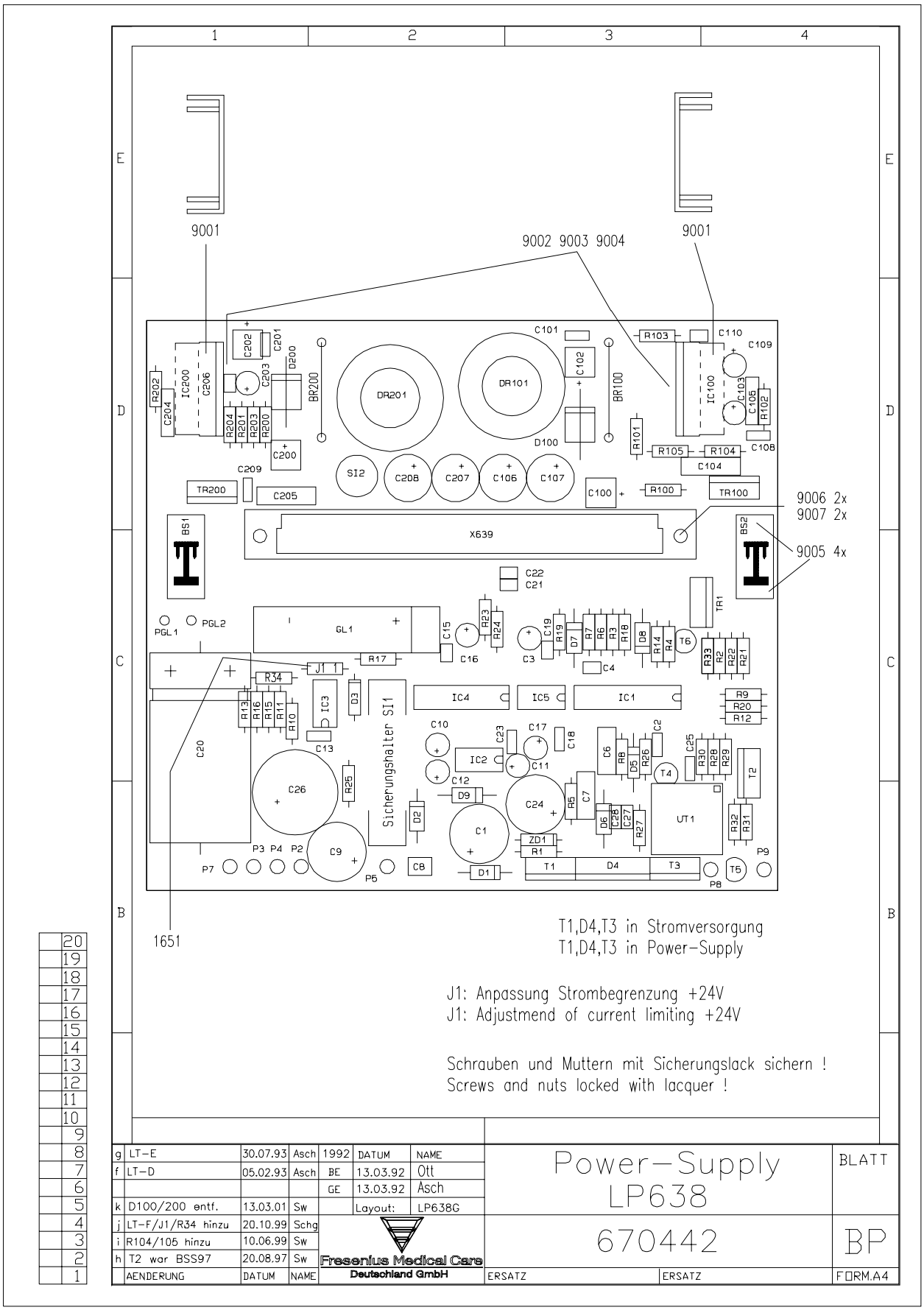
  The input voltage for the +24-V regulator is applied across P5 and P3. Supplied with clock pulses, this input voltage is connected to the storage reactor L1 via the transistor T3. With T3 being closed, the difference between input voltage and output voltage is above L1 (P8 more positive than P9). This fact causes the current in the reactor to increase linearly. The freewheeling diode D4 is inhibited during this phase.
  As soon as T3 opens, D4 becomes conductive. The output voltage plus the on-state voltage of D4 is now applied across the reactor (P9 more positive than P8). The current in the coil, which now also flows via D4, decreases linearly until switch T3 closes again. The output voltage for the duration of the individual switching cycles is held constant by means of C20.
  The output voltage is regulated by IC 1, which is provided as pulse width modulator. This modulator is supplied with pulses at a frequency of approx. 70 kHz (determined by R3 and C4). For regulation, the pulse width of the output signal (IC 1, pins 12 and 13) is modified.
  In order to make a regulation possible, the output voltage is fed back to IC 1, pin 1, via R9 and R2. Pin 2 of IC 1 is connected to the reference voltage of the module (IC 1, pin 16). IC 1 regulates the pulse width such that the difference between the voltages of IC 1, pin 1, and IC 1, pin 2, becomes zero. Via an inverter stage (R31, R32, and T5), the pulse-width-modulated signal switches transistor T2 which, in turn, is used to activate the transmitter UT1. At the output of UT1, diode D6 cuts off the negative portions of the transmitter voltage. Via D5 and R26, the signal reaches the gate of the main switching transistor T3. Transistor T4 is used to short-circuit the gate-source capacity of T3 at its cutoff moment. This leads to the achievement of as short a switching time as possible and a considerable reduction in switch dissipations.
  The current which flows through T3 is constantly monitored via a current multiplier (between P4 and P7). To this end, the voltage at the current multiplier is amplified via IC 3, pin 6, R10, and R13 and delivered to a comparator (IC 3, pin 3). The amplification factor has been selected such that the comparator output (IC 3, pin 1) changes its state at a current of approx. 15 A. The signal of the comparator output is directed to IC 1, pin 4, thus causing the current clock pulse to be interrupted immediately and the pulse width to be limited to a lesser degree (quasi-soft start) upon the next clock pulse due to the connection to IC 1, pin 9 (T6, R4, and C3). As a consequence, the output voltage (+24 V) drops.
  In order to protect the following circuits and solder tracks, the +24-V voltage is monitored via IC 5. Should the output voltage fall below +22 V (R20, R22) for more than approx. 200 ms (determined by C17), IC 5 cuts PWM IC 1 off via the latter's pin 10 (low-voltage recognition). The regulator then starts again only after the machine has been switched off and on.
  Should an excess voltage develop at the output (> +26 V, R12 and R21), IC 5 activates the triac TR1, which short-circuits the output voltage, thus releasing the fuse SI1 (at P5).

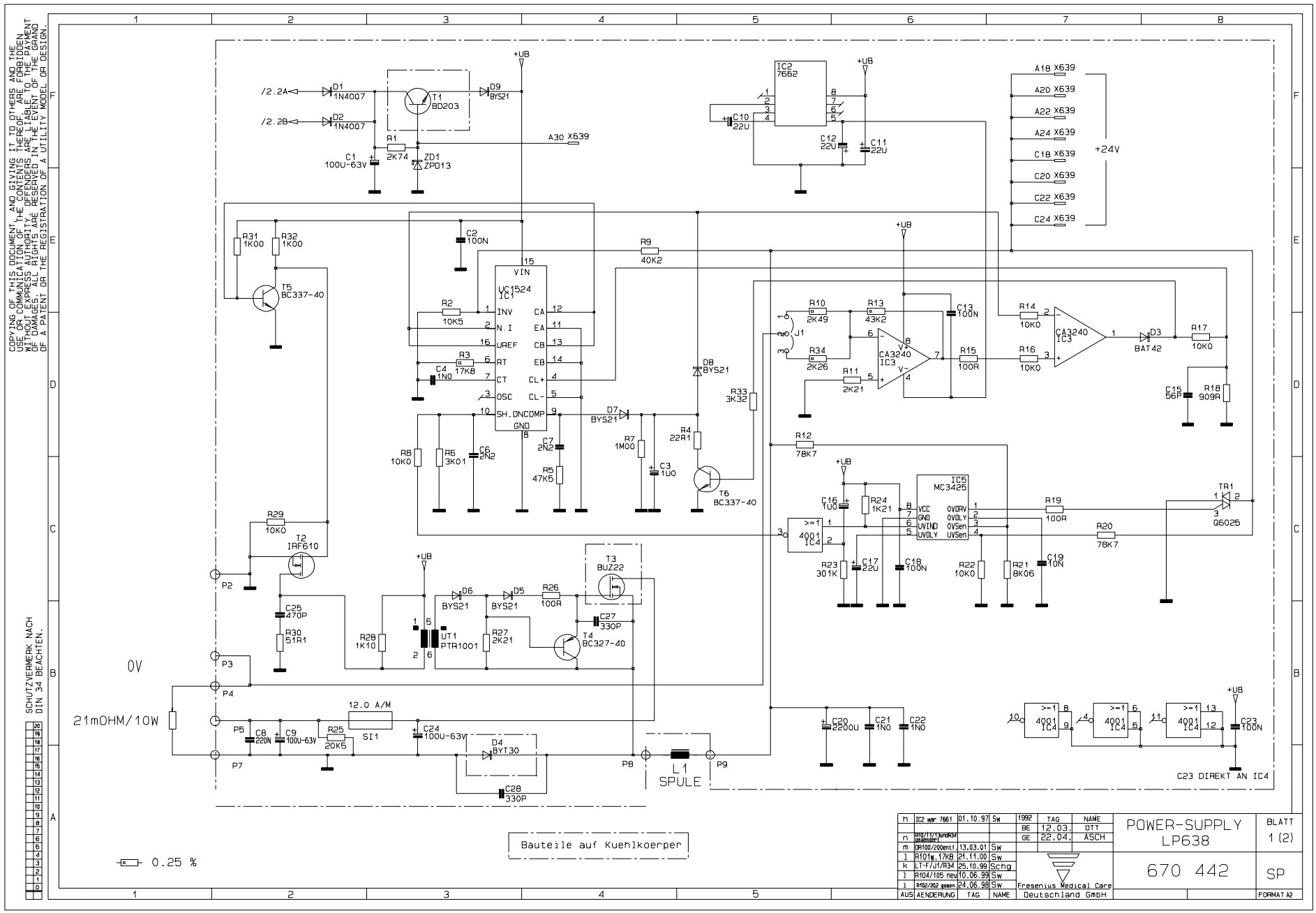
Fig.: Block diagram P.C.B. LP 638 Power supply



P.C.B. LP 638

Component layout diagram





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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

h	IC2 war 7661	01.10.97	Sw	1992	TAG	NAME	POWER-SUPPLY LP638
				BE	12.03.	DTT	
n	R12/11/13und34 geändert			GE	22.04.	ASCH	
m	DM100/200entf.	13.03.01	Sw				
l	R101w.17K8	21.11.00	Sw				
k	LT-F/J1/R34	25.10.99	Schg				670 442
j	R104/105 neu	10.06.99	Sw				
i	R402/202 geaen.	24.06.98	Sw	Fresenius Medical Care Deutschland GmbH			
AUS	AENDERUNG	TAG	NAME				

BLATT 1 (2)

SP

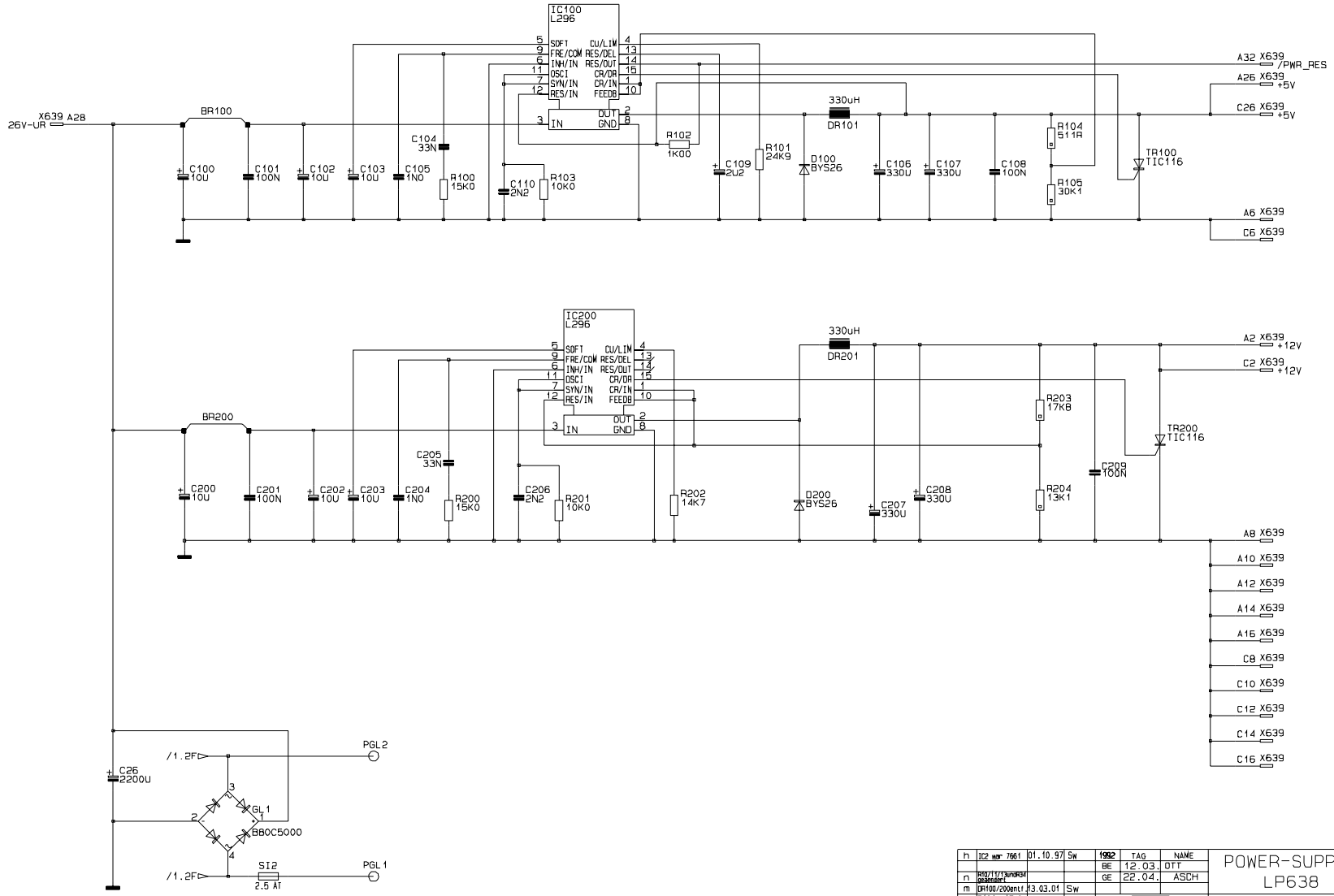
FORMAT A2

P.C.B. LP 638

Circuit diagram 1/2

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



h	IC2 war 7661	01.10.97	Sw	1992	TAG	NAME
				BE	12.03.	OTT
n	R12/11/12und34 geändert			GE	22.04.	ASCH
m	DR100/200entf	13.03.01	Sw			
l	R101w.17KB	21.11.00	Sw			
k	LT-F/J1/R34	25.10.99	Schg			
j	R104/105 neu	10.06.99	Sw			
i	R402/202 gesen.	24.06.98	Sw			
AUS	AENDERUNG	TAG	NAME			

Fresenius Medical Care Deutschland GmbH

POWER-SUPPLY LP638

670 442

BLATT 2 (2)

SP

FORMAT A2

P.C.B. LP 638

Circuit diagram 2/2

8.18 P.C.B. LP 639 Power logic

8.18.1 Description

- **Standby voltage generation**

  Via transformer TR2, rectifier GL1 and smoothing capacitor C1, a non-regulated DC voltage of approx. 15 V is directed to the input of the linear regulator T1. The connected circuits are supplied by the regulated 12-V output voltage via diode D3. Since the power logic must be supplied even in case of power failures, the battery voltage is used to generate a voltage of approx. 10 V via the regulator IC 13. Via D2, this voltage is wire-ORed with the output voltage of T1.

- **Power-on logic**

  Upon actuation of the On/Off key on the front panel of the hemodialysis machine, an L level is applied to IC 4, pin 1, via X639A, Pin 7. This causes a rising edge to develop at IC 8, pin 3, via the interposed gates of IC 4 and IC 5. The output of FF IC 8 turns to H level, while the power relay RL3 is activated via T8. Simultaneously, the bistable relay RL4 is activated via the differentiating network C28, R3 and the transistor T10. As a result, the contact RL4, pin 4 and pin 5, is made so that, via the linear regulator comprising T6, R55, and ZD1, a voltage of approx. 10 V is available at X639A, pin 13, for the generation of the power failure alarm tone.

  In order to switch the machine off, a high edge is applied to IC 8, pin 11, via DI10. By means of this edge, the level at the D input of IC 8, pin 9, is stored (H level with the machine switched on). Simultaneously, a time element for the switchoff delay starts running (R54, C27). After approx. 1 sec (I5/2 – 3) or 2.5 sec (I5/1 – 2), IC4 , pin 6, turns to L level, and IC 6, pin 3, is supplied with H level, depending on the position of jumper 5. This H level can be read in by the CPUs via X639A, pin 22, and is used as switchoff recognition. After another approx. 150 ms, the second time element R41, C26 has elapsed. The machine is switched off via IC 8, pin 4. At the same time, the bistable relay RL4 is activated via the differentiating network C29, R2, and the transistor T11. As a result, the power failure alarm tone is suppressed (contact 4/5 RL4).

- **Power failure recognition and battery connection**

  Since it must be possible to recognize a power failure, the secondary voltage of the standby transformer is monitored. The pulsating DC voltage is directed to the Schmitt trigger IC 4, pin 9, via diodes DI3, DI4, and the voltage divider R6, R7. Via IC6, pin 4, the 100-Hz square-wave signal developing at the output of IC 4, pin 8, is applied to the reset input of the counter module IC 9, pin 11 (4040). The pulse width of the square-wave signal is designed such that only short-time LOW pulses (approx. 3 ms wide) can develop at IC 9, pin 11; i.e. counter IC 9 is usually inhibited.
  A 5-V square-wave signal with a frequency of 2 kHz is applied to pin 23 of plug X639A. After a level conversion to 12 V (R62, T13, R63), this signal reaches the clock pulse input of the counter module IC 9, pin 10, via several interposed gates. During the L level phase of the reset signal, the counter IC 9, is counted up. Should this L level phase become impermissibly long because the supply voltage is dropping or the power failing, the counter module IC 9, pin 3, achieves H level after 8 ms. Via IC 6, pin 6, the reset signal for IC 9 is inhibited, the clock pulse separated from the counter by means of IC 6, pin 12, and the condition thus stored. The H level, which is now applied to X639A, pin 6, is a signal of power failure for the CPUs. Via IC 7, pin 8, the output signal of the counter module IC 9, pin3, is directed to the gate IC 7, pin 12. There it is linked to the watchdog signal of CPU 1, so that a H level is transmitted to T9 with the watchdog being intact only. T9 then supplies the coil of the battery relay RL5 with current, and the contact 7/8 RL5 is closed. The slide-in modules (24V_EM; X639A, pins 29, 39, 31) and both the 5-V and the 12-V switching regulators (26V_UR; X639C A28) are supplied with the battery voltage via diodes D4 and D6.

  As soon as IC 9, pin 3, has achieved H level, the square-wave signal, which is obtained from the standby voltage (see above), is applied to the clock pulse input of counter IC 9, pin 10, via IC 6, pin 9. After the power has returned, the counter is supplied with pulses by the square-wave signal, thus starting to count up again. Since the counter frequency has decreased (100 Hz instead of 2 kHz), it takes 160 ms until IC 9, pin 3, turns to L level and the battery is separated from the circuit again. This delayed switch-off and the rapid connection (8 ms) is intended to guarantee a smooth transition between normal operation and standby operation. The L level at IC 9, pin 3, serves to reestablish the initial situation (2-kHz pulse at IC 9, pin 3, and 100-Hz square-wave signal at IC 9, pin 11).

- **Battery test**

  In order to obtain information on the condition of the battery, it is briefly loaded during the T1 test. The resultant battery voltage is read in.

  By applying a H level to X639A, pin 10, a pulse of approx. 100 ms width is generated via the differentiating network C39, R24 and the two gates (IC 4, pin 13; IC 7, pins 1 and 2). This pulse is used as signal for activating the transistor T4. The battery is then loaded with a current of approx. 3 A by resistor R39.

- **Heater relay control and monitoring**

  The heater relay is controlled by CPU 2. Relay RL2 is switched on by an L level at X639A, pin 17. For reasons of safety, the voltage supply for the relay has been looped via the sensor cable (HEAT_REL_24V), so that it is impossible to operate the heater relay with the sensor cable pulled off.

  The voltage applied across the coil of the heater relay is monitored by comparator IC 1, pins 2 and 3. Resistors R64 to R67 are laid out such that the condition of the relay is recognized as being cut off (IC 1, pin 1, at L level) only with a voltage lower than 3 V being applied across RL2. The output signal of the comparator can be read in by CPU 2 via X639A, pin 12. In addition, this signal also acts upon the heater control (see below).

- **Battery charging circuit**

  Emergency operation of the machine is guaranteed by three series lead-acid storage batteries of 6 V each. An integrated module IC 10 of type UC2906 is used to charge the batteries. The connected 24-V voltage (24V_SW) is used to provide for the supply voltage required for charging. The charging procedure is accomplished in three stages.

  Up to a battery voltage of 15 V, the minimum charging current amounts to 14 mA (in this charging state, the charging current decreases with increasing battery voltage).

  After a voltage of 15 V has been reached, the batteries are charged with a constant current. The charging current amounts to approx. 250 mA. This state is maintained until the battery voltage has risen to 22.4 V.
  Then the charging current starts dropping. As soon as the current has fallen below 14 mA, IC 10 switches over to the float state, maintaining the battery voltage at approx. 21 V.

  As soon as current consumption causes the battery voltage to drop, the above procedure starts again.

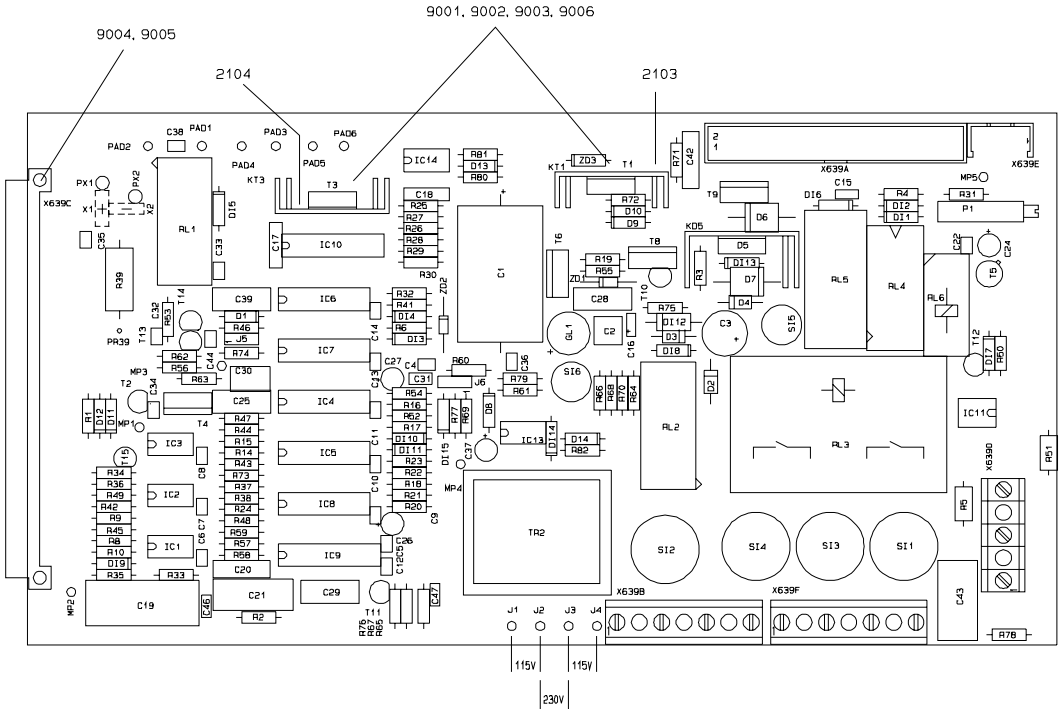
- **Heater control**

  The temperature sensor is applied to X639A, pin 21. Together with R33 and R10, it constitutes a voltage divider. The rated-value specification for heater control is superimposed over this voltage divider via X639A, pin 18, and resistors R32 and R34.

  The resultant voltage is directed to the operational amplifier IC 2, pin 6 (LM358), which is connected as PID regulator. The regulator is compensated via a voltage, which is applied to X639A, pin 24, and is converted to the requisite range of voltage via IC 1, pin 6. This voltage cannot be influenced by means of the hardware (software compensation!). The output voltage of the PID regulator is superimposed over a delta signal, which is generated by IC 3, and is then delivered to comparator IC 2, pin 3. This comparator activates transistor stages T2 and T12. The L level applied to IC 11, pin 2, causes the triac coupler IL410 to ignite its internal triac, thus delivering the gate trigger voltage for the external main triac (1, 2, 3 X639D) via R51. The external main triac accepts the load current of the heater rod. The triac is fired with each zero crossover of the 230-V supply voltage only. To this end, a zero voltage recognition is integrated in the coupler IC 11. As a result, any disturbances, which are caused by the steep build-up of current under load in case of phase-shift control, are avoided.

  The heater control can be inhibited by an H level at X639A, pin 19. As already mentioned above, the heater relay monitoring unit also acts upon the control. Via diodes D9 and D10 as well as the RC element R72, R71, C42, the heater relay IC 1, pin 1, if de-energized (IC 1, pin 1, at H level), causes the heater control to be inhibited. The heater control is released only after the heater relay (IC 1, pin 1, turns to L level) has been activated and C42 has been unloaded via R71. This measure is intended to protect the triac coupler IC 11 and to treat the relay contacts of the heater relay gently.

  For hot rinsing, an H level at X639A, pin 20, causes R34 and R10 to be connected in parallel, resulting in a rated temperature value of approx. 84 °C.



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JUMPER 6
UNTERSPANNUNG
LOWER VOLTGE

POS.	GRENZE / LIMIT
2 - 3	VARIABLE (P1)
1 - 2	FIXED/FEST

= STANDARD

JUMPER 5
AUSSCHALTVERZOEGERUNG
SWITCH OFF DELAY TIME

POS.	ZEIT / TIME
2 - 3	1 SEC.
1 - 2	2 SEC.

= STANDARD

n	Verb.X639A33	29.03.95	ASCH	1992	TAG	NAME
m	AD16uR1neu	24.03.94	ASCH	BE	02.04.	DTT
				GE	24.04.	ASCH.
r	KD5 neu LT-I					
q	keine Serie					
p	Verb.PAD1-D5	16.04.96	Schg			
o	IC14/PlneuLT-F	17.07.95	Scheu			
AUS	AENDERUNG	TAG	NAM			

FRESENIUS AG
WERK
SCHWEINFURT

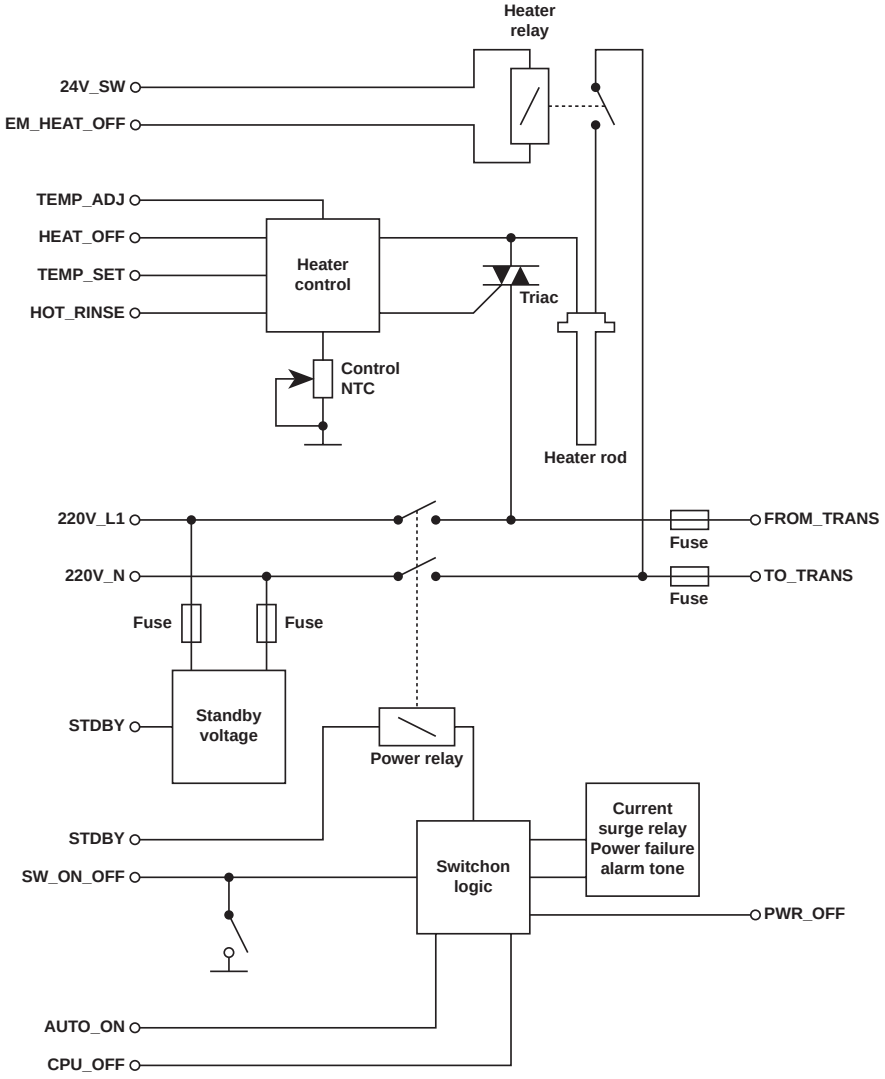
POWER-LOGIC
LP639

676035

BP

FORMAT A3

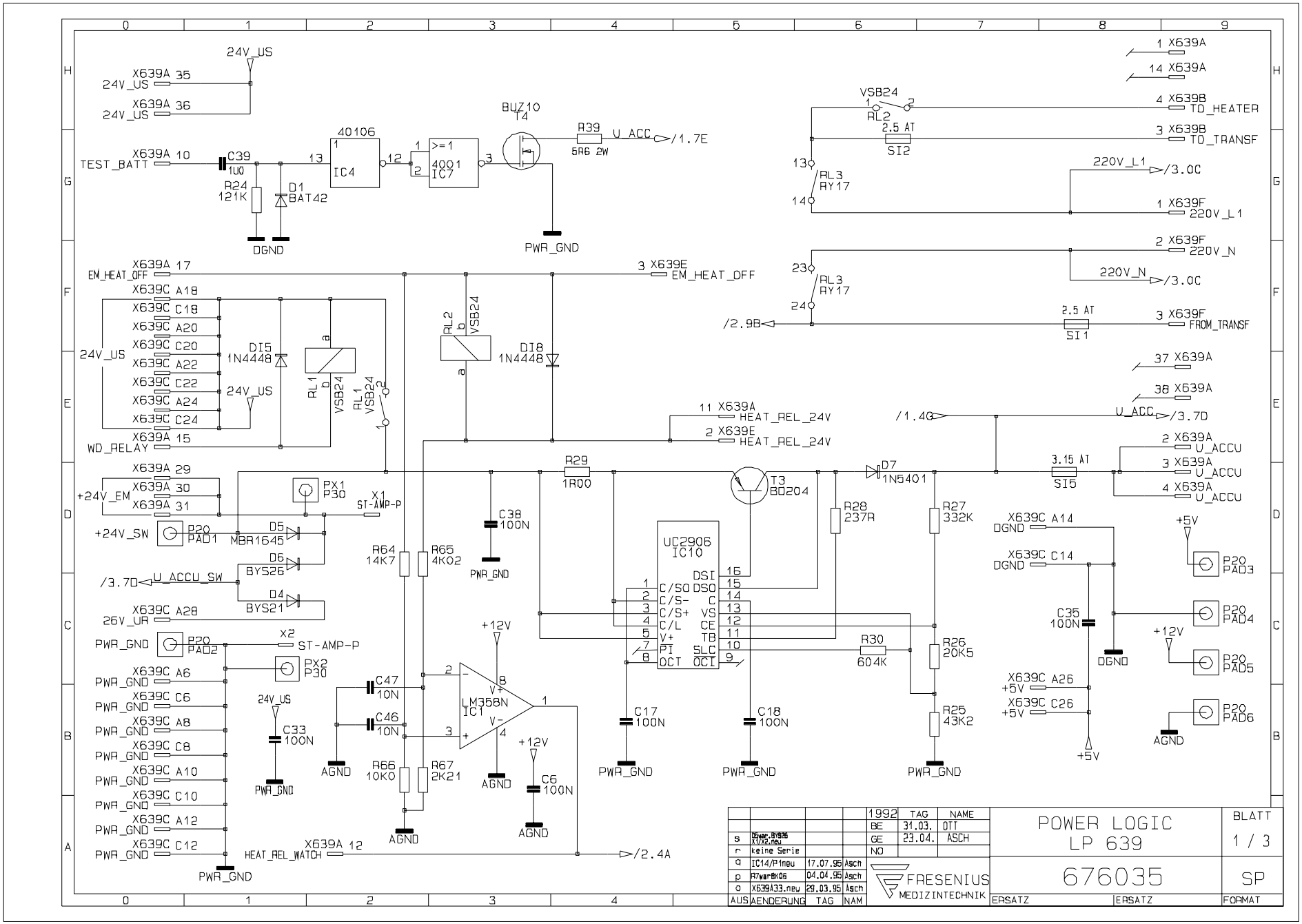
Fig.: Block diagram P.C.B. LP 639 Power logic



8.18.2 Circuit diagram and component layout diagram
P.C.B. LP 639 Power logic

P.C.B. LP 639

Component layout diagram



				1992	TAG	NAME
				BE	31.03.	OTT
5	Dwgs.BYS26 X1/X2.neu			GE	23.04.	ASCH
n	keine Serie			NO		
q	IC14/P1neu	17.07.95	Asch			
p	R7warBK06	04.04.95	Asch			
o	X639A33.neu	29.03.95	Asch			
AUS	AENDERUNG	TAG	NAM			

FRESENIUS MEDIZINTECHNIK

POWER LOGIC
LP 639

676035

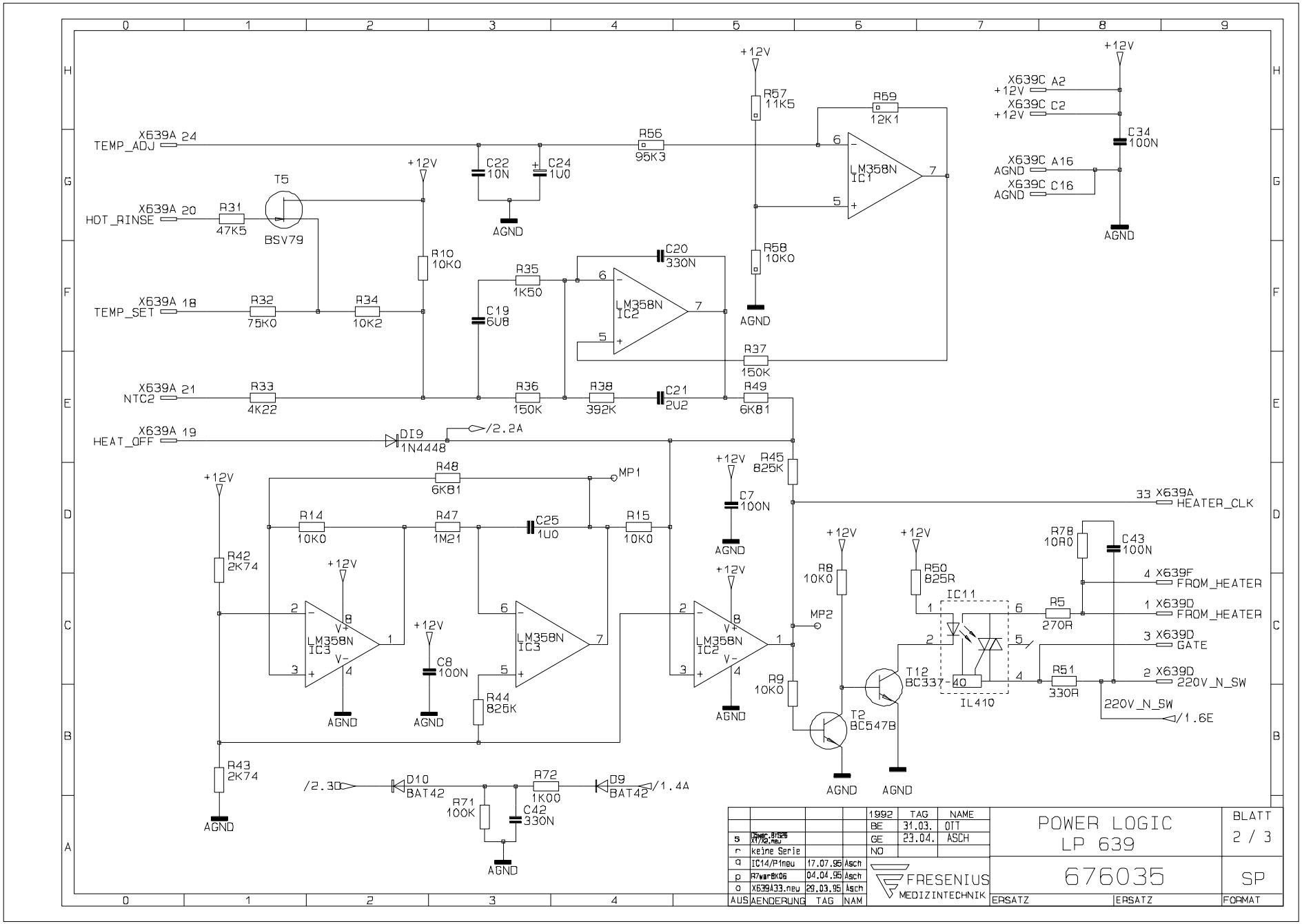
BLATT 1 / 3

SP

ERSATZ | ERSATZ | FORMAT

P.C.B. LP 639

Circuit diagram 1/3



				1992	TAG	NAME
				BE	31.03.	OTT
5	[illegible]			GE	23.04.	ASCH
n	keine Serie			NO		
q	IC14/P1neu	17.07.95	Asch			
p	R7warBK06	04.04.95	Asch			
o	X639A33.neu	29.03.95	Asch			
AUS	AENDERUNG	TAG	NAM			

FRESENIUS MEDIZINTECHNIK

POWER LOGIC
LP 639

676035

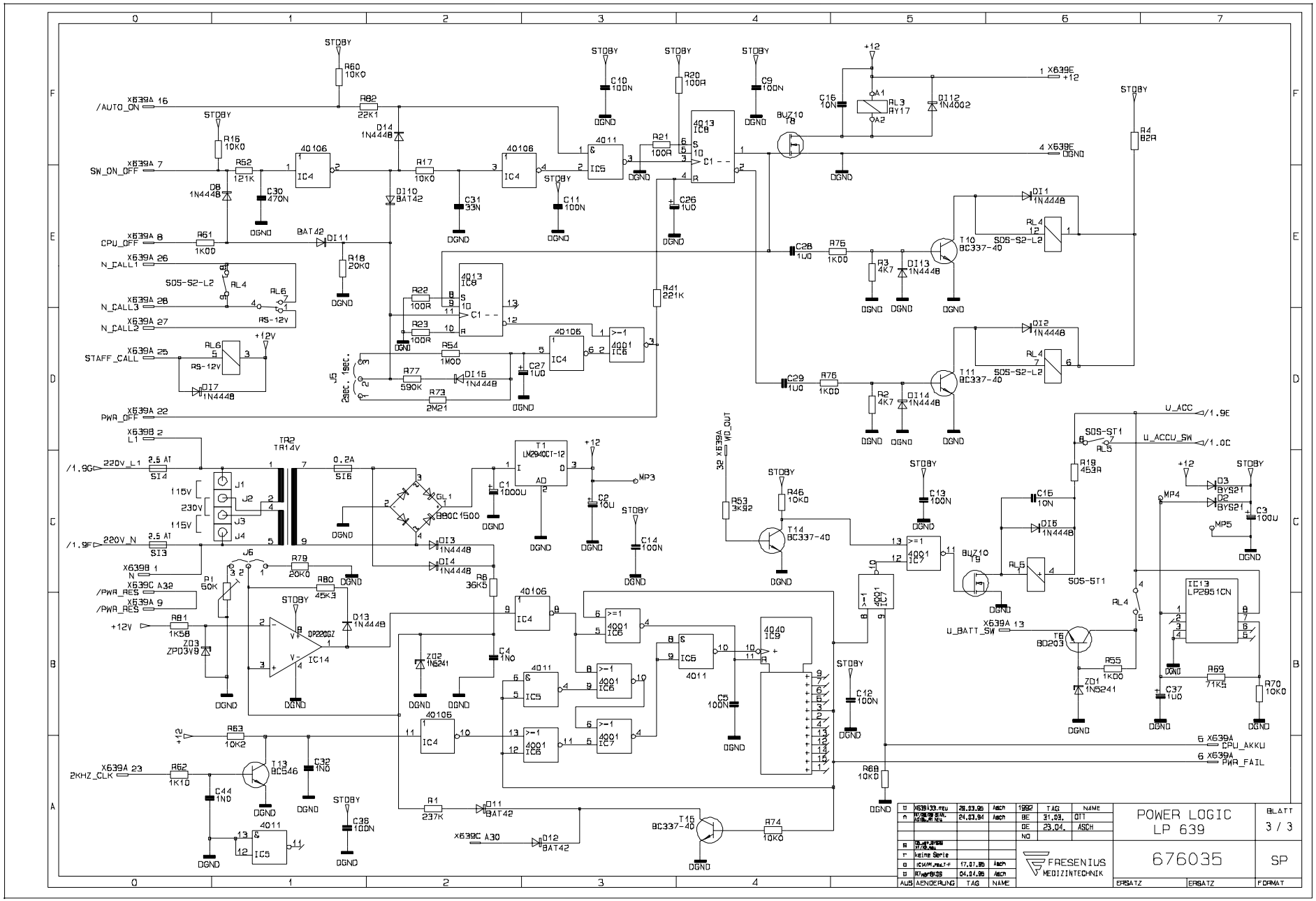
BLATT 2 / 3

SP

ERSATZ | ERSATZ | FORMAT

P.C.B. LP 639

Circuit diagram 2/3


POWER LOGIC
LP 639
BLATT
3 / 3
676035
SP
FRESENIUS
MEDIZINTECHNIK
ERSATZ
ERSATZ
FORMAT
/AUTO_ON
SW_ON_OFF
CPU_OFF
N_CALL1
N_CALL3
N_CALL2
STAFF_CALL
PWR_OFF
/1.9G
/1.9F
/PWR_RES
/PWR_RES
+12V
2KHZ_CLK
U_BATT_SW
U_ACC
U_ACCU_SW
CPU_AKKU
PWR_FAIL
WD_OUT
220V_L1
220V_N
115V
230V
115V
RS-12V
+12
STDBY
DGND
IC4
IC5
IC6
IC7
IC8
IC9
IC13
IC14
T1
T8
T9
T10
T11
T12
T13
T14
T15
T16
RL3
RL4
RL5
RL6
SDS-ST1
SDS-S2-L2
TR2
TR14V
GL1
J1
J2
J3
J4
J5
J6
P1
ZD1
ZD2
ZD3
MP3
MP4
MP5

P.C.B. LP 639

Circuit diagram 3/3

8.19 P.C.B. LP 950 Control board (HEP)

8.19.1 Description

- **General information**

The electronics of the pump consists of two printed circuit boards:
– LP 950 (control and power part)
– LP 644-4 (display and keyboard)

Both boards are connected by means of a 20-pin printed circuit board connectors (ST4).

- **P.C.B. LP 950**

P.C.B. LP 950 comprises the digital control unit.

This unit communicates with the dialysis monitor via a 14-pin connector (ST1). In addition, an RS232 interface to the monitor is accomplished via the IC 18.
Further plugs are ST2 leading to the stepper motor, ST3 leading to the optical sensor, and ST5 leading to the position sensor (Hall switch).

Two switches are used to set different modes of operation. SW1 is used to determine the syringe type used. SW2 is used to simulate the blood detector in case service work becomes necessary.

The circuit can be divided into four functional groups, which will be described in detail below.

- **Voltage supply**

The 25-V capacity supply (LV) is directly applied to the switching power supply unit for 5-V supply and to the switching transistor T1.
During normal operation, transistor T1 is permanently conductive, so that the +25-V voltage is then applied to the stepper-motor power driver.
In case of an error, the microcontroller (MC) or the GATE-ARRAY (GA) can inhibit the FET T2 and, thus, the transistor T1. As a result, the stepper motor control (SMS) can be separated fromthe LV.

The +5-V supply voltage (VCC) for the digital components is generated from the 25-V power supply by means of a switching-power-supply-unit regulator LM3578. As a result, a better efficiency is achieved than would be the case if use were made of a horizontal control unit.

When the hemodialysis machine is being switched off or if the voltage fails, the operating data of the slide-in module is filed and stored in the NOVRAM IC 17.

Together with the monostable circuit IC 15, comparator IC 13 constitutes a power-on-reset circuit.

- **Stepper motor activation**

  IC 14 constitutes the stepper motor controller module (SMC). IC 14 allows both full-step and eighth-step operation, reversal of the direction of rotation, as well as a controllable motor current limitation.

  The digital control inputs and control outputs of the stepper motor controller are TTL-compatible and can be directly connected to the microcontroller or to the gate array.

  The step frequency, the direction of rotation and the module release of the stepper motor controller are applied to port 1.1 to port 1.3.

  Selection of full-step or eighth-step operation respectively is made by port A.7 of the gate array. Via port A.4, the phase current can be set to two different rated values.

- **Interface to the hemodialysis machine**

  A level of 12 V is applied to the signals emitted by the hemodialysis machine. The signals are limited to a voltage of 5 V by the optocoupler.

  A 12-V level is applied to the ALARM and PUMP output signals by means of a transistor and a pull-up resistor.

  A serial interface is provided for expansions.

  The voltage level of these signals amounts to 5 V.

- **Digital control unit**

  Together with the dual operational amplifier and various passive components, the microcontroller and the gate array constitute the digital control unit.

  The gate array is provided as customer-specific integrated circuit. It comprises the following functions:

  8-byte RAM
  This RAM is provided as expansion of the microcontroller RAM.

  4-digit display controller
  Up to four seven-segment displays can be energized by the integrated display controller.
  Three displays with a joint cathode are incorporated in the heparin pump.
  The cathodes of the three displays are connected to the column connectors S00 to S20 of the gate array; the segments are connected to the respective connectors via variable multipliers. S30 is connected to the cathodes of the two control LEDs. The anodes are applied to the segment connectors SA0 to SF0 via variable multipliers. The display controller uses the multiplex method to operate the displays.

  8-bit port A and 4-bit port B
  Port A and port B are bidirectional ports with integrated data directional registers.
  The hexagonal coding switch (selection of syringe type) is connected to ports PA.0 to PA.3; the Hall sensor provided for end position recognition is connected to port PA.5.
  Port PA.7 emits the signal for reversal from full-step to eighth-step operation and vice versa to the stepper motor controller.
  Port B is not utilized.

  Key encoder for 16 keys
  By means of the key encoder, 16 keys can be operated in a 4x4 matrix in multiplex operation.
  In addition to a 64-mS debouncing unit, an N-key-rollover function and a multiple key recognition, the key encoder is provided with an integrated permanent-key-depression recognition.
  The code of the key recognized is stored in a register, causing an interrupt to be released in the internal interrupt controller.
  Connectors X0 to X1 for columns and connectors Y0 to Y2 for lines are used for the heparin pump.

  Two external interrupt inputs
  Connectors EX0 to EX1 are provided as additional interrupt inputs. The respective interrupt is released by a rising edge at the inputs.
  The pulses of the optical sensor are converted into a digital signal by comparator IC 13 and connected to input EX0.
  A signal, which is reduced to 5 V, is applied to input EX1. This signal is emitted by the optical detector in the hemodialysis machine. This signal is a square-wave oscillation of approx. 100 Hz with air or a clear fluid being recognized by the sensor; with blood, L level is applied to the signal.

  Interrupt controller
  In addition to the interrupts released by the key encoder and the external interrupt inputs, a 1-MS and a 16-MS interrupt is made available by the gate array.
  All interrupts can be individually inhibited or released in a corresponding control register.
  After an interrupt request, the releasing interrupt must be determined in the respective routine by reading the interrupt flag register. In order to erase the respective interrupt flag, it suffices to read the pertinent address (see gate array data sheet).

Oscillator

For being externally connected, the oscillator requires a crystal, two capacitors and a resistor. Since the internal timers must be set to the correct basic pulse rate, the frequency of the crystal must amount to 4.096 MHz.

Watchdog

The watchdog output of the gate array is set to H level by the program. Every millisecond, the integrated watchdog circuit sets the output level to L level, thus releasing an interrupt at the input INT0 of the microcontroller.

During normal operation, the microcontroller now performs an interrupt routine, during which the watchdog circuit in the gate array is reset. This causes the watchdog output to return to H level.

The base of FET T2 is likewise connected to the watchdog signal via a lowpass filter. The time constant of the lowpass filter is selected such that, during normal operation, the 1-ms interrupt causes transistor T1 to become conductive.

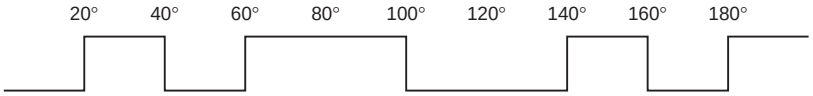
- **Speed and rotational direction monitoring unit**

A slotted disc is attached to the drive shaft of the syringe holder for monitoring the speed and the direction of rotation. This slotted disc penetrates the range of the optical sensor, where it generates a specific light-dark pattern.

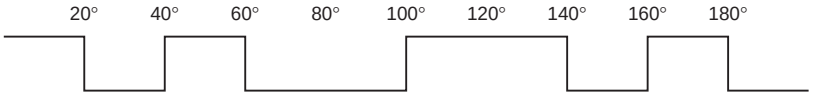
The optical sensor converts this pattern into corresponding voltage levels.

The slotted disc is divided in three sections of 120 degrees each, each of them having the same window configuration:
20-degree opening, 20-degree tie, 40-degree opening, 40-degree tie.

The division of the disc in two sections leads to the following speed-related pattern:



In the opposite direction of rotation, the pattern is as follows:



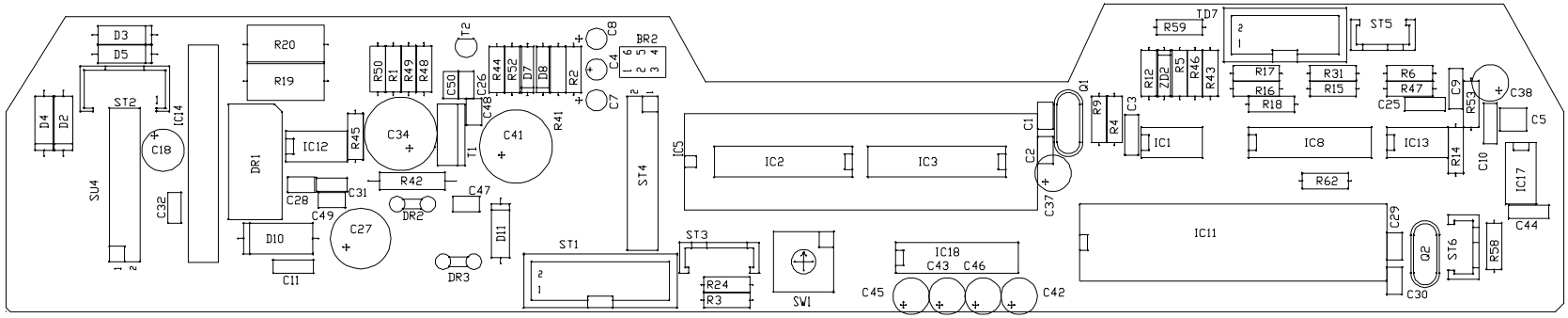
These patterns are evaluated by the software, so that a stop of the shaft is recognized not later than after 20°, because an L/H level or an H/L level respectively must be achieved after the corresponding period of time, if the shaft is rotating.

After a 80-degree rotation, an error in the direction of rotation of the shaft is recognized.

8.19.2 Circuit diagram and component layout diagram
P.C.B. LP 950 Control board (HEP)

P.C.B. LP 950

Component layout diagram



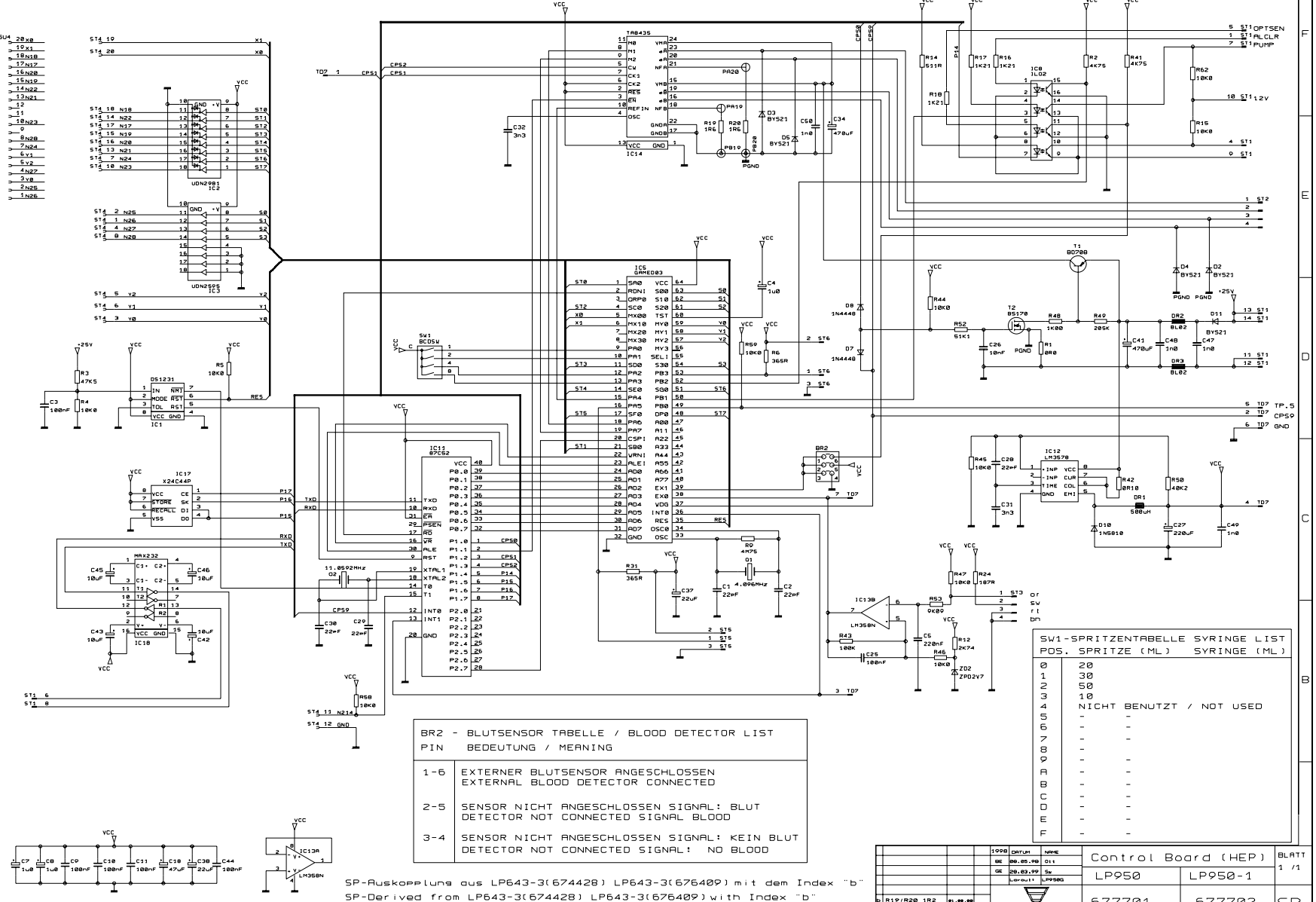
Bestückung / Fitted	ST4U	ST4
677701	–	X
677702	X	–

BP–Auskopplung aus LP643–3(674428) und LP643–4(676409) Index "b"
BP–Dervired from LP643–3(674428) and LP643–4(676409) with Index "b"

				1998	DATUM	NAME	CONTROL–BOARD (HEP)		BLATT
				BE	06.10.98	Ott			
				GE	08.03.99	Sw			
					Layout:	LP950G	LP950	LP950–1	
c	IC14/C34 verschoben								
a	Textfeld geä.	23.06.99	Sw	Fresenius Medical Care Deutschland GmbH			677701	677702	BP
	ÄNDERUNG	DATUM	NAME				ERSATZ	ERSATZ	FORM.A3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



		1998	DATUM	NAME	Control Board (HEP)
		GE	00.05.99	Dit	LP950 / LP950-1
		GE	20.03.99	Su	BLATT 1 /1
			Layout	LP950G	
B	R19/R20 1R2	01.06.00			677701 / 677702 SP
A	SO4 beleguna	14.07.99	Su	Fresenius Medical Care Deutschland GmbH	
	ANDERUNG	DATUM	NAME		FORM.A3

P.C.B. LP 950

Circuit diagram

8.20 P.C.B. LP 644-4 Display board (HEP)

8.20.1 Description

- **General information**

  P.C.B. LP 644-4 comprises the elements for the display and the operating keyboard. All elements are covered by a water-tight foil.

- **Display**

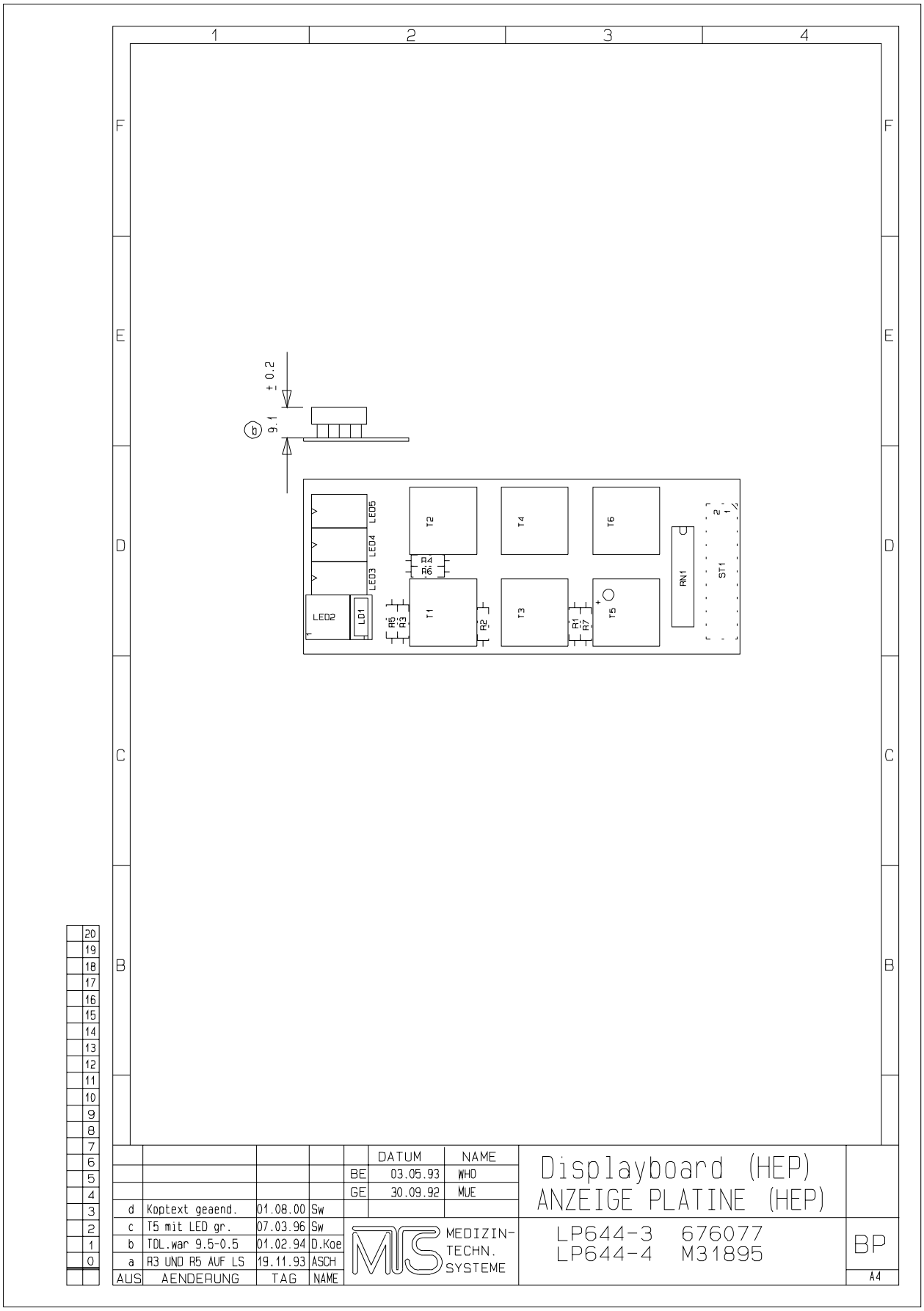
  All of the information required is given to the user by three seven-segment displays and two LEDs.

- **Keyboard**

  The keyboard comprises six individual make contacts, which are directly soldered into the printed circuit board. These contacts constitute a 2x3 matrix and are directly connected to and evaluated by the gate array.

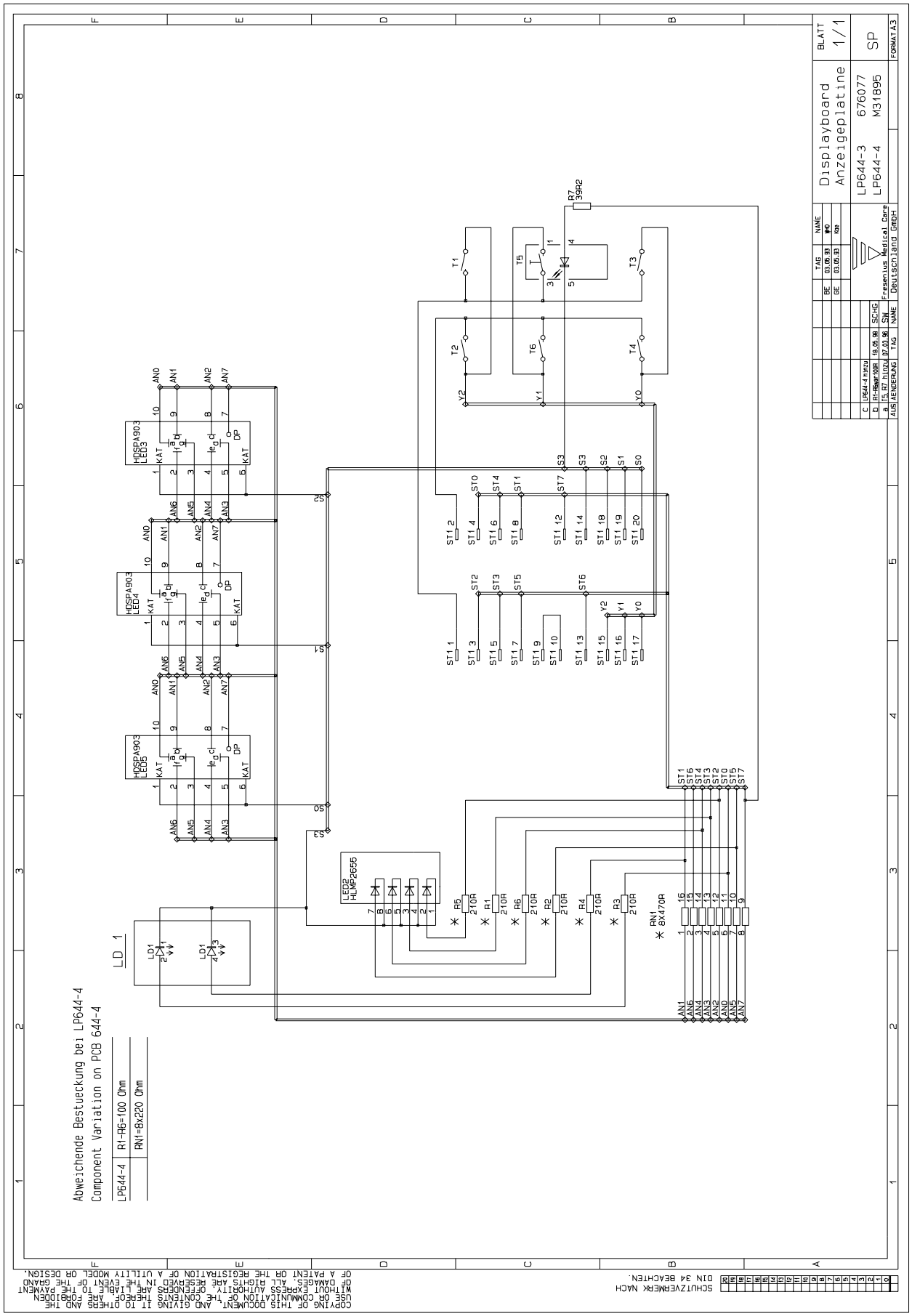
P.C.B. LP 644-4

Component layout diagram



P.C.B. LP 644-4

Circuit diagram



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8.21 P.C.B. LP 645 Position sensor membrane pump

8.21.1 Description

- **General information**

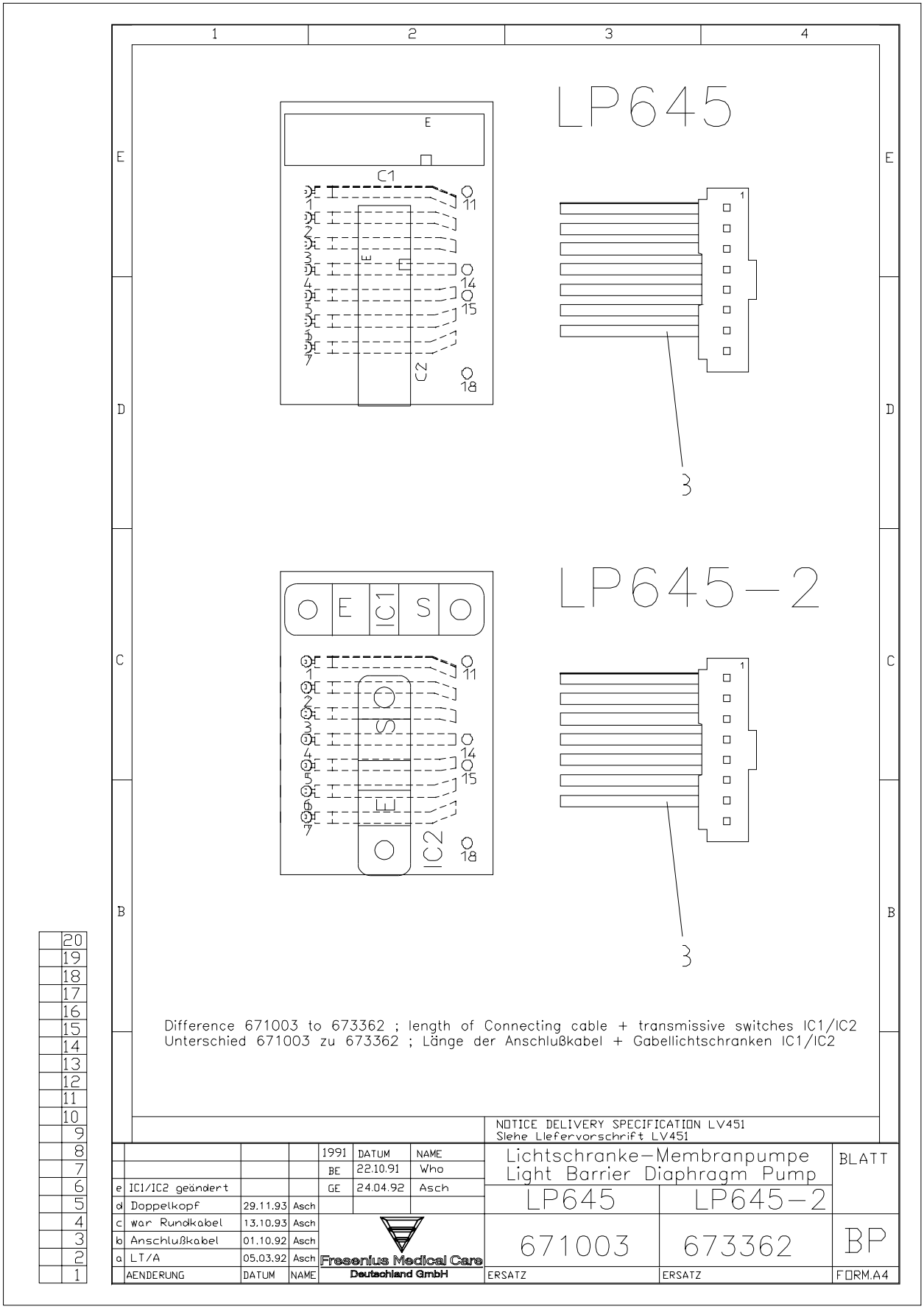
  P.C.B. LP 645 comprises the circuit for the recognition of the end position of the membrane pump.

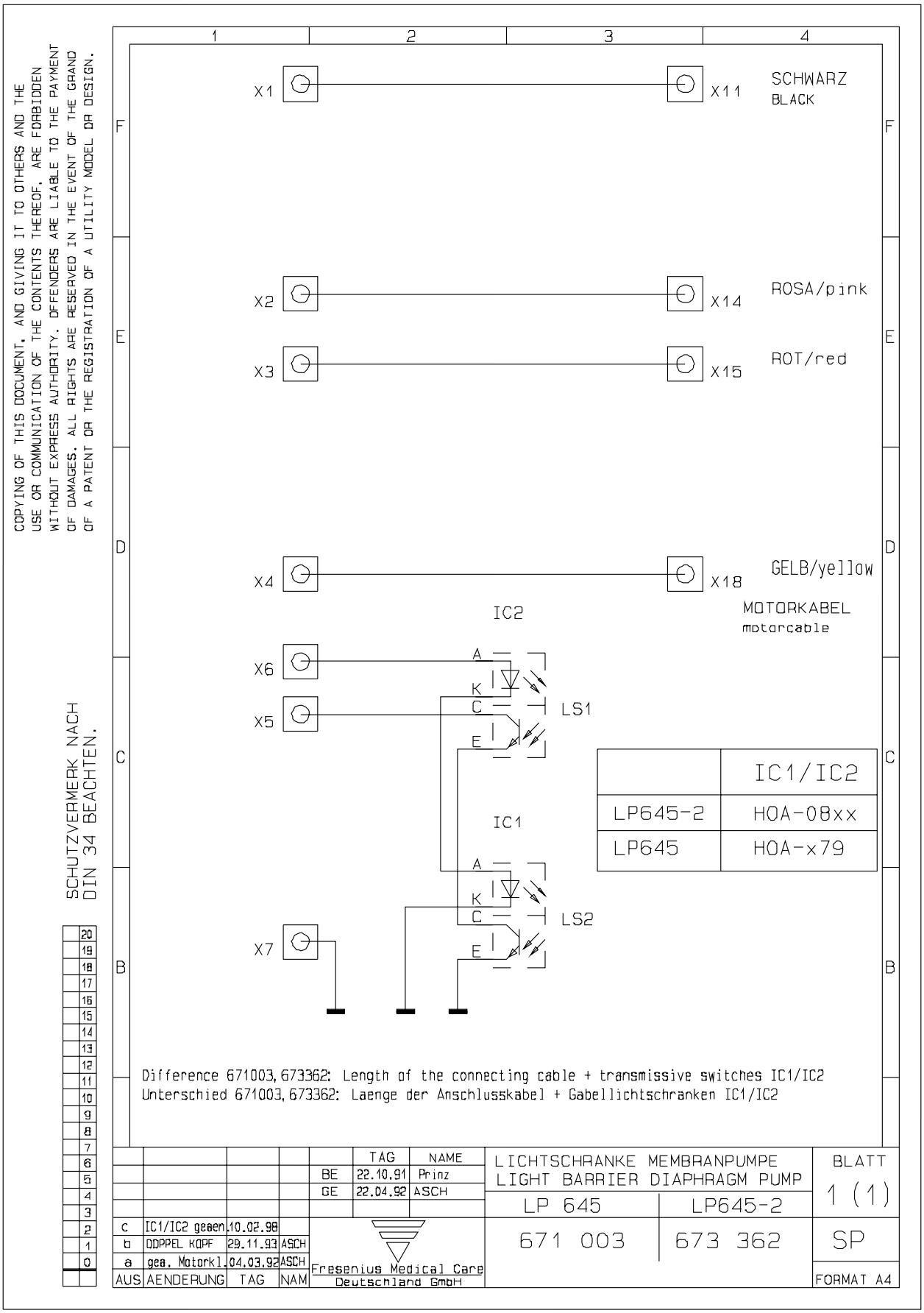
- **Description**

  IC 1 and IC 2 are provided as two optical sensors. They are configurated such that their outputs are linked by an AND operation. Their joint output is X5.
  X5 turns to L level when the membrane of the pump is located in the front dead point (initialization point).

P.C.B. LP 645

Component layout diagram




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SCHUTZVERMERK NACH DIN 34 BEACHTEN.
X1
X11
SCHWARZ
BLACK
X2
X14
ROSA/pink
X3
X15
ROT/red
X4
X18
GELB/yellow
MOTORKABEL
motorcable
IC2
X6
X5
A
K
C
E
LS1
IC1
X7
LS2
IC1/IC2
LP645-2
HOA-08xx
LP645
HOA-x79
Difference 671003, 673362: Length of the connecting cable + transmissive switches IC1/IC2
Unterschied 671003, 673362: Laenge der Anschlusskabel + Gabellichtschranken IC1/IC2
TAG
NAME
BE 22.10.91 Prinz
GE 22.04.92 ASCH
c IC1/IC2 geaen 10.02.98
b DOPPEL KOPF 29.11.93 ASCH
a gea. Motorkl. 04.03.92 ASCH
AUS AENDERUNG TAG NAM
Fresenius Medical Care Deutschland GmbH
LICHTSCHRANKE MEMBRANPUMPE
LIGHT BARRIER DIAPHRAGM PUMP
LP 645
LP645-2
671 003
673 362
BLATT
1 (1)
SP
FORMAT A4

8.22 P.C.B. LP 647 Power logic A (4008 B/S)

8.22.1 Description

- **General notes**

  In order to save space, power logic has been divided in 3 components in the 4008 B/S:
  – P.C.B. LP 647 Power logic A
  – P.C.B. LP 743 Power control 2
  – P.C.B. LP 744 Power control 1

- **Standby voltage generation**

  Via the transformer TR2, the rectifier GL1 and the filter capacitors C1, C3, C4, an unstabilized DC voltage of approx. 15 V is supplied to the input of the linear regulator (1/T1). The connected circuits are supplied by the stabilized output voltage of 12 V via the diode D3. In order that the turnon logic can be supplied in case of power failure, a voltage of approx. 10 V is generated from the battery voltage via the regulator IC13 and ORed to the output voltage of T1 via D2.

- **Power turnon logic**

  When the ON/OFF key on the front panel of the machine is pressed, a LOW level is applied to IC4, pin 1, via X647A, pin 7. By means of the interconnected gates of IC4 and IC5, this causes a rising edge at IC8, pin 3. The output of the FF (IC8, pin 1) becomes HIGH, and the power relay RL3 is activated via T8. At the same time, the bistable relay RL4 is energized via the differentiator C28, R3 and the transistor T10. Among others, this causes the contact RL4, pin 4 and pin 5 to close, so that a voltage of approx. 10 V is available at X647A, pin 16, via the linear regulator consisting of T6, R55 and ZD1, for generation of the audible power failure alarm.

  To turn the system off, a high edge is applied to IC8, pin 11, via DI10. This edge is used to store the level at input D (IC8, pin 9) (H level with the system turned on). At the same time, a time function element for the turnoff delay starts to run (R54, C27). After approx. 1 sec, IC4, pin 6, becomes LOW and IC6, pin 3, is supplied with HIGH level. This level can be read by the CPUs via X647A, pin 22, and is provided as turnoff detection. After another approx. 150 ms, the second time function element (R41, C26) has elapsed as well, and the system is turned off via IC8, pin 4. At the same time, the differentiator C29 and R2 and the transistor T11 are used to energize the bistable relay RL4, so that the audible power failure alarm is suppressed (contact 4/5 RL4).

- **Power failure detection and battery connection**

  In order that a power failure can be detected, the secondary voltage of the standby transformer is monitored. The pulsating DC voltage is supplied to the Schmitt trigger IC4, pin 9, via the diodes DI3, DI4 and the voltage divider R6, R7. The rectangular signal (100 Hz) developing at the output (8/IC4) is applied to the reset input of the counter module 4040 (IC9, pin 11) via IC6, pin 4. The pulse width of the rectangular signal has been designed such that only short LOW pulses (approx. 3 ms wide) can develop at IC9, pin 11, i.e. the counter IC9 is inhibited most of the time.

  A 5-V rectangular signal with a frequency of 2 kHz is applied to pin 23 of connector X647A. After the level has been converted to 12 V (R62, R13, R63), this signal is supplied to the clock pulse input of the counter module (IC9, pin 10) via several intermediate gates. The counter is then incremented in the LOW phase of the reset signal (IC9 pin 9). Should this LOW phase take impermissibly long, caused by a decreasing supply voltage or a power failure, the counter module at IC9, pin 3, turns to HIGH level after 8 ms. Via IC6, pin 6, the reset signal for IC9 is inhibited, the clock pulse disconnected from the counter by means of IC6, pin 12, and the state thus stored. The HIGH level now applied to X647A, pin 6, signals power failure to the CPUs. Via IC7, pin 8, the output signal of the counter module (IC9, pin 3) is supplied to the gate IC7, pin 12. There, the signal is linked to the watchdog signal of CPU1 so that a HIGH level is transferred to T9 with an intact watchdog only. T9 then supplies the coil of the battery relay RL5 with current, and the contact 7/8 RL5 is closed. The slide-in modules (24 V EM; X647A, pins 29, 39, 31) and the switching regulators for 5 V and 12 V (26 V UR; X647C A28) are supplied with the battery voltage via the diodes D4 and D6.

  As soon as the HIGH level at IC9, pin 3, is achieved, the rectangular signal obtained from the standby voltage (see above) is applied to the clock pulse input of the counter (IC9, pin 9). After power has returned, this rectangular signal supplies the counter with clock pulses, and the counter continues to decrement. Caused by the lower counter frequency (100 Hz instead of 2 kHz), it will now take 160 ms until IC9, pin 3, becomes LOW again and the battery is again disconnected from the circuit. This delayed turnoff and the fast connection (8 ms) are intended to ensure a smooth transition from normal operation to power failure operation and vice versa. The initial situation (2 kHz clock pulse at IC9, pin 3, and 100 Hz rectangular signal at IC9, pin 11) is restored by means of the LOW level at 3/IC9.

- **Battery test**

  The battery is briefly charged during the T1 test and the resulting battery voltage read in, so that a statement on the state of the battery can be obtained.

  Upon applying HIGH level to pin 10 of X647A, a pulse of approx. 100 ms length is generated via the differentiator C39, R24 and the two gates (IC4, pin 13; IC7, pins 1 and 2). This pulse serves as the activation signal for the transistor T4. The resistor R39 then loads the battery with a current of approx. 3 A.

- **Battery charging circuit**

  Emergency operation of the system is ensured by three series lead batteries, each of 6 V. An incorporated module of type UC2906 (IC10) is used for charging the batteries. The connected 24 V voltage (24V SW) serves as the supply voltage for the charging procedure, which is performed in different stages.

  Up to a battery voltage of 15 V, the minimum charging current is 14 mA (on this charging level, the charging current decreases with increasing battery voltage).

After 15 V has been reached, the system switches over to charging with a constant current. Here, the charging current is approx. 240 mA. This level is maintained as long as necessary for the battery voltage to rise to 22.4 V.

Subsequently, the charging current starts to decrease. As soon as the current falls below 14 mA, IC10 switches to the float state and maintains the battery voltage at approx. 21 V.

If the battery voltage is decreased because power is consumed, the above procedure is restarted.

- **Heater relay control and monitoring**

The heater relay is controlled by CPU 2. Relay RL2 is switched on by an L level at X647A, pin 17. For reasons of safety, the voltage supply for the relay has been looped via the sensor cable (HEAT_REL_24V), so that it is impossible to operate the heater relay with the sensor cable pulled off.

The voltage applied across the coil of the heater relay is monitored by comparator IC 1, pins 2 and 3. Resistors R64 to R67 are laid out such that the condition of the relay is recognized as being cut off (IC 1, pin 1, at L level) only with a voltage lower than 3 V being applied across RL2. The output signal of the comparator can be read in by CPU 2 via X647A, pin 12. In addition, this signal also acts upon the heater control.

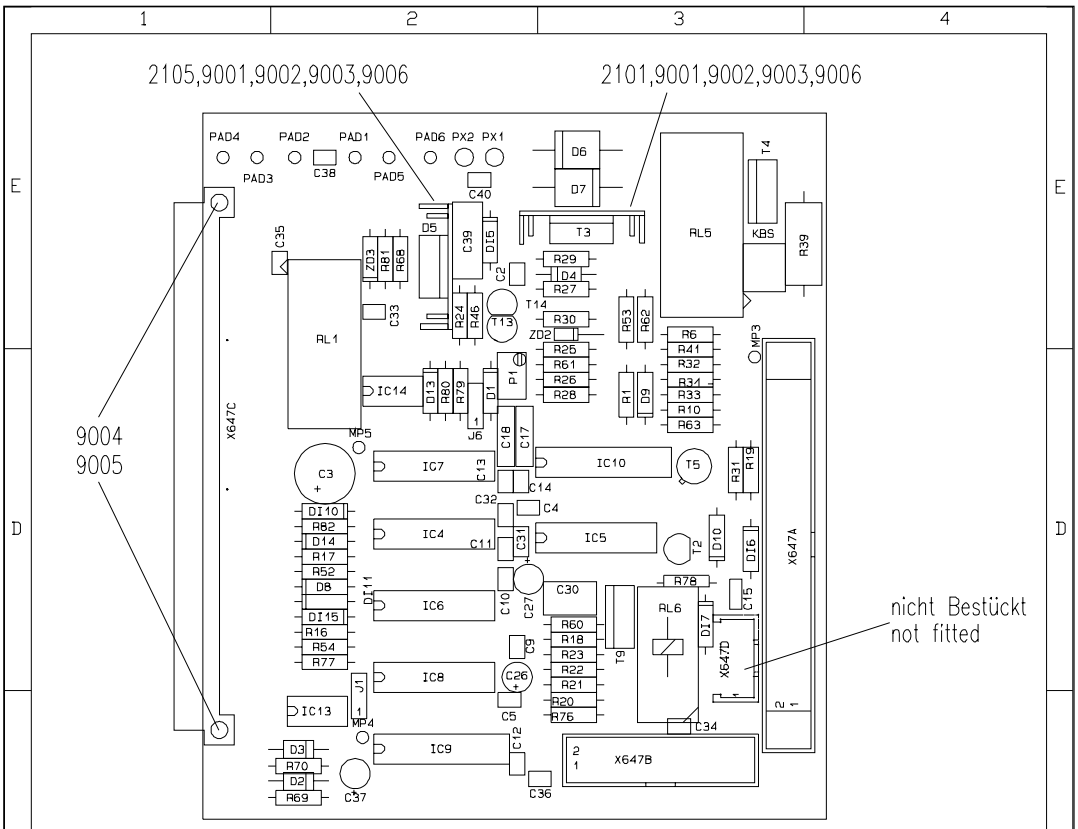
- **Heater control**

The temperature sensor is applied to X647A, pin 2. Together with R33 and R10, it constitutes a voltage divider. The rated-value specification for heater control is superimposed over this voltage divider via X647A, pin 18, and resistors R32 and R34.

The resultant voltage is directed to the operational amplifier IC 2, pin 6 (LM358), which is connected as PID regulator. The regulator is compensated via a voltage, which is applied to X647A, pin 24, and is converted to the requisite range of voltage via IC 1, pin 6. This voltage cannot be influenced by means of the hardware (software compensation!). The output voltage of the PID regulator is superimposed over a delta signal, which is generated by IC 3, and is then delivered to comparator IC 2, pin 3. This comparator activates transistor stages T2 and T12. The L level applied to IC 11, pin 2, causes the triac coupler IL410 to ignite its internal triac, thus delivering the gate trigger voltage for the external main triac (1, 2, 3 X744D) via R51. The external main triac accepts the load current of the heater rod. The triac is fired with each zero crossover of the 220-V supply voltage only. To this end, a zero voltage recognition is integrated in the coupler IC 11. As a result, any disturbances, which are caused by the steep build-up of current under load in case of phase-shift control, are avoided.

The heater control can be inhibited by an H level at X647A, pin 19. As already mentioned above, the heater relay monitoring unit also acts upon the control. Via diodes D9 and D10 as well as the RC element R72, R71, C42, the heater relay IC 1, pin 1, if de-energized (IC 1, pin 1, at H level), causes the heater control to be inhibited. The heater control is released only after the heater relay (IC 1, pin 1, turns to L level) has been activated and C42 has been unloaded via R71. This measure is intended to protect the triac coupler IC 11 and to treat the relay contacts of the heater relay gently.

For hot rinsing, an H level at X647A, pin 20, causes R34 and R10 to be connected in parallel, resulting in a rated temperature value of approx. 84 °C.



Jumper J1: Ausschaltverzögerung / Jumper J1: Switch off delay time

Pos.	Zeit/Time	
2-3	1 sec.	= Standard
1-2	2 sec.	

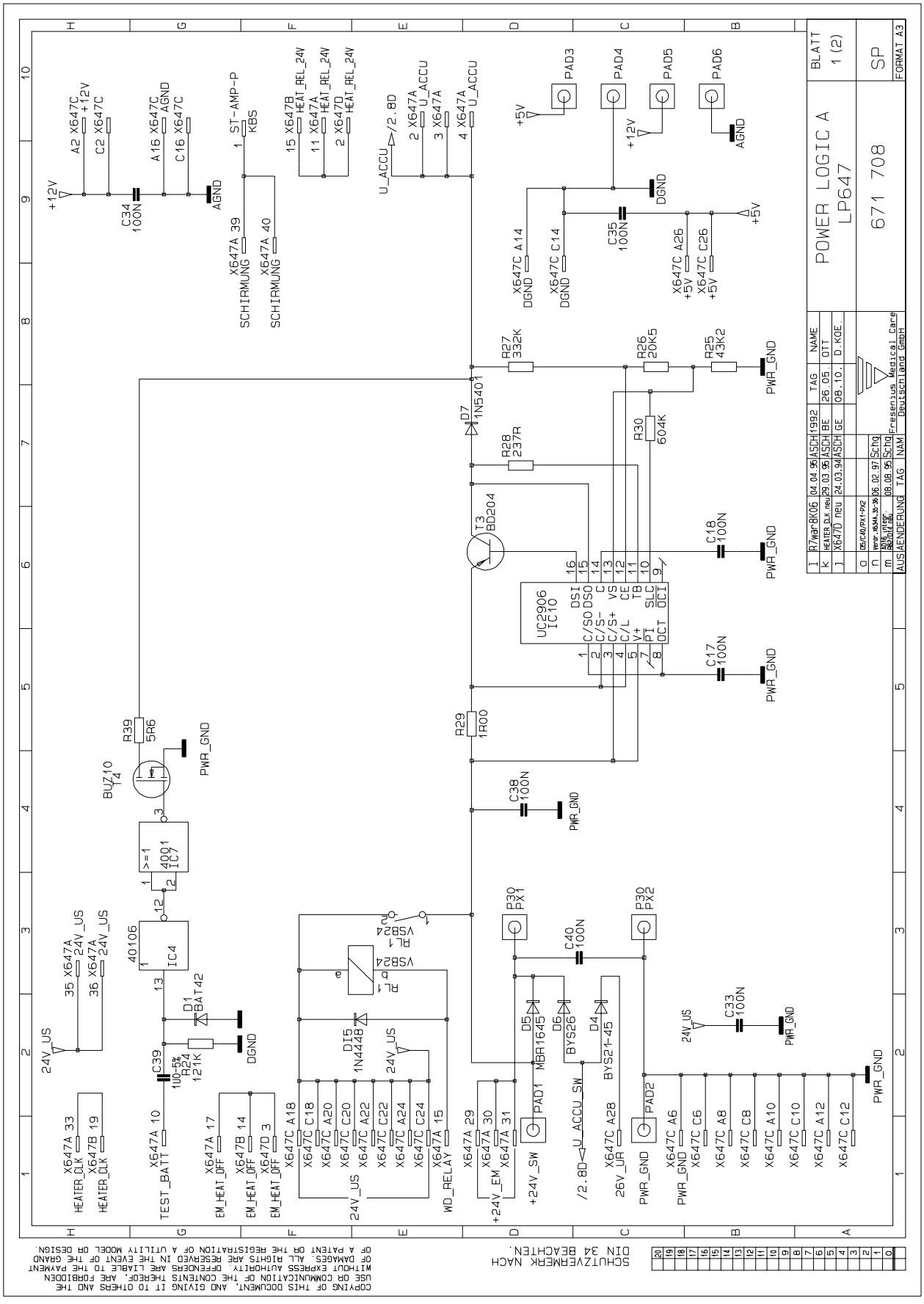
Jumper J6: Unterspannung / Jumper J6: Lower voltage

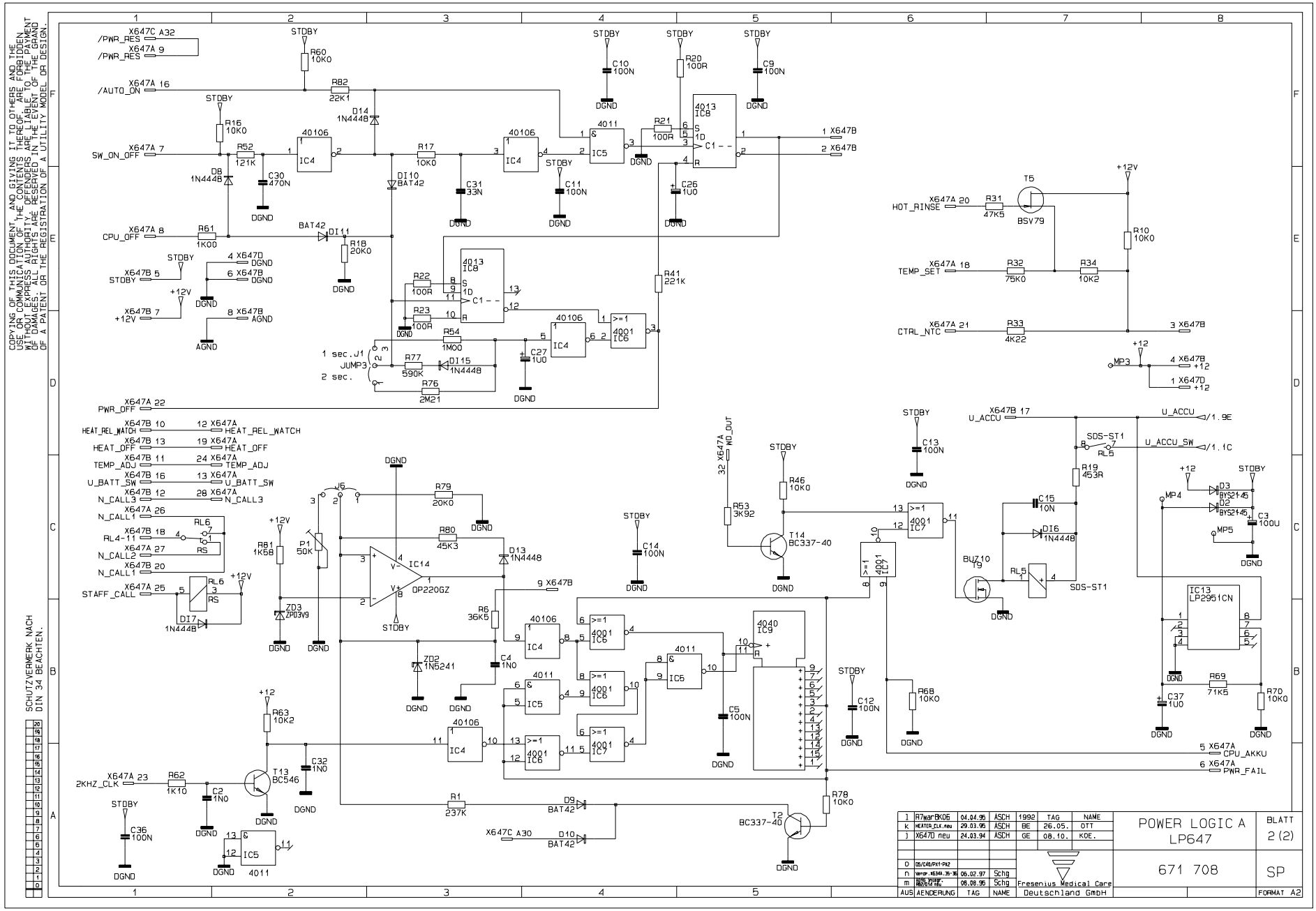
Pos.	Grenze/Limit	
2-3	Variable(P1)	
1-2	Fest/Fixed	= Standard

l	Pos.2105 geändert	28.09.98	Schg		DATUM	NAME	
k	D5/KD5 X647AJ5-36 an 24V_US	26.03.97	Neub	BE	26.05.92	Ott	Power Logic A LP647
j	Nachverdr.X647A.35-36	06.02.97	Schg	GE	26.05.92	D. Köhler	BLATT
i	IC14/P1/J6/D14/RB2	08.08.95	Schg				
m	X647D nicht Bestückt			Fresenius Medical Care Deutschland GmbH			671708 BP
	AENDERUNG	DATUM	NAME		ERSATZ	ERSATZ	FORM.A4

P.C.B. LP 647

Circuit diagram 1/2




POWER LOGIC A
LP647
671 708
SP
BLATT 2 (2)
Fresenius Medical Care Deutschland GmbH
FORMAT A2
/PWR_RES
/AUTO_ON
SW_ON_OFF
CPU_OFF
PWR_OFF
HOT_RINSE
TEMP_SET
CTRL_NTC
U_ACCU
U_ACCU_SW
CPU_AKKU
PWR_FAIL
2KHZ_CLK
STAFF_CALL
N_CALL1
N_CALL2
N_CALL3
HEAT_OFF
HEAT_REL_WATCH
TEMP_ADJ
U_BATT_SW
STDBY
+12V
AGND
DGND
IC4
IC5
IC6
IC7
IC8
IC9
IC13
IC14
T2
T5
T9
T13
T14
BSV79
BC546
BC337-40
OP220GZ
LP2951CN
SDS-ST1
BUZ10
JUMP3
1 sec.
2 sec.

P.C.B. LP 647

Circuit diagram 2/2

8.23 P.C.B. LP 649-2 Display board (4008 B/S)

8.23.1 Description

4008 B/S hemodialysis machine – monitor

- **Address decoder (circuit diagram 1/7)**

  IC 1 and IC 2 are provided as 4-in-16-bit address decoders. The 22 chip selects are generated by means of addresses X635/12 – 15 for IC 1 as well as X635/16 inverted for IC 2. The 22 chip selects are generated only if, via IC 9, either X635/24 (write) or X635/25 (read) is at L level. IC 71 is used as data bus driver and is reversed in its direction by WR and RD.

- **Keyboard matrix (circuit diagram 5/7)**

  Pin 1 and Pin 2 of connector X2 are directly connected to the circuits in the power supply unit. These circuits are used to switch the machine on and off. Via IC 54, the remaining keys on the touch sensitive keyboard are connected to a matrix at the keyboard encoder IC 53 and an additional data line DB4. After the CPU has received the interrupt, the keyboard code is read out by means of CS20 via buffer IC 73.

- **Brightness control (circuit diagram 1+2/7)**

  The clock pulses required for the bar graph displays are delivered by IC 74 and the following gate logic. Superimposed by a higher frequency and a variable mark-to-space ratio, these clock pulses are used to modulate the brightness. To this end, the shift register IC 6 is provided. IC 6 is loaded in parallel with the desired mark-to-space ratio via a data word at DB0 to DB7 (CS21) and is cyclically rotating due to clock generator IC 7/11, 12, 13, and IC 7/8, 9, 10.

- **Status indicator LEDs (circuit diagram 6/7)**

  All status indicators as well as the condition indicator are switched by means of IC 21 and IC 25 (CS16, CS17).
  Both modules are
  – connected to the brightness control by R1,
  – not connected to the brightness control by R2 (maximum brightness).
  The I/O LED (power on/off) is directly connected to the 5-V supply network.

- **Alarm indicator LEDs (circuit diagram 7/7)**

  All alarm indicators are energized by IC 23 and CS18. The key click is controlled by output 19 of IC 23 via IC 46/5.14.

- **Text display (circuit diagram 1/7)**

  The text display is an intelligent, 20-character display module. It is provided with its own processor. Texts are loaded by means of ASCII characters via the data bus CS22. By means of CS20 and via IC 73 and the data bus, the CPU can read whether the text display is ready to receive new data.

- **UF displays (circuit diagram 2/7)**

  IC 3 and IC 4 are 8-digit seven-segment display drivers. IC 3 operates the UF Volume and UF Rate displays. IC 4 operates the UF Goal and UF Time Left displays. These display drivers are loaded in series with 16-bit packets upon each rising clock edge. This information is buffered into the digit registers by means of the IC 5/16 and IC 5/18 load signals. Included in this information is the brightness control.

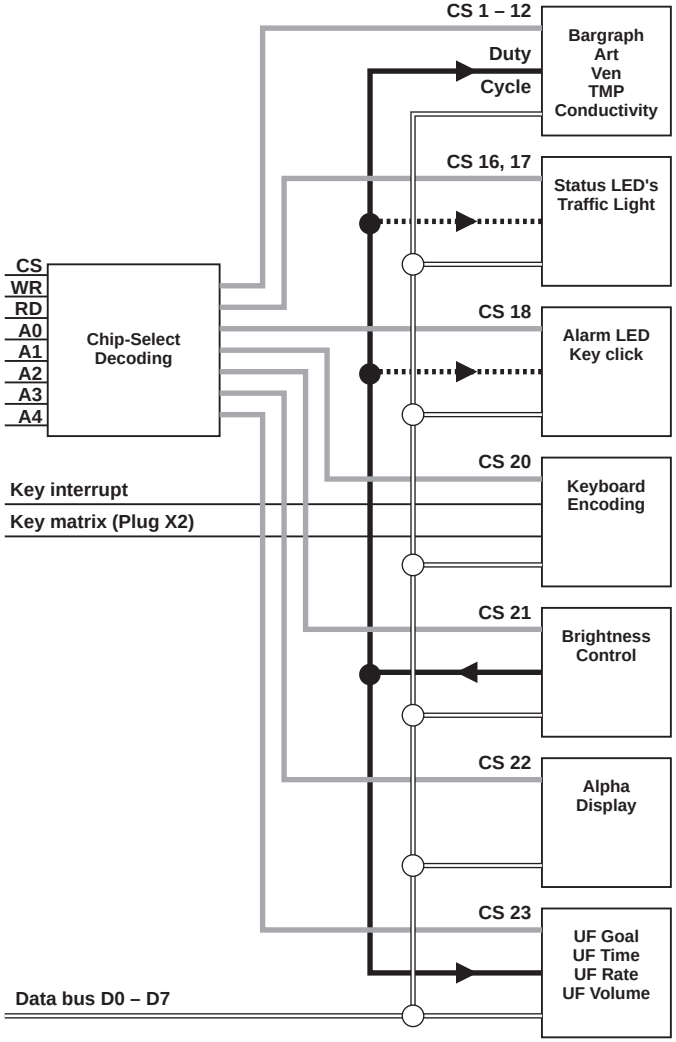
- **Bar graph displays for arterial and venous pressure (circuit diagram 3/7)**

  IC 19, IC 16, and IC 22 are provided as data memories for the arterial bar graph display, whose 30 LEDs are organized within a 5x6 matrix. The 5 lines are directly energized via the emitter sequence IC 30 by data buses DB0 to DB4 and the 6 columns via the 8-out-of-3 decoder IC 28 by data buses DB5 to DB7. The actual value (CS1) is stored by IC 19; the upper limit (CS2) is stored by IC 16; and the lower limit (CS3) is stored by IC 22. The respectively applicable LED is energized via the clock inputs CL1, CL2, and CL3. The clock pulses of CL1 have a longer duration than those of CL2 and CL3. Thus, the actual value is represented more brightly than the limits.
  IC 18 (CS4), IC 15 (CS5), and IC 20 (CS6) are provided as data memories for the venous bar graph display. The working method of this display is identical with that of the arterial bar graph display.

- **Bar graph displays for TMP and conductivity (circuit diagram 4/7)**

  IC 59 (CS7), IC 55 (CS8), and IC 57 (CS9) are provided as data memories for the TMP bar graph display.
  IC 58 (CS10), IC 60 (CS11), and IC 56 (CS12) are provided as data memories for the CD bar graph display.
  The working method of both displays is identical with that of the arterial bar graph displays.

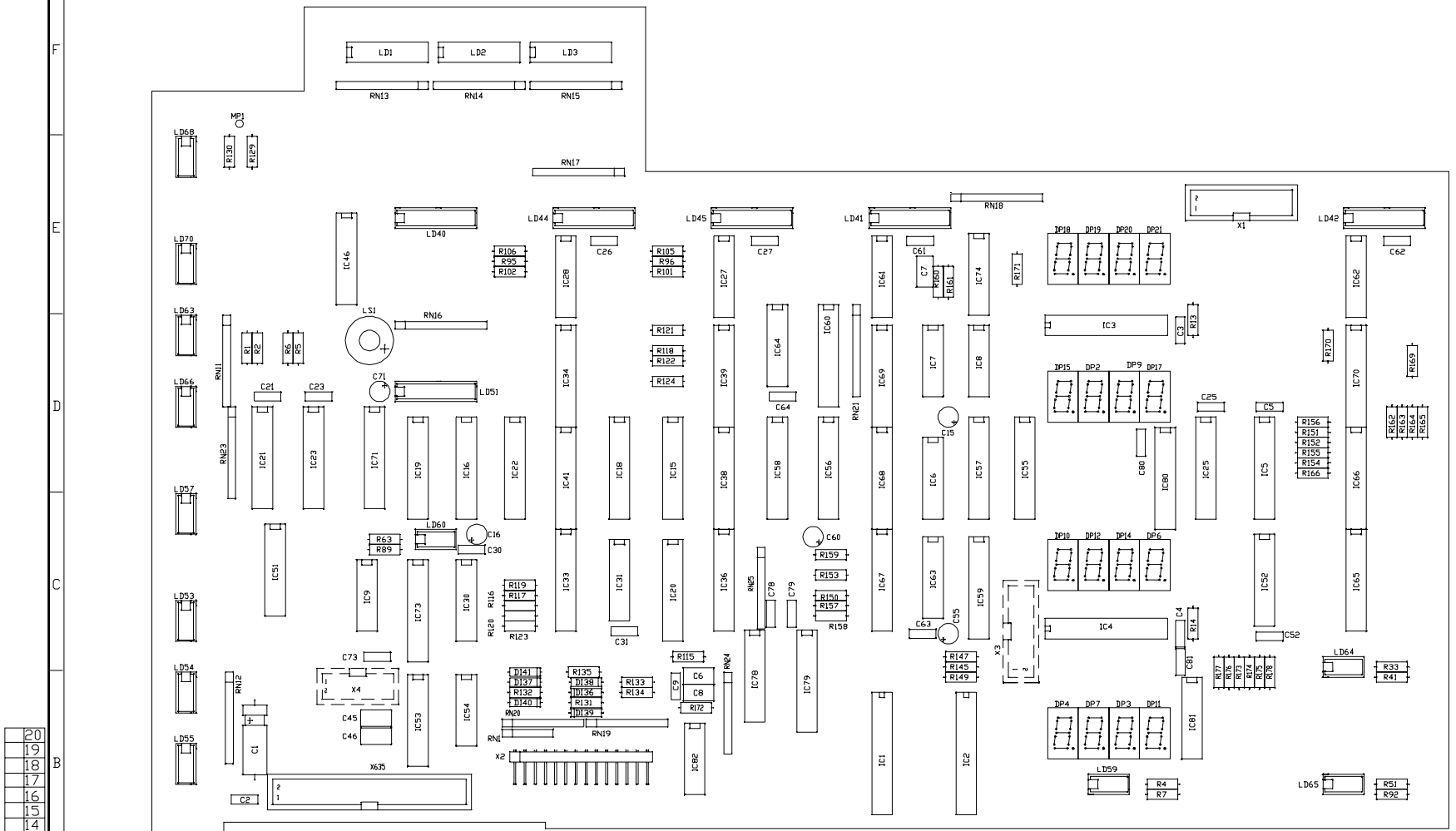
Fig.: Block diagram P.C.B. LP 649-2 Display board



8.23.2 Circuit diagram and component layout diagram
P.C.B. LP 649-2 Display-Board

P.C.B. LP 649-2

Component layout diagram



g	Neues LT-f/Fernbed.	27.10.97	Schl	1992	DATUM	NAME	
f	C75 entfällt	12.08.96	Schg	BE	01.07.92	Ott	
e	Neues LT-E	27.06.96	Schg	GE	12.10.92	Asch	
d	RN1 4x hinzu	03.11.94	Asch				
c	LT-D/LD64	10.10.94	Asch				
h	LT-GC/X4 verschoben						
	ANDERUNG	DATUM	NAME				

Fresenius Medical Care Deutschland GmbH

LP649-2

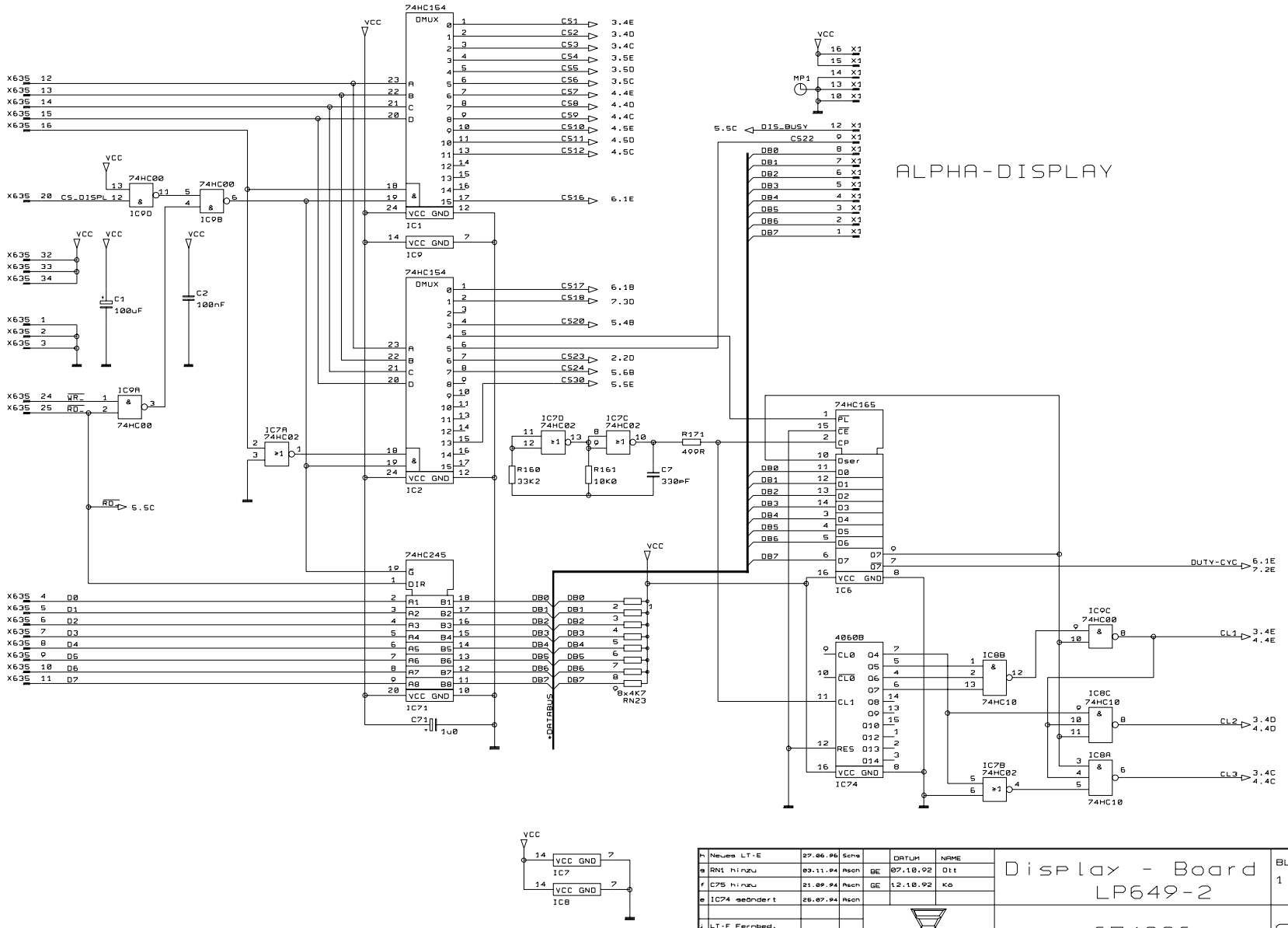
674226

BLATT BP

ERSATZ ERSATZ FORM.A3

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



h	Neues LT-E	27.06.96	Schä		DATUM	NAME
g	RN1 hinzu	03.11.94	Asch	BE	07.10.92	Ott
f	C75 hinzu	21.09.94	Asch	GE	12.10.92	Ka
e	IC74 geändert	26.07.94	Asch			
d	LT-F Fernbed.					
c	C75 entfällt	12.08.96	Schä			
	ÄNDERUNG	DATUM	NAME			

Fresenius Medical Care Deutschland GmbH

Display - Board LP649-2

674226

BLATT 1 /7

SP

ERSATZ ERSATZ

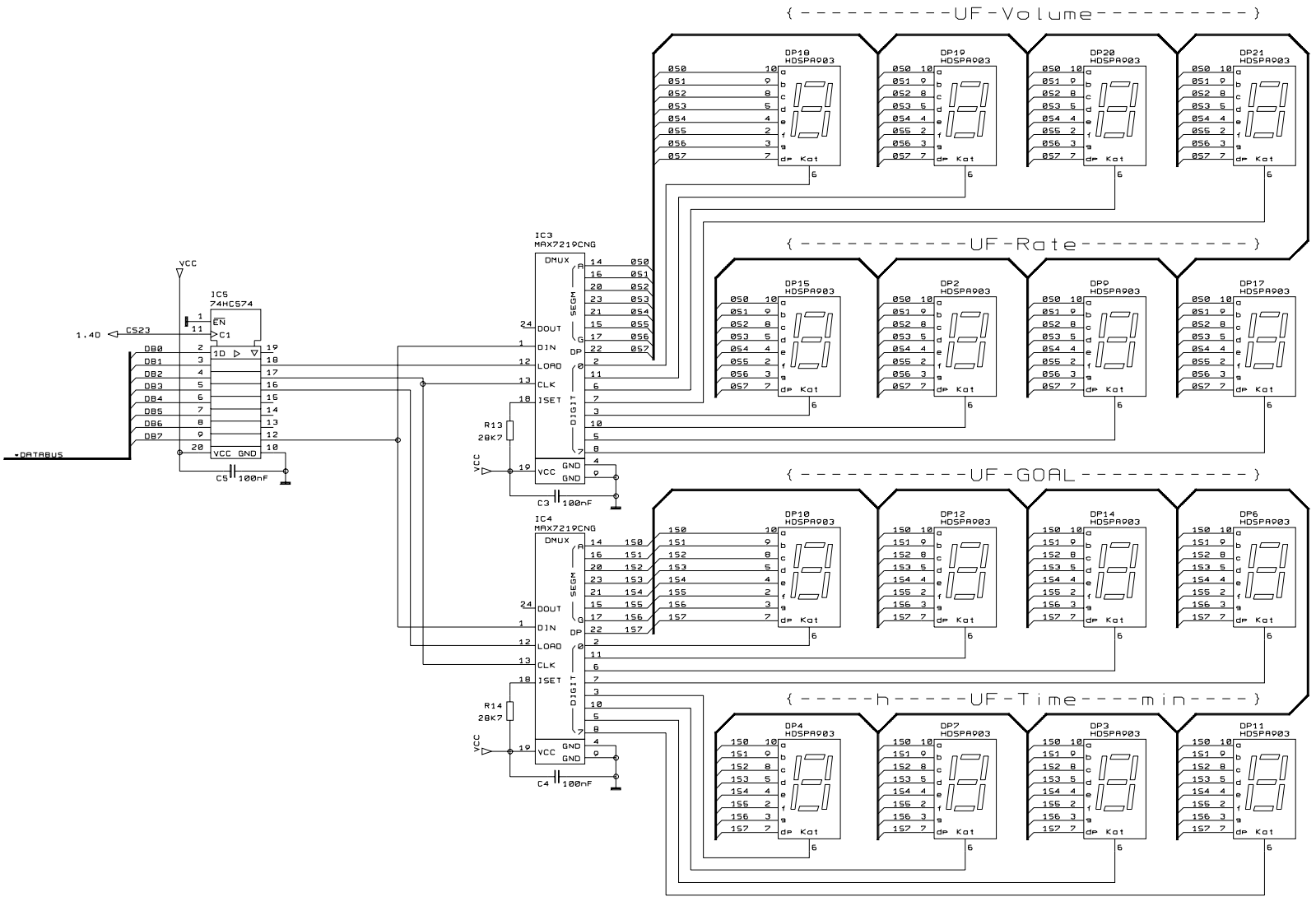
FORM.A3

P.C.B. LP 649-2

Circuit diagram 1/7

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN.



h	Neues LT-E		DATUM	NAME	Display - Board LP649-2
g	RN1 hinzu	BE	07.10.92	Ott	
f	C75 hinzu	GE	12.10.92	Ke	
e	IC74 geändert				
d	LT-F Fernbed.	Fresenius Medical Care Deutschland GmbH			674226
i	C75 entfällt				
	ÄNDERUNG		ERSATZ	ERSATZ	FORM.A3

BLATT 2 17

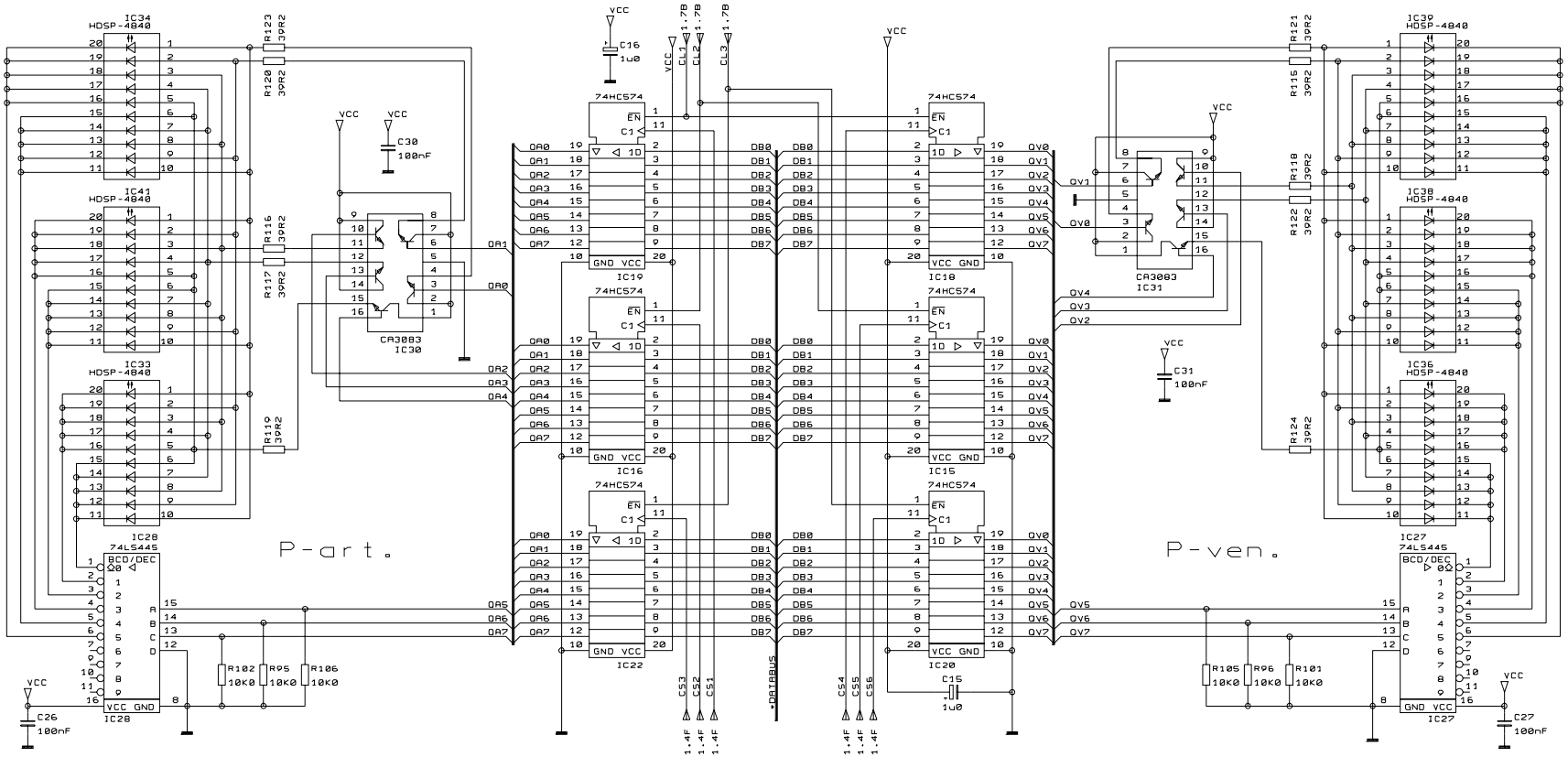
SP

P.C.B. LP 649-2

Circuit diagram 2/7

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

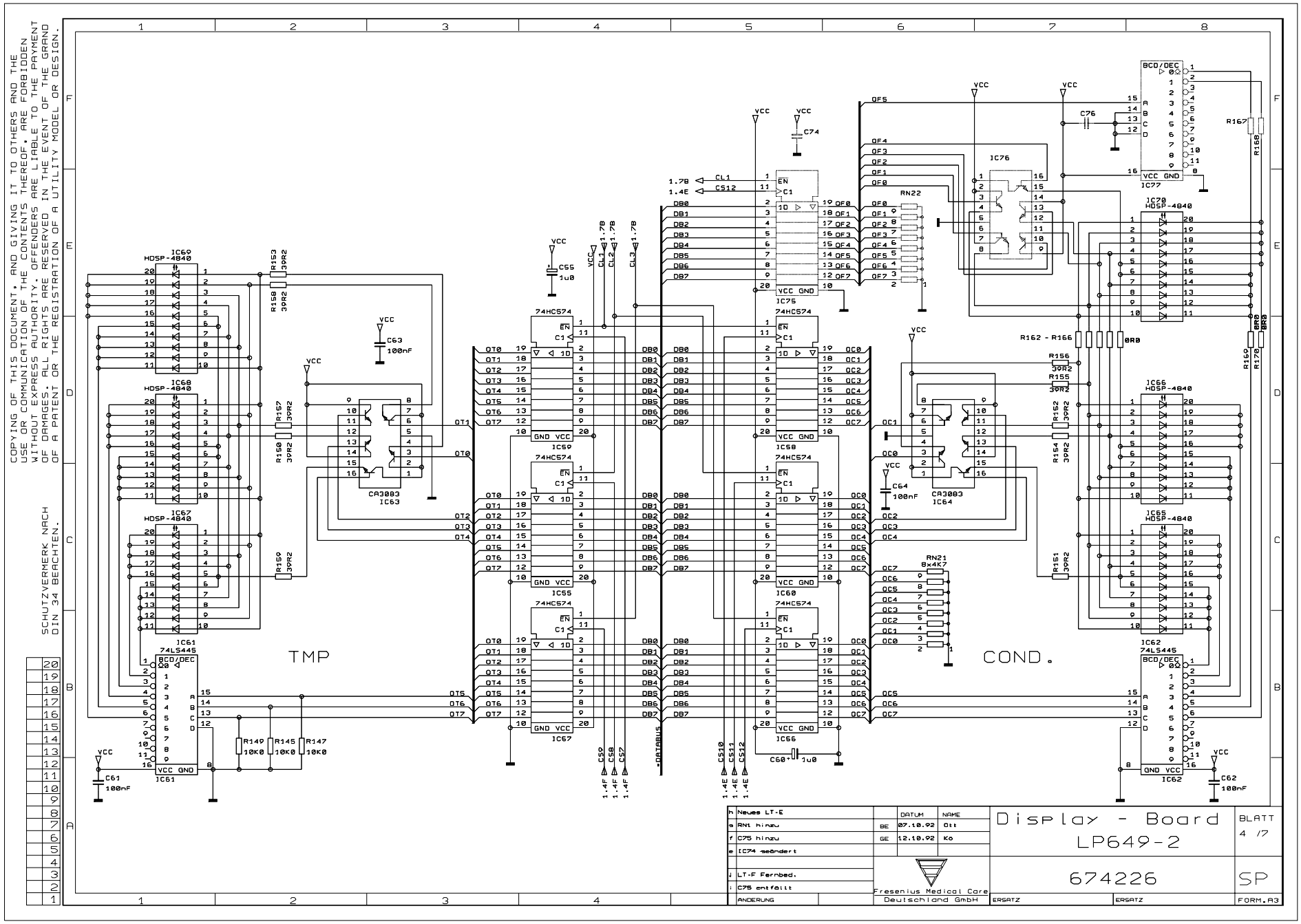


h	Neues LT-E		DATUM	NAME	Display - Board LP649-2 BLATT 3 /7
g	RN1 hinzu	BE	07.10.92	Ott	
f	C75 hinzu	GE	12.10.92	Ke	
e	IC74 geändert				
d	LT-F Fernbed.				674226 SP
c	C75 entfällt				
	ÄNDERUNG				ERSATZ ERSATZ FORM.A3

Fresenius Medical Care Deutschland GmbH

P.C.B. LP 649-2

Circuit diagram 3/7



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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

		DATUM	NAME
h	Neues LT-E		
g	RN1 hinzu	BE 07.10.92	Ott
f	C75 hinzu	GE 12.10.92	Ko
e	IC74 geändert		
d			
c	LT-F Fernbed.		
b	C75 entfällt		
a	ÄNDERUNG		

Fresenius Medical Care Deutschland GmbH

Display - Board LP649-2

674226

BLATT 4 / 7

SP

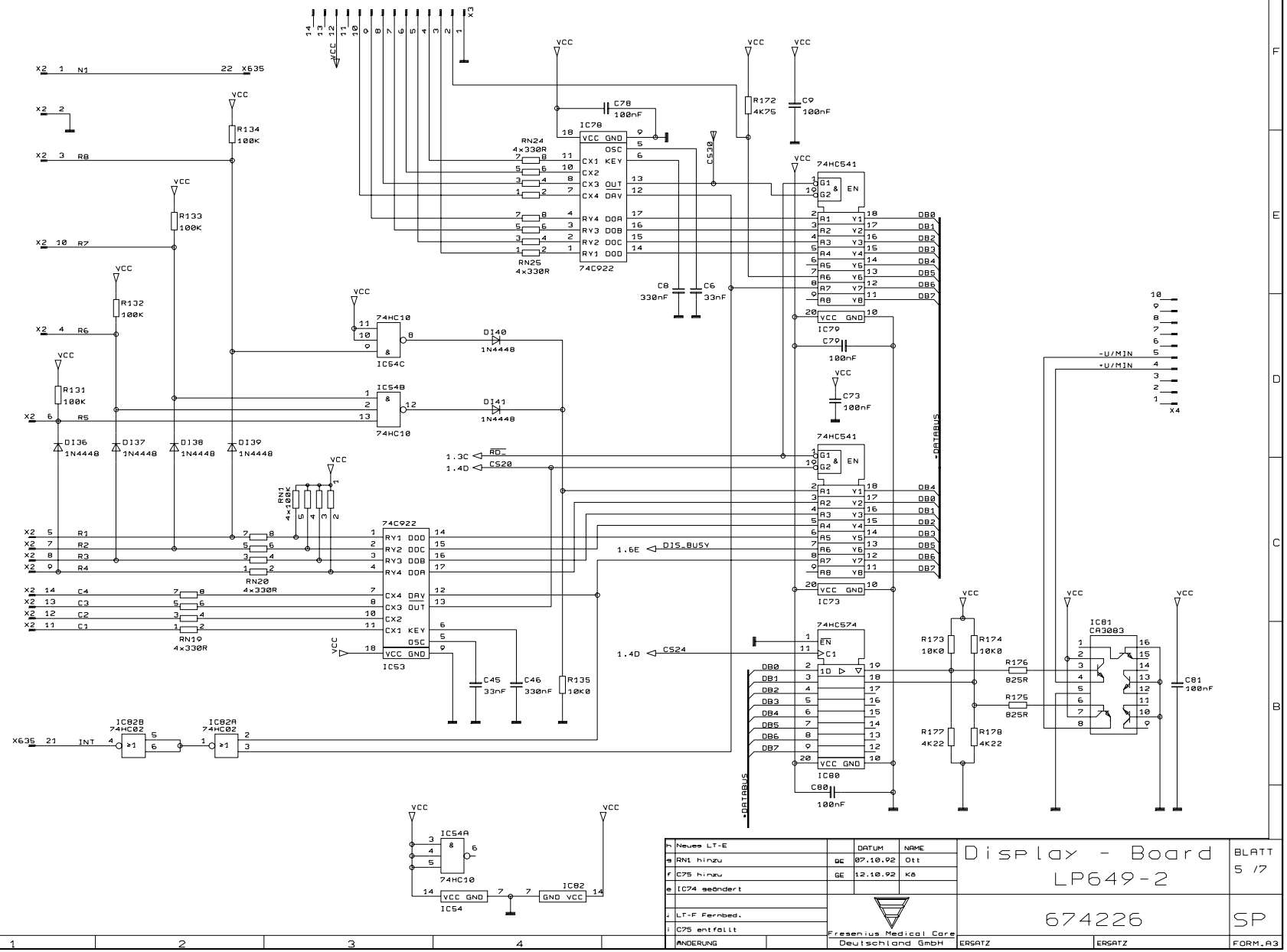
ERSATZ | ERSATZ | FORM. A3

P.C.B. LP 649-2

Circuit diagram 4/7

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN.



h	Neues LT-E		DATUM	NAME
g	RN1 hinzu	DE	07.10.92	Ott
f	C75 hinzu	GE	12.10.92	KS
e	IC74 geändert			
d	LT-F Fernbed.			
c	C75 entfällt			
	ÄNDERUNG			

Fresenius Medical Care
Deutschland GmbH

Display - Board
LP649-2

674226

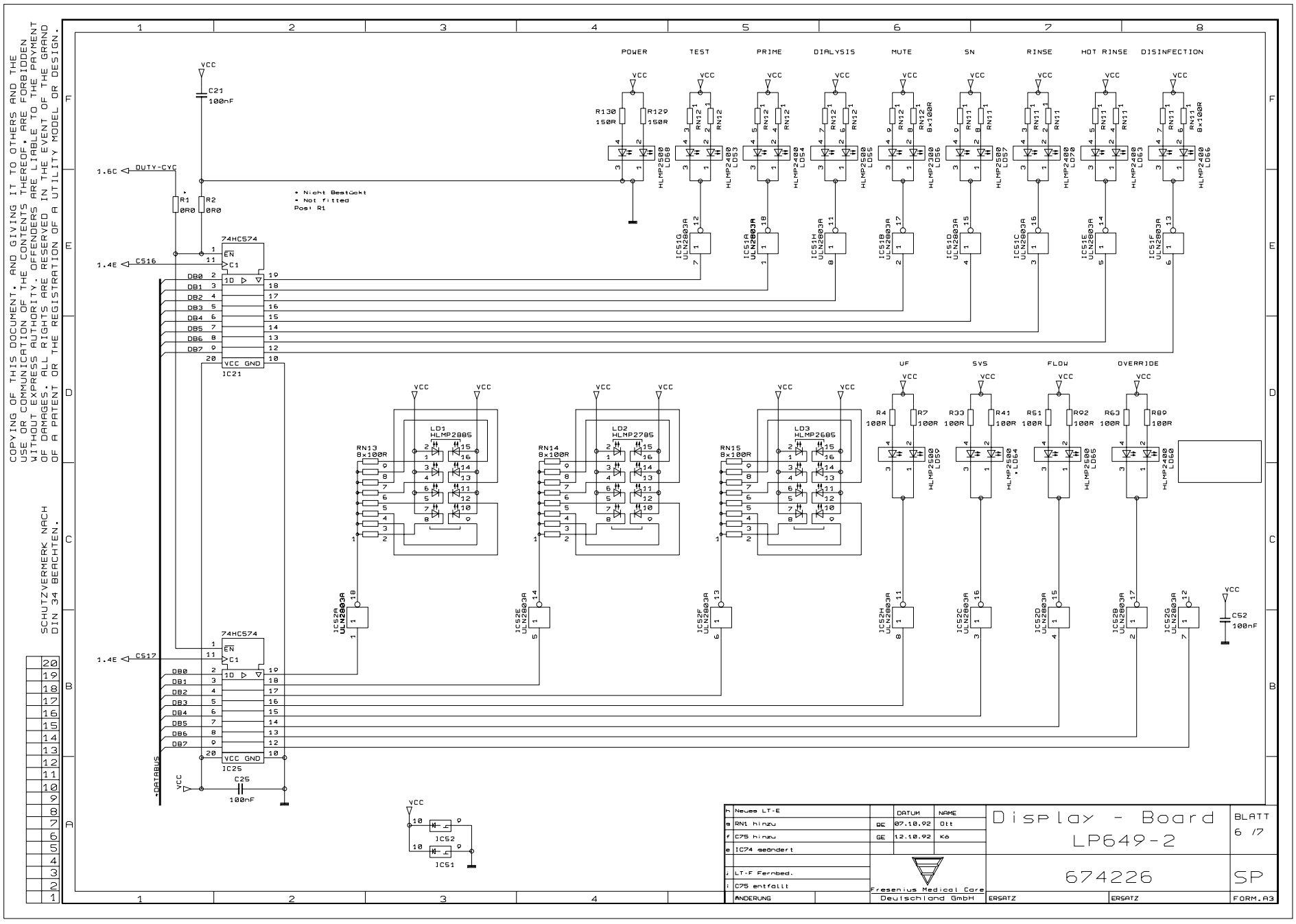
BLATT 5 / 7

SP

ERSATZ | ERSATZ | FORM.A3

P.C.B. LP 649-2

Circuit diagram 5/7

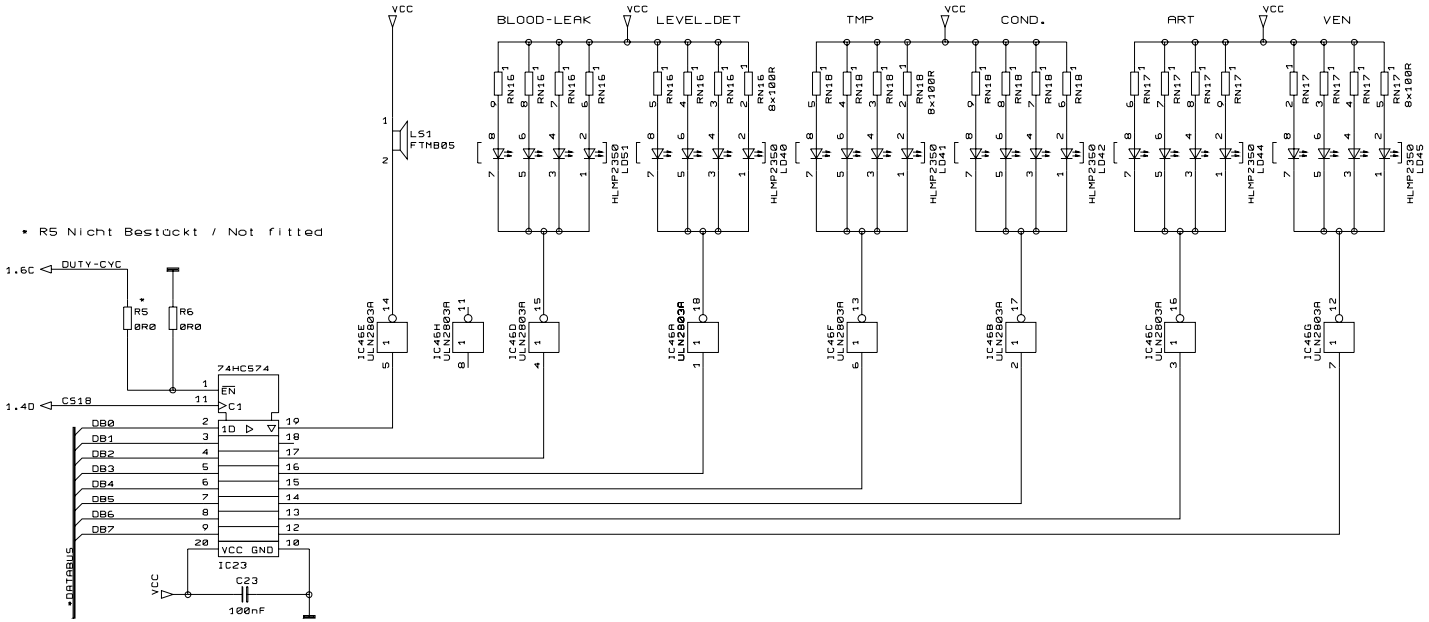

POWER
TEST
PRIME
DIALYSIS
MUTE
SN
RINSE
HOT RINSE
DISINFECTION
UF
SVS
FLOW
OVERRIDE
VCC
C21
100nF
1.6C
DUTY-CYC
R1
0R0
R2
0R0
• Nicht Bestückt
• Not fitted
Post R1
74HC574
EN
C1
1D
VCC GND
IC21
1.4E
CS16
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
R130
150R
R129
150R
RN11
RN12
HLMP2500
LD68
HLMP2400
LD53
LD54
LD55
LD56
LD57
LD70
LD63
LD66
IC51G
IC51A
IC51H
IC51B
IC51D
IC51C
IC51E
IC51F
ULN2803A
LD1
HLMP2885
LD2
HLMP2785
LD3
HLMP2685
RN13
8x100R
RN14
8x100R
RN15
8x100R
IC52A
IC52E
IC52F
R4
100R
R7
100R
R33
100R
R41
100R
R51
100R
R92
100R
R63
100R
R89
100R
HLMP2500
LD59
LD64
LD65
HLMP2400
LD60
IC52H
IC52C
IC52D
IC52B
IC52G
IC25
C25
100nF
C52
100nF
IC52
IC51
•DATABUS
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h Neues LT-E
g RN1 hinzu
f C75 hinzu
e IC74 geändert
i LT-F Fernbed.
i C75 entfällt
ÄNDERUNG
DATUM
NAME
BE 07.10.92 Ott
GE 12.10.92 Ke
Fresenius Medical Care Deutschland GmbH
Display - Board
LP649-2
674226
BLATT 6 17
SP
ERSATZ
ERSATZ
FORM.A3

P.C.B. LP 649-2

Circuit diagram 6/7

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN.



f	Neues LT-E		DATUM	NAME	Display - Board LP649-2
a	RN1 hinzu	BE	07.10.92	Ott	BLATT 7 17
f	C75 hinzu	GE	12.10.92	Ka	
e	IC74 geändert				
					674226 SP
	LT-F Fernbed.				
	C75 entfällt	Fresenius Medical Care Deutschland GmbH			
	ANDERUNG		ERSATZ	ERSATZ	FORM.A3

P.C.B. LP 649-2

Circuit diagram 7/7

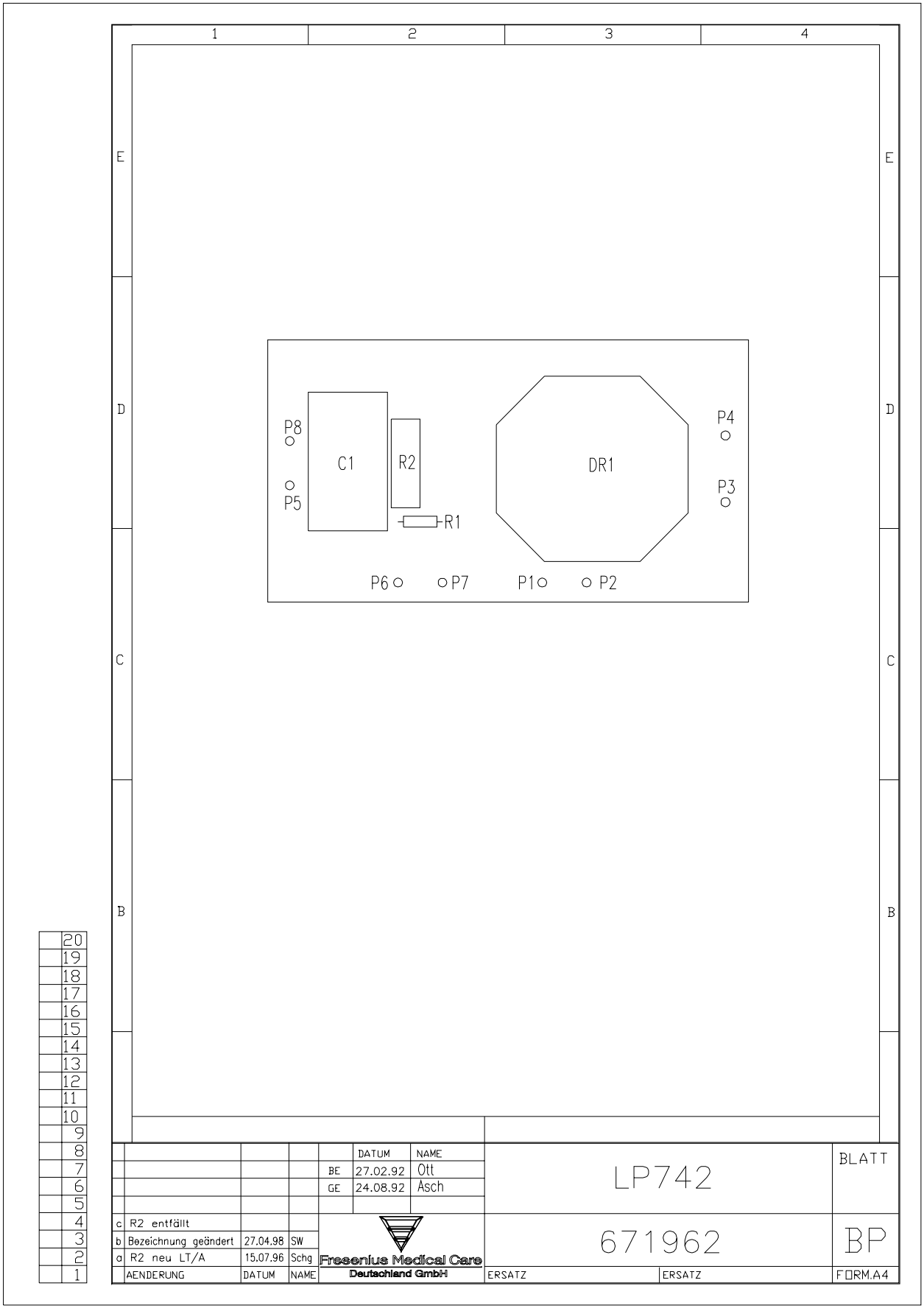
8.24 P.C.B. LP 742 Interference filter

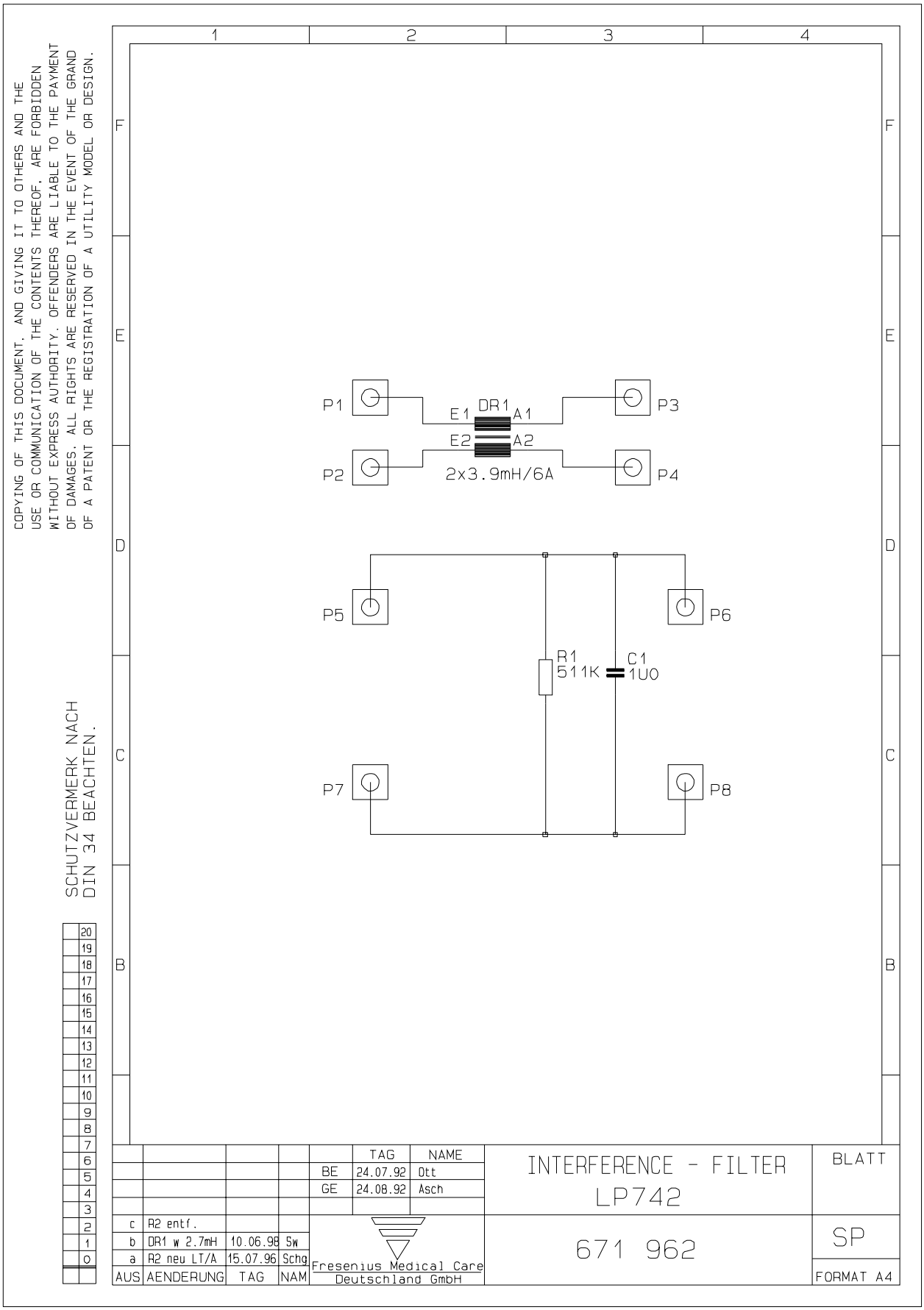
8.24.1 Description

P.C.B. LP 742 is provided for radio interference protection.

P.C.B. LP 742

Component layout diagram




1
2
3
4
A
B
C
D
E
F
P1
P2
P3
P4
P5
P6
P7
P8
E1
E2
DR1
A1
A2
2x3.9mH/6A
R1
511K
C1
1U0
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SCHUTZVERMERK NACH DIN 34 BEACHTEN.
TAG
NAME
BE 24.07.92 Ott
GE 24.08.92 Asch
INTERFERENCE - FILTER
LP742
BLATT
c R2 entf.
b DR1 w 2.7mH 10.06.98 Sw
a R2 neu LT/A 15.07.96 Schg
AUS AENDERUNG TAG NAM
Fresenius Medical Care
Deutschland GmbH
671 962
SP
FORMAT A4

8.25 P.C.B. LP 743 Power control 2 (4008 B/S)

8.25.1 Description

- **General information**

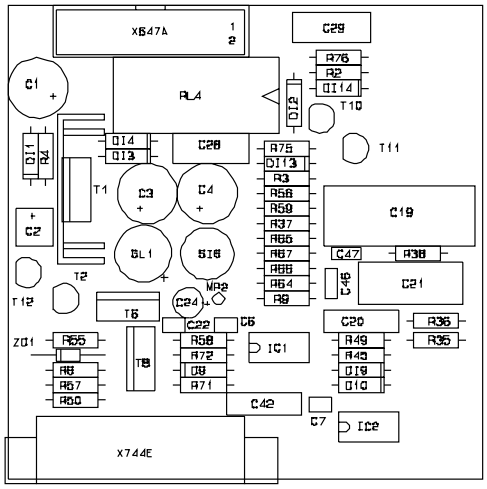
  The power logic in the 4008 B/S comprises 3 components:
  – P.C.B. LP 647 Power logic A
  – P.C.B. LP 743 Power control 2
  – P.C.B. LP 744 Power control 1

- **Function**

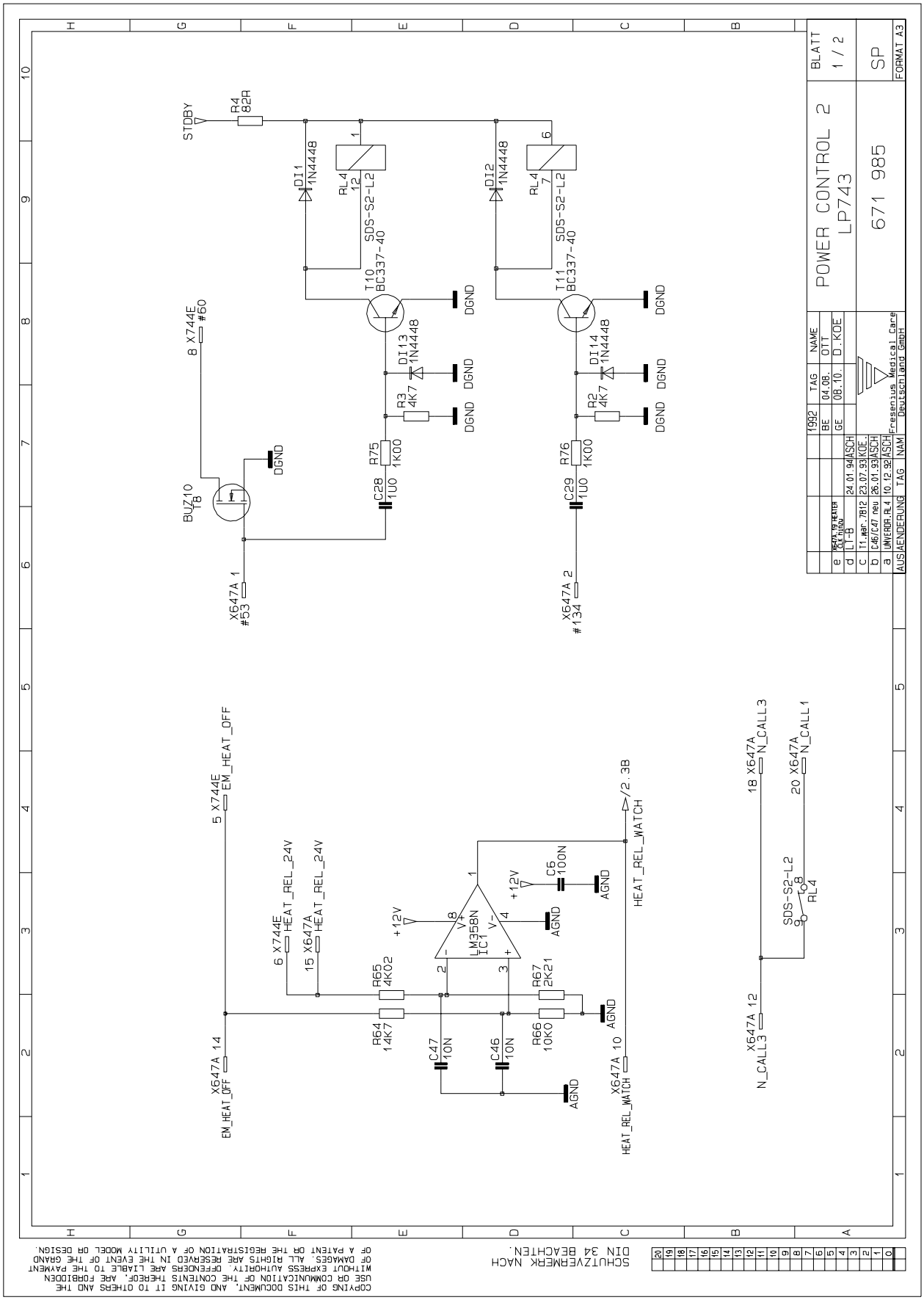
  ➜ P.C.B. LP 647 Power logic A description (chapter 8.22.1)

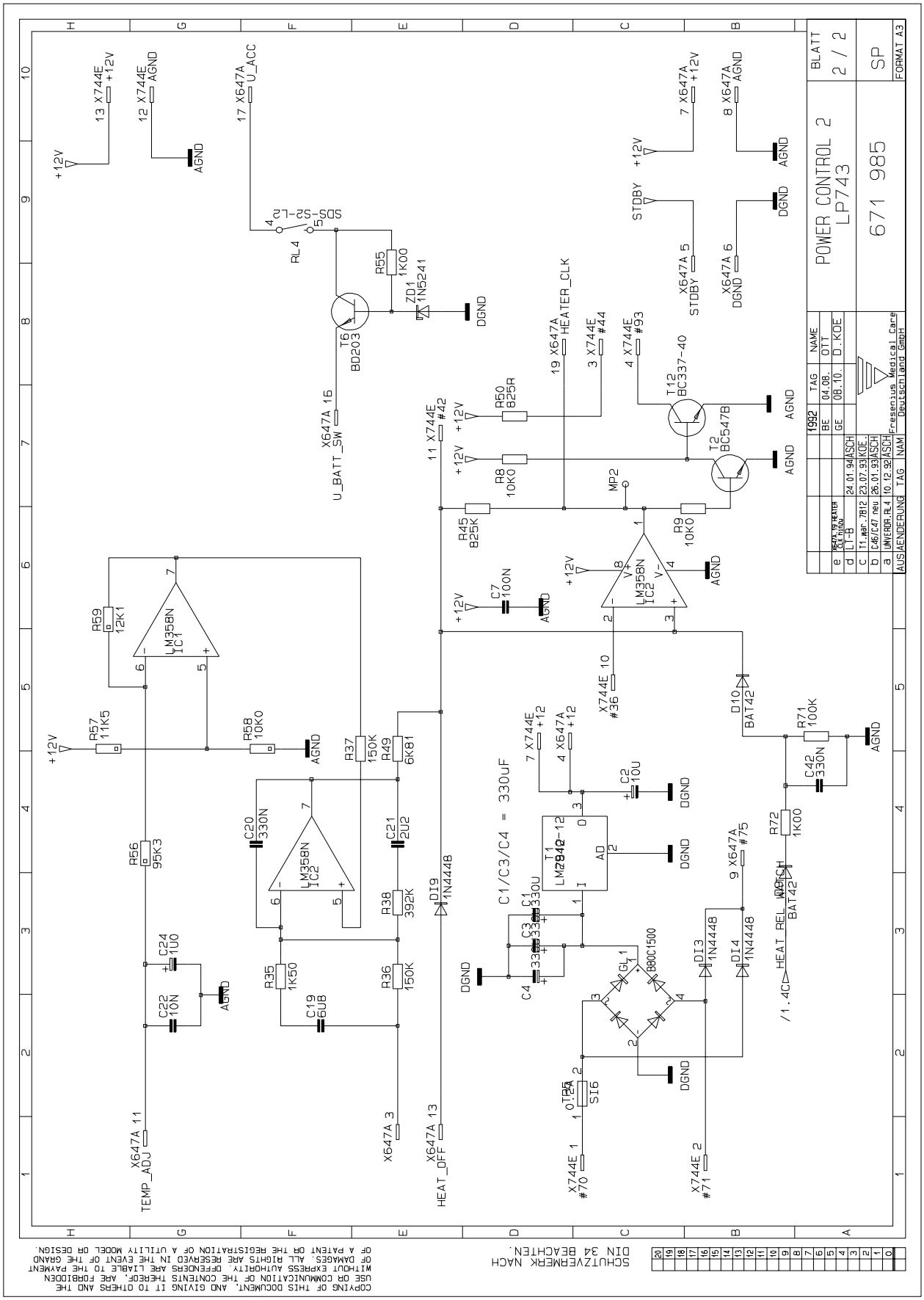
P.C.B. LP 743

Component layout diagram



g	NEU LT-C				DATUM	NAME	POWER CONTROL 2 LP743
f	VERB.X647A.19	29.03.95	ASCH	BE	06.08.92	DIT	
e	NEU LT-B	24.01.94	ASCH	GE	10.08.92	ASCH	
d	T1 war 7812	23.07.93	KOE.				
c	C46/C47 NEU	26.01.93	ASCH	MEDIZIN-TECHN. SYSTEME			671 985 BP
b	NEU LT-D	05.07.95					
AUS	AENDERUNG	TAG	NAME				A4


STDBY
R4 82R
DI1 1N4448
DI2 1N4448
RL4 12
RL4 7
SDS-S2-L2
T10 BC337-40
T11 BC337-40
DI13 1N4448
DI14 1N4448
R3 4K7
R2 4K7
R75 1K00
R76 1K00
C28 1U0
C29 1U0
DGND
8 X744E #60
BUZ10 T8
X647A 1 #53
X647A 2 #134
5 X744E EM_HEAT_OFF
6 X744E HEAT_REL_24V
15 X647A HEAT_REL_24V
+12V
LM358N IC1
C6 100N
AGND
R65 4K02
R67 2K21
R64 14K7
R66 10K0
C47 10N
C46 10N
X647A 14 EM_HEAT_OFF
X647A 10 HEAT_REL_WATCH
/2.3B
18 X647A N_CALL3
20 X647A N_CALL1
X647A 12 N_CALL3
POWER CONTROL 2
LP743
671 985
SP
BLATT 1 / 2
FORMAT A3
Fresenius Medical Care Deutschland GmbH
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POWER CONTROL 2
LP743
671 985
SP
BLATT 2 / 2
FORMAT A3
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8.26 P.C.B. LP 744 Power control 1 (4008 B/S)

8.26.1 Description

- **General information**

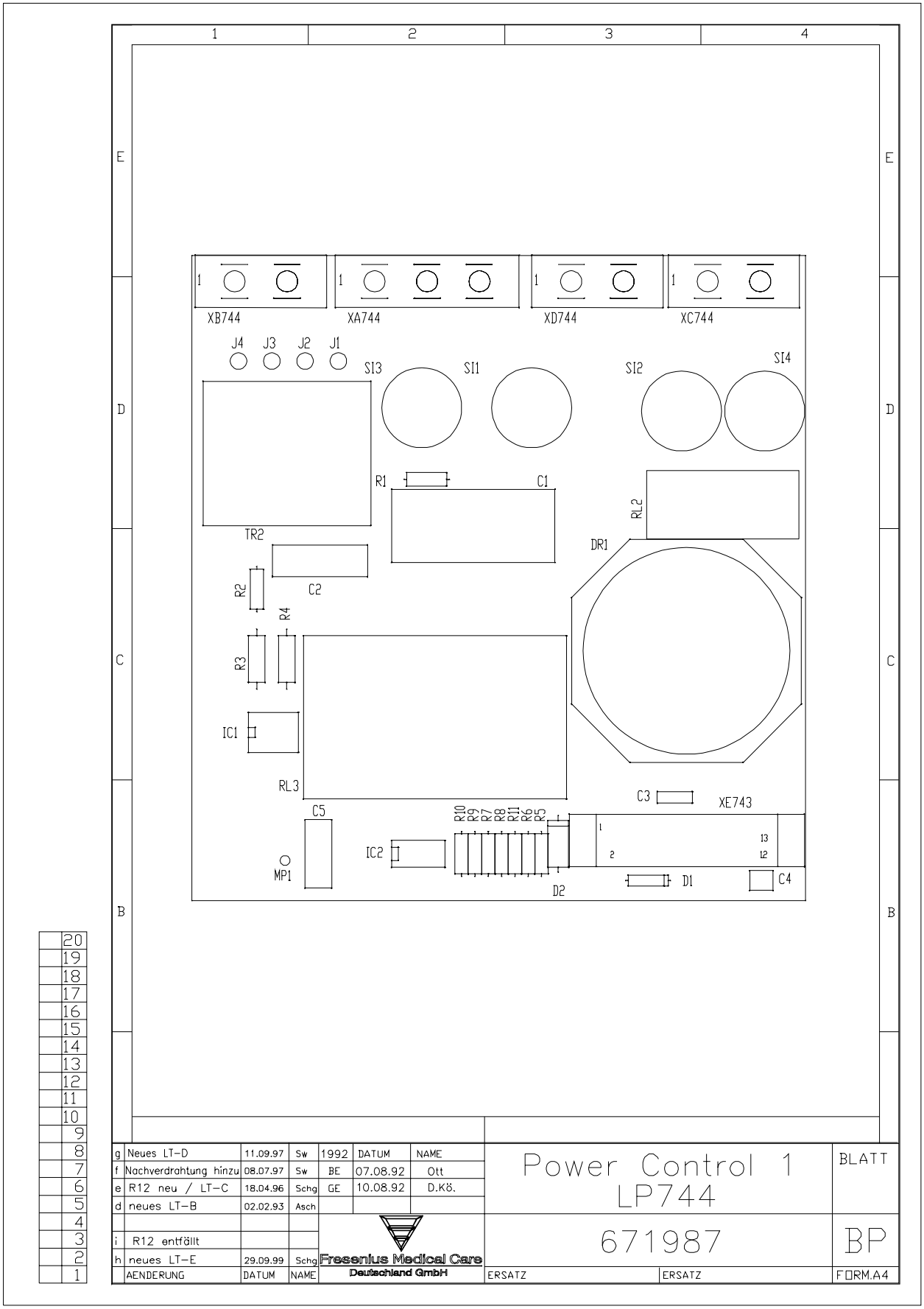
  The power logic in the 4008 B/S comprises 3 components:
  – P.C.B. LP 647 Power logic A
  – P.C.B. LP 743 Power control 2
  – P.C.B. LP 744 Power control 1

- **Function**

  ➜ P.C.B. LP 647 Power logic A description (chapter 8.22.1)

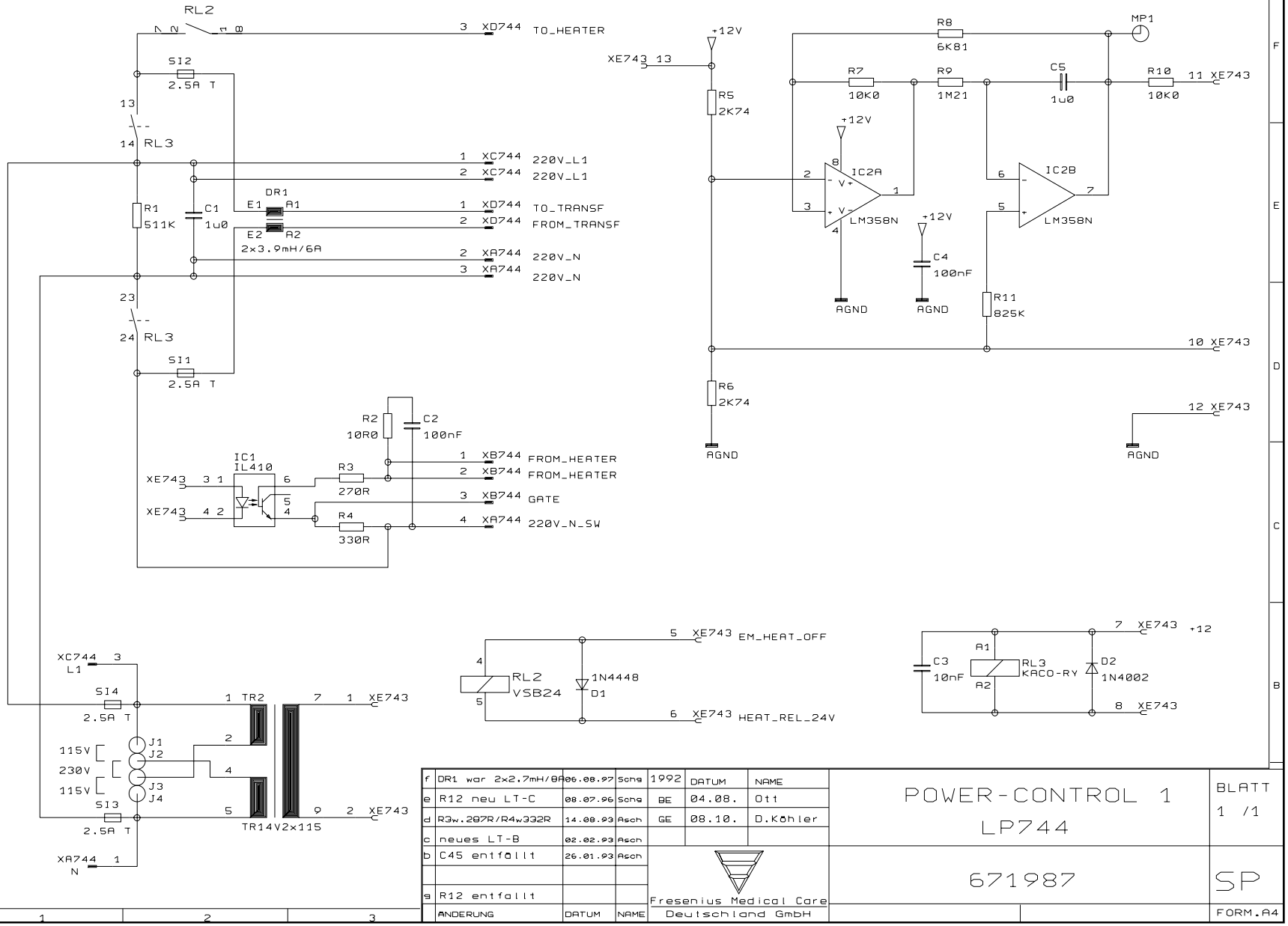
P.C.B. LP 744

Component layout diagram



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f	DR1 war 2x2.7mH/8A	06.08.97	Schg	1992	DATUM	NAME	POWER-CONTROL 1 LP744
e	R12 neu LT-C	08.07.96	Schg	BE	04.08.	Ott	BLATT 1 /1
d	R3w.297R/R4w332R	14.08.93	Asch	GE	08.10.	D.Köhler	
c	neues LT-B	02.02.93	Asch				
b	C45 entfällt	26.01.93	Asch				671987 SP
g	R12 entfallt			Fresenius Medical Care Deutschland GmbH			
	ANDERUNG	DATUM	NAME				FORM.A4

P.C.B. LP 744

Circuit diagram

8.27 P.C.B. LP 747 Distribution board

8.27.1 Description

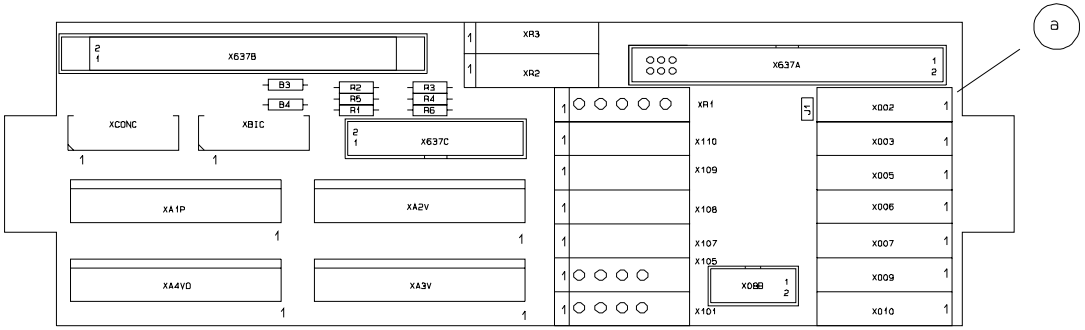
The connectors of the low-voltage consumers provided in the hydraulic unit of the hemodialysis machine (up to 24 V) are plugged onto and supplied with the respective voltage on P.C.B. LP 747.
The connection to the monitor is established by means of three ribbon cables:

– Sensors up to no more than 12 V
– Actuators (pumps) up to no more than 24 V
– Actuators (valves) up to no more than 24 V

8.27.2 Circuit diagram and component layout diagram P.C.B. LP 747 Distribution board

P.C.B. LP 747
Component layout diagram

MONITOR X 637 B (PUMPENSTECKER - PUMP PLUG)
MONITOR X 637 C (VENTILSTEUERUNG - VALVE CONTROL)
MONITOR X 637 A (SENSORENSTECKER - SENSOR PLUG)



XA1P --- UF-PUMPE / UF-PUMP
. LUFTABSCHEIDER / AIR SEPERATOR
. ENTGASUNGSPUMPE / DEGASSING PUMP
. FLUSSPUMPE / FLOW PUMP

XA2V --- VENTILE / VALVE
. V24; V24B; V26; V30
. V41; V43; V84; V91;
. V-DIASAVE

XA3V --- V31 - V38 BILANZKAMMERVENTILE / BALANCING CHAMBER VALVE
. V86 UND V87

XA4V0 --- ZENTRALE KONZENTRATVERSORGUNG / CENTRAL CONCENTRATE SUPPLY
. V99; V100; V102; V104;
. V27; V40; V45;

XR2 --- UEBERWACHUNG V84 / MONITORING V84
X08B --- BLUTLECKDETEKTOR / BLOOD LEAK DETECTOR
X002 --- NTC 2
X003 --- NTC 3
X005 --- SCHWIMMERSCHALTER / FLOAT SWITCH
X006 --- NIVEAUFUEHLER / LEVEL SENSOR
X007 --- LF-ZELLE / CD CELL
X009 --- TMP - DRUCKAUFNEHMER / TMP PRESSURE TRANSDUCER
X010 --- SCHALTKONTAKT SPUELKAMMER / SWITCH CONTACT RINSE CHAMBER
X101 --- DRUCKSCHALTER ZENTR.KONZ.VERS. / PRESSURE SWITCH CENTR.CONC.SUP
X105 --- BIBAG KONTAKT / BIBAG CONTACT
XR3 --- BIBAG DRUCKSENSOR / PRESSURE SENSOR
XR1 --- NTC_4
X110 --- LF - ZELLE 2 / CD CELL 2
X109 :
X108 :
X107 :
J1 : BESTUECKT FUER GERAETE OHNE BICARBONAT / FITTED FOR MACHINES WITHOUT BIC

PIN ROW STIFTLEISTE	KODIERUNG CODE PIN-NR. DER STIFTLEISTE ABZWICKEN
XA1P	12
XA2V	1
XA4V0	11
XA3V	2

					DATUM	NAME
				BE	22.07.92	Ott
e	NTC_4 und LF 2			GE	27.07.92	Asch
d	J1 HINZU	30.09.94	Asch			
c	+12V an X101	27.08.93	Asch			
b	LT-B	26.04.93	Asch			
a	KODIERUNG	23.10.92	Asch			
	ANDERUNG	DATUM	NAME			

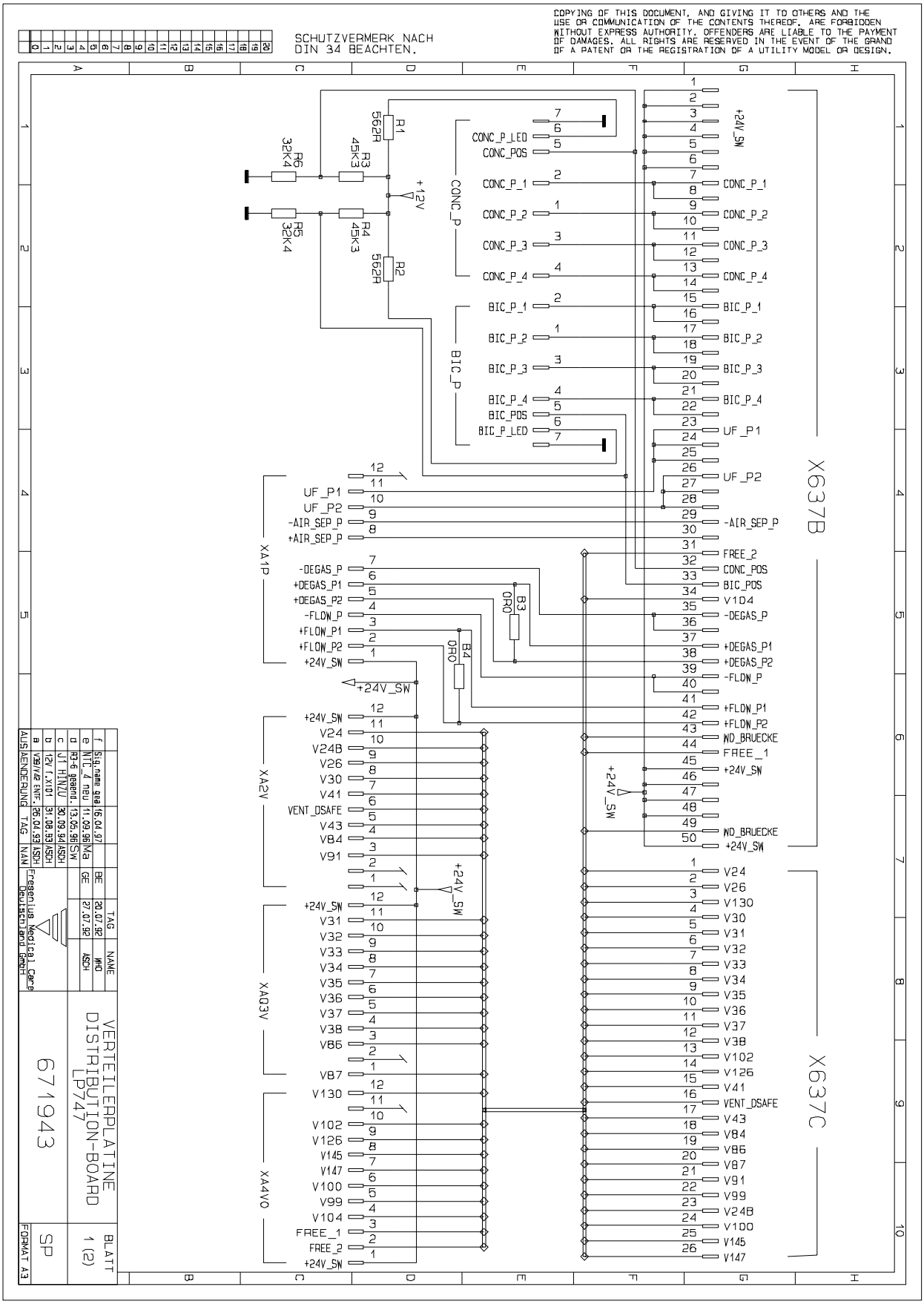
Fresenius Medical Care Deutschland GmbH

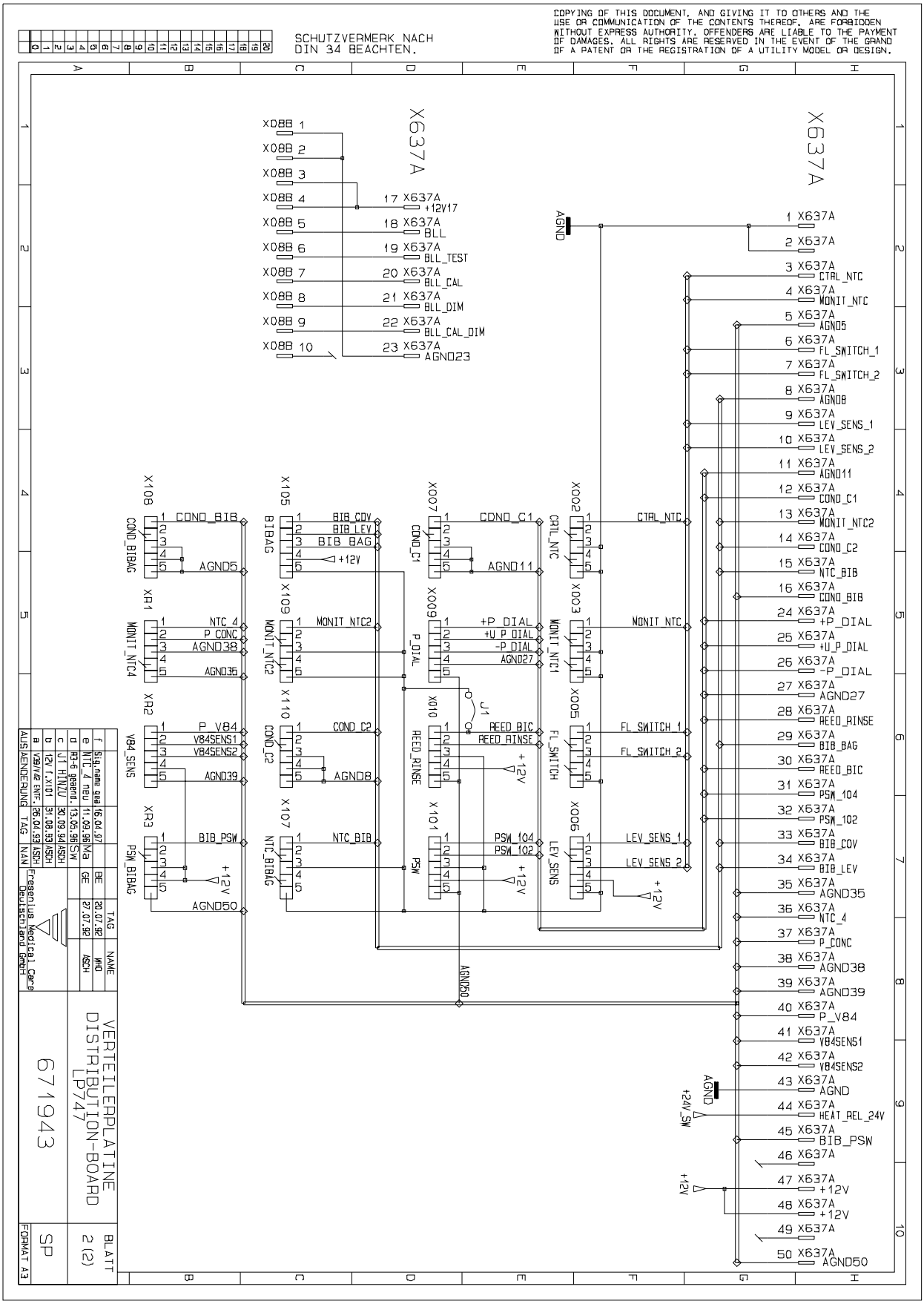
VERTEILER DISTRIBUTION BOARD LP747

671943

BLATT BP

ERSATZ ERSATZ FORM.A3


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X637B
X637C
XA1P
XA2V
XA03V
XA4V0
R1 562R
R2 562R
R3 45K3
R4 45K3
R5 32K4
R6 32K4
B3 0R0
B4 0R0
+12V
+24V_SW
CONC_P
BIC_P
CONC_P_LED
CONC_POS
CONC_P_1
CONC_P_2
CONC_P_3
CONC_P_4
BIC_P_1
BIC_P_2
BIC_P_3
BIC_P_4
BIC_POS
BIC_P_LED
UF_P1
UF_P2
-AIR_SEP_P
+AIR_SEP_P
-DEGAS_P
+DEGAS_P1
+DEGAS_P2
-FLDN_P
+FLDN_P1
+FLDN_P2
WD_BRUECKE
FREE_1
FREE_2
BIC_POS
CONC_POS
V1D4
V24
V24B
V26
V30
V41
VENT_DSAFE
V43
V84
V91
V31
V32
V33
V34
V35
V36
V37
V38
V86
V87
V130
V102
V126
V145
V147
V100
V99
V104
Fresenius Medical Care Deutschland GmbH
VERTEILERPLATINE DISTRIBUTION-BOARD LP747
671943
SP
BLATT 1 (2)
FORMAT A3


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X637A
X08B 1
X08B 2
X08B 3
X08B 4
X08B 5
X08B 6
X08B 7
X08B 8
X08B 9
X08B 10
17 X637A +12V17
18 X637A BLL
19 X637A BLL_TEST
20 X637A BLL_CAL
21 X637A BLL_DIM
22 X637A BLL_CAL_DIM
23 X637A AGND23
AGND
1 X637A
2 X637A
3 X637A CTRL_NTC
4 X637A MONIT_NTC
5 X637A AGND5
6 X637A FL_SWITCH_1
7 X637A FL_SWITCH_2
8 X637A AGND8
9 X637A LEV_SENS_1
10 X637A LEV_SENS_2
11 X637A AGND11
12 X637A COND_C1
13 X637A MONIT_NTC2
14 X637A COND_C2
15 X637A NTC_BIB
16 X637A COND_BIB
24 X637A +P_DIAL
25 X637A +U_P_DIAL
26 X637A -P_DIAL
27 X637A AGND27
28 X637A REED_RINSE
29 X637A BIB_BAG
30 X637A REED_BIC
31 X637A PSW_104
32 X637A PSW_102
33 X637A BIB_COV
34 X637A BIB_LEV
35 X637A AGND35
36 X637A NTC_4
37 X637A P_CONC
38 X637A AGND38
39 X637A AGND39
40 X637A P_V84
41 X637A V84SENS1
42 X637A V84SENS2
43 X637A AGND
44 X637A HEAT_REL_24V
45 X637A BIB_PSW
46 X637A
47 X637A +12V
48 X637A +12V
49 X637A
50 X637A AGND50
+24V_SW
+12V
X109 COND_BIB
X105 BIB_COV BIB_LEV BIB_BAG +12V
X007 COND_C1 AGND11
X002 CTRL_NTC
X003 MONIT_NTC
X005 FL_SWITCH_1 FL_SWITCH_2
X006 LEV_SENS_1 LEV_SENS_2 +12V
XR1 NTC_4 P_CONC AGND38 AGND35
X009 +P_DIAL +U_P_DIAL -P_DIAL AGND27
X010 REED_BIC REED_RINSE +12V
X101 PSW_104 PSW_102 +12V
XR2 P_V84 V84SENS1 V84SENS2 AGND39
X110 COND_C2 AGND8
XR3 BIB_PSW +12V AGND50
X107 NTC_BIB
J1
AGND50
AUS/AENDERUNG
a V84/V42 ENTF.
b 12V f.X101
c J1 HINZU
d R3-6 überm.
e NTC_4 neu
f Sig.name ges.
TAG
NAM
BE
GE
SW
Ma
NAME
MHD
ASCH
Fresenius Medical Care Deutschland GmbH
671943
VERTEILERPLATINE DISTRIBUTION-BOARD LP747
SP
FORMAT A3
BLATT 2 (2)

8.28 P.C.B. LP 748 Display board (BP)

8.28.1 Description

- **General information**

  P.C.B. LP 748 comprises the display and the keyboard.

  Plug connection on P.C.B. LP 748:
  – X189, connection to P.C.B. LP 624

- **Display**

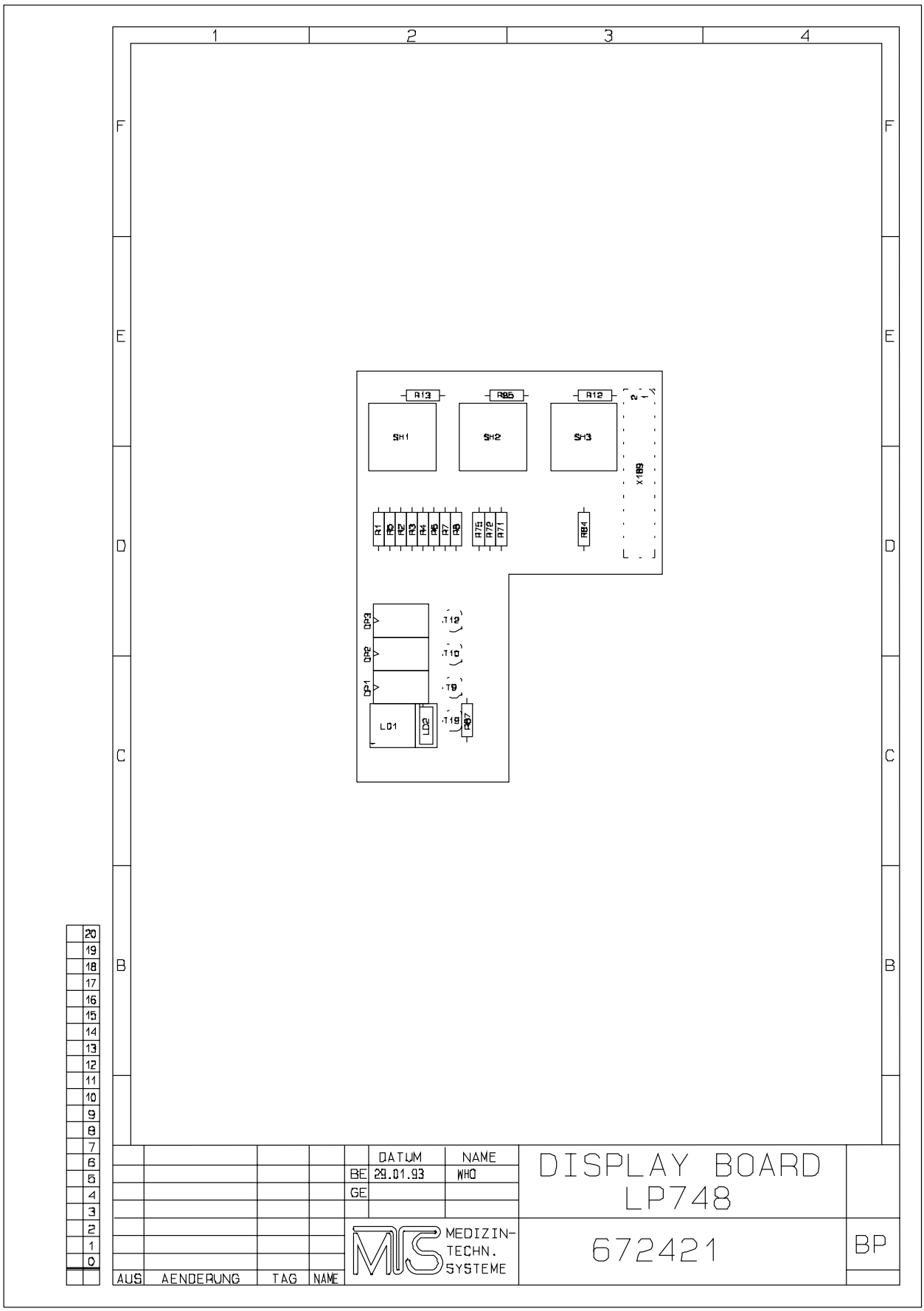
  All of the information required is given to the user by three multiplexed seven-segment displays and two LEDs, which are fitted on bases.

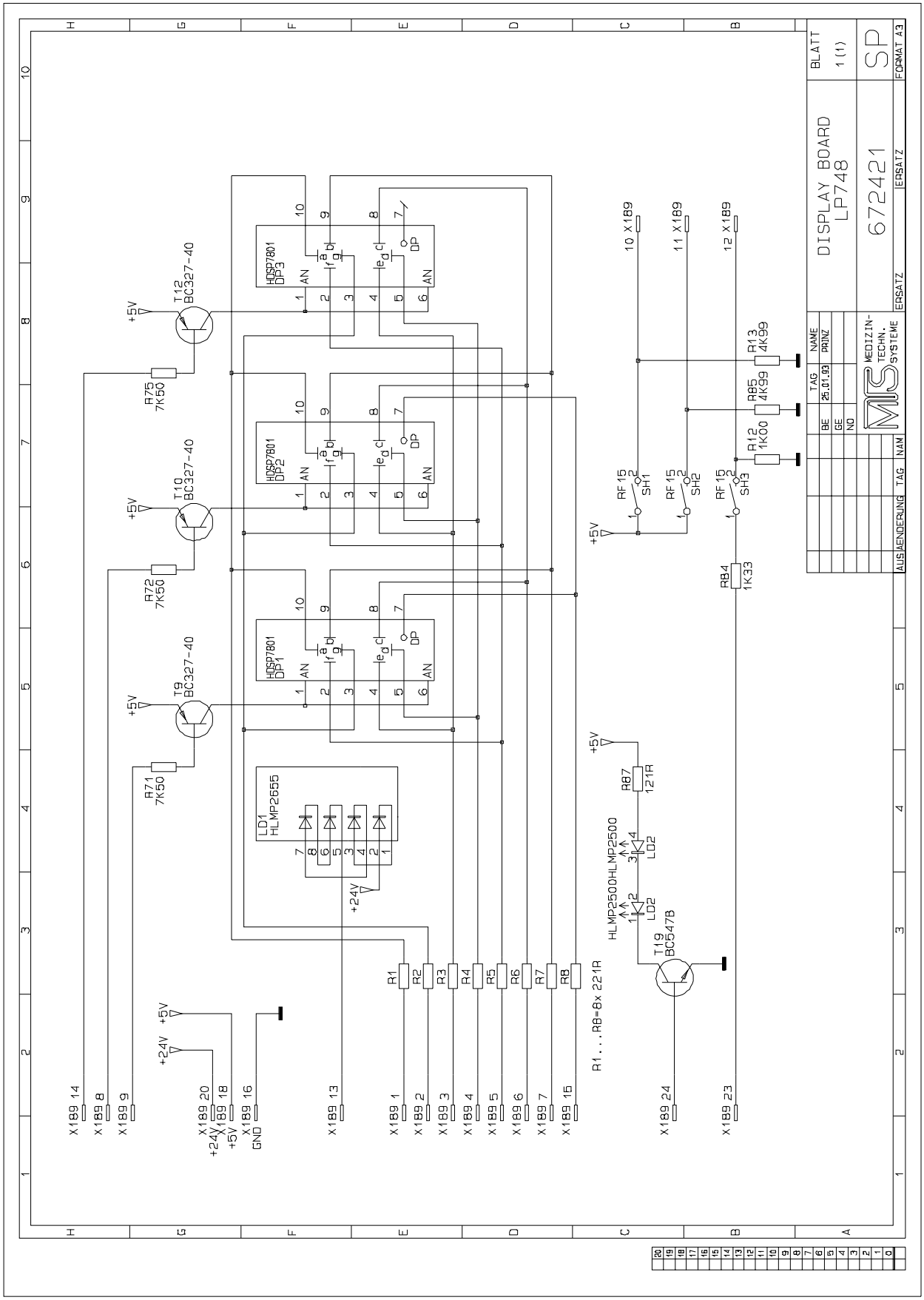
- **Keyboard**

  The keyboard comprises three individual unlighted keys, which are directly soldered into the printed circuit board.
  If he presses a key, the user obtains an acknowledgement by a snap contact.

P.C.B. LP 748

Component layout diagram




DISPLAY BOARD
LP748
672421
SP
BLATT 1 (1)
FORMAT A3
ERSATZ
MEDIZIN-TECHN. SYSTEME
NAME PRINZ
TAG 25.01.99
BE
GE
NO
AUSLAENDERUNG TAG NAM
T9 BC327-40
T10 BC327-40
T12 BC327-40
T19 BC547B
HDSP7801 DP1
HDSP7801 DP2
HDSP7801 DP3
LD1 HLMP2655
LD2 HLMP2500
R1...R8=8x 221R
R71 7K50
R72 7K50
R75 7K50
R84 1K33
R85 4K99
R87 121R
SH1
SH2
SH3
+5V
+24V
GND
X189 1
X189 2
X189 3
X189 4
X189 5
X189 6
X189 7
X189 8
X189 9
X189 13
X189 14
X189 15
X189 16
X189 18
X189 20
X189 23
X189 24

8.29 P.C.B. LP 763 Multi interface board (COMMCO III)

8.29.1 Description

P.C.B. LP 763 is an interface board for the 4008 series. It is provided for communiction and data exchange of the CPU1 with other components inside or outside the machine.

- **Functional groups**

The P.C.B. LP763 comprises six functional units:
– RS232 interface
– Keybox interface
– CAN interface
– COMMCO-1 interface
– BTM interface
– DIP switch array (4)

- **RS232 interface description**

The interface chip SCC2698 (IC1) has eight serial channels. To obtain an RS232 level for transmission, each channel is connected across a RS232 level converter (MAX238 IC2, IC3). Each serial channel is wired to its own 10-pin plug. The plug connectors can be used to adapt new components or as monitoring connections.
If nothing is stated to the contrary, the transfer parameters are 9600Bd, 8 data bits, 1 stop bit, no parity.
The channels are either already allocated or reserved:

Conn.	*Channel*	*Use*	*Special feature*
X2	a	Standard HDF scale	Baud rate 1200 Bd
X3	b	Standard HDF HDF pump	
X4	c	Test channel	
X5	d	Heparin pump	
X6	e	FINESSE interface (CAMUS protocol)	decoupled 2400/9600Bd
X7	f	Blood pressure monitor BPM (in 4008H)	
X8	g	Blood volume monitor BVM (in 4008H)	
X9	h	not used	

The serial channel e among these channels is specially designed. It is decoupled and has a dielectric strength of 4 kV. Decoupling is ensured by two specially designed level converters (MAX250, IC8 and MAX251, IC9) in conjunction with the optocouplers (OC1, OC3) and the transformer UT1. Channel e is provided for the connection of a FINESSE system or a service PC.

- **Keybox interface (connector X11)**

  An interface for the keybox (patient key interface) installed on P.C.B. LP763 is controlled by the digital inputs and outputs of the SCC2698 (IC1). When the FINESSE system and the keybox is connected, this interface is used to read the patient keys and to transfer the data via channel e to the FINESSE system.
  The key contents are read by a bit-serial process in a non RS232-compatible format. The yellow LED in the keybox is activated via a digital output in conjunction with transistor T1.

- **CAN interface (connector X12)**

  The INTEL CAN controller 82527 (IC5) is used for data exchange with modules that themselves have a CAN connection. The connections of the CAN bus are routed to the CAN bus A of the 4008 via a driver (Si9200 or 75LBC031, IC6). Transmission parameters are 500 kBaud for CAN V2.0b-active-standard. The levels comply with the requirements of ISO 11898.

- **COMMCO-1 interface**

  To ensure compatibility the plug connections on P.C.B. LP763 for the connection of the COMM-CO-I board were not changed.

  Connector assignment**:**

Conn.	*Connection to*
X1	X1 on P.C.B. LP 729 (COMMCO-1)
X10	X4 on P.C.B. LP 729 (COMMCO-1)
X14	X1 on P.C.B. LP 751 (decoupled for FINESSE interface of COMMCO-1)

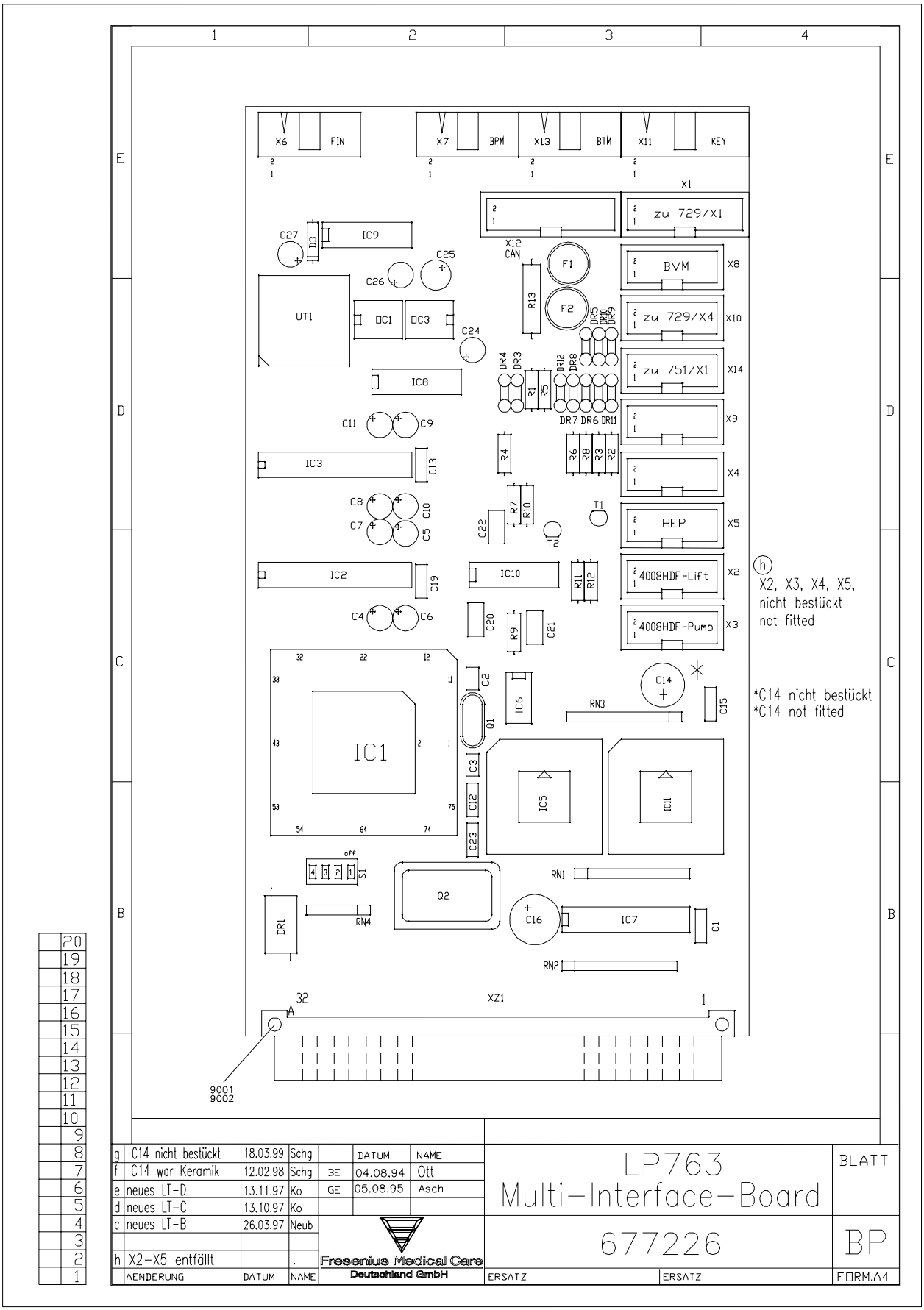
- **BTM interface (connector X13)**

  To permit connection of a BTM module in dialysis machines with software 4.0 or higher (without using a COMMCO board), a BTM interface has been integrated on P.C.B. LP 763. The major parts of this interface are a programmable PLD (lattice ispLSI1016, IC11), several driver gates (IC10) and a driver transistor (T2) with circuitry.
  The function of the BTM interface is to ensure bidirectional communication between the dialysis machine and the BTM module.

- **DIP switch array (S1)**

  As certification of the FINESSE system is still pending, it was decided to integrate the download function (possibility to preset the machine data via the FINESSE system) in the software, but to bar access to this function by a switch until certification of the FINESSE.
  For this purpose a 4x switch array was integrated on the board. The state of the switches can be read via digital input pins of the interface chip (IC1).

Switch	*Use*	*Function in ON position*	*Function in OFF position*
1	Enable FINESSE download	enabled	disabled
2	not used		
3	not used		
4	not used		



	AENDERUNG	DATUM	NAME
g	C14 nicht bestückt	18.03.99	Schg
f	C14 war Keramik	12.02.98	Schg
e	neues LT-D	13.11.97	Ko
d	neues LT-C	13.10.97	Ko
c	neues LT-B	26.03.97	Neub
h	X2-X5 entfällt		.

	DATUM	NAME
BE	04.08.94	Ott
GE	05.08.95	Asch

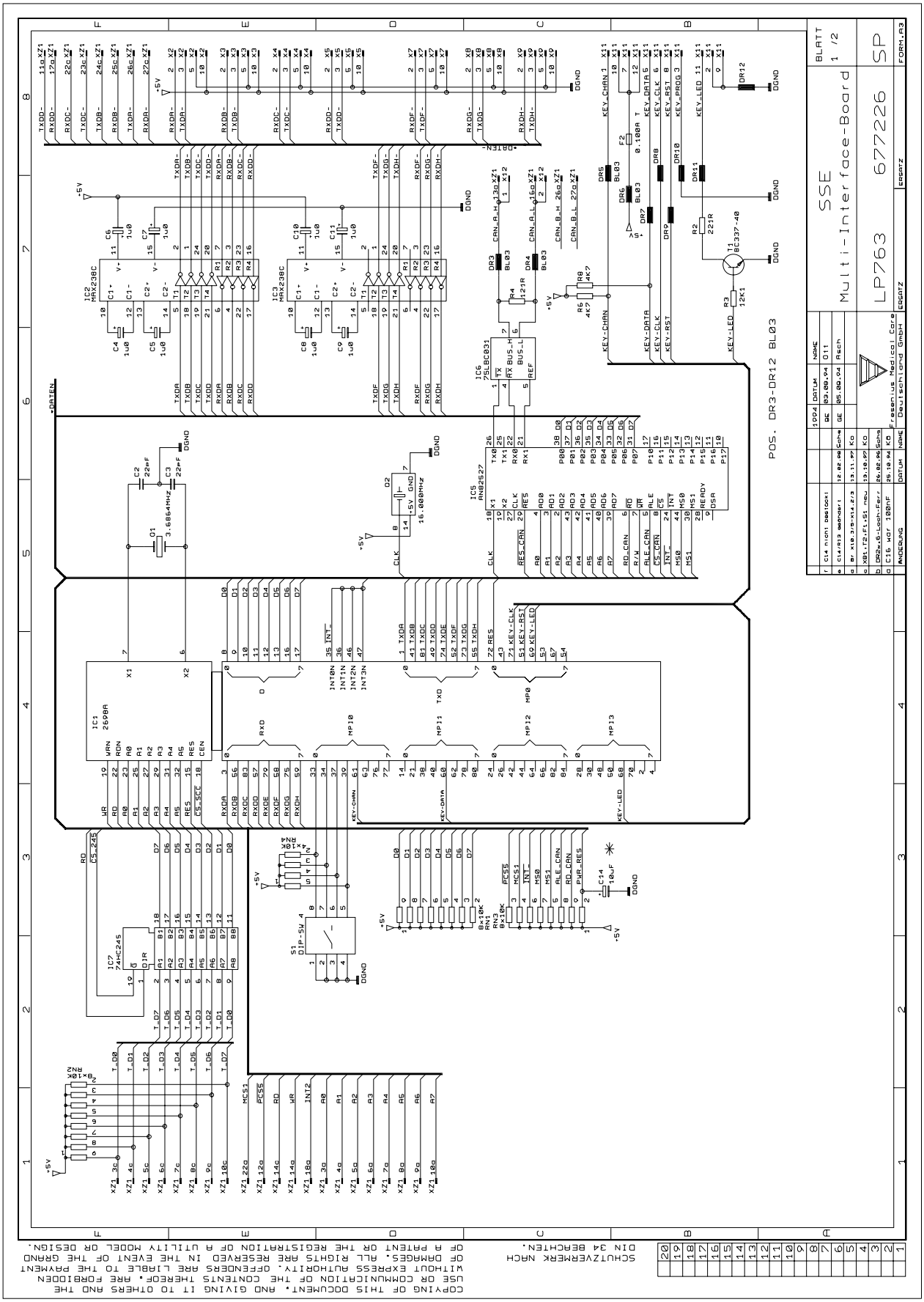
Fresenius Medical Care
Deutschland GmbH

LP763
Multi-Interface-Board

677226

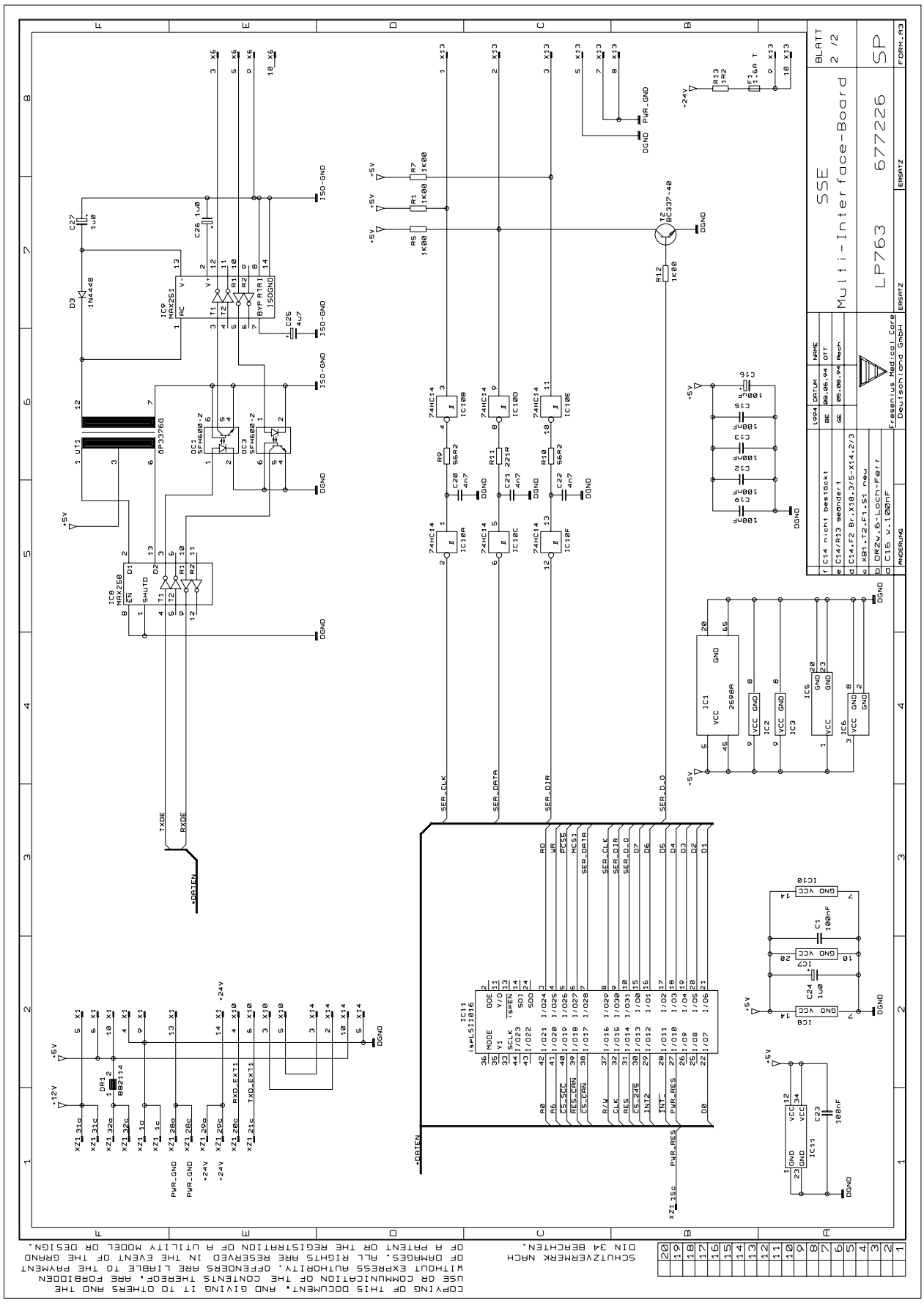
BLATT
BP

ERSATZ | ERSATZ | FORM.A4



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SCHUTZVERMERK NACH DIN 34 BEACHTEN.


SSE
Multi-Interface-Board
LP763 677226 SP
BLATT 2 /2
Fresenius Medical Care Deutschland GmbH
SCHUTZVERMERK NACH DIN 34 BEACHTEN.
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8.30 P.C.B. LP 922 Display board (4008 S)

8.30.1 Description

4008 E hemodialysis machine – monitor

- **Address decoder (circuit diagram 1/6)**

  IC 1 and IC 2 are provided as 4-in-16-bit address decoders. The 22 chip selects are generated by means of addresses A0 to A3 and A4 for IC 1 as well as A4 inverted for IC 2. The 22 chip selects are generated only if, via IC 25, either X1/24 (write) or X1/25 (read) is at L level. IC 20 is used as data bus driver and is reversed in its direction by WR and RD.

- **Keyboard matrix (circuit diagram 1/6)**

  Pins N1 and GND of plug X2 are directly connected to the circuits in the power supply unit. These circuits are used to switch the machine on and off. Via IC 33, the remaining keys on the touch sensitive keyboard are connected to a matrix at the keyboard encoder IC 27 and an additional data line DB4. After the CPU has received the interrupt, the keyboard code is read out by means of CS20 via buffer IC 36.

- **Brightness control (circuit diagram 1/6)**

  The clock pulses required for the bar graph displays are delivered by IC 74 and the following gate logic. Superimposed by a higher frequency and a variable mark-to-space ratio, these clock pulses are used to modulate the brightness. To this end, the shift register IC 6 is provided. IC 6 is loaded in parallel with the desired mark-to-space ratio via a data word at DB0 to DB7 (CS21).

- **Status indicator LEDs (circuit diagram 4/6)**

  All status indicators as well as the condition indicator are switched by means of IC 6 (CS16, CS17).
  Both modules are
  – connected to the brightness control by R9,
  – not connected to the brightness control by R8 (maximum brightness).
  The I/O LED (power on/off) is directly connected to the 5-V supply network.

- **Alarm indicator LEDs (circuit diagram 5/6)**

  All alarm indicators are energized by IC 12 and CS18. The key click is controlled by output 19 of IC 12 via IC 32/5.6.

- **Bar graph displays for arterial and venous pressure (circuit diagram 2/6)**

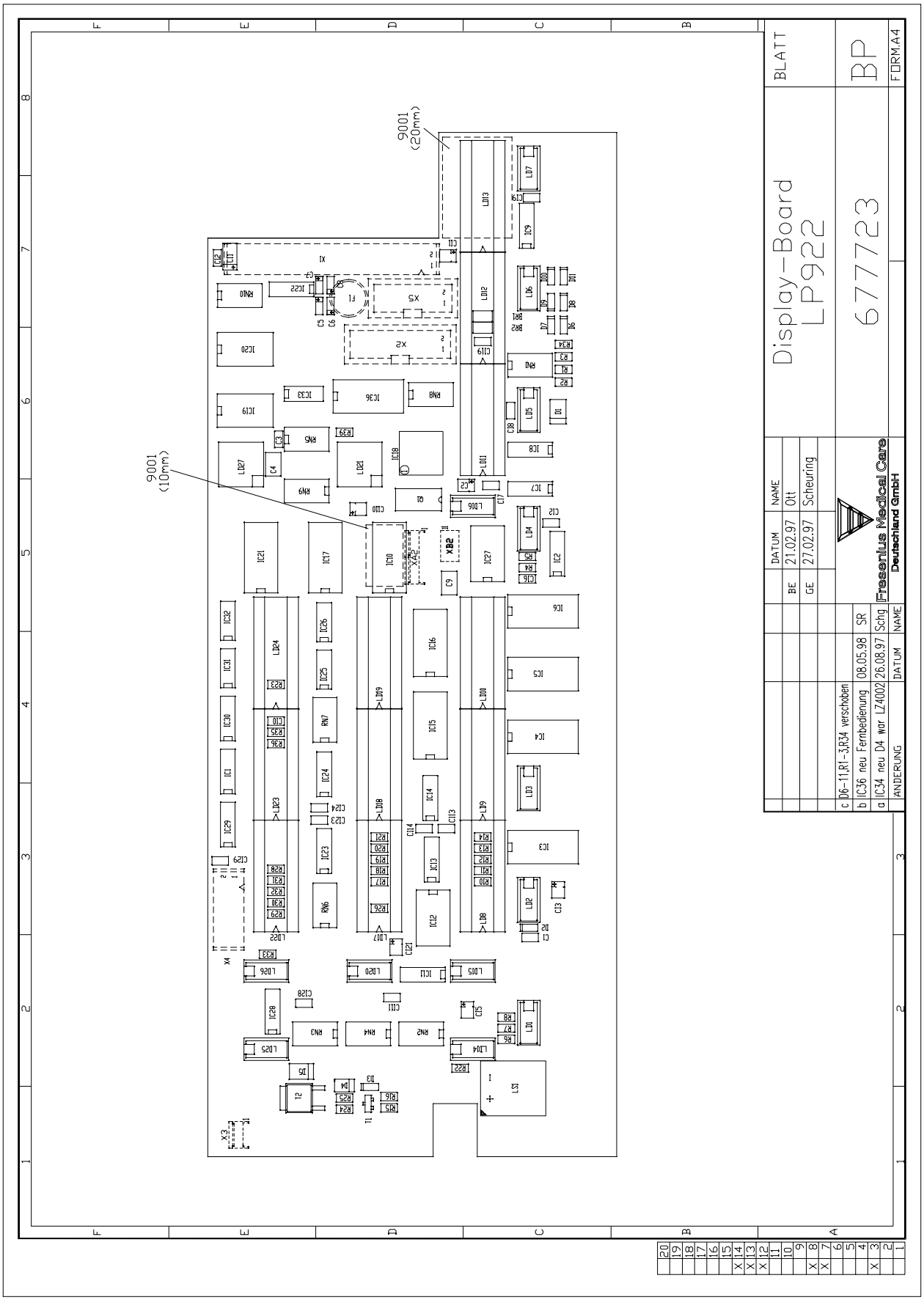
  IC 3 and IC 4 are provided as data memories for the arterial bar graph display, whose 30 LEDs are organized within a 5x6 matrix. The 5 lines are directly energized via the emitter sequence IC 13 by data buses DB0 to DB4 and the 6 columns via the 8-out-of-3 decoder IC 14 by data buses DB5 to DB7. The actual value (CS1) and the upper limit (CS2) are stored by IC 3 and the lower limit (CS3) is stored by IC 4. The respectively applicable LED is energized via the clock inputs CL1, CL2, and CL3. The clock pulses of CL1 have a longer duration than those of CL2 and CL3. Thus, the actual value is represented more brightly than the limits.
  IC 10 (CS4, CS5), and IC 11 (CS6) are provided as data memories for the venous bar graph display. The working method of this display is identical with that of the arterial bar graph display.

- **Bar graph displays for TMP and conductivity (circuit diagram 3/6)**

  IC 17 (CS7) and IC 21 (CS8, CS9) are provided as data memories for the TMP bar graph display.
  IC 4 (CS10) and IC 5 (CS12) are provided as data memories for the CD bar graph display.
  The working method of both displays is identical with that of the arterial bar graph displays.

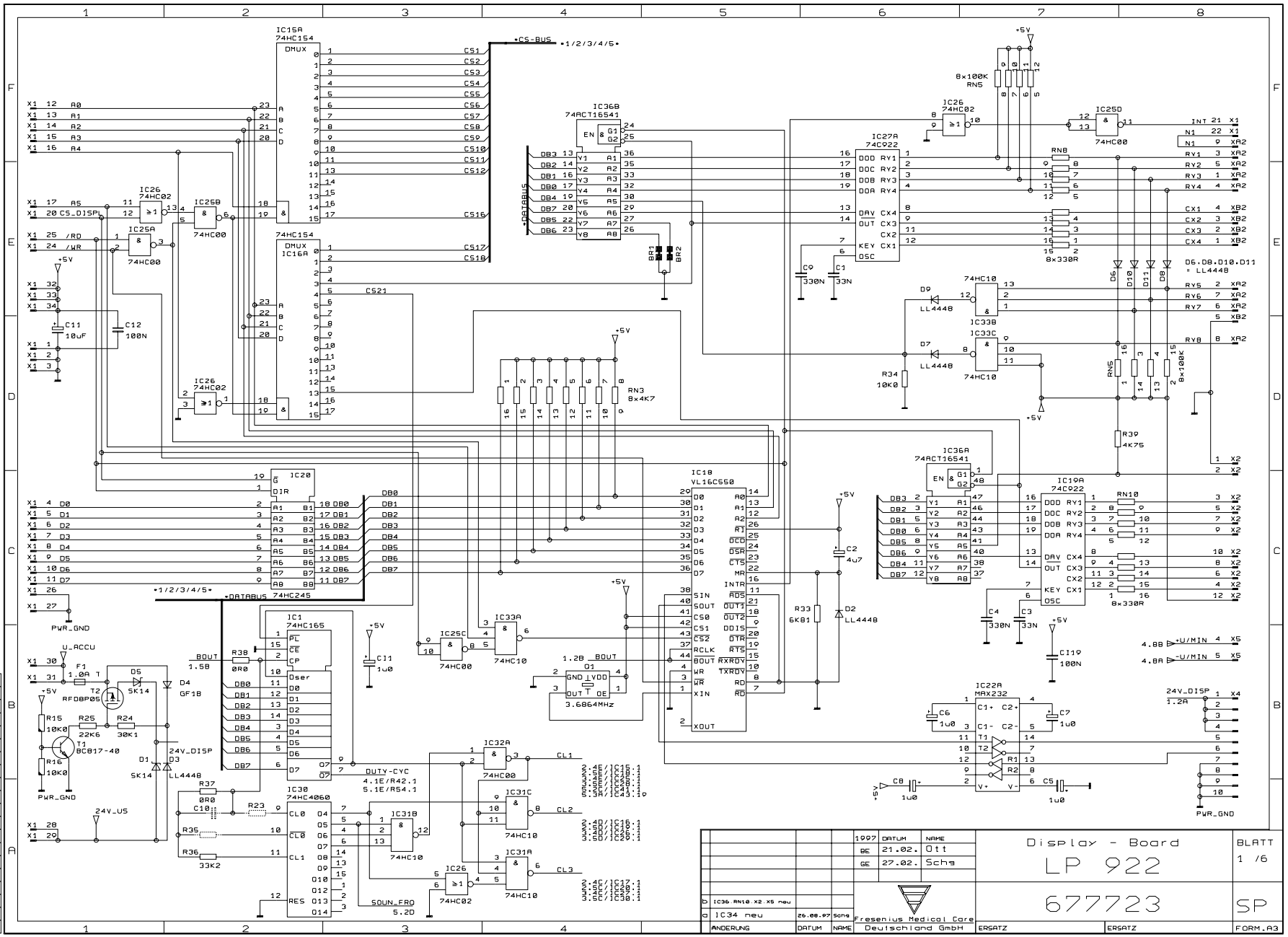
P.C.B. LP 922

Component layout diagram



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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



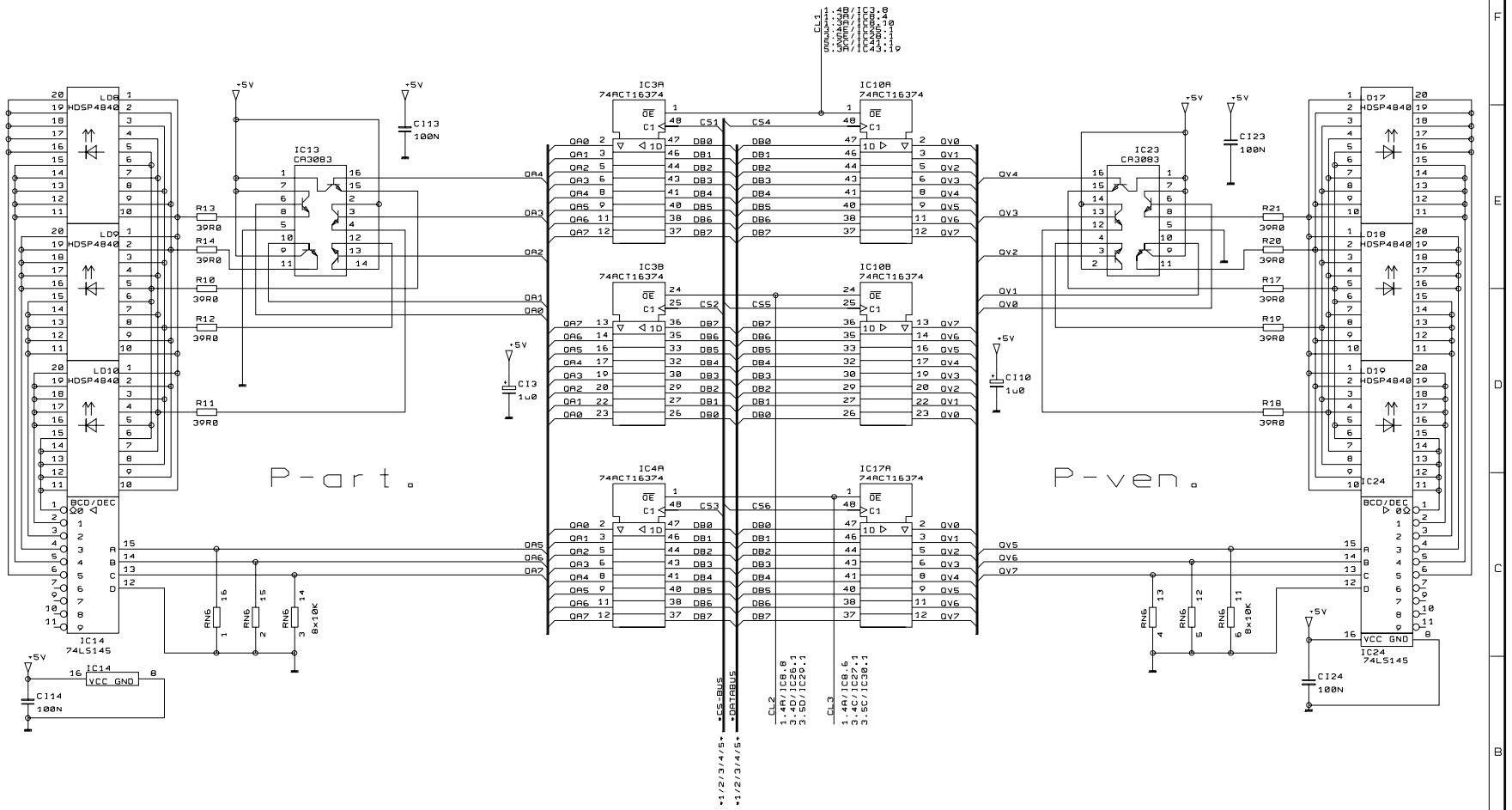
				1997	DATUM	NAME	Display - Board LP 922
				BE	21.02.	Ott	BLATT 1 /6
				GE	27.02.	Schs	
b	IC36 RN10 X2 X5 neu						677723 SP
a	IC34 neu	26.08.97	Schs	Fresenius Medical Care Deutschland GmbH			
ÄNDERUNG		DATUM	NAME	ERSATZ		ERSATZ	FORM.A3

P.C.B. LP 922

Circuit diagram 1/6

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



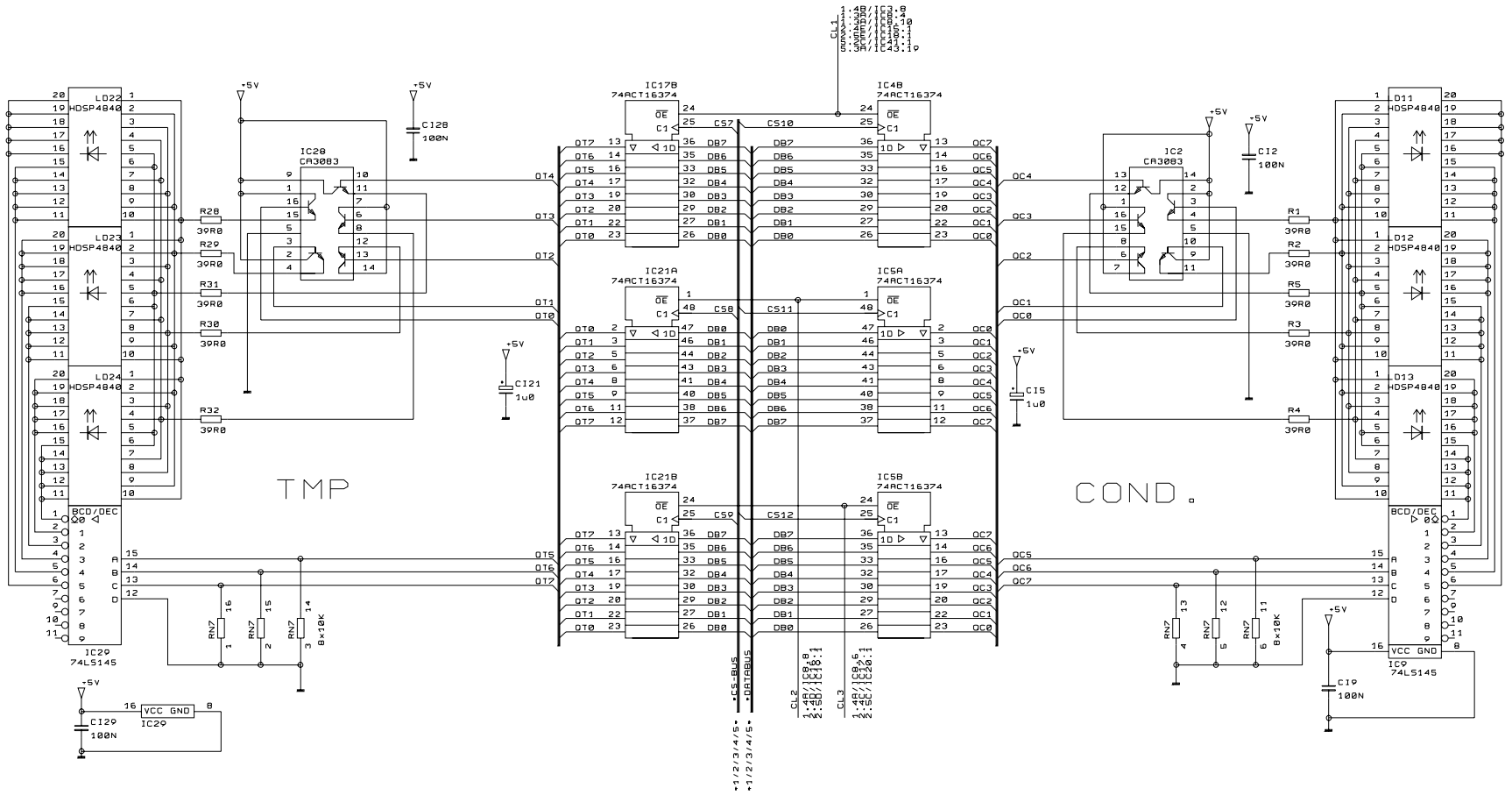
		1997	DATUM	NAME	Display - Board LP 922
		BE	21.02.	Ott	BLATT 2 /6
		GE	27.02.	Schg	
b	siehe Bl.1				677723 SP
a	IC34 neu				
ANDERUNG		Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 922

Circuit diagram 2/6

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



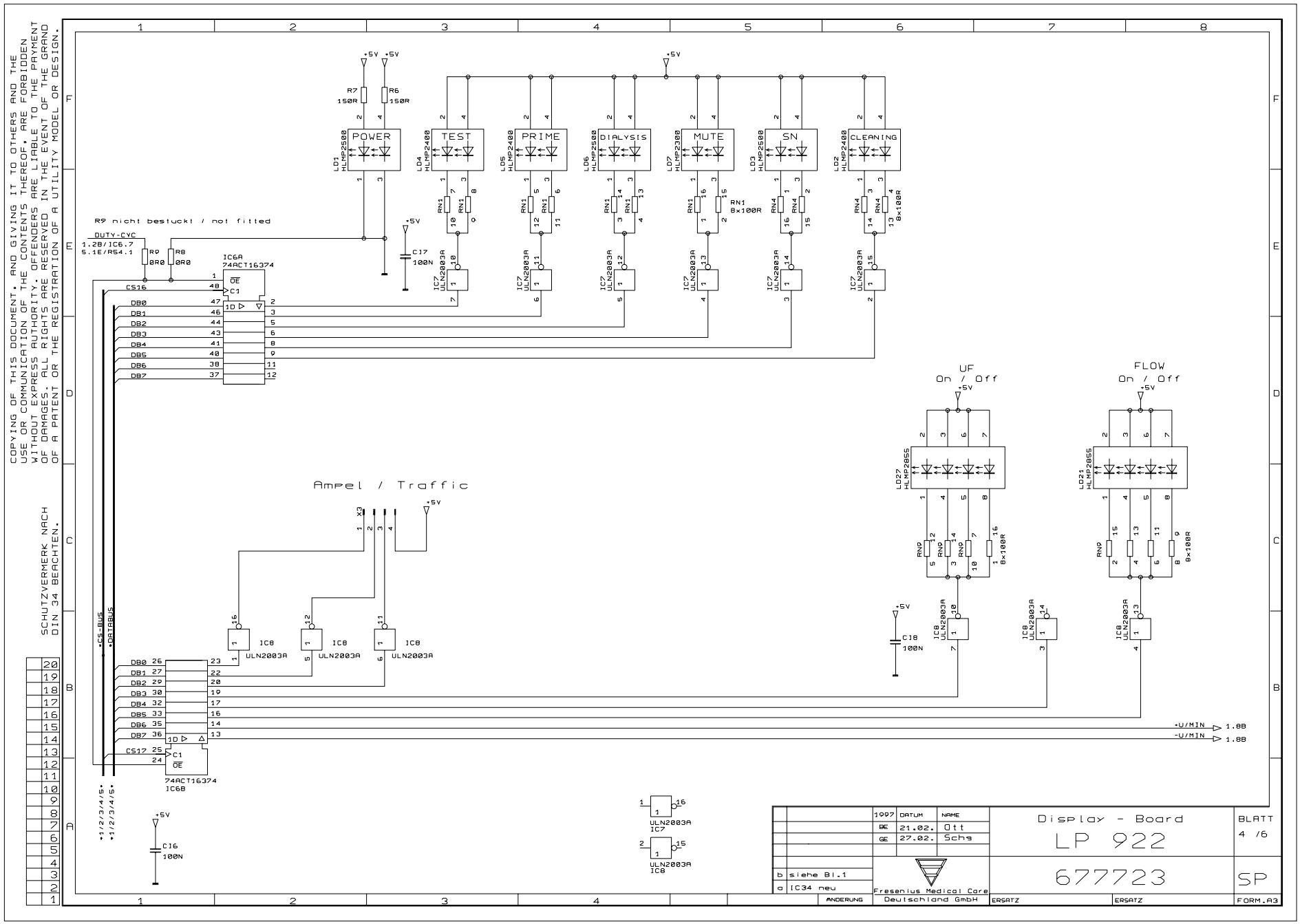
	1997	DATUM	NAME	Display - Board LP 922
	BE	21.02.	Ott	BLATT 3 /6
	GE	27.02.	Schg	
b siehe Bl.1				677723 SP
a IC34 neu				
ANDERUNG	Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 922

Circuit diagram 3/6

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



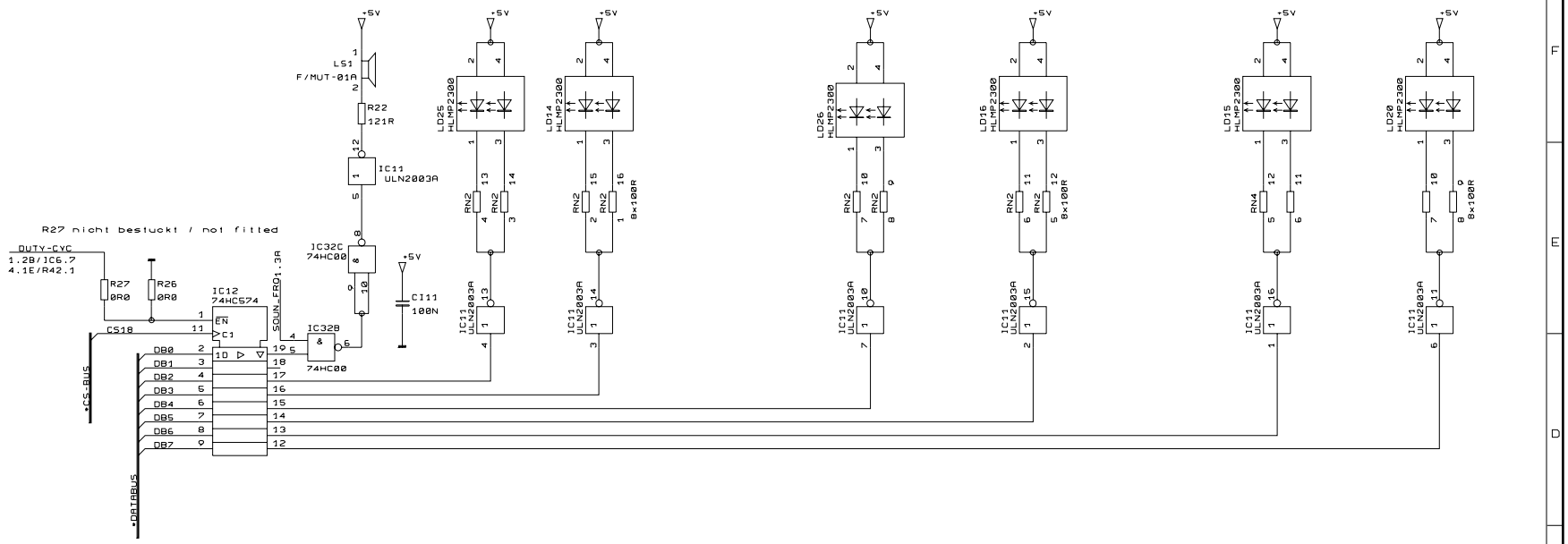
		1997	DATUM	NAME	Display - Board LP 922
		BE	21.02.	Ott	BLATT 4 /6
		GE	27.02.	Schg	
b	siehe Bl.1				677723 SP
a	IC34 neu				
ÄNDERUNG		Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 922

Circuit diagram 4/6

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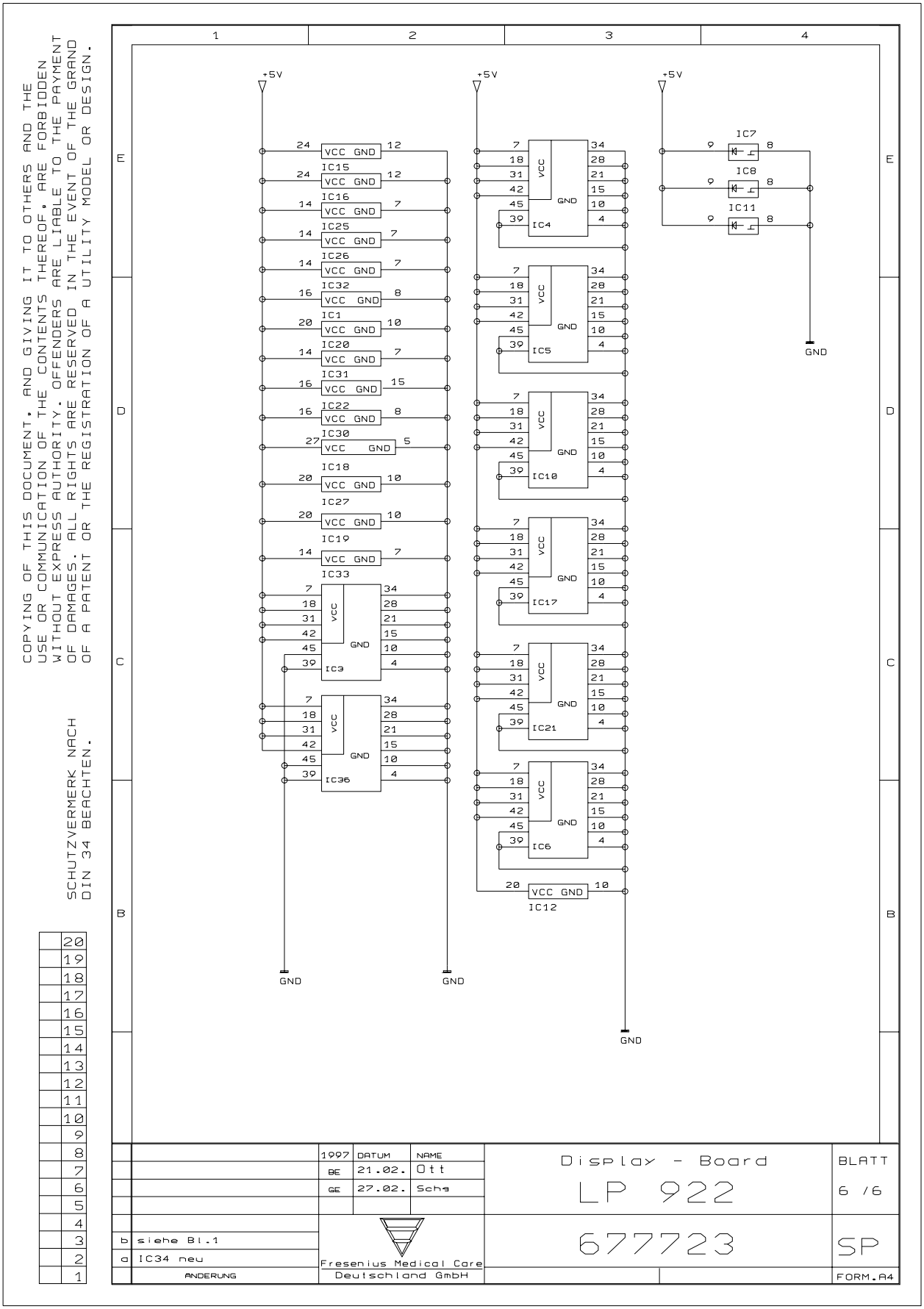
SCHUTZVERMERK NACH DIN 34 BEACHTEN.



		1997	DATUM	NAME	Display - Board LP 922
		BE	21.02.	Ott	BLATT 5 /6
		GE	27.02.	Schg	
b	siehe Bl.1				677723 SP
a	IC34 neu				
ÄNDERUNG		Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ / FORM.A3

P.C.B. LP 922

Circuit diagram 5/6


1
2
3
4
E
D
C
B
+5V
GND
VCC
IC15
IC16
IC25
IC26
IC32
IC1
IC20
IC31
IC22
IC30
IC18
IC27
IC19
IC33
IC3
IC36
IC4
IC5
IC10
IC17
IC21
IC6
IC12
IC7
IC8
IC11
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SCHUTZVERMERK NACH DIN 34 BEACHTEN.
1997 DATUM NAME
BE 21.02. Ott
GE 27.02. Schg
Display - Board
LP 922
BLATT 6 /6
b siehe Bl.1
a IC34 neu
ANDERUNG
Fresenius Medical Care Deutschland GmbH
677723
SP
FORM.A4

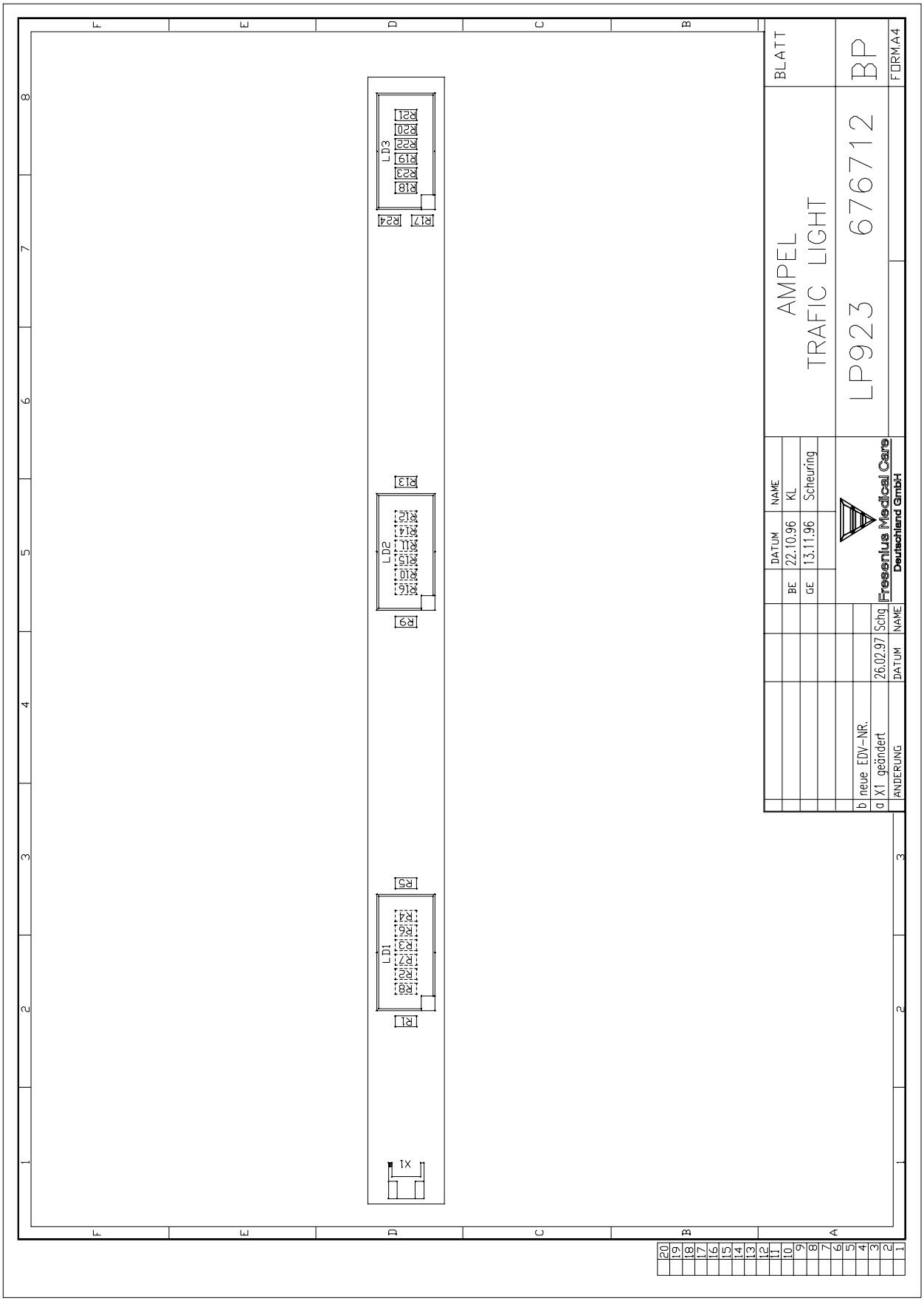
8.31 P.C.B. LP 923 Traffic light (4008 H/S)

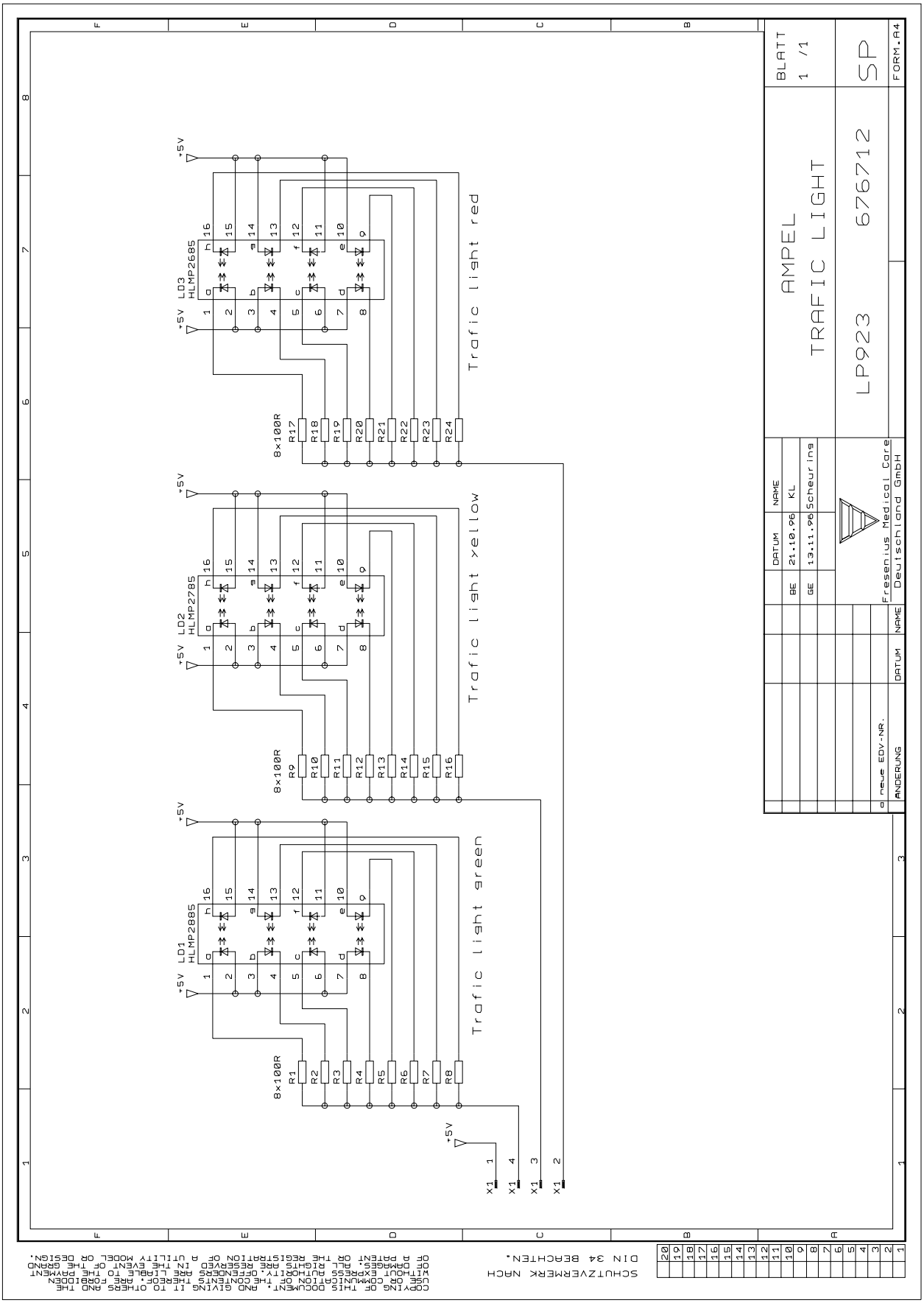
8.31.1 Description

P.C.B. LP 923 indicates the machine status:
– green status indicator (operation)
– yellow status indicator (warning/info)
– red status indicator (alarm)

P.C.B. LP 923

Component layout diagram




AMPEL
TRAFIC LIGHT
LP923 676712 SP
BLATT 1 /1
FORM.A4
Fresenius Medical Care Deutschland GmbH
Trafic light red
Trafic light yellow
Trafic light green
LD1 HLMP2885
LD2 HLMP2785
LD3 HLMP2685
8x100R
+5V
X1
SCHUTZVERMERK NACH DIN 34 BEACHTEN.

8.32 P.C.B. LP 924 Display board (4008 H)

8.32.1 Description

4008 H hemodialysis machine – monitor

- **Address decoder (circuit diagram 1/6)**

  IC 1 and IC 2 are provided as 4-in-16-bit address decoders. The 22 chip selects are generated by means of addresses A0 to A3 and A4 for IC 1 as well as A4 inverted for IC 2. The 22 chip selects are generated only if, via IC 3, either X1/24 (write) or X1/25 (read) is at L level. IC 7 is used as data bus driver and is reversed in its direction by WR and RD.

- **Keyboard matrix (circuit diagram 1/6)**

  Pins N1 and GND of plug X3 are directly connected to the circuits in the power supply unit. These circuits are used to switch the machine on and off. Via IC 14, the remaining keys on the touch sensitive keyboard are connected to a matrix at the keyboard encoder IC 12 and an additional data line DB4. After the CPU has received the interrupt, the keyboard code is read out by means of CS20 via buffer IC 13.

- **Brightness control (circuit diagram 1+2/6)**

  The clock pulses required for the bar graph displays are delivered by IC 10 and the following gate logic. Superimposed by a higher frequency and a variable mark-to-space ratio, these clock pulses are used to modulate the brightness. To this end, the shift register IC 6 is provided. IC 6 is loaded in parallel with the desired mark-to-space ratio via a data word at DB0 to DB7 (CS21) and is cyclically rotating due to clock generator IC 5/4,5,6 and IC 5/8,9,10.

- **Status indicator LEDs (circuit diagram 4/6)**

  All status indicators as well as the condition indicator are switched by means of IC 35 and IC 38 (CS16, CS17).
  Both modules are
  – connected to the brightness control by R42,
  – not connected to the brightness control by R43 (maximum brightness).
  The I/O LED (power on/off) is directly connected to the 5-V supply network.

- **Alarm indicator LEDs (circuit diagram 5/6)**

  All alarm indicators are energized by IC 39 and CS18. The key click is controlled by output 19 of IC 39 via IC 40/5.14.

- **Bar graph displays for arterial and venous pressure (circuit diagram 2/6)**

  IC 15, IC 16, and IC 17 are provided as data memories for the arterial bar graph display, whose 30 LEDs are organized within a 5x6 matrix. The 5 lines are directly energized via the emitter sequence IC 21 by data buses DB0 to DB4 and the 6 columns via the 8-out-of-3 decoder IC 31 by data buses DB5 to DB7. The actual value (CS1) is stored by IC 15; the upper limit (CS2) is stored by IC 16; and the lower limit (CS3) is stored by IC 17. The respectively applicable LED is energized via the clock inputs CL1, CL2, and CL3. The clock pulses of CL1 have a longer duration than those of CL2 and CL3. Thus, the actual value is represented more brightly than the limits.
  IC 18 (CS4), IC 19 (CS5), and IC 20 (CS6) are provided as data memories for the venous bar graph display. The working method of this display is identical with that of the arterial bar graph display.

- **Bar graph displays for TMP and conductivity (circuit diagram 3/6)**

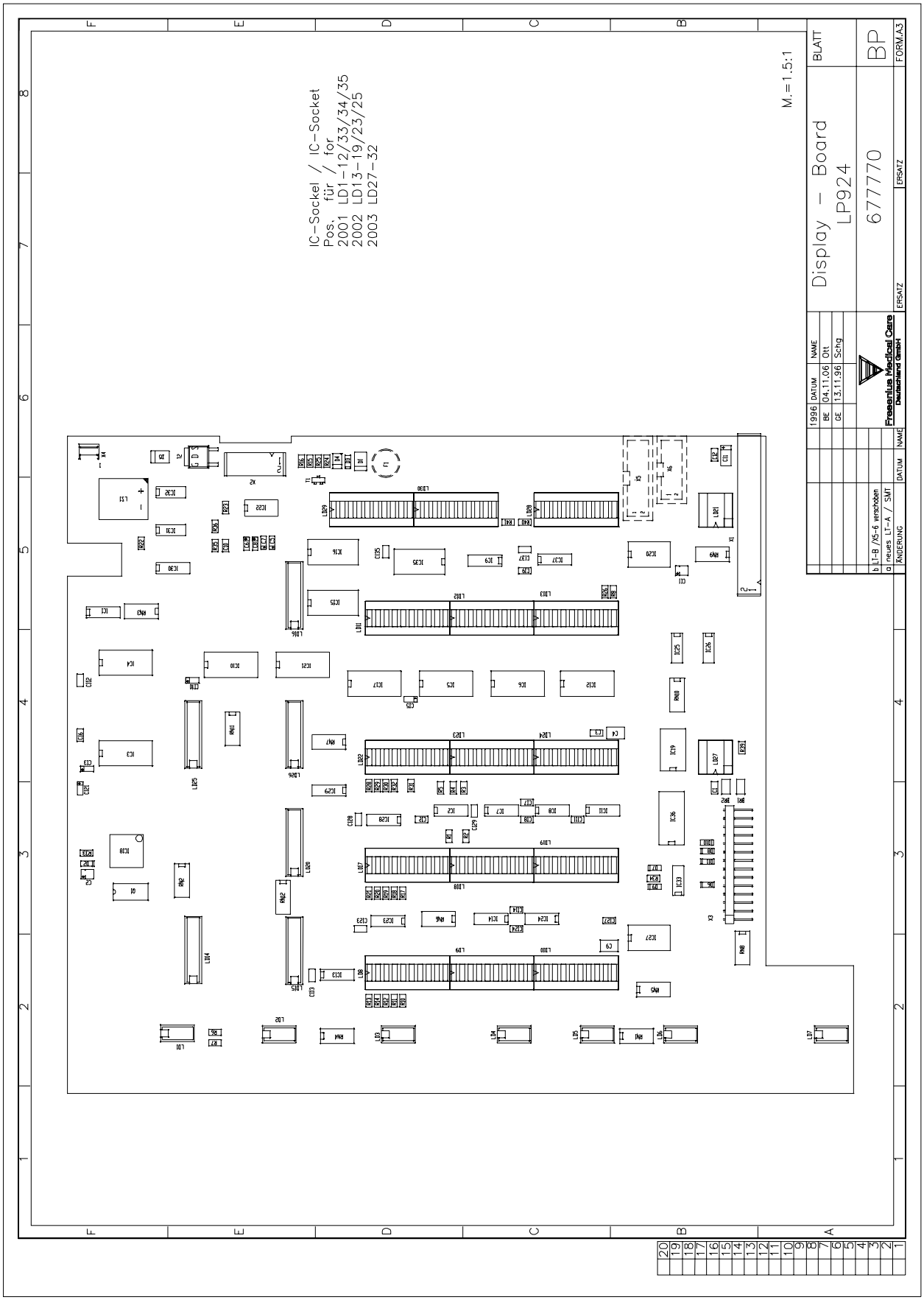
  IC 25 (CS7), IC 26 (CS8), and IC 27 (CS9) are provided as data memories for the TMP bar graph display.
  IC 28 (CS10), IC 24 (CS11), and IC 30 (CS12) are provided as data memories for the CD bar graph display.
  The working method of both displays is identical with that of the arterial bar graph displays.

- **Bar graph displays for temperature and flow (circuit diagram 6/6)**

  The bar graph displays for temperature and flow are energized by of IC 41 via CS19. The data lines DB4 to DB7 contain the value for the temperature display. Only 16 of the 20 LEDs of the temperature display are actually visible. The data lines D0 to D3 contain the value for the flow display. IC 72/19 is responsible for the brightness control of the temperature display, and IC 72/18 serves to turn the display dark.

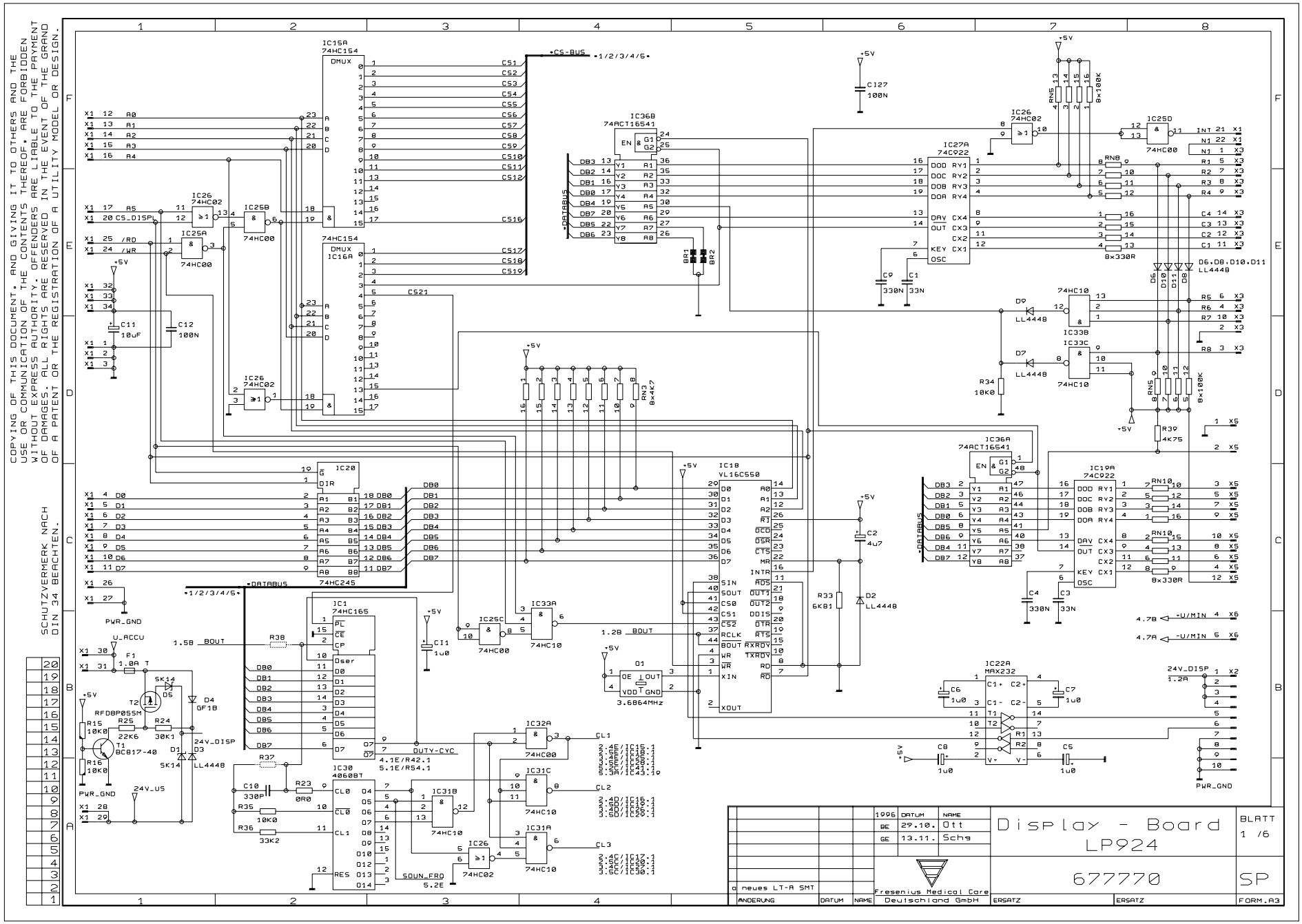
P.C.B. LP 924

Component layout diagram



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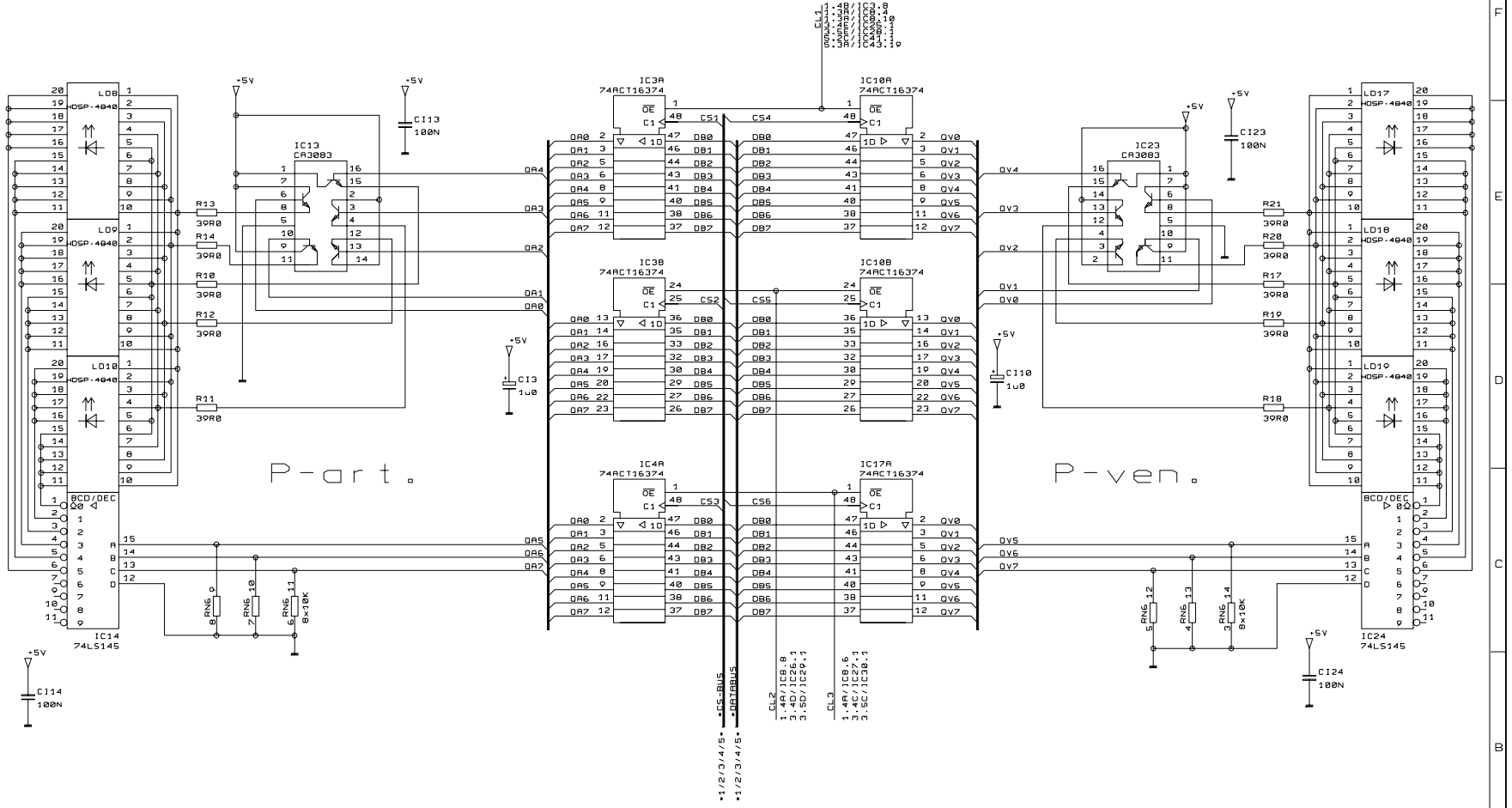


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Circuit diagram 1/6

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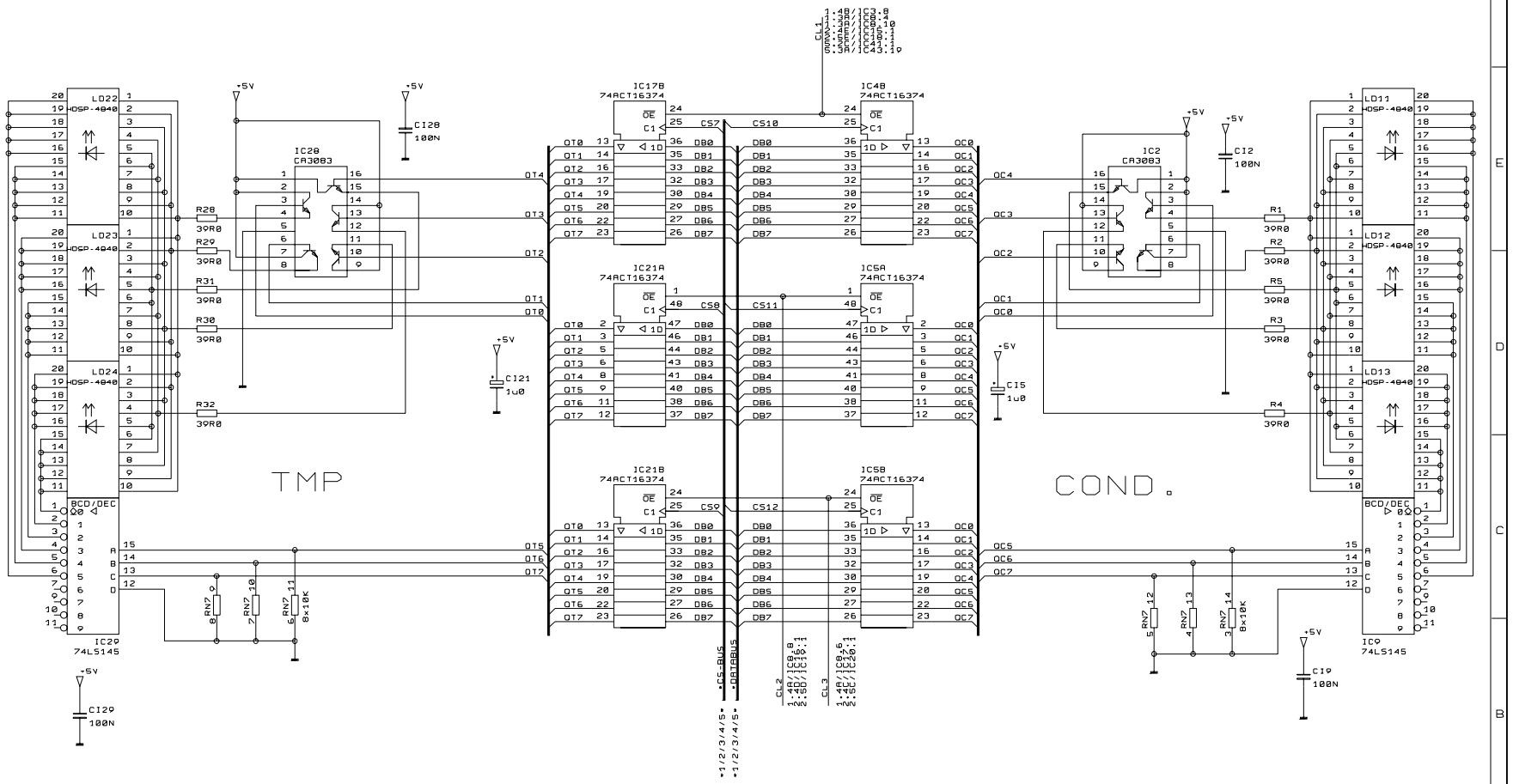
	1996	DATUM	NAME	Display - Board LP924	BLATT 2 /6
	BE	29.10.	Ott		
	GE	13.11.	Schs		
a siehe Bl.1				677770	SP
ANDERUNG	Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ	FORM.A3

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Circuit diagram 2/6

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SCHUTZVERMERK NACH
DIN 34 BEACHTEN.



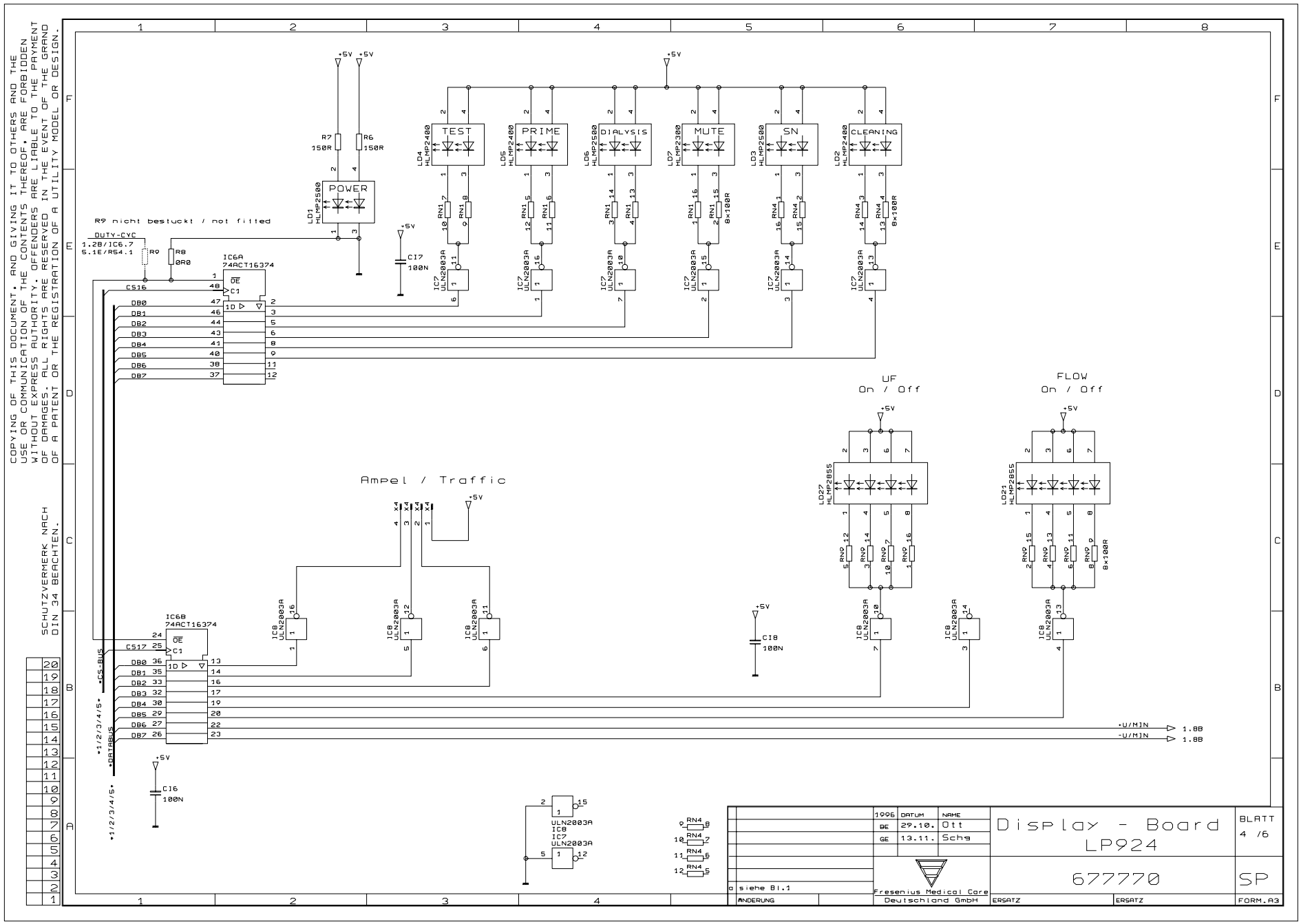
1996	DATUM	NAME	Display - Board LP924
BE	29.10.	Ott	BLATT 3 /6
GE	13.11.	Schs	
			677770 SP
a siehe Bl.1			Fresenius Medical Care Deutschland GmbH
ANDERUNG			ERSATZ ERSATZ FORM.A3

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Circuit diagram 3/6

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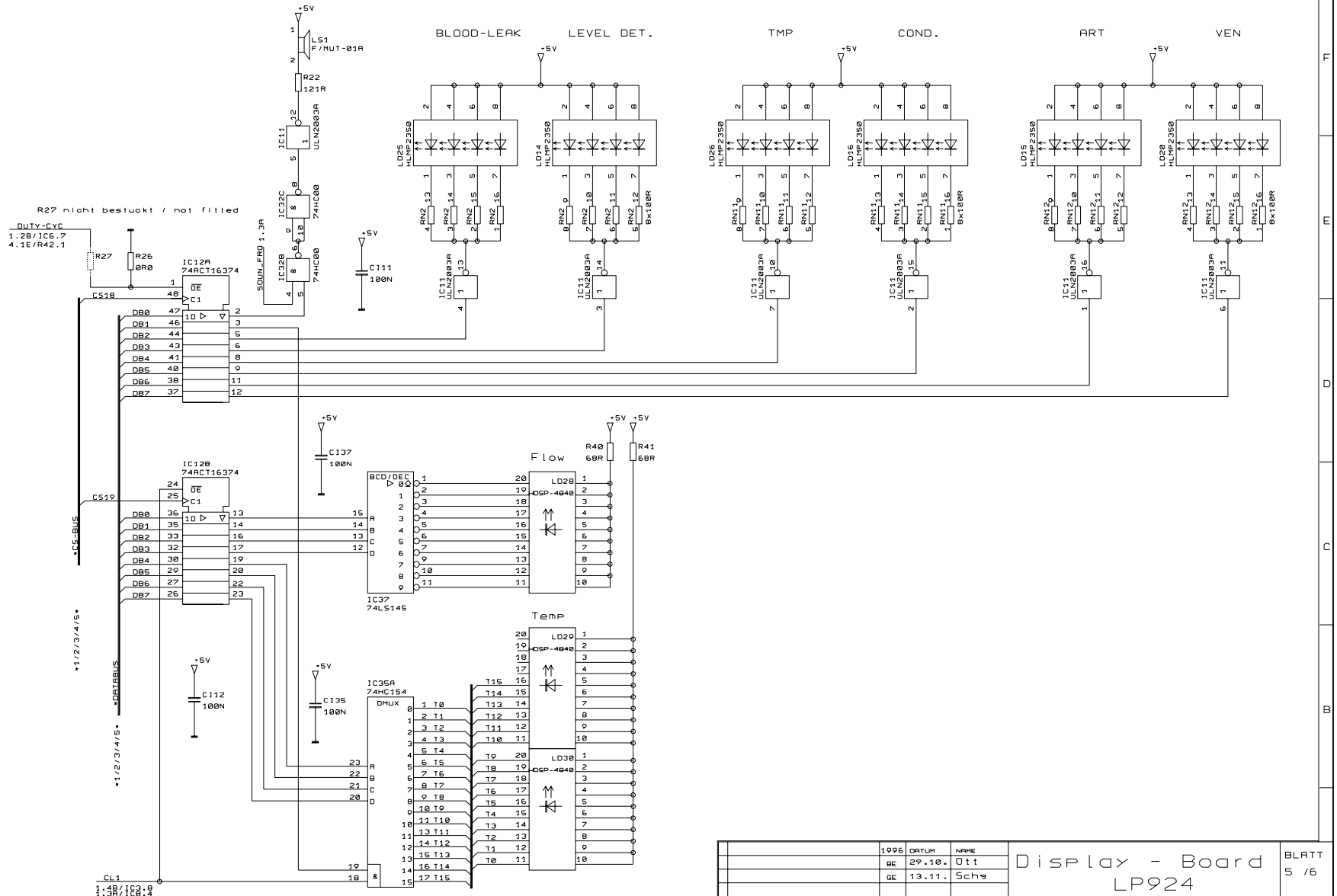
1996	DATUM	NAME	Display - Board LP924
BE	29.10.	Ott	BLATT 4 /6
GE	13.11.	Schä	
a siehe Bl.1			677770 SP
ANDERUNG	Fresenius Medical Care Deutschland GmbH	ERSATZ	ERSATZ FORM.A3

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Circuit diagram 4/6

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SCHUTZVERMERK NACH DIN 34 BEACHTEN.



	1996	DATUM	NAME	Display - Board LP924
	BE	29.10.	Ott	
	GE	13.11.	Schs	
a siehe Bl.1				677770
ANDERUNG	Fresenius Medical Care Deutschland GmbH			ERSATZ / ERSATZ

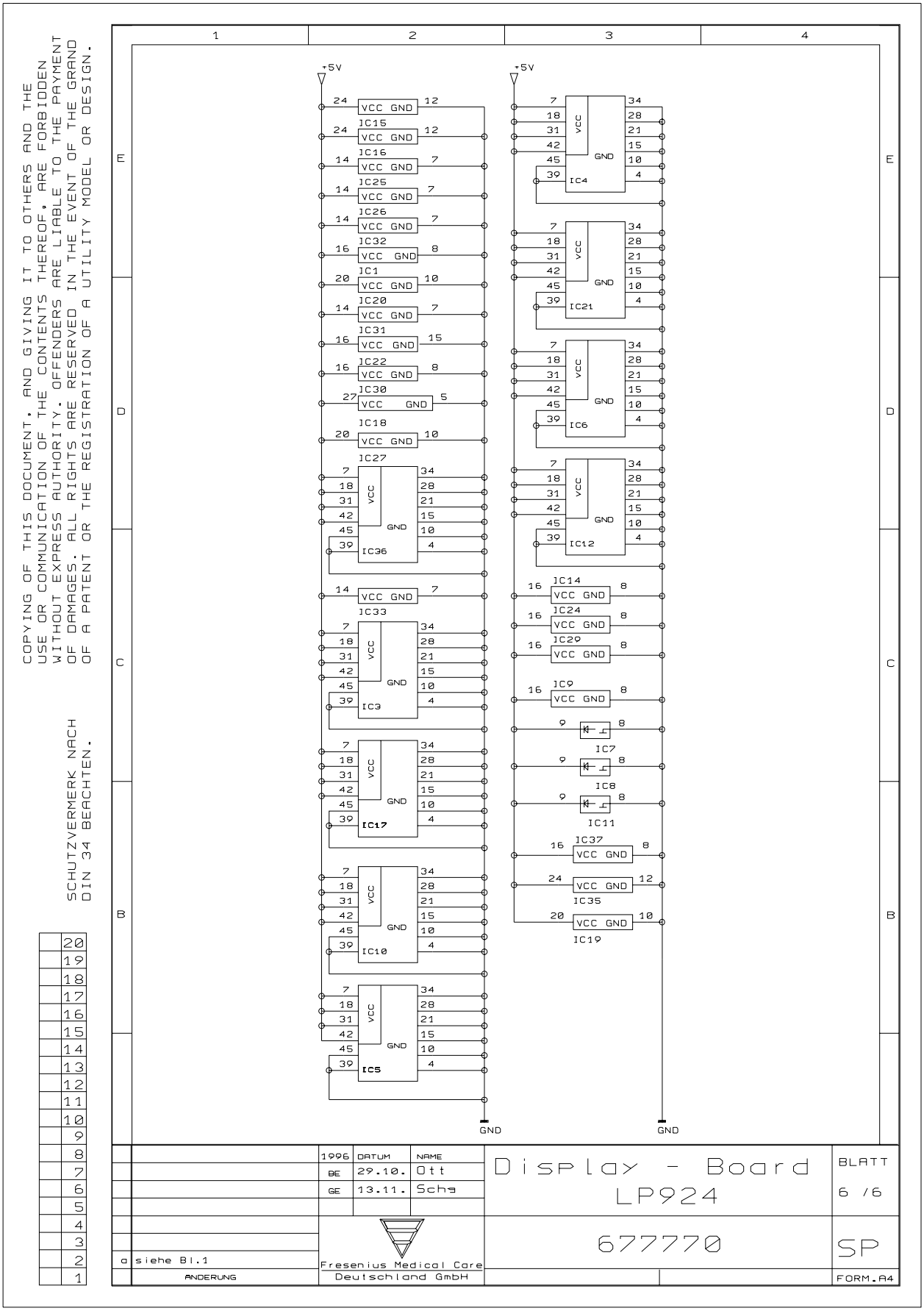
BLATT 5 /6

SP

FORM.A3

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Circuit diagram 5/6



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SCHUTZVERMERK NACH DIN 34 BEACHTEN.

8.33 P.C.B. LP 941 Hydraulics processor

8.33.1 Description

- **Voltage supply**

The PC board is supplied with 24 V via plug X1. The switching regulator IC 10 which generates +5 V max., 1 A from 24 V is used for operation of the processor system. The IC 18 serves to supply the operation amplifier with +12 V. The reference voltage is generated with the reference voltage source IC 15 as well as IC 5. T8 is a power driver for the bridge voltage of the pressure transducers. IC 23 and T 7 are used to generate the negative supply voltage for the instrument amplifier. For delayed switching on of the operating voltages, the two switch controllers IC 10 and IC 18 are enabled with the low level on pin 5 only when the applied +24 V has become stable for approx. 5 seconds (watchdog test of 4008). Evaluation is performed with the comparator circuit associated with IC 13a.

- **Processor system**

Technical data:
– SAF C515C processor
– 6 MHz clock frequency
– 64 KByte EPROM
– 32 KByte RAM
– 16 KBit serial EEPROM
– 10 Bit A/D converter with 8 multiplexed inputs
– 6 ports
– CAN controller AN 82527
– Interface for CAN driver
– 2 ports

The SAF C515C IC 8 receives its 6 Mhz clock frequency via GAL IC 7, pin 19. The oscillator frequency of 12 Mhz is generated by a separate oscillator component Q1 and divided by 2 for operation of the controller in the GAL.
The 12 Mhz clock from the oscillator component is directly applied to CAN controller IC 9, pin 18.
The SAF C515C operates with a multiplexed data/address bus. The upper addresses A8 to A15 are available at port 2 and the lower addresses A0 to A7 are available at address latch IC 3 after decoding.

The IC 22 is a 64k x 8 Eprom.
The RAM, IC 6, has a memory area of 32k x 8.
The GAL IC 7 is used for the decoding of the chip select signals. The EEPROM IC 11 is provided for the backup of calibration data and treatment data during a power failure.

The integrated CAN controller of SAF C 515 C is used for communication with the CPU 1. Data exchange is performed via pins 79/80 of IC 8 and via the CAN bus driver IC 12.

The additional CAN controller IC 9 with the CAN bus driver IC 25 is provided for future options in the hydraulics system which will be performed with intelligent actuators.
Data exchange with the additional CAN controller is performed via the data bus in conjunction with the signals /CS_CAN und /INT_CAN for transmitting and receiving as well as /RD and/WR for reading and writing the ports of IC 9.

- **Watchdog**

  The watchdog circuit consists of the watchdog timer IC 17 with integrated voltage regulator, a dual optoelectronic coupler IC 16 and the transistors T 6 and T 9. The time-determining resistor is R 51. The time window is set to a basis of 500 ms. The open trigger window (TOW) of the component internally ranges from -20% to + 20% (400...600 ms). Each time three successive pulses arrive from IC8, pin 60, via T 6 and IC 16b on pin 3 of the watchdog timer within this time window, the output pin 1 goes to L level and switches the 24 V_SW on via IC 16a and T 9. When a pulse is missing or triggering is too low or too fast, i.e. outside the time window, pin 1 of IC 17 goes to H level and the 24 V_WD is switched off.

- **Peripherals**

  T 1 to T 5 are the power drivers for the respective actuators. The current status of the respective actuator is read back via P 0.0 to P 1.1 via the voltage dividers R67 to R86 connected to the outputs.

  T11a represents a hardware lock of the air pump and the test valve when V 43 F (G54) is closed.

  T 11a enables the supply voltage for the AIR_PUMP and the V_TEST only when valve 43 is open. Acknowledgement for V43 status is via P1.4 (G19)

  IC 4 is an instrument amplifier for the pressure transducers with series connected voltage follower IC 2a. IC 2b is used for offset setting. The circuits associated with IC 19 and IC 21 have a corresponding set-up and serve for later extensions.

  IC 1a generates the processor and CAN controller reset.
  IC 1b detects power failure via a control line from the CAN distributor.
  IC 12 is a bidirectional CAN bus driver.

  IC 28 is a serial dual-channel 8-bit DA converter. Channel 1, in combination with IC 27b, is used to calibrate the temperature measuring circuit associated with IC 31a (not used at present).
  Channel 2 and IC 27a are available for options.

- **Level detection**

  The circuit with the Opamps 32 to 34 serves for level detection in the air separation chambers of the acetate and bicarbonate inlet line. IC 32a decouples the circuit from the 2KHz sinus signal. The measuring branch with IC 33b and IC 34b is applied at the analog input AN6, the reference branch with IC 33a and IC 34a is applied at the analog input AN7 of the controller.
  The activation threshold can be set by changing R137 in the measuring branch or via the software.

- **Conductivity Measurement**

  The portion of the circuit with IC 20,21,30 as well as the transistors T12 and T13 are used for CD measurement and are identical with the circuits on LP 633. The measured value is read in at the analog input of the controller on AN4.

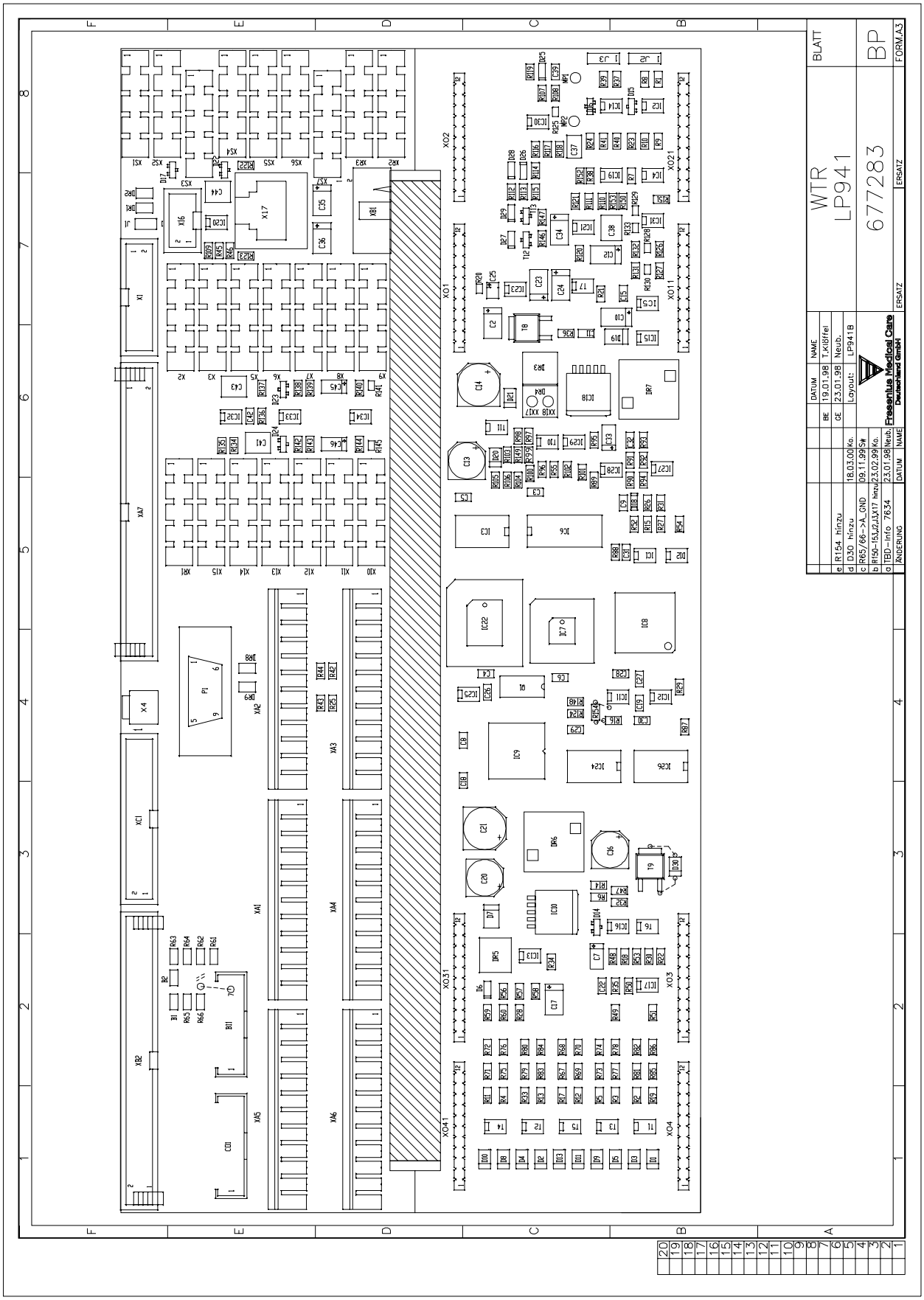
  Both the level and the conductivity circuits are supplied via line COND_C1 from the oscillator on LP 633.
  This means that these circuits can also be tested during detuning in the T1-test.

- **Reference Voltage Monitoring**

  D 12 is an additional reference voltage diode of 1.235 Volt which is connected with the analog input AN2 of the controller. U_REF and REF_ADC are compared with each other. A deviation of the voltage of REF_ADC from the nominal value by > ±2.5% is considered to be an error.
  In addition, the hardware monitors the reference value for ± 5 % via comparators IC 29a and 92b and the voltage divider, consisting of the resistors R 99, 100, 101 and 149.
  After switching on, the comparators are T1 tested.
  The transistors T 10b and 10a are alternatively switched to conducting via the signals TEST_H and TEST_L, and the upper or lower part of the voltage divider is bridged with R97 or R98. This is equivalent to a U_REF change of ± 10 %. The respective open collector output of each comparator goes to L level and the REF_FAIL signal is read in on P1.5 of IC 9.
  The reading in of the REF_ADC voltage and the testing of the comparator stage is performed only once after switching on (T1-test !!). The output of the comparator stage is continuously monitored (refer to T0 monitoring „2.2.4.3 Reference Voltage Monitoring“).

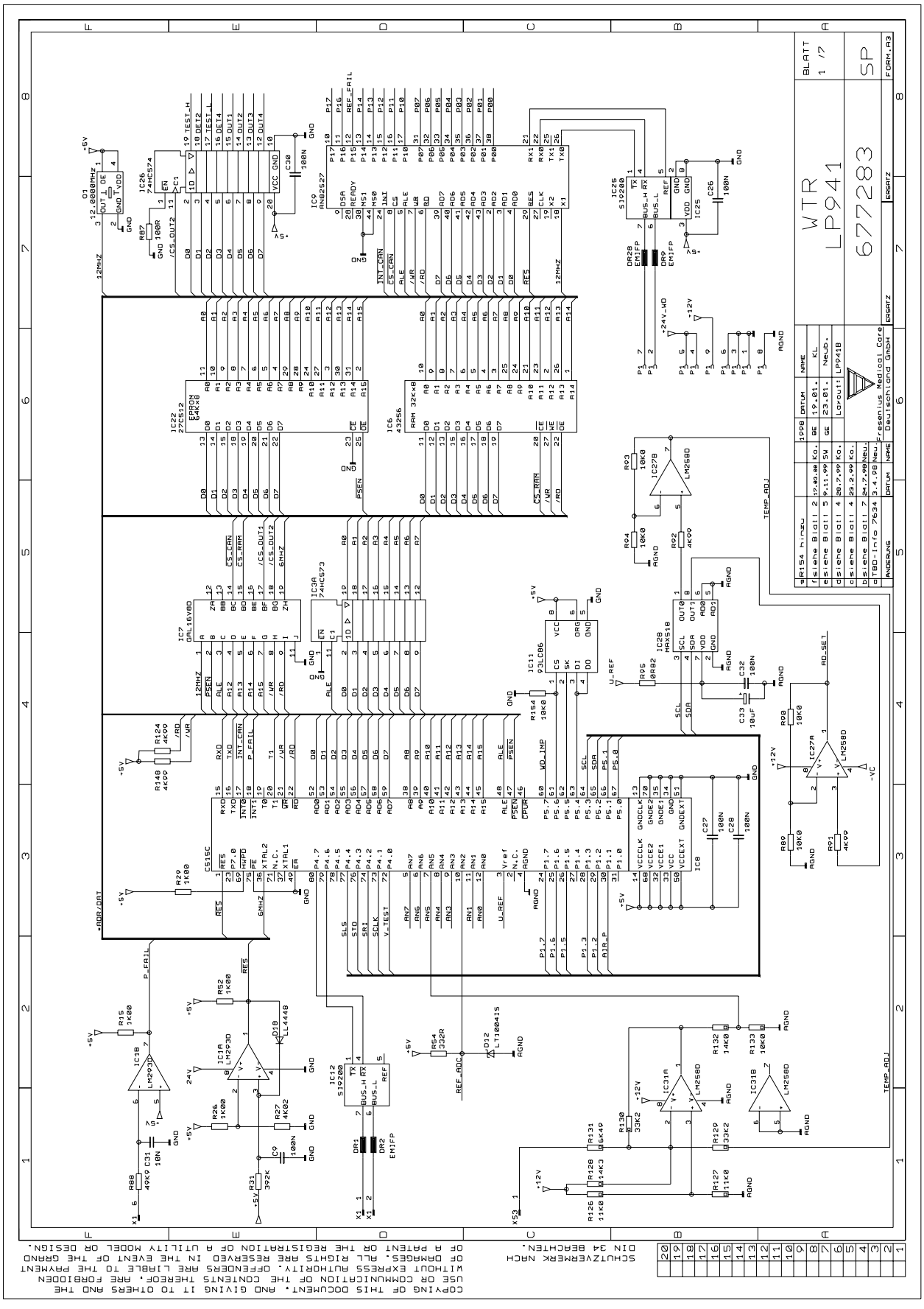
P.C.B. LP 941

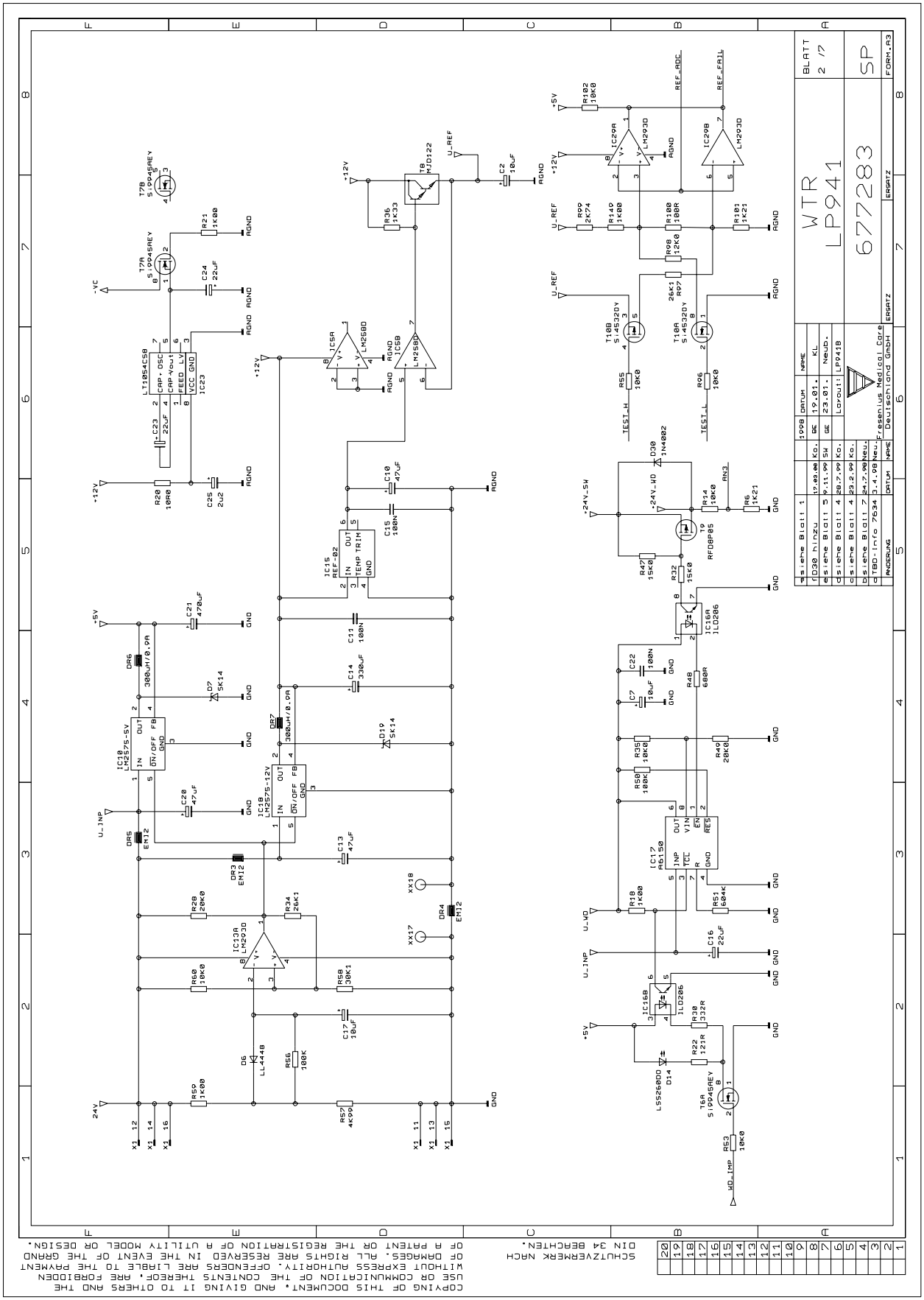
Component layout diagram

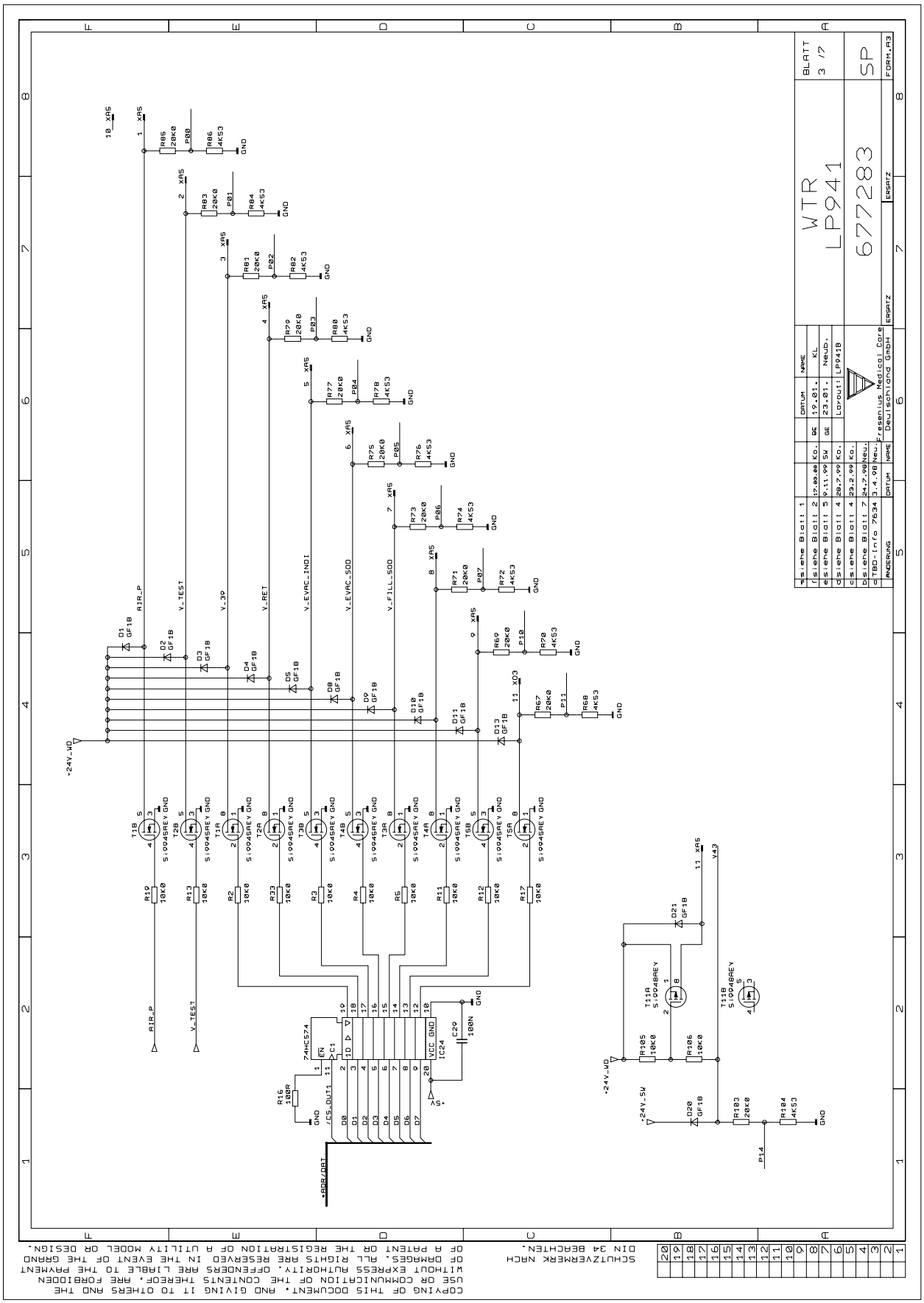


P.C.B. LP 941

Circuit diagram 1/7

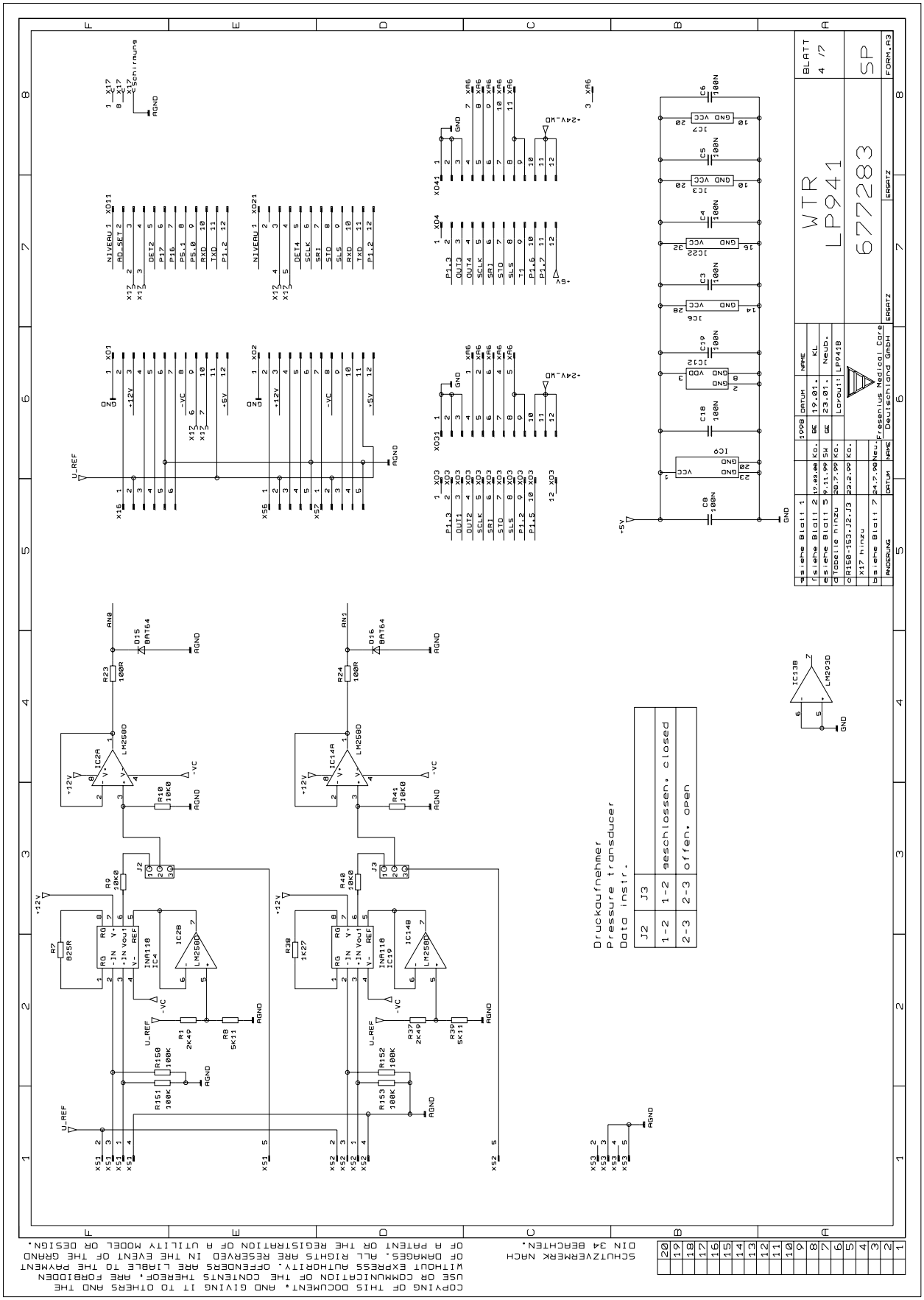


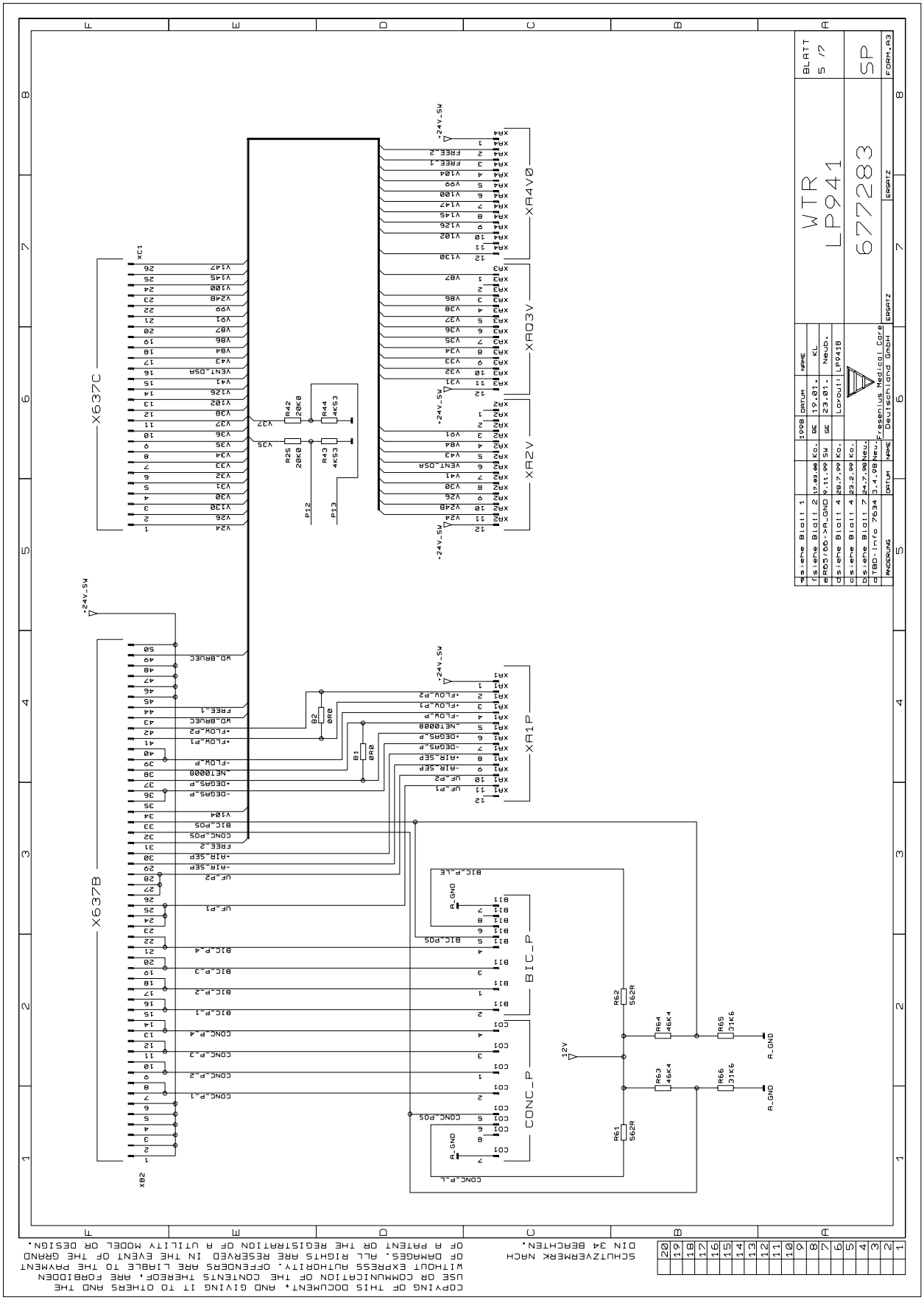

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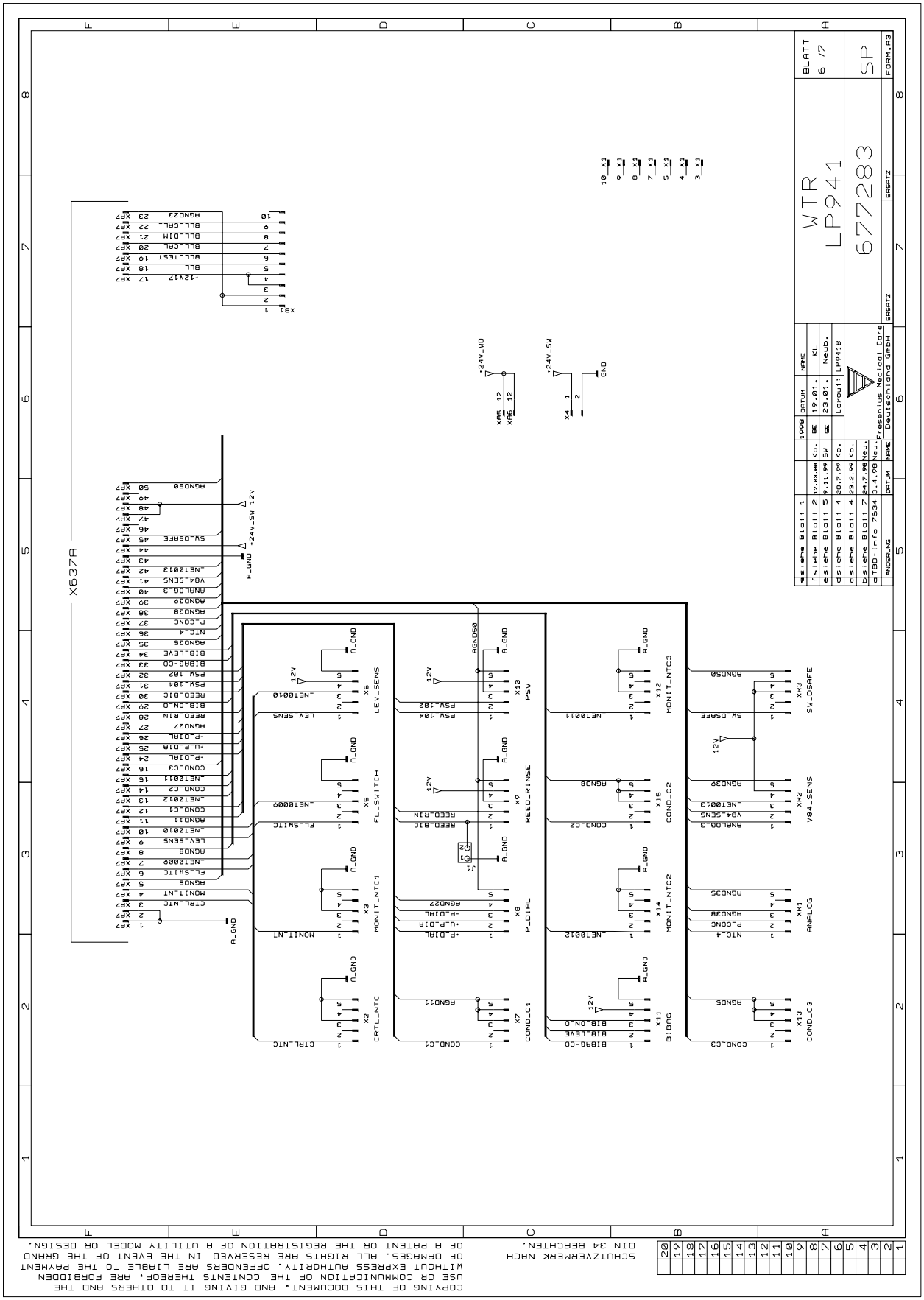
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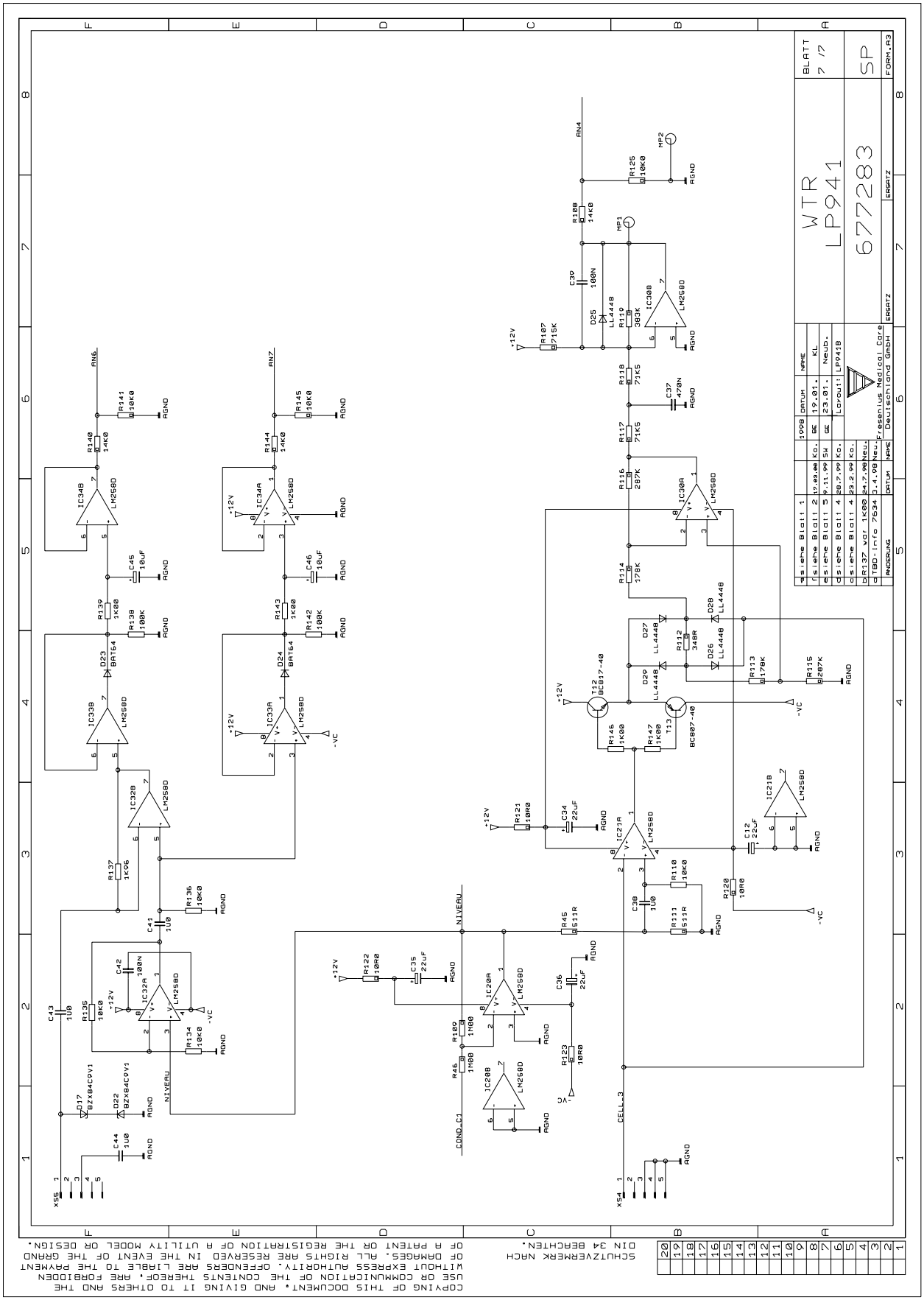

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2-3 offen. open
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